Financial Econometric Modelling

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Part I

Fundamental Concepts and Methods

Chapter 1

Financial Asset Prices and Returns

1.1 Introduction

What is financial econometrics? As pointed out by Fan (2004), this simple question is quite difficult to answer. Financial econometrics is an interdisciplinary area that integrates the fields of finance, economics, probability, statistics and applied mathematics and any attempt to give a formal definition is unlikely to be successful. Crudely speaking, therefore, financial econometrics may be regarded as the the examination and modelling of financial data using the tools provided by its constituent disciplines, with the aim of developing a deeper understanding of the way in which financial markets work. Central to this process is establishing a reliable data set for the econometric investigation. The financial data which are of primary of interest are usually the prices financial assets and particularly the yields or returns to investments in a financial assets. The first logical step in the study of financial econometrics, therefore, is to become familiar with what a financial asset is, how prices for these assets are quoted and reported and how yields or returns to the investment are constructed.

One feature of financial econometrics which differs substantial from traditional macro-econometrics is that there is hardly ever a paucity of data on which to test the hypothesis of interest. This does not mean, however, that financial data are not as prone to the problems of measurement error and revisions as are macro-econometric data. The downside of financial data is that very often a lot of work is required in order to get it into the kind of shape necessary for empirical analysis. Furthermore, these problems go far beyond the trivial ones of typographical errors and measurement error. This chapter cannot cover all the interesting data twists that an applied financial econometrician will encounter, but it will highlight some of the issues in the hope that this will stimulate an awareness of the old adage that the results are only as

good as the data.

1.2 Financial Assets

Although they have no intrinsic physical worth, financial assets derive their value from the fact that they represent a contractual claim on a stream of services or cash flows. The major categories of financial assets that will be used in this book are cash, fixed-income securities, equity securities and derivative securities.

Cash
Cash represents a claim on the stream of services that it can secure by virtue of its role as a medium of exchange. One particularly important transaction that may be regarded as a cash investment is dealing in the foreign exchange market. As the exchange rate represents the price of one currency in terms of another, trading in currencies may be regarded as investments in cash.

Fixed-Income Securities
Fixed-income securities provide a return in the form of the eventual return of principal at maturity and also interest payments at fixed, regular intervals. Although the original distinguishing feature of this class of financial asset was that the periodic payment was known in advance, recent developments have the emergence of payments linked to a short-term interest rate.
Money market fixed-income securities are short-term assets whose markets are particularly active (or liquid). There are a bewildering array of money market instruments, but only two that will feature in this book.

- *Treasury Bills* are the simplest form of government debt. The government sells Treasury Bills in the money market and redeems them at the maturity of the bill. No interest is payable during the life of the bill and so they trade at discount to the face value that will be paid at maturity. The most common maturities are 3, 6 and 9 months.

- *Eurodollar Deposits* are the deposits of United States banks which are denominated in US$ but held with banks outside the United States. Most of these deposits have a relatively short maturity (less than 6 months) and as a result the Eurodollar deposit rate is used as a representative short-term interest rate.

The *bond market* is where the longer term borrowing of governments or corporations is conducted. A bond is security which promises to pay the owner of the bond its face value at the time of the maturity of the bond and usually a coupon payment. There are also zero-coupon bonds that pay no regular interest and are therefore traded at prices that are below their face value. In recent times this distinction has become less important because zero-coupon bonds may be created from coupon paying bonds by separating the coupons from

principal and trading each of these components independently. This is process known as "stripping".
Another common way in which the fixed-income securities market is classified is by the issuer of the securities. The distinction is sometimes made between bonds issued by financial intermediates (FI bonds) and non-financial intermediaries (NFI bonds). Financial intermediaries are entities that facilitate financial transactions between two or more parties and include commercial banks, investment banks and insurance companies.

Equity Securities
Equities or common stocks give the owner an equity stake in a company and the claim on company's assets and earnings and can bought and sold on stock markets. Stocks give the owner the right to a payment which represents the distribution of some of the company's earnings, which is known as a dividend. The dividend is usually expressed as the amount each share receives or as percentage of the current market price, which is referred to as dividend yield.

Derivative Securities
Derivative securities provide a payoff based on the value(s) of other assets such as commodity, bond stock prices, so in effect they derive their value from the behaviour of their underlying assets. Derivatives started out as over-the-counter trades where interested parties made mutually beneficial trades but in recent years have been traded actively on exchanges such as the Chicago Board of Exchange, as more standardised contracts emerged.
There are two major classes of derivative securities that will be discussed in this book.

- *Options* contracts offer the buyer the right, but not the obligation, to buy (call option) or sell (put option) a financial asset at a particular price during a certain period of time or on a specific date.

- *Futures* contracts specify the delivery of either an asset or a cash value at a time known as the maturity for an agreed price which is payable at maturity. The entity who commits to purchase the asset on delivery takes a long position. The entity who commits to delivering the asset takes a short position.

Outstanding Value of Global Financial Assets
Figure 1.1 gives a snapshot of the total value of outstanding financial assets worldwide in 2010. The total value of the outstanding assets, which include the value of outstanding global stocks, government bonds and the bonds issued by financial and non-financial intermediaries, is approximately US$ 150 trillion. The stock market is the largest contributor to this total although it is interesting to note that the value of government bonds outstanding is roughly equal to the value of outstanding bonds issued by financial intermediaries. The effect of the global financial crisis from late 2007 to 2009 is clearly evident;

the value of outstanding global stocks almost halved from US$ 65 trillion in 2007 to US$ 34 trillion in 2008 and by 2010 had still not reached its 2007 peak. One of the most significancy developments in financial markets in recent years has been the growth of derivatives markets and what Figure 1.1 does not show is the fact that the outstanding value of stocks and bonds is completely dwarfed by the size of the derivatives market. The problem with measuring the size of the derivatives market stems from the fact that there is large volume of over-the-counter trade which makes it difficult to quantity exactly what the volume of derivatives trade is. The Bank for International Settlements estimates that outstanding over-the-counter derivatives amounted to US$ 707 trillion in June 2011 and estimates from the World Federation of Exchanges puts the value of exchange-traded derivatives only slightly lower than this amount. The combined outstanding value of derivatives is therefore a staggering figure which has been estimated as 20 times larger than world gross domestic product.

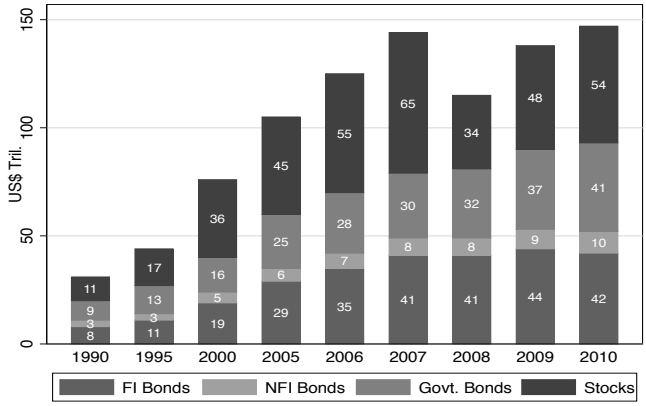


Figure 1.1: Total outstanding stock of global financial assets 2000 - 2010.
Source: McKinsey & Company

1.3 Equity Prices and Returns

The discussion of prices of financial assets and the returns to holding financial assets dealt with in this section will be couched in terms of common stocks which represent an equity claim on the company and which received a dividend payment. Prices and returns to other financial assets are determined in a similar way.

1.3.1 Prices

Arguably the fundamental type of data that financial econometrics is interested in is the price of a financial asset. The price of an equity security is de-

fined as the amount at which a transaction can occur (quoted price) or has occurred (historical price). When dealing with high-frequency data the appropriate prices are usually quoted prices. An illustration is provided in Table 1.1 of quoted prices obtained from Yahoo Finance for common stock in the United States company Boeing.

Table 1.1

Quoted prices on Yahoo Finance for The Boeing Company (BA): 12 September 2014.

The Boeing Company (BA) - NYSE			
127.64 ↓ 0.58(0.45%)			
Prev Close:	128.22	Day's Range:	127.20 - 127.99
Open	127.82	52wk Range:	109.14 - 144.57
Bid:	127.50	Volume:	1,988,616
Ask:	127.84	Market Cap:	91,98 Bil.

Source: `https://au.finance.yahoo.com`

It is clear that recording a "price" for the purposes of doing econometric analysis is not entirely straightforward as a number of alternatives are available. In addition to the previous day's closing price and the current day's opening price there are also the current bid and ask prices. The bid price is the maximum price that buyers are willing to pay for the stock and the ask price is the minimum price that sellers are willing to accept for the stock. Many studies that use intra-day data, known as high-frequency data, often involve using the midpoint of the bid and ask prices as the best estimate of the current price. This convention does, however, result in some interesting problems for the econometric analysis, which are known as issues in market microstructure.

When dealing with historical prices at lower frequencies the situation is less complex. Table 1.2 reports the historical daily prices for the United States stock Microsoft for the month of August 2014. The choice for the researcher is now between the opening price, the closing price, an average of the two and the adjusted closing price. In most cases the closing price adjusted for stock splits and dividends, Close*, is chosen.

The effect of a dividend is to lower the price by the amount of the dividend so that the closing price on 18 August is greater than the opening price on 19 August. In order to ensure that the effect of the dividend is smoothed out in historical prices, the correction is to subtract the dividend from the closing price on the previous day compute the factor $(P_{t-1} - D_t)/P_{t-1}$ and then multiply all previous prices by this factor. On the 18 August the closing price and the adjustment factor are

$$\$44.83 = 45.11 - 0.28 \quad \text{and} \quad \frac{45.11 - 0.28}{45.11} = 0.9938\,,$$

Table 1.2

Daily prices for the U.S. stock Microsoft (MSFT) for the month of August 2014. All prices are quoted in US$. The column, Close*, gives the closing price adjusted for dividends and stock splits. A dividend of US$ 0.28 per share was paid on 19 August 2014.

Date	Open	High	Low	Close	Volume	Close*
29 Aug 2014	45.09	45.44	44.86	45.43	21607600	45.43
28 Aug 2014	44.75	44.98	44.61	44.88	17657600	44.88
27 Aug 2014	44.90	45.00	44.76	44.87	20823000	44.87
26 Aug 2014	45.31	45.40	44.94	45.01	14873100	45.01
25 Aug 2014	45.40	45.44	45.04	45.17	16898100	45.17
22 Aug 2014	45.35	45.47	45.07	45.15	18294500	45.15
21 Aug 2014	44.84	45.25	44.83	45.22	22272000	45.22
20 Aug 2014	45.34	45.40	44.90	44.95	24750700	44.95
19 Aug 2014	44.97	45.34	44.83	45.33	28115600	45.33
Dividend 0.28						
18 Aug 2014	44.94	45.11	44.68	45.11	26891100	44.83
15 Aug 2014	44.58	44.90	44.40	44.79	41611300	44.51
14 Aug 2014	44.08	44.42	44.01	44.27	19313200	44.00
13 Aug 2014	43.68	44.18	43.52	44.08	22889500	43.81
12 Aug 2014	43.04	43.59	43.00	43.52	21431100	43.25
11 Aug 2014	43.26	43.45	43.02	43.20	20351600	42.93
8 Aug 2014	43.23	43.32	42.91	43.20	28942,700	42.93
7 Aug 2014	42.84	43.45	42.65	43.23	30314900	42.96
6 Aug 2014	42.74	43.17	42.21	42.74	24634000	42.47
5 Aug 2014	43.31	43.46	42.83	43.08	26266400	42.81
4 Aug 2014	42.97	43.47	42.81	43.37	34277400	43.10
1 Aug 2014	43.21	43.25	42.60	42.86	31170300	42.59

Source: `https://au.finance.yahoo.com`

respectively. As a consequence, the adjusted closing price on the 15 August is

$$\$44.51 = 44.79 \times 0.9938\,.$$

Note that the process of adjustment means that the historical prices do not necessarily reflect the actual prices at which trades took place. The adjustment process for a stock split is similar. Say, for examples, a stock stock splits 2-for-1 so that the price is suddenly half of what it used to be . To avoid this kind of discontinuity, all historical prices need to be divided by 2 and all the historical volume multiplied by 2 so that the price after the split and the price before the split are comparable.

Another problem to contend with is that a close look at the calendar days in the first column will reveal a number of missing days corresponding, in this particular instance, to weekends and public holidays. Of course there may be days other than public holidays and weekends when a stock does not trade

and this is something which needs to be guarded against. In addition, when comparing series from different countries, the public holidays do not always fall on the same days.

1.3.2 Returns

The return to a financial asset probably receives more attention in financial econometrics than does the price of an asset. Broadly speaking a financial return is a measure of the results of the decision to invest in a financial asset, a measure which accounts for the capital gain or loss due to the price change over the holding period of the asset and also the impact of the contractual stream of cash flows that take place over the course of the holding period. In principle, a financial asset can be held for any amount of time. In recent times, high-frequency data has become more readily available so that returns can be computed for most holding periods, even very short ones. Historically historical data on prices was usually at the daily, weekly or monthly frequencies and the holding period of the investment is limited to a multiple of this frequency.

Dollar Returns

The simplest possible measure of return on holding an asset for k periods between time t and $t-k$ is the dollar return, denoted $\$R_{kt}$, given by

$$\$R_{kt} = P_t - P_{t-k}\,.$$

Although this a very intuitive response to the problem of computing the return to an investment its major drawback is that it is not a scale-free measure. In other words, the measure of return depends on the unit in which prices (and dividends) are quoted. To make returns comparable across international financial markets scale-free measures of returns are required.

Simple Returns

The simple return on an asset between time $t-1$ and t is given by

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1\,. \tag{1.1}$$

The relative price ratio P_t/P_{t-1} also known as the price relative (or prel) for short is a useful quantity to compute. If the ratio is greater than 1 then returns are positive and if it is less than 1 returns are negative. Equation 1.1 may be rearranged as

$$1 + R_t = \frac{P_t}{P_{t-1}}\,,$$

in which $1 + R_t$ is known as the simple gross return. The usefulness of the simple gross return is that it represents the value at time t of investing of \$1 at time $t-1$.

The return to holding the asset for k periods, $R_t(k)$, is given by

$$\begin{aligned} R_t(k) &= \frac{P_t}{P_{t-k}} - 1 \\ &= \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}} \times \cdots \times \frac{P_{t-k+2}}{P_{t-k+1}} \times \frac{P_{t-k+1}}{P_{t-k}} - 1 \\ &= (1 + R_t) \times (1 + R_{t-1}) \times \cdots \times (1 + R_{t-k+2}) \times (1 + R_{t-k+1}) - 1 \\ &= \prod_{j=0}^{k-1} (1 + R_{t-j}) - 1 \,. \end{aligned} \tag{1.2}$$

The important result to be emphasised is that simple returns are not additive when computing multi-period returns.
If the data frequency is monthly, then the simple return for a holding period of one year is given by

$$R_t(12) = \left[\prod_{j=0}^{11} (1 + R_{t-j}) \right] - 1 \,. \tag{1.3}$$

The most common period over which a return is quoted is one year and returns data are commonly presented in per annum terms. This means that the current monthly return needs to be scaled so that it is interpretable as an annual return, that is expressed on a per annum basis. In the case of monthly returns, the associated annualised simple return is computed as

$$\text{Annualised } R_t(12) = (1 + R_t)^{12} - 1 \,. \tag{1.4}$$

The expression (1.4) is obtained from (1.3) by making the assumption that the best guess of the per annum return is that the current monthly return will persist for the next 12 months. In this case, all the terms in the product expansion (square brackets) of equation (1.3) will be identical.

Log Returns

The log return of an asset is defined as

$$r_t = \log(1 + R_t) = \log P_t - \log P_{t-1} \,. \tag{1.5}$$

Log returns are also referred to as continuously compounded returns. To understand why this is so use will be made of Euler's number which is defined as

$$e \equiv \lim_{s \to \infty} \left(1 + \frac{1}{s}\right)^s \approx 2.71828 \,.$$

This definition was used the Swiss mathematician Jacob Bernoulli (1655-1705) to study the effect of compound interest. The formula represents the value of an account at the end of the year which started with \$1.00 and paid 100%

interest per year but with the interest compounded continuously instead of at discrete intervals.
If m is the compounding period and r_t the return, then it follows that

$$P_t = P_{t-1}\Big(1 + \frac{r_t}{m}\Big)^m,$$

and continuous compounding is the case in which $m \to \infty$

$$P_t = P_{t-1} \lim_{m\to\infty} \Big(1 + \frac{r_t}{m}\Big)^m. \tag{1.6}$$

Let $s = m/r_t$, then the expression in (1.6) is rewritten as

$$\begin{aligned} P_t &= P_{t-1} \lim_{s\to\infty} \Big[\Big(1 + \frac{1}{s}\Big)^{s\,r_t}\Big] \\ &= P_{t-1}\Big[\lim_{s\to\infty} \Big(1 + \frac{1}{s}\Big)^{s}\Big]^{r_t} \\ &= P_{t-1}e^{r_t}. \end{aligned} \tag{1.7}$$

Taking logarithms of expression (1.7) yields the definition of the log returns given in equation (1.5).
Log returns are particularly useful because of the simplification they allow in dealing with multi-period returns.For example, the 2-period return is given by

$$\begin{aligned} r_t(2) &= \log P_t - \log P_{t-2} \\ &= (\log P_t - \log P_{t-1}) + (\log P_{t-1} - \log P_{t-2}) \\ &= r_t + r_{t-1}\,, \end{aligned} \tag{1.8}$$

so that, by extension, the k-period return is

$$\begin{aligned} r_t(k) &= \log P_t - \log P_{t-k} \\ &= (\log P_t - \log P_{t-1}) + (\log P_{t-1} - \log P_{t-2}) + \cdots + (\log P_{t-k+1} - \log P_{t-k}) \\ &= r_t + r_{t-1} + \cdots + r_{t-(k-1)} \\ &= \sum_{j=0}^{k-1} r_{t-j}\,, \end{aligned} \tag{1.9}$$

In other words, the n-period log return is simply the sum of the single period log returns over the pertinent period.
For the case of data observed monthly, the annual log return is

$$r_t(12) = \log P_t - \log P_{t-12} = \sum_{j=0}^{11} r_{t-j}\,. \tag{1.10}$$

Once again, expression (1.9) may be used to obtain the returns expressed on a per annum basis by simply multiplying all monthly returns by 12, making the

implicit assumption being that the best guess of the per annum return is that the current monthly return will persist for the next 12 months.
By analogy, if prices are observed quarterly, then the individual quarterly returns can be annualised by multiplying the quarterly returns by 4. Similarly, if prices are observed daily, then the daily returns are annualised by multiplying the daily returns by the number of trading days 252. The choice of 252 for the number of trading days is an approximation as a result of holidays and leap years etc. Other choices are 250 and, very rarely, the number of calendar days, 365, is used.

Table 1.3

Monthly prices for the U.S. stock Microsoft for the years 2012 and 2013. Also shown are alternative measures of the one-month return to holding Microsoft. Prices are month-end closing prices adjusted for splits and dividends quoted in US$.

Date	Price	Prel	Monthy Dollar Return	Monthly Simple Return	Monthly Log Return	Annual Simple Return	Annual Log Return
Jan 2012	29.530	...	...	...	...	...	...
Feb 2012	31.740	1.075	2.210	0.075	0.072	1.378	0.866
Mar 2012	32.250	1.016	0.510	0.016	0.016	0.211	0.191
Apr 2012	32.020	0.993	−0.230	−0.007	−0.007	−0.082	−0.086
May 2012	29.190	0.912	−2.830	−0.088	−0.093	−0.671	−1.110
Jun 2012	30.590	1.048	1.400	0.048	0.047	0.754	0.562
Jul 2012	29.470	0.963	−1.120	−0.037	−0.037	−0.361	−0.448
Aug 2012	30.820	1.046	1.350	0.046	0.045	0.712	0.537
Sep 2012	29.780	0.966	−1.040	−0.034	−0.034	−0.338	−0.412
Oct 2012	28.530	0.958	−1.245	−0.042	−0.043	−0.401	−0.512
Nov 2012	26.620	0.933	−1.915	−0.067	−0.069	−0.566	−0.834
Dec 2012	26.730	1.004	0.110	0.004	0.004	0.051	0.049
Jan 2013	27.470	1.028	0.740	0.028	0.027	0.388	0.328
Feb 2013	27.800	1.012	0.330	0.012	0.012	0.154	0.143
Mar 2013	28.610	1.029	0.810	0.029	0.029	0.411	0.345
Apr 2013	33.100	1.157	4.490	0.157	0.146	4.751	1.749
May 2013	34.880	1.054	1.780	0.054	0.052	0.875	0.629
Jun 2013	34.530	0.990	−0.350	−0.010	−0.010	−0.114	−0.121
Jul 2013	31.830	0.922	−2.700	−0.078	−0.081	−0.624	−0.977
Aug 2013	33.400	1.049	1.570	0.049	0.048	0.782	0.578
Sep 2013	33.310	1.097	−0.090	−0.003	−0.003	−0.032	−0.032
Oct 2013	35.350	1.061	2.040	0.061	0.059	1.041	0.713
Nov 2013	38.130	1.079	2.780	0.079	0.076	1.480	0.908
Dec 2013	37.430	0.982	−0.700	−0.018	−0.019	−0.199	−0.222

Source: Bloomberg.

Table 1.3 demonstrates some calculations based on historical monthly prices for the United States stock Microsoft to provide examples of the mechanics

of computing returns from the price of a stock. Note that no returns figures are reported for January 2012. This emphasises that an observation is lost at the beginning of the sample when computing returns because the price of the stock before the start of the sample period is not available. The monthly dollar, simple and log returns to Microsoft for February 2012 are respectively

$$\begin{aligned} \$R_t &= 31.74 - 29.53 = \$2.210, \\ R_t &= \frac{31.74 - 29.53}{29.53} = 0.075 = 7.5\%, \\ r_t &= \log(1 + 0.075) = 0.072 = 7.2\%. \end{aligned}$$

This demonstrates that continuously compounded returns are very similar to simple returns as long as the return is relatively small, which it generally will be for monthly or daily returns. Indeed, it is only really at the third decimal place that the differences between the two definitions of returns become readily apparent.
Despite the similarities in the two measures of returns, appreciable differences emerge when the returns are annualised. For the simple return in February 2012 the calculation is

$$R_t(12) = (1 + 0.075)^{12} - 1 = 1.378 = 137.8\%\,.$$

By contrast the annualised log return is

$$r_t(12) = 12 \times 0.072 = 0.866 = 86.6\%\,.$$

Note that the practise of quoting figures as annual rates is usually related to scaling the data. Returns, when computed over the short time intervals of a day and even shorter, can be relatively small in value and this may lead to arithmetic errors when doing complex computations involving the returns. Annualising the returns scales can help to alleviate this problem.

Dealing with Dividends

Adjusting the computation of returns for the payment of a dividend, D_t, between time $t - 1$ and t, is relatively straightforward. The dollar return becomes

$$\$R_t = P_t + D_t - P_{t-1}\,,$$

in which P_t and P_{t-1} are the unadjusted prices. The simple and gross returns are then given by

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}} - 1, \tag{1.11}$$

$$(1 + R_t) = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}\,, \tag{1.12}$$

respectively. It is apparent from (1.11) and (1.12) that the simple and gross returns to a stock in the presence of a dividend payment are easily computed in terms of the price relative and the dividend yield.
Adjusting log returns for a dividend payment simply requires using the correct definition of gross simple returns when taking logarithms

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t + D_t - P_{t-1}}{P_{t-1}}\right) = \log\left(\frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}\right).$$

Most of the discussion relating to the computation of returns has reflected common practise and ignored the issue of dividends. This practise stems from dividends being paid relatively infrequently and constituting a minor proportion of the return relative to price movements.

Excess Returns

The difference between the return on a risky financial asset and a risk free interest rate, denoted r_{ft}, and usually taken be the interest rate on a government bond, is known as the excess return. The simple and log excess returns on an asset are therefore defined, respectively, as

$$Z_t = R_t - r_{ft}, \qquad z_t = r_t - r_{ft}\,. \tag{1.13}$$

In computing the excess returns it is important to ensure that the risk-free interest rate is expressed in same unit of time as the return on the risky financial asset. For example, interest rates are normally quoted as annual rates so in the case of monthly returns the quoted annual risk-free interest rate would need to be divided by 12.

1.3.3 Portfolio Returns

Very often in financial econometrics it is not the return to a single asset that is the object of the investigation but rather the return to a portfolio of financial assets. In order to deal with this problem the aggregation of the returns of the assets in the portfolio needs to be addressed.
Consider a portfolio with only two assets with portfolio shares w_1 and w_2 respectively. The portfolio shares represent the fraction of the total portfolio value allocated to each of the assets with the normalisation condition

$$w_1 + w_2 = 1.$$

Using the definition of simple gross returns for each asset, the value of the portfolio between $t-1$ and t may be calculated as

$$P_t = P_{t-1}w_1(1 + R_{1t}) + P_{t-1}w_2(1 + R_{2t}) = P_{t-1}\big(w_1(1 + R_{1t}) + w_2(1 + R_{2t})\big).$$

Rearranging slightly this expression becomes

$$(1 + R_{Pt}) \equiv \frac{P_t}{P_{t-1}} = w_1(1 + R_{1t}) + w_2(1 + R_{2t}), \tag{1.14}$$

or, in other words, the one-period gross return to a portfolio, $1 + R_{Pt}$, is given by the sum of the gross returns to each of the assets. Expanding the right-hand side of equation (1.14) gives

$$1 + R_{Pt} = w_1 + w_1 R_{1t} + w_2 + w_2 R_{2t},$$

which yields the important result that for simple returns, the portfolio rate of return is equal to weighted average of the returns to the assets

$$R_{Pt} = w_1 R_{1t} + w_2 R_{2t}\,,$$

so long as $w_1 + w_2 = 1$. For N assets the simple portfolio return is given by

$$R_{Pt} = \sum_{i=1}^{N} w_i R_{it}, \qquad \sum_{i=1}^{N} w_i = 1. \tag{1.15}$$

This result does not extend to the case of log returns. From equation (1.5) and using the result in (1.15) it follows that

$$r_{Pt} = \log(1 + R_{Pt}) = \log\left(1 + \sum_{i=1}^{N} w_i R_{it}\right) \neq \sum_{i=1}^{N} w_i r_{it}\,. \tag{1.16}$$

In most practical situations the fact that the log return to the portfolio is not the weighted sum of the log returns to the constituent assets is simply ignored. This is acceptable when the log returns are small, as is likely for short holding periods, in which case log return on the portfolio is negligibly different to the weighted sum of the logarithm of the constituent asset returns because $r_{Pt} = \log(1 + R_{Pt}) \approx R_{Pt}$.
The result in equation (1.16) then begs the question as to exactly how log returns may be combined to give the portfolio return. Consider again the case of two assets. Using the definition of log returns for each asset and expression (1.7), the value of the portfolio between $t-1$ and t may be calculated as

$$P_t = P_{t-1}\, w_1\, e^{r_{1t}} + P_{t-1}\, w_2\, e^{r_{2t}}\,,$$

so that

$$\log\left(\frac{P_t}{P_{t-1}}\right) \equiv r_{Pt} = \log\left(w_1\, e^{r_{1t}} + w_2\, e^{r_{2t}}\right).$$

For N assets the log portfolio return is then

$$r_{Pt} = \log\Big(\sum_{i=1}^{N} w_i\, e^{r_{it}}\Big). \tag{1.17}$$

More often than not, financial econometric studies use log returns and simply aggregate these returns in terms of a weighted average to obtain portfolio returns. This approach will also be used in Chapter 3 where simple portfolios are constructed using linear regression. Strictly speaking, the results of

this section show that this procedure is not correct. Once returns, either simple of log returns, are available then equations (1.2) and (1.9) may be used for temporal aggregation of the portfolio returns. The situation is summarised in Table 1.4.

Table 1.4

Summary of expressions for computing portfolio returns using simple and log returns and how to aggregate portfolio returns to obtain the k period portfolio return.

Aggregation	Simple Returns	Log Returns
Portfolio Return	$R_{Pt} = \sum_{i=1}^{N} w_i R_{it}$	$r_{Pt} = \log\left(\sum_{i=1}^{N} w_i \, e^{r_{it}}\right)$
K-Period Return	$R_{Pt}(k) = \prod_{i=0}^{k-1}(1 + R_{Pt-i}) - 1$	$r_{Pt}(k) = \sum_{i=0}^{k-1} r_{Pt-i}$

1.4 Stock Market Indices

A problem of particular importance is the return to a portfolio that comprises all or at least a selection of prominent stocks on a stock exchange. An aggregate summary measure of the performance of the stock market as whole is known as a stock market index. Indices are nothing more than a selection of (a large number of) stocks that are combined in a particular way to create a portfolio. The index then represents the value of the portfolio and is expressed in terms of an average price that has been normalised in some way. Because stock market indices are price indices, the computation of returns to the index can be performed in exactly the same way as if it were a single stock.

The major stock market indices are constructed in one of two ways. *Price-weighted indices* construct a portfolio of all the stocks in the index in which one share of each of the stocks in the index. In other words the total monetary value invested in each share is only proportional to the price of that share. *Value-weighted indices* construct a portfolio of all the stocks in the index in which the weight given to each stock is proportional to the total market value of its outstanding equity.

The six indices used most often in financial econometric work are plotted in Figure 1.2.

- *Deutscher Aktien Index* (DAX) comprises the 30 largest German companies that trade on the Frankfurt Stock Exchange. It is a value weighted-index although the weights are computed in a slightly more complex way than in a simple value weighting scheme.

- *Dow Jones Industrial Average Index* (DJIA) computed using 30 prominent United State corporations. The DJIA is a price-weighted index.
- *Financial Times Stock Exchange 100 Index* (FTSE) is a value-weighted index computed using the 100 largest companies listed on the London Stock Exchange.
- *Hang Seng Index* (HSX) comprises 40 of the largest companies that trade on the Hong Kong Exchange. It is a value-weighted index.
- *Nikkei 225 Index* (Nikkei or NKX) is a price-weighted index made up of 225 prominent companies listed on the Tokyo Stock Exchange.
- *Standard and Poors Composite 500* (S&P 500) is a market-value weighted index. The index is computed by summing the market value of the outstanding equity in each firm in the index.

The falls in the indices around the collapse of the dot-com bubble in the early 2000s and the global financial crisis of 2008-2009 are evident.

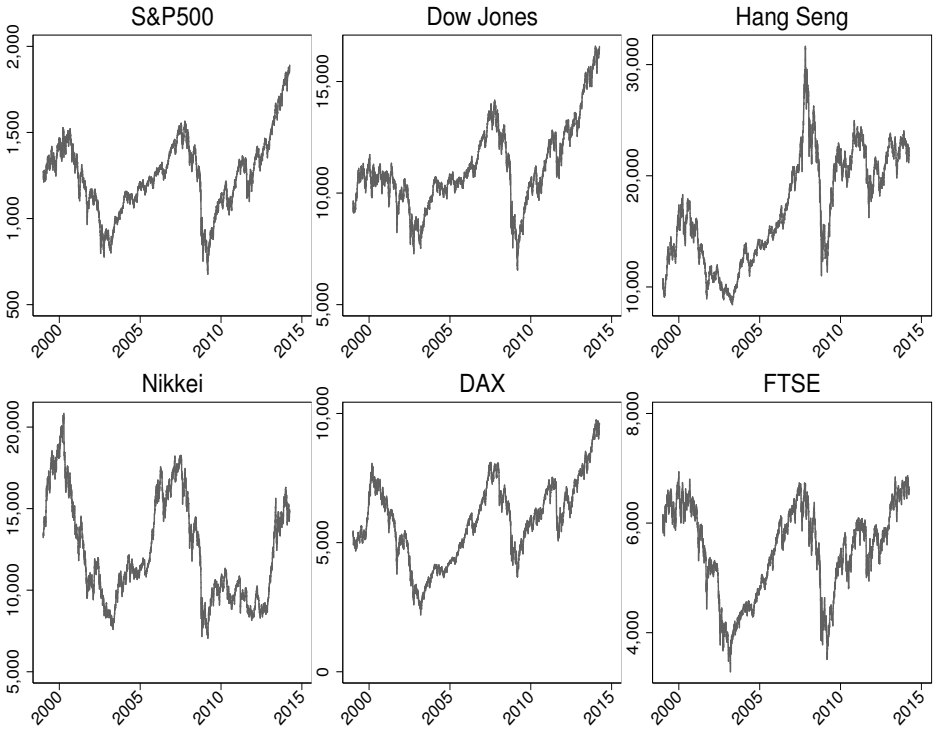


Figure 1.2: Daily observations on six international stock market indices for the period 4 January 1999 to 2 April 2014.

In addition to these six indices, another commonly encountered index is the *NASDAQ Composite Index*, which is a market-value weighted index of all the stocks listed on the NASDAQ stock exchange. It usually regarded as an index of the performance of technology companies and is particularly associated with the dot-com bubble of the early 2000s.

Table 1.5

The 30 United States stocks used in the construction of the Dow Jones Index. Month-end closing prices adjusted for splits and dividends and quoted in US$ are shown for the month of December 2013 together with total outstanding value of the company's shares (US$B).

Company	Ticker	Closing Price (Dec.2013)	Market Cap. (Dec.2013)	Mkt Cap. Share
3M Co.	MMM	140.250	93.300	0.020
American Express Co.	AXP	90.730	97.196	0.021
AT&T Inc.	T	35.150	97.196	0.021
The Boeing Co.	BA	136.490	102.566	0.022
Caterpillar Inc.	CAT	90.810	57.787	0.012
Chevron Corp.	CVX	124.910	240.224	0.051
Cisco Systems Inc.	CSCO	22.450	120.032	0.025
The Coca-Cola Co./	KO	41.310	182.422	0.039
EI du Pont de Nemours & Co.	DD	64.970	60.169	0.013
Exxon Mobil Corp.	XOM	101.200	442.094	0.094
General Electric Co.	GE	28.030	283.590	0.060
The Goldman Sachs Group Inc.	GS	177.260	83.353	0.018
The Home Depot Inc.	HD	82.340	115.953	0.025
Intel Corp.	INTC	25.960	129.047	0.027
International Business Machine	IBM	187.570	203.674	0.043
Johnson & Johnson	JNJ	91.590	258.415	0.055
JPMorgan Chase & Co.	JPM	58.480	219.837	0.047
McDonaldś Corp.	MCD	97.030	96.548	0.020
Merck & Co. Inc.	MRK	50.050	146.242	0.031
Microsoft Corp.	MSFT	37.430	312.464	0.066
NIKE Inc.	NKE	78.640	69.955	0.015
Pfizer Inc.	PFE	30.630	198.515	0.042
The Procter & Gamble Co.	PG	81.410	221.291	0.047
The Travelers Companies Inc.	TRV	90.540	32.963	0.007
United Technologies Corp.	UTX	113.800	104.421	0.022
United Health Group Inc.	UNH	75.300	75.809	0.016
Verizon Communications Inc.	VZ	49.140	140.626	0.030
Visa Inc.	V	222.680	141.756	0.030
Wal-Mart Stores Inc.	WMT	78.690	254.623	0.054
The Walt Disney Co.	DIS	76.400	134.256	0.028

Source: Bloomberg.

Table 1.5 lists the 30 component stocks of the Dow Jones Index obtained from Bloomberg in September 2014. The monthly closing price for December 2013 is also listed together with the market capitalisation (US$ bill.) of the component stocks (price of share $\times$ number of outstanding shares). Despite the fact that the DJIA is a price-weighted index, Table 1.5 also shows the notional share that each stock would have in a value-weighted index.
The DJIA is computed as

$$DJIA_t = \frac{1}{D}\sum_{j=1}^{30} P_{jt}\,.$$

where, D, which is known as the Dow Jones divisor. The divisor started out as the number of stocks in the index so the DJIA was a simple average, but subsequent adjustment due to stock splits and structural changes required the divisor to be adjusted in order to preserve the continuity of the index. The appropriate value of the divisor in December 2013 was 0.15571590501117 so that the DJIA is now larger than the sum of the prices of the components. Using the prices in Table 1.5, the DJIA for December 2013 is computed as

$$\begin{aligned}\text{DJIA}_{\text{Dec13}} &= \frac{140.25 + 90.73 + 35.15 + \cdots + 222.68 + 78.69 + 76.40}{0.15571590501117}\\ &= \frac{2581.25}{0.15571590501117}\\ &= 16576.662\,,\end{aligned}$$

which is identical to the value of the index, 16576.66, quoted by Bloomberg for December 2013. The DJIA is a price-weighted average. The main advantage of price weighting is its simplicity but its primary disadvantage is that stocks with the highest prices, like Visa ($222.68), IBM ($187.57) and Goldman Sachs ($177.26), have perhaps a greater relative impact on the index than perhaps they should have.
The other major type of weighting scheme employed is to weight the stocks by market capitalisation. As a consequence, stocks like Exxon (0.094), Microsoft (0.066) and General Electric (0.060), would have the largest weights in the index if it were value-weighted. The primary disadvantage of value weighting is that constituent securities whose prices have risen the most (or fallen the most) have a greater (or lower) weight in the index. This weighting method can potentially lead to overweighting stocks that have risen in price (and may be overvalued) and underweighting stocks that have declined in price (and may be undervalued).
The differences between price weighting and value weighting are illustrated in Figure 1.3 in which the 30 constituent stocks of the Dow Jones are combined to form two hypothetical indices, one based on simple price weighting and the other using shares constructed from market capitalisation as shown in Table 1.5. Both indices are normalised to take the value 100 in January 1990. While the price-weighted and value-weighted indices track each other fairly

closely over the period, the price-weighted index seems to over-emphasise both the dot-com bubble and also the speed of the recovery from the global financial crisis.

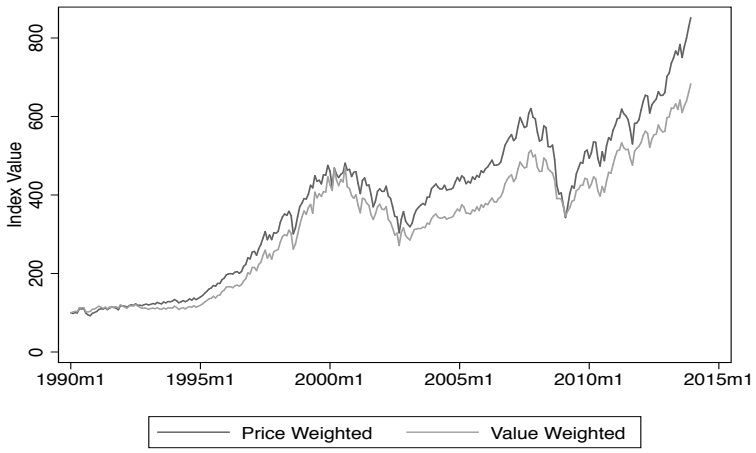


Figure 1.3: The effect of price weighting and value weighting on an index comprising 30 stocks that make up the Dow Jones Industrial Average. Index is computed using monthly data on prices and market capitalisation for the period January 1990 to December 2013 and scaled to start from 100.

1.5 Bond Yields

As noted in Section 1.2, zero-coupon bonds may be created from coupon paying bonds by separating the coupons from the principal and trading each of these components independently in a process known as "stripping". Consequently much of the econometric analysis of the bond markets uses data based on zero-coupon bonds. The critical concept when dealing with bonds, which relates to the return on a stock, is the yield to maturity. If a zero coupon bond has a face value of \$1 paid at maturity, n, the price of the bond purchased at time t equals the discounted present value of the principal, given by

$$P_{nt} = 1 \times \exp\left(-n y_{nt}\right) , \tag{1.18}$$

in which y_{nt} is the discount rate also known as the yield, commonly expressed in per annum terms. The yield on a bond is therefore the discount rate that equates the present value of the bond's face value to its price. Taking natural logarithms and rearranging equation (**??**) to gives

$$y_{n,t} = -\frac{1}{n} p_{n,t} . \tag{1.19}$$

This expression shows that the yield is inversely proportional to the natural logarithm of the price of the bond, where the proportionality constant is $-1/n$. Moreover as the price of the bond P_{nt} is always less than \$1 then from the properties of logarithms, p_{nt} is a negative number and the yield in equation (1.19) will always be positive.
Governments issue bonds of differing lengths to maturity. Bonds at the shorter end of the maturity spectrum (maturity less than 12 months) are generally zero-coupon bonds, while the coupon bonds can have a maturity as long as 30 years. The *term structure* of interest rates is the relationship between time to maturity and yield to maturity and the *yield curve* is a plot of the term structure of yield to maturity against time to maturity at a specific time. Figure 1.4 presents scatter plots of observed United States zero-coupon bond yield curves for the months of March, May, July and August 1989, for yields yields ranging from 1 to 120 months. The yields are computed from the end-of-month price quotes taken from the CRSP government bonds files and is the data used by Diebold and Li (2006).

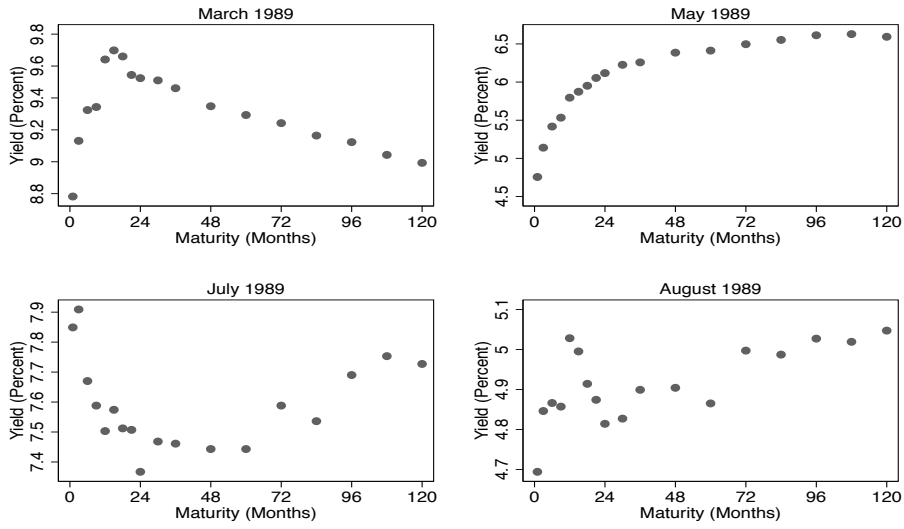


Figure 1.4: Observed yield curves for the months of March, May, July and August 1989 for United States zero coupon bonds. The data are taken from CRSP government bonds files and is the data used by Diebold and Li (2006)

The plots of the yield curve in Figure 1.4 reveal a few well-known features.

1. At any one time when the yield curve is observed, all the maturities may not be represented. This is particularly true at longer maturities where the number of observed yields is much sparser than at the short end of the maturity spectrum.

2. The yields at longer maturities tend to be less volatile than the yields at the shorter end of the maturity spectrum.

3. Based on the assumption that longer-term financing should carry a risk premium, the expectation may be that the yield curve slopes upward. shows however that the yield curve assumes a variety of shapes, including upward sloping, downward sloping, humped and even inverted humped.

Modelling bond yields and the term structure are important problems in financial econometrics and various aspects relating to the modelling of bond yields will be addressed in Chapters 6, 9 and 14.

1.6 Exercises

1. **Equity Prices, Dividends and Returns**

   (a) Log on to Yahoo Finance (`https://au.finance.yahoo.com`) and load the current quoted prices for for The Boeing Company (BA). Compare the current situation with that reported in Table 1.1.

   (b) Observe the historical daily prices for Boeing. What do you notice about the order in which they are presented.

   (c) Examine the daily prices and scrutinise the days on which dividend payments are made. Verify that the dividend adjustments made to the historical price series are correct.

   (d) Obtain monthly price data on Boeing. Are the quoted monthly prices beginning or end of month quotes?

2. **Simple and Logarthmic Returns**

   capm.wf1, capm.dta, capm.xlsx

   The data are monthly prices on five United States stocks and the commodity gold for the period April 1990 to July 2004.

   (a) Plot the price indices and comment on the results.

   (b) Compute simple and logarithmic returns to each of the assets. For each asset plot the two returns series and comment on any differences.

   (c) Compute the simple and logarithmic returns to each of the assets over the entire sample period and comment on the difference.

(d) Assume that you hold each of the stocks in a portfolio. Compute the portfolio returns in both simple and logarithmic form for the first seven months of 2004.

3. **Returns**

DJindexstocks.wf1, DJindexstocks.dta, DJindexstocks.xlsx

(a) Consider the historical prices for Microsoft for the years 2012 and 2013. For these two years, compute the price relative, simple and logarithmic monthly returns, and simple and logarithmic annualised returns. Compare your results with Table 1.3.

(b) Compute the logarithmic and simple returns to holding each of the 30 stocks in the Dow Jones for the month of December 2012.

(c) Assuming equal shares compute the simple and logarithmic returns to holding a portfolio comprising each of the 30 Dow Jones stocks for the month of December 2012.

4. **Stock Indices**

stockindices.wf1, stockindices.dta, stockindices.xlsx

The data are daily observations on the Dow Jones, SP500, Hang Seng, Nikkei, Dax and FTSE stock indices for the period 4 January 1999 to 2 April 2014.

(a) Plot the indices. Compare your results with Figure 1.2.

(b) Compute the daily logarithmic and simple returns of each of the indices and plot them. Comment on any differences.

(c) Express the daily logarithmic and simple returns in annualised form and plot the resultant series. Comment on your results.

(d) Compute the returns to holding each of the indices over the entire sample period in both logarithmic and simple form. Comment on the results.

5. **Dow Jones Index**

DJindexstocks.wf1, DJindexstocks.dta, DJindexstocks.xlsx

The data file contains the prices and market capitalisation of 30 stocks which made up the Dow Jones Industrial Average in September 2014.

(a) Compute the Dow Jones Industrial Average for December 2013 using

$$DJIA_t = \frac{1}{D}\sum_{j=1}^{30} P_{jt}\,.$$

where the Dow Jones divisor, D, is taken to be 0.15571590501117. Verify that your result is identical to quoted value of the Dow for that month.

(b) Construct portfolio shares for each of the Dow Jones stocks based on market capitalisation for the month of December 2013. Comment on the which stocks receive the most weight in the Dow under the price and market capitalisation weighting schemes, respectively.

(c) Combine the 30 constituent stocks of the Dow Jones to form two indices, one based on simple price weighting and the other using shares constructed from market capitalisation. Plot the indices over the sample period and comment on the differences.

6. **Australian Stocks**

AusFirms.wf1, AusFirms.dta, AusFirms.xlsx

The data are monthly observations on the prices of the largest 136 stocks in Australia from December 1999 to June 2014. Consider a portfolio constructed by holding one share in every one of the N stocks in the dataset that records a price, P_{jt} for at every time t in the sample period.

(a) Compute the simple and log returns to the portfolio over the sample period using the formulae

$$R(P) = \frac{P_T}{P_1} - 1\,, \qquad r(P) = \log\left(\frac{P_T}{P_1}\right),$$

in which

$$P_t = \sum_{j=1}^{N} P_{jt}$$

Comment on the results.

(b) Compute the portfolio weights of each stock in the portfolio for every time t using the formula

$$w_{it} = \frac{P_{it}}{\sum_{i=1}^{N} P_{it}}\,,$$

in which N is the number of stocks in the portfolio.

(c) Compute the simple return and log returns to the portfolio in each time period, respectively,

$$R_{Pt} = \sum_{i=1}^{N} w_{i\,t-1} R_{it}, \qquad r_{Pt} = \log\Big(\sum_{i=1}^{N} w_{i\,t-1}\, e^{r_{it}}\Big),$$

remembering to use the weight at the beginning of the holding period.

(d) Compare the results obtained in (a) and (c).

Chapter 2

Properties of Financial Data

2.1 Introduction

The financial pages of newspapers and magazines, online financial sites, and academic journals all routinely report a plethora of financial statistics. Even within a specific financial market, the data may be recorded at different observation frequencies and the same data may be presented in various ways. As will be seen, the time series based on these representations have very different statistical properties and reveal different features of the underlying phenomena relating to both long run and short run behaviour. The characteristics of financial data may also differ across markets. For example, there is no reason to expect that equity markets behave the same way as currency markets, or for commodity markets to behave the same way as bond markets. In some cases, like currency markets, trading is a nearly continuous activity, while other markets open and close in a regulated manner according to specific times and days. Options markets have their own special characteristics and offer a wide and growing range of financial instruments that relate to other financial assets and markets.

One important preliminary role of statistical analysis is to find stylised facts that characterise different types of financial data and particular markets. Such analysis is primarily descriptive and helps us to understand the prominent features of the data and the differences that can arise from basic elements like varying the sampling frequency and implementing various transformations. Accordingly, the primary aim of this chapter is to highlight the main characteristics of financial data and establish a set of stylised facts for financial time series. These characteristics will be used throughout the book as important inputs in the building and testing of financial models.

2.2 A First Look at the Data

This section identifies the key empirical characteristics of financial data. Special attention is devoted to establishing a set of stylised empirical facts that characterise financial data. These empirical characteristics are important for building financial models.

2.2.1 Prices

Figure 2.1 gives a plot of the monthly United States equity price index (S&P500) for the period January 1933 to December 1990. The time path of equity prices shows long-run growth over this period whose general shape is well captured by an exponential trend. This observed exponential pattern in the equity price index may be expressed formally as

$$P_t = P_{t-1} \exp(r_t)\,, \tag{2.1}$$

where P_t is the current equity price, P_{t-1} is the previous month's price and r_t is the rate of the increase between month $t-1$ and month t.

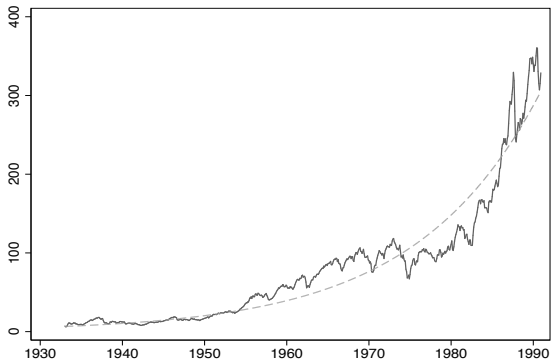


Figure 2.1: Monthly equity price index for the United States from January 1933 to December 1990. Fitted values (dashed line) are obtained from an exponential model as in equation (2.3).

If r_t in (2.1) is restricted to take the same constant value, r, in all time periods, then equation (2.1) becomes

$$P_t = P_{t-1} \exp(r)\,. \tag{2.2}$$

The relationship between the current price, P_t and the price two months earlier, P_{t-2}, is

$$P_t = P_{t-1} \exp(r) = P_{t-2} \exp(r) \exp(r) = P_{t-2} \exp(2r)\,.$$

By continuing this recursion, the relationship between the current price, P_t, and the price t months earlier, P_0, is given by

$$P_t = P_0 \exp(rt). \tag{2.3}$$

It is this exponential function that is plotted in Figure 2.1 in which $P_0 = 7.09$ is the equity price in January 1933 and constant growth rate in monthly prices is taken to be $r = 0.0055$.

The exponential function in equation (2.3) provides a predictive relationship based on long-run growth behaviour. It shows that in January 1933 an investor who wished to know the price of equities in December 1990. There are $1990 - 1993 + 1 = 58$ years in the period which corresponds to $t = (58 \times 12) - 1 = 695$ months. The forecast price for December 1990 is therefore

$$P\,(Dec.1990) = 7.09 \times \exp\,(0.0055 \times 695) = 324.143.$$

The actual equity price in December 1990 for the data plotted in Figure 2.1 is 328.75 so that the percentage forecast error is

$$100 \times \frac{324.143 - 328.75}{328.75} = -1.401\%.$$

Of course, equation (2.3) is based on information over the intervening period that would not be available to an investor in 1933. So, the prediction is called *ex post*, meaning that it is performed after the event. If we wanted to use this relationship to predict the equity price in December 2000, then the prediction would be *ex ante* or forward looking and the suggested trend price would be

$$P\,(Dec.2000) = 7.09 \times \exp\,(0.0055 \times 815) = 627.15.$$

In contrast to the *ex post* prediction, the predicted share price of 627.15 now grossly underestimates the actual equity price of 1330.93. The fundamental reason for this is that the information between 1990 and 2000 has not been used to inform the choice of the value of the crucial parameter r.

An alternative way of analysing the long run time series behaviour of asset prices is to plot the logarithm of prices over time. An example is given in Figure 2.2 where the natural logarithm of the equity price given in Figure 2.1 is presented. Comparing the two series shows that while prices increase at an increasing rate (Figure 2.1) the logarithm of price increases at a constant rate (Figure 2.2). To see why this is the case, we take natural logarithms of equation (2.3) to yield

$$p_t = p_0 + rt\,, \tag{2.4}$$

where lowercase letters now denote the natural logarithms of the variables, namely, $\log P_t$ and $\log P_0$. This is a linear equation between p_t and t in which the slope is equal to the constant r. This equation also forms the basis of the definition of log returns, a point that is now developed in more detail.

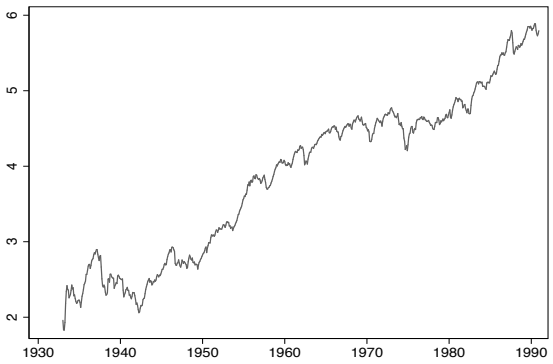


Figure 2.2: The natural logarithm of the monthly equity price index for the United States from January 1933 to December 1990.

2.2.2 Returns

Figure 2.3 plots monthly logarithmic equity returns for the United States over the period January 1933 to December 1990. The returns are seen to hover around a return value that is near zero over the sample period. This value is in fact $r = 0.0055$, which is the estimate used in the earlier computations. In fact, we often consider data on financial asset returns to be distributed about a mean return value of zero. This feature of equity returns contrasts dramatically with the trending character of the corresponding equity prices presented in Figure 2.1.

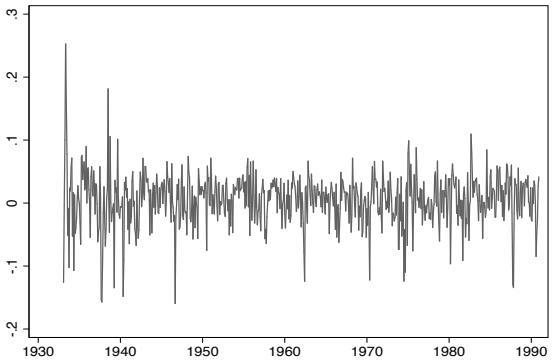


Figure 2.3: Monthly United States equity returns for the period January1933 to December 1990.

The empirical differences in the two series for prices and returns reveals an interesting aspect of stock market behaviour. It is often emphasised in the financial literature that investment in equities should be based on long run

considerations rather than the prospect of short run gains. The reason is that stock prices can be very volatile in the short run. This short run behaviour is reflected in the high variability of the stock returns shown in Figure 2.3. Yet, although stock returns hover around a value of approximately zero, stock prices (which accumulate these returns) tend to trend noticeably upwards over time, as is apparent in Figure 2.1. This tendency of stock prices to drift upwards over time is taken up again in Chapter 5. For present purposes, it is sufficient to remark that when returns are measured over very short periods of time, any tendency of prices to drift upwards is virtually imperceptible because that effect is so small and is swamped by the apparent volatility of the returns. This interpretation puts emphasis on the fact that returns generally focus on short run effects whereas price movements can trend noticeably upwards over long periods of time.

2.2.3 Dividends

In many applications in finance, as in economics, the focus is on understanding the relationships among two or more series. For instance, in present value models of equities, the price of an equity is equal to the discounted future stream of dividend payments

$$P_t = \mathrm{E}_t \left[\frac{D_{t+1}}{(1+\delta_{t+1})} + \frac{D_{t+2}}{(1+\delta_{t+2})^2} + \frac{D_{t+3}}{(1+\delta_{t+n})^3} + \cdots \right], \tag{2.5}$$

where $\mathrm{E}_t(D_{t+n})$ represents the expectation of dividends in the future at time $t+n$ given information available at time t and δ_{t+n} is the corresponding discount rate.
The relationship between equity prices and dividends is highlighted in Figure 2.4 which plots United States equity prices and dividend payments from January 1933 to December 1990. There appears to be a relationship between the two series as both series exhibit positive exponential trends. To analyse the relationship between equity prices and dividends more closely, consider the dividend yield,

$$\delta_t = \frac{D_t}{P_t}, \tag{2.6}$$

which is presented in Figure 2.5 based on the data in Figure 2.4. The dividend yield exhibits no upward trend and instead wanders randomly around the level of 0.05. This behaviour is in stark contrast to the equity price and dividend series which both exhibit strong upward trending behaviour.
The example of the dividend yield illustration of how combining two or more series can change the time series properties of the data - in the present case by apparently eliminating the strong upward trending behaviour. The process of combining trending financial variables into new variables that do not exhibit trends is a form of trend reduction. An extremely important case of trend reduction by combining variables is known as cointegration, a concept that is discussed in detail in Chapter 6.

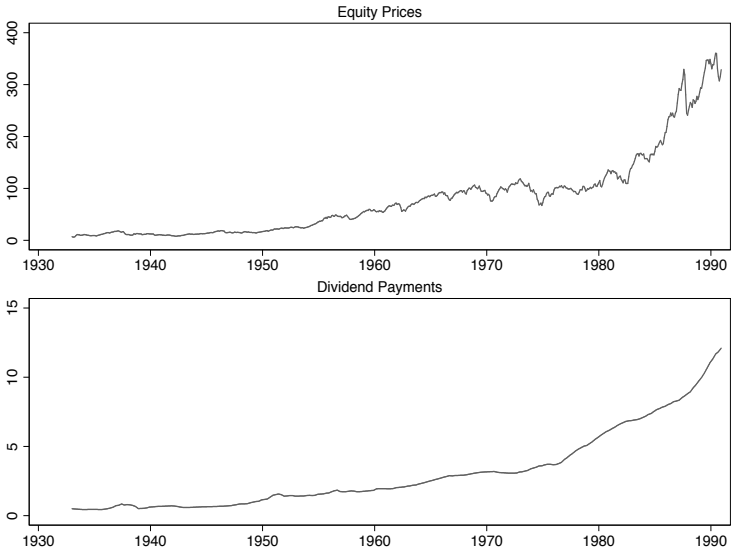


Figure 2.4: Monthly United States equity prices and dividend payments for the period January1933 to December 1990.

The computation of the dividend yield can be motivated from the present value equation in (2.5), by adopting two simplifying assumptions. First, expectations of future dividends are given by present dividends $\mathrm{E}_t(D_{t+n}) = D$. Second, the discount rate is assumed to be fixed at δ. Using these two assumptions in (2.5) gives

$$\begin{aligned} P_t &= D\left(\frac{1}{(1+\delta)} + \frac{1}{(1+\delta)^2} + ...\right) \\ &= \frac{D}{1+\delta}\left(1 + \frac{1}{(1+\delta)} + \frac{1}{(1+\delta)^2} + ...\right) \\ &= \frac{D}{1+\delta}\left(\frac{1}{1-1/(1+\delta)}\right) \\ &= \frac{D}{\delta}, \end{aligned}$$

where the penultimate step uses the sum of a geometric progression.[1] Rear-

[1] An infinite geometric progression is summed as follows

$$1 + \lambda + \lambda^2 + \lambda^3 + ... = \frac{1}{1-\lambda}, \qquad |\lambda| < 1,$$

where in the example $\lambda = 1/(1+\delta)$.

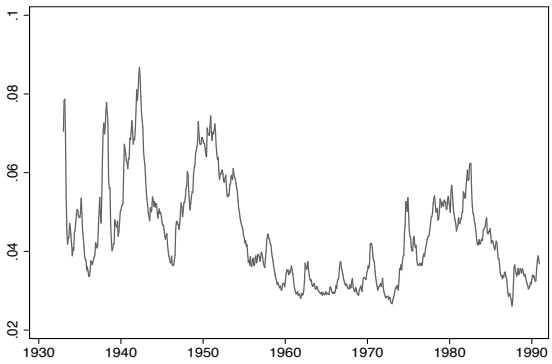


Figure 2.5: Monthly United States dividend yield for the period December 1946 to February 1987.

ranging this expression gives

$$\delta = \frac{D}{P_t}, \tag{2.7}$$

which shows that the discount rate, δ, is equivalent to the dividend yield in equation (2.6).
An alternative representation of the present value model suggested by equation (2.6) is take natural logarithms and rearrange for $\log(P_t)$ to give

$$\log(P_t) = -\log(\delta_t) + \log(D_t).$$

Assuming equities are priced according to the present value model, this equation shows that there is a one-to-one relationship between $\log P_t$ and $\log D_t$. This relationship is another example of trending variables that move together, which is explored in Chapter 6.

2.2.4 Bond Yields

Figure 2.6 gives plots of yields on United States zero coupon bonds for maturities of 3, 6 and 9 months. The yields to the different maturities are not distinguished from one another as the primary purpose is merely to illustrate the general time series behaviour of bond yields.
The time series behaviour of bond yields as shown in Figure 2.6 illustrates three important properties.

1. The yields are increasing over time, so they exhibit trending behaviour. This feature of financial time series is the subject matter of Chapter 5.

2. The variance of the yields tends to grow as the levels of the yields increase. This is called the levels effect and is investigated in more detail in Chapter 9.

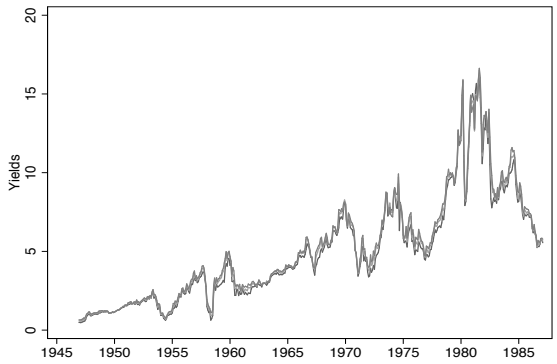


Figure 2.6: Monthly United States zero coupon bond yields for maturities of 3, 6 and 9 months the period December 1946 to February 1987. The different maturities are not distinguished in order to emphasise their general time series properties.

3. The yields of different maturities follow one another very closely and indeed can hardly be distinguished from each other in Figure 2.6. Variables that exhibit trending behaviour but which also move together over time are dealt with in Chapter 6.

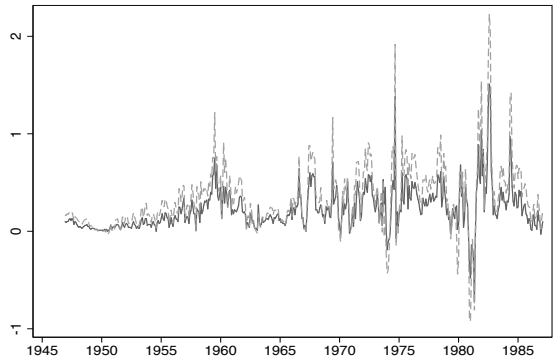


Figure 2.7: Monthly United States 6-month (solid line) and 9-month (dashed line) zero coupon spreads computed relative to the 3-month zero coupon yield for the period January1933 to December 1990.

Property 3 of bond yields may be highlighted by compute the spread between the yields on a long maturity bond and a short maturity bond. Figure 2.7 gives the 6 and 9 month spreads relative to the 3 month zero coupon yield. Comparison of Figures 2.6 and 2.7 reveals that yields exhibit vastly different time series patterns to spreads, with the latter showing no evidence of trends.

However, there is still evidence that the variance of the spreads is not constant over the sample period.

2.2.5 Financial Distributions

An important assumption underlying many theoretical and empirical models in finance is that returns are normally distributed. This assumption is widely used in portfolio allocation models, in Value-at-Risk (VaR) calculations, in pricing options, and in many other applications. An example of an empirical returns distribution is given in Figure 2.8 which gives the histogram of hourly United States exchange rate returns computed relative to the British pound. Even though this distribution exhibits some characteristics that are consistent with a normal distribution such as symmetry, the distribution differs from normality in two important ways, namely, the presence of heavy tails and a sharp peak in the centre of the distribution.

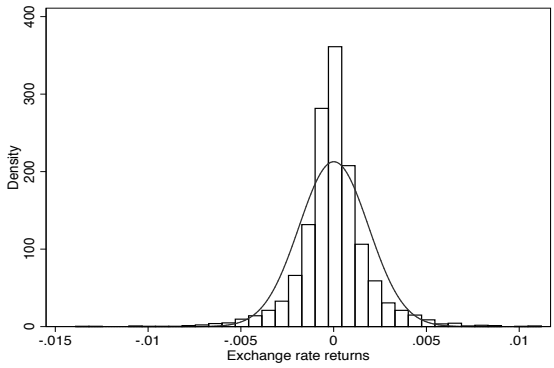


Figure 2.8: Empirical distribution of hourly \$/£ exchange rate returns for the period 1 January 1986 00:00 to 15 July 1986 11:00 with a normal distribution overlaid.

Distributions exhibiting these properties are known as leptokurtic distributions. As the empirical distribution exhibits tails that are much thicker than those of a normal distribution, the actual probability of observing excess returns is higher than that implied by the normal distribution. The empirical distribution also exhibits some peakedness at the centre of the distribution around zero, and this peakedness is sharper than that of a normal distribution. This feature suggests that there are many more observations where the exchange rate hardly moves and for which there are a greater number of smaller returns than there would be in the case of draws from a normal population.
The example given in Figure 2.8 is for exchange rate returns. But the property of heavy tails and peakedness of the distribution of returns is common for other asset markets including equities, commodities and real estate mar-

kets. All of these empirical distributions are therefore inconsistent with the assumption of normality and financial models that are based on normality, therefore, may result in financial instruments such as options being incorrectly priced or measures of risk being underestimated.

2.2.6 Transactions

A property of all of the financial data analysed so far is that observations on a particular variable are recorded at discrete and regularly spaced points in time. The data on equity prices and dividend payments in Figure 2.4 and the data on zero coupon bond yields in Figure 2.6, are all recorded every month. In fact, higher frequency data are also available at regularly spaced time intervals, including daily, hourly and even 10-15 minute observations.
More recently, transactions data have become available which records the price of every trade conducted during the trading day. An example is given in Table 2.1 which gives a snapshot of the trades recorded on American Airlines on August 1, 2006. The variable Trade, x_t, is a binary variable signifying whether a trade has taken place at time t so that

$$x_t = \begin{cases} 1 & : \text{ Trade occurs} \\ 0 & : \text{ No trade occurs.} \end{cases}$$

Table 2.1

A snapshot of American Airlines (AMR) transactions data on August 1 2006, at 9 hours and 42 minutes.

Sec.	Trade (x)	Duration (u)	Price (P)
5	1	1	\$21.58
6	0	·	\$21.58
7	0	·	\$21.58
8	0	·	\$21.58
9	0	·	\$21.58
10	0	·	\$21.58
11	1	6	\$21.59
12	1	1	\$21.59
13	0	·	\$21.59
14	1	2	\$21.59

The duration between trades, u, is measured in seconds, and the corresponding price of the asset at the time of the trade, P, is also recorded. The table shows that there is a trade at the 5 second mark where the price is \$21.58. The next trade occurs at the 11 second mark at a price of \$21.59, so the duration

between trades is $u = 6$ seconds. There is another trade straight away at the 12 second mark at the same price of \$21.59, in which case the duration is just $u = 1$ second. There is no trade in the following second, but there is one two seconds later at the 14 second mark, again at the same price of \$21.59, so the duration is $u = 2$ seconds.

The time differences between trades of American Airlines (AMR) shares is further highlighted by the histogram of the duration times, u, given in Figure 2.9. This distribution has an exponential shape with the duration time of $u = 1$ second, being the most common. However, there are a number of durations in excess of $u = 25$ seconds, and there are some times even in excess of 50 seconds.

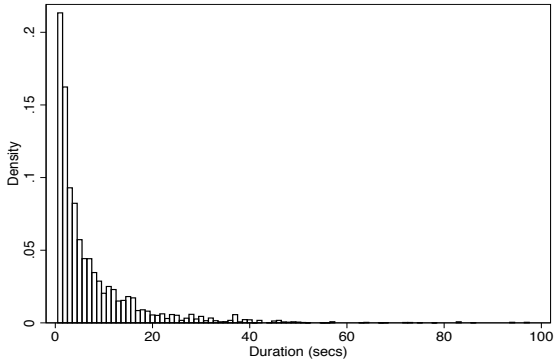


Figure 2.9: Empirical distribution of durations (in seconds) between trades of American Airlines (AMR) on 1 August 2006 from 09:30 to 04:00 (23 401 observations).

The important feature of transactions data that distinguishes it from the time series data discussed above, is that the time interval between trades is not regular or equally spaced. In fact, if high frequency data are used, such as 1 minute data, there will be periods where no trades occur in the window of time and the price will not change. This is especially so in thinly traded markets. The implication of using such transactions data is that the models specified in econometric work need to incorporate those features, including the apparent randomness in the observation interval between trades. Correspondingly, the appropriate statistical techniques are expected to be different from the techniques used to analyse regularly spaced financial time series data. These issues for high frequency irregularly spaced data are investigated further in Chapter 15 on financial microstructure effects.

2.3 Summary Statistics

In the previous section, the time series properties of financial data are explored using graphical tools, specifically line charts and histograms. In this section a number of statistical methods are used to summarise financial data. While these methods are general summary measures of financial data, a few important cases will be highlighted in which it is inappropriate to summarise the data using these simple measures.

2.3.1 Univariate

Intuitively, there are at least four stylised facts about the returns to an asset that an investor would like to know about when considering investing in an asset.

1. The expected return from investing in the asset.
2. The risk associated with the investment, where risk refers to the uncertainty surrounding the value of, or payoff from, the investment in the asset. In other words, risk reflects the chance that the actual return on an investment may be very different than the expected return.
3. A more subtle summary statistic would be whether or not the extreme returns are above the expected value, meaning that the distribution of the returns is positively skewed. Obviously, investors prefer large positive extreme returns to large negative extreme returns.
4. Finally, the relative likelihood of occurrence of extreme returns is important as investors prefer returns closer to expected returns.

Sample Mean

A measure of the expected return is given by the sample mean

$$\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t.$$

The returns to monthly S&P 500 data, r_t, are plotted for the period January 1933 to December 1990 in Figure 2.3. The sample mean of these data is $\bar{r} = 0.005568$. Expressed in annual terms, the mean return is $0.005568 \times 12 = 0.0668$ so that the average return over the period 1933 to 1990 is 6.68% per annum. The sample mean represents the level around which r_t fluctuates and therefore represents a summary measure of the *location* of the data.
An example where the sample mean is an inappropriate summary measure is where data are trending. Figure 2.10 plots the equity price index (as opposed to returns to the index) with the sample mean of prices, $\overline{P} = 80.253$, superimposed. $\overline{P}$ no longer represents the long-run level around which P_t is located and therefore does not represent an appropriate summary measure of the location of the data.

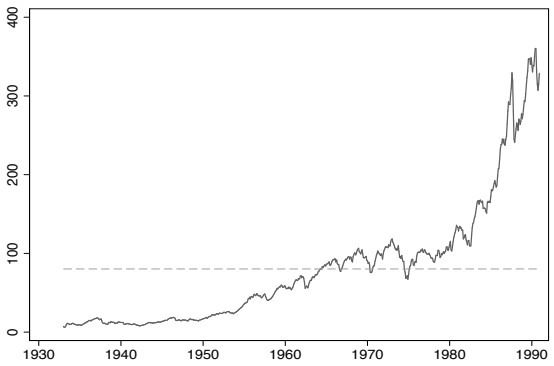


Figure 2.10: Monthly United States equity price index for the period January 1933 to December 1990 with the sample mean (dashed line) superimposed.

Sample Variance and Standard Deviation

A measure of the expected deviation of the actual return on an asset around its expected return is given by the sample variance.

$$s^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2 ,$$

This form of the sample variance is a biased estimator of the population variance. An unbiased estimator is to replace the T in the denominator with $T-1$ which is known as a degrees of freedom or small sample correction. In most financial econometric applications, the sample size T is large enough for this difference to negligible. In the case of the returns data, the sample variance is $s^2 = 0.040260^2 = 0.00162$.
In finance, the sample standard deviation, which is the square root of the sample variance,

$$s = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2},$$

is usually used as the measure of the riskiness of an investment and is called the *volatility* of a financial return. The standard deviation has the same scale as returns (the variance has the scale of returns squared) and is therefore easily interpretable.

Sample Skewness

If the extreme returns in any sample are mainly positive (negative), the distribution of r_t is positively (negatively) skewed. A measure of skewness in the

sample is

$$\text{SK} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{s}\right)^3.$$

If the sample skewness is zero, then the distribution is said to be symmetric. Figure 2.11 gives a histogram of the United States equity returns previously plotted in Figure 2.3, which shows that there is a larger concentration of returns below the sample mean of $\bar{r} = 0.005568$ (left tail) than there is for returns above the sample mean (right tail). The sample skewness is computed to be SK $= -0.299$, where the sign of the statistic emphasises negative skewness.

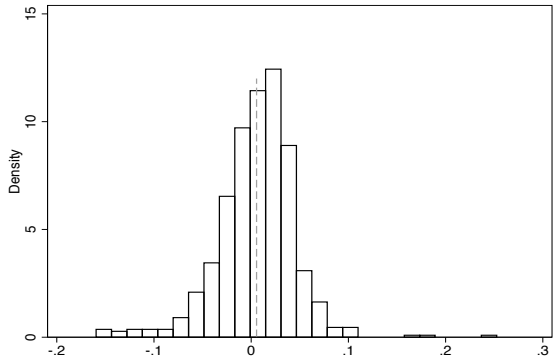


Figure 2.11: Empirical distribution of monthly United States equity returns for the period January 1933 to December 1990. The vertical dashed line represents the sample mean of returns.

Sample Kurtosis

If there are extreme returns relative to a benchmark distribution (usually the normal distribution), the distribution of r_t exhibits excess kurtosis. A measure of kurtosis in the sample is

$$\text{KT} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{s}\right)^4.$$

Comparing this value to KT $= 3$, which is the kurtosis value of a normal distribution, gives a measure of excess kurtosis

$$\text{EXCESS KT} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{s}\right)^4 - 3.$$

In the case of the United States log equity returns, the sample kurtosis is KT $=$ 7.251. This value is greater than 3 and there are more extreme returns in the data that that predicted by the normal distribution.

2.3.2 Bivariate

The statistical measures discussed so far summarise the characteristics of the returns to a single asset. Perhaps even more important in finance is understanding the interrelationships between two or more financial assets. For example, in constructing a diversified portfolio, the aim is to include assets whose fluctuations in returns do not match each other perfectly. In this way the value of the portfolio is protected even though there will be certain assets in the portfolio that are performing poorly.

Covariance

A measure of co-movements between the returns on two assets, r_{it} and r_{jt}, is the sample covariance given by

$$s_{ij} = \frac{1}{T}\sum_{t=1}^{T}\left(r_{it} - \bar{r}_i\right)\left(r_{jt} - \bar{r}_j\right),$$

in which $\bar{r}_i$ and $\bar{r}_j$ are the respective sample means of the returns on assets i and j. A positive covariance, $s_{ij} > 0$, shows that the returns of asset i and asset j have a tendency to move together. That is, when return on asset i is above its mean, the return on asst j is also likely to be above its mean. A negative covariance, $s_{ij} < 0$, indicates that when the returns of asset i are above its sample mean, on average, the returns on asset j are likely to be below its sample mean. Covariance has a particularly important role to play in empirical finance, as will become clear in Chapter 3.

Correlation

Another measure of association is the correlation coefficient given by

$$c_{ij} = \frac{s_{ij}}{\sqrt{s_{ii}s_{jj}}},$$

in which

$$s_{ii} = \frac{1}{T}\sum_{t=1}^{T}\left(r_{it} - \bar{r}_i\right)^2, \qquad s_{jj} = \frac{1}{T}\sum_{t=1}^{T}\left(r_{jt} - \bar{r}_j\right)^2,$$

represent the respective variances of the returns of assets i and j. The correlation coefficient is the covariance scaled by the standard deviations of the two returns. The correlation has the property that is has the same sign as the covariance, as well as the additional property that it lies in the range $-1 \leq c_{ij} \leq 1$ and is therefore not unit dependent.

2.4 Percentiles and Computing Value-at-Risk

The percentiles of a distribution are a set of summary statistics that summarise both the location and the spread of a distribution. Formally, a percentile is a measure that indicates the value of a given random variable below which a given percentage of observations fall. So the important measure of the location of a distribution, the median, below which 50% of the observations of the random variable fall, is also the 50th percentile. The median is an alternative to the sample mean as a measure of location and can be very important in financial distributions in which large outliers are encountered. The difference between the 25th percentile (or first quartile) and the 75th percentile (or third quartile) is known as the inter-quartile range. which provides an alternative to the variance as a measure of the dispersion of the distribution. It transpires that the percentiles of the distribution, particularly the 1st and 5th percentiles are important statistics in the computation of an important risk measure in finance known as Value-at-Risk or VaR.

Losses faced by financial institutions have the potential to be propagated through the financial system and undermine its stability. The onset of heightened fears for the riskiness of the banking system can be rapid and have widespread ramifications. The potential loss faced by banks is therefore a crucial measure of the stability of the financial sector. Pérignon and Smith (2010) examine the daily trading revenues, a measure of a bank's fundamental soundness, for Bank of America. Summary measures and percentiles of the daily trading revenues from 2001 to 2004 are presented in Table 2.2.

Table 2.2

Descriptive statistics and percentiles for daily trading revenue in millions of Bank of America for the period 2 January 2001 to 31 December 2004.

Statistics		Percentiles	
Observations	1008	1%	−24.82
Mean	13.87	5%	−9.45
Std. Dev.	14.91	10%	−2.72
Skewness	0.12	25%	4.84
Kurtosis	4.93	50%	13.15
Maximum	84.33	75%	22.96
Minimum	−57.39	90%	30.86
		95%	36.44
		99%	57.10

Following a wave of banking collapses in the 1990s financial regulators, in the guise of the Basel Committee on Banking Supervision (1996), started requiring banks to hold capital to buffer against possible losses, measured using a method called Value-at-Risk (VaR). VaR quantifies the loss that a bank

can face on its trading portfolio within a given period and for a given confidence interval. In the context of a bank, VaR is defined in terms of the lower tail of the distribution of trading revenues. Specifically, the 1% VaR for the next h periods conditional on information at time T is the 1st percentile of expected trading revenue at the end of the next h periods. For example, if the daily 1% h-period VaR is \$30million, then there is a 1% chance the bank will lose \$30 million or more. Although \$30 million is a loss, by convention the VaR is quoted as a positive amount.

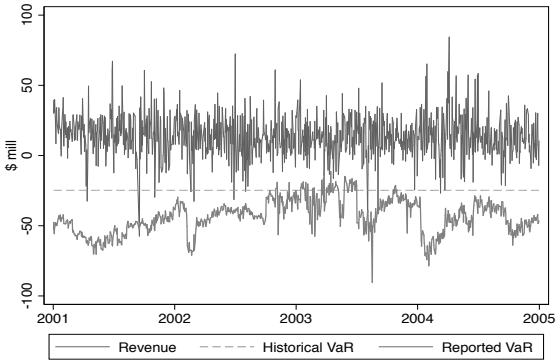


Figure 2.12: Time series plot of the daily 1% Value-at-Risk reported by Bank of America from 2 January 2001 to 31 December 2004.

There are three common ways to compute VaR.

(i) Historical Simulation

The historical method simply computes the percentiles of the distribution from historical data. Based on the sample percentiles. in Table 2.2 the 1% daily VaR for Bank of America using all available historical data (2001 - 2004) is

$$VaR\,(1\%, daily) = \$24.8214\ m\,.$$

(ii) The Variance-Covariance Method

This method assumes that the trading revenues are normally distributed. Since 1% of the distribution lies in the tail delimited by -2.33 then, using the summary statistics reported in Table 2.2,

$$VaR\,(1\%, daily) = 13.8699 - 2.33 \times 14.9089 = -\$20.8679\ m\,.$$

This value is slightly lower (in absolute value) than that provided by historical simulation because the assumption of normality ignores the slightly fatter tails exhibited by the empirical distribution of daily trading revenues.

(iii) Monte Carlo Simulation

The third method involves simulating a model for daily trading revenues several times and constructing simulated percentiles. This approach is revisited in Chapter 7.

Figure 2.12 plots the daily trading revenue of the Bank of America together with the 1% daily VaR reported by the bank. Even to the naked eye it is apparent that Bank of America had only four violations of the 1% daily reported VaR during the period 2001-2004 ($T = 1008$), amounting to only 0.4%. The daily VaR computed from historical simulation is also shown and it suggests that the Bank of America was over-conservative in its estimation of daily VaR during this period.

2.5 The Efficient Markets Hypothesis

An important and controversial theory in finance is the efficient markets hypothesis, which in its most general form, theorises that all available information concerning the value of a risky asset is factored into the current price of the asset (Fama, 1965; Samuelson, 1965). There are two ways to examine this proposition; the first is to test asset returns for autocorrelation and the second is to compare the variance of asset returns over different time horizons.

2.5.1 Return Predictability

The autocorrelation statistic, defined by

$$acf(k) = \frac{T^{-1}\sum_{t=k+1}^{T}(r_t - \overline{r})(r_{t-k} - \overline{r})}{T^{-1}\sum_{t=1}^{T}(r_t - \overline{r})^2}, \tag{2.8}$$

measures the strength of movements in current returns, r_t with returns on the same asset k periods earlier r_{t-k}. In equation (2.8), the numerator represents the autocovariance of returns k periods apart and the denominator represents the variance of returns. If returns exhibit no autocorrelation then future movements in returns are unpredictable. If, however, returns exhibit positive or negative autocorrelation, then successive values of returns tend to have the same sign and this pattern can be exploited in predicting the future behaviour of returns.

Table 2.3 gives the first 10 autocorrelations of hourly DM/$ exchange rate returns in column 2. All the autocorrelations appear close to zero, suggesting that exchange rate returns are not predictable and the foreign exchange market is efficient.

Autocorrelations of returns reveals information on the mean of returns. Applying this approach to squared returns in terms of the statistic

$$acf^2(k) = \frac{T^{-1}\sum_{t=k+1}^{T}\left(r_t^2 - \overline{r^2}\right)\left(r_{t-k}^2 - \overline{r^2}\right)}{T^{-1}\sum_{t=1}^{T}\left(r_t^2 - \overline{r^2}\right)^2},$$

Table 2.3

Autocorrelation properties of returns and functions of returns for the hourly DM/$ exchange rate for the period 1 January 1986 00:00 to 15 July 1986 11:00.

Lag	r_t	r_t^2	$\|r_t\|$	$\|r_t\|^{0.5}$
1	−0.061	0.134	0.204	0.200
2	−0.001	0.085	0.124	0.130
3	0.020	0.016	0.068	0.071
4	−0.032	0.051	0.078	0.065
5	0.032	0.033	0.032	0.022
6	−0.058	0.009	0.024	0.031
7	0.015	−0.026	−0.033	−0.024
8	−0.018	−0.026	−0.013	0.006
9	−0.001	−0.030	−0.039	−0.038
10	0.019	−0.021	−0.038	−0.034

reveals information about the variance of returns. Column 3 of Table 2.3 suggests that while the level of returns are not predictable, the same cannot be said of the variance of returns. Note, however, that this conclusion does not violate the efficient markets hypothesis, which is solely concerned only with the expected value of the level of returns. The application of autocorrelations to squared returns represents an important diagnostic tool in models of time-varying volatility which is the subject matter of Part IV.

Autocorrelations can also be computed for various transformations of returns, such as

$$r_t^3, \quad r_t^4, \quad |r_t|, \quad |r_t|^\alpha .$$

The first two transformations provide evidence of autocorrelations in skewness and kurtosis respectively. The third transformation provides an alternative measure of the presence of autocorrelation in the variance. The last case simply represents a general transformation. For example, setting $\alpha = 0.5$ computes the autocorrelation of the standard deviation (the square root of the variance). The presence of stronger autocorrelation in squared returns than returns, suggests that other transformations of returns may reveal even stronger autocorrelation patters and this conjecture is born out by the results reported in Table 2.3.

2.6 The Variance Ratio

An alternative way to examine the efficient markets hypothesis is to compare the variance on returns over different time horizons. Consider the variances

of the 1-period returns and the n-period returns

$$s_1^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2 \,, \qquad s_n^2 = \frac{1}{T}\sum_{t=1}^{T}(r_{nt} - n\bar{r})^2 \,,$$

in which

$$\begin{aligned} r_t &= \log P_t - \log P_{t-1} \\ r_{nt} &= \log P_t - \log P_{t-n} = r_t + r_{t-1} + \cdots + r_{t-(n-1)} \,, \end{aligned}$$

and $n\bar{r}$ represents the sample mean of n-period returns.
If there is no autocorrelation the variance of n-period returns should equal n times the variance of the 1-period returns. The ratio

$$VR_n = \frac{s_n^2}{n\, s_1^2} \,,$$

is known as the variance ratio and has the following implications for the properties of excess returns:

$$VR_n = \begin{cases} = 1 & \text{[No autocorrelation]} \\ > 1 & \text{[Positive autocorrelation]} \\ < 1 & \text{[Negative autocorrelation].} \end{cases}$$

The first of these results is easily demonstrated. Consider a $n = 3$ period return

$$r_{3t} = r_t + r_{t-1} + r_{t-2} \,,$$

which is the sum of the three 1-period returns. Let the sample mean for the 1-period returns be $\bar{r}$. Subtracting $\bar{r}$ from both sides three times gives

$$(r_{3t} - 3\bar{r}) = (r_t - \bar{r}) + (r_{t-1} - \bar{r}) + (r_{t-2} - \bar{r}) \,.$$

Squaring both sides and averaging over a sample of size T gives

$$\begin{aligned} &\frac{1}{T}\sum_{t=1}^{T}(r_{3t} - 3\bar{r})^2 \\ &\quad = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2 && \text{[Variance of } r_t\text{]} \\ &\quad + \frac{1}{T}\sum_{t=1}^{T}(r_{t-1} - \bar{r})^2 && \text{[Variance of } r_{t-1}\text{]} \\ &\quad + \frac{1}{T}\sum_{t=1}^{T}(r_{t-2} - \bar{r})^2 && \text{[Variance of } r_{t-2}\text{]} \\ &\quad + \frac{2}{T}\sum_{t=1}^{T}(r_t - \bar{r})(r_{t-1} - \bar{r}) && \text{[Autocovariance of } r_t, r_{t-1}\text{]} \\ &\quad + \frac{2}{T}\sum_{t=1}^{T}(r_t - \bar{r})(r_{t-2} - \bar{r}) && \text{[Autocovariance of } r_t, r_{t-2}\text{]} \\ &\quad + \frac{2}{T}\sum_{t=1}^{T}(r_{t-1} - \bar{r})(r_{t-2} - \bar{r}) && \text{[Autocovariance of } r_{t-1}, r_{t-2}\text{].} \end{aligned}$$

This expansion requires values for r_0 and r_{-1}. To implement this formulation in practice, the range of the summations are suitably adjusted. In the case of zero autocovariances (no autocorrelation) the relationship simplifies to

$$s_3^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2 + \frac{1}{T}\sum_{t=1}^{T}(r_{t-1} - \bar{r})^2 + \frac{1}{T}\sum_{t=1}^{T}(r_{t-2} - \bar{r})^2 .$$

Assuming that the variance for r_t is the same as the variance for r_{t-1} and r_{t-2} then in the case of no autocorrelation in $n = 3$ period returns

$$s_3^2 = 3s_1^2 .$$

A more detailed discussion of the autocorrelation function is provided in Chapter 4. The assumption of equal variance for r_{t-1} and r_{t-2} is known as stationarity and is addressed in detail in Chapters 4 and 5. Modelling with variables that do not satisfy this assumption is dealt with in Chapter 6.

2.7 Exercises

1. **Equity Prices, Dividends and Returns**

pv.wf1, pv.dta, pv.xlsx

   (a) Plot the equity price over time and interpret its time series properties. Compare the result with Figure 2.1.

   (b) Plot the natural logarithm of the equity price over time and interpret its time series properties. Compare this graph with Figure 2.2.

   (c) Plot the return on equities over time and interpret its time series properties. Compare this graph with Figure 2.3.

   (d) Plot the price and dividend series using a line chart and compare the result in Figure 2.4.

   (e) Compute the dividend yield and plot this series using a line chart. Compare the graph with Figure 2.5.

   (f) Compare the graphs in parts (a) and (b) and discuss the time series properties of equity prices, dividend payments and dividend yields.

   (g) The present value model predicts a one-to-one relationship between the logarithm of equity prices and the logarithm of dividends. Use a scatter diagram to verify this property and comment on your results.

   (h) Compute the returns on United States equities and then calculate the sample mean, variance, skewness and kurtosis of these returns. Interpret the statistics.

2. **Yields**

zero.wf1, zero.dta, zero.xlsx

(a) Plot the 2, 3, 4, 5, 6 and 9 months United States zero coupon yields using a line chart and compare the result in Figure 2.6.

(b) Compute the spreads on the 3-month, 5-month and 9-month zero coupon yields relative to the 2-month yield and and plot these spreads using a line chart. Compare the graph with Figure 2.6.

(c) Compare the graphs in parts (a) and (b) and discuss the time series properties of yields and spreads.

3. **Duration Times Between American Airline (AMR) Trades**

amr.wf1, amr.dta, amr.xlsx

(a) Use a histogram to graph the empirical distribution of the duration times between American Airline trades. Compare the graph with Figure 2.9.

(b) Interpret the shape of the distribution of durations times.

4. **Exchange Rates**

hour.wf1, hour.dta, hour.xlsx

(a) Draw a line chart of the $\$/£$ exchange rate and discuss its time series characteristics.

(b) Compute the returns on $\$/£$ pound exchange rate. Draw a line chart of this series and discuss its time series characteristics.

(c) Compare the graphs in parts (a) and (b) and discuss the time series properties of exchange rates and exchange rate returns.

(d) Use a histogram to graph the empirical distribution of the returns on the $\$/£$. Compare the graph with Figure 2.11.

(e) Compute the first 10 autocorrelations of the returns, squared returns, absolute returns and the square root of the absolute returns.

(f) Repeat parts (a) to (e) using the $\mathbb{DM}/\$$ exchange rate and comment on the time series characteristics, empirical distributions and patterns of autocorrelation for the two series. Discuss the implications of these results for the efficient markets hypothesis.

5. **Value-at-Risk**

bankamerica.wf1, bankamerica.dta, bankamerica.xlsx

(a) Compute summary statistics and percentiles for the daily trading revenues of Bank of America. Compare the results with Table 2.2.

(b) Draw a histogram of the daily trading returns and superimpose a normal distribution on top of the plot. What do you deduce about the distribution of the daily trading revenues.

(c) Plot the trading revenue together with the historical 1% VaR and the reported 1% Var. Compare the results with Figure 2.12.

(d) Now assume that a weekly VaR is required. Repeat parts (a) to (c) for weekly trading revenues.

Chapter 3

Linear Regression Models

3.1 Introduction

One of the most widely used models in empirical finance is the linear regression model. This model provides a framework in which to explain the movements of one financial variable in terms of one, or many explanatory variables. Important examples include estimating the weights on assets included in a minimum variance portfolio and the capital asset pricing model (CAPM). Although these basic models stipulate linear relationships between the variables, the framework is easily extended to a range of nonlinear relationships as well. The model can be extended to capture sharp changes in returns caused by stock market crashes, day-of-the-week effects, policy announcements and important events by means of qualitative response variables or dummy variables.

3.2 A Minimum Variance Portfolio

The aim is to choose a portfolio of assets to minimise the overall risk of the portfolio, as measured by its (squared) volatility, or its variance. To derive the minimum variance portfolio, consider a portfolio consisting of two assets with returns r_{1t} and r_{2t}, respectively. The return on the portfolio is given by

$$r_{pt} = w_1 r_{1t} + w_2 r_{2t}, \tag{3.1}$$

in which $w_1 + w_2 = 1$ are weights that define the relative contributions of each asset in the portfolio and that satisfy an adding-up restriction.
The returns have the following properties

Mean:	$\mu_1 = \mathrm{E}(r_{1t}),$	$\mu_2 = \mathrm{E}(r_{2t}),$
Variance:	$\sigma_1^2 = \mathrm{E}[(r_{1t} - \mu_1)^2],$	$\sigma_2^2 = \mathrm{E}[(r_{2t} - \mu_2)^2],$
Covariance:	$\sigma_{12} = \mathrm{E}[(r_{1t} - \mu_1)(r_{2t} - \mu_2)].$	

The expected return on this portfolio is

$$\mu_p = \mathrm{E}(w_1 r_{1t} + w_2 r_{2t}) = w_1 \mathrm{E}(r_{1t}) + w_2 \mathrm{E}(r_{2t}) = w_1 \mu_1 + w_2 \mu_2, \tag{3.2}$$

and a measure of the portfolio's risk is given by

$$\begin{aligned} \sigma_p^2 &= \mathrm{E}[(r_{pt} - \mu_p)^2] \\ &= \mathrm{E}[(w_1(r_{1t} - \mu_1) + w_2(r_{2t} - \mu_2))^2] \\ &= w_1^2 \mathrm{E}[(r_{1t} - \mu_1)^2] + w_2^2 \mathrm{E}[(r_{2t} - \mu_2)^2] + 2w_1 w_2 \mathrm{E}[(r_{1t} - \mu_1)(r_{2t} - \mu_2)] \\ &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12}. \end{aligned} \tag{3.3}$$

Using the adding-up restriction imposed by equation (**??**), the risk of the portfolio is equivalent to

$$\sigma_p^2 = w_1^2 \sigma_1^2 + (1 - w_1)^2 \sigma_2^2 + 2w_1(1 - w_1)\sigma_{12}. \tag{3.4}$$

To find the optimal portfolio that minimises risk, the following optimisation problem is solved

$$\min_{w_1} \sigma_p^2.$$

Differentiating (3.4) with respect to w_1 gives

$$\frac{d\sigma_p^2}{dw_1} = 2w_1 \sigma_1^2 - 2(1 - w_1)\sigma_2^2 + 2(1 - 2w_1)\sigma_{12}.$$

Setting this derivative to zero and rearranging for w_1 gives the optimal portfolio weight on the first asset as

$$w_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}. \tag{3.5}$$

Upon using (**??**) gives the optimal weight on the other asset as

$$w_2 = 1 - w_1 = \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}. \tag{3.6}$$

To estimate the optimal portfolio from a sample of returns, the population parameters are replaced by their estimators

$$\widehat{w}_1 = \frac{\widehat{\sigma}_2^2 - \widehat{\sigma}_{12}}{\widehat{\sigma}_1^2 + \widehat{\sigma}_2^2 - 2\widehat{\sigma}_{12}}, \qquad \widehat{w}_2 = \frac{\widehat{\sigma}_1^2 - \widehat{\sigma}_{12}}{\widehat{\sigma}_1^2 + \widehat{\sigma}_2^2 - 2\widehat{\sigma}_{12}}.$$

The estimate of the mean and the risk of the portfolio are, respectively,

$$\begin{aligned} \widehat{\mu}_p &= \widehat{w}_1 \widehat{\mu}_1 + \widehat{w}_2 \widehat{\mu}_2, \\ \widehat{\sigma}_p^2 &= \widehat{w}_1^2 \widehat{\sigma}_1^2 + (1 - \widehat{w}_1)^2 \widehat{\sigma}_2^2 + 2\widehat{w}_1(1 - \widehat{w}_1)\widehat{\sigma}_{12}. \end{aligned}$$

Consider constructing a portfolio consisting of Microsoft and Walmart stocks. Using monthly equity prices from April 1990 to July 2004 ($T = 172$) log returns are computed as

$$\begin{array}{lcl} \text{Microsoft} & : & r_{1t} = \log P_{1t} - \log P_{1\,t-1}\,, \\ \text{Walmart} & : & r_{2t} = \log P_{2t} - \log P_{2\,t-1}\,, \end{array}$$

and are plotted in Figure 3.1.

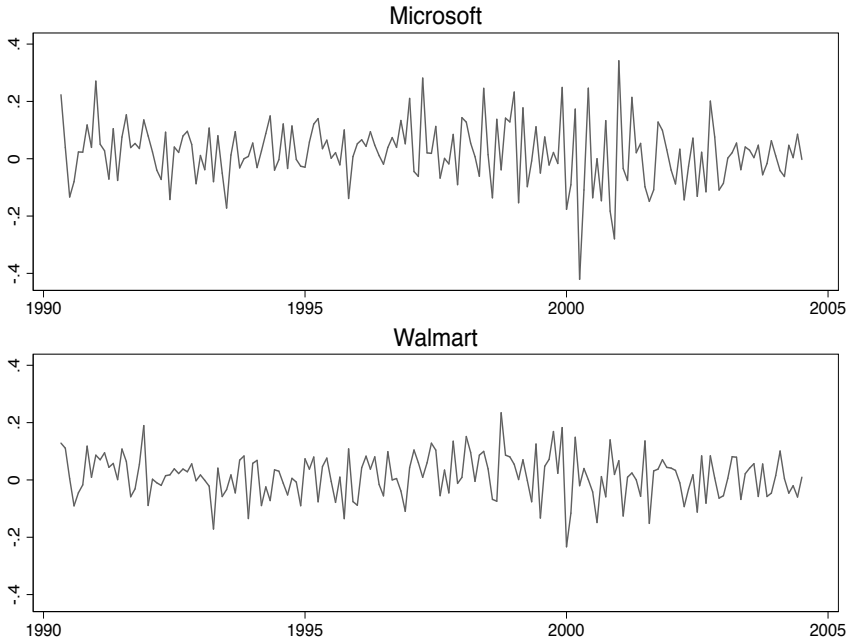


Figure 3.1: The returns to United States stocks Microsoft and Walmart computed using monthly data for the period April 1990 to July 2004.

The estimated means returns are

$$\widehat{\mu}_1 = 0.020877, \qquad \widehat{\mu}_2 = 0.013496\,,$$

and the estimated covariance matrix of the returns is

$$\begin{bmatrix} \widehat{\sigma}_1^2 & \widehat{\sigma}_{12} \\ \widehat{\sigma}_{12} & \widehat{\sigma}_2^2 \end{bmatrix} = \begin{bmatrix} 0.011333 & 0.002380 \\ 0.002380 & 0.005759 \end{bmatrix}.$$

In computing the elements of the covariance matrix the biased form presented in Chapter 2 is used in which T is used in the denominator instead of $T - 1$.

The estimates on the optimal weights are

$$\begin{aligned}\widehat{w}_1 &= \frac{\widehat{\sigma}_2^2 - \widehat{\sigma}_{12}}{\widehat{\sigma}_1^2 + \widehat{\sigma}_2^2 - 2\widehat{\sigma}_{12}} \\ &= \frac{0.005759 - 0.002380}{0.011333 + 0.005759 - 2 \times 0.002380} = 0.274, \\ \widehat{w}_2 &= 1 - \widehat{w}_1 = 1 - 0.274 = 0.726.\end{aligned}$$

The optimal portfolio to minimise risk is to allocate 0.274 of the investor's wealth to Microsoft and 0.726 to Walmart.
An estimate of the mean of the minimum variance portfolio is

$$\begin{aligned}\widehat{\mu}_p &= \widehat{w}_1\widehat{\mu}_1 + \widehat{w}_2\widehat{\mu}_2 \\ &= 0.274 \times 0.020877 + 0.726 \times 0.013496 \\ &= 0.015519.\end{aligned}$$

An estimate of the risk of the minimum variance portfolio is

$$\begin{aligned}\widehat{\sigma}_p^2 &= \widehat{w}_1^2\widehat{\sigma}_1^2 + (1-\widehat{w}_1)^2\widehat{\sigma}_2^2 + 2\widehat{w}_1(1-\widehat{w}_1)\widehat{\sigma}_{12} \\ &= 0.274^2 \times 0.011333 + (1-0.274)^2 \times 0.005759 + \\ &\quad 2 \times 0.274 \times (1-0.274) \times 0.002380 \\ &= 0.004833.\end{aligned}$$

Comparing this estimate to the individual risks on Microsoft and Walmart shows that the risk on the portfolio is reduced

$$\begin{aligned}\widehat{\sigma}_p^2 &= 0.004833 < \widehat{\sigma}_1^2 = 0.011333, \\ \widehat{\sigma}_p^2 &= 0.004833 < \widehat{\sigma}_2^2 = 0.005759.\end{aligned}$$

3.3 Specification of the Linear Regression Model

An alternative way of expressing the minimum variance portfolio model is to consider the population linear regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

where y_t is the dependent variable, x_t is the explanatory variable,and β_0 and β_1 are the intercept and slope parameters, respectively.
The disturbance term, u_t, represents the movements in y_t that are not explained by the linear regression model. The disturbances, u_t, are assumed to have the following properties.

(i) Mean – The disturbances have zero mean, $\mathrm{E}(u_t) = 0$.

(ii) Homoskedasticity – The variance of the disturbances is constant for all observations, $\mathrm{var}(u_t) = \sigma^2$.

(iii) No autocorrelation – Disturbances corresponding to different observations are independent, $\mathrm{E}(u_t\, u_{t-j}) = 0$, $j \neq 0$.

(iv) Independence – The disturbances are uncorrelated with the explanatory variable, $\mathrm{E}(u_t\, x_t) = 0$.

(v) Normality – The disturbances follow a normal distribution.

Choosing the parameters β_0 and β_1 to minimise the disturbance variance $\sigma^2 = \mathrm{E}(u_t^2)$ gives

$$\begin{aligned} \beta_1 &= \frac{\mathrm{cov}(y_t, x_t)}{\mathrm{var}(x_t)}\,, \\ \beta_0 &= \mathrm{E}(y_t) - \beta_1\, \mathrm{E}(x_t)\,. \end{aligned}$$

To show the relationship between the linear regression model and the minimum variance portfolio, define the following variables:

$$y_t = r_{2t}, \qquad x_t = r_{2t} - r_{1t}\,.$$

Now

$$\mathrm{cov}(y_t, x_t) = \mathrm{cov}(r_{2t}, r_{2t} - r_{1t}) = \mathrm{var}(r_{2t}) - \mathrm{cov}(r_{2t}, r_{1t}) = \sigma_2^2 - \sigma_{12}$$

$$\mathrm{var}(x_t) = \mathrm{var}(r_{2t} - r_{1t}) = \mathrm{var}(r_{2t}) + \mathrm{var}(r_{1t}) - 2\,\mathrm{cov}(r_{2t}, r_{1t}) = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12},$$

so that the slope parameter β_1 is

$$\beta_1 = \frac{\mathrm{cov}(y_t, x_t)}{\mathrm{var}(x_t)} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} = w_1.$$

This demonstrates that the regression slope parameter is in that the optimal portfolio weight, w_1, associated with Microsoft. The optimal weight on Walmart is

$$w_2 = 1 - \beta_1\,.$$

The intercept in the population regression model is

$$\beta_0 \;=\; \mathrm{E}(y_t) - \beta_1\, \mathrm{E}(x_t) = \mathrm{E}(r_{2t}) - w_1\, \mathrm{E}(r_{2t} - r_{1t}) = w_1\mu_1 - (1 - w_1)\mu_2 = \mu_p,$$

which is the expected return on the minimum variance portfolio.
The disturbance term u_t for the minimum variance portfolio model has a precise interpretation. For this model, because $y_t = r_{2t}$, $x_t = r_{2t} - r_{1t}$ and $\beta_0 = \mu_p$, $\beta_1 = w_1$, the linear regression model is

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t\,, \\ r_{2t} &= \mu_p + w_1(r_{2t} - r_{1t}) + u_t\,. \end{aligned}$$

Upon rearranging this expression

$$u_t + \mu_p = w_1 r_{1t} + (1 - w_1) r_{2t}\,,$$

which shows that the sum of the disturbance term, u_t, and the mean of the portfolio, μ_p, represent the returns on the optimal portfolio. By construction, as $\mathrm{E}(u_t) = 0$, the mean return on the portfolio is

$$\mathrm{E}(u_t + \mu_p) = \mathrm{E}(u_t) + \mu_p = \mu_p \,,$$

which agrees with the definition of μ_p.
The sample counterpart of the bivariate population regression model is

$$y_t = \widehat{\beta}_0 + \widehat{\beta}_1 x_t + \widehat{u}_t \,,$$

where $\widehat{\beta}_i$ is the estimator of the population parameter β_i and $\widehat{u}_t$ is the residual. The estimators $\{\widehat{\beta}_0, \widehat{\beta}_1\}$ are obtained by replacing the population quantities in the formulae for $\{\beta_0, \beta_1\}$, by their sample estimates

$$\begin{aligned} \widehat{\beta}_1 &= \frac{\widehat{\sigma}_{yx}}{\widehat{\sigma}_{x^2}} = \frac{\frac{1}{T}\sum_{t=1}^{T}(y_t - \overline{y})(x_t - \overline{x})}{\frac{1}{T}\sum_{t=1}^{T}(x_t - \overline{x})^2} \\ \widehat{\beta}_0 &= \overline{y} - \widehat{\beta}_1 \overline{x}, \end{aligned}$$

where $\overline{y}$ and $\overline{x}$ are the sample means

$$\overline{y} = \frac{1}{T}\sum_{t=1}^{T} y_t, \qquad \overline{x} = \frac{1}{T}\sum_{t=1}^{T} x_t.$$

Formally, the least squares formulae are derived by choosing $\{\widehat{\beta}_0, \widehat{\beta}_1\}$ to minimise the residual sum of squares

$$RSS = \sum_{t=1}^{T} \widehat{u}_t^2 \,.$$

The algebra of this simple two-variable least squares problem is outlined in an Appendix to this Chapter.
Using the returns on Microsoft, r_{1t}, and Walmart, r_{2t}, to construct the variables $y_t = r_{2t}$ and $x_t = r_{2t} - r_{1t}$, the sample statistics of y_t and x_t are

$$\overline{y} = 0.013496, \quad \overline{x} = -0.007380, \quad \widehat{\sigma}_x^2 = 0.012331, \quad \widehat{\sigma}_{yx} = 0.003379 \,,$$

and the parameter estimates of the regression model are

$$\begin{aligned} \widehat{\beta}_1 &= \frac{\widehat{\sigma}_{yx}}{\widehat{\sigma}_{x^2}} = \frac{0.003379}{0.012331} = 0.274 \\ \widehat{\beta}_0 &= \overline{y} - \widehat{\beta}_1 \overline{x} = 0.013496 - 0.274 \times (-0.007380) = 0.015519 \,. \end{aligned}$$

These are same as the minimum variance estimates where $\widehat{\beta}_1 = 0.274$ is the estimated optimal weight on Microsoft and $\widehat{\beta}_0 = 0.015519$ is the estimate of the return on the portfolio.

The slope estimate, $\widehat{\beta}_1$, shows that a 1-unit increase in x_t causes a 0.274-unit increase in y_t on average. In the context of the minimum variance model, an increase in the spread of the returns on Walmart over Microsoft by 1 basis point results in the weight attached to Walmart in the portfolio increasing by 0.274 on average.
The intercept estimate, $\widehat{\beta}_0$, shows that a value of zero for the explanatory variable, $x_t = 0$, results in a value of 0.015519 for y_t on average. In the context of the minimum variance model, a zero value of x_t occurs when either there is no change in the price of each asset resulting in zero returns for each asset, or the returns on Microsoft and Walmart equal each other. In this case the estimate of the intercept corresponds to the return that would be obtained from having a minimum variance portfolio.
The bivariate linear regression model of the population is extended to include multiple explanatory variables as follows:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_K x_{K,t} + u_t\,, \tag{3.7}$$

in which y_t is the dependent variable which is a function of a constant, a set of K explanatory variables given by $x_{1t}, x_{2t}, \cdots, x_{K,t}$ and a disturbance term, u_t. The sample counterpart of (3.7) is

$$y_t = \widehat{\beta}_0 + \widehat{\beta}_1 x_{1t} + \widehat{\beta}_2 x_{2t} + \cdots + \widehat{\beta}_K x_{K,t} + \widehat{u}_t, \tag{3.8}$$

where $\widehat{\beta}_k$ is the estimator of the population parameters β_k, and $\widehat{u}_t$ represents the regression residual.
As with the bivariate regression model, the estimators $\{\widehat{\beta}_0, \widehat{\beta}_1, \cdots, \widehat{\beta}_K\}$ in the multiple regression model are chosen to minimise the residual sum of squares

$$RSS = \sum_{t=1}^{T} \widehat{u}_t^2\,.$$

For more details, see the Appendix to the Chapter.

3.4 Diagnostics

The estimated regression model is based on the assumption that the model is correctly specified. To test this assumption a number of diagnostic procedures are performed. These diagnostics are divided into three categories which relate to the key variables that summarise the model, namely, the dependent variable Y_t, the explanatory variables X_t and the disturbances u_t.

3.4.1 Diagnostics on the Dependent Variable

The fundamental aim of the linear regression model is to explain the movements in the dependent variable y_t. A natural measure of the success of an

estimated model is given by the proportion of the variation in the dependent variable explained by the model, given by the coefficient of determination

$$R^2 = \frac{\text{Explained sum of squares}}{\text{Total sum of squares}} = \frac{\sum_{t=1}^{T}(y_t - \overline{y})^2 - \sum_{t=1}^{T}\widehat{u}_t^2}{\sum_{t=1}^{T}(y_t - \overline{y})^2}.$$

The coefficient of determination satisfies the inequality $0 \leq R^2 \leq 1$. Values close to unity suggest a very good model fit and values close to zero representing a poor fit. For the minimum variance portfolio problem, the coefficient of determination is $R^2 = 0.1608$, showing that 16.08% of the variations in the returns on Microsoft, r_{2t}, are explained by variations in the spread of returns between Microsoft and Walmart, $r_{2t} - r_{1t}$.
A potential drawback with R^2 is that it never decreases when another variable is added to the model. From a statistical point of view, what is important in selecting explanatory variables is to include just those variables which significantly help to improve the explanatory power of the model. This is achieved by penalising the R^2 statistic for the addition of extra parameters. The adjusted coefficient of determination is given by

$$\overline{R}^2 = 1 - (1 - R^2)\frac{T-1}{T-K-1},$$

in which the adjustment factor $(T-1)/(T-K-1)$ becomes smaller as K the number of explanatory grows. This correction therefore represents a degrees of freedom correction to penalise the addition of additional variables that do not significantly help to raise R^2. For the minimum variance portfolio model $K = 1$, and the adjusted coefficient of determination is $\overline{R}^2 = 0.1558$. This means that 15.58% of the variations in the returns on Microsoft, r_{2t}, are explained by variations in the spread of returns between Microsoft and Walmart, $r_{2t} - r_{1t}$.
A related measure to the coefficient of determination is the standard error of the regression

$$\widehat{\sigma}_u = \sqrt{\frac{\sum_{t=1}^{T}\widehat{u}_t^2}{T-K-1}}, \tag{3.9}$$

which is simply the standard deviation of the ordinary least squares residuals. The standard error of the minimum variance portfolio regression is $\widehat{\sigma}_u = 0.069931$ and this represents an estimate of the volatility of the portfolio.
Computing the variance of this quantity results in a value of

$$\widehat{\sigma}_u^2 = 0.069931^2 = 4.8903 \times 10^{-3},$$

which is similar to the estimate of the risk computed for the minimum variance portfolio. The difference in the two risk estimates is due to the degrees of

freedom correction used to compute $\widehat{\sigma}_u^2$. Readjusting the estimate of risk with $K = 1$, gives a value of

$$\frac{T-2}{T}\widehat{\sigma}_u^2 = \frac{171-2}{171} \times 0.069931^2 = 0.004833,$$

which agrees with the risk estimate of $\widehat{\sigma}_p^2$ from the minimum variance calculations.

3.4.2 Diagnostics on the Explanatory Variables

As the aim of the regression model is to explain movements in the dependent variable over and above its mean $\overline{y}$, using information on the explanatory variables $x_{1t}, x_{2t}, \cdots, x_{Kt}$, this implies that for this information to be important the slope parameters $\beta_1, \beta_2, \cdots, \beta_K$ associated with these explanatory variables must be non-zero. To investigate this proposition tests are performed on these parameters individually and jointly.
To test the importance of a single explanatory variable in the regression equation, the associated parameter estimate is tested to see if it is zero using a t test. The null and alternative hypotheses are respectively

$$\begin{array}{lcll} H_0 & : & \beta_k = 0 & [x_{kt} \text{ is does not contribute to explaining } y_t] \\ H_1 & : & \beta_k \neq 0 & [x_{kt} \text{ is does contribute to explaining } y_t]. \end{array}$$

The t statistic to perform this test is

$$t_{T-K-1} = \frac{\widehat{\beta}_k}{\text{se}(\widehat{\beta}_k)}, \tag{3.10}$$

where $\widehat{\beta}_k$ is the estimated coefficient of β_k and $\text{se}(\widehat{\beta}_k)$ is the corresponding standard error. his test statistic follows the t distribution with $T - K - 1$ degrees of freedom, denoted t_{T-K-1}. The null hypothesis is rejected at the α significance level if the test yields a smaller p value

$$\begin{array}{lcl} p \text{ value} < \alpha & : & \text{Reject } H_0 \text{ at the } \alpha \text{ level of significance} \\ p \text{ value} > \alpha & : & \text{Fail to reject } H_0 \text{ at the } \alpha \text{ level of significance.} \end{array} \tag{3.11}$$

It is typical to choose $\alpha = 0.05$ as the significance level, which means that there is a 5% chance of rejecting the null hypothesis when it is actually true.
To perform a t test on the slope coefficient of the minimum variance regression model, the following quantities are needed

$$\begin{aligned} \widehat{\sigma}_u &= 0.069931, \\ \sum_{t=1}^{T} (x_t - \overline{x})^2 &= T\widehat{\sigma}_x^2 = 171 \times 0.012331 = 2.1086. \end{aligned}$$

The standard error of $\widehat{\beta}_1$ is computed as

$$\text{se}(\widehat{\beta}_1) = \sqrt{\frac{\widehat{\sigma}_u^2}{\sum_{t=1}^{T}(x_t - \overline{x})^2}} = \sqrt{\frac{0.069931^2}{2.1086}} = 4.8158 \times 10^{-2}.$$

The t statistic is

$$t = \frac{\widehat{\beta}_1}{\text{se}(\widehat{\beta}_1)} = \frac{0.274}{4.8158 \times 10^{-2}} = 5.6896.$$

This yields a p value of $0.000 < 0.05$ showing that the variable x_t is significant at the 5% level in determining movements in the dependent variable y_t.
As $w_1 = \beta_1$ for the minimum variance portfolio model, this test is also a test of portfolio diversification

$$\begin{array}{lcll} H_0 & : & w_1 = 0 & \text{[no gains from portfolio diversification]} \\ H_1 & : & w_1 \neq 0 & \text{[gains from portfolio diversification].} \end{array}$$

The result of the test suggests there are (statistical) gains from portfolio diversification.
The t test presented so far is designed to determine the importance of an explanatory variable by determining if the slope parameter is zero. More general tests can be performed to test for non-zero values of β_1 by using the t statistic

$$t = \frac{\widehat{\beta}_1 - \beta_1}{\text{se}(\widehat{\beta}_1)}.$$

For the minimum variance portfolio model a test of an equally weighted portfolio is given by the hypotheses

$$\begin{array}{lcll} H_0 & : & w_1 = 0.5 & \text{[equally weighted portfolio]} \\ H_1 & : & w_1 \neq 0.5 & \text{[non-equally weighted portfolio].} \end{array}$$

The t statistic is now

$$t = \frac{\widehat{\beta}_1 - \beta_1}{\text{se}(\widehat{\beta}_1)} = \frac{0.274 - 0.5}{4.8158 \times 10^{-2}} = -4.6929.$$

The p value is 0.000 showing that an equally-weighted portfolio is rejected at the 5% level.
More generally, tests of several restrictions can be examined jointly. In the case of the multiple regression model with K explanatory variables, a joint test of the significance of all of the explanatory variables is based on the null and alternative hypotheses

$$\begin{array}{ll} H_0: & \beta_1 = \beta_2 = ... = \beta_K = 0 \\ H_1: & \text{at least one } \beta_k \text{ is not zero.} \end{array}$$

Notice that this test does not include the intercept parameter β_0, so the total number of restrictions is K. Two tests of the null hypothesis are the χ^2 test

$$\chi^2_K = \frac{R^2}{(1-R^2)/(T-K-1)}, \tag{3.12}$$

which is distributed as χ^2 with K degrees of freedom and is a large sample test (that is, it works well when the sample size is large). Applying a small sample correction to the χ^2 statistic and dividing by K, yields the F test

$$F_{K,T-K-1} = \frac{R^2/K}{(1-R^2)/(T-K-1)}, \tag{3.13}$$

which follows an F distribution with K degrees of freedom in the numerator and $T-K-1$ degrees of freedom in the denominator.

3.4.3 Diagnostics on the Disturbance Term

The third set of diagnostic tests are based on the disturbance term, u_t. For the regression model to represent a well specified model there should be no information contained in the disturbance term. That is, u_t should be random for a correctly specified model

$$\begin{array}{lcll} H_0 & : & u_t \text{ is random} & \text{[model is specified correctly]} \\ H_1 & : & u_t \text{ is non-random} & \text{[model is misspecified].} \end{array}$$

If this condition is not satisfied, not only does this represent a violation of the assumptions underlying the linear regression model, but it also suggests that there are some arbitrage opportunities which can be used to improve predictions of the dependent variable.
This set of diagnostics is especially helpful in those situations where, for example, the fit of the model is poor as given by a small value of the coefficient of determination. In this situation, the specified model is only able to explain a small proportion of the overall movements in the dependent variable. But if it is the case that u_t is random, this suggests that the model cannot be improved despite a relatively large proportion of variation in the dependent variable is unexplained. In empirical finance this type of situation is perhaps the norm particularly in the case of modelling financial returns because the volatility tends to dominate the mean. In this noisy environment it is difficult to identify the signal in the data.

Residual Plots

A plot of $\widehat{u}_t$ over the sample provides an initial descriptive tool to identify potential patterns and abnormal returns. A sequence of positive (negative) residuals suggests that the model continually underestimates (overestimates) the dependent variable.

A plot of the residuals from the minimum variance regression defined by

$$\widehat{u}_t = y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t \tag{3.14}$$

is given in Figure 3.2.

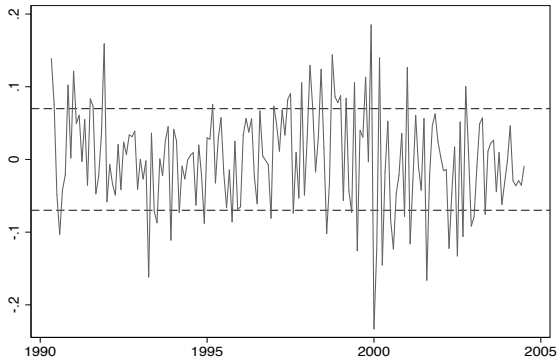


Figure 3.2: A plot of the Walmart residuals, u_t, computed using equation (3.14).

In this particular example, the adjusted residual $\widehat{u}_t + \widehat{\beta}_0$ would be the return at t if the minimum variance portfolio had been adopted. The plot is suggestive that u_t is not random as there appears to be a cycle in the data and volatility tends to vary over the sample. It follows that there is a possibility of further portfolio diversification by the inclusion of additional assets in the portfolio.

LM Test of Autocorrelation

This test is very important when using time series data. The aim of the test is to detect if the disturbance term is related to previous disturbance terms. The null and alternative hypotheses are respectively

$$\begin{array}{ll} H_0: & \text{No autocorrelation} \\ H_1: & \text{Autocorrelation.} \end{array}$$

If there is no autocorrelation this provides support for the model, whereas rejection of the null hypothesis suggests that the model excludes important information. The test consists of using the least squares residuals $\widehat{u}_t$ in the following equation

$$\widehat{u}_t = \gamma_0 + \gamma_1 x_{1t} + \gamma_2 x_{2t} + \cdots + \gamma_K x_{K,t} + \rho_1 \widehat{u}_{t-1} + v_t, \tag{3.15}$$

where v_t is a disturbance term. This equation is similar to the linear regression model (3.7) with the exception that y_t is replaced by $\widehat{u}_t$ and there is an

additional explanatory variable given by the lagged residual $\widehat{u}_{t-1}$. Note that in running this test regression, the missing value created by the need the lagged term $\widehat{u}_t$ is replaced with a zero (see, Davidson and MacKinnon, 1993). The test statistic is

$$TR^2 \sim \chi^2_1\,, \tag{3.16}$$

where T is the sample size, R^2 is the coefficient of determination from estimating (3.15) and the test statistic follows the chi-square distribution with one degree of freedom, χ^2_1.
This test of autocorrelation using (3.15) constitutes a test of first order autocorrelation. Extensions to higher order autocorrelation is straightforward. For example, a test for second order autocorrelation is based on the regression equation

$$\widehat{u}_t = \gamma_0 + \beta_1 x_{1t} + \gamma_2 x_{2t} + \cdots + \gamma_K x_{K,t} + \rho_1 \widehat{u}_{t-1} + \rho_2 \widehat{u}_{t-2} + v_t, \tag{3.17}$$

in which all the missing values due to the lagged terms $\widehat{u}_{t-i}$ are set equal to zero. The test statistic is still (3.16) and the test is now distributed as χ^2_2.
A test for first order autocorrelation of the residuals from the minimum variance model yields the statistic

$$TR^2 = 171 \times 0.007377 = 1.261\,.$$

Using the χ^2_1 distribution the p value is 0.2614 showing that the null hypothesis of no first order autocorrelation is not rejected at the 5% level.

White Test of Heteroskedasticty

White's test of heteroskedasticity (White, 1980) is important when using cross-section data or when modelling time-varying volatility, a topic that is dealt with in Part IV of the book. The aim of the test is to determine the constancy of the disturbance variance σ^2. The null and alternative hypotheses are respectively

$$\begin{array}{lll} H_0: & \text{Homoskedasticity} & [\sigma^2 \text{ is constant}] \\ H_1: & \text{Heteroskedasticity} & [\sigma^2 \text{ is time-varying}]. \end{array}$$

In the case of $K = 2$ explanatory variables, the test is based on the regression equation

$$\widehat{u}^2_t = \gamma_0 + \gamma_1 x_{1t} + \gamma_2 x_{2t} + \alpha_{11} x^2_{1t} + \alpha_{12} x_{1t} x_{2t} + \alpha_{22} x^2_{2t} + v_t \tag{3.18}$$

where v_t is a disturbance term. The test statistic is

$$TR^2 \sim \chi^2_5\,,$$

where T is the sample size and R^2 comes from the estimated equation and the distribution is χ^2_5 because there are 5 coefficients on possible explanatory

variables in equation (3.18) whose influence is being tested. Under the null hypothesis

$$\gamma_1 = \gamma_2 = \alpha_{11} = \alpha_{12} = \alpha_{22} = 0$$

and the population disturbance variance reduces to a constant given by $\sigma_u^2 = \gamma_0$.
The White test for heteroskedasticity of the residuals from the minimum variance model, in which there is only one explanatory variable, is based on the regression equation

$$\widehat{u}_t^2 = \gamma_0 + \gamma_1 x_{1t} + \alpha_{11} x_{1t}^2 + v_t. \tag{3.19}$$

The value of the White statistic is

$$TR^2 = 171 \times 0.0193 = 3.304,$$

which in this case follows the χ_2^2 distribution because there are only two potential explanatory variables in equation (3.19). Using the χ_2^2 distribution the p value is 0.1916, meaning that the null hypothesis of homoskedasticity is not rejected at the 5% level.

Normality Test

The assumption that u_t is normally distributed is important in performing hypothesis tests. A common way to test this assumption is the Jarque-Bera test . The null and alternative hypotheses are respectively:

$$\begin{array}{ll} H_0: & \text{Normality} \\ H_1: & \text{Nonnormality.} \end{array}$$

The test statistic is

$$\chi_2^2 = T\left(\frac{\text{SK}^2}{6} + \frac{(\text{KT}-3)^2}{24}\right), \tag{3.20}$$

where T is the sample size, and SK and KT are skewness and kurtosis, respectively, of the least squares residuals

$$\text{SK} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\widehat{u}_t}{\widehat{\sigma}_u}\right)^3, \qquad \text{KT} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\widehat{u}_t}{\widehat{\sigma}_u}\right)^4.$$

and $\widehat{\sigma}$ is the standard error of the regression in (3.9). The JB statistic is distributed as χ^2 with 2 degrees of freedom.
The value of the JB statistic of the residuals from the minimum variance model is $JB = 0.9771$. Using the χ_2^2 distribution, the p value is computed to be 0.6135, which leads to the conclusion that the null hypothesis of normality cannot be rejected at the 5% level.

3.5 Estimating the CAPM

An important application of the linear regression model is the capital asset pricing model (CAPM) which encapsulates the risk characteristics of an asset in terms of its beta risk,

$$\beta = \frac{\text{cov}(r_{it} - r_{ft}, r_{mt} - r_{ft})}{\text{var}(r_{mt} - r_{ft})}.$$

This quantity is a measure of the exposure of the returns on the asset to movements in the market, relative to a risk-free rate of interest. Individual stocks, or indeed portfolios of stocks, may be classified as follows in terms of their degree of beta risk:

Aggressive	:	$\beta > 1$,
Tracks the Market	:	$\beta = 1$,
Conservative	:	$0 < \beta < 1$,
Independence	:	$\beta = 0$,
Imperfect Hedge	:	$-1 < \beta < 0$,
Perfect Hedge	:	$\beta = -1$.

The CAPM is equivalent to the linear regression model

$$r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + u_t,$$

in which u_t is a disturbance term. The slope parameter β represents the asset's beta risk. The intercept parameter α represents the abnormal return to the asset over and above the asset's exposure to the excess return on the market. In terms of the model, the total risk of an asset can be decomposed as

$$\underbrace{\text{E}[(r_{it} - r_{ft})^2]}_{\text{Total risk}} = \underbrace{\text{E}[(\alpha + \beta(r_{mt} - r_{ft}))^2]}_{\text{Systematic risk}} + \underbrace{\text{E}(u_t^2)}_{\text{Idiosyncratic risk}},$$

since $\text{E}[(r_{mt} - r_{ft}), u_t] = 0$. The systematic risk is also known as non-diversifiable risk while the idiosyncratic risk represents the diversifiable risk. This analysis suggests that the standard error of the regression $\widehat{\sigma}_u$ provides an estimate of the idiosyncratic risk of the asset. Furthermore, the model R^2 or $\overline{R}^2$ provides an estimate of the proportion of the total risk of an asset that is non-diversifiable, and $1 - R^2$ represents the proportion that is diversifiable.

To illustrate the general ideas involved in estimating the CAPM, a data set comprising monthly returns to 10 United States industry portfolios for the period January 1927 to December 2013 is used, together with a benchmark monthly returns to the market and the monthly return on a risk free rate of interest. The industry portfolios are: consumer nondurables (ind1), consumer durables (ind2), manufacturing (ind3), energy (ind4), technology (ind5), telecommunications (ind6), wholesale and retail (ind7), healthcare (ind8), utilities (ind9) and a catch-all that includes mining, construction, entertainment and finance (ind10). The return on the market is constructed as the value-weight

return of all CRSP firms incorporated in the United States and listed on the NYSE, AMEX, or NASDAQ and the risk free rate is the 1-month U.S. Treasury Bill rate.

Table 3.1

Ordinary least squares estimates of the CAPM using 10 industry portfolios using monthly data for the U.S. beginning January 1927 and ending December 2013. Standard errors are given in parentheses and p values in square brackets.

Industry		α	β	$\overline{R}^2$	$\widehat{\sigma}_u$
Nondurables	(ind1)	0.2054	0.7577	0.7779	2.1962
		(0.0684)	(0.0125)		
Durables	(ind2)	0.0032	1.2443	0.7467	3.9320
		(0.1225)	(0.0224)		
Manufacturing	(ind3)	0.0081	1.1280	0.9234	1.7619
		(0.0549)	(0.0100)		
Energy	(ind4)	0.2307	0.8558	0.5949	3.8306
		(0.1193)	(0.0218)		
Technology	(ind5)	0.0094	1.2362	0.8250	3.0885
		(0.0962)	(0.0176)		
Telecommunications	(ind6)	0.1520	0.6573	0.5908	2.9675
		(0.0924)	(0.0169)		
Retail and Wholesale	(ind7)	0.1070	0.9694	0.7884	2.7245
		(0.0849)	(0.0155)		
Health	(ind8)	0.2549	0.8413	0.6498	3.3504
		(0.1044)	(0.0191)		
Utilities	(ind9)	0.0892	0.7820	0.5759	3.6399
		(0.1134)	(0.0207)		
Other	(ind10)	−0.1030	1.1261	0.8756	2.3026
		(0.0717)	(0.0131)		

The results obtained by estimating the CAPM for the 10 industry portfolios are given in Table 3.1. The aggressive portfolios ($\widehat{\beta}_1 > 1$) are durables, manufacturing, technology and other. The remaining 6 portfolios are conservative portfolios ($0 < \widehat{\beta}_1 < 1$). As expected none of the industry portfolios provide a perfect hedge against systematic risk.

For the retail and wholesale portfolio (ind7), the estimate of beta risk is 0.9694, indicating that this portfolio tracks the market closely. They following hypotheses are therefore of interest

$$\begin{array}{lcll} H_0 & : & \beta_1 = 1 & \text{[portfolio tracks the market perfectly]} \\ H_1 & : & \beta_1 \neq 1 & \text{[portfolio does not track the market perfectly].} \end{array}$$

The t statistic is testing this hypothesis is

$$t = \frac{\widehat{\beta}_1 - 1}{\text{se}(\widehat{\beta}_1)} = \frac{0.9694 - 1}{0.0155} = -1.9665\,.$$

The p value is 0.0495, which is just statistically significant at the 5% level, but not at the 1% level.

Manufacturing has the highest proportion of systematic (non-diversifiable) risk in terms of total risk with a $\overline{R}^2$ of 0.9234. The industry with the largest (absolute) level of idiosyncratic (diversifiable) risk is durables, followed by energy and utilities.

The CAPM has been extended in a number of ways to allow for additional determinants of excess returns. In a seminal paper, Fama and French (1993) augment the CAPM by including two additional risk factors to explain the return on a risky investment. These factors are: the performance of small stocks relative to big stocks (SMB), known as a 'Size' factor; and the performance of value stocks relative to growth stocks (HML), known as a 'Value' factor. In addition, Carhart (1997) suggests a further extension based on 'Momentum' (MOM) which captures the returns to a portfolio constructed by buying stocks with high returns over the past three to twelve months and selling stocks with low returns over the same period. The factor captures the herding behaviour of investors. The four factors are plotted in Figure 3.3.

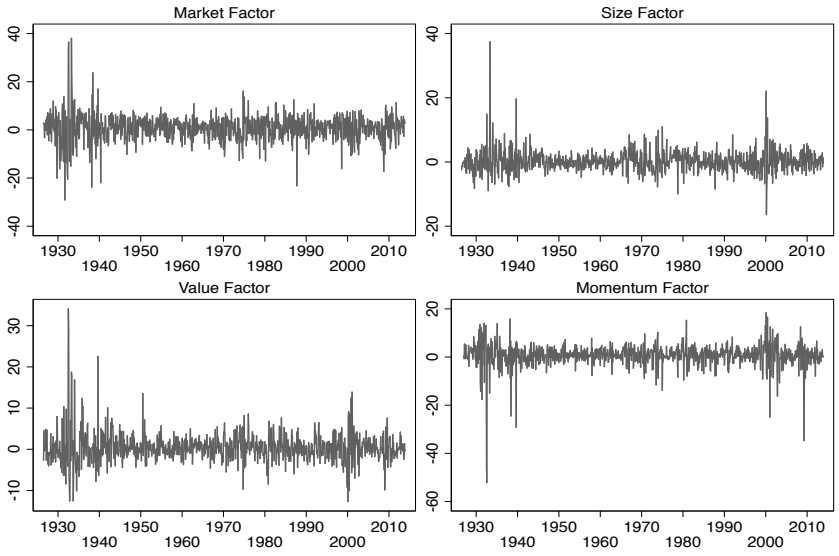


Figure 3.3: Monthly data for market, size, value and momentum factors of the extended CAPM model for the period January 1927 to December 2012.

The multi-factor CAPM is

$$r_{it} - r_{ft} = \alpha + \beta_1(r_{mt} - r_{ft}) + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{MOM}_t + u_t,$$

where u_t is a disturbance term. The contributions of SMB, HML and MOM are determined by the parameters β_2, β_3 and β_4 respectively. In the special case where these additional factors do not explain movements in the excess return on the asset $r_{it} - r_{ft}$, that is

$$\beta_2 = \beta_3 = \beta_4 = 0,$$

the model reduces to the standard CAPM regression equation.

Table 3.2

Ordinary least squares estimates of the multi-factor CAPM using 10 industry portfolios using monthly data for the U.S. beginning January 1927 and ending December 2013. Standard errors are given in parentheses.

Variable	Constant α	EMKT β_1	SMB β_2	HML β_3	MOM β_4	CAPM Test $(\beta_2 = \beta_3 = \beta_4)$
Nondurables	0.1786	0.7674	-0.0315	0.0284	0.0246	5.1183
	(0.0707)	(0.0140)	(0.0223)	(0.0211)	(0.0161)	(0.1633)
Durables	0.0889	1.1723	0.0338	0.1451	-0.1516	69.6072
	(0.1228)	(0.0243)	(0.0388)	(0.0367)	(0.0281)	(0.0000)
Manufacturing	-0.0117	1.1041	-0.0013	0.1276	-0.0213	83.3447
	(0.0547)	(0.0108)	(0.0173)	(0.0163)	(0.0125)	(0.0000)
Energy	0.0718	0.8949	-0.2103	0.2655	0.1156	89.0035
	(0.1186)	(0.0235)	(0.0374)	(0.0354)	(0.0271)	(0.0000)
Technology	0.1901	1.2424	0.0978	-0.3716	-0.0892	193.399
	(0.0915)	(0.0181)	(0.0289)	(0.0273)	(0.0209)	(0.0000)
Telecommunications	0.2582	0.6779	-0.1398	-0.0949	-0.0700	38.7436
	(0.0940)	(0.0186)	(0.0297)	(0.0281)	(0.0215)	(0.0000)
Retail and wholesale	0.1701	0.9596	0.0752	-0.1207	-0.0395	28.6710
	(0.0867)	(0.0172)	(0.0274)	(0.0259)	(0.0198)	(0.0000)
Health	0.2968	0.8925	-0.1049	-0.1662	0.0231	48.7621
	(0.1056)	(0.0209)	(0.0333)	(0.0316)	(0.0242)	(0.0000)
Utilities	0.0074	0.7768	-0.1798	0.3114	0.0082	119.0311
	(0.1112)	(0.0220)	(0.0351)	(0.0332)	(0.0255)	(0.0000)
Other	-0.1411	1.0425	0.0642	0.3349	-0.0805	539.8381
	0.0602	0.0119	0.0190	0.0180	0.0138	(0.0000)

The results obtained by estimating the multi-factor CAPM for the 10 industry portfolios are given in Table 3.2. As expected $\widehat{\beta}_1$ is highly significant, indicating that the market factor is still the dominant explanation of industry portfolio returns. The signs on the additional factors $\widehat{\beta}_2$, $\widehat{\beta}_3$ and $\widehat{\beta}_4$ change suggesting that different industries have vastly differing exposures to these factors. The last column of Table Table 3.2 gives the results of a test of the hypotheses of the validity of the multi-factor CAPM

$$\begin{array}{lcll} H_0 & : & \beta_2 = \beta_3 = \beta_4 = 0 & \text{[CAPM preferred]} \\ H_1 & : & \text{at least one restriction fails} & \text{[Multi-factor CAPM preferred].} \end{array}$$

Under the null hypothesis the χ^2 version of the joint test of significance will have 3 degrees of freedom.[1] In the case of durables, the value of the test statistic is $\chi^2_3 = 5.1183$. The p value computed using the χ^2 distribution is 0.1633 showing that the one-factor CAPM is not rejected at the 5% level for this portfolio. In other words, the additional risk factors are not priced in this portfolio. For the remaining industry portfolios, the joint tests of the significance of the additional factors are all statistically significant at the 5% level, indicating that the multi-factor CAPM is the preferred model and that the additional factors (SMB,HML,MOM) are factored into the price of risk for these portfolios.

3.6 Measuring Portfolio Performance

Define μ_p as the expected return on the portfolio, μ_m as the expected return on the market, r_f is the risk-free rate, σ_p as the risk of a portfolio and β as the beta risk obtained from CAPM. There are three commonly used metrics to measure portfolio performance.

Sharpe Ratio (Sharpe, 1966)

The Sharpe ratio is a measure of average excess return to the portfolio per unit of total portfolio risk. It is defined as

$$\mathcal{S} = \frac{\mu_p - r_f}{\sigma_p}.$$

The Sharpe ratio demonstrates how well the return of an asset compensates the investor for the risk taken. In particular, when comparing two risky assets the one with a higher Sharpe ratio provides better return for the same risk. The Sharpe ratio has proved very popular in empirical finance because it may be computed directly from any observed time series of returns.

Treynor Index (Treynor, 1966).

The Treynor ratio is defined as

$$\mathcal{T} = \frac{\mu_p - r_f}{\beta}.$$

Like the Sharpe ratio, this measure also gives a measure of excess returns per unit of risk, but is uses beta risk as the denominator and not total portfolio risk as in the Sharpe ratio.

Jensen's Alpha (Jensen, 1968)

Jensen's alpha is obtained from the CAPM regression as

$$\alpha = \mu_p - \beta(\mu_m - r_f).$$

[1] The χ^2 version of test is obtained by multiplying the F test by the degrees of freedom.

The performance of the 10 industry portfolios used in the CAPM application are now examined. In practice, the performance measures are estimated by replacing the population parameters by their sample estimates. The sample descriptive statistics for the key variables which are used to calculate the performance measures are given in Table 3.3.

Table 3.3

Summary statistics for monthly data on the excess returns to market portfolio, risk free rate of interest and returns to 10 United States industry portfolios for the period January 1927 and ending December 2013.

Variable	Mean	Std. Dev.	Skewness	Kurtosis
Excess market	0.6449	5.4265	0.1589	10.3545
Risk-free	0.2873	0.2547	1.0386	4.2202
Nondurables	0.9814	4.6608	−0.0427	8.7468
Durables	1.0930	7.7941	1.1298	17.0680
Manufacturing	1.0230	6.3553	0.8529	14.7770
Energy	1.0700	6.0093	0.1838	6.0085
Technology	1.0941	7.3712	0.2684	8.9494
Telecommunications	0.8632	4.6393	−0.0169	6.0209
Retail and wholesale	1.0195	5.9141	−0.0322	9.0310
Health	1.0848	5.6582	0.0991	9.5581
Utilities	0.8808	5.5909	0.0613	10.6150
Other	0.9105	6.5228	0.8470	15.8170

For the nondurables portfolio an illustration of computations of the performance measures is follows.

1. The Sharpe ratio is computed as

$$\widehat{\mathcal{S}}=\frac{\widehat{\mu}_p - r_f}{\widehat{\sigma}_p} = \frac{0.9814 - 0.2873}{4.6608} = 0.1489,$$

   where the risk-free rate is based on the sample mean given in the descriptive statistics table.

2. The Treynor Index is computed as

$$\widehat{\mathcal{T}} = \frac{\widehat{\mu}_p - r_f}{\widehat{\beta}} = \frac{0.9814 - 0.2873}{0.7577} = 0.9161,$$

   where $\widehat{\beta} = \widehat{\beta}_1$ which is the beta risk estimate from the CAPM regression equation for Nondurables.

3. Jensen's Alpha is computed as

$$\widehat{\alpha} = \widehat{\mu}_p - r_f - \widehat{\beta}(\widehat{\mu}_m - r_f),$$

which is obtained directly from the estimate of the intercept α in the CAPM regression equation.

The relative performance ranking of the 10 United States portfolios over the sample period based on the estimates of the performance measures are given in Table 3.4.

Table 3.4

Performance measures and rankings for 10 United States industry portfolios computed using monthly returns data for the period January 1927 to December 2013.

Variable	Sharpe Ratio	Treynor Index	Jensen's Alpha	Rank Sharpe	Rank Treynor
Nondurables	0.1489	0.9161	−0.2054	1	2
Durables	0.1034	0.6475	−0.0032	9	9
Manufacturing	0.1158	0.6522	−0.0081	6	8
Energy	0.1303	0.9146	−0.2307	3	3
Technology	0.1095	0.6527	−0.0094	7	7
Telecom.	0.1241	0.8762	−0.1520	4	4
Retail	0.1238	0.7553	−0.1070	5	6
Health	0.1410	0.9479	−0.2549	2	1
Utilities	0.1062	0.7590	−0.0892	8	5
Other	0.0955	0.5534	−0.1030	10	10

The correct treatment of risk in evaluating portfolio models has been the subject of much research. While it is well understood that adjusting the portfolio for risk is important, the exact nature of this adjustment is more problematic. The results in Table 3.4 highlight a feature that is commonly encountered in practical performance evaluation, namely, that the Sharpe and Treynor measures rank performance differently. Of course, this is not surprising because the Sharpe ratio accounts for total portfolio risk, while the Treynor measure adjusts excess portfolio returns for systematic risk only. The similarity between the rankings provided by Treynor's index and Jensen's alpha is also to be expected given that the alpha measure is derived from a CAPM regression which explicitly accounts for systematic risk via the inclusion of the market factor.

All of the rankings are consistent in one respect, namely that a positive alpha is a necessary condition for good performance and hence alpha is probably the most commonly used measure. The 'other' industry portfolio is the only portfolio to yield a negative estimate of alpha and hence is ranked last by all three metrics.

3.7 Qualitative Explanatory Variables

In all of the applications and examples investigated so far the explanatory variables are all quantitative whereby each variable takes on a different value for each sample observation. However, there are a number of applications in financial econometrics where it is appropriate to allow some of the explanatory variables to exhibit qualitative movements. Formally this is achieved by using a dummy variable which is 1 for an event and 0 for a non-event

$$I_t = \begin{cases} 0: & \text{[non-event]} \\ 1: & \text{[event].} \end{cases}$$

3.7.1 Stock Market Crashes

Consider the augmented present value model

$$P_t = \beta_0 + \beta_1 D_t + \beta_2 I_t + u_t,$$

where P_t is the stock market price, D_t is the dividend payment and u_t is a disturbance term. The variable I_t is a dummy variable that captures the effects of a stock market crash on the price of the asset

$$I_t = \begin{cases} 0: & \text{[pre-crash period]} \\ 1: & \text{[post-crash period].} \end{cases}$$

The effect of the dummy variable is to change the intercept in the regression

$$\begin{array}{ll} P_t = \beta_0 + \beta_1 D_t + u_t & : \text{[pre-crash period]} \\ P_t = (\beta_0 + \beta_2) + \beta_1 \mathrm{D}_t + u_t & : \text{[post-crash period].} \end{array}$$

For a stock market crash $\beta_2 < 0$,which represents a downward shift in the present value relationship between the asset price and dividend payment. An important stock market crash that began on 10 March 2000 is known at the dot-com crash because the stocks of technology companies fell sharply. The effect on one of the largest tech stocks, Microsoft, is highlighted in Figure 3.4 by the large falls in its share price over 2000. The biggest movement is in April 2000 where there is a negative return of -42.07% for the month. Modelling of Microsoft is also complicated by the unfavourable ruling of its antitrust case at the same time which would have exacerbated the size of the fall in April. Further inspection of the returns shows that there is a further fall in December of -27.94%, followed by a positive return of $+34.16\%$ in January of the next year.
To capture this phenomenon, consider the augmented CAPM

$$\begin{aligned} r_{it} - r_{ft} &= \beta_0 + \beta_1 \left(r_{mt} - r_{ft}\right) \\ &\quad + \beta_2 \ \mathrm{Apr00}_t + \beta_3 \ \mathrm{Dec00}_t + \beta_4 \ \mathrm{Jan01}_t + u_t, \end{aligned}$$

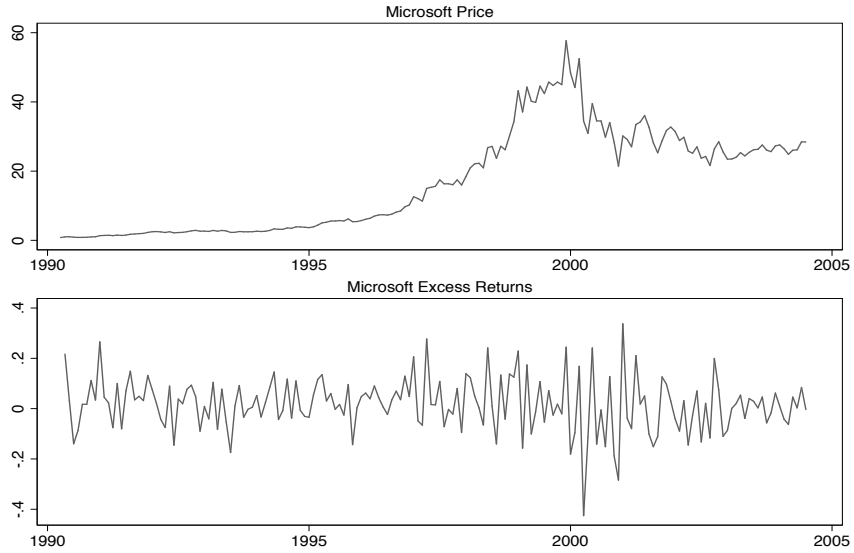


Figure 3.4: Monthly Microsoft price and returns for the period April 1990 to July 2004.

where Apr00_t, Dec00_t, and Jan01_t are dummy variables. The effect of the dummy variables is to change the intercept in the regression

$$\begin{array}{ll} r_{it} - r_{ft} = \beta_0 + \beta_1 \left(r_{mt} - r_{ft}\right) + \beta_2 \, \text{Apr00}_t + u_t & \text{[April 2000]} \\ r_{it} - r_{ft} = \beta_0 + \beta_1 \left(r_{mt} - r_{ft}\right) + \beta_3 \, \text{Dec00}_t + u_t & \text{[Dec. 2000]} \\ r_{it} - r_{ft} = \beta_0 + \beta_1 \left(r_{mt} - r_{ft}\right) + \beta_4 \, \text{Jan01}_t + u_t & \text{[Jan. 2001]} \\ r_{it} - r_{ft} = \beta_0 + \beta_1 \left(r_{mt} - r_{ft}\right) + u_t & \text{[Other months].} \end{array}$$

Introducing dummy variables for each of these three months into a CAPM model yields

$$\begin{aligned} r_{it} - r_{ft} \quad = \quad & 0.015 + 1.370 \left(r_{mt} - r_{ft}\right) - 0.391 \, \text{Apr00}_t \\ & -0.298 \, \text{Dec00}_t + 0.282 \, \text{Jan01}_t + \widehat{u}_t. \end{aligned}$$

The parameter estimates $\widehat{\beta}_2$ and $\widehat{\beta}_3$ are negative to reflect the falls in returns on these months, and $\widehat{\beta}_4$ is positive to reflect the (positive) market correction in January 2001.
Figure 3.5 gives histograms without and with these three dummy variables and show that the dummy variables are successful in purging the outliers from the tails of the distribution. This result is confirmed by the Jarque-Bera test because the JB statistic is found to have a p value of 0.651 for the augmented model, indicating that the null hypothesis of normality of the residuals cannot be rejected.

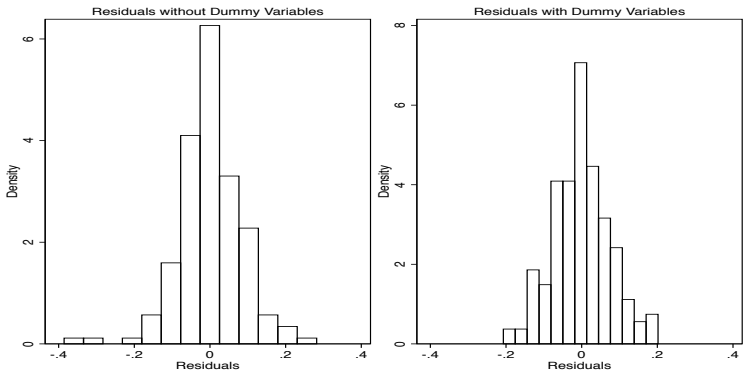


Figure 3.5: Histograms of residuals from a CAPM regression using Microsoft returns for the period April 1990 to July 2004, both with and without dummy variables for the dot-com crash.

3.7.2 Day-of-the-week Effects

Sometimes share prices exhibit greater movements on Monday than during the week. One reason for this extra volatility arises from the build up of information over the weekend when the stock market is closed. To capture this behaviour consider the regression model

$$r_t = \beta_0 + \beta_1 \text{Mon}_t + \beta_2 \text{Tue}_t + \beta_3 \text{Wed}_t + \beta_4 \text{Thu}_t + u_t,$$

where the data are daily. The dummy variables are defined as

$$\begin{aligned}
\text{Mon}_t &= \begin{cases} 0: & \text{[not Monday]} \\ 1: & \text{[Monday]} \end{cases} \\
\text{Tue}_t &= \begin{cases} 0: & \text{[not Tuesday]} \\ 1: & \text{[Tuesday]} \end{cases} \\
\text{Wed}_t &= \begin{cases} 0: & \text{[not Wednesday]} \\ 1: & \text{[Wednesday]} \end{cases} \\
\text{Thu}_t &= \begin{cases} 0: & \text{[not Thursday]} \\ 1: & \text{[Thursday]} \end{cases}
\end{aligned}$$

Notice that there are just 4 dummy variables to explain the 5 days of the week. This is because the setting of all dummy variables to zero

$$\text{Mon}_t = \text{Tue}_t = \text{Wed}_t = \text{Thu}_t = 0$$

defines the regression model on the Friday.

$$\begin{aligned}
r_{it} - r_{ft} &= \beta_0 + \beta_1 (r_{mt} - r_{ft}) + \beta_2 + u_t \quad \text{[Monday]} \\
r_{it} - r_{ft} &= \beta_0 + \beta_1 (r_{mt} - r_{ft}) + \beta_3 + u_t \quad \text{[Tuesday]} \\
r_{it} - r_{ft} &= \beta_0 + \beta_1 (r_{mt} - r_{ft}) + \beta_4 + u_t \quad \text{[Wednesday]} \\
r_{it} - r_{ft} &= \beta_0 + \beta_1 (r_{mt} - r_{ft}) + \beta_5 + u_t \quad \text{[Thursday]} \\
r_{it} - r_{ft} &= \beta_0 + \beta_1 (r_{mt} - r_{ft}) + u_t \quad \text{[Friday].}
\end{aligned}$$

3.7.3 Event Studies

Event studies are widely used in empirical finance to model the effects of qualitative changes arising from a particular event on financial variables. Typically events arise from some announcement caused by for example, a change in the CEO of a company, an unfavourable antitrust decision, or the effects of monetary policy announcements on the market. In fact, the stock market crash and day-of-the-week effects examples of dummy variables given above also constitute event studies. A typical event study involves specifying a regression equation based on a particular model to represent 'normal' returns, and then defining separate dummy variables at each point in time over the event window to capture the 'abnormal' returns, positive or negative. The parameter on a particular dummy is the 'abnormal' return at that point in time as it represents the return over and above the 'normal' return.
An example with a 5-day event window is

$$\begin{aligned}
r_t &= \underbrace{\beta_0 + \beta_1 r_{mt}}_{\text{'Normal' return}} \\
&+ \underbrace{\delta_{-2} I_{T-4} + \delta_{-1} I_{T-3} + \delta_0 I_{T-2} + \delta_1 I_{T-1} + \delta_2 I_T}_{\text{'Abnormal' return}} + u_t.
\end{aligned}$$

The abnormal return on the day of the announcement is δ_0, on the days prior to the announcement given by δ_{-2} and δ_{-1}, and on the days after the announcement given by δ_1 and δ_2.
The abnormal return for the whole of the event window is

$$\text{Total abnormal return } = \delta_{-2} + \delta_{-1} + \delta_0 + \delta_1 + \delta_2.$$

This suggests that a test of the statistical significance of the event and its effect on generating abnormal returns over the event window period is based on the restrictions

$$\begin{aligned}
H_0: &\quad \delta_{-2} = \delta_{-1} = \delta_0 = \delta_1 = \delta_2 = 0 \quad \text{[Normal returns]} \\
H_1: &\quad \text{at least one restriction is not valid} \quad \text{[Abnormal returns].}
\end{aligned}$$

A χ^2 test of the joint restrictions can be used which will have 5 degrees of freedom.

3.8 Exercises

1. **Minimum Variance Portfolios**

capm.wf1, capm.dta, capm.xlsx

Consider the equity prices of the United States companies Microsoft and Walmart for the period April 1990 to July 2004 ($T = 172$).

(a) Compute the continuously compounded (log) returns on Microsoft and Walmart.

(b) Compute the variance-covariance matrix of the returns on these two stocks. Verify that the covariance matrix of the returns is

$$\begin{bmatrix} 0.011332 & 0.002380 \\ 0.002380 & 0.005759 \end{bmatrix},$$

where the diagonal elements are the variances of the individual asset returns and the off-diagonal elements are the covariances. Note that the off-diagonal elements are in fact identical because the covariance matrix is a symmetric matrix. n computing the elements of the covariance matrix use the biased form presented in Chapter 2 in which T is used in the denominator instead of $T - 1$.

(c) Use the expressions in (3.5) and (3.6) to verify that the minimum variance portfolio weights between these two assets are

$$\begin{aligned} w_1 &= \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} = \frac{0.005759 - 0.002380}{0.011332 + 0.005759 - 2 \times 0.002380} = 0.274 \\ w_2 &= 1 - w_1 = 1 - 0.274 = 0.726. \end{aligned}$$

(d) Using the computed weights in part (c), compute the return on the portfolio as well as its mean and variance (without any degrees of freedom adjustment).

(e) Estimate the regression equation

$$r_{\text{Wmart},t} = \beta_0 + \beta_1(r_{\text{Wmart},t} - r_{\text{Msoft},t}) + u_t,$$

where u_t is a disturbance term.

i. Interpret the estimate of β_1 and discuss how it is related to the optimal portfolio weights computed in part (c).

ii. Interpret the estimate of β_0.

iii. Compute the variance of the least squares residuals, without any degrees of freedom adjustment, and interpret the result.

(f) Using the results in part (e)

i. Construct a test of an equal weighted portfolio, $w_1 = w_2 = 0.5$.
ii. Construct a test of portfolio diversification.

(g) Repeat parts (a) to (f) for Exxon and GE.

(h) Repeat parts (a) to (f) for gold and IBM.

2. **Estimating the CAPM**

capm.wf1, capm.dta, capm.xlsx

(a) Compute the monthly excess returns on Exxon, General Electric, Gold, IBM, Microsft and Walmart. Be particularly carefully when computing the correct risk free rate to use. [*Hint: the variable* TBILL *is quoted as an annual rate.*]

(b) Estimate the CAPM for each asset and interpret the estimated beta risk.

(c) For each asset, test the restrictions $\beta_0 = 0$ and $\beta_1 = 1$ individually and then test these restrictions jointly. Provide an interpretation of the CAPM if the restriction $\beta_0 = 0$ is valid.

3. **Fama-French Three Factor Model**

fama_french.wf1, fama_french.dta, fama_french.xlsx

The data set contains monthly the Fama-French market, risk free, size, book-to-market and momentum factors for the period January 1927 to December 2013. The return on the market is constructed as the value-weight return of all CRSP firms incorporated in the United States and listed on the NYSE, AMEX, or NASDAQ and the risk free rate is the 1-month U.S. Treasury Bill rate. The file also contains the monthly returns to 25 United States portfolios formed by sorting on size and book-to-market. The data is available for download from Ken French's webpage, `http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/`.

(a) For each of the 25 portfolios in the data set, estimate the CAPM and interpret the beta risk.

(b) Estimate the Fama-French three factor model for each portfolio and interpret the estimate of the beta risk and compare the estimate obtained in part (a).

(c) Perform a joint test of the size (SMB) and value (HML) risk factors in explaining excess returns in each portfolio.

4. **Present Value Model**

pv.wf1, pv.dta, pv.xlsx

The data file contains monthly United States data on equity prices and dividends for the period January 1871 to December 2013. Recall from Chapter 2 that the present value model for the price of an equity is equal to the discounted future stream of dividend payments

$$\begin{aligned} P_t &= D\left(\frac{1}{(1+\delta)} + \frac{1}{(1+\delta)^2} + ...\right) \\ &= \frac{D}{1+\delta}\left(1 + \frac{1}{(1+\delta)} + \frac{1}{(1+\delta)^2} + ...\right) \\ &= \frac{D}{1+\delta}\left(\frac{1}{1-1/(1+\delta)}\right) \\ &= \frac{D}{\delta}, \end{aligned}$$

where the penultimate step uses the sum of a geometric progression. This result implies that test of the present value model will be a test of the hypothesis $\beta_1 = 1$ in the linear regression model

$$p_t = \beta_0 + \beta_1 d_t + u_t,$$

where lowercase denotes logarithms and u_t is a disturbance term. The coefficient β_0 will be the estimate of δ.

(a) Estimate the model and interpret the parameter estimates.

(b) Examine the properties of the model by:

   i. plotting the ordinary least squares residuals;
   ii. testing for heteroskedasticity;
   iii. testing for autocorrelation; and
   iv. testing for normality.

(c) Test the restriction $\beta_1 = 1$ and interpret the result. In particular, interpret the estimate of β_0 when $\beta_1 = 1$.

5. **Fisher Hypothesis**

fisher.wf1, fisher.dta, fisher.xlsx

The data file contains United States quarterly data for the period 1954:Q3 to 2007:Q4 on the nominal interest rate, r_t, the price level, p_t, and inflation, π_t. The Fisher hypothesis states that nominal interest rates, r, fully reflect long-run movements in expected inflation, $\mathrm{E}(\pi)$, or

$$r = i + \mathrm{E}(\pi)$$

where i is the real interest rate. If the real interest rate is assumed to be constant then there will be a one-for-one adjustment of the nominal interest rate to the expected inflation rate.

To test this model in a linear regression setting consider model

$$r_t = \beta_0 + \beta_1 \mathrm{E}(\pi_t) + u_t,$$

where u_t is a disturbance term. Note that the expected rate of inflation is unobservable and so the model as it stands cannot be estimated. On the assumption that expectations are formed in a rational way so that, on average, expected inflation is equal to realised inflation, gives

$$\pi_t = \mathrm{E}(\pi_t) + w_t$$

where w_t is the zero-mean error in the expectation. A test of the Fisher hypothesis can be formulated as a test of $\beta_1 = 1$ in the linear regression model

$$r_t = \beta_0 + \beta_1 \pi_t + e_t,$$

in which e_t is now the composite error term $u_t - w_t$.

(a) Draw a scatter plot of r_t and π_t and superimpose a line of best fit in order to get a visual appreciation of the relationship between nominal interest rates and actual inflation.

(b) Estimate the linear regression version of the Fisher equation and interpret the parameter estimates.

(c) Test the restriction $\beta_1 = 1$ and interpret the result. In particular, interpret the estimate of β_0 when $\beta_1 = 1$.

(d) Draw a histogram of the residuals with a normal distribution overlaid on it. Do a Jarque-Bera test for normality of the residuals. Are the results consistent with your interpretation of the histogram.

6. **Measuring Portfolio Performance**

famafrench.wf1, famafrench.dta, famafrench.xlsx

The data set contains monthly the Fama-French market, risk free, size, book-to-market and momentum factors for the period January 1927 to December 2013. The file also contains the monthly returns to 10 United States industry portfolios, namely, nondurables, durables, manufacturing, energy, technology, telecommunications, retail/wholesale, health, utilities and other.

(a) Compute summary statistics for the monthly returns on the market portfolio, the risk free rate of interest and the 10 industry portfolios. Compare your results with Table 3.3.

(b) If μ_p is the expected return on the portfolio, μ_m is the expected return on the market, r_f is the risk-free rate, σ_p as the risk of a portfolio and β as the beta risk obtained from CAPM, then compute the following estimated portfolio performance measures for the nondurables portfolio.

i. The Sharpe ratio

$$\widehat{\mathcal{S}}=\frac{\widehat{\mu}_p - r_f}{\widehat{\sigma}_p}.$$

ii. The Treynor Index

$$\widehat{\mathcal{T}} = \frac{\widehat{\mu}_p - r_f}{\widehat{\beta}}.$$

iii. Jensen's Alpha

$$\widehat{\alpha} = \widehat{\mu}_p - r_f - \widehat{\beta}(\widehat{\mu}_m - r_f).$$

which is obtained directly from the estimate of the intercept α in the CAPM regression equation.

Compare your results with Table 3.4.

(c) Recompute Jensen's Alpha but this time use a three factor CAPM. Is there any substantive change in the results?

(d) Repeat steps (b) and (c) for the technology portfolio.

7. **Microsoft and the Dot-Com Crisis**

capm.wf1, capm.dta, capm.xlsx

The dot-com crash began on 10 March 2000.

(a) Plot the price of Microsoft shares and the associated log returns. Verify that the biggest falls in the share price occurs in April 2000 where there is a negative return of 42.07% for the month and that the large negative return of 27.94% in December 2000 is followed by a correction of 34.16% in January 2001.

(b) Estimate the CAPM model for Micosoft

$$r_{it} - r_{ft} = \beta_0 + \beta_1(r_{mt} - r_{ft}) + u_t,$$

in which r_{ft} and r_{mt} are the risk free and market returns, respectively. Draw a line time series plot the residuals and see if these large returns are evident.

(c) Draw a histogram of the residuals with a normal distribution overlaid. Test the residuals for normality using a Jarque-Bera test. Comment on your results.

(d) Construct the dummy variables

$$D_{1t} = \begin{cases} 1: & \text{Apr. 2000} \\ 0: & \text{Otherwise} \end{cases},$$

$$D_{2t} = \begin{cases} 1: & \text{Dec. 2000} \\ 0: & \text{Otherwise} \end{cases},$$

$$D_{3t} = \begin{cases} 1: & \text{Jan. 2001} \\ 0: & \text{Otherwise} \end{cases}.$$

Estimate the regression

$$r_{it} - r_{ft} = \beta_0 + \beta_1(r_{mt} - r_{ft}) + \gamma_1 D_{1t} + \gamma_2 D_{2t} + \gamma_3 D_{3t} + u_t.$$

Comment on the results.

(e) Test the null hypothesis that all the γ_i coefficients are zero.

(f) Draw a histogram of the residuals with a normal distribution overlaid. Test the residuals for normality using a Jarque-Bera test. Compare these results with those in part (c).

8. **The Retirement of Lee Raymond as the CEO of Exxon**

capm.wf1, capm.dta, capm.xlsx

In December of 2005, Lee Raymond retired as the CEO of Exxon receiving the largest retirement package ever recorded of around $400m. How did the markets view this event?

(a) Estimate the market model for Exxon from January 1970 to September 2005

$$r_t = \beta_0 + \beta_1 r_{mt} + u_t,$$

where r_t is the log return on Exxon and r_{mt} is the market return computed from the S&P500. Verify that the result is

$$r_t = 0.009 + 0.651\ r_{mt} + \widehat{u}_t,$$

where $\widehat{u}_t$ is the residual.

(b) Construct the dummy variables

$$\begin{aligned} D_{2005:10,t} &= \begin{cases} 1: & \text{Oct. 2005} \\ 0: & \text{Otherwise} \end{cases}, \\ D_{2005:11t} &= \begin{cases} 1: & \text{Nov. 2005} \\ 0: & \text{Otherwise} \end{cases}, \\ &\vdots \\ D_{2006:2t} &= \begin{cases} 1: & \text{Feb. 2006} \\ 0: & \text{Otherwise} \end{cases}, \end{aligned}$$

(c) Re-estimate the market model including the 5 dummy variables constructed in part (b) over the extended sample from January 1970 to February 2006. Verify that the estimated regression equation is

$$\begin{aligned} r_t &= 0.009 + 0.651\, r_{mt} - 0.121\, \text{Oct05}_t + 0.007\, \text{Nov05}_t - 0.041\, \text{Dec05}_t \\ &\quad + 0.086\, \text{Jan06}_t - 0.059\, \text{Feb06}_t + \widehat{u}_t\,. \end{aligned}$$

i. What is the relationship between the parameter estimates of β_0 and β_1 computed in parts (a) and (c)?

ii. Do you agree that the total estimated abnormal return on Exxon from October 2005 to February 2006 is

$$\begin{aligned} \text{Total abnormal return} &= -0.121 + 0.007 - 0.041 + 0.086 - 0.059 \\ &= -0.128. \end{aligned}$$

(d) An alternative way to compute abnormal returns is to use the estimated model in part (a) and substitute in the values of r_{mt} for the event window. As the monthly returns on the market for this period are

$$\{-0.0179, 0.0346, -0.0009, 0.0251, 0.0004\}\,,$$

recompute the abnormal returns. Compare these estimates with the estimates obtained in part (c).

(e) Perform the following tests of abnormal returns.

i. There was no abnormal return at the time of retirement on Decemberv2005.

ii. There were no abnormal returns before retirement.

iii. There were no abnormal returns after retirement.

iv. There were no abnormal returns at all.

APPENDIX to Chapter 3: Some Results for the Linear Regression Model

This Appendix provides a limited derivation of the ordinary least squares estimators of the multiple linear regression model and also the sampling distributions of the estimators. Attention is focussed on a model with one independent variable and two explanatory variables in order to give some insight into the general result.
Consider the linear regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t\,, \qquad u_t \sim iid\, N(0, \sigma^2)\,. \tag{3.21}$$

The residual sum of squares is given by

$$RSS(\widehat{\beta}) = \sum_{t=1}^{T} \widehat{u}_t^2 = y_t = \sum_{t=1}^{T} (\widehat{\beta}_0 + \widehat{\beta}_1 x_t)^2 \tag{3.22}$$

Differentiating RSS with respect to $\{\beta_0$ and β_1 and setting the results equal to zero yields

$$\begin{array}{rcl} \dfrac{\partial RSS}{\partial \beta_0} & = & \sum\limits_{t=1}^{T} (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t) = 0 \\ \dfrac{\partial RSS}{\partial \beta_1} & = & \sum\limits_{t=1}^{T} (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t) x_t = 0\,. \end{array} \tag{3.23}$$

This system of first-order conditions can be written in matrix form as

$$\begin{bmatrix} \sum_{t=1}^{T} y_t \\ \sum_{t=1}^{T} y_t x_t \end{bmatrix} - \begin{bmatrix} T & \sum_{t=1}^{T} x_t \\ \sum_{t=1}^{T} x_t & \sum_{t=1}^{T} x_t^2 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and solving for $[\ \widehat{\beta}_0,\ \widehat{\beta}_1\ ]'$ gives

$$\begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} T & \sum_{t=1}^{T} x_t \\ \sum_{t=1}^{T} x_t & \sum_{t=1}^{T} x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^{T} y_t \\ \sum_{t=1}^{T} x_t y_t \end{bmatrix}, \tag{3.24}$$

which is the ordinary least squares estimator $\widehat{\beta} = [\ \widehat{\beta}_0 \quad \widehat{\beta}_1\ ]'$.
Equation (3.24) can be expressed slightly differently by defining the vector $\boldsymbol{x}_t = [\ 1 \quad x_t\ ]'$. Now given that

$$\boldsymbol{x}_t \boldsymbol{x}_t' = \begin{bmatrix} 1 \\ x_t \end{bmatrix} \begin{bmatrix} 1 & x_t \end{bmatrix} = \begin{bmatrix} 1 & x_t \\ x_t & x_t^2 \end{bmatrix}$$

and

$$\boldsymbol{x}_t y_t = \begin{bmatrix} 1 \\ x_t \end{bmatrix} y_t = \begin{bmatrix} y_t \\ x_t y_t \end{bmatrix},$$

the ordinary least squares estimator of β in (3.24) can be re-expressed as

$$\widehat{\beta} = \Big[\sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{x}_t'\Big]^{-1} \sum_{t=1}^{T} \boldsymbol{x}_t y_t \,. \tag{3.25}$$

The inverse of the 2×2 matrix is given by

$$\begin{bmatrix} T & \sum_{t=1}^{T} x_t \\ \sum_{t=1}^{T} x_t & \sum_{t=1}^{T} x_t^2 \end{bmatrix}^{-1} = \frac{1}{T\sum_{t=1}^{T} x_t^2 - \big(\sum_{t=1}^{T} x_t\big)^2} \begin{bmatrix} \sum_{t=1}^{T} x_t^2 & \sum_{t=1}^{T} x_t \\ -\sum_{t=1}^{T} x_t & T \end{bmatrix}.$$

Using this matrix inverse and rules for matrix multiplication the solution for $\widehat{\beta}_1$ becomes

$$\begin{aligned} \widehat{\beta}_1 &= \frac{T\sum_{t=1}^{T} x_t y_t - \sum_{t=1}^{T} x_t \sum_{t=1}^{T} y_t}{T\sum_{t=1}^{T} x_t^2 - \big(\sum_{t=1}^{T} x_t\big)^2} \\ &= \frac{T\sum_{t=1}^{T} x_t y_t - T\overline{x}\, T\overline{y}}{T\sum_{t=1}^{T} x_t^2 - \big(T\overline{x}\big)^2} \\ &= \frac{\sum_{t=1}^{T} x_t y_t - T\overline{x}\,\overline{y}}{\sum_{t=1}^{T} x_t^2 - T\overline{x}^2} \\ &= \frac{\mathrm{cov}(y_t\, x_t)}{\mathrm{var}(x_t)} \,. \end{aligned}$$

Once the ordinary least squares estimates have been computed, the ordinary least squares estimator, s^2, of the variance, σ^2, is obtained from

$$s^2 = \frac{1}{T} \sum_{t=1}^{T} (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t)^2 \,. \tag{3.26}$$

In computing s^2 in equation (3.26) it is common to express the denominator in terms of the degrees of freedom, $T - K - 1$ instead of merely T, where $K = 1$ is the number of explanatory variables excluding the constant. of course, if $K > 1$, estimation of σ^2 proceeds exactly as in equation (3.26) where, of course, the appropriate number of regressors and coefficients are now included in the computation.

Distribution of the Estimators

The advantage of this notation is that it is completely general and is not limited to the case of $K = 1$. In the event of K regressors the relevant matrix and vector are defined accordingly and the estimation is straightforward. The ordinary least squares estimator of the parameters of the K variable regression model, $\beta = [\beta_0 \quad \beta_1 \quad \cdots \beta_K]'$ given in equation (3.25) may be written as

$$\widehat{\beta} = \Big[\sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{x}_t'\Big]^{-1} \sum_{t=1}^{T} \boldsymbol{x}_t y_t = \beta + \Big[\sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{x}_t'\Big]^{-1} \sum_{t=1}^{T} \boldsymbol{x}_t u_t \,, \tag{3.27}$$

where the last term is obtained by substituting for y_t from regression equation (3.21). It is usual to write equation (3.27) in the form

$$\sqrt{T}(\widehat{\beta} - \beta) = \Big[\frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{x}_t'\Big]^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \boldsymbol{x}_t u_t \,, \tag{3.28}$$

The distribution of the estimator $\widehat{\beta}$, is therefore going to depend crucially on the behaviour of the two terms on the right hand side of (3.28).
The distribution of the estimator is established in three steps.

Law of Large Numbers The law of large numbers states that for very weak conditions on $\boldsymbol{x}_t = [1 \quad x_t]'$, the sample covariance matrix of $\boldsymbol{x}_t$ converges, as the sample size gets infinitely large, to the true covariance matrix of $\boldsymbol{x}_t$, denoted M_{xx}. In other words

$$\text{plim} \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{x}_t' = \text{E}(\boldsymbol{x}_t \boldsymbol{x}_t') = M_{xx}.$$

Central Limit Theorem The central limit theorem is a statement about the limiting distribution of scaled sums of random variables. Because the variables u_t and $\boldsymbol{x}_t$ are independently and identically distributed, the Lindberg-Levy Central limit theorem can be used to claim that[2]

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} \boldsymbol{x}_t u_t \xrightarrow{d} N(0, M_{xx}\,\sigma^2\,).$$

In other words, the limiting distribution of $\widehat{\beta}$ is a normal distribution with mean 0 and variance $\sigma^2 M_{xx}$. The mean and the variance of the distribution follow from two results which rely critically on the independence assumption, $\text{E}(\boldsymbol{x}_t u_t) = 0$. These results are

$$\text{E}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} \boldsymbol{x}_t u_t\right) = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} \text{E}(\boldsymbol{x}_t u_t) = 0$$
$$\text{var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} \boldsymbol{x}_t u_t\right) = \frac{1}{T}\left(\sum_{t=1}^{T} \text{var}(\boldsymbol{x}_t u_t)\right) = \frac{1}{T}\left(\sum_{t=1}^{T} \text{E}[\boldsymbol{x}_t \boldsymbol{x}_t']\sigma^2\right) = M_{xx}\,\sigma^2.$$

Slutsky's Theorem Slutsky's theorem says that if a variable Y_t converges in probability to a distribution, D, and X_t converges in probability to a constant c then

$$X_t Y_t \xrightarrow{d} cD.$$

[2]For a more detailed discussion of these conditions and the appropriate choice of central limit theorem see, Hamilton (1994) or Martin, Hurn and Harris (2013).

Using Slutsky's theorem with the two results established previously in terms of the law of large numbers and the central limit theorem, it follows immediately that

$$\underbrace{\left[\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{x}_t\boldsymbol{x}_t'\right]^{-1}}_{X_t}\underbrace{\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\boldsymbol{x}_t u_t}_{Y_t} \xrightarrow{d} \underbrace{M_{xx}^{-1}}_{c}\ \underbrace{N(0, M_{xx}\,\sigma^2)}_{D}.$$

Distribution of $\widehat{\beta}$ The final step is to move the term in M_{xx}^{-1} into the variance of the distribution (remembering to square it), so that the final result is

$$\sqrt{T}(\widehat{\beta}-\beta) \xrightarrow{d} N(0, M_{xx}^{-1}\,\sigma^2),$$

or

$$\widehat{\beta} \overset{a}{\sim} N(\beta, \frac{1}{T}M_{xx}^{-1}\,\sigma^2).$$

Chapter 4

Modelling with Stationary Variables

4.1 Introduction

An important feature of the linear regression model discussed in Chapter 3 is that all variables are designated at the same point in time. To allow for financial variables to adjust to shocks over time the linear regression model is extended to allow for a range of dynamics. The first class of dynamic models developed is univariate whereby a single financial variable is modelled using its own lags as well as lags of our financial variables. Then multivariate specifications are developed in which several financial variables are jointly modelled.

An important characteristic of the multivariate class of models investigated in the chapter is that each variable in the system is expressed as a function of its own lags as well as the lags of all of the other variables in the system. This model is known as a vector autoregression (VAR), model that is characterised by the important feature that every equation has the same set of explanatory variables. This feature of a VAR has several advantages. First, estimation is straightforward, being simply the application of ordinary least squares applied to each equation one at a time. Second, the model provides the basis of performing causality tests which can be used to quantify the value of information in determining financial variables. These tests can be performed in three ways beginning with Granger causality tests, impulse response functions and variance decompositions. Third, multivariate tests of financial theories can be undertaken as these theories are shown to impose explicit restrictions on the parameters of a VAR which can be verified empirically. Fourth, the VAR provides a very convenient and flexible forecasting tool to compute predictions of financial variables, a topic that is investigate further in Chapter 7.

4.2 Stationarity

The models in this chapter, which use standard linear regression techniques, require that the variables involved satisfy a condition known as stationarity. Stationarity, or more correctly, its absence is the subject matter of Chapters 5 and 6. For the present a simple illustration will indicate the main idea. Consider Figures 4.1 and 4.2 which show the daily S&P500 index and associated log returns, respectively.

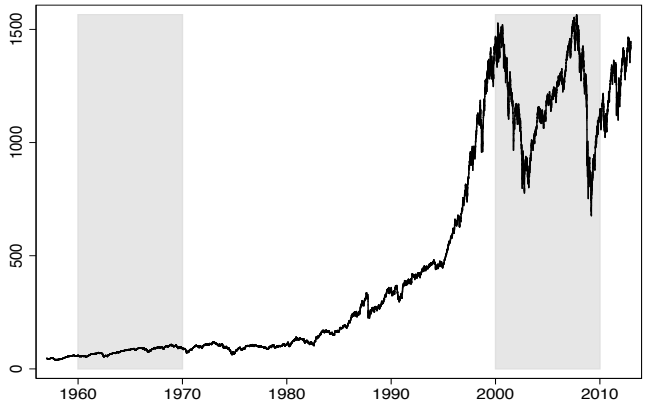


Figure 4.1: Snapshots of the time series of the S&P500 index comprising daily observations for the period January 1957 to December 2012.

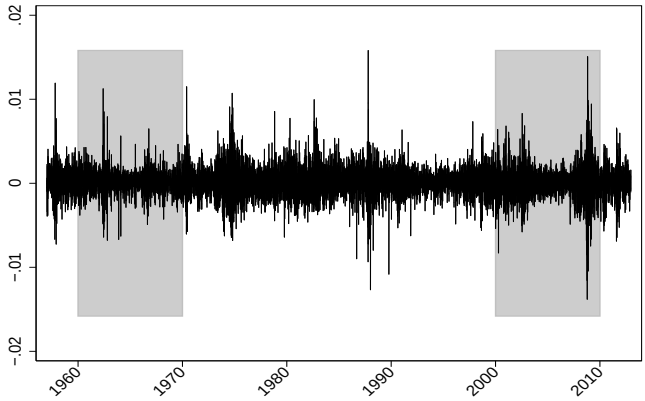


Figure 4.2: Snapshots of the time series of S&P500 log returns computed from daily observations for the period January 1957 to December 2012.

Assume that an observer is able to take a snapshot of the two series at differ-

ent points in time; the first snapshot shows the behaviour of the series for the decade of the 1960s and the second shows their behaviour from 2000-2010. It is clear that the behaviour of the series in Figure 4.1 is completely different in these two time periods. What the impartial observer sees in 1960-1970 looks nothing like what happens in 2000-2010. The situation is quite different for the log returns plotted in Figure 4.2. To the naked eye the behaviour in the two shaded areas is remarkable similar given that the intervening time span is 30 years.
In both this chapter and the next chapter it will simply be assumed that the series we deal with exhibit behaviour similar to that in Figures 4.2. This assumption is needed so that past observations can be used to estimate relationships, interpret the relationships and forecast future behaviour by extrapolating from the past. In practice, of course, stationarity must be established using the techniques described in Chapter 5. It is not sufficient merely to assume that the condition is satisfied.

4.3 Univariate Autoregressive Models

4.3.1 Specification

The simplest specification of a dynamic model of the dependent variable y_t is where the explanatory variables are the own lags of the dependent variable

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + u_t, \tag{4.1}$$

in which $\phi_0, \phi_1, \cdots, \phi_p$ are unknown parameters and u_t is a disturbance term with zero mean and variance σ^2. This equation shows that the information used to explain movements in y_t are the own lags with the longest lag being the pth lag. This property is formally represented by the conditional expectations operator which gives the predictor of y_t based on information available at time $t-1$

$$\mathrm{E}_{t-1}(y_t) = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}. \tag{4.2}$$

Equation (4.1) is referred to as an autoregressive model with p lags, or simply AR(p). Estimation of the unknown parameters is achieved by using ordinary least squares. These parameter estimates can also be used to identify the role of past information by performing tests on the parameters.

4.3.2 Properties

To understand the properties of AR models, consider the AR(1) model

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t, \tag{4.3}$$

where $|\phi_1| < 1$, a condition which ensures that y_t is stationary. One of the important implications of stationarity is that $\mathrm{E}(y_t) = \mathrm{E}(y_{t-1})$, so that by applying the unconditional expectations operator to both sides of (4.3) gives

$$\mathrm{E}(y_t) = \mathrm{E}(\phi_0 + \phi_1 y_{t-1} + u_t) = \phi_0 + \phi_1 \mathrm{E}[y_{t-1}].$$

The unconditional mean of y_t is therefore

$$\mathrm{E}(y_t) = \frac{\phi_0}{1-\phi_1}.$$

The unconditional variance is defined as

$$\gamma_0 = \mathrm{E}\{[y_t - \mathrm{E}(y_t)]^2\}.$$

Now

$$y_t - \mathrm{E}(y_t) = (\phi_0 + \phi_1 y_{t-1} + u_t) - [\phi_0 + \phi_1 \mathrm{E}(y_{t-1})] = \phi_1[y_{t-1} - \mathrm{E}(y_{t-1})] + u_t.$$

Squaring both sides and taking unconditional expectations gives

$$\begin{aligned} \mathrm{E}\{[y_t - \mathrm{E}(y_t)]^2\} &= \phi_1^2 \mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]^2\} + \mathrm{E}(u_t^2) + 2\,\mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]u_t\} \\ &= \phi_1^2 \mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]^2\} + \mathrm{E}(u_t^2), \end{aligned}$$

using the fact that $\mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]u_t\} = 0$. Moreover, because

$$\gamma_0 = \mathrm{E}\{[y_t - \mathrm{E}(y_t)]^2\} = \mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]^2\},$$

it follows that

$$\gamma_0 = \phi_1^2 \gamma_0 + \sigma^2,$$

which upon rearranging gives

$$\gamma_0 = \frac{\sigma^2}{1-\phi_1^2}.$$

The first order autocovariance is

$$\begin{aligned} \gamma_1 &= \mathrm{E}\{[y_t - \mathrm{E}(y_t)][y_{t-1} - \mathrm{E}(y_{t-1})]\} \\ &= \mathrm{E}\{[\phi_1 y_{t-1} - \phi_1 \mathrm{E}(y_{t-1}) + u_t][y_{t-1} - \mathrm{E}(y_{t-1})]\} \\ &= \phi_1 \mathrm{E}\{[y_{t-1} - \mathrm{E}(y_{t-1})]^2\} \\ &= \phi_1 \gamma_0. \end{aligned}$$

It follows that the kth autocovariance is

$$\gamma_k = \phi_1^k \gamma_0. \tag{4.4}$$

It immediately follows from this result that the autocorrelation function (ACF) of the AR(1) model is

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \phi_1^k.$$

For $0 < \phi_1 < 1$, the autocorrelation function declines for increasing k so that the effects of previous values on y_t gradually diminish. For higher order AR models the properties of the ACF are in general more complicated.
To compute the ACF, the following sequence of AR models are estimated by ordinary least squares one equation at a time:

$$\begin{aligned} y_t &= \phi_{10} + \rho_1 y_{t-1} + u_{1t}, \\ y_t &= \phi_{20} + \rho_2 y_{t-2} + u_{2t}, \\ &\vdots \quad\quad \vdots \quad\quad\quad \vdots \\ y_t &= \phi_{30} + \rho_k y_{t-k} + u_{kt}, \end{aligned}$$

where the estimated ACF is given by $\{\widehat{\rho}_1, \widehat{\rho}_2, \cdots, \widehat{\rho}_k\}$. The notation adopted for the constant term emphasises that this term will be different for each equation.
Another measure of the dynamic properties of AR models is the partial autocorrelation function (PACF), which measures the relationship between y_t and y_{t-k} but now with the intermediate lags included in the regression model. The PACF at lag k is denoted as ϕ_{kk}. By implication the PACF for an AR(p) model is zero for lags greater than p. For example, in the AR(1) model the PACF has a spike at lag 1 and thereafter is $\phi_{kk} = 0, \ \forall \ k > 1$. This is in contrast to the ACF which in general has non-zero values for higher lags. Note that by construction the ACF and PACF at lag 1 are equal to each other.
To compute the PACF the following sequence of AR models are estimated by ordinary least squares, again one equation at a time:

$$\begin{aligned} y_t &= \phi_{10} + \phi_{11} y_{t-1} + u_{1t} \\ y_t &= \phi_{20} + \phi_{21} y_{t-1} + \phi_{22} y_{t-2} + u_{2t} \\ y_t &= \phi_{30} + \phi_{31} y_{t-1} + \phi_{32} y_{t-2} + \phi_{33} y_{t-3} + u_{3t} \\ &\vdots \quad\quad \vdots \quad\quad\quad \vdots \quad\quad\quad \vdots \\ y_t &= \phi_{k0} + \phi_{k1} y_{t-1} + \phi_{k2} y_{t-2} + \cdots + \phi_{kk} y_{t-k} + u_{kt}, \end{aligned}$$

where the estimated PACF is therefore given by $\{\widehat{\phi}_{11}, \widehat{\phi}_{22}, \cdots, \widehat{\phi}_{kk}\}$.
Consider United States monthly data on real equity returns expressed as a percentage, r_t, from February 1871 to June 2004. The ACF and PACF of the equity returns are computed by means of a sequence of regressions. The ACF for lags 1 to 3 is computed using the following three regressions (standard errors in parentheses):

$$\begin{aligned} r_t &= \underset{(0.099)}{0.247} + \underset{(0.024)}{0.285}\ r_{t-1} + \widehat{v}_{1t}, \\ r_t &= \underset{(0.103)}{0.342} + \underset{(0.025)}{0.008}\ r_{t-2} + \widehat{v}_{2t}, \\ r_t &= \underset{(0.103)}{0.361} - \underset{(0.025)}{0.053}\ r_{t-3} + \widehat{v}_{3t}\,. \end{aligned}$$

The estimated ACF is

$$\{\widehat{\rho}_1 = 0.285, \widehat{\rho}_2 = 0.008, \widehat{\rho}_3 = -0.053\}\,.$$

By contrast, the PACF for lags 1 to 3 is computed using the following three regressions (standard errors in parentheses):

$$\begin{aligned}
r_t &= \underset{(0.099)}{0.247} + \underset{(0.024)}{0.285}\, r_{t-1} + \widehat{v}_{1t}, \\
r_t &= \underset{(0.098)}{0.266} + \underset{(0.025)}{0.308}\, r_{t-1} - \underset{(0.025)}{0.080}\, r_{t-2} + \widehat{v}_{2t}, \\
r_t &= \underset{(0.099)}{0.274} + \underset{(0.025)}{0.305}\, r_{t-1} - \underset{(0.026)}{0.070}\, r_{t-2} - \underset{(0.025)}{0.035}\, r_{t-3} + \widehat{v}_{3t}\,.
\end{aligned}$$

The estimated PACF is

$$\{\widehat{\phi}_{11} = 0.285,\ \widehat{\phi}_{22} = -0.080,\ \widehat{\phi}_{33} = -0.035\}\,.$$

The significance of the estimated coefficients in the regressions required to compute the ACF and PACF suggest that a useful starting point for a dynamic model of real equity returns is a simple univariate autoregressive model. The parameter estimates obtained by estimating an AR(6) model by ordinary least squares are as follows (standard errors in parentheses):

$$\begin{aligned}
r_t = \underset{(0.099)}{0.243} + \underset{(0.025)}{0.303}\, r_{t-1} - \underset{(0.026)}{0.064}\, r_{t-2} - \underset{(0.026)}{0.041}\, r_{t-3} \\
+ \underset{(0.026)}{0.019}\, r_{t-4} + \underset{(0.026)}{0.056}\, r_{t-5} + \underset{(0.025)}{0.022}\, r_{t-6} + \widehat{v}_t\,,
\end{aligned}$$

in which $\widehat{v}_t$ is the least squares residual. The first lag is the most important both economically, having the largest point estimate (0.303) and statistically, having the largest t statistic $(0.303/0.025 = 12.12)$. The second and fifth lags are also statistically important at the 5% level. The insignificance of the parameter estimate on the sixth lag suggests that an AR(5) model may be a more appropriate and parsimonious model or real equity returns.

4.3.3 Mean Aversion and Reversion in Returns

There is evidence that returns on assets exhibit positive autocorrelation for shorter maturities and negative autocorrelation for longer maturities. Positive autocorrelation represents mean aversion as a positive shock in returns in one period results in a further increase in returns in the next period, whereas negative autocorrelation arises when a positive shock in returns leads to a decrease in returns in the next period.

An illustration of mean aversion and reversion in autorcorrelations is provided by the NASDAQ share index. Using monthly, quarterly and annual frequencies for the period 1989 to 2009 the following results are obtained from

estimating a simple AR(1) model (standard errors in parentheses):

$$\begin{array}{lcl} \text{Monthly} & : & r_t = \underset{(0.438)}{0.599} + \underset{(0.063)}{0.131}\ r_{t-1} + v_{1t}, \\ \text{Quarterly} & : & r_t = \underset{(1.520)}{1.950} + \underset{(0.111)}{0.058}\ r_{t-1} + v_{2t}, \\ \text{Annual} & : & r_t = \underset{(7.363)}{8.974} - \underset{(0.238)}{0.131}\ r_{t-1} + v_{3t}\ . \end{array}$$

There appears to be mean aversion in returns for time horizons less than a year as the first order autocorrelation is positive for monthly and quarterly returns. By contrast, there is mean reversion for horizons of at least a year as the first order autocorrelation is now negative with a value of -0.131 for annual returns.

To understand the change in the autocorrelation properties of returns over different maturities, consider the following model of prices, P_t, in terms of fundamentals, F_t

$$\begin{array}{rcll} p_t & = & f_t + u_t & u_t \sim iid\, N(0, \sigma_u^2), \\ f_t & = & f_{t-1} + v_t & v_t \sim iid\, N(0, \sigma_v^2), \end{array}$$

where lower case letters denote logarithms and v_t and u_t are disturbance terms assumed to be independent of each other. Note that u_t represents transient movements in the actual price from its fundamental price.

The 1-period return is

$$r_t = p_t - p_{t-1} = v_t + u_t - u_{t-1}.$$

and the h-period return is

$$\begin{array}{rcl} r_t(h) & = & p_t - p_{t-h} = r_t + r_{t-1} + \cdots + r_{t-h+1} \\ & = & (v_t + u_t - u_{t-1}) + (v_{t-1} + u_{t-1} - u_{t-2}) + \cdots \\ & & \quad +(v_{t-h+1} + u_{t-h+1} - u_{t-h}) \\ & = & v_t + v_{t-1} + \cdots v_{t-h+1} + u_t - u_{t-h}. \end{array}$$

The autocovariance is

$$\begin{array}{rcl} \gamma_h & = & \mathrm{E}[(p_t - p_{t-h})(p_{t-h} - p_{t-2h})] \\ & = & \mathrm{E}[(v_t + v_{t-1} \cdots v_{t-h+1} + u_t - u_{t-h}) \\ & & \quad \times (v_{t-h} + v_{t-h-1} + \cdots v_{t-2h+1} + u_{t-h} - u_{t-2h})] \\ & = & \mathrm{E}(u_t u_{t-h}) - \mathrm{E}(u_t u_{t-2h}) - \mathrm{E}(u_{t-h}^2) + \mathrm{E}(u_{t-h} u_{t-2h}) \\ & = & 2\,\mathrm{E}(u_t u_{t-h}) - \mathrm{E}(u_t u_{t-2h}) - \mathrm{E}(u_{t-h}^2). \end{array}$$

As u_t is iid by assumption, for any $h > 1$, $\mathrm{E}(u_t u_{t-h}) = \mathrm{E}(u_t u_{t-2h}) = 0$, and $\gamma_h = -\sigma_u^2$, implying that the autocovariance is negative. However, if we assume u_t is positively serially correlated with autocovariance decaying towards zero, when h is small, γ_h may be positive. When h becomes large enough, γ_h must eventually becomes negative as $\lim_{h\to\infty} \gamma_h = -\sigma_u^2$.

4.4 Univariate Moving Average Models

4.4.1 Specification

An alternative way to introduce dynamics into univariate models is to allow the lags in the dependent variable y_t to be implicitly determined via the disturbance term u_t. The specification of the model is

$$y_t = \psi_0 + u_t, \tag{4.5}$$

with u_t specified as

$$u_t = v_t + \psi_1 v_{t-1} + \psi_2 v_{t-2} + \cdots + \psi_q v_{t-q}, \tag{4.6}$$

where v_t is a disturbance term with zero mean and constant variance σ_v^2, and $\psi_0, \psi_1, \cdots, \psi_q$ are unknown parameters. As u_t is a weighted sum of current and past disturbances, this model is referred to as a moving average model with q lags, or more simply MA(q). Estimation of the unknown parameters is more involved for this class of models than it is for the autoregressive model as it requires a nonlinear least squares algorithm.

4.4.2 Properties

To understand the properties of MA models, consider the MA(1) model

$$y_t = \psi_0 + v_t + \psi_1 v_{t-1}, \tag{4.7}$$

where $|\psi_1| < 1$. Applying the unconditional expectations operator to both sides gives the unconditional mean

$$\mathrm{E}(y_t) = \mathrm{E}(\psi_0 + v_t + \psi_1 v_{t-1}) = \psi_0 + \mathrm{E}(v_t) + \psi_1 \mathrm{E}(v_{t-1}) = \psi_0.$$

The unconditional variance is

$$\gamma_0 = \mathrm{E}\{[y_t - \mathrm{E}(y_t)]^2\} = \mathrm{E}[(v_t + \psi_1 v_{t-1})^2] = \sigma_v^2(1 + \psi_1^2).$$

The first-order autocovariance is

$$\begin{aligned} \gamma_1 &= \mathrm{E}\{[y_t - \mathrm{E}(y_t)][y_{t-1} - \mathrm{E}(y_{t-1})]\} \\ &= \mathrm{E}[(v_t + \psi_1 v_{t-1})(v_{t-1} + \psi_1 v_{t-k})] \\ &= \psi_1 \sigma_v^2, \end{aligned}$$

while for autocovariances of $k > 1$, $\gamma_k = 0$. The ACF of a MA(1) model is summarised as

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \begin{cases} \dfrac{\psi_1}{1 + \psi_1^2} & : \quad k = 1 \\ 0 & : \quad \text{otherwise.} \end{cases} \tag{4.8}$$

This result is in contrast to the ACF of the AR(1) model as now there is a spike in the ACF at lag 1. As this spike corresponds to the lag length of the model, it follows that the ACF of a MA(q) model has non-zero values for the first q lags and zero thereafter.
To understand the PACF properties of the MA(1) model, consider rewriting (4.7) using the lag operator

$$y_t = \psi_0 + (1 + \psi_1 L)v_t,$$

whereby $Lv_t = v_{t-1}$. As $|\psi_1| < 1$, this equation is rearranged by multiplying both sides by $(1 + \psi_1 L)^{-1}$

$$\begin{aligned} (1 + \psi_1 L)^{-1} y_t &= (1 + \psi_1 L)^{-1}\psi_0 + v_t \\ (1 - \psi_1 L + \psi_1^2 L^2 + \cdots) y_t &= (1 + \psi_1 L)^{-1}\psi_0 + v_t. \end{aligned}$$

As this is an infinite AR model, the PACF is non-zero for higher order lags in contrast to the AR model which has just non-zero values up to an including lag p.

4.5 Autoregressive-Moving Average Models

The autoregressive and moving average models are now combined to yield an autoregressive-moving average model

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + u_t\,, \\ u_t &= v_t + \psi_1 v_{t-1} + \psi_2 v_{t-2} + \cdots + \psi_q v_{t-q}\,, \end{aligned}$$

where v_t is a disturbance term with zero mean and constant variance σ_v^2. This model is denoted as ARMA(p,q). As with the MA model, the ARMA model requires a nonlinear least squares procedure to estimate the unknown parameters.

4.6 Regression Models

A property of the regression models discussed in the previous chapter is that the dependent and explanatory variables all occur at time t. To allow for dynamics into this model, the autoregressive and moving average specifications discussed above can be used. Some ways that dynamics are incorporated into this model are as follows.

1. Including lagged autoregressive disturbance terms:

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t\,, \\ u_t &= \rho_1 u_{t-1} + v_t. \end{aligned}$$

2. Including lagged moving average disturbance terms:

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t\,, \\ u_t &= v_t + \theta_1 v_{t-1}. \end{aligned}$$

3. Including lagged dependent variables:

$$y_t = \beta_0 + \beta_1 x_t + \lambda y_{t-1} + u_t.$$

4. Including lagged explanatory variables:

$$y_t = \beta_0 + \beta_1 x_t + \gamma_1 x_{t-1} + \gamma_2 x_{t-2} + \beta_2 z_{t-1} + u_t.$$

5. Joint specification:

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + \lambda_1 y_{t-1} + \gamma_1 x_{t-1} + \gamma_2 x_{t-2} + \beta_2 z_{t-1} + u_t\,, \\ u_t &= \rho_1 u_{t-1} + v_t + \theta_1 v_{t-1}. \end{aligned}$$

A natural specification of dynamics in the linear regression model arises in the case of models of forward market efficiency. Lags here are needed for two reasons. First, the forward rate acts as a predictor of future spot rates. Second, if the data are overlapping whereby the maturity of the forward rate is longer than the frequency of observations, the disturbance term will have a moving average structure. This point is taken up in Exercise 2.
An important reason for including dynamics into a regression model is to correct for potential misspecification problems that arise from incorrectly excluding explanatory variables. In Chapter 3, misspecification of this type is detected using the LM autocorrelation test applied to the residuals of the estimated regression model.

4.7 Vector Autoregressive Models

Once a decision is made to move into a multivariate setting, it becomes difficult to delimit one variable as the 'dependent' variable to be explained in terms of all the others. It may be that all the variables are in fact jointly determined.

4.7.1 Specification and Estimation

This problem was first investigated by Sims (1980) using United States data on the nominal interest rate, money, prices and output. He suggested that to start with it was useful to treat all variables as determined by the system of equations. The model will therefore have an equation for each of the variables under consideration. The most important distinguishing feature of the system of equations, however, is each equation will have exactly the same the set of

explanatory variables. This type of model is known as a vector autoregressive model (VAR).
An example of a bivarate VAR(q) is

$$y_{1t} = \phi_{10} + \sum_{i=1}^{q} \phi_{11,i} y_{1t-i} + \sum_{i=1}^{q} \phi_{12,i} y_{2t-i} + u_{1t}\,, \tag{4.9}$$

$$y_{2t} = \phi_{20} + \sum_{i=1}^{q} \phi_{21,i} y_{1t-i} + \sum_{i=1}^{q} \phi_{22,i} y_{2t-i} + u_{2t}, \tag{4.10}$$

where y_{1t} and y_{2t} are the dependent variables, q is the lag length which is the same for all equations and u_{1t} and u_{2t} are disturbance terms.
Interestingly, despite being a multivariate system of equations with lagged values of the each variable potentially influencing all the others, estimation of a VAR is performed by simply applying ordinary least squares to each equation one at a time. Despite the model being a system of equations, ordinary least squares applied to each equation is appropriate because the set of explanatory variables is the same in each equation.
Higher dimensional VARs containing k variables $\{y_{1t}, y_{2t}, \cdots, y_{kt}\}$, are specified and estimated in the same way as they are for bivariate VARs. For example, in the case of a trivariate model with $k = 3$, the VAR is specified as

$$\begin{aligned}
y_{1t} &= \phi_{10} + \sum_{i=1}^{q} \phi_{11,i} y_{1t-i} + \sum_{i=1}^{q} \phi_{12,i} y_{2t-i} + \sum_{i=1}^{q} \phi_{13,i} y_{3t-i} + u_{1t}\,, \\
y_{2t} &= \phi_{20} + \sum_{i=1}^{q} \phi_{21,i} y_{1t-i} + \sum_{i=1}^{q} \phi_{22,i} y_{2t-i} + \sum_{i=1}^{q} \phi_{23,i} y_{3t-i} + u_{2t}\,, \\
y_{3t} &= \phi_{30} + \sum_{i=1}^{q} \phi_{31,i} y_{1t-i} + \sum_{i=1}^{q} \phi_{32,i} y_{2t-i} + \sum_{i=1}^{q} \phi_{33,i} y_{3t-i} + u_{3t}\,.
\end{aligned} \tag{4.11}$$

Estimation of the first equation involves regressing y_{1t} on a constant and all of the lagged variables. This is repeated for the second equation where y_{2t} is the dependent variable, and for the third equation where y_{3t} is the dependent variable.
In matrix notation the VAR is conveniently represented as

$$y_t = \Phi_0 + \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \cdots + \Phi_q y_{t-q} + u_t, \tag{4.12}$$

where the parameters are given by

$$\Phi_0 = \begin{bmatrix} \phi_{10} \\ \phi_{20} \\ \vdots \\ \phi_{k0} \end{bmatrix}, \qquad \Phi_i = \begin{bmatrix} \phi_{11,i} & \phi_{12,i} & \cdots & \phi_{1k,i} \\ \phi_{21,i} & \phi_{22,i} & & \phi_{2k,i} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{k1,i} & \phi_{k2,i} & \cdots & \phi_{kk,i} \end{bmatrix}.$$

The disturbances $u_t = \{u_{1t}, u_{2t}, ..., u_{kt}\}$, have zero mean with covariance matrix

$$\Omega = \begin{bmatrix} \text{var}(u_{1t}) & \text{cov}(u_{1t}, u_{2t}) & \cdots & \text{cov}(u_{1t}, u_{kt}) \\ \text{cov}(u_{2t}, u_{1t}) & \text{var}(u_{2t}) & \cdots & \text{cov}(u_{2t}, u_{kt}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(u_{kt}, u_{1t}) & \text{cov}(u_{kt}, u_{2t}) & \cdots & \text{var}(u_{kt}) \end{bmatrix}. \tag{4.13}$$

This matrix has two properties. First, it is a symmetric matrix so that the upper triangular part of the matrix is the mirror of the lower triangular part

$$\text{cov}(u_{it}, u_{jt}) = \text{cov}(u_{jt}, u_{it}), \qquad i \neq j.$$

Second, the disturbance terms in each equation are allowed to be correlated with the disturbances of other equations

$$\text{cov}(u_{it}, u_{jt}) \neq 0, \qquad i \neq j.$$

This last property is important when undertaking impulse response analysis and computing variance decompositions, topics which are addressed at a later stage.

Now consider extending the AR(6) model for real equity returns, r_t, to include lagged real dividend yields, y_t, as possible explanatory variables. This seems like a reasonable course of action given that the present value model established a theoretical link between equity prices and dividends. Equally important, however, is a model to explain real dividend yields and a natural specification of a model of real dividend yields is to include as explanatory variables both own lags and lags of real equity returns. Treating both real equity returns, r_t, and real dividend payments, y_t, as potentially endogenous, a VAR(6) model is estimated for monthly United States data from 1871 to 2004. The parameter estimates (with standard errors in parentheses) are given in Table 4.1. Notice that the parameter estimates of the effects of the dividend yield on equity returns are significant at the lags 2 and 6. Furthermore, the parameter estimates of the effects of real equity returns on dividends at lags 2, 3, 5 and 6 are also statistically significant. A joint test of the parameters of the lags of r_t in the y_t equation yields a Chi-square statistic of 60.395. The p value is 0.000, showing that the restrictions are easily rejected and that lagged values of r_t are important in explaining the behaviour of y_t.

The estimate of the covariance matrix of the residuals given in equation (4.13) which is constructed using the parameter estimates reported in Table 4.1 is given by

$$\widehat{\Omega} = \begin{bmatrix} 15.0522 & -0.1099 \\ -0.1099 & 0.2537 \end{bmatrix}. \tag{4.14}$$

This shows that there is a negative covariance between the residuals of the equity return equation and residuals of the dividend yield equation. The matrix $\widehat{\Omega}$ is an important input into methods for selecting the lag length of the VAR.

Table 4.1

Parameter estimates of a bivariate VAR(6) model for United States monthly real equity returns and real dividend payments for the period 1871 to 2004.

Lag	Equity Returns		Dividend Returns	
	r	y	r	y
1	0.296 (0.025)	−0.019 (0.193)	0.001 (0.003)	0.918 (0.025)
2	−0.064 (0.026)	0.504 (0.262)	0.008 (0.003)	0.015 (0.034)
3	−0.040 (0.026)	−0.296 (0.258)	0.007 (0.003)	−0.282 (0.033)
4	0.021 (0.026)	0.395 (0.257)	0.001 (0.003)	0.250 (0.033)
5	0.053 (0.026)	−0.259 (0.263)	0.012 (0.003)	0.015 (0.034)
6	0.013 (0.025)	−0.350 (0.191)	0.014 (0.003)	−0.030 (0.025)
Constant	0.254 (0.102)		0.016 (0.013)	

4.7.2 Lag Length Selection

An important part of the specification of a VAR is the choice of the lag structure p. If the lag length is too short important parts of the dynamics are excluded from the model. If the lag structure is too long then there are redundant lags which can reduce the precision of the parameter estimates, thereby raising the standard errors and yielding t statistics that are relatively too small. Moreover, in choosing a lag structure in a VAR, care needs to be exercised as degrees of freedom can quickly diminish for even moderate lag lengths.

An important practical consideration in estimating the parameters of a VAR(p) model is the optimal choice of lag order. A common data-driven way of selecting the lag order is to use information criteria. An information criterion is a scalar that is a simple but effective way of balancing the improvement in the fit of the equations with the loss of degrees of freedom which results from increasing the lag order of a time series model.

The three most commonly used information criteria for selecting a parsimonious time series model are the Akaike information criterion (AIC) (Akaike, 1974, 1976), the Hannan information criterion (HIC) (Hannan and Quinn, 1979; Hannan, 1980) and the Schwarz information criterion (SIC) (Schwarz, 1978). If k is the number of parameters estimated in the model, these informa-

tion criteria are given by

$$\begin{aligned} AIC &= \log|\widehat{\Omega}| + \frac{2k}{T-q}, \\ HIC &= \log|\widehat{\Omega}| + \frac{2k\log(\log(T-q))}{T-q}, \\ SIC &= \log|\widehat{\Omega}| + \frac{k\log(T-q)}{T-q}. \end{aligned} \tag{4.15}$$

in which q is the maximum lag order being tested for and $\widehat{\Omega}$ is the ordinary least squares estimate of the matrix in equation (4.13) which is reported in (4.14)In the scalar case, the determinant of the estimated covariance matrix, $|\widehat{\Omega}|$, is replaced by the estimated residual variance, $\widehat{\sigma}^2$.
Choosing an optimal lag order using information criteria requires the following steps.

Step 1: Choose a maximum number of lags for the VAR model. This choice is informed by the ACFs and PACFs of the data, the frequency with which the data are observed and also the sample size.

Step 2: Estimate the model sequentially for all lags up to and including q. For each regression, compute the relevant information criteria.

Step 3: Choose the specification of the model corresponding to the minimum values of the information criteria. In some cases there will be disagreement between different information criteria and the final choice is then an issue of judgement.

The bivariate VAR(6) for equity returns and dividend yields in Table 4.1 arbitrarily chose $q = 6$. In order to verify this choice the information criteria outlined in Section 4.7.2 should be used. For example, the Hannan-Quinn criterion (HIC) for this VAR for lags from 1 to 8 is as follows:

Lag:	1	2	3	4	5	6	7	8
HIC:	7.155	7.148	7.146	7.100	7.084	7.079*	7.086	7.082

It is apparent that the minimum value of the statistic is $HIC = 7.079$, which corresponds to an optimal lag structure of 6. This provides support for the choice of the number of lags used to estimate the VAR reported in Table 4.1.

4.7.3 Granger Causality Testing

In a VAR model, all lags are assumed to contribute to information on each dependent variable, but in most empirical applications are large number of the estimated coefficients are statistically insignificant. It is then a question of crucial importance to determine if at least one of the parameters on the lagged values of the explanatory variables in any equation are are not zero. In the

bivariate VAR case, this suggests that a test of the information content of y_{2t} on y_{1t} in equation (4.9) is given by testing the joint restrictions

$$\phi_{21,1} = \phi_{21,2} = \phi_{21,3} = \cdots = \phi_{21,q} = 0.$$

These restrictions can be tested jointly using a χ^2 test.
If y_{2t} is important in predicting future values of y_{1t} over and above lags of y_{1t} alone, then y_{2t} is said to cause y_{1t} in Granger's sense (Granger, 1969). It is important to remember, however, that Granger causality is based on the presence of predictability. Evidence of Granger causality and the lack of Granger causality from y_{2t} to y_{1t}, are denoted, respectively, as

$$y_{2t} \rightarrow y_{1t} \qquad y_{2t} \nrightarrow y_{1t}\,.$$

It is also possible to test for Granger causality in the reverse direction by performing a joint test of the lags of y_{1t} in the y_{2t} equation. Combining both sets of causality results can yield a range of statistical causal patterns:

Unidirectional:	$y_{2t} \rightarrow y_{1t}$
(from y_{2t} to y_{1t})	$y_{1t} \nrightarrow y_{2t}$
Bidirectional:	$y_{2t} \rightarrow y_{1t}$
(feedback)	$y_{1t} \rightarrow y_{2t}$
Independence:	$y_{2t} \nrightarrow y_{1t}$
	$y_{1t} \nrightarrow y_{2t}$

Table 4.2 gives the results of the Granger causality tests based on the χ^2 statistic. Both p values are less than 0.05 showing that there is bidirectional Granger causality between real equity returns, r_t, and real dividend yields, y_t. Note that the results of the Granger causality test for $y \nrightarrow r$ reported in Table 4.2 may easily be verified using the estimation results obtained from the univariate model where real equity returns are a function of lags 1 to 6 of r_t and y_t, a test of the information value of real dividend yields is given by the statistic $\chi^2 = 20.288$. There are 6 degrees of freedom resulting in a p value is 0.0025, suggesting real dividend yields are statistically important in explaining real equity returns at the 5% level. This is in complete agreement with the results of the Granger causality tests concerning the information content of dividends.

4.7.4 Impulse Response Analysis

The Granger causality test provides one method for understanding the overall dynamics of lagged variables. An alternative, but related approach, is to track the effects of shocks through the model on the dependent variables. In this way the full dynamics of the system are displayed and how the variables

Table 4.2

Results of Granger causality tests based on the estimates of a bivariate VAR(6) model for United States monthly real equity returns, r_t, and real dividend payments, y_t, for the period 1871 to 2004.

Null Hypothesis:	Chi-square	Degrees of Freedom	p value
$y \nrightarrow r$	20.288	6	0.0025
$r \nrightarrow y$	60.395	6	0.0000

interact with each other over time. This approach is formally called impulse response analysis.

In performing impulse response analysis a natural candidate to represent a shock is the disturbance term $u_t = \{u_{1t}, u_{2t}, ..., u_{kt}\}$ in the VAR as it represents that part of the dependent variables that is not predicted from past information. The problem though is that the disturbance terms are correlated as highlighted by the fact that the covariance matrix in (4.13) in general has non-zero off-diagonal terms. The approach in impulse response analysis is to transform u_t into another disturbance term which has the property that it has a covariance matrix with zero off-diagonal terms. Formally the transformed residuals are referred to as orthogonal shocks which have the property that u_{2t} to u_{kt} do not have an immediate effect on u_{1t}, u_{3t} to u_{kt} do not have an immediate effect on u_{2t}, etc.

Figure 4.3 gives the impulse responses of the VAR equity-dividend model. In order to generate the impulse responses it is necessary to make some assumptions about the ordering of the variables in the VAR so that there is an implicit constraint on the contemporaneous relationship between equity returns and dividend yields. The ordering used here is one in which places r_t first and y_t second. There are four figures to capture the four sets of impulses. The first column gives the response of equity returns and dividend yields to a shock in returns, whereas the second column shows how equity returns and dividend yields are affected by a shock to yields. A positive shock to returns has a damped oscillatory effect on returns which quickly dissipates. The effect on yields is initially negative which quickly becomes positive, reaching a peak after 8 months, before decaying monotonically. The effect of a positive shock to yields slowly dissipates approaching zero after nearly 30 periods. The immediate effect of this shock on returns is zero by construction in the first period, which reflects the ordering assumption alluded to previously, and then hovers near zero exhibiting a damped oscillatory pattern.

4.7.5 Variance Decomposition

The impulse response analysis provides information on the dynamics of the VAR system of equations and how each variable responds and interacts to

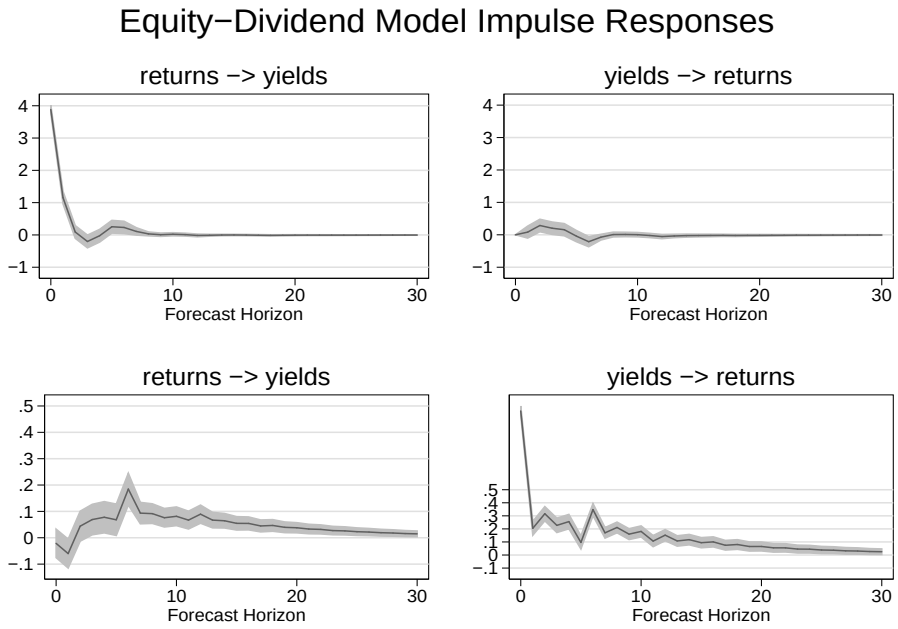


Figure 4.3: Impulse responses for the VAR(6) model of equity returns and dividend yields. Data are monthly for the period January 1871 to June 2004.

shocks in the other variables in the system. To gain insight into the relative importance of shocks on the movements in the variables in the system a variance decomposition is performed. In this analysis, movements in each variable over the horizon of the impulse response analysis are decomposed into the separate relative effects of each shock with the results expressed as a percentage of the overall movement. It is because the impulse responses are expressed in terms of orthogonalized shocks that it is possible to carry out this decomposition.

Consider again the bivariate VAR(6) model of real equity returns, r_t, and real dividend yields, y_t, estimated using month United States data for the period February 1871 to June 2004, whose parameter estimates are reported in Table 4.1. The 10-period variance decomposition of the VAR, based on the same contemporaneous ordering of variables as the impulse responses, is reported in Table 4.3.

The dividend shocks contribute very little to equity returns with the maximum contribution still less than 2%. In contrast, equity returns shocks after 15 periods contribute more than 10% of the variance in dividends. These results suggest that the effects of shocks to equity returns on dividend yields are relatively more important that the reverse case.

Table 4.3

Variance decomposition computed from a VAR(6) estimated using monthly data on United States equity returns and dividend yields for the period for the period February 1871 to June 2004.

Period	Decomposition of r		Decomposition of y	
	r	y	r	y
1	100.000	0.000	0.316	99.684
5	98.960	1.040	1.114	98.886
10	98.651	1.348	8.131	91.869
15	98.593	1.406	10.698	89.302
20	98.554	1.445	11.686	88.313
25	98.539	1.460	11.996	88.004
30	98.535	1.465	12.081	87.919

4.7.6 Diebold-Yilmaz Spillover Index

An important application of the variance decomposition of a VAR is the spillover index proposed by Diebold and Yilmaz (2009) where the aim is to compute the total contribution of shocks on an asset market arising from all other markets. Table 4.4 gives the volatility decomposition for a 10 week horizon of the weekly asset returns of 19 countries based on a VAR with 2 lags and a constant. The sample period begins December 4th 1996, and ends November 23rd 2007.

The first row of the table gives the contributions to the 10-week forecast variance of shocks in all 19 asset markets on United States weekly returns. By excluding own shocks, which equal 93.6%, the total contribution of the other 18 asset markets is given in the last column and equals

$$1.6 + 1.5 + \cdots + 0.3 = 6.4\%.$$

Similarly, for the United Kingdom, the total contribution of the other 18 asset markets to its forecast variance is

$$40.3 + 0.7 + \cdots + 0.5 = 44.3\%.$$

Of the 19 asset markets, the United States appears to be the most independent of all international asset markets as it has the lowest contributions from other asset markets, equal to just 6.4%. The next lowest is Turkey with a contribution of 14%. Germany's asset market appears to be the most affected by international asset markets where the contribution of shocks from external markets to its forecast variance is 72.4%.

By adding up the separate contributions to each asset market in the last column gives the total contributions of non-own shocks on all 19 asset markets

$$6.4 + 44.3 + \cdots + 14.2 = 675.0\%.$$

Table 4.4

Diebold-Yilmaz spillover index of global stock market returns. Based on a VAR with 2 lags and a constant with the variance decomposition based on a 10 week horizon.

To	US	UK	FRA	GER	HKG	JPN	AUS	IDN	KOR	MYS	PHL	SGP	TAI	THA	ARG	BRA	CHL	MEX	TUR	Others
US	93.6	1.6	1.5	0	0.3	0.2	0.1	0.1	0.2	0.3	0.2	0.2	0.3	0.2	0.1	0.1	0	0.5	0.3	6.4
UK	40.3	55.7	0.7	0.4	0.1	0.5	0.1	0.2	0.2	0.3	0.2	0	0.1	0.1	0.1	0.1	0	0.4	0.5	44.3
FRA	38.3	21.7	37.2	0.1	0	0.2	0.3	0.3	0.3	0.2	0.2	0.1	0.1	0.3	0.1	0.1	0.1	0.1	0.3	62.8
GER	40.8	15.9	13	27.6	0.1	0.1	0.3	0.4	0.6	0.1	0.3	0.3	0	0.2	0	0.1	0	0.1	0.1	72.4
HKG	15.3	8.7	1.7	1.4	69.9	0.3	0	0.1	0	0.3	0.1	0	0.2	0.9	0.3	0	0.1	0.3	0.4	30.1
JPN	12.1	3.1	1.8	0.9	2.3	77.7	0.2	0.3	0.3	0.1	0.2	0.3	0.3	0.1	0.1	0	0	0.1	0.1	22.3
AUS	23.2	6	1.3	0.2	6.4	2.3	56.8	0.1	0.4	0.2	0.2	0.2	0.4	0.5	0.1	0.3	0.1	0.6	0.7	43.2
IDN	6	1.6	1.2	0.7	6.4	1.6	0.4	77	0.7	0.4	0.1	0.9	0.2	1	0.7	0.1	0.3	0.1	0.4	23
KOR	8.3	2.6	1.3	0.7	5.6	3.7	1	1.2	72.8	0	0	0.1	0.1	1.3	0.2	0.2	0.1	0.1	0.7	27.2
MYS	4.1	2.2	0.6	1.3	10.5	1.5	0.4	6.6	0.5	69.2	0.1	0.1	0.2	1.1	0.1	0.6	0.4	0.2	0.3	30.8
PHL	11.1	1.6	0.3	0.2	8.1	0.4	0.9	7.2	0.1	2.9	62.9	0.3	0.4	1.5	1.6	0.1	0	0.1	0.2	37.1
SGP	16.8	4.8	0.6	0.9	18.5	1.3	0.4	3.2	1.6	3.6	1.7	43.1	0.3	1.1	0.8	0.5	0.1	0.3	0.4	56.9
TAI	6.4	1.3	1.2	1.8	5.3	2.8	0.4	0.4	2	1	1	0.9	73.6	0.4	0.8	0.3	0.1	0.3	0	26.4
THA	6.3	2.4	1	0.7	7.8	0.2	0.8	7.6	4.6	4	2.3	2.2	0.3	58.2	0.5	0.2	0.1	0.4	0.3	41.8
ARG	11.9	2.1	1.6	0.1	1.3	0.8	1.3	0.4	0.4	0.6	0.4	0.6	1.1	0.2	75.3	0.1	0.1	1.4	0.3	24.7
BRA	14.1	1.3	1	0.7	1.3	1.4	1.6	0.5	0.5	0.7	1	0.8	0.1	0.7	7.1	65.8	0.1	0.6	0.7	34.2
CHL	11.8	1.1	1	0	3.2	0.6	1.4	2.3	0.3	0.3	0.1	0.9	0.3	0.8	2.9	4	65.8	2.7	0.4	34.2
MEX	22.2	3.5	1.2	0.4	3	0.3	1.2	0.2	0.3	0.9	1	0.1	0.3	0.5	5.4	1.6	0.3	56.9	0.6	43.1
TUR	3	2.5	0.2	0.7	0.6	0.9	0.6	0.1	0.6	0.3	0.6	0.1	0.9	0.8	0.5	1.1	0.6	0.2	85.8	14.2
Others	291.9	84.1	31	11.2	80.8	19.2	11.5	31.4	13.6	16.2	9.9	8.2	5.9	11.8	21.4	9.4	2.6	8.4	6.7	675
Own	385.5	139.8	68.2	38.8	150.6	96.9	68.3	108.3	86.4	85.4	72.8	51.2	79.5	70	96.7	75.2	68.4	65.4	92.4	Index = 35.5%

As the contributions to the total forecast variance by construction are normalized to sum to 100% for each of the 19 asset markets, the percentage contribution of external shocks to the 19 asset market is given by the spillover index

$$SPILLOVER = \frac{675.0}{19} = 35.5\%.$$

This value shows that approximately one-third of the forecast variance of asset returns is the result of shocks from external asset markets with the remaining two-thirds arising from internal shocks on average.

4.8 Exercises

1. **Computing the ACF and PACF**

   pv.wf1, pv.dta, pv.xlsx

   (a) Compute the percentage monthly return on equities and dividends.
   (b) Compute the ACF of real equity returns for up to 6 lags. Compare a manual procedure with an automated version provided by econometric software.
   (c) Compute the PACF of real equity returns for up to 6 lags.Compare a manual procedure with an automated version provided by econometric software.
   (d) Repeat parts (b) and (c) for real dividend yields.

2. **Forward Market Efficiency**

   spot.wf1, spot.dta, spot.xlsx

   The forward market is efficient if the lagged forward rate is an unbiased predictor of the current spot rate.

   (a) Estimate the following model of the spot and the lagged 1-month forward rate

   $$s_t = \beta_0 + \beta_1 f_{t-4} + u_t,$$

   where the forward rate is lagged four periods (the data are weekly). Verify that weekly data on the \$/AUD spot exchange rate and the 1 month forward rate yields

   $$s_t = 0.066 + 0.916 f_{t-4} + e_t,$$

   where a lag length of four is chosen as the data are weekly and the forward contract matures in one month. Test the restriction $\beta_1 = 1$ and interpret the result.

(b) Compute the ACF and PACF of the least squares residuals, e_t, for the first 8 lags. Verify that the results are as follows.

Lag:	1	2	3	4	5	6	7	8
ACF	0.80	0.54	0.29	0.07	0.07	0.09	0.13	0.15
PACF	0.80	-0.28	-0.14	-0.07	0.40	-0.11	-0.04	-0.02

(c) There is evidence to suggest that the ACF decays quickly after 3 lags. Interpret this result and use this information to improve the specification of the model and redo the test of $\beta_1 = 1$.

(d) Repeat parts (a) to (c) for the 3-month and the 6-month forward rates.

3. **Estimating AR and MA Models**

pv.wf1, pv.dta, pv.xlsx

(a) Compute the percentage monthly return on equities and dividends. Plot the two returns and interpret their time series patterns.

(b) Estimate an AR(6) model of equity returns. Interpret the parameter estimates.

(c) Estimate an AR(6) model of equity returns but now augment the model with 6 lags on dividend yields. Perform a test of the information value of dividend yields in understanding equity returns.

(d) Repeat parts (b) and (c) for real dividend yields.

(e) Estimate a MA(3) model of real equity returns.

(f) Estimate a MA(6) model of equity returns.

(g) Perform a test that the parameters on lags 4 to 6 are zero.

(h) Repeat parts (e) to (g) using real dividend yields.

4. **Mean Aversion and Reversion in Stock Returns**

int_yr.wf1, int_yr.dta, int_yr.xlsx
int_qr.wf1, int_qr.dta, int_qr.xlsx
int_mn.wf1,int_mn.dta, int_mn.xlsx

(a) Estimate the following regression equation using returns on the NASDAQ (r_t) for each frequency (monthly, quarterly, annual)

$$r_t = \phi_0 + \phi_1 r_{t-1} + u_t,$$

where u_t is a disturbance term. Interpret the results.

(b) Repeat part (a) for the Australian share price index.

(c) Repeat part (a) for the Singapore Straits Times stock index.

5. **Poterba-Summers Pricing Model**

Poterba and Summers (1988) assume that the logarithm of the price of an asset p_t, behaves according to

$$\begin{aligned} p_t &= f_t + u_t, \\ f_t &= f_{t-1} + v_t, \\ u_t &= \phi_1 u_{t-1} + w_t, \end{aligned}$$

where f_t is the logarithm of the fundamental price, u_t represents transient price movements, and v_t and w_t are independent disturbance terms with zero means and constant variances, σ_v^2 and σ_w^2 respectively.

(a) Show that the kth order autocorrelation of the one period return

$$r_t = p_t - p_{t-1} = v_t + u_t - u_{t-1},$$

is

$$\rho_k = \frac{\sigma_w^2}{\sigma_v^2} \frac{\phi_1^{k-1}(\phi_1 - 1)}{(1 + \phi_1 + 2\sigma_w^2/\sigma_v^2)} < 0.$$

(b) Show that the first order autocovariance function of the h-period return

$$r_t(h) = p_t - p_{t-h} = r_t + r_{t-1} + \cdots + r_{t-h+1},$$

is

$$\gamma_h = \frac{\sigma_w^2}{1 - \phi_1^2}(2\phi_1^h - \phi_1^{2h} - 1) < 0.$$

6. **Roll Model of Bid-Ask Bounce**

spot.wf1, spot.dta, spot.xlsx

Roll (1984) assumes that the logarithm of the price, p_t, of an asset follows

$$p_t = f + \frac{s}{2} I_t,$$

where f is a constant fundamental price, s is the bid-ask spread and I_t is a binary indicator variable given by

$$I_t = \begin{cases} +1: & \text{with probability 0.5 (buyer)} \\ -1: & \text{with probability 0.5 (seller).} \end{cases}$$

(a) Derive $\mathrm{E}[I_t]$, $\mathrm{var}(I_t)$, $\mathrm{cov}(I_t, I_{t-1})$, $\mathrm{cor}(I_t, I_{t-1})$.

(b) Derive $\mathrm{E}[\Delta I_t]$, $\mathrm{var}(\Delta I_t)$, $\mathrm{cov}(\Delta I_t, \Delta I_{t-1})$, $\mathrm{cor}(\Delta I_t, \Delta I_{t-1})$.

(c) Show that the autocorrelation function of Δp_t is

$$\begin{aligned} \mathrm{cor}(\Delta p_t, \Delta p_{t-1}) &= -\frac{1}{2}, \\ \mathrm{cor}(\Delta p_t, \Delta p_{t-k}) &= 0, \qquad k > 1. \end{aligned}$$

(d) Suppose that the price is now given by

$$p_t = f_t + \frac{s}{2} I_t,$$

where the fundamental price f_t is now assumed to be random with zero mean and variance σ^2. Derive the autocorrelation function of Δp_t.

7. **An Equity-Dividend VAR**

pv.wf1, pv.dta, pv.xlsx

(a) Compute the percentage monthly return on equities and dividends and estimate a bivariate VAR for these variables with 6 lags.

(b) Test for the optimum choice of lag length using the Hannan-Quinn criterion and specifying a maximum lag length of 12. If required, re-estimate the VAR.

(c) Test for Granger causality between equity returns and dividends and interpret the results.

(d) Compute the impulse responses for 30 periods and interpret the results.

(e) Compute the variance decomposition for 30 periods and interpret the results.

8. **Campbell-Shiller Present Value Model**

cam_shiller.wf1, cam_shiller.dta, cam_shiller.xlsx

Let y_t be real dividend yields (expressed in percentage terms) and let v_t be deviations from the present value relationship between equity prices and dividends computed from the linear regression

$$p_t = \beta + \alpha d_t + v_t.$$

Campbell and Shiller (1987) develop a VAR model for y_t and v_t given by

$$\begin{bmatrix} y_t \\ v_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ v_{t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}.$$

The data set contains the equity prices, STOCK_t, and dividend payments, DIV_t. Use these series to do the following tasks.

(a) Estimate the parameter α and compute the least squares residuals $\widehat{v}_t$.

(b) Estimate a VAR(1) containing the dividend yields and $\widehat{v}_t$.

(c) Campbell and Shiller show that

$$\phi_{22} = \delta^{-1} - \alpha\phi_{12},$$

where δ represents the discount factor. Use the parameter estimate of α obtained in part (a) and the parameter estimates of ϕ_{12} and ϕ_{22} obtained in part (b), to estimate δ. Interpret the result.

9. **Causality Between Stock Returns and Output Growth**

stock_out.wf1, stock_out.dta, stock_out.xlsx

(a) For the United States, compute the percentage continuous stock returns and output growth rates, respectively.

(b) It is hypothesised that stock returns lead output growth but not the reverse. Test this hypothesis by performing a test for Granger causality between the two series using 1 lag.

(c) Test the robustness of these results by using higher order lags up to a maximum of 4. What do you conclude about the causal relationships between stock returns and output growth in the United States?

(d) Repeat parts (a) to (c) for Japan, Singapore and Taiwan.

10. **Volatility Linkages**

diebold.wf1, diebold.dta, diebold.xlsx

Diebold and Yilmaz (2009) construct spillover indexes of international real asset returns and volatility based on the variance decomposition of a VAR. The data file contains weekly data on real asset returns, *rets*, and volatility, *vol*, of 7 developed countries and 12 emerging countries from the first week of January 1992 to the fourth week of November 2007.

(a) Compute descriptive statistics of the 19 real asset market returns given in *rets*. Compare the estimates with the results reported in Table 1 of Diebold and Yilmaz.

(b) Estimate a VAR(2) containing a constant and the 19 real asset market returns.

(c) Estimate VD_{10}, the variance decomposition for horizon $h = 10$, and compare the estimates with the results reported in Table 3 of Diebold and Yilmaz.

(d) Using the results in part (c) compute the 'Contribution from Others' by summing each row of VD_{10} excluding the diagonal elements, and the 'Contribution to Others' by summing each column of VD_{10} excluding the diagonal elements. Interpret the results.

(e) Repeat parts (a) to (d) with the 19 series in *rets* replaced by *vol*, and the comparisons now based on Tables 2 and 4 in Diebold and Yilmaz.

Part II

Addressing Nonstationarity

Chapter 5

Nonstationarity in Financial Time Series

5.1 Introduction

An important property of asset prices identified in Chapter 2 is that they exhibit strong trends. Financial series exhibiting no trending behaviour are referred to as being stationary and are the subject matter of Chapter 4, while series that are characterised by trending behaviour are referred to as being nonstationary. This chapter focuses on identifying and testing for nonstationarity in financial time series. The identification of nonstationarity will hinge on a test for $\rho = 1$ in a model of the form

$$y_t = \rho y_{t-1} + u_t \,,$$

in which u_t is a disturbance term. This test is commonly referred to as a test for a unit root. This situation is different from the kinds of hypothesis tests performed under null hypothesis of a stationary processes conducted in Chapter 4 because the process is now nonstationary under the null hypothesis, $\rho = 1$, and as a consequence the test statistic does not have a normal distribution in large samples.

The classification of variables as either stationary or nonstationary has important implications in both finance and econometrics. From a finance point of view, the presence of nonstationarity in the price of financial asset is consistent with the efficient markets hypothesis which states that all of the information in the price of an asset is contained in its most recent price. If the nonstationary process is explosive, $\rho > 1$ then this may be taken as evidence of a bubble in the price of the asset.

5.2 Characteristics of Financial Data

In Chapter 2 the efficient markets hypothesis was introduced which theorises that all available information concerning the value of a risky asset is factored into the current price of the asset. The return to a risky asset may be written as

$$r_t = p_t - p_{t-1} = \alpha + v_t\,, \qquad v_t \sim iid\,(0, \sigma^2)\,, \tag{5.1}$$

where p_t is the logarithm of the asset price. The parameter α represents the average return on the asset. From an efficient markets point of view, provided that v_t is not autocorrelated, then r_t is unpredictable using information at time t.
An alternative representation of equation (5.1) is to rearrange it in terms of p_t as

$$p_t = \alpha + p_{t-1} + v_t\,. \tag{5.2}$$

This representation of p_t is known as a random walk with drift, where the mean parameter α represents the drift. From an efficient market point of view this equation shows that in predicting the price of an asset in the next period, all of the relevant information is contained in the current price.
To understand the properties of the random walk with drift model of asset prices in (5.2), Figure 5.1 provides a plot of a simulated random walk with drift. In simulating equation (5.2), the drift parameter α is set equal to the mean return on the S&P500 while the volatility, σ^2 corresponds to the variance of the logarithm of S&P500 returns. The simulated price is has similar time series characteristics to the observed logarithm of the price index given in Figure 2.2 in Chapter 2 and Figure fig::transformations.
In particular, the simulated price exhibits two important characteristics, namely, an increasing mean and an increasing variance. These characteristics may be demonstrated formally as follows. Lag the random walk with drift model in equation (5.2) by one period yields

$$p_{t-1} = \alpha + p_{t-2} + v_{t-1},$$

and then substituting this expression for p_{t-1} in (5.2) gives

$$p_t = \alpha + \alpha + p_{t-2} + v_t + v_{t-1}\,.$$

Repeating this recursive substitution process for t-steps in total gives

$$p_t = p_0 + \alpha t + v_t + v_{t-1} + v_{t-2} + \cdots + v_1\,,$$

in which p_t is fully determined by its initial value, p_0, a deterministic trend component and the summation of the complete history of disturbances. Taking expectations of this expression and using the property that $\mathrm{E}[v_t] = \mathrm{E}[v_{t-1}] \cdots = 0$, gives the mean of p_t

$$\mathrm{E}[p_t] = p_0 + \alpha t\,.$$

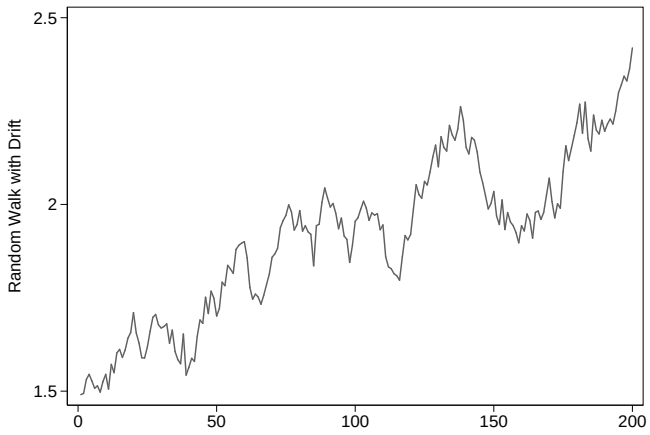


Figure 5.1: Simulated random walk with drift model using equation (5.2). The initial value of the simulated data is the natural logarithm of the S&P500 equity price index in February 1871 and the drift and volatility parameters are estimated from the returns to the S&P500 index. The distribution of the disturbance term is taken to be the normal distribution.

This demonstrates that the mean of the random walk with drift model increases over time provided that $\alpha > 0$. The variance of p_t in the random walk model is defined as

$$\text{var}(p_t) = \text{E}[(p_t - \text{E}[p_t])^2] = t\sigma^2$$

by using the property that the disturbances are independent. As with the expression for the mean the variance also is an increasing function over time, that is p_t exhibits fluctuations with increasing amplitude as time progresses. It is now clear that the efficient market hypothesis has implications for the time series behaviour of financial asset prices. Specifically in an efficient market asset prices will exhibit trending behaviour.
In Chapter 4 the idea was developed of an observer who observes snapshots of a financial time series at different points in time. If the snapshots exhibit similar behaviour in terms of the mean and variance of the observed series, the series is said to be *stationary*, but if the observed behaviour in either the mean or the variance of the series (or both) is completely different then it is *non-stationary*. More formally, a variable y_t is stationary if its distribution, or some important aspect of its distribution, is constant over time. There are two commonly used definitions of stationarity known as weak (or covariance) and strong (or strict) stationarity[1] and it is the former that will be of primary interest.

[1]Strict stationarity is a stronger requirement than that weak stationarity pertains to all of the moments of the distribution not just the first two.

Definition: Weak (or Covariance) Stationarity
A process is is weakly stationary if both the population mean and the population variance are constant over time and if the covariance between two observations is a function only of the distance between them and not of time.

The efficient markets hypothesis requires that financial asset returns have a non-zero (positive) mean and variance that are independent of time as in equation (5.1). Formally this means that returns are weakly or covariance stationary. By contrast, the logarithm of prices is a random walk with drift, (5.2), in which the mean and the variance are functions of time. It follows, therefore, that a series with these properties is referred to as being non stationary.

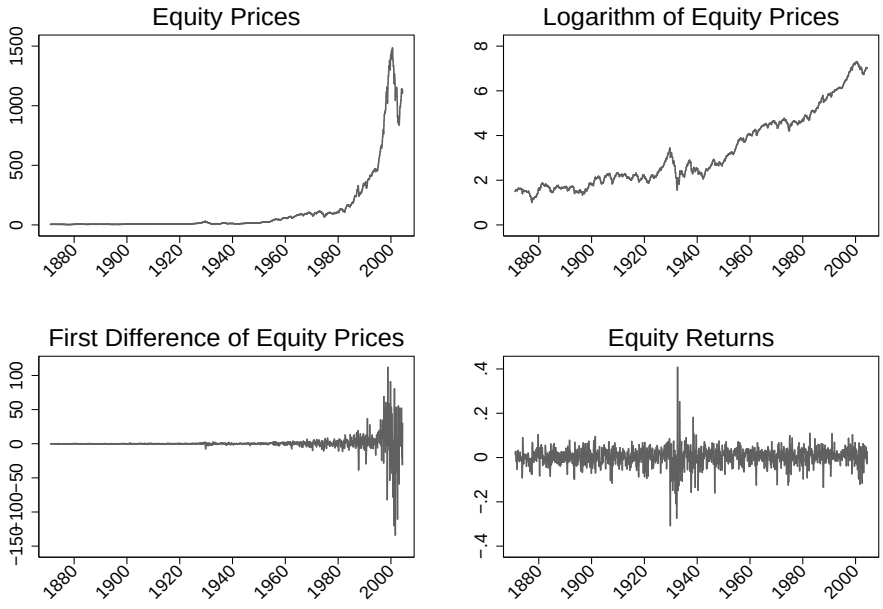


Figure 5.2: Different transformations of monthly United States equity prices for the period January 1871 to June 2004.

Figure 5.2 highlights the time series properties of the real United States equity price and various transformations of this series, from January 1871 to June 2004. The transformed equity prices are the logarithm of the equity price, the first difference of the equity price and and the first difference of the logarithm of the equity price (log returns).
A number of conclusions may be drawn from the behaviour of equity prices in Figure 5.2 which both reinforce and extend the ideas developed previously. Both the equity price and its logarithm are nonstationary in the mean as both

exhibit positive trends. Furthermore, a simple first difference of the equity price renders the series stationary in the mean, which is now constant over time, but the variance is still increasing with time. The implication of this is that simply first differencing of the equity price does not yield a stationary series. Finally, equity returns defined as the first difference of the logarithm of prices is stationary in both mean and variance. The appropriate choice of filter to detrend the data is the subject matter of the next section.

5.3 Deterministic and Stochastic Trends

While the term 'trend' is deceptively easy to define, being the persistent long-term movement of a variable over time, in practice it transpires that trends are fairly tricky to deal with and the appropriate choice of filter to detrend the data is therefore not entirely straightforward. The main reason for this is that there are two very different types of trending behaviour that are difficult to distinguish between.

(i) Determimistic trend

A deterministic trend is a nonrandom function of time

$$y_t = \alpha + \delta t + u_t \,,$$

in which t is a simple time trend taking integer values from 1 to T. In this model, shocks to the system have a transitory effect in that the process always reverts to its mean of $\alpha + \delta t$. This suggests the removing the deterministic trend from y_t will give a series that does not trend. That is

$$y_t - \widehat{\alpha} - \widehat{\delta} t = \widehat{u}_t \,,$$

in which ordinary least squares has been used to estimate the parameters, is stationary. Another approaches to estimating the parameters of the deterministic elements, generalised least squares, is considered at a later stage.

(ii) Stochastic trend

By contrast, a stochastic trend is random and varies over time, for example,

$$y_t = \alpha + y_{t-1} + u_t \,, \tag{5.3}$$

which is known as a random walk with drift model. In this model, the best guess for the next value of series is the current value plus some constant, rather than a deterministic mean value. As a result, this kind of models is also called 'local trend' or 'local level' models. The appropriate filter here is to difference the data to obtain a stationary series as follows

$$\Delta y_t = \alpha + u_t \,.$$

Distinguishing between deterministic and stochastic trends is important as the correct choice of detrending filter depends upon this distinction. The deterministic trend model is stationary once the deterministic trend has been removed (and is called a **trend-stationary process**) whereas a stochastic trend can only be removed by differencing the series (a **difference-stationary process**).

Most financial econometricians would agree that the behaviour of many financial time series is due to stochastic rather than deterministic trends. It is hard to reconcile the predictability implied by a deterministic trend with the complications and surprises faced period-after-period by financial forecasters. Consider the simple AR(1) regression equation

$$y_t = \alpha + \rho y_{t-1} + u_t \, .$$

The results obtained by fitting this regression to monthly data on United States zero coupon bonds with maturities ranging from 2 months to 9 months for period January 1947 to February 1987 are given in Table 5.1

Table 5.1

Ordinary least squares estimates of an AR(1) model estimated using monthly data on United States zero coupon bonds with maturities ranging from 2 months to 9 months for period January 1947 to February 1987

Maturity	Intercept		Slope	
(mths)	$(\widehat{\alpha})$	se$(\widehat{\alpha})$	$(\widehat{\rho})$	se$(\widehat{\rho})$
2	0.090	0.046	0.983	0.008
3	0.087	0.045	0.984	0.008
4	0.085	0.044	0.985	0.007
5	0.085	0.044	0.985	0.007
6	0.087	0.045	0.985	0.007
9	0.088	0.046	0.985	0.007

The major result of interest in the results in Table 5.1 is that in all the estimated regressions estimate of the slope coefficient, $\widehat{\rho}$ is very close to unity and indicative of a stochastic trend in the data along the lines of equation (5.3). This empirical result is quite consistent one for all the maturities and, furthermore, the pattern is a fairly robust one that applies to other financial markets such as currency markets (spot and forward exchange rates) and equity markets (share prices and dividends) as well.

The behaviour under simulation of series with deterministic (dashed lines) and stochastic trend models (solid lines) is demonstrated in Figure 5.3 using simulated data. The nonstationary series look similar, both showing clear evidence of trending. The key difference between a deterministic trend and a

stochastic trend however is that removing a deterministic trend from the difference stationary process, illustrated by the solid line in panel (b) of Figure 5.3, does not result in a stationary series. The longer the series is simulated for, the more the evidence reveals the more erratic behaviour of the difference stationary process which has been detrended incorrectly.
It is in fact this feature of the makeup of y_t that makes its behaviour very different to the simple deterministic trend model because simply removing the deterministic trend will not remove the nonstationarity in the data that is due to the summation of the disturbances.
The element of summation of the disturbances in nonstationarity is the origin of an important term, *the order of integration* of a series.

Definition: Order of Integration
A process is integrated of order d, denoted by $I(d)$, if it can be rendered stationary by differencing d times. That is, y_t is non-stationary, but $\Delta^d(y_t - y_{t-1})$ is stationary.

Accordingly a process is said to be integrated of order one, denoted by $I(1)$, if it can be rendered stationary by differencing once, that is y_t is non-stationary, but $\Delta y_t = y_t - y_{t-1}$ is stationary. If $d = 2$, then y_t is $I(2)$ and needs to be differenced twice to achieve stationarity as follows

$$\Delta^2(y_t - y_{t-1}) = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) = y_t - 2y_{t-1} + y_{t-2}.$$

By analogy, a stationary process is integrated of zero, $I(0)$, if it does not require any differencing to achieve stationarity.
There is one final important point that arises out of the simulated behaviour illustrated in Figure 5.3. At first sight panel (c) may suggest that differencing a financial time series, irrespective of whether it is trend of difference stationary, may be a useful strategy because both the resultant series in panel (c) appear to be stationary. The logic of the argument then becomes, if the series has a stochastic trend then this is the correct course of action and if it is trend stationary then a stationary series will result in any event. This is not, however, a strategy to be recommended. Consider again the deterministic trend model

$$y_t = \alpha + \delta t + u_t$$

In first-difference form this becomes

$$\Delta y_t = \delta + u_t - u_{t-1},$$

so that the process of taking the first difference has introduced a moving average error term which has a unit root. This is known as over-differencing and it can have treacherous consequences for subsequent econometric analysis, should the true data generating process actually be trend-stationary. In fact for the simple problem of estimating the coefficient δ in the differenced model it produces an estimate that is tantamount to using only the first and last data points in estimation process.

5.3.1 Unit Roots†

A series that is $I(1)$ is also said to have a unit root and tests for nonstationarity are called tests for unit roots. The reason for this is easily demonstrated. Consider the general n - th order autoregressive process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_n y_{t-n} + u_t.$$

This may be written in a different way by using the lag operator, L, which is defined as

$$y_{t-1} = Ly_t\,, \qquad y_{t-2} = L^2 y_t \qquad \cdots \qquad y_{t-n} = L^n y_t\,,$$

so that

$$y_t = \phi_1 L y_t + \phi_2 L^2 y_t + \ldots + \phi_n L^n y_t + u_t,$$

or

$$\Phi(L)\, y_t = u_t,$$

where

$$\Phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_n L^n,$$

is called a polynomial in the lag operator. The roots of this polynomial are the values of L which satisfy the equation

$$1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_n L^n = 0.$$

If all of the roots of this equation are greater in absolute value than one, then y_t is stationary. If, on the other hand, any of the roots is equal to one (a unit root) then y_t is non-stationary.
The AR(1) model is

$$(1 - \phi_1 L)\, y_t = u_t,$$

and the roots of the equation

$$1 - \phi_1 L = 0$$

are of interest. The single root of this equation is given by $L^* = 1/\phi_1$ and the root is greater than unity only if $|\phi_1| < 1$. If this is the case then the AR(1) process is stationary. If, on the other hand, the root of the equation is unity, then $|\phi_1| = 1$ and the AR(1) process is non-stationary.
In the AR(2) model

$$\left(1 - \phi_1 L - \phi_2 L^2\right) y_t = u_t,$$

it is possible that there are two unit roots, corresponding to the roots of the equation

$$1 - \phi_1 L - \phi_2 L^2 = 0.$$

A solution is obtained by factoring the equation yield

$$(1 - \varphi_1 L)(1 - \varphi_2 L) = 0,$$

in which $\varphi_1 + \varphi_2 = \phi_1$ and $\varphi_1 \varphi_2 = -\phi_2$. The roots of this equation are $1/\varphi_1$ and $1/\varphi_2$, respectively, and y_t will have a unit root if either of the roots is unity. In the event of $\phi_1 = 2$ and $\phi_2 = -1$ then both absolute roots of the equation are one and y_t has two unit roots and is therefore $I(2)$.

5.4 The Dickey-Fuller Testing Framework

The original testing procedures for unit roots were developed by Dickey and Fuller (1979, 1981) and this framework remains one of the most popular methods to test for nonstationarity in financial time series.

5.4.1 Dickey-Fuller (DF) Test

Consider again the AR(1) regression equation

$$y_t = \alpha + \rho y_{t-1} + u_t\,, \tag{5.4}$$

in which u_t is a disturbance term with zero mean and constant variance σ^2. The null and alternative hypotheses are respectively

$$\begin{array}{lll} H_0: & \rho = 1 & (\text{Variable is nonstationary}) \\ H_1: & \rho < 1 & (\text{Variable is stationary}). \end{array} \tag{5.5}$$

To carry out the test, equation (5.4) is estimated by ordinary least squares and a t statistic is constructed to test that $\rho = 1$

$$t_\rho = \frac{\widehat{\rho} - 1}{\text{se}(\widehat{\rho})}. \tag{5.6}$$

This is all correct up to this stage: the estimation of (5.4) by ordinary least squares and the use of the t statistic in (5.6) to test the hypothesis are both sound procedures. The problem is that the distribution of the statistic in (5.6) is not a Student t distribution. In fact the distribution of this statistic under the null hypothesis of nonstationarity is non-standard and is known as the Dickey-Fuller distribution. Consequently, the t statistic given in (5.6) is commonly known as the Dickey-Fuller unit root test to recognize that even though it is a t statistic by construction its distribution is not.
In practice, equation (5.4) is transformed in such a way to convert the t statistic in (5.6) to a test that the slope parameter of the transformed equation is zero. This has the advantage that the t statistic commonly reported in standard regression packages directly yields the Dickey-Fuller statistic. Subtract y_{t-1} from both sides of (5.4) and collect terms to give

$$y_t - y_{t-1} = \alpha + (\rho - 1)y_{t-1} + u_t, \tag{5.7}$$

or by defining $\beta = \rho - 1$, so that

$$y_t - y_{t-1} = \alpha + \beta y_{t-1} + u_t. \tag{5.8}$$

Equations (5.4) and (5.8) are exactly the same models with the connection being that $\beta = \rho - 1$.
Consider again the monthly data on United States zero coupon bonds with maturities ranging from 2 months to 9 months for period January 1947 to

February 1987 used in the estimation of the AR(1) regressions reported in Table 5.1. Estimating equation (5.4) yields the following results (with standard errors in parentheses)

$$y_t = \underset{(0.046)}{0.090} + \underset{(0.008)}{0.983}\, y_{t-1} + e_t, \tag{5.9}$$

On the other hand, estimating the transformed equation (5.8) yields

$$y_t - y_{t-1} = \underset{(0.046)}{0.090} - \underset{(0.008)}{0.017}\, y_{t-1} + \widehat{u}_t. \tag{5.10}$$

Comparing the estimated equations in (5.9) and (5.10) shows that they differ only in terms of the slope estimate on y_{t-1}. The differences in the two slope estimates is easily reconciled as the slope estimate of (5.9) is $\widehat{\rho} = 0.983$, whereas an estimate of β may be recovered as

$$\widehat{\beta} = \widehat{\rho} - 1 = 0.983 - 1 = -0.017.$$

This is also the slope estimate obtained in (5.10). To perform the test of H_0 : $\rho = 1$, the relevant t statistics are

$$\begin{aligned} t_\rho &= \frac{\widehat{\rho} - 1}{\text{se}(\widehat{\rho})} = \frac{0.983 - 1}{0.008} = -2.120\,, \\ t_\beta &= \frac{\widehat{\beta} - 0}{\text{se}(\widehat{\beta})} = \frac{-0.017 - 0}{0.008} = -2.120\,, \end{aligned}$$

which demonstrates that the two methods are indeed equivalent.
The Dickey-Fuller test regression must now be extended to deal with the possibility that under the alternative hypothesis, the series may be stationary around a deterministic trend. As established in Sections **??** and **??**, financial data often exhibit trends and one of the problems faced by the empirical researcher is distinguishing between stochastic and deterministic trends. If the data are trending and if the null hypothesis of nonstationarity is rejected, it is imperative that the model under the alternative hypothesis is able to account for the major characteristics displayed by the series being tested. If the test regression in equation (5.8) is used and the null hypothesis of a unit root rejected, the alternative hypothesis is that of a process which is stationary around the constant mean α. In other words, the model under the alternative hypothesis contains no deterministic trend. Consequently, the important extension of the Dickey-Fuller framework is to include a linear time trend, t, in the test regression so that the estimated equation becomes

$$y_t - y_{t-1} = \alpha + \beta y_{t-1} + \delta t + u_t\,. \tag{5.11}$$

The Dickey-Fuller test still consists of testing $\beta = 0$. Under the alternative hypothesis, y_t is now a stationary process with a deterministic trend.

Once again using the monthly data on United States zero coupon bonds, the estimated regression including the time trend gives the following results (with standard errors in parentheses)

$$\Delta y_t = \underset{(0.052)}{0.030} - \underset{(0.014)}{0.046}\ y_{t-1} + \underset{(0.001)}{0.001}\ t + \widehat{u}_t.$$

The value of the Dickey-Fuller test is

$$t_\beta = \frac{\widehat{\beta} - 0}{\text{se}(\widehat{\beta})} = \frac{-0.046 - 0}{0.014} = -3.172.$$

Finally, the Dickey-Fuller test can be performed without a constant and a time trend by setting $\alpha = 0$ and $\delta = 0$ in (5.11). This form of the test, which assumes that the process has zero mean, is only really of use when testing the residuals of a regression for stationarity as they are known to have zero mean, a problem that is returned to in Chapter 6.
There are therefore three forms of the Dickey-Fuller test, namely,

$$\begin{array}{ll} \textbf{Model 1:} & \Delta y_t = \beta y_{t-1} + u_t\,, \\ \textbf{Model 2:} & \Delta y_t = \alpha + \beta y_{t-1} + u_t\,, \\ \textbf{Model 3:} & \Delta y_t = \alpha + \delta t_t + \beta y_{t-1} + u_t\,. \end{array} \tag{5.12}$$

For each of these three models the form of the Dickey-Fuller test is still the same, namely the test of $\beta = 0$. The pertinent distribution in each case, however, is not the same because the distribution of the test statistic changes depending on whether a constant and or a time trend is included. The distributions of different versions of Dickey-Fuller tests are shown in Figure 5.4. The key point to note is that all three Dickey Fuller distributions are skewed to the left with respect to the standard normal distribution. In addition, the distribution becomes less negatively skewed as more deterministic components (constants and time trends) are included.
The monthly United States zero coupon bond data have been used to estimate Model 2 and Model 3. Using the Dickey-Fuller distribution the p-value for the Model 2 Dickey-Fuller test statistic (-2.120) is 0.237 and because $0.237 > 0.05$ the null hypothesis of nonstationarity cannot be rejected at the 5% level of significance. This is evidence that the interest rate is nonstationary. For Model 3, using the Dickey-Fuller distribution reveals that the p-value of the test statistic (-3.172) is 0.091 and because $0.091 > 0.05$, the null hypothesis cannot be rejected at the 5% level of significance. This result is qualitatively the same result as the Dickey-Fuller test based on Model 2, although there is quite a large reduction in the p-value from 0.237 in the case of Model 2 to 0.091 in Model 3.

5.4.2 Augmented Dickey-Fuller (ADF) Test

In estimating any one of the test regressions in equation (5.12), there is a real possibility that the disturbance term will exhibit autocorrelation. One reason for the presence of autocorrelation will be that many financial series are

interact with each other and because the test regressions are univariate equations the effects of these interactions are ignored. One common solution to correct for autocorrelation is to proceed as in Chapter 4 and include lags of the dependent variable Δy_t in the test regressions (5.12). These equations then become

$$\begin{aligned}
&\textbf{Model 1:} \quad \Delta y_t = \beta y_{t-1} + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + u_t\,, \\
&\textbf{Model 2:} \quad \Delta y_t = \alpha + \beta y_{t-1} + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + u_t\,, \\
&\textbf{Model 3:} \quad \Delta y_t = \alpha + \delta t + \beta y_{t-1} + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + u_t\,,
\end{aligned} \tag{5.13}$$

in which the lag length p is chosen to ensure that u_t does not exhibit autocorrelation. The unit root test still consists of testing $\beta = 0$.
The inclusion of lagged values of the dependent variable represents an augmentation of the Dickey-Fuller regression equation so this test is commonly referred to as the Augmented Dickey-Fuller (ADF) test. Setting $p = 0$ in any version of the test regressions in (5.13) gives the associated Dickey-Fuller test. The distribution of the ADF statistic in large samples is also the Dickey-Fuller distribution.
For example, using Model 2 in (5.13) to construct the augmented Dickey-Fuller test with $p = 2$ lags for the United States zero coupon 2-month bond yield, the estimated regression equation is

$$\Delta y_t = \underset{(0.046)}{0.092} - \underset{(0.008)}{0.017}\, y_{t-1} + \underset{(0.045)}{0.117}\, \Delta y_{t-1} - \underset{(0.046)}{0.080}\, \Delta y_{t-2} + \widehat{u}_t.$$

The value of the Augmented Dickey-Fuller test is

$$t_\beta = \frac{\widehat{\beta} - 0}{\text{se}(\widehat{\beta})} = \frac{-0.017 - 0}{0.008} = -2.157.$$

Using the Dickey-Fuller distribution the p-value is 0.223. Since $0.223 > 0.05$ the null hypothesis is not rejected at the 5% level of significance This result is qualitatively the same result as the Dickey-Fuller test with $p = 0$ lags.
The selection of p affects both the size and power properties of a unit root test. If p is chosen to be too small, then substantial autocorrelation will remain in the error term of the test regressions (5.13) and this will result in distorted statistical inference because the large sample distribution under the null hypothesis no longer applies in the presence of autocorrelation. However, including an excessive number of lags will have an adverse effect on the power of the test.
To select the lag length p to use in the ADF test, a common approach is to base the choice on information criteria as discussed in in Chapter 4. Two commonly used criteria are the Akaike Information criteria (AIC) and the Schwarz information criteria (SIC). A lag-length selection procedure that has good

properties in unit root testing is the modified Akaike information criterion (MAIC) method proposed by Ng and Perron (2001). The lag length is chosen to satisfy

$$\widehat{p} = \arg\min_{p} \text{MAIC}(p) = \log(\widehat{\sigma}^2) + \frac{2(\tau_p + p)}{T - p_{\max}}, \tag{5.14}$$

in which

$$\tau_p = \frac{\widehat{\beta}^2}{\widehat{\sigma}^2} \sum_{t=p_{\max}+1}^{T} \widehat{u}_{t-1}^2,$$

and the maximum lag length is chosen as $p_{\max} = \text{int}[12(T/100)^{1/4}]$. In estimating $\widehat{p}$, it is important that the sample over which the computations are performed is held constant.

There are two other more informal ways of choosing the length of the lag structure p. The first of these is to include lags until the t statistic on the lagged variable is statistically insignificant using the t distribution. Unlike the ADF test, the distribution of the t statistic on the lagged dependent variables has a standard distribution based on the Student t distribution. The second informal approach dealing with the need to choose the lag length p is effectively to circumvent making a decision at all. The ADF test is performed for a range of lags, say $p = 0, 1, 2, 3, 4, \cdots$. If all of the tests show that the series is nonstationary then the conclusion is clear. If four of the 5 tests show evidence of nonstationarity then there is still stronger evidence of nonstationarity than there is of stationarity.

5.5 Beyond the Dickey-Fuller Framework†

A number of extensions and alternatives to the Dickey-Fuller and Augmented Dickey-Fuller unit roots tests have been proposed. A number of developments, some of which are commonly available in econometric software packages, are considered briefly.

5.5.1 Structural Breaks

The form of the nonstationarity emphasised so far is based on the series following a random walk. An alternative form of nonstationarity discussed earlier is based on a deterministic linear time trend. Another form of nonstationarity is when the series exhibits a structural break as this represents a shift in the mean and hence by definition is non-mean reverting. The simplest approach is where the timing of the structural break is known. The approach is to include a dummy variable in (5.13) to capture the structural break according to

$$\Delta y_t = \alpha + \beta y_{t-1} + \delta t + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + \gamma \text{BREAK}_t + u_t, \tag{5.15}$$

where the structural break dummy variable is defined as

$$\text{BREAK}_t = \begin{cases} 0 & : \quad t \leq \tau \\ 1 & : \quad t > \tau \end{cases}, \tag{5.16}$$

and τ is the observation where there is a break. The unit root test is still based on testing $\beta = 0$, however the p-values are now also a function of the timing of the structural break τ, so even more tables are needed. The correct p-values for a unit roots test with a structural break are available in Perron (1989). For a review of further extensions of unit root tests with structural breaks, see Maddala and Kim (1998).
An example of a possible structural break is highlighted in Figure 5.2 where there is a large fall in the share price at the time of the 1929 stock market crash.

5.5.2 Generalised Least Squares Detrending

Consider the following model

$$y_t = \alpha + \delta t + u_t\,, \tag{5.17}$$
$$u_t = \phi u_{t-1} + v_t\,, \tag{5.18}$$

in which u_t is a disturbance term with zero mean and constant variance σ^2. This is the fundamental equation from which Model 3 of the Dickey-Fuller test is derived. If the aim is still to test for a unit root in y_t the null and alternative hypotheses are

$$\begin{array}{lcll} H_0 & : & \phi = 1 & \text{[Nonstationary]} \\ H_1 & : & \phi < 1\,. & \text{[Stationary]} \end{array} \tag{5.19}$$

Instead of proceeding in the manner described previously and using Model 3 in either (5.12) or (5.13), an alternative approach is to use a two-step procedure.

Step 1: Detrending

Estimate the parameters of equation (5.17) by ordinary least squares and then construct a detrended version of y_t given by

$$y_t^* = y_t - \widehat{\alpha} - \widehat{\delta} t\,.$$

Step 2: Testing

Test for a unit root using the deterministically detrended data, y_t^*, from the first step, using the Dickey-Fuller or augmented Dickey-Fuller test. Model 1 will be the appropriate model to use because, by construction, y_t^* will have zero mean and no deterministic trend.

It turns out that in large samples (or asymptotically) this procedure is equivalent to the single-step approach based on Model 3.
Elliott, Rothenberg and Stock (1996) suggest an alternative detrending step which proceeds as follows. Define a constant $\phi^* = 1 + \bar{c}/T$ in which the value of the $\bar{c}$ depends upon the whether the detrending equation has only a constant or both a constant and a time trend. The proposed values of $\bar{c}$ are

$$\begin{cases} \bar{c} = -7 & [\text{Constant } (\alpha \neq 0, \delta = 0)] \\ \bar{c} = -13.5 & [\text{Trend } (\alpha \neq 0, \delta \neq 0)]. \end{cases}$$

and use this constant to rewrite the detrending regression as

$$y_t^* = \gamma_0 \alpha^* + \gamma_1 t^* + u_t^* \,, \tag{5.20}$$

in which u_t^* is a composite disturbance term,

$$\begin{aligned} y_t^* &= y_t - \phi^* y_{t-1} \,, & (5.21) \\ \alpha^* &= 1 - \phi^* \,, & (5.22) \\ t^* &= t - \phi^*(t-1) \,, & (5.23) \end{aligned}$$

and the starting values for each of the series at $t = 1$ are taken to by $y_1^* = y_1$ and $\alpha_1^* = t_1^* = 1$, respectively. The starting values are important because if $\bar{c} = -T$ the detrending equation reverts to the simple detrending regression (5.17). If, on the other hand, $\bar{c} = 0$ then the detrending equation is an equation in first-differences. It is for this reason that this method, which is commonly referred to as generalised least squares detrending, is also known as quasi-differencing and partial generalised least squares (Phillips and Lee, 1995). Once the ordinary least squares estimates $\widehat{\gamma}_0$ and $\widehat{\gamma}_1$ are available, the detrended data

$$\widehat{u}_t^* = y_t^* - \widehat{\gamma}_0 \alpha^* - \widehat{\gamma_1} t^* \,,$$

is tested for a unit root. If Model 1 of the Dickey-Fuller framework is used then the test is referred to as the GLS-DF test. Note, however, that because the detrended data depend on the value of $\bar{c}$ the critical value are different to the Dickey-Fuller critical values which rely on simple detrending. The generalised least squares (or quasi-differencing) approach was introduced to try and overcome one of the important shortcomings of the Dickey-Fuller approach, namely that the Dickey-Fuller tests have low power. What this means is that the Dickey-Fuller tests struggle to reject the null hypothesis of non-stationarity (a unit root) when it is in fact false. The modified detrending approach proposed by Elliott, Rothenberg and Stock (1996) is based on the premise that the test is more likely to reject the null hypothesis of a unit root if under the alternative hypothesis the process is very close to being nonstationary. The choice of value for $\bar{c} = 0$ in the detrending process ensures that the quasi-differenced data have an autoregressive root that is very close to one. For example, based on a sample size of $T = 200$, the quasi difference parameter $\phi^* = 1 + \bar{c}/T$ is 0.9650 for a regression with only a constant and 0.9325 for a regression with a constant and a time trend.

5.5.3 Nonparametric Adjustment for Autocorrelation

Phillips and Perron (1988) propose an alternative method for adjusting the Dickey-Fuller test for autocorrelation. Their test is based on estimating the Dickey-Fuller regression equation, either (5.8) or (5.11), by ordinary least squares but using a nonparametric approach to correct for the autocorrelation. The Phillips-Perron statistic is

$$\widetilde{t}_\beta = t_\beta \left(\frac{\widehat{\gamma}_0}{\widehat{f}_0} \right)^{1/2} - \frac{T(\widehat{f}_0 - \widehat{\gamma}_0)\mathrm{se}(\widehat{\beta})}{2\widehat{f}_0^{1/2} s}, \tag{5.24}$$

where t_β is the ADF statistic, s is the standard error of the regression, $\widehat{f}_0$ is known as the long-run variance which is computed as

$$\widehat{f}_0 = \widehat{\gamma}_0 + 2 \sum_{j=1}^{p} (1 - \frac{j}{p}) \widehat{\gamma}_j, \tag{5.25}$$

where p is the length of the lag, and $\widehat{\gamma}_j$ is the jth estimated autocovariance function of the ordinary least squares residuals obtained from estimating either (5.8) or (5.11)

$$\widehat{\gamma}_j = \frac{1}{T} \sum_{t=j+1}^{T} \widehat{u}_t \widehat{u}_{t-j}. \tag{5.26}$$

The critical values are the same as the Dickey-Fuller critical values when the sample size is large.

5.5.4 Unit Root Test with Null of Stationarity

The Dickey-Fuller testing framework for unit root testing, including the generalised least squares detrending and Phillips-Perron variants, are for the null hypothesis that a time series y_t is nonstationary or $I(1)$. There is, however, a popular test that is often reported in the empirical literature which has a null hypothesis of stationarity or $I(0)$. Consider the regression model

$$y_t = \alpha + \delta t + z_t \,,$$

where z_t is given by

$$z_t = z_{t-1} + \varepsilon_t \,, \qquad \varepsilon_t \sim iid\, N(0, \sigma_\varepsilon^2) \,.$$

The null hypothesis that y_t is a stationary $I(0)$ process is tested in terms of the null hypothesis $H_0 : \sigma_\varepsilon^2 = 0$ in which case z_t is simply a constant. Define $\{\widehat{z}_1, \cdots, \widehat{z}_T\}$ as the ordinary least squares residuals from regression of y_t on a constant and a deterministic trend. Now define the standardised test statistic

$$\mathrm{S} = \frac{\sum_{t=1}^{T} (\sum_{j=1}^{t} \widehat{z}_j)^2}{T^2 \widehat{f}_0} \,,$$

in which $\widehat{f}_0$ is a consistent estimator of the long-run variance of z_t. This test statistic can is most commonly known as the KPSS test, after Kwiatkowski, Phillips, Schmidt and Shin (1992). This can also be regarded as a test for over-differencing following the earlier discussion of over-differencing.

5.5.5 Higher Order Unit Roots

A failure to reject the null hypothesis of nonstationarity suggests that the series needs to be differenced at least once to render it stationary ie $d \geq 1$. The question is how many times does the series have to be differenced to achieve stationarity. To identify the value of d, the unit root tests discussed above are performed sequentially as follows.

1. Test the level of the series for a unit root.
   (a) If the null is rejected, stop and conclude that the series is $I(0)$.
   (b) If you fail to reject the null, conclude that the process is at least $I(1)$ and move to the next step.
2. Test the first difference of the series for a unit root.
   (a) If the null is rejected, stop and conclude that the series is $I(1)$.
   (b) If you fail to reject the null, conclude that the process is at least $I(2)$ and move to the next step.
3. Test the second difference of the series for a unit root.
   (a) If the null is rejected, stop and conclude that the series is $I(2)$.
   (b) If you fail to reject the null, conclude that the process is at least $I(3)$ and move to the next step.

As it is very rare for financial series to exhibit orders of integration higher than $I(2)$, it is safe to stop at this point. The pertinent p-values vary at each stage of the sequential unit root testing procedure.

5.6 Price Bubbles

During the 1990s, led by Dot-Com stocks and the internet sector, the United States stock market experienced a spectacular rise in all major indices, especially the NASDAQ index. Figure 5.5 plots the monthly NASDAQ index, expressed in real terms, for the period February 1973 to January 2009. The series grows fairly steadily until the early 1990s and begins to surge. The steep upward movement in the series continues until the late 1990s as investment in Dot-Com stocks grew in popularity. Early in the year 2000 the Index drops abruptly and then continues to fall to the mid-1990s level. In summary, over

the decade of the 1990s, the NASDAQ index rose to the historical high on 10 March 2000. Concomitant with this striking rise in stock market indices, there was much popular talk among economists about the effects of the internet and computing technology on productivity and the emergence of a new economy associated with these changes. What caused the unusual surge and fall in prices, whether there were bubbles, and whether the bubbles were rational or behavioural are among the most actively debated issues in macroeconomics and finance in recent years.

A recent series of papers places empirical tests for bubbles and rational exuberance is an interesting new development in the field of unit root testing (Phillips and Yu, 2011; Phillips, Wu and Yu (PWY hereafter), 2011). Instead of concentrating on performing a test of a unit root against the alternative of stationarity (essentially using a one-sided test where the critical region is defined in the left-hand tail of the distribution of the unit root test statistic), they show that the process having an explosive unit root (the right tail of the distribution) is appropriate for asset prices exhibiting price bubbles. The null hypothesis of interest is still $\rho = 1$ but the alternative hypothesis is now $\rho > 1$ in (5.4), or

$$\begin{array}{ll} H_0: & \rho = 1 \quad (\text{Variable is nonstationary, No price bubble}) \\ H_1: & \rho > 1 \quad (\text{Variable is explosive, Price bubble}). \end{array} \tag{5.27}$$

To motivate the presence of a price bubble, consider the following model

$$P_t(1+R) = \mathrm{E}_t\left[P_{t+1} + D_{t+1}\right], \tag{5.28}$$

where P_t is the price of an asset, R is the risk-free rate of interest assumed to be constant for simplicity, D_t is the dividend and $\mathrm{E}_t[\cdot]$ is the conditional expectations operator. This equation highlights two types of investment strategies. The first is given by the left hand-side which involves investing in a risk-free asset at time t yielding a payoff of $P_t(1+R)$ in the next period. Alternatively, the right hand-side shows that by holding the asset the investor earns the capital gain from owning an asset with a higher price the next period plus a dividend payment. In equilibrium there are no arbitrage opportunities so the two two types of investment are equal to each other. Now write the equation as

$$P_t = \beta\ \mathrm{E}_t\left[P_{t+1} + D_{t+1}\right], \tag{5.29}$$

where $\beta = (1+R)^{-1}$ is the discount factor. Now writing this expression at $t+1$

$$P_{t+1} = \beta\ \mathrm{E}_t\left[P_{t+2} + D_{t+2}\right], \tag{5.30}$$

which can be used to substitute out P_{t+1} in (5.29)

$$P_t = \beta\ \mathrm{E}_t\left[\beta\ \mathrm{E}_t\left[P_{t+2} + D_{t+2}\right] + D_{t+1}\right] = \beta\ \mathrm{E}_t\left[D_{t+1}\right] + \beta^2\ \mathrm{E}_t\left[D_{t+2}\right] + \beta^2\ \mathrm{E}_t\left[P_{t+2}\right].$$

Repeating this approach $N-$times gives the price of the asset in terms of two components

$$P_t = \sum_{j=1}^{N} \beta^j \mathrm{E}_t\left[D_{t+j}\right] + \beta^N \mathrm{E}_t\left[P_{t+N}\right]. \tag{5.31}$$

The first term on the right-hand side is the standard present value of an asset whereby the price of an asset equals the discounted present value stream of expected dividends. The second term represents the price bubble

$$B_t = \beta^N \mathrm{E}_t \left[P_{t+N} \right], \tag{5.32}$$

as it is an explosive nonstationary process. Consider the conditional expectation of the bubble the next period discounted by β and using the property $\mathrm{E}_t \left[\mathrm{E}_{t+1} \left[\cdot \right] \right] = \mathrm{E}_t \left[\cdot \right]$:

$$\beta \, \mathrm{E}_t \left[B_{t+1} \right] = \beta \, \mathrm{E}_t \left[\beta^N \mathrm{E}_{t+1} \left[P_{t+N+1} \right] \right] = \beta^{N+1} \mathrm{E}_t \left[P_{t+N+1} \right]. \tag{5.33}$$

However, this expression would also correspond to the bubble in (5.32) if the N forward iterations that produced (5.31) actually went for $N+1$ iterations. In which case

$$B_t = \beta \mathrm{E}_t \left[B_{t+1} \right],$$

or, as $\beta = (1+R)^{-1}$

$$\mathrm{E}_t \left[B_{t+1} \right] = (1+R) B_t,$$

which represents a random walk in B_t but with an explosive parameter $1+R$. Interestingly enough, if we were to follow the convention and apply the ADF test to the full sample (February 1973 to January 2009), the unit root test would not reject the null hypothesis $H_0 : \rho = 1$ in favour of the right-tailed alternative hypothesis $H_1 : \rho > 1$ at the 5 % level of significance. One would conclude that there is no significant evidence of exuberance in the behaviour of the NASDAQ index over the sample period. This result would sit comfortably with the consensus view that there is little empirical evidence to support the hypothesis of explosive behaviour in stock prices (see, for example, Campbell, Lo and MacKinlay, 1997, p260).

On the other hand, Evans (1991) argues that explosive behaviour is only temporary in the sense that economic eventually bubbles collapse and that therefore the observed trajectories of asset prices may appear rather more like an I(1) or even a stationary series than an explosive series, thereby confounding empirical evidence. Evans demonstrates by simulation that standard unit root tests have difficulties in detecting such periodically collapsing bubbles.

To address the lack of power of the full sample-based unit root test in detecting periodically collapsing bubbles, PWY (2011) suggest implementing recursively a unit root test based on expanding windows of observations, starting with $T_0 = [Tr_0]$ observations in the first regression and ending with T observations in the final regression, where T is the full sample size, $r_0 \in (0,1)$ is the fraction of T. The test statistic is the maximum of the t-statistics based on the expanding windows. The asymptotic distribution of the test statistic is the supremum of the Dickey-Fuller distributions whose quantiles can be obtained by simulations. By matching the recursive t-statistics with the right tailed critical value of the Dickey-Fuller distributions based on the first crossing principle, one can obtain estimates of the origination and conclusion dates

of bubbles. The use of recursive unit root testing proves to an invaluable approach in the detection and dating of bubbles.
Figure 5.6 plots the ADF statistic with 1 lag computed from forward recursive regressions by fixing the start of the sample period and progressively increasing the sample size observation by observation until the entire sample is being used. Interestingly, the NASDAQ shows no evidence of rational exuberance until June 1995. In July 1995, the test detects the presence of a bubble, $\widehat{\rho} > 0$, with the supporting evidence becoming stronger from this point until reaching a peak in February 2000. The bubble continues until February 2001 and by March 2001 the bubble appears to have dissipated and $\widehat{\rho} < 0$. Interestingly, the first occurrence of the bubble is July 1995, which is more than one year before the remark by Greenspan (1996) on 5 December 1996, coining the phrase of irrational exuberance, to characterise herding behaviour in stock markets.
To check the robustness of the results Figure 5.7 plots the ADF statistic with 1 lag for a series of rolling window regressions. Each regression is based on a subsample of size $T = 77$ with the first sample period from February 1973 to June 1979. The fixed window is then rolled forward one observation at a time. The general pattern to emerge is completely consistent with the results reported in Figure 5.6.
Of course these results do not have any causal explanations for the exuberance of the 1990s in internet stocks. Several possibilities exist, including the presence of a rational bubble, herding behaviour, or explosive effects on economic fundamentals arising from time variation in discount rates. Identification of the explicit economic source or sources of will involve more explicit formulation of the structural models of behaviour. What this recursive methodology does provide, however, is support of the hypothesis that the NASDAQ index may be regarded as a mildly explosive propagating mechanism. This methodology can also be applied to study recent phenomena in real estate, commodity, foreign exchange, and equity markets, which have attracted attention.

5.7 Exercises

1. **Unit Root Properties of Commodity Price Data**

commodity.wf1, commodity.dta, commodity.xlsx

   (a) For each of the commodity prices in the dataset, compute the natural logarithm and use the following unit root tests to determine the stationarity properties of each series. Where appropriate test for higher orders of integration.

      i. Dickey-Fuller test with a constant and no time trend.

ii. Augmented Dickey-Fuller test with a constant and no time trend, and $p = 2$ lags.

iii. Phillips-Perron test with a constant and no time trend.

(b) Perform a panel unit root test on the 7 commodity prices with a constant and no time trend and with $p = 2$ lags. (**We have not yet introduced Panel Unit Root Test. –Jun**)

2. **Equity Market Data**

pv.wf1, pv.dta, pv.xlsx

(a) Use the equity price series to construct the following transformed series; the natural logarithm of equity prices, the first difference of equity prices and log returns of equity prices. Plot the series and discuss the stationarity properties of each series. Compare the results with Figure 5.2.

(b) Construct similarly transformed series for dividend payments and discuss the stationarity properties of each series.

(c) Construct similarly transformed series for earnings and and discuss the stationarity properties of each series.

(d) Use the following unit root tests to test for stationarity of the natural logarithms of prices, dividends and earnings:

i. Dickey-Fuller test with a constant and no time trend.

ii. Augmented Dickey-Fuller test with a constant and no time trend and $p = 1$ lag.

iii. Phillips-Perron test with a constant and no time trend and $p = 1$ lags.

In performing these tests it may be necessary to test for higher orders of integration.

(e) Repeat part (d) where the lag length for the ADF and PP tests is based on the automatic bandwidth selection procedure.

3. **Unit Root Tests of Bond Market Data**

zero.wf1, zero.dta, zero.xlsx

(a) Use the following unit root tests to determine the stationarity properties of each yield

i. Dickey-Fuller test with a constant and no time trend.
ii. Augmented Dickey-Fuller test with a constant and no time trend, and $p = 2$ lags.
iii. Phillips-Perron test with a constant and no time trend.

In performing these tests it is necessary to test for higher orders of integration.

4. **Fisher Hypothesis**

fisher.wf1, fisher.dta, fisher.xlsx

Under the Fisher hypothesis the nominal interest rate fully reflects the long-run movements in the inflation rate.

(a) Construct the percentage annualised inflation rate, π_t.
(b) Plot the nominal interest rate and inflation.
(c) Perform unit root tests to determine the level of integration of the nominal interest rate and inflation. In performing the unit root tests, test the sensitivity of the results by using a model with a constant and no time trend, and a model with a constant and a time trend. Let the lags be determined by the automatic lag length selection procedure. Discuss the results in terms of the level of integration of each series.
(d) Compute the real interest rate as

$$r_t = i_t - \pi_t,$$

where i_t is nominal interest rate and π_t is the inflation rate. Test the real interest rate r_t for stationarity using a model with a constant but no time trend. Does the Fisher hypothesis hold? Discuss.

5. **Price Bubbles in the Share Market**

bubbles.wf1, bubbles.dta, bubbles.xlsx

The data represents a subset of the **equity_us.*** data in order to focus on the 1987 stock market crash. The present value model predicts the following relationship between the share price P_t, and the dividend D_t

$$p_t = \beta_0 + \beta_1 d_t + u_t,$$

where u_t is a disturbance term. A rational bubble occurs when the actual price persistently deviates from the present value price $\beta_0 + \beta_1 d_t$. The null and alternative hypotheses are

$$\begin{array}{lll} H_0: & \text{Bubble} & (u_t \text{ is nonstationary}) \\ H_1: & \text{Cointegration} & (u_t \text{ is stationary}). \end{array}$$

(a) Create the logarithms of real equity prices and real dividends and use unit root tests to determine the level of integration of the series.

(b) Estimate a bivariate VAR with a constant and use the SIC lag length criteria to determine the optimal lag structure.

(c) Test for a bubble by performing a cointegration between p_t and d_t using Model 3 with the number of lags based on the optimal lag length obtained form the estimated VAR.

(d) Are United States equity prices driven solely by market fundamentals or do bubbles exist.

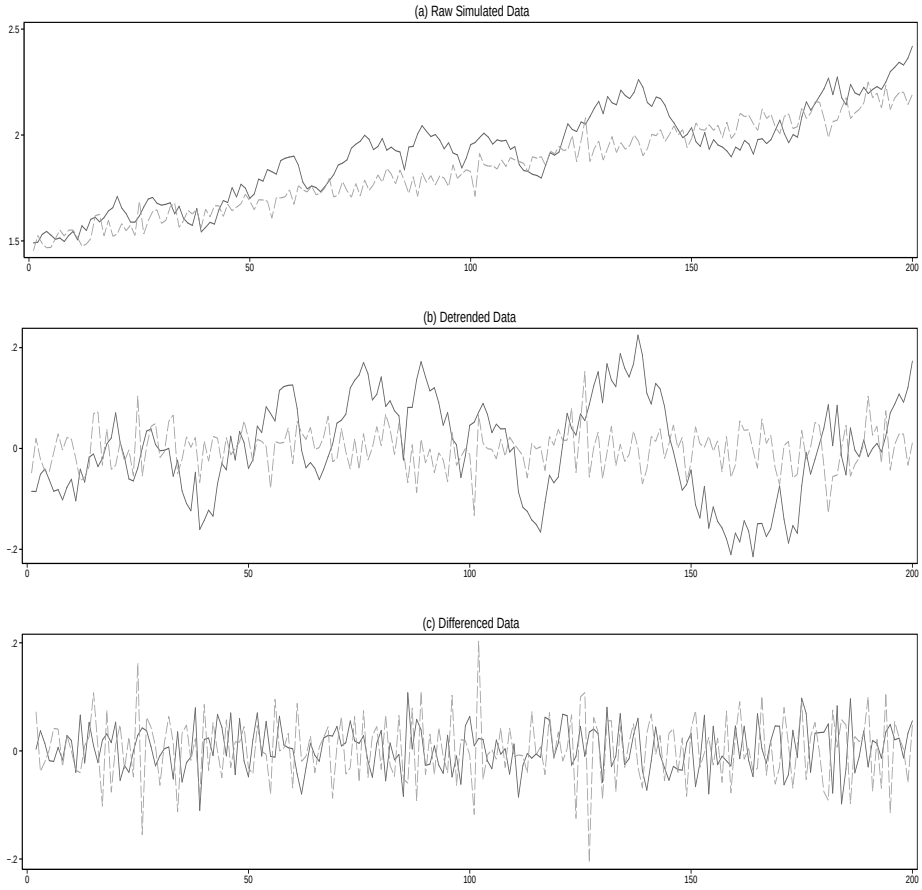


Figure 5.3: Panel (a) comparing a process with a deterministic time trend (dashed line) to a process with a stochastic trend (solid line). In panel (b) the estimated deterministic trend is used to detrend both time series data. The deterministically trending data (dashed line) is now stationary, but the model with a stochastic trend (solid line) is still not stationary. In panel (c) both series are differenced.

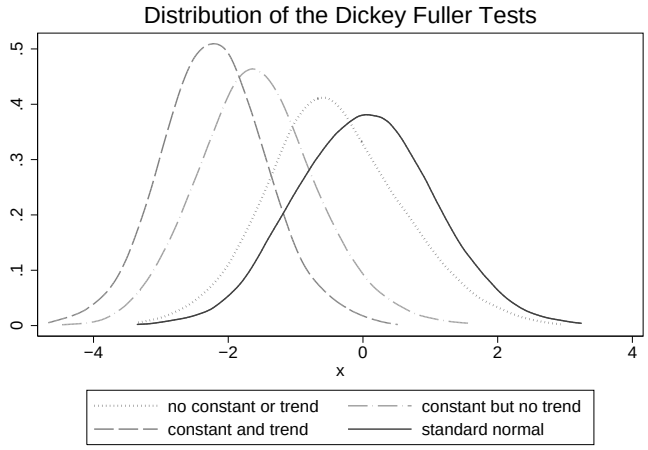


Figure 5.4: Comparing the standard normal distribution (solid line) to the simulated Dickey-Fuller distribution without an intercept or trend (dashed line), with and intercept but without a trend (dot-dashed line) and with both intercept and trend (dotted line).

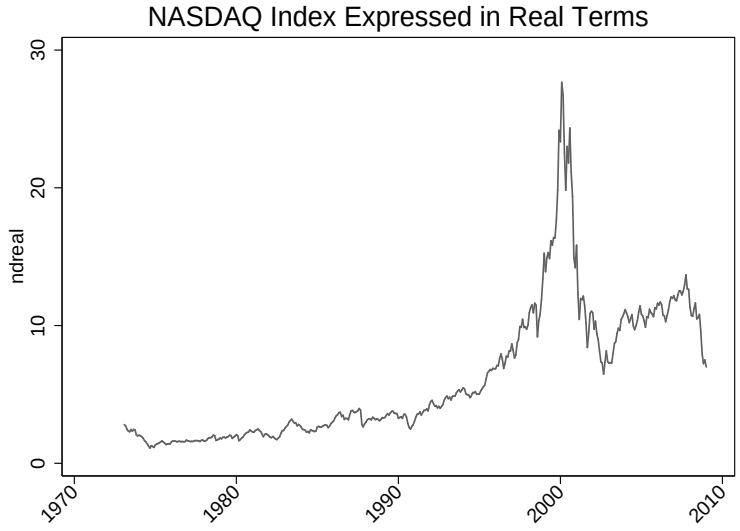


Figure 5.5: The monthly NASDAQ index expressed in real terms for the period February 1973 to January 2009.

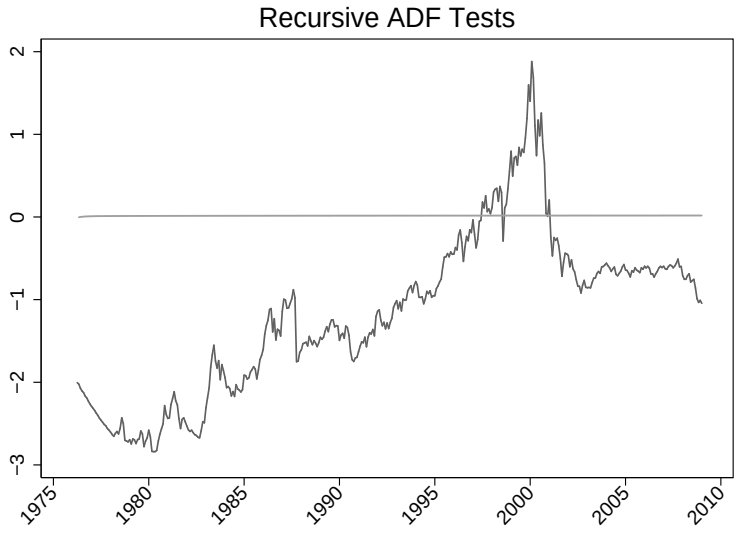


Figure 5.6: Testing for price bubbles in the monthly NASDAQ index expressed in real terms for the period February 1973 to January 2009 by means of recursive Augmented Dickey Fuller tests with 1 lag. The startup sample is 39 observations from February 1973 to April 1976. The approximate 5% critical value is also shown.

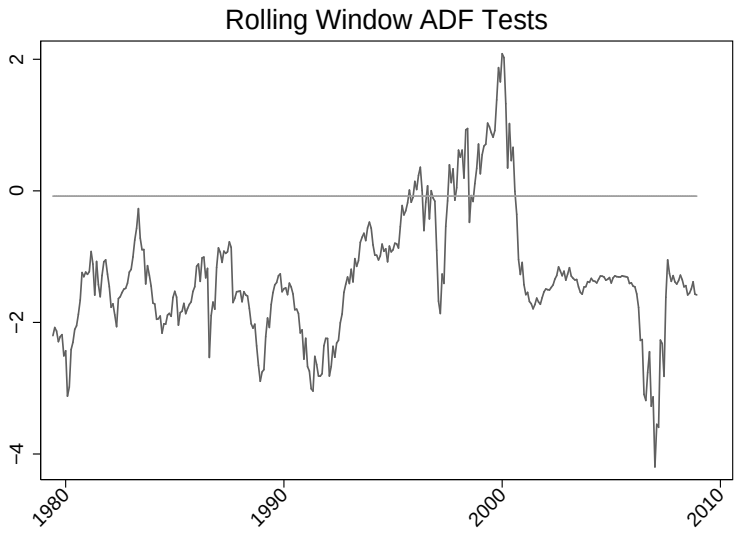


Figure 5.7: Testing for price bubbles in the monthly NASDAQ index expressed in real terms for the period February 1973 to January 2009 by means of rolling window Augmented Dickey Fuller tests with 1 lag. The size of the window is set to 77 observations so that the starting sample is February 1973 to June 1979. The approximate 5% critical value is also shown.

Chapter 6

Cointegration

6.1 Introduction

An important implication of the analysis of stochastic trends and the unit root tests discussed in Chapter 5 is that nonstationary time series can be rendered stationary through differencing the series. This use of the differencing operator represents a univariate approach to achieving stationarity since the discussion of nonstationary processes so far has concentrated on a single time series. In the case of $N > 1$ nonstationary time series $y_t = \{y_{1t}, y_{2t}, \cdots, y_{N,t}\}$, an alternative method of achieving stationarity is to form linear combinations of the series. The ability to find stationary linear combinations of nonstationary time series is known as cointegration (Engle and Granger, 1987).

Cointegration provides a basis for interpreting a number of models in finance in terms of long-run relationships. Having uncovered the long-run relationships between two or more variables by establishing evidence of cointegration, the short-run properties of financial variables are modelled by combining the information from the lags of the variables with the long-run relationships obtained from the cointegrating relationship. This model is known as a vector error-correction model (VECM) which is shown to be a restricted form of the vector autoregression models (VAR) discussed in Chapter 4.

The existence of cointegration among sets of nonstationary time series has three important implications.

1. Cointegration implies a set of dynamic long-run equilibria where the weights used to achieve stationarity represent the parameters of the equilibrium relationship.

2. The estimates of the weights to achieve stationarity (the long-run parameter estimates) converge to their population values at a super-consistent rate of T compared to the usual $\sqrt{T}$ rate of convergence for stationary variables.

3. Modelling a system of cointegrated variables allows for specification of both long-run and short-run dynamics in terms of the VECM.

6.2 Equilibrium Relationships

An important property of asset prices identified in both Chapters 2 and 5 is that they exhibit strong trends. This is indeed the case for United States as seen in Figure 6.1 which shows that the logarithm of monthly real equity prices, $p_t = \log P_t$, exhibit a strong positive trend over the period 1871 to 2004. The same is true for the logarithms of real dividends, $d_t = \log D_t$, and real earnings per share, $s_t = \log Y_t$, also illustrated in Figure 6.1. As discussed in Chapter 5, many important financial time series exhibit trending behaviour and are therefore nonstationary.

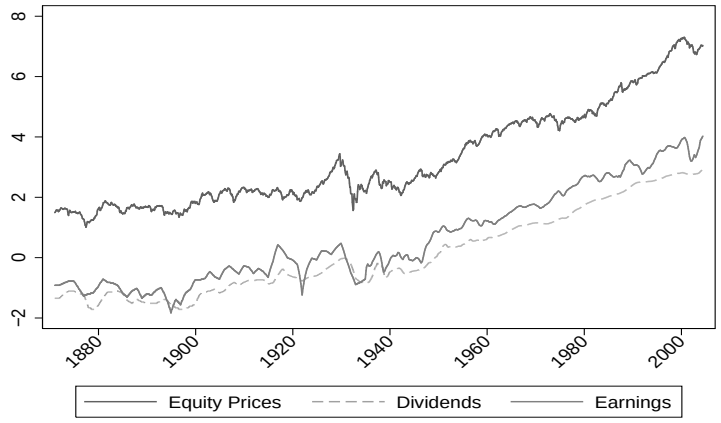


Figure 6.1: Time series plots of the logarithms of monthly United States real equity prices, real dividends and real earnings per share for the period February 1871 to June 2004.

It may be an empirical fact that the financial variables, illustrated in Figure 6.1 are $I(1)$, but theory suggests some theoretical link between the behaviour of prices, dividends and earnings. An early influential paper in this area is by Gordon (1959), who outlines two views of asset price determination. In the dividend view, the investor purchases as stock to acquire the entire future stream of dividend payments. This path of future dividends is approximated by the current dividend and the expected growth in the dividend. If the expected growth of dividends are assumed constant then there is a long-run relationship between prices and dividends given by

$$p_t = \mu_d + \beta_d d_t + u_{d,t}\,. \qquad \text{[Dividend model]} \qquad (6.1)$$

Important feature is that both p_t and d_t are I(1) but if $\mu_d + \beta_d d_t$ truly does represent the expected value of p_t, then it must follow that the disturbance term, $u_{d,t}$ is stationary or I(0).
Alternatively, in the earnings view of the world, the investor buys equity in order to obtain the income per share and is indifferent as to whether the returns are packaged in terms of the fraction of earnings distributed as a dividend or in terms of the rise in the share's value. This suggests a relationship of the form

$$p_t = \mu_s + \beta_s s_t + u_{s,t}\,, \qquad \text{[Earnings model]} \tag{6.2}$$

where once again $u_{s,t}$ must be $I(0)$ if this represents a valid long-run relationship vector.
In other words, in either view of the world, p_t can be decomposed into a long-run component and a short-run component which represents temporary deviations of p_t from its long-run. This can be represented as

$$\underbrace{p_t}_{\text{Actual}} = \underbrace{\mu_d + \beta_d d_t}_{\text{Long-run}} + \underbrace{u_{d,t}}_{\text{Short-run}}\,,$$

or in the case of the earnings model

$$\underbrace{p_t}_{\text{Actual}} = \underbrace{\mu_s + \beta_s s_t}_{\text{Long-run}} + \underbrace{u_{y,t}}_{\text{Short-run}}\,.$$

A linear combination of nonstationary variables generates a new variable that is stationary is a result known as cointegation. Furthermore, the concept of cointegration is not limited to the bivariate case. If the growth of dividends is driven by retained earnings, then the path of future dividends is approximated by the current dividend and the expected growth in the dividend given by retained earnings. This suggests an equilibrium relationship of the form

$$p_t = \mu + \beta_d d_t + \beta_s s_t + u_t\,, \qquad \text{[Combined model]}$$

where as before p_t, d_t and s_t are $I(1)$ and u_t is $I(0)$. If the owner of the share is indifferent to the fraction of earnings distributed, then cointegrating parameters, β_d and β_s will be identical. Of course, all dividends are paid out of retained earnings so there will be a relationship between these two variables as well, a fact which raises the interesting question of more than one cointegrating relationship being present in multivariate contexts. This is issue is taken up again in Section 6.8.

6.3 Equilibrium Adjustment

Assume that we have two variables y_{1t} and y_{2t} who share a long-run equilibrium relationship given by

$$y_{1t} = \mu + \beta y_{2t} + u_t\,,$$

in which u_t is a mean-zero disturbance term and although the equation is normalised with respect to y_{1t} the notation is deliberately chosen to reflect the fact that both variables are possibly endogenously determined. This relationship is presented in Figure 6.2 for $\beta > 0$.

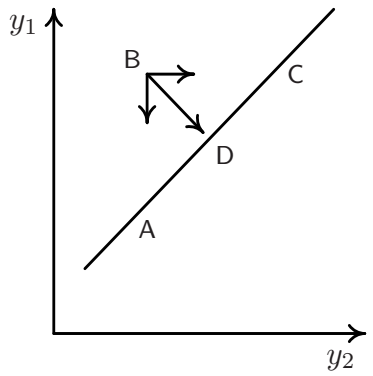


Figure 6.2: Phase diagram to demonstrate the equilibrium adjustment if two variables are cointegrated.

The system is in equilibrium anywhere along the line ADC. Now suppose there is shock to the system such that $y_{1t-1} > \mu + \beta y_{2t-1}$ or equivalently $u_{t-1} > 0$ and the system is displaced to point B. An equilibrium relationship implies necessarily that any shock to the system will result in an adjustment taking place in such a way that equilibrium is restored. There are three cases.

1. **The adjustment is done by y_{1t}:**

$$\Delta y_{1t} = \alpha_1(y_{1t-1} - \mu - \beta y_{2t-1}) + v_{1t}\,. \tag{6.3}$$

   Since $y_{1t-1} - \mu - \beta y_{2t-1} > 0$, inspection of equation (6.3) reveals that Δy_{1t} should be negative, which in turn suggests the restriction $\alpha_1 < 0$. In Figure 6.2 this adjustment is represented by a perpendicular move down from B towards A.

2. **The adjustment is done by y_{2t}:**

$$\Delta y_{2t} = \alpha_2(y_{1t-1} - \mu - \beta y_{2t-1}) + v_{2t}\,. \tag{6.4}$$

   Since $y_{1t-1} - \mu - \beta y_{2t-1} > 0$, inspection of equation (6.4) reveals that Δy_{2t} should be positive, which in turn suggests the restriction $\alpha_2 > 0$. In Figure 6.2 this adjustment is represented by a horizontal move from B towards C.

3. **Both y_{1t} and y_{2t} adjust:**
   In this case both equations (6.3) and (6.4) operate with y_{1t} decreasing and y_{2t} increasing. The strength of the movements in the two variables

is determined by the relative magnitudes of the parameters α_1 and α_2. If both variables bear an equal share of the adjustment the movement back to equilibrium is from point B to point D as shown in Figure 6.2.

Prima facie evidence of equilibrium relationships between equity prices and dividends, and equity prices and earnings is presented in panels (a) and (b), respectively, of Figure 6.3. Scatter plots of these relationships together with lines of best fit demonstrate that both these relationships are similar to the equilibrium represented in Figure 6.2. Furthermore, casual inspection of the equilibrium relationships suggests that the values of β_d and β_s are both close to 1.

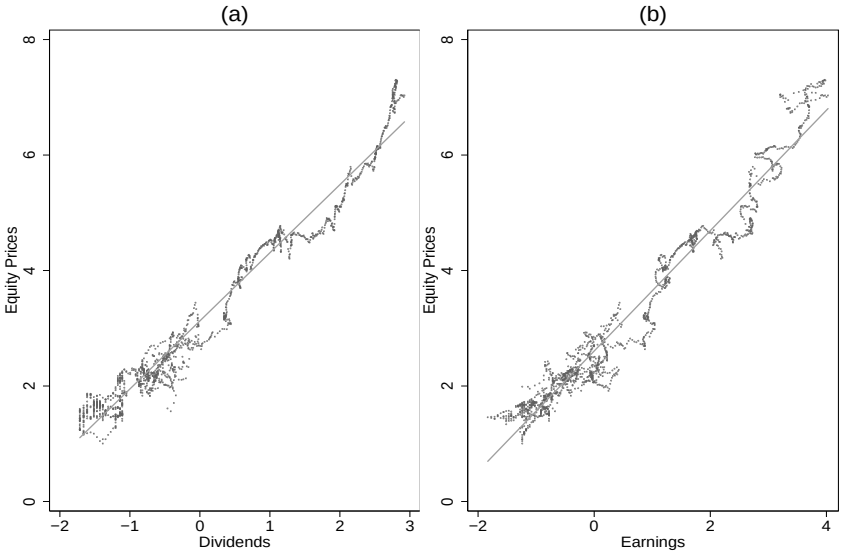


Figure 6.3: Scatter plots of the logarithms of month United States real equity prices and real dividends, panel (a), and real equity prices and real earnings per share, panel (b), for the period February 1871 to June 2004.

In order to explore which of the variables do the adjusting in the event of a shock which forces the system away from equilibrium, equations (6.3) and (6.4) must be estimated. Particularising these equations to the prices/dividends and prices/earnings relationships and estimating by sequential application of ordinary least squares yields the following results. For the dividend model the estimates are

$$\begin{aligned} \Delta p_t &= -0.0009 \left(p_{t-1} - 1.1787\, d_{t-1} - 3.128 \right) + \widehat{v}_{1t}\,, \\ \Delta d_t &= 0.0072 \left(p_{t-1} - 1.1787\, d_{t-1} - 3.128 \right) + \widehat{v}_{2t}\,, \end{aligned}$$

while for the prices/earnings model the results are

$$\begin{aligned} \Delta p_t &= -0.0053 \left(p_{t-1} - 1.0410\, s_{t-1} - 2.6073 \right) + \widehat{v}_{1t}\,, \\ \Delta s_t &= 0.0035 \left(p_{t-1} - 1.0410\, s_{t-1} - 2.6073 \right) + \widehat{v}_{2t}\,. \end{aligned}$$

It appears that the equilibrium adjustment predicted by equations (6.3) and (6.4) is confirmed for these two relationships. In particular, the signs on the adjustment parameters satisfy the conditions required for there to be equilibrium adjustment.

6.4 Vector Error Correction Models

Taken together equations (6.3) and (6.4) are known as a vector error correction model or VECM. In practice, the specification of a VECM requires the inclusion of more complex short-run dynamics, introduced through the addition of lags in dependent variables, and also the inclusion of constants and time trends in the same way that these deterministic variables are included in unit root tests. Here the situation is slightly more involved because these deterministic variables can appear in either the long-run cointegrating equation or in the short-run dynamics, or VAR, part of the equation. There are five different models to consider all of which are listed below. For simplicity the short-run dynamics or VAR part of the VECM are not included in this listing of the models.

Model 1 (No Constant or Trend):

No intercept and no trend in the cointegrating equation and no intercept and no trend in the VAR:

$$\begin{aligned} \Delta y_{1t} &= \alpha_1 (y_{1t-1} - \beta y_{2t-1}) + v_{1t}\,, \\ \Delta y_{2t} &= \alpha_2 (y_{1t-1} - \beta y_{2t-1}) + v_{2t}\,. \end{aligned}$$

This specification is included for completeness but, in general, the model will only rarely be of any practical use as most empirical specifications will require at least a constant whether or in the long-run or short-run or both.

Model 2 (Restricted Constant):

Intercept and no trend in the cointegrating equation and no intercept and no trend in the VAR

$$\begin{aligned} \Delta y_{1t} &= \alpha_1 (y_{1t-1} - \beta y_{2t-1} - \mu) + v_{1t}\,, \\ \Delta y_{2t} &= \alpha_2 (y_{1t-1} - \beta y_{2t-1} - \mu) + v_{2t}\,. \end{aligned}$$

This model is referred to as the restricted constant model as there is only one intercept term μ in the long-run equation which acts as the intercept for both dynamic equations.

Model 3 (Unrestricted Constant):

Intercept and no trend in the cointegrating equation and intercept and no trend in the VAR

$$\begin{aligned}\Delta y_{1t} &= \delta_1 + \alpha_1(y_{1t-1} - \beta y_{2t-1} - \mu) + u_{1t}\,,\\ \Delta y_{2t} &= \delta_2 + \alpha_2(y_{1t-1} - \beta y_{2t-1} - \mu) + u_{2t}\,.\end{aligned}$$

Model 4 (Restricted Trend):

Intercept and trend in the cointegrating equation and intercept and no trend in the VAR

$$\begin{aligned}\Delta y_{1t} &= \delta_1 + \alpha_1(y_{1t-1} - \beta y_{2t-1} - \mu - \phi\text{TREND}) + u_{1t}\,,\\ \Delta y_{2t} &= \delta_2 + \alpha_2(y_{1t-1} - \beta y_{2t-1} - \mu - \phi\text{TREND}) + u_{2t}\,.\end{aligned}$$

Similar to Model 2, this model is called the restricted trend model because there is only one trend term in the long-run equation.

Model 5 (Unrestricted Trend):

Intercept and trend in the cointegrating equation and intercept and trend in the VAR

$$\begin{aligned}\Delta y_{1t} &= \delta_1 + \theta_1\text{TREND} + \alpha_1(y_{1t-1} - \beta y_{2t-1} - \mu - \phi\text{TREND}) + u_{1t}\,,\\ \Delta y_{2t} &= \delta_2 + \theta_2\text{TREND} + \alpha_2(y_{1t-1} - \beta y_{2t-1} - \mu - \phi\text{TREND}) + u_{2t}\,.\end{aligned}$$

As with the unit root tests lagged values of all of the dependent variables (VAR terms) are included as additional regressors to capture the short-run dynamics. As the system is multivariate, the lags of all dependent variables are included in all equations. For example, a VECM based on Model 2 (restricted constant) with p lags on the dynamic terms becomes

$$\begin{aligned}\Delta y_{1t} &= \alpha_1(y_{1t-1} - \beta y_{2t-1} - \mu) + \sum_{i=1}^{p}\pi_{11,i}\Delta y_{1t-i} + \sum_{i=1}^{p}\pi_{12,i}\Delta y_{2t-i} + u_{1t}\,,\\ \Delta y_{2t} &= \alpha_2(y_{1t-1} - \beta y_{2t-1} - \mu) + \sum_{i=1}^{p}\pi_{21,i}\Delta y_{1t-i} + \sum_{i=1}^{p}\pi_{22,i}\Delta y_{2t-i} + u_{2t}\,.\end{aligned}$$

Exogenous variables determined outside of the system are also allowed. Finally, the system can be extended to include more than two variables. In this case there is the possibility of more than a single cointegrating equation which means that the system adjusts in general to several shocks, a theme taken up again in Section 6.8.

6.5 Relationship between VECMs and VARs

The VECM represents a restricted form of a VAR. Instead of the VAR format where all variables are stationary (first differences in this instance), the VECM

specifically includes the long-run equilibrium relationship in which the variables enter in levels. To highlight this relationship consider a simple VECM given by

$$\begin{array}{rcl} y_{1t} - y_{1t-1} & = & \alpha_1(y_{1t-1} - \beta y_{2t-1}) + v_{1t}\,, \\ y_{2t} - y_{2t-1} & = & \alpha_2(y_{1t-1} - \beta y_{2t-1}) + v_{2t}\,, \end{array} \tag{6.5}$$

in which there is one cointegrating equation and no lagged difference terms on the right hand side. There are three parameters to be estimated, namely, the cointegating parameter β and the two error correction parameters α_1 and α_2.
Now re-express each equation in terms of the levels of the variables as

$$\begin{array}{rcl} y_{1t} & = & (1+\alpha_1)y_{1t-1} - \alpha_1\beta y_{2t-1} + v_{1t}\,, \\ y_{2t} & = & \alpha_2 y_{1t-1} + (1-\alpha_2\beta)y_{2t-1} + v_{2t}\,. \end{array} \tag{6.6}$$

Not that the VAR is a VAR(1) which has one lag of the levels of the variables on the right hand side. This is a general relationship between a VAR and a VECM. If the underlying VAR is specified to be a VAR(n) then the VECM will have $n-1$ lagged difference terms, that is a VECM($n-1$).

$$\begin{array}{rcl} y_{1t} & = & \phi_{11}y_{1t-1} + \phi_{12}y_{2t-1} + v_{1t}\,, \\ y_{2t} & = & \phi_{21}y_{1t-1} + \phi_{22}y_{2t-1} + v_{2t}\,, \end{array} \tag{6.7}$$

where the parameters in (6.7) are related to those in (6.6) by the restrictions

$$\phi_{11} = 1+\alpha_1, \quad \phi_{12} = -\alpha_1\beta \quad \phi_{21} = \alpha_2, \quad \phi_{22} = 1-\alpha_2\beta.$$

Equation (6.7) is a VAR in the levels of the variables discussed in Chapter 4. Estimating the VAR yields estimates of ϕ_{11}, ϕ_{12}, ϕ_{21} and ϕ_{22}.
A comparison of equations (6.6) and (6.7) shows that cointegration imposes one cross-equation restriction on this system, which accounts for the difference in the number of parameters in the VAR and the VECM. This restriction arises as both variables are determined by the same underlying long-run relationship which involves the parameter β. The form of the restriction is recovered by noting that

$$\alpha_1 = \phi_{11} - 1, \qquad \alpha_2 = \phi_{21}, \qquad \beta = (1-\phi_{22})\phi_{21}^{-1}\,.$$

The additional VAR parameter can be expressed as a function of the other three VAR parameters as

$$\phi_{12} = (1-\phi_{11})(1-\phi_{22})\phi_{21}^{-1}\,.$$

This result suggests that if there is cointegration, estimating the unrestricted VAR in levels produces an estimate of ϕ_{12} that is close to the value that would be obtained from substituting the remaining VAR parameters estimates into this expression.
Alternatively, if there is no cointegration then there is nothing for the system to error-correct to and the error-correction parameters in (6.5) are simply $\alpha_1 =$

$\alpha_2 = 0$. The VECM is now a VAR in first differences. It is recognition of a second-best strategy whereby if no long-run relationship exists, then the next strategy is to model just the short-run relationships amongst the variables. This discussion touches on the old problem in time-series modelling of when to difference variables in order to address the problem of nonstationarity. The solution is to know whether there is cointegration or not. If there is cointegration, a VAR in levels is the correct specification. If there is no cointegration a VAR in first differences is required. Of course, if there is cointegartion an VECM can be specified, but in large samples this would be equivalent to estimating the VAR in levels. This result also highlights the importance of VECMs in modelling financial variables because it demonstrates that the old practice of automatically differencing variables to render them stationary and then estimating a VAR on the differenced data, rules out the possibility of a long-run relationship and hence any role for an error-correction term in modelling the dynamics.

6.6 Estimation

To illustrate the estimation of a VECM, consider a very simple specification based on Model 3 (unrestricted constant) with one lag on all the dynamic terms. The full VECM consists of the following three equations

$$y_{1t} = \mu + \beta y_{2t} + u_t\,, \tag{6.8}$$

$$\Delta y_{1t} = \delta_1 + \phi_{11}\Delta y_{1t-1} + \phi_{12}\Delta y_{2t-1} + \alpha_1(y_{1t-1} - \beta y_{2t-1}) + v_{1t}\,, \tag{6.9}$$

$$\Delta y_{2t} = \delta_2 + \phi_{21}\Delta y_{1t-1} + \phi_{22}\Delta y_{2t-1} + \alpha_2(y_{1t-1} - \beta y_{2t-1}) + v_{2t}\,, \tag{6.10}$$

whose parameters must be estimated. Two estimators are discussed initially, namely, the the Engle-Granger two-step procedure that provides estimates of the cointegrating equation without considering the dynamics from the VECM or the potential endogeneity of y_{2t}, and the the Johansen estimator that provides estimates of the cointegrating equation that takes into account all of the dynamics of the model. For this reason, the Johansen procedure is referred to as an efficient estimation procedure and the Engle-Granger method as the inefficient estimation procedure.

The Engle and Granger estimator (Engle and Granger, 1987)
The Engle Granger two stage procedure is implemented by estimating equations (6.8), (6.9) and (6.10) by ordinary least squares in two steps.

Long-run:
Regress y_{1t} on a constant and y_{2t} and compute the residuals $\widehat{u}_t$.

Short-run:
Estimate each equation of the error correction model in turn by ordinary least squares as follows

1. Regress Δy_{1t} on a constant, $\widehat{u}_{t-1}$, Δy_{1t-1} and Δy_{2t-1}.
2. Regress Δy_{2t} on a constant, $\widehat{u}_{t-1}$, Δy_{1t-1} and Δy_{2t-1}.

The error correction parameter estimates, $\widehat{\alpha}_1$ and $\widehat{\alpha}_2$, are the slope parameter estimates on $\widehat{u}_{t-1}$ in these two equations, respectively.

This estimator yields super-consistent estimates of the cointegrating vector (Stock, 1987; Phillips, 1987). Nevertheless the Engle-Granger estimator does not necessarily produce estimates that are asymptotically efficient, except under very strict conditions which are, in practice, unlikely to be satisfied. This results in the estimates having nonstandard distributions which invalidates the use of standard inferential methods.
The econometric problems with the Engle-Granger procedure arise from the potential endogeneity of y_t and autocorrelation in the disturbances u_t when simply estimating equation (6.8) by ordinary least squares. Thus, while it is not necessary to take into account short-run dynamics to obtain super-consistent estimates of the long-run parameters, it is necessary to model the short-run dynamics to obtain an efficient estimator with t statistics that have asymptotic distributions based on the normal distribution.

The Johansen estimator (Johansen, 1988, 1991, 1995).
In estimating the cointegrating regression in the two-step procedure none of the dynamics from the VECM are included in the estimation. A way to correct for this is to estimate all the parameters of the model jointly, a procedure known as the Johansen estimator This estimator provides more efficient estimates of the cointegrating parameters but the second stage still involves the same sequence of least squares regression but the $\widehat{u}_{t-1}$ will be different.
The Engle-Granger and Johansen estimators are now compared by estimating VECM model specified in equations (6.8) to (6.10) using the United States data on equity prices, dividends and earnings. Two separate cointegrating regressions are estimated, one for prices/dividends relationship (the dividend model) and one for prices/earnings relationship (the earnings model). The Engle-Granger two stage estimates are reported in Table 6.1. Table 6.2 gives the estimates of the VECM specified in equations (6.8) - (6.10) for the United States data on equity prices and earnings using the Johansen estimator.
The cointegration parameters returned by the Engle-Granger procedure in both cases are slightly greater than unity. Although it is tempting to look at the standard errors and claim that they are in fact significantly different from unity, this conclusion is premature as will be come apparent later. The signs of the error-correction parameters are consistent with the system converging to its long-run equilibrium as given by the cointegating equation because in both dynamic equations $\widehat{\alpha}_1 < 0$ and $\widehat{\alpha}_2 > 0$, respectively. Finally, one really interesting result concerns the estimate of the intercept μ in the cointegration equation for dividends. Equation (2.7) in Chapter 2 establishes that this intercept is related to the factor at which future dividends are discounted, δ. The relationship is

$$\delta = \exp(-\mu) = \exp(-3.129) = 0.044\,.$$

Table 6.1

Engle-Granger two-stage estimates of the price-dividend and price-earnings models. Estimates are for Model 3 (unrestricted constant) with 1 lag. The sample period is January 1871 to June 2004.

	Dividend Model			Earnings Model		
Parameter	Long Run	Δp_t	Δd_t	Long Run	Δp_t	Δy_t
β	1.179 (0.005)			1.042 (0.005)		
μ	3.129 (0.008)			2.607 (0.009)		
δ_i		0.002 (0.001)	0.000 (0.000)		0.002 (0.001)	0.000 (0.000)
ϕ_{i1}		0.291 (0.024)	0.000 (0.003)		0.286 (0.024)	0.011 (0.007)
ϕ_{i2}		0.148 (0.087)	0.877 (0.012)		0.074 (0.042)	0.8781 (0.012)
α_i		−0.007 (0.003)	0.002 (0.000)		−0.008 (0.003)	0.004 (0.001)

This estimate lines up nicely with the rough estimate of 0.05 obtained from Figure 2.5 in Chapter 2.

Not surprisingly there are few changes to the dynamic parameters of the VAR. The major changes, however, are in the standard errors of the parameter estimates of the cointegrating vector. The β estimates are 1.169 as opposed to 1.179 for dividends and 1.079 as opposed to 1.042 for earnings. These results are fairly similar which is to be expected given the super-consistency property of the estimators. The real difference is in the estimates of the standard errors with the Johansen estimates being about ten times larger than those yielded by the Engle-Granger procedure. This appreciable difference in standard errors illustrates very clearly that inference using the standard errors obtained from the Engle-Granger procedure cannot be relied on. Consider a standard t test of $H_0 : \beta = 1$.

Engle-Granger

Price/dividend model	Price/earnings model
$\frac{1.179-1}{0.005} = 35.8$	$\frac{1.042-1}{0.005} = 8.4$

Johansen

Price/dividend model	Price/earnings model
$\frac{1.169-1}{0.039} = 4.3$	$\frac{1.079-1}{0.039} = 2.0$

Although all these tests indicate a rejection of the null hypothesis, the rejection in the case of the Johansen procedure is less clear cut by far indicating

Table 6.2

Estimates of the of the price-dividend and price-earnings models using the Johansen estimator. Estimates are based on Model 3 (unrestricted constant) with 1 lag. The sample period is January 1871 to June 2004.

	Dividend Model			Earnings Model		
Parameter	Long Run	Δp_t	Δd_t	Long Run	Δp_t	Δy_t
β	1.169 (0.039)			1.079 (0.039)		
μ	3.390 (—)			2.791 (—)		
δ_i		0.002 (0.001)	0.000 (0.000)		0.001 (0.001)	0.001 (0.000)
ϕ_{i1}		0.291 (0.024)	0.000 (0.003)		0.286 (0.024)	0.012 (0.007)
ϕ_{i2}		0.148 (0.087)	0.877 (0.012)		0.072 (0.042)	0.871 (0.012)
α_i		−0.007 (0.003)	0.002 (0.000)		−0.008 (0.003)	0.004 (0.001)

that standard errors obtained via the Engle-Granger procedure should not be relied upon for inference.

6.7 Fully Modified Estimation†

The ordinary least squares estimator of β in (6.8) is super-consistent but inefficient. Solutions to the efficiency problem and bias introduced by possible endogeneity of the right-hand-side variables and serial correlation in u_t have also been addressed within single equation framework as opposed to the the system framework adopted by the Johansen estimator.
Consider the following system of equations

$$\begin{bmatrix} 1 & -\beta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}, \tag{6.11}$$

in which it should be apparent that both y_{1t} and y_{2t} are I(1) variables and u_{1t} and u_{2t} are I(0) disturbances. The first equation in the system is the cointegrating regression between y_{1t} and y_{2t} with the constant term taken to be zero for simplicity. The second equation is the nonstationary generating process for y_{2t}. In order to complete the system fully it is still necessary to specify the properties of the disturbance vector $u_t = [u_{1t}\ u_{2t}]'$. The most simple generating process that allows for serial correlation in u_t and possible endogeneity of

y_{2t} is the following simple autoregressive scheme of order 1

$$\begin{array}{rcl} u_{1t} & = & b_{11,1}u_{1t-1} + b_{12,0}u_{2t} + b_{12,1}u_{2t-1} + \epsilon_{1t} \\ u_{2t} & = & b_{21,0}u_{1t} + b_{21,1}u_{1t-1} + b_{22,1}u_{2t-1} + \epsilon_{2t} \end{array} \tag{6.12}$$

in which $\epsilon_t = [\epsilon_{1t}\ \epsilon_{2t}]' \sim iid(0, \Sigma)$ with

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}.$$

The notation in equation (6.12) can be simplified by using the lag operator L, defined as

$$L^0 z_t = z_t, \quad L^1 z_t = z_{t-1}, \quad L^2 z_t = z_{t-2}, \quad \cdots \quad L^n z_t = z_{t-n}\,.$$

For more information on the lag operator see, for example, Hamilton (1994) and Martin, Hurn and Harris (2013).
Using the lag operator, the system of equations (6.12) can be written as

$$B(L)u_t = \epsilon_t\,,$$

where

$$B(L) = \begin{bmatrix} 1 - b_{11,1}L & -b_{12,0} - b_{12,1}L \\ -b_{21,0} - b_{21,1}L & 1 - b_{22,1}L \end{bmatrix} = \begin{bmatrix} b_{11}(L) & b_{12}(L) \\ b_{21}(L) & b_{22}(L) \end{bmatrix}. \tag{6.13}$$

Once $B(L)$ is written in the form of the second matrix on the right-hand side of (6.13), then the matrix polynomials in the lag operator $b_{ij}(L)$ can be specified to have any order and, in addition, leads as well as lags of u_t can be entertained in the specification. In other words, the assumption of a simple autoregressive model of order 1 at the outset can be generalised without any additional effort.
In order to express the system (6.11) in terms of ϵ_t and not u_t and hence remove the serial correlation, it is necessary to premultiply by $B(L)$. The result is

$$\begin{bmatrix} b_{11}(L) & -\beta b_{11}(L) + b_{12}(L) \\ b_{21}(L) & -\beta b_{21}(L) + b_{22}(L) \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} 0 & b_{12}(L) \\ 0 & b_{22}(L) \end{bmatrix} \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}, \tag{6.14}$$

The problem with single equation estimation of the cointegrating regression is now obvious: the cointegrating parameter β appears in both equations of (6.14). This suggests that to estimate the cointegrating vector, a systems approach is needed which takes into account this cross-equation restriction, the solution provided by Johansen estimator (Johansen, 1988, 1991, 1995).
It follows from (6.14) that for a single equation approach to produce asymptotically efficient parameter estimates two requirements that need to be satisfied.

1. There should be no cross equation restrictions so that $b_{21}(L) = 0$.

2. There should be no contemporaneous correlation between the disturbance term in the equation used to estimate β and the ϵ_{2t}, the error term in the equation generating y_{2t}. If this condition is not satisfied, the second equation in (6.14) cannot be ignored in the estimation of β.

Assuming now that $b_{21}(L) = 0$, adding and subtracting $(y_{1t} - \beta y_{2t})$ from the first equation in (6.14) and rearranging yields

$$y_{1t} - \beta y_{2t} + [b_{11}(L) - 1](y_{1t} - \beta y_{2t}) + b_{12}(L)(y_{2t} - y_{2t-1}) = \epsilon_{1t}\,. \tag{6.15}$$

The problem remains that $\mathrm{E}[\epsilon_{1t}\,\epsilon_{2t}] = \sigma_{12} \neq 0$ so that the second condition outlined earlier is not yet satisfied. The remedy is to multiply the second equation by $\rho = \sigma_{12}/\sigma_{22}$ and subtract the result from the first equation in (6.14). The result is

$$y_{1t} - \beta y_{2t} + [b_{11}(L) - 1](y_{1t} - \beta y_{2t}) + (b_{12}(L) - \rho b_{22}(L))(y_{2t} - y_{2t-1}) = v_t\,, \tag{6.16}$$

in which $v_t = \epsilon_{1t} - \rho\epsilon_{2t}$. As a result of this restructuring it follows that

$$\mathrm{E}[v_t\,\epsilon_{2t}] = \mathrm{E}[(\epsilon_{1t} - \rho\epsilon_{2t})\,\epsilon_{2t}] = \sigma_{12} - \rho\sigma_{22} = \sigma_{12} - \frac{\sigma_{12}}{\sigma_{22}}\,\sigma_{22} = 0\,,$$

so that the second condition for efficient single equation estimation of the cointegrating parameter β is now satisfied.
Equation (6.16) provides a relationship between y_{1t} and its long-run equilibrium level, βy_{2t}, with the dynamics of the relationship being controlled by the structure of the polynomials in the lag operator, $b_{11}(L)$, $b_{12}(L)$ and $b_{22}(L)$. A very general specification of these lag polynomials will allow for different lag orders and also leads as well as lags. In other words, the a general version of (6.16) will allow for both the leads and lags of the cointegrating relationship, $(y_{1t} - \beta y_{2t})$ and the leads and lags of Δy_{2t}. A reduced form version of this equation is

$$y_{1t} = \beta y_{2t} + \sum_{\substack{k=-q \\ k\neq 0}}^{q} \pi_k(y_{1t-k} - \beta y_{2t-k}) + \sum_{k=-q}^{q} \alpha_k \Delta y_{2t-k} + \eta_t\,, \tag{6.17}$$

where for the sake of simplicity the lag length in all cases has been set at q.
As noted by Lim and Martin (1995), this approach to obtaining asymptotically efficient parameter estimates of the cointegrating vector can be interpreted as a parametric filtering procedure, in which the filter expresses u_{1t} in terms of observable variables which are then included as regressors in the estimation of the cointegrating vector.The intuition behind this approach is that improved estimates of the long-run parameters can be obtained by using information on the short-run dynamics.

The Phillips and Loretan estimator (Phillips and Loretan, 1991)

The Phillips and Loretan (1991) estimator excludes the leads of the cointegrating vector from equation (6.17) are excluded. The equation is

$$y_{1t} = \beta y_{2t} + \sum_{k=1}^{q} \pi_k (y_{1t-k} - \beta y_{2t-k}) + \sum_{k=-q}^{q} \alpha_k \Delta y_{2t-k} + \eta_t \,, \tag{6.18}$$

which is estimated by non-linear least squares. This procedure yields super-consistent and asymptotically efficient estimates of the cointegrating vector if all the restrictions in moving from (6.14) to (6.18) are satisfied.

Dynamic least squares (Saikkonen, 1991; Stock and Watson, 1993)
The dynamic least squares estimator excludes the lags and leads of the cointegrating vector from equation (6.17). The equation is

$$y_{1t} = \beta y_{2t} + \sum_{k=-q}^{q} \alpha_k \Delta y_{2t-k} + \eta_t \,, \tag{6.19}$$

which has the advantage of being estimated by ordinary least squares. This procedure yields super-consistent and asymptotically efficient estimates of the cointegrating vector if all the restrictions in moving from (6.14) to (6.19) are satisfied.

Fully modified least squares (Phillips and Hansen, 1990)
The fully modified estimator excludes the lags and leads of the cointegrating vector and limits the terms in Δy_{2t} to the contemporaneous difference with coefficient ρ. The resulting model is

$$y_{1t} = \beta y_{2t} + \rho \Delta y_{2t} + \eta_t \,. \tag{6.20}$$

Comparison of the first equation in (6.11) and (6.20) implies that

$$u_{1t} = \rho \Delta y_{2t} + \eta_t \,. \tag{6.21}$$

The fully modified ordinary least squares approach is now implement in three steps.

1. Estimate first equation in (6.11) by ordinary least squares to obtain $\widehat{\beta}$ and $\widehat{u}_{1t}$.
2. Estimate (6.21) by ordinary least squares to obtain estimates of $\widehat{\rho}$ of $\widehat{\sigma}^2_\eta$.
3. Regress the constructed variable $y_{1t} - \widehat{\rho} \Delta y_{2t}$ on y_{2t} and get a revised estimate of $\widehat{\beta}$. Use the estimate of $\widehat{\sigma}^2_\eta$ to construct standard errors.

The Engle and Yoo estimator (Engle and Yoo, 1991)

The Engle and Yoo estimator starts by formulating the error correction version of equation (6.20) by adding and subtracting y_{1t-1} from the left-hand-side and adding and subtracting βy_{2t-1} from the right-hand-side and rearranging to yield

$$\Delta y_{1t} = -(y_{1t-1} - \beta y_{2t-1}) + (\beta + \rho)\Delta y_{2t} + \eta_t \,. \tag{6.22}$$

Given an estimate $\widehat{\beta}$, a reduced form version of (6.22) is

$$\Delta y_{1t} = -\delta(y_{1t-1} - \widehat{\beta} y_{2t-1}) + \alpha \Delta y_{2t} + w_t \,. \tag{6.23}$$

in which

$$w_t = \alpha\delta y_{2t-1} + \eta_t \,, \qquad \alpha = \beta - \widehat{\beta} \,. \tag{6.24}$$

The Engle and Yoo estimator is implemented in three steps.

1. Estimate the first equation in (6.11) by ordinary least squares to obtain $\widehat{\beta}$ and $\widehat{u}_{1t}$.
2. Estimate (6.23) by ordinary least squares to obtain estimates of $\widehat{w}_t$ and $\widehat{\delta}$.
3. Regress the residuals $\widehat{w}_t$ on y_{2t-1} and in order to obtain $\widehat{\alpha}$. The revised estimate of β is given by $\widehat{\beta} + \widehat{\alpha}$.

Table 6.3

Single equation estimates of the cointegration regression between stock prices and dividends and stock prices and earnings, respectively. The dynamic ordinary least squares estimates use one forward lead and one backward lag. The sample period is January 1871 to June 2004.

	Dividend Model			Earnings Model		
	OLS	DOLS	FMOLS	OLS	DOLS	FMOLS
β	1.179	1.174	1.191	1.042	1.043	1.065
	(0.005)	(0.040)	(0.038)	(0.005)	(0.039)	(0.038)
μ	3.129	3.117	3.143	2.607	2.607	2.612
	(0.008)	(0.056)	(0.053)	(0.009)	(0.065)	(0.064)

Table 6.3 compares the ordinary least squares estimator of the cointegrating regression with the fully modified and dynamic ordinary least squares estimators. Comparison with the results in Table 6.2 shows that the fully modified ordinary least squares estimator works particularly well in the case of the earnings model, which previously was identified as the more problematic of the two models in terms of potential endogeneity. The dynamic least squares estimator is less impressive in this situation, although there may be scope for improvement by considering a longer lead/lag structure. Interestingly, the standard errors on the fully modified and dynamic least squares

approaches are similar to those of the Johansen approach. The results suggest that modified single equation approaches can help to improve inference in the cointegrating regression. The limitation of these approaches remains that the dimension of the cointegration space is always limited to unity.

6.8 Testing for Cointegration

Up to this point the existence of a cointegrating relationship has merely been posited or assumed. Of course, the identification of cointegration is a crucial step in modelling with nonstationary variables and is, in fact, the place where the modelling procedure actually begins. Yule (1926) first drew attention to the problems of modelling with unrelated nonstationary variables and Granger and Newbold (1974) later showed that regression involving non stationary variables can lead to spurious correlations. Spurious regressions arise when unrelated nonstationary variables are found to have a statistically significant relationship. Suppose y_t and x_t are unrelated $I(1)$ variables, the chance of getting a nonzero estimate of a regression coefficient of x_t on y_t, even though the true value is zero, is substantial. Banerjee, Dolado, Galbraith and Hendry (1993) showed that in a sample size of 100 a rejection probability of 75.3% was obtained. Morevoer, the problem does not go away in large samples, in fact the opposite is true which the rejection probability of a zero coefficient going up the larger the sample gets. To guard against spurious regressions it is critically important that cointegration can be identified reliably.

6.8.1 Residual-based tests

A natural way to test for cointegration is a two-step procedure consisting of estimating the cointegrating equation by least squares in the first step and testing the residuals for stationarity in the second step. As the unit root test treats the null hypothesis as nonstationary, in applying the unit root procedure to test for cointegration the null hypothesis is no cointegration whereas the alternative hypothesis of stationarity represents cointegration:

$$\begin{array}{lll} H_0: & \text{No Cointegration} & (u_t \text{ is nonstationary}) \\ H_1: & \text{Cointegration} & (u_t \text{ is stationary}). \end{array}$$

This is a sensible strategy given that the estimator of the cointegrating equation is super-consistent and converges at the faster rate of T to its population value compared to the usual rate of $\sqrt{T}$ for stationary variables. However, in applying a unit root test to the ordinary least squares residuals the critical values must take into account the loss of degrees of freedom in estimating the cointegrating equation. The critical values of the tests depend on the sample size and the number of deterministic terms and other regressors in the first stage regression. Tables are provided by Engle and Granger (1987) and Engle and Yoo (1987). MacKinnon (1991) provides response surface estimates of the critical values that are now used in most computer packages.

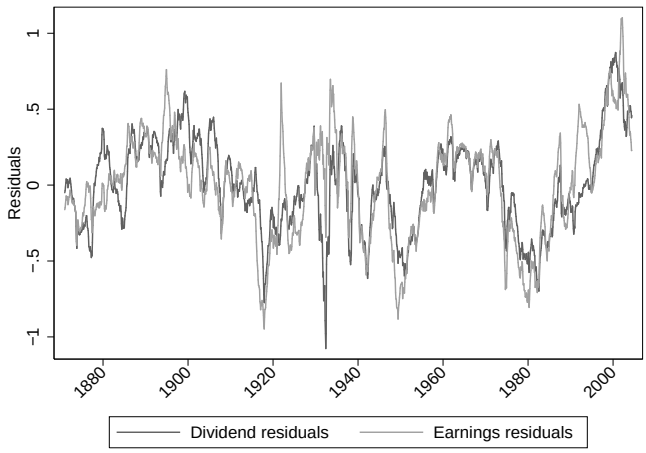


Figure 6.4: Plot of the residuals from the first stage of the Engle-Granger two stage procedure applied to the dividend model and the earnings model, respectively. Data are monthly observations from February 1871 to June 2004 on United States equity prices, dividends and earnings per share.

The residuals obtained by estimating the cointegrating regressions for the dividend model, (6.1), and the earnings model, (6.2), respectively, by ordinary least squares are plotted in Figure 6.4. The series appear to have mean zero and there is no trend apparent giving the appearance of stationarity. Formal tests of the stationarity of the residuals are carried out using the Dickey-Fuller framework, based on a test regression with no constant or trend. The results are shown in Table 6.4 for up to four lags used to augment the test regression. Despite the aberration of the Dickey-Fuller test (0 lags) failing to reject the null hypothesis of nonstationarity, the results from the augmented Dickey-Fuller test are unequivocal. The null hypothesis of nonstationarity is rejected and the residuals are $I(0)$. This confirms the intuition provided by Figure 6.4 and allows the conclusion that both the dividend model and the earnings model represent valid long-run relationships between equity prices and dividends and equity prices and earnings per share, respectively.

Although residual-based tests of cointegration are a natural way to think about the problem of testing for cointegration they suffer from the same problem as all single equation approaches to cointegration, namely, that the number of cointegrating relationships is necessarily limited to one. This is not problematic in the case of two variables, but it is severely limiting when wanting to consider the multivariate case.

Table 6.4

Residual-based test of cointegration between United States equity prices and dividends and equity prices and earnings. The test regression has no constant term and number of lags as shown. Critical values are from MacKinnon (1991).

Lags	Dividend Model Statistic	Earnings Model Statistic
0	-2.654	-2.674
1	-3.890	-4.090
2	-3.630	-3.921
3	-3.576	-3.936
4	-3.814	-4.170

Note that the 5% critical value of the test is -3.340

6.8.2 Johansen Reduced-rank Tests

Consider the following simple model

$$\begin{bmatrix} \Delta y_{1t} \\ \Delta y_{2t} \end{bmatrix} = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \end{bmatrix} + \begin{bmatrix} v_{1t} \\ v_{2t} \end{bmatrix}, \tag{6.25}$$

which is a bivariate VAR rearranged to look like a VECM but with no long-run equilibrium relationships imposed. In other words, the matrix

$$\Pi = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix},$$

is an unrestricted matrix in which the rows and columns of the matrix are not related in a linear fashion. This condition is referred to as the matrix having full rank. As this model is simply a VAR model written in a particular way for this to be a correct representation of the data both y_{1t} and y_{2t} must be stationary.

Now consider the situation when y_{1t} and y_{2t} share a long-run relationship with cointegrating parameter β with speed of adjustment parameters α_1 and α_2 in the first and second equations, respectively. Equation (6.25) must be restricted to reflect this long-run relationship to yield the familiar VECM

$$\begin{bmatrix} \Delta y_{1t} \\ \Delta y_{2t} \end{bmatrix} = \begin{bmatrix} \alpha_1 & \alpha_1\beta \\ \alpha_2 & \alpha_2\beta \end{bmatrix} \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \end{bmatrix} + \begin{bmatrix} v_{1t} \\ v_{2t} \end{bmatrix}. \tag{6.26}$$

so that

$$\Pi = \begin{bmatrix} \alpha_1 & \alpha_1\beta \\ \alpha_2 & \alpha_2\beta \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \begin{bmatrix} 1 & \beta \end{bmatrix}.$$

The effect of the long-run relationship is to restrict the elements of the matrix Π. In particular the second column of Π is simply the first column multiplied

by β so that there is now dependence between the columns of the matrix. The matrix Π is now referred to as having reduced rank, in this case rank one.
If the matrix Π has rank zero then the system becomes

$$\begin{bmatrix} \Delta y_{1t} \\ \Delta y_{2t} \end{bmatrix} = \begin{bmatrix} v_{1t} \\ v_{2t} \end{bmatrix}, \tag{6.27}$$

in which both y_{1t} and y_{2t} are nonstationary.
It is now apparent from equations (6.25) to (6.25) that testing for cointegration is equivalent to testing the validity of restrictions on the matrix Π, or determining the rank of this matrix. In other words, testing for cointegration amounts to testing if the matrix Π has reduced rank. As the rank of the matrix is determined from the number of significant eigenvalues, Johansen provides two tests of cointegration based on the eigenvalues of the matrix Π, known as the maximal eigenvalue test and the trace test respectively (Johansen, 1988, 1991, 1995). Testing for cointegration based on the eigenvalues of Π is now widely used because it has two advantages over the two-step residual based test, namely, the tests generate the correct p values and the tests are easily applied in a multivariate context where testing for several cointegrating equations jointly is required.
The Johansen cointegration test proceeds sequentially. If there are two variables being tested for cointegration the maximum number of hypotheses considered is two. If there are N variables being tested for possible cointegration the maximum number of hypotheses considered is N.

Stage 1:

H_0 : No cointegrating equations
H_1 : One or more cointegrating equations .

Under the null hypothesis all of the variables are $I(1)$ and there is no linear combination of the variables that achieves cointegration. Under the alternative hypothesis there is (at least) one linear combination of the $I(1)$ variables that yields a stationary disturbance and hence cointegration. If the null hypothesis is not rejected then the hypothesis testing stops. Alternatively, if the null hypothesis is rejected it could be the case that there is more than one linear combination of the variables that achieves stationarity so the process continues.

Stage 2:

H_0 : One cointegrating equation
H_1 : Two or more cointegrating equations .

If the null hypothesis is not rejected the testing procedure stops and the conclusion that there are two cointegrating equations. Otherwise proceed to the next stage.

Stage N:

H_0 : $N-1$ cointegrating equations
H_1 : All variables are stationary .

At the final stage, the alternative hypothesis is that all variables are stationary and not that there are N cointegating equations. For there to be N linear stationary combinations of the variables, the variables need to be stationary in the first place.

Large values of the Johansen cointegration statistic relative to the critical value result in rejection of the null hypothesis. Alternatively, small p values less than 0.05 for example, represents a rejection of the null hypothesis at the 5% level. In performing the cointegration test, it is necessary to specify the VECM to be used in the estimation of the matrix Π. The deterministic components (constant and time trend) as well as the number of lagged dependent variables to capture autocorrelation in the residuals must be specified.

Table 6.5

Johansen tests of cointegration between United States equity prices, dividends and earnings. Testing is based on Model 3 (unrestricted constant) with 2 lags in the underlying VAR.

		Dividend Model			
		Trace Test		Max Test	
Rank	Eigenvalue	Statistic	5% CV	Statistic	5% CV
0	·	32.2643	15.41	30.8132	14.07
1	0.01907	1.4510	3.76	1.4510	3.76
2	0.00091	·	·	·	·
		Earnings Model			
		Trace Test		Max Test	
Rank	Eigenvalue	Statistic	5% CV	Statistic	5% CV
0	·	33.1124	15.41	32.1310	14.07
1	0.01988	0.9814	3.76	0.9814	3.76
2	0.00061	·	·	·	·
		Combined Model			
		Trace Test		Max Test	
Rank	Eigenvalue	Statistic	5% CV	Statistic	5% CV
0	·	109.6699	29.68	83.0022	20.97
1	0.05055	26.6677	15.41	25.4183	14.07
2	0.01576	1.2495	3.76	1.2495	3.76
3	0.00078	·	·	·	·

The results of the Johansen cointegration test applied to the United States equity prices, dividends and earnings data is given in Table 6.5. Results are provided for the dividend model, the earnings model and a combined model which tests all three variables simultaneously. For the first two models, $N = 2$, so the maximum rank of the Π matrix is 2. Inspection of the first null hy-

pothesis of zero rank or no cointegration shows that the null hypothesis is easily rejected at the 5% level for both the dividend and earnings models. There is therefore at least one cointegrating vector in both of these specifications. The next hypothesis corresponds to Π having rank one or there being one cointegating equation. The null hypothesis is not rejected at the 5% level for both models, so the conclusion is that there is one cointegrating equation that combines prices and dividends and one cointegrating equation that combines prices and earnings into stationary series.
The results of the Johansen cointegration test applied to the combined model consisting of real equity prices, real dividends and earnings per share are given in the bottom panel of Table 6.5. The body of the table contains three rows as there are now $N = 3$ variables being examined. The first null hypothesis of zero rank or no cointegration is easily rejected at the 5% level so there is at least one linear combination of these variables that is stationary. The next hypothesis corresponds to Π having rank one or there being one cointegating equation. The null hypothesis is again rejected at the 5% level so there are at least two cointegrating relationships between these three variables. The null hypothesis of a rank of two cannot be rejected at the 5% level, so the conclusion is that there are two linear combinations of these three variables that produce a stationary residual.

6.9 Multivariate Cointegration

The results of the Johansen cointegration test applied to the the three variable system of real equity prices, real dividends and earnings per share in the previous section indicated that there are two cointegrating vectors. There are thus two combinations of these three nonstationary variables that yield stationary residuals. The next logical step is to estimate a VECM which takes all three variables as arguments and imposes a cointegrating rank of two on the estimation. The results of this estimation are shown in Table 6.6.
The interpretation of the results in Table 6.6 proceeds as follows.

(1) Cointegrating equations:
The first cointegrating equation estimates the long-run relationship between price and earnings and is normalised with respect to price. The second cointegrating relationship is between dividends and earnings, normalised with repeat to dividends.

(2) Speed of adjustment parameters:
The signs and significance of the speed of adjustment parameters on the error correction terms help to establish the stability of the estimated relationships. Stability requires that the coefficient of adjustment on the error correction term in the equation for Δp_t be negative. This is indeed the case and the estimate is also significant, although marginally so. The coefficient of adjustment in the earnings equation is positive and significant which is also required by theory. Interestingly, the adjustment coef-

Table 6.6

Estimates of a three-variable VECM(1) for equity prices, p_t, dividends, d_t and earnings per share, s_t, using the Johansen estimator based on Model 3 (unrestricted constant). The sample period is January 1871 to June 2004.

The two estimated cointegrating equations are

$$p_t = \underset{(0.042)}{1.072} s_t + 2.798 + \widehat{u}_{1t}$$

$$d_t = \underset{(0.012)}{0.910} s_t - 0.445 + \widehat{u}_{2t}$$

Variable	Δp_t	Δd_t	Δs_t
$\widehat{u}_{1t-1}$	−0.0082	0.0017	0.0029
	(0.0034)	(0.0004)	(0.0010)
$\widehat{u}_{2t-1}$	0.0014	−0.0072	0.0049
	(0.0069)	(0.0009)	(0.0020)
Δp_{t-1}	0.2868	−0.0020	0.0134
	(0.0242)	(0.0032)	(0.0070)
Δd_{t-1}	0.3674	0.8194	0.0542
	(0.1015)	(0.0133)	(0.0292)
Δs_{t-1}	0.0699	0.0235	0.8748
	(0.0465)	(0.0061)	(0.0133)
Constant	0.0005	0.0006	0.0009
	(0.0012)	(0.0001)	(0.0004)

ficient in the dividend equation is also significant. This is to be expected because earnings and dividends are closely related as demonstrated by the second cointegrating equation. What this suggests is that dividends and earnings adjust more aggressively than prices do to correct any deviation from long-run equilibrium.

As expected the adjustment parameter on the second error-correction term is negative and significant in the dividend equation and positive and significant in the dividend equation. Notice however that the coefficient of adjustment on $\widehat{u}_{2t-1}$ in the Δp_t equation is insignificant which is to be expected given that price is not expected to adjust to a divergence from long-run equilibrium between dividends and earnings.

(3) Dynamic parameters:

The first test of interest on the parameters of the VECM relates to the significance of the constant terms in the short-run dynamic specification of the system. This relates to the choice of Model 3 (unrestricted constant) as opposed to Model 2 (restricted constant) where the constant term only appears in the cointegrating equations. Although the constants are all small in absolute size at least two of them appear to be

estimated fairly precisely. The joint hypothesis that they are all zero, or equivalently that Model 2 is preferable to Model 3, is therefore unlikely to be accepted.

An important issue in estimating multivariate systems in which there are cointegrating relationships is that the estimates of the cointegrating vectors are not unique, but depend on the normalisation rules which are adopted. For example, the results obtained when estimating this three variable system but imposing the normalisation rule that both cointegrating equations are normalised on p_t are reported in Table 6.7.

Table 6.7

Estimates of the three-variable VECM for equity prices, dividends and earnings per share using the Johansen estimator. Estimates are based on Model 3 (unrestricted constant) with 1 lag of the differenced variables. The sample period is January 1871 to June 2004.

The two estimated cointegrating equations are

$$p_t = \underset{(0.039)}{1.072} s_t + 2.798 + \widehat{u}_{1t}$$
$$p_t = \underset{(0.039)}{1.777} d_t + 3.323 + \widehat{u}_{2t}$$

Variable	Δp_t	Δd_t	Δs_t
$\widehat{u}_{1t-1}$	−0.0070	−0.0045	0.0071
	(0.0051)	(0.0007)	(0.0015)
$\widehat{u}_{2t-1}$	0.0012	0.0062	−0.0042
	(0.0059)	(0.0008)	(0.0017)
Δp_{t-1}	0.2868	−0.0020	0.0134
	(0.0242)	(0.0032)	(0.0070)
Δd_{t-1}	0.3674	0.8194	0.0542
	(0.1015)	(0.0133)	(0.0292)
Δs_{t-1}	0.0699	0.0235	0.8748
	(0.0465)	(0.0061)	(0.0133)
Constant	0.0005	0.0006	0.0009
	(0.0012)	(0.0001)	(0.0004)

The two cointegrating regressions reported in Table 6.7 are now the familiar expressions that have been dealt with in the bivariate cases throughout the chapter (see for example, Table 6.2). While this seems to contradict the results reported in Table 6.6 the two sets of long-run relationships are easily reconciled. It follows directly from the results in Table 6.7 that

$$p_t = 1.777 d_t = 1.072 s_t \Rightarrow d_t = 1.072/1.777 s_t = 0.9107 s_t \,,$$

which corresponds to the second cointegrating equation in Table 6.6. Alterna-

tively, subtracting the first equation from the second gives

$$p_t - p_t = 0 = (1.777d_t + 3.323) - (1.072s_t + 2.798)\,,$$

and rearranging gives

$$d_t = 0.910y_t - 0.445\,,$$

which is the same as the result reported in Table 6.6.
One final interesting point to note is that Table 6.7 confirms the rather weak adjustment by prices to disequilibrium. Both the adjustment parameters on $\widehat{u}_{1t-1}$ and $\widehat{u}_{2t-1}$ in this specification are insignificantly different from zero. What this suggests is that dividends and earnings per share tend to pick up most of the adjustment in relation to shocks which disturb the long-run equilibrium.

6.10 Cointegration and the Yield Curve

In Chapter 2 the concept of the yield to maturity of a bond was introduced. Yield to maturity allows bonds of different maturing and coupon to be compared because it represents the compound average yield to maturity if the bond does not default. A plot of bond yields as a function of maturity is known as the yield curve and the empirical verification of theories of the determination of the yield curve has been a fertile research area in empirical finance. One of the most tested theories of the yield curve is the expectations hypothesis which posits that the yield on a long-maturity bond is simply the average of the expected future yields on a short maturity bond. For the same of argument take $n = 3$ month maturity bond, y_{3t} and $m = 1$ month maturity bond, then the investor at time $t = 0$ can either invest \$1 for 3 months at the 3-month yield or invest \$1 in a1-month bond at the current yield, y_t and reinvest at the prevailing one month yield in months two and three. The expectations hypothesis requires that these two strategies to yield the same return, so that

$$y_{3t} = \frac{1}{3}\mathrm{E}_t[y_t + y_{t+1} + y_{t+2}]. \tag{6.28}$$

Equation (6.28) may be rearranged so that it is the spread between the long and short interest rate that appears on the left hand side. Subtracting r_t from both sides, adding and subtracting $1/3r_{t+1}$ from the right hand side rearranging yields

$$\begin{aligned} y_{3t} - r_t &= \mathrm{E}_t[-\frac{2}{3}r_t + \frac{1}{3}r_{t+1} + \frac{1}{3}r_{t+1} - \frac{1}{3}r_{t+1} + \frac{1}{3}r_{t+2}] \\ &= \mathrm{E}_t[-\frac{2}{3}r_t + \frac{2}{3}r_{t+1} - \frac{1}{3}r_{t+1} + \frac{1}{3}r_{t+2}] \\ &= \mathrm{E}_t[\frac{2}{3}\Delta r_{t+1} + \frac{1}{3}\Delta r_{t+2}], \end{aligned} \tag{6.29}$$

in which the spread between the long and the short yields is expressed as a weighted sum of changes in the short yield. In fact equation (6.29) generalises very nicely to all n-maturity yields, y_{nt}, and m-maturity short yields, y_{mt}, such that $k = n/m$ is an integer in the following way

$$y_{nt} - y_{mt} = \mathrm{E}_t \sum_{j=1}^{k-1} \left(1 - \frac{j}{k}\right) \Delta^m y_{m,\, t+jm}. \tag{6.30}$$

The determinants of the spread between long and short yields in equations (6.29) and (6.30) lends itself to empirical analysis within the cointegration framework developed in this chapter. Unit root tests applied to bond yields invariably result in the conclusion that they can be regarded as nonstationary $I(1)$ variables, at least for the specific sample being tested. If yields are generally $I(1)$, then their first differences will be stationary $I(0)$ variables. Equations (6.29) and (6.30) provide an empirical test of the theory because, if the expectations hypothesis is true then the spread, can be expressed as a sum of stationary variables, and must therefore be stationary. Put differently, the linear combination of a long and short yield, which are both $I(1)$, has the interpretation of being a cointegrating relationship which gives rise to a stationary spread.

Consider monthly data from December 1946 to February 1987 on United States zero coupon bond yields for maturities of 3, 6 and 9 months. The matrix scatter plot of the yields against each other shown in Figure 6.5 clearly indicates that there is some common dynamic between the yields.

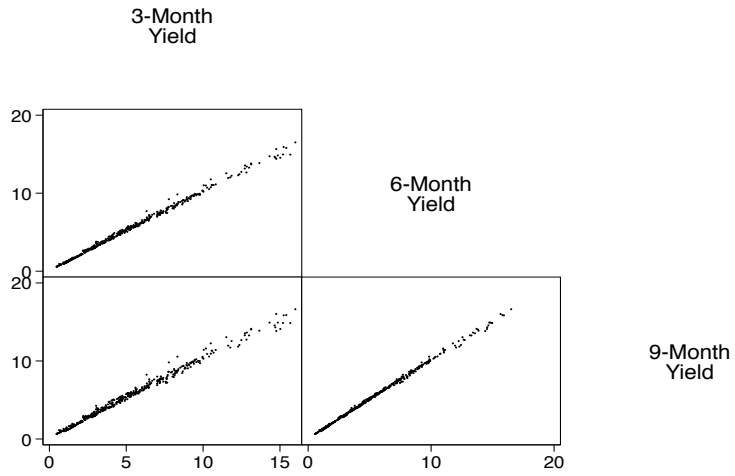


Figure 6.5: Scatter plots of the 3-month, y_{3t}, 6-month, y_{6t}, and 9-month, y_{9t}, zero coupon bond yields. The data are monthly for the period December 1946 to February 1987.

Using equation (6.30), the following relationships can be deduced

$$y_{9t} - y_{3t} = u_t, \qquad u_t = \mathrm{E}_t \sum_{j=1}^{2} \Delta^3 y_{3\,t+j3} \tag{6.31}$$

$$y_{6t} - y_{3t} = v_t, \qquad u_t = \mathrm{E}_t\, \Delta^2 y_{3\,t+3}, \tag{6.32}$$

in which both u_t and v_t are $I(0)$. Note that the spread $y_{9t} - y_{6t}$ is not an independent quantity because simply subtracting (6.32) from (6.31) gives

$$y_{9t} - y_{6t} = u_t - v_t,$$

which is also $I(0)$. It follows, therefore, that testing for the cointegrating rank between the 3, 6 and 9 month zero coupon bond yields should give $r = 2$ cointegrating vectors.

Table 6.8

Johansen trace test of for the cointegrating rank of a VECM using 3-month, 6-month and 9-month zero coupon yields. The VECM has a restricted constant specification and 1 lag in the dynamic equations.

H_0	Statistic	Critical Value (5%)	p-value
None	164.351	35.193	0.000
One	73.917	20.262	0.000
Two	5.434	9.165	0.239

The results of the Johansen cointegration trace test applied to the 3-month, 6-month and 9-month zero coupon interest rates are given in Table 6.8. Inspection of the first null hypothesis of no cointegration shows that the null is easily rejected at the 5% level. A similar result holds for the next null hypothesis of one cointegrating equation which is also easily rejected at the 5% level. Moving to the third null hypothesis of two cointegration equations, the results show that this null hypothesis is not rejected at the 5% level. As the null hypothesis is not rejected at the third stage, the conclusion is that there are two cointegrating equations which combine the three interest rates into two stationary series. The Johansen maximal eigenvalue test gives the exactly the same conclusion.
A VECM is estimated for the 3-month, y_{3t}, 6-month, y_{6t} and 9-month, y_{9t}, zero coupon bond yields, using the restricted constant specification with one lag. The estimated cointegrating equations are

$$\begin{aligned} y_{9t} &= 0.213 + 1.025y_{3t} + \widehat{u}_{1t}, \\ y_{6t} &= 0.123 + 1.021y_{3t} + \widehat{u}_{2t}, \end{aligned}$$

and the estimated VECM is

$$\begin{aligned} \Delta y_{9t} &= 0.289\,(y_{9\,t-1} - 1.025y_{3\,t-1} - 0.213) - 0.849\,(y_{6\,t-1} - 1.021y_{3\,t-1} - 0.123) \\ &\quad + 0.214\Delta y_{9\,t-1} + 0.064\Delta y_{6\,t-1} - 0.144\Delta y_{3\,t-1} + \widehat{v}_{1t} \\ \Delta y_{6t} &= 0.543\,(y_{9\,t-1} - 1.025y_{3\,t-1} - 0.213) - 1.072\,(y_{6\,t-1} - 1.021y_{3\,t-1} - 0.123) \\ &\quad + 0.360\Delta y_{9\,t-1} - 0.190\Delta y_{6\,t-1} - 0.149\Delta y_{3\,t-1} + \widehat{v}_{2t} \\ \Delta y_{3t} &= 0.216\,(y_{9\,t-1} - 1.025y_{3\,t-1} - 0.213) - 0.308\,(y_{6\,t-1} - 1.021y_{3\,t-1} - 0.123) \\ &\quad + 0.471\Delta y_{9\,t-1} - 0.235\Delta y_{6\,t-1} - 0.091\Delta y_{3\,t-1} + \widehat{v}_{3t}. \end{aligned}$$

A number of tests may now be carried out on the VECM to help understand its dynamic properties.

Dynamic Stability

The presence of two sets of error-correction parameters makes it difficult to determine whether or not the VECM is stable simply by inspecting the co-

efficient estimates. A useful tool for evaluating the dynamic stability of the estimated VECM is to shock the system and observe whether or not it behaves as expected. Figure 6.6 plots the estimated impulse responses for y_{9t} and y_{6t} when v_{3t}, the residual in the equation for the dynamics of the short term yield, y_{3t}, is shocked. The number of impulses is 24 representing a 2 years horizon. Stable VECM dynamics should ensure that both y_{9t} and y_{6t} converge to their long-run equilibrium values within this period. The impulse responses certainly converge to a steady state value within 2 years, but it remains to be checked that the values to which they converge are indeed the correct long-run values implied by the VECM.

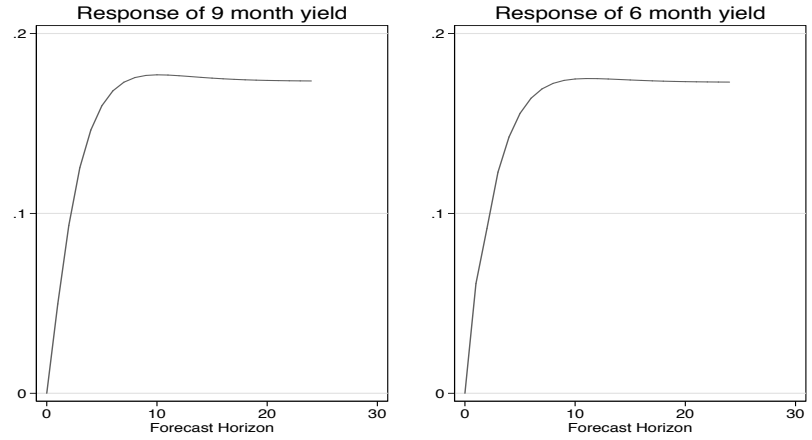


Figure 6.6: Impulse responses of the VECM for the 3-month, y_{3t}, 6-month, y_{6t}, and 9-month, y_{9t}, zero coupon bond interest rates, using the restricted constant specification with with one lag. The impulse is to y_{3t} and the responses shown are for y_{9t} and y_{6t}.

To work out the long-run parameter estimates from the impulse responses select the following final values from the set of impulses

$$\frac{\partial y_{9\,t+24}}{\partial v_{3t}} = 0.173496, \quad \frac{\partial y_{6\,t+24}}{\partial v_{3t}} = 0.172859, \quad \frac{\partial y_{3\,t+24}}{\partial v_{3,t}} = 0.169266.$$

The long-run response of y_{9t} to y_{3t} is

$$\frac{\partial y_{9\,t+24}}{\partial y_{3\,t+24}} = \frac{\partial y_{9\,t+24}}{\partial v_{3t}} \frac{\partial v_{3t}}{\partial y_{3\,t+24}} = 0.173496 \times \frac{1}{0.169266} = 1.025.$$

The long-run response of y_{6t} to y_{3t} is

$$\frac{\partial y_{6\,t+24}}{\partial y_{3\,t+24}} = \frac{\partial y_{6\,t+24}}{\partial v_{3t}} \frac{\partial v_{3t}}{\partial y_{3\,t+24}} = 0.172859 \times \frac{1}{0.169266} = 1.021.$$

These estimates agree with the long-run parameter estimates of the two cointegrating equations or the VECM of this system. It seems reasonable to conclude that the dynamics of the VECM are stable.

Tests on the Cointegrating Parameters

Of particular interest in testing the expectations hypothesis of the term structure of interest rates, is a test that two cointegrating parameter vectors corresponding to y_{9t}, y_{6t} and y_{3t} are, respectively, $(1, 0, -1)$ and $(0, 1, -1)$ for the first and second cointegrating equations. In performing these three tests, under general conditions the parameter estimates in large sample have standard $N(0, 1)$ distributions, so one way to proceed is to do a simple t-test on the long-run coefficients on y_{3t} in each of these equations.
In the first cointegrating equation, the relevant test statistic is given by

$$t = \frac{1.025 - 1.000}{0.0104} = 2.3549,$$

with a p value of 0.009. Notwithstanding the fact that the point estimate of the coefficient is very close to 1, on statistical grounds the null hypothesis that it is equal to one is rejected. Similarly in the second cointegrating equation, the test statistic is

$$t = \frac{1.021 - 1.000}{0.0067} = 3.1006,$$

with a p value of .001 and the null hypothesis is also rejected.

Weak Exogeneity

The expectations hypothesis argues that it is the short-term yield that drives the longer yields. One possible implication of this is that y_{3t} does not adjust when the system is out of equilibrium, leaving all of the adjustment to y_{9t} and y_{6t}. This conjecture can be tested by testing whether or not the adjustment parameters on the two cointegrating equations in the dynamic equation for Δy_{3t} are both zero. If this null hypothesis cannot be rejected then y_{3t} is *weakly exogenous* with respect to the system.
The F tests of the null hypothesis that the adjustment parameters on the two cointegrating equations are zero in each of the dynamic equations, respectively, are as follows:

$$\begin{aligned} \Delta y_{9t} &: F(2, 416) = 5.67 \qquad \text{p value} = 0.0037 \\ \Delta y_{6t} &: F(2, 416) = 3.57 \qquad \text{p value} = 0.0289 \\ \Delta y_{3t} &: F(2, 416) = 0.24 \qquad \text{p value} = 0.7855. \end{aligned}$$

The results are very much as expected. The null hypothesis of no adjustment is strongly rejected in the Δy_{9t} and Δy_{6t} equations, while the hypothesis cannot be rejected for the Δy_{3t} equation. This means that y_{3t} may be regarded as weakly exogenous in the system.

To conclude, the expectations theory of the term structure of interest rates stand up pretty well in this example. The estimated VECM exhibits stable dynamics and the short run yield y_{3t} appears to drive the longer rates in the sense of being weakly exogenous. The only minor blemish is that that the long-run cointegrating parameters are marginally, but statistically significantly different from those values that would generate the pure spread as the error correction terms in the dynamic equations.

6.11 Exercises

1. **Simulating a VECM**

   Consider a simple bivariate VECM

$$\begin{aligned} y_{1t} - y_{1t-1} &= \delta_1 + \alpha_1(y_{2t-1} - \beta y_{1t-1} - \mu)\,, \\ y_{2t} - y_{2t-1} &= \delta_2 + \alpha_2(y_{2t-1} - \beta y_{1t-1} - \mu)\,. \end{aligned}$$

   (a) Using the initial conditions for the endogenous variables $y_1 = 100$ and $y_2 = 110$ simulate the model for 30 periods using the parameters

$$\delta_1 = \delta_2 = 0; \alpha_1 = -0.5; \alpha_2 = 0.1; \beta = 1; \mu = 0\,.$$

   Compare the two series. Also check to see that the long-run value of y_2 is given by $\beta y_1 + \mu$.

   (b) Simulate the model using the following parameters:

$$\delta_1 = \delta_2 = 0; \alpha_1 = -1.0; \alpha_2 = 0.1; \beta = 1; \mu = 0\,.$$

   Compare the resultant series with the those in (a) and hence comment on the role of the error correction parameter α_1.

   (c) Simulate the model using the following parameters:

$$\delta_1 = \delta_2 = 0; \alpha_1 = 1.0; \alpha_2 = -0.1; \beta = 1; \mu = 0\,.$$

   Compare the resultant series with the previous ones and hence comment on the relationship between stability and cointegration.

   (d) Simulate the model using the following parameters:

$$\delta_1 = \delta_2 = 0; \alpha_1 = -1.0; \alpha_2 = 0.1; \beta = 1; \mu = 10\,.$$

   Comment on the role of the parameter μ. Also check to see that the long-run value of y_2 is given by $\beta y_1 + \mu$.

   (e) Simulate the model using the following parameters:

$$\delta_1 = \delta_2 = 1; \alpha_1 = -1.0; \alpha_2 = 0.1; \beta = 1; \mu = 0\,.$$

   Comment on the role of the parameters δ_1 and δ_2.

(f) Explore a richer class of models which also includes short-run dynamics. For example, consider the model

$$\begin{aligned} y_{1t} - y_{1t-1} &= \delta_1 + \alpha_1(y_{2t-1} - \beta y_{1t-1} - \mu) + \phi_{11}(y_{1t-1} - y_{1t-2}) \\ &\quad + \phi_{12}(y_{2t-1} - y_{2,t-2}) , \\ y_{2t} - y_{2t-1} &= \delta_2 + \alpha_2(y_{2t-1} - \beta y_{1t-1} - \mu) + \phi_{21}(y_{1t-1} - y_{1,t-2}) \\ &\quad + \phi_{22}(y_{2t-1} - y_{2t-2}) . \end{aligned}$$

2. **The Present Value Model**

pv.wf1, pv.dta, pv.xlsx

The present value model predicts the following relationship between the two series

$$p_t = \beta_0 + \beta_1 d_t + u_t ,$$

where p_t is the natural logarithm of real price of equities, d_t is the natural logarithm of real dividend payments, u_t is a disturbance term and β_1 is the discount rate and $\beta_1 = 1$.

(a) Test for cointegration between p_t and d_t using Model 3 and $p = 1$ lags.

(b) Given the results in part (a) estimate a bivariate ECM for p_t and d_t using Model 3 with $p = 1$ lag. Interpret the results paying particular attention to the long-run parameter estimates, β_0 and β_1 and the error correction parameter estimates, $\widehat{\alpha}_i$.

(c) Derive an estimate of the long-run real discount rate from $R = \exp(-\beta_0)$ and interpret the result.

(d) Test the restriction $H_0 : \beta_1 = 1$.

(e) Discuss whether the empirical results support the present value model.

3. **Forward Market Efficiency**

spot.wf1, spot.dta, spot.xlsx

The data for this question were obtained from Corbae, Lim and Ouliaris (1992) who test for speculative efficiency by considering the equation

$$s_t = \beta_0 + \beta_1 f_{t-n} + u_t,$$

where s_t is the natural logarithm of the spot rate, f_{t-n} is the natural logarithm of the forward rate lagged n periods and u_t is a disturbance term. In the case of weekly data and the forward rate is the 1-month rate, f_{t-4} is an unbiased estimator of s_t if $\beta_1 = 1$.

(a) Use unit root tests to determine the level of integration of s_t, f_{t-1}, f_{t-2} and f_{t-3}.

(b) Test for cointegration between s_t and f_{t-4} using Model 2 with $p = 0$ lags.

(c) Provided that the two rates are cointegrated, estimate a bivariate VECM for s_t and f_{t-4} using Model 2 with $p = 0$ lags.

(d) Interpret the coefficients β_0 and β_1. In particular, test that $\beta_1 = 1$.

(e) Repeat these tests for the 3 month and 6 month forward rates. *Hint: remember that the frequency of the data is weekly.*

4. **Spurious Regression Problem**

A spurious relationship occurs when two independent variables are incorrectly identified as being related. A simple test of independence is based on the estimated correlation coefficient, $\widehat{\rho}$.

(a) Consider the following bivariate models

(i)	$y_{1t} = v_{1t}\,,$	$y_{2t} = v_{2t}\,,$
(ii)	$y_{1t} = y_{1t-1} + v_{1t}\,,$	$y_{2t} = y_{2t-1} + v_{2t}\,,$
(iii)	$y_{1t} = y_{1t-1} + v_{1,t}\,,$	$y_{2t} = 2y_{2t-1} - y_{2t-2} + v_{2t}\,,$
(iv)	$y_{1t} = 2y_{1t-1} - y_{1t-2} + v_{1t}\,,$	$y_{2t} = 2y_{2t-1} - y_{2t-2} + v_{2t}\,,$

in which v_{1t}, v_{2t} are *iid* $N(0, \sigma^2)$ with $\sigma^2 = 1$. Simulate each bivariate model 10000 times for a sample of size $T = 100$ and compute the correlation coefficient, $\widehat{\rho}$, of each draw. Compute the sampling distributions of $\widehat{\rho}$ for the four sets of bivariate models and discuss the properties of these distributions in the context of the spurious regression problem.

(b) Repeat part (a) with $T = 500$. What do you conclude?

(c) Repeat part (a), except for each draw estimate the regression model

$$y_{2t} = \beta_0 + \beta_1 y_{1t} + u_t\,, \qquad u_t \sim iid\,(0, \sigma^2)\,.$$

Compute the sampling distributions of the least squares estimator $\widehat{\beta}_1$ and its t statistic for the four sets of bivariate models. Discuss the properties of these distributions in the context of the spurious regression problem.

5. **Fisher Hypothesis**

fisher.wf1, fisher.dta, fisher.xlsx

Under the Fisher hypothesis the nominal interest rate fully reflects the long-run movements in the inflation rate. The Fisher hypothesis is represented by

$$i_t = \beta_0 + \beta_1 \pi_t + u_t,$$

where u_t is a disturbance term and the slope parameter is $\beta_1 = 1$.

(a) Construct the percentage annualised inflation rate, π_t.

(b) Perform unit root tests to determine the level of integration of the nominal interest rate and inflation. In performing the unit root tests, test the sensitivity of the results by using a model with a constant and no time trend, and a model with a constant and a time trend. Let the lags be determined by the automatic lag length selection procedure. Discuss the results in terms of the level of integration of each series.

(c) Estimate a bivariate VAR with a constant and use the SIC lag length criteria to determine the optimal lag structure.

(d) Test for cointegration between i_t and π_t using Model 2 with the number of lags based on the optimal lag length obtained form the estimated VAR. Remember if the optimal lag length of the VAR is p, the lag structure of the VECM is $p - 1$.

(e) Redo part (d) subject to the restriction that $\beta_1 = 1$.

(f) Does the Fisher hypothesis hold in the long-run? Discuss.

6. **Purchasing Power Parity**

ppp.wf1, ppp.dta, ppp.xlsx

Under the assumption of purchasing power parity (PPP), the nominal exchange rate adjusts in the long-run to the price differential between foreign and domestic countries

$$S = \frac{P}{F}.$$

This suggests that the relationship between the nominal exchange rate and the prices in the two countries is given by

$$s_t = \beta_0 + \beta_1 p_t + \beta_2 f_t + u_t,$$

where lower case letters denote natural logarithms and u_t is a disturbance term which represents departures from PPP with $\beta_2 = -\beta_1$.

(a) Construct the relevant variables, s, f, p and the foreign price differential $p - f$.

(b) Use unit root tests to determine the level of integration of all of these series. In performing the unit root tests, test the sensitivity of the results by using a model with a constant and no time trend, and a model with a constant and a time trend. Let the lags be $p = 12$. Discuss the results in terms of the level of integration of each series.

(c) Test for cointegration between s p and f using Model 3 with $p = 12$ lags.

(d) Given the results in part (c) estimate a trivariate VECM for s, p and f using Model 3 and $p = 12$ lags.

(e) Interpret the long-run parameter estimates. *Hint: if the number of cointegrating equations is greater than one, it is helpful to rearrange the cointegrating equations so one of the equations expresses s as a function of p and f.*

(f) Interpret the error correction parameter estimates.

(g) Interpret the short-run parameter estimates.

(h) Test the restriction $H_0 : \beta_2 = -\beta_1$.

7. **The Term Structure of Interest Rates**

zero.wf1, zero.dta, zero.xlsx

The term expectations hypothesis of the term structure of interest rates predicts the following relationship between a long-term interest rate of maturity n and a short-term rate of maturity $m < n$

$$y_{n,t} = \beta_0 + \beta_1 y_{m,t} + u_t,$$

where u_t is a disturbance term and β_0 is represents the term premium and $\beta_1 = 1$ for the pure expectations hypothesis.

(a) Test for cointegration between $y_{9,t}$ and $y_{3,t}$ using Model 2 and $p = 1$ lags.

(b) Given the results in part (a) estimate a bivariate ECM for $y_{9,t}$ and $y_{3,t}$ using Model 2 with $p = 1$ lags. Write out the estimated model (the cointegrating equation(s) and the ECM). In estimating the VECM order the yields from the longest maturity to the shortest.

(c) Interpret the long-run parameter estimates of β_1 and β_2.

(d) Interpret the error correction parameter estimates of γ_1 and γ_1.

(e) Interpret the short-run parameter estimates of $\pi_{i,j}$.

(f) Test the restriction $\beta_1 = 1$.

(g) Repeat parts (a) to (f) for the 6-month $(y_{6,t})$ and 3-month $(y_{3,t})$ yields.

(h) Repeat parts (a) to (f) for the 9-month $(y_{9,t})$, 6-month $(y_{6,t})$ and 3-month $(y_{3,t})$ yields.

(i) Repeat parts (a) to (f) for all 6 yields $(y_{9,t}, y_{6,t}, y_{5,t}, y_{4,t}, y_{3,t}, y_{2,t})$.

(j) Discuss whether the empirical results support the term structure of interest rate model.

(k) Questions (a) to (k) are all based on specifying Model 2 as the ECM. Reestimate the VECM where Model 3 is chosen. As the difference between Model 2 and Model 3 is the inclusion of intercepts in each equation of the VECM, perform a test that each intercept is zero. Interpret the results of this test.

(l) In estimating the VECM in the previous question, the order of the yields consists of choosing the longest maturity first and the shortest maturity last ie

$$y_{9,t}, y_{6,t}, y_{3,t}.$$

Now reestimate the VECM choosing the ordering

$$y_{9,t}, y_{3,t}, y_{6,t}.$$

Show that the estimated cointegrating equation(s) from this system can be obtained from the previous system based on an alternative ordering. Hence show that the estimates of the cointegrating equation(s) is (are) not unique.

(m) Test for weak exogeneity in the bivariate system containing $y_{9,t}$ and $y_{3,t}$. To perform the test that $y_{9,t}$ is weakly exogenous. Repeat the test for a system that contains the interest rates $y_{6,t}$ and $y_{3,t}$ and then for the trivariate system $y_{9,t}$, $y_{6,t}$ and $y_{3,t}$.

Chapter 7

Forecasting

7.1 Introduction

The future values of variables are important inputs into the current decision making of agents in financial markets and forecasting methods, therefore, are widely used in financial markets. Formally, a forecast is a quantitative estimate about the most likely value of a variable based on past and current information and where the relationship between variables is embodied in an estimated model. In the previous chapters a wide variety of econometric models have been introduced, ranging from univariate to multivariate time series models, from single equation regression models to multivariate vector autoregressive models. The specification and estimation of these financial models provides a mechanism for producing forecasts that are objective in the sense that the forecasts can be recomputed exactly by knowing the structure of the model and the data used to estimate the model. This contrasts with back-of-the-envelope methods which are not reproducible. Forecasting can also serve as a method for comparing alternative models. Forecasting methods not only provide an important way to choose between alternative models, but also a way of combining the information contained in forecasts produced by different models.

7.2 Types of Forecasts

Illustrative examples of forecasting in financial markets abound.

i The determination of the price of an asset based on present value methods requires discounting the present and future dividend stream at a discount rate that potentially may change over time.

ii Firms are interested in forecasting the future health of the economy when making decisions about current capital outlays because this investment earns a stream of returns over time.

iii In currency markets, forward exchange rates provide an estimate, forecast, of the future spot exchange rate.

iv In options markets, the Black-Scholes method for pricing options is based on the assumption that the volatility of the underlying asset that the option is written on is constant over the life of the option.

v In futures markets, buyers and sellers enter a contract to buy and sell commodities at a future date.

vi Model-based computation of Value-at-Risk requires repeated forecasting of the value of a portfolio over a given time horizon.

Although all these examples are vastly different, the forecasting principles in each case are identical. Before delving into the actual process of generating forecasts it is useful to establish some terminology.
Consider an observed sample of data $\{y_1, y_2, \cdots, y_T\}$ and an econometric model is to be used to generate forecasts of y over an horizon of H periods. The forecasts of y which are denoted $\widehat{y}$ are of two main types.

Ex Ante Forecasts: The entire sample $\{y_1, y_2, \cdots, y_T\}$ is used to estimate the data and the task is to forecast the variable over the horizon $T + H$.

Ex Post Forecasts: The model is estimated over a restricted sample period that excludes the last H observations, $\{y_1, y_2, \cdots, y_{T-H}\}$. The model is then forecasted out-of-sample over these H observations, but as the actual value of these observations have already been observed and it is therefore possible to compare the accuracy of the forecasts with the actual values.

Ex post and ex ante forecasts may be illustrated as follows:

Sample	$y_1, y_2, \cdots, y_{T-H}, y_{T-H+1}, y_{T-H+2} \cdots y_T$
Ex Post	$y_1, y_2, \cdots, y_{T-H}, \widehat{y}_{T-H+1}, \widehat{y}_{T-H+2} \cdots \widehat{y}_T$
Ex Ante	$y_1, y_2, \cdots, y_{T-H}, y_{T-H+1}, y_{T-H+2} \cdots y_T \quad \widehat{y}_{T+1}, \cdots \widehat{y}_{T+H}$

It is clear therefore that forecasting ex ante for H periods ahead requires the successive generation of $\widehat{y}_{T+1}, \widehat{y}_{T+2}$ up to and including $\widehat{y}_{T+H}$. This is referred to a *multi-step* forecast. On the other hand, ex post forecasting allows some latitude for choice. The forecast $\widehat{y}_{T-H+1}$ is based on data up to and including y_{T-H}. In generating the forecast $\widehat{y}_{T-H+2}$ the observation y_{T-H+1} is available for use. Forecasts that use this observation are referred to as a *one-step ahead* or *static* forecast. Ex post forecasting also allows multi-step forecasting using data up to and including y_{T-H} and this is known as *dynamic* forecasting.
There is a distinction between forecasting based on dynamic time series models and forecasts based on broader linear or nonlinear regression models. Forecasts based on dynamic univariate or multivariate time series models

developed in Chapter 4 are referred to as *recursive* forecasts. Forecasts that are based on econometric models that related one variable to another as in the linear regression model outlined in Chapter 3 are known as *structural* forecasts. It should be noted, however, that that the distinction between these two types of forecasts is often unclear as econometric models often contain both structural and dynamic time series features. An area in forecasting that has attracted a lot of recent interest which incorporates both recursive and structural elements is the problem of or *predictive regressions*, dealt with in Section 7.9.

Finally, forecasts in which only a single figure, say $\widehat{y}_{T+H}$, is reported for period $T+H$ is known as a *point* forecast. The point forecast represents the best guess of the value of y_{T+H}. Even if this guess is a particularly good one and it is known that on average the forecast is correct, or more formally $\mathrm{E}(\widehat{y}_{T+H}) = y_{T+H}$, there is some uncertainty associated with every forecast. *Interval* forecasts encapsulate this uncertainty by providing a range of forecast values for $\widehat{y}_{T+H}$ within which the actual value y_{T+H} is expected to be found at some given level of confidence.

7.3 Forecasting with Univariate Time Series Models

To understand the basic principles of forecasting financial econometric models, the simplest example namely a univariate autoregressive model with one lag, AR(1), model, is sufficient to demonstrate the key elements. Extending the model to more complicated univariate and multivariate models only increases the complexity to the computation but not the underlying fundamental technique of how the forecasts are generated.

Consider the AR(1) model

$$y_t = \phi_0 + \phi_1 y_{t-1} + v_t. \tag{7.1}$$

Suppose that the data consist of T sample observations $y_1, y_2, \cdots, y_T$. Now consider using the model to forecast the variable one period into the future, at $T+1$. The model at time $T+1$ is

$$y_{T+1} = \phi_0 + \phi_1 y_T + v_{T+1}. \tag{7.2}$$

To be able to compute a forecast of y_{T+1} it is necessary to know everything on the right-hand side of equation (7.2). Inspection of this equation reveals that some of these terms are known and some are unknown at time T:

Observations:	y_T	Known,
Parameters:	ϕ_0, ϕ_1	Unknown,
Disturbance:	v_{T+1}	Unknown.

The aim of forecasting is to replace the unknowns with the best guess of these quantities. In the case of the parameters, the best guess is simply to replace

them with their point estimates, $\widehat{\phi}_0$ and $\widehat{\phi}_1$, where all the sample data are used to obtain the estimates. Formally this involves using the mean of the sampling distribution to replace the population parameters ϕ_0, ϕ_1 by their sample estimates. Adopting the same strategy, the unknown disturbance term v_{T+1} in (7.2) is replaced by using the mean of its distribution, namely $\mathrm{E}[v_{T+1}] = 0$. The resulting forecast of y_{T+1} based on equation (7.2) is given by

$$\widehat{y}_{T+1} = \widehat{\phi}_0 + \widehat{\phi}_1 y_T + 0 = \widehat{\phi}_0 + \widehat{\phi}_1 y_T, \tag{7.3}$$

where the replacement of y_{T+1} by $\widehat{y}_{T+1}$ emphasizes the fact that the latter is a forecast quantity.
Now consider extending the forecast range to $T + 2$, the second period after the end of the sample period. The strategy is the same as before with the first step being expressing the model at time $T + 2$ as

$$y_{T+2} = \phi_0 + \phi_1 y_{T+1} + v_{T+2}, \tag{7.4}$$

in which all that all terms are now unknown at the end of the sample at time T:

Parameters:	ϕ_0, ϕ_1	Unknown,
Observations:	y_{T+1}	Unknown,
Disturbance:	v_{T+2}	Unknown.

As before, replace the parameters ϕ_0 and ϕ_1 by their sample estimators, $\widehat{\phi}_0$ and $\widehat{\phi}_1$, and the disturbance v_{T+2} by its mean $\mathrm{E}[v_{T+2}] = 0$. What is new in equation (7.4) is the appearance of unknown quantity y_{T+1} on the right-hand side of the equation Again, adopting the strategy of replacing unknowns by a best guess requires that the forecast of this variable obtained in the previous step, $\widehat{y}_{T+1}$ be used. Accordingly, the forecast for the second period is

$$\widehat{y}_{T+2} = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+1} + 0 = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+1}.$$

Clearly extending this analysis to H implies a forecasting equation of the form

$$\widehat{y}_{T+H} = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+H-1} + 0 = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+H-1}.$$

The need to use the forecast from the previous step to generate a forecast in the next step is commonly referred to as recursive forecasting. Moreover, as all of the information embedded in the forecasts $\widehat{y}_{T+1}, \widehat{y}_{T+2}, \cdots \widehat{y}_{T+H}$ is based on information up to and including the last observation in the sample at time T, the forecasts are commonly referred to as conditional mean forecasts where the conditioning is based on information at time T.
Extending the AR(1) model to an AR(2) model

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + v_t,$$

involves the sample strategy to forecast y_t. Writing the model at time $T + 1$ gives

$$y_{T+1} = \phi_0 + \phi_1 y_T + \phi_2 y_{T-1} + v_{T+1}.$$

Replacing the parameters $\{\phi_0, \phi_1, \phi_2\}$ by their sample estimators $\{\widehat{\phi}_0, \widehat{\phi}_1, \widehat{\phi}_2\}$ and the disturbance v_{T+1} by its mean $\mathrm{E}[v_{T+1}] = 0$, the forecast for the first period into the future is

$$\widehat{y}_{T+1} = \widehat{\phi}_0 + \widehat{\phi}_1 y_T + \widehat{\phi}_2 y_{T-1}.$$

To generate the forecasts for the second period, the AR(2) model is written at time $T+2$

$$y_{T+2} = \phi_0 + \phi_1 y_{T+1} + \phi_2 y_T + v_{T+2}.$$

Replacing all of the unknowns on the right-hand side by their appropriate best guesses, gives

$$\widehat{y}_{T+2} = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+1} + \widehat{\phi}_2 y_T.$$

To derive the forecast of y_t at time $T+3$ the AR(2) model is written at $T+3$

$$y_{T+3} = \phi_0 + \phi_1 y_{T+2} + \phi_2 y_{T+1} + v_{T+3}.$$

Now all terms on the right-hand side are unknown and the forecasting equation becomes

$$\widehat{y}_{T+3} = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{y}_{T+2} + \widehat{\phi}_2 \widehat{y}_{T+1}.$$

This univariate recursive forecasting procedure is easily demonstrated. Consider the logarithm of monthly United States equity index, p_t, for which data are available from February 1871 to June 2004, and associated returns, $r_t = p_t - p_{t-1}$, expressed as percentages. To generate **ex ante forecasts** of returns using a simple AR(1) model, the parameters are estimated using the entire available sample period and these estimates, together with the actual return for June 2004 are used to generate the recursive forecasts. Consider the case where ex ante forecasts are required for July and August 2004. The estimated model is

$$r_t = 0.2472 + 0.2853\, r_{t-1} + \widehat{v}_t,$$

where $\widehat{v}_t$ is the least squares residual. Given that the actual return for June 2004 is 2.6823% the forecasts for July and August are, respectively,

$$\begin{array}{lcrcrcl} \text{July} & : & \widehat{r}_{T+1} & = & 0.2472 + 0.2853\, r_T & & \\ & & & = & 0.2472 + 0.2853 \times 2.6823 & = & 1.0122, \\ \text{August} & : & \widehat{r}_{T+2} & = & 0.2472 + 0.2853\, \widehat{r}_{T+1} & & \\ & & & = & 0.2472 + 0.2853 \times 1.0120 & = & 0.5359. \end{array}$$

Suppose now that **ex post forecasts** are required for the period January 2004 to June 2004. The model is estimated over the period February 1871 to December 2013 to yield

$$r_t = 0.2459 + 0.2856\, r_{t-1} + \widehat{v}_t,$$

where $\widehat{v}_t$ is the least squares residual. The forecasts are now generated recursively using the estimated model and also the fact that the equity return in

December 2003 is 2.8858%:

$$
\begin{array}{lcrcrcl}
\text{January} & : & \widehat{r}_{T+1} & = & 0.2459 + 0.2856\, r_T & & \\
 & & & = & 0.2459 + 0.2856 \times 2.8858 & = & 1.0701\%, \\
\text{February} & : & \widehat{r}_{T+2} & = & 0.2459 + 0.2856\, \widehat{r}_{T+1} & & \\
 & & & = & 0.2459 + 0.2856 \times 1.0701 & = & 0.5515\%, \\
\text{March} & : & \widehat{r}_{T+3} & = & 0.2459 + 0.2856\, \widehat{r}_{T+2} & & \\
 & & & = & 0.2459 + 0.2856 \times 0.5515 & = & 0.4034\%, \\
\text{April} & : & \widehat{r}_{T+4} & = & 0.2459 + 0.2856\, \widehat{r}_{T+3} & & \\
 & & & = & 0.2459 + 0.2856 \times 0.4034 & = & 0.3611\%, \\
\text{May} & : & \widehat{r}_{T+5} & = & 0.2459 + 0.2856\, \widehat{r}_{T+4} & & \\
 & & & = & 0.2459 + 0.2856 \times 0.3611 & = & 0.3490\%, \\
\text{June} & : & \widehat{r}_{T+6} & = & 0.2459 + 0.2856\, \widehat{r}_{T+5} & & \\
 & & & = & 0.2459 + 0.2856 \times 0.3490 & = & 0.3456\%.
\end{array}
$$

The forecasts are illustrated in Figure 7.1. It is readily apparent how quickly the forecasts are driven toward the unconditional mean of returns. This is typical of time series forecasts based on stationary data.

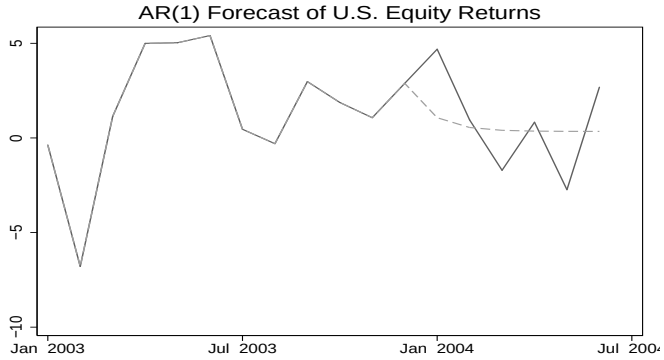


Figure 7.1: Forecasts (dashed line) of United States equity returns generated by an AR(1) model. The estimation sample period is February 1871 to December 2003 and the forecast period is from January 2004 to June 2004.

7.4 Forecasting with Multivariate Time Series Models

The recursive method used to generate the forecasts of a univariate time series model is easily generalised to multivariate models.

7.4.1 Vector Autoregressions

Consider a bivariate vector autoregression with one lag, VAR(1), given by

$$\begin{array}{rcl} y_{1t} & = & \phi_{10} + \phi_{11} y_{1t-1} + \phi_{12} y_{2t-1} + v_{1t}, \\ y_{2t} & = & \phi_{20} + \phi_{21} y_{1t-1} + \phi_{22} y_{2t-1} + v_{2t}. \end{array} \tag{7.5}$$

Given data up to time T, a forecast one period ahead is obtained by writing the model at time $T+1$

$$\begin{array}{rcl} y_{1T+1} & = & \phi_{10} + \phi_{11} y_{1T} + \phi_{12} y_{2T} + v_{1T+1}, \\ y_{2T+1} & = & \phi_{20} + \phi_{21} y_{1T} + \phi_{22} y_{2T} + v_{2T+1}. \end{array}$$

The knowns on the right-hand side are the last observations of the two variables, y_{1T} and y_{2T} and the unknowns are the the disturbance terms v_{1T+1} and v_{2T+1} and the parameters $\{\phi_{10}, \phi_{11}, \phi_{12}, \phi_{20}, \phi_{21}, \phi_{22}\}$. Replacing the unknowns by the best guesses, as in the univariate AR model, yields the following forecasts for the two variables at time $T+1$:

$$\begin{array}{rcl} \widehat{y}_{1T+1} & = & \widehat{\phi}_{10} + \widehat{\phi}_{11} y_{1T} + \widehat{\phi}_{12} y_{2T}, \\ \widehat{y}_{2T+1} & = & \widehat{\phi}_{20} + \widehat{\phi}_{21} y_{1T} + \widehat{\phi}_{22} y_{2T}. \end{array}$$

To generate forecasts of the VAR(1) model in (7.5) in two periods ahead, the model is written at time $T+2$

$$\begin{array}{rcl} y_{1T+2} & = & \phi_{10} + \phi_{11} y_{1T+1} + \phi_{12} y_{2T+1} + v_{1T+2}, \\ y_{2T+2} & = & \phi_{20} + \phi_{21} y_{1T+1} + \phi_{22} y_{2T+1} + v_{2T+2}. \end{array}$$

Now all terms on the right-hand side are unknown. As before the parameters are replaced by the estimators and the disturbances are replaced by their means, while y_{1T+1} and y_{2T+1} are replaced by their forecasts from the previous step, resulting in the two-period ahead forecasts

$$\begin{array}{rcl} \widehat{y}_{1T+2} & = & \widehat{\phi}_{10} + \widehat{\phi}_{11} \widehat{y}_{1T+1} + \widehat{\phi}_{12} \widehat{y}_{2T+1}, \\ \widehat{y}_{2T+2} & = & \widehat{\phi}_{20} + \widehat{\phi}_{21} \widehat{y}_{1T+1} + \widehat{\phi}_{22} \widehat{y}_{2T+1}. \end{array}$$

In general, the forecasts of the VAR(1) model for $H-$periods ahead are

$$\begin{array}{rcl} \widehat{y}_{1T+H} & = & \widehat{\phi}_{10} + \widehat{\phi}_{11} \widehat{y}_{1T+H-1} + \widehat{\phi}_{12} \widehat{y}_{2T+H-1}, \\ \widehat{y}_{2T+H} & = & \widehat{\phi}_{20} + \widehat{\phi}_{21} \widehat{y}_{1T+H-1} + \widehat{\phi}_{22} \widehat{y}_{2T+H-1}. \end{array}$$

An important feature of this result is that even if forecasts are required for just one of the variables, say y_{1t}, it is necessary to generate forecasts of the other variables as well.

To illustrate forecasting using a VAR consider in addition to the logarithm of the equity index, p_t and associated returns, r_t, consider also the log returns to dividends d_t. As before data are available for the period February 1871

to June 2004 and suppose ex ante forecasts are required for July and August 2004. The estimated bivariate VAR model is

$$\begin{aligned} r_t &= 0.2149 + 0.2849\, r_{t-1} + 0.1219\, d_{t-1} + \widehat{v}_{1t}, \\ d_t &= 0.0301 + 0.0024\, r_{t-1} + 0.8862\, d_{t-1} + \widehat{v}_{2t}, \end{aligned}$$

where $\widehat{v}_{1t}$ and $\widehat{v}_{2t}$ are the residuals from the two equations. The forecasts for equity and dividend returns in July are

$$\begin{aligned} \widehat{r}_{T+1} &= 0.2149 + 0.2849\, r_T + 0.1219\, d_T \\ &= 0.2149 + 0.2849 \times 2.6823 + 0.1219 \times 1.0449 \\ &= 1.1065\%, \end{aligned}$$

$$\begin{aligned} \widehat{d}_{T+1} &= 0.0301 + 0.0024\, r_T + 0.8862\, d_T \\ &= 0.0301 + 0.0024 \times 2.6823 + 0.8862 \times 1.0449 \\ &= 0.9625\%. \end{aligned}$$

The corresponding forecasts for August are

$$\begin{aligned} \widehat{r}_{T+2} &= 0.2149 + 0.2849\, \widehat{r}_{T+1} + 0.1219\, \widehat{d}_{T+1} \\ &= 0.2149 + 0.2849 \times 1.1065 + 0.1219 \times 0.9625 \\ &= 0.6475\%, \end{aligned}$$

$$\begin{aligned} \widehat{d}_{T+2} &= 0.0301 + 0.0024\, \widehat{r}_{T+1} + 0.8862\, \widehat{d}_{T+1} \\ &= 0.0301 + 0.0024 \times 1.1065 + 0.8862 \times 0.9625 \\ &= 0.6475\%. \end{aligned}$$

7.4.2 Vector Error Correction Models

An important relationship between vector autoregressions and vector error correction models discussed in Chapter 6 is that a VECM represents a restricted VAR. This suggests that a VECM can be re-expressed as a VAR which, in turn, can be used to forecast the variables of the model.
Consider the following bivariate VECM containing one lag

$$\begin{aligned} \Delta y_{1t} &= \gamma_1 \left(y_{2t-1} - \beta y_{1t-1} - \mu\right) + \pi_{11}\Delta y_{1t-1} + \pi_{12}\Delta y_{2t-1} + v_{1t}, \\ \Delta y_{2t} &= \gamma_2 \left(y_{2t-1} - \beta y_{1t-1} - \mu\right) + \pi_{21}\Delta y_{1t-1} + \pi_{22}\Delta y_{2t-1} + v_{2t}. \end{aligned}$$

Rearranging the VECM as a (restricted) VAR(2) in the levels of the variables, gives

$$\begin{aligned} y_{1t} = &-\gamma_1\mu + (1 + \pi_{11} - \gamma_1\beta)y_{1t-1} \\ &- \pi_{11}y_{1t-2} + (\gamma_1 + \pi_{12})y_{2t-1} - \pi_{12}y_{2t-2} + v_{1t}, \\ y_{2t} = &-\gamma_2\mu + (\pi_{21} - \gamma_2\beta)y_{1t-1} \\ &- \pi_{21}y_{1t-2} + (1 + \gamma_2 + \pi_{22})y_{2t-1} - \pi_{22}y_{2t-2} + v_{2t}. \end{aligned}$$

Alternatively, it is possible to write

$$\begin{aligned} y_{1t} &= \phi_{10} + \phi_{11} y_{1t-1} + \phi_{12} y_{1t-2} + \phi_{13} y_{2t-1} + \phi_{14} y_{2t-2} + v_{1t}, \\ y_{2t} &= \phi_{20} + \phi_{21} y_{1t-1} + \phi_{22} y_{1t-2} + \phi_{23} y_{2t-1} + \phi_{24} y_{2t-2} + v_{2t}, \end{aligned} \tag{7.6}$$

in which the VAR and VECM parameters are related as follows

$$\begin{array}{ll} \phi_{10} = -\gamma_1 \mu & \phi_{20} = -\gamma_2 \mu \\ \phi_{11} = 1 + \pi_{11} - \gamma_1 \beta & \phi_{21} = \pi_{21} - \gamma_2 \beta \\ \phi_{12} = -\pi_{11} & \phi_{22} = -\pi_{21} \\ \phi_{13} = \gamma_1 + \pi_{12} & \phi_{23} = 1 + \gamma_2 + \pi_{22} \\ \phi_{14} = -\pi_{12} & \phi_{24} = -\pi_{22}. \end{array} \tag{7.7}$$

Now that the VECM is re-expressed as a VAR in the levels of the variables in equation (7.6), the forecasts are generated for a VAR as discussed in Section 7.4.1 with the VAR parameter estimates computed from the VECM parameter estimates based on the relationships in (7.7).
Using the same dataset as that used in producing ex ante VAR forecasts, the procedure is easily repeated for the VECM. The estimated VECM model with a restricted constant (Model 3) and with two lags in the underlying VAR model is [1]

$$\begin{aligned} r_t &= 0.2056 - 0.0066(p_{t-1} - 1.1685\, d_{t-1} - 312.9553) \\ &\quad +0.2911\, r_{t-1} + 0.1484\, d_{t-1} + \widehat{v}_{1t}, \\ d_t &= 0.0334 + 0.0023(p_{t-1} - 1.1685\, d_{t-1} - 312.9553) \\ &\quad +0.0002\, r_{t-1} + 0.8768\, d_{t-1} + \widehat{v}_{2t}, \end{aligned}$$

where $\widehat{v}_{1t}$ and $\widehat{v}_{2t}$ are the residuals from the two equations. Writing the VECM as a VAR in levels gives

$$\begin{aligned} p_t &= (0.2056 + 0.0066 \times 312.9553) \\ &\quad + (1 - 0.0066 + 0.2911)\, p_{t-1} - 0.2911\, p_{t-2} \\ &\quad + (0.0066 \times 1.1685 + 0.1484) d_{t-1} - 0.1484\, d_{t-2} + \widehat{v}_{1t}, \\ d_t &= (0.0334 - 0.0023 \times 312.9553) \\ &\quad + (0.0023 + 0.0002)\, p_{t-1} - 0.0002\, p_{t-2} \\ &\quad + (1 - 0.0023 \times 1.1685 + 0.8768)\, d_{t-1} - 0.8768\, d_{t-2} + \widehat{v}_{2t}, \end{aligned}$$

or

$$\begin{aligned} p_t &= 2.2711 + 1.2845\, p_{t-1} - 0.2911\, p_{t-2} + 0.1561\, d_{t-1} - 0.1484\, d_{t-2} + \widehat{v}_{1t}, \\ d_t &= -0.6864 + 0.0025\, p_{t-1} - 0.0002\, p_{t-2} + 1.8741\, d_{t-1} - 0.8768\, d_{t-2} + \widehat{v}_{2t}. \end{aligned}$$

The forecast for July log equities is

$$\begin{aligned} \widehat{p}_{T+1} &= 2.2711 + 1.2845\, p_T - 0.2911\, p_{T-1} + 0.1561\, d_T - 0.1484\, d_{T-1} \\ &= 704.0600, \end{aligned}$$

[1] These estimates are the same as the estimates reported in Chapter 6 with the exception that the intercepts now reflect the fact that the variables are scaled by 100.

and for July log dividends is

$$\begin{aligned}\widehat{d}_{T+1} &= -0.6864 + 0.0025\, p_T - 0.0002\, p_{T-1} + 1.8741\, d_T - 0.8768\, d_{T-1} \\ &= 293.3700.\end{aligned}$$

Similar calculations reveal that the forecasts for August log equities and dividends are:

$$\begin{aligned}\widehat{p}_{T+2} &= 704.3400, \\ \widehat{d}_{T+1} &= 294.4300.\end{aligned}$$

Based on these forecasts of the logarithms of equity prices and dividends, the forecasts for the percentage equity returns in July and August 2004 are, respectively,

$$\begin{aligned}\widehat{r}_{T+1} &= 704.0600 - 703.2412 = 0.8188\%, \\ \widehat{r}_{T+2} &= 704.3400 - 704.0600 = 0.2800\%,\end{aligned}$$

and the corresponding forecasts for dividend returns are, respectively,

$$\begin{aligned}\widehat{d}_{T+1} &= 293.3700 - 292.3162 = 1.0538\%, \\ \widehat{d}_{T+2} &= 294.4300 - 293.3700 = 1.0600\%.\end{aligned}$$

7.5 Forecast Evaluation Statistics

The discussion so far has concentrated on forecasting a variable or variables over a forecast horizon H, beginning after the last observation in the dataset. This of course is the most common way of computing forecasts. Formally these forecasts are known as *ex ante* forecasts. However, it is also of interest to be able to compare the forecasts with the actual value that are realised to determine their accuracy. One approach is to wait until the future values are observed, but this is not that convenient if an answer concerning the forecasting ability of a model is required immediately.

A common solution adopted to determine the forecast accuracy of a model is to estimate the model over a restricted sample period that excludes the last H observations. The model is then forecasted out-of-sample over these observations, but as the actual value of these observations have already been observed it is possible to compare the accuracy of the forecasts with the actual values. As the data are already observed, forecasts computed in this way are known as *ex post* forecasts.

There are a number of simple summary statistics that are used to determine the accuracy of forecasts. Define the forecast error in period $T + h$ as the difference between the actual and forecast value over the forecast horizon

$$y_{T+1} - \widehat{y}_{T+1},\, y_{T+2} - \widehat{y}_{T+2},\, \cdots,\, y_{T+H} - \widehat{y}_{T+H},$$

then it follows immediately that the smaller the forecast error the better is the forecast. The most commonly used summary measures of overall closeness of

the forecasts to the actual values are:

$$\begin{array}{lll}
\text{Mean Absolute Error:} & MAE = & \frac{1}{H}\sum_{h=1}^{H}|y_{T+h}-\widehat{y}_{T+h}|, \\
\text{Mean Absolute Percentage Error:} & MAPE = & \frac{1}{H}\sum_{h=1}^{H}\left|\frac{y_{T+h}-\widehat{y}_{T+h}}{y_{T+h}}\right|, \\
\text{Mean Square Error:} & MSE = & \frac{1}{H}\sum_{h=1}^{H}(y_{T+h}-\widehat{y}_{T+h})^2, \\
\text{Root Mean Square Error:} & RMSE = & \sqrt{\frac{1}{H}\sum_{h=1}^{H}(y_{T+h}-\widehat{y}_{T+h})^2}.
\end{array}$$

The use of these statistics is easily demonstrated in the context of the United States equity returns, r_t. To allow the generation of *ex post* forecasts an AR(1) model is estimated using data for the period February 1871 to December 2003. Forecasts for the period January to June of 2004 for are then used with the observed monthly percentage return on equities to generate the required summary statistics.
To compute the MSE for the forecast period the actual sample observations of equity returns from January 2004 to June 2004 are required. These are

$$4.6892\%, 0.9526\%, -1.7095\%, 0.8311\%, -2.7352\%, 2.6823\%.$$

The MSE is

$$\begin{aligned}
MSE &= \frac{1}{6}\sum_{h=1}^{6}(y_{t+h}-f_{t+h})^2 \\
&= \frac{1}{6}\Big((4.6892-1.0701)^2+(0.9526-0.5515)^2+(-1.7095-0.4034)^2 \\
&+ (0.8311-0.3611)^2+(-2.7352-0.3490)^2+(2.6823-0.3456)^2\Big) \\
&= 5.4861.
\end{aligned}$$

The RMSE is

$$RMSE = \sqrt{\frac{1}{6}\sum_{h=1}^{6}(y_{t+h}-f_{t+h})^2} = \sqrt{5.4861} = 2.3423.$$

Taken on its own, the root mean squared error of the forecast, 2.3423, does not provide a descriptive measure of the relative accuracy of this model per se, as its value can easily be changed by simply changing the units of the data. For example, expressing the data as returns and not percentage returns results in the RMSE falling by a factor of 100. Even though the RMSE is now smaller that does not mean that the forecasting performance of the AR(1) model has

improved in this case. The way that the RMSE and the MSE are used to evaluate the forecasting performance of a model is to compute the same statistics for an alternative model: the model with the smaller RMSE or MSE, is judged as the better forecasting model.
The forecasting performance of several models are now compared. The models are an AR(1) model of equity returns, a VAR(1) model containing equity and dividend returns, and a VECM(1) based on Model 3, containing log equity prices and log dividends. Each model is estimated using a reduced sample on United States monthly percentage equity returns from February 1871 to December 2003, and the forecasts are computed from January to June of 2004. The forecasts are then compared using the MSE and RMSE statistics.

Table 7.1

Forecasting performance of models of United States monthly percentage equity returns. All models are estimated over the period January 1871 to December 2003 and the forecasts are computed from January to June of 2004.

Forecast/Statistic	AR(1)	VAR(1)	VECM(1)
January 2004	1.0701%	1.2241%	0.9223%
February 2004	0.5515%	0.7333%	0.3509%
March 2004	0.4034%	0.5780%	0.1890%
April 2004	0.3611%	0.5200%	0.1474%
May 2004	0.3490%	0.4912%	0.1411%
June 2004	0.3456%	0.4721%	0.1447%
MSE	5.4861	5.4465	5.5560
RMSE	2.3422	2.3338	2.3571

The results in Table 7.1 show that the VAR(1) is the best forecasting model as it yields the smallest MSE and RMSE. The AR(1) is second best followed by the VECM(1).
Probably the most widely used formal test for comparing two different forecasts is the Diebold and Mariano (1995). Suppose we have two competing forecasts and we are able to compute the RMSE for every forecast period t for each of these competing forecasts, which may be denoted $RMSE_{1t}$ and $RMSE_{2t}$ respectively. Now define the difference

$$d_t = RMSE_{1t} - RMSE_{2t}$$

then the Diebold-Mariano test of equal predictive accuracy is the simple t test that $\mathrm{E}(d_t) = 0$.
There is an active research area in financial econometrics at present in which these statistical (or *direct*) measures of forecast performance are replaced by problem-specific (or *indirect*) measures of forecast performance in which the

evaluation relates specifically to an economic decision (Elliot and Timmerman, 2008; Patton and Sheppard, 2009). Early examples of the indirect approach to forecast evaluation are Engle and Colacito (2006) evaluate forecast performance in terms of portfolio return variance, while Fleming, Kirby and Ostdiek (2001, 2003) apply a quadratic utility function that values one forecast relative to another. Becker, Clements, Doolan and Hurn (2013) provide a survey and comparison of these different approaches to forecast evaluation.

7.6 Evaluating the Density of Forecast Errors

The discussion of generating forecasts of financial variables thus far focusses on either the conditional mean (point forecasts) or the conditional variance (interval forecasts) of the forecast distribution. A natural extension is also to forecast higher order moments, including skewness and kurtosis. In fact, it is of interest in the area of risk management to forecast all moments of the distribution and hence forecast the entire probability density of key financial variables.
As is the case with point forecasts where statistics are computed to determine the relative accuracy of the forecasts, the quality of the density forecasts are also evaluated to determine their relative accuracy in forecasting all moments of the distribution. However, the approach is not to try and evaluate the forecasts properties of each moment separately, but rather test all moments jointly by using the probability integral transformation (PIT).

7.6.1 Probability Integral Transform

Consider a very simple model of a data generating process for the

$$\begin{aligned} y_t &= \mu + v_t \\ v_t &\sim iid\, N(0, \sigma^2), \end{aligned}$$

in which $\mu = 0.0$ and $\sigma^2 = 1.0$. Now denote the cumulative distribution function of the standard normal distribution evaluated at any point z as $\Phi(z)$, then if a sample of observed values y_t are indeed generated correctly, then

$$u_t = \Phi\Big(\frac{y_t - \mu}{\sigma}\Big) \qquad t = 1, 2, \cdots, T,$$

results in the transformed time series u_t having an *iid* uniform distribution. This transformation is known as the probability integral transform.
Figure 7.2 contains an example of how the transformed times series u_t is obtained from the actual time series y_t where the specified model is $N(0, 1)$. This result is a reflection of the property that if the cumulative distribution is indeed the correct distribution, transforming y_t to u_t means that each y_t has the same probability of being realised as any other value of y_t.

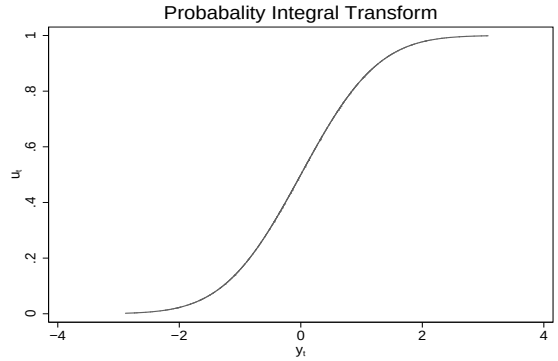


Figure 7.2: Probability integral transform showing how the the time series y_t is transformed into u_t based on the distribution $N(0, 1)$.

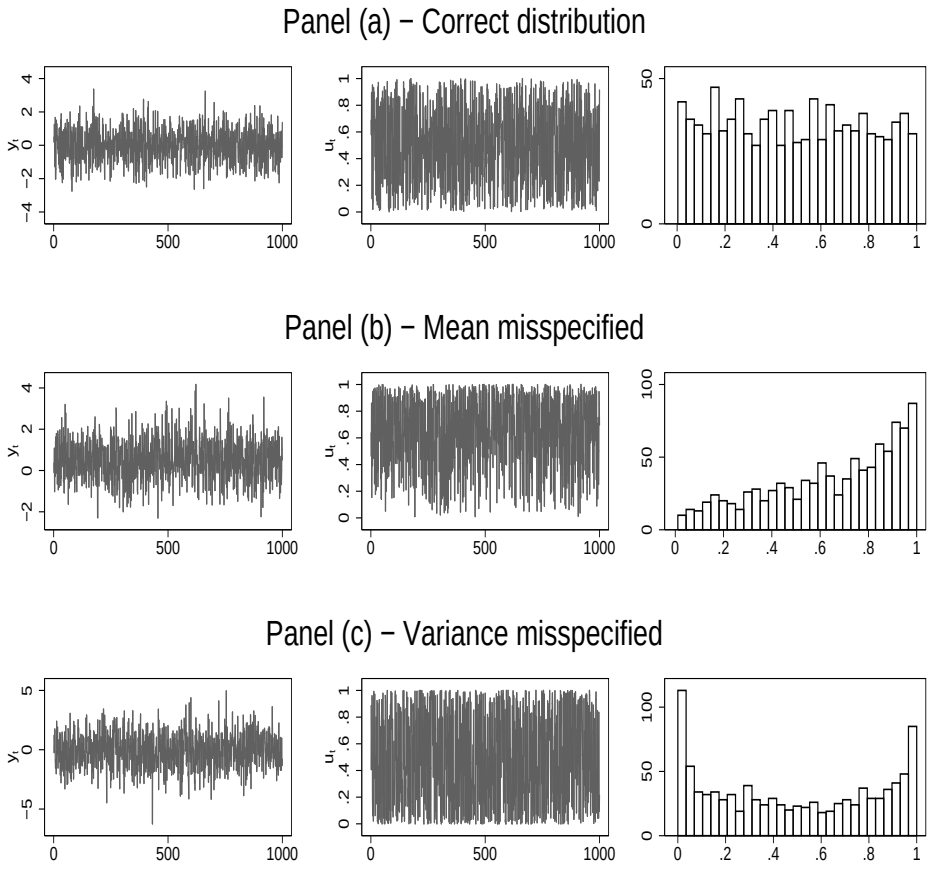


Figure 7.3: Simulated time series to show the effects of misspecification on the probability integral transform. In panel (a) there is no misspecification while panels (b) and (c) demonstrate the effect of misspecification in the mean and variance of the distribution respectively.

The probability integral transform in the case where the specified model is chosen correctly is highlighted in panel (a) of Figure 7.3. A time series plot of 1000 simulated observations, y_t, drawn from a $N(0,1)$ distribution is transformed into via the cumulative normal distribution to u_t. Finally the histogram of the transformed time series, u_t is shown. Inspection of this histogram confirms that the distribution of u_t is uniform and that the distribution used in transforming y_t is indeed the correct one.
Now consider the case where the true data generating process for y_t is the $N(0.5,1)$ distribution, but the incorrect distribution, $N(0,1)$, is used as the forecast distribution to perform the PIT. The effect of misspecification of the mean on the forecasting distribution is illustrated in panel (b) of Figure 7.3. A time series of 1000 simulated observations from a $N(0.5,1.0)$ distribution, y_t, is transformed using the incorrect distribution, $N(0,1)$, and the histogram of the transformed time series, u_t is plotted. The fact that u_t is not uniform in this case is a reflection of a misspecified model. The histogram exhibits a positive slope reflecting that larger values of y_t have a relatively higher probability of occurring than small values of y_t.
Now consider the case where the variance of the model is misspecified. If the data generating process is a $N(0,2)$ distribution, but the forecast distribution used in the PIT is once again $N(0,1)$ then it is to be expected that the forecast distribution will understate the true spread of the data. This is clearly visible in panel (c) of Figure 7.3. The histogram of u_t is now U-shaped implying that large negative and large positive values have a higher probability of occurring than predicted by the $N(0,1)$ distribution.

7.6.2 Equity Returns

The models used to forecast United States equity returns r_t in Section 7.3 are all based on the assumption of normality. Consider the AR(1) model

$$r_t = \phi_0 + \phi_1 r_{t-1} + v_t\,, \qquad v_t \sim N(0,\sigma^2)\,.$$

Assuming the forecast is ex post so that r_t is available, the one-step ahead forecast error is given by

$$\widehat{v}_t = r_t - \widehat{\phi}_0 - \widehat{\phi}_1 r_{t-1}\,,$$

with distribution

$$f(\widehat{v}_t) \sim N(r_t - \widehat{\phi}_0 - \widehat{\phi}_1 r_{t-1}, \sigma^2)\,. \tag{7.8}$$

Using monthly data from January 1871 to June 2004, this distribution is

$$f(\widehat{v}_t) \sim N(r_t - 0.2472 - 0.2853\, r_{t-1}, 3.929^2)\,.$$

The PIT corresponding to the estimated distribution in (7.8) the transformed time series are computed as

$$u_t = \Phi\left(\frac{\widehat{v}_t}{\widehat{\sigma}}\right)\,,$$

in which $\widehat{\sigma}$ is the standard error of the regression. A histogram of the transformed time series, u_t, is given in Figure 7.4. It appears that the AR(1) forecasting model of equity returns is misspecified because the distribution of u_t is non-uniform. The interior peak of the distribution of u_t suggests that the distribution of y_t is more peaked than that predicted by the normal distribution. Also, the pole in the distribution at zero suggests that there are some observed negative values of y_t that are also not consistent with the specification of a normal distribution. These two properties combined suggest that the specified model fails to take into account the presence of higher order moments such as skewness and kurtosis. The analysis of the one-step ahead AR(1) forecasting model can easily be extended to the other estimated models of equity returns including the VAR and the VECM investigated in Section 7.4 to forecast equity returns.

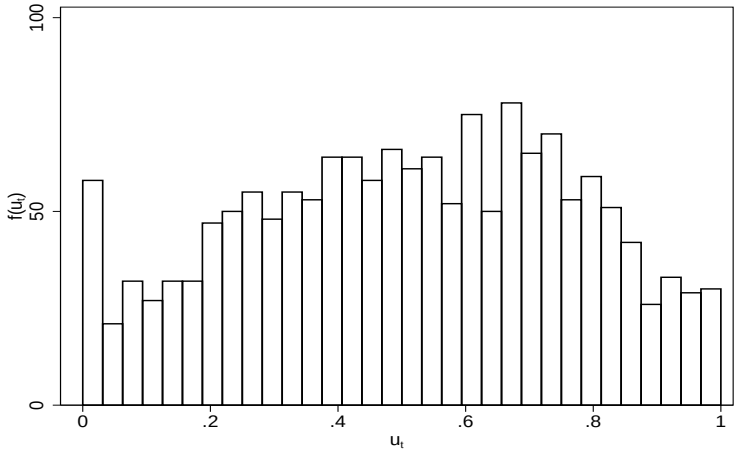


Figure 7.4: Probability integral transform applied to the estimated one-step ahead forecast errors of the AR(1) model of United States equity returns, January 1871 to June 2004.

As applied here, the PIT is *ex post* as it involves using the within sample one-step ahead prediction errors to perform the analysis and it is also a simple graphical implementation in which misspecification is detected by simple inspection of the histogram of the transformed time series, u_t. It is possible to relax both these assumptions. Diebold, Gunther and Tay (1998) discuss an alternative *ex ante* approach, while Ghosh and Bera (2005) propose a class of formal statistical tests of the null hypothesis that u_t is uniformly distributed.

7.7 Combining Forecasts

Given that all models are wrong but some are useful, it is not surprising that the issue of combining forecasts has generated a great deal of interest (Timmerman, 2006; Elliott and Timmerman, 2008) and very often the financial press will report consensus forecasts which are essentially averages of different forecasts of the same quantity. This raises an important question in forecasting: is it better to rely on the best individual forecast or is there any gain to averaging the competing forecasts?
Suppose that two unbiased forecasts of a variable y_t are available, given by $\widehat{y}_t^1$ and $\widehat{y}_t^2$, with respective variances σ_1^2 and σ_2^2 and covariance σ_{12}. A weighted average of these two forecasts is

$$\widehat{y}_t = \omega \widehat{y}_t^1 + (1-\omega)\widehat{y}_t^2,$$

and the variance of average is

$$\sigma^2 = \omega^2\sigma_1^2 + (1-\omega)^2\sigma_2^2 + 2\omega(1-\omega)\sigma_{12}.$$

A natural approach is to choose the weight ω in order to minimise the variance of the forecast. The first-order condition for a minimum is given by

$$\frac{\partial \sigma^2}{\partial \omega} = 2\omega\sigma_1^2 - 2(1-\omega)\sigma_2^2 + 2\sigma_{12} - 4\omega\sigma_{12}.$$

Setting this expression to zero and solving gives

$$\omega = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}\,.$$

It is clear therefore that the weight attached to $\widehat{y}_t^1$ varies inversely with its variance. In passing, these weights are of course identical to the optimal weights for the minimum variance portfolio derived in Chapter 3.
This point can be illustrated more clearly if the forecasts are assumed to be uncorrelated, $\sigma_{12} = 0$. In this case,

$$\omega = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}, \qquad 1-\omega = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2},$$

and it is clear that both forecasts have weights varying inversely with their variances. By rearranging the expression for ω as follows

$$\omega = \Big(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}\Big)\Big(\frac{\sigma_2^{-2}\sigma_1^{-2}}{\sigma_2^{-2}\sigma_1^{-2}}\Big) = \frac{\sigma_1^{-2}}{\sigma_1^{-2} + \sigma_2^{-2}}\,, \tag{7.9}$$

the inverse proportionality is now manifestly clear in the numerator of expression (7.9). This simple intuition in the two forecast case translates into a

situation in which there are N forecasts $\{\widehat{y}_t^1, \widehat{y}_t^2, \cdots, \widehat{y}_t^N\}$ of the same variable y_t. If these forecasts are all unbiased and uncorrelated and if the weights satisfy

$$\sum_{i=1}^{N} \omega_i = 1 \qquad \omega_i \geq 0 \quad i = 1, 2, \cdots, N,$$

then from (7.9) the optimal weights are

$$\omega_i = \frac{\sigma_i^{-2}}{\sum_{j=1}^{N} \sigma_j^{-2}},$$

and the weight on forecast i is inversely proportional to its variance.
While the weights in expression (7.9) are intuitively appealing as they are based on the principle of producing a minimum variance portfolio. Important questions remain, however, about how best to implement the combination of forecasts approach in practice. Bates and Granger (1969) suggested using (7.9) to construct the weights with the required estimates of the forecast variances, $\widehat{\sigma}_i^2$, given by the the forecast mean square error. All this approach requires then is an estimate of the mean square error of all the competing forecasts in order to compute the optimal weights, $\widehat{\omega}_i$. Granger and Ramanathan (1984) later show that this method is numerically equivalent to weights constructed from running the restricted regression

$$y_t = \omega_1 \widehat{y}_t^1 + \omega_2 \widehat{y}_t^2 + \cdots + \omega_N \widehat{y}_t^N + v_t,$$

in which the coefficients are constrained to be non-negative and to sum to one. Of course enforcing these restrictions in practice can be tricky and sometimes *ad hoc* methods need to be adopted. One method is the sequential elimination of forecasts with weights estimated to be negative until all the remaining forecasts in the proposed combination forecast have positive weights. This is sometimes referred to as forecast encompassing because all the forecasts that eventually remain in the regression encompass all the information in those that are left out.
Yet another approach to averaging forecasts is based on the use of information criteria (Buckland, Burnham and Augustin, 1997; Burnham and Anderson, 2002), which may be interpreted as the relative quality of an econometric model. Suppose there are N different models each with an estimated Akaike information criterion $AIC_1, AIC_2, \cdots, AIC_N$, then the model that returns the minimum value of the information criterion is usually the model of choice. Denote the minimum value of the information criterion for this set of models as $AIC_{\min}$, then the expression[2]

$$\exp\left[\Delta I_i/2\right] = \exp\left[(AIC_i - AIC_{\min})/2\right]$$

[2]The exact form of this expression derives from the likelihood principle which is discussed in Chapter 10. The AIC is an unbiased estimate of -2 times the log-likelihood function of model i, so the after dividing by -2 and exponentiating the result is a measure of the likelihood that model i actually generated the observed data.

may be interpreted as a relative measure of the loss of information from using model i instead of the model that produces $AIC_{\min}$. due to using model i instead of the model yielding $I_{\min}$. It is therefore natural to allow the forecast combination to reflect this relative information by computing the weights

$$\widehat{\omega}_i = \frac{\exp\left[\Delta I_i/2\right]}{\sum\limits_{j=i}^{N} \exp\left[\Delta I_i/2\right]}.$$

The Schwarz (Bayesian) Information Criterion (SIC) has also been suggested as an alternative information criterion to use in this context.[3]
Of course the simplest idea would be assign equal weight to these forecasts construct the simple average

$$\widehat{y}_t = \frac{1}{N}\sum_{i=}^{N} \widehat{y}_{1it}\,.$$

Interestingly enough, simulation studies and practical work generally indicated that this simplistic strategy often works best, especially when there are large numbers of forecasts to be combined, notwithstanding all the subsequent work on the optimal estimation of weights (Stock and Watson, 2001). Two possible explanations of why averaging might in practice work better than constructing the optimal combination focus are as follows.

i There may be significant error in the estimation of the weights, due either to parameter instability (Clemen, 1989; Winkler and Clemen, 1992, Smith and Wallis, 2009) or structural breaks (Hendry and Clements, 2004)).

ii The fact that the variances of the competing forecasts may be very similar and their covariances positive suggests that large gains obtained by constructing optimal weights are unlikely (Elliott, 2011).

7.8 Regression Model Forecasts

The forecasting of univariate and multivariate models discussed so far are all based on time series models as each dependent variable is expressed as a function of own lags and lags of other variables. Now consider forecasting the linear regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

[3]When the SIC is is used to construct the optimal weights have the interpretation of a Bayesian averaging procedure. Illustrative examples may be found in Garratt, Koop and Vahey, (2008) and Kapetanios, Vabhard and Price (2008).

where y_t is the dependent variable, x_t is the explanatory variable, u_t is a disturbance term, and the sample period is $t = 1, 2, \cdots, T$. To generate a forecast of y_t at time $T+1$, as before, the model is written at $T+1$ as

$$y_{T+1} = \beta_0 + \beta_1 x_{T+1} + u_{T+1}.$$

The unknown values on the right hand-side are y_{T+1} and u_{T+1}, as well as the parameters $\{\beta_0, \beta_1\}$. As before, u_{T+1} is replaced by its expected value of $\mathrm{E}[u_{T+1}] = 0$, while the parameters are replaced by their sample estimates, $\{\widehat{\beta}_0, \widehat{\beta}_1\}$. However, it is not clear how to deal with x_{T+1}, the future value of the explanatory variable. One strategy is to specify hypothetical future values of the explanatory variable that in some sense capture scenarios the researcher is interested in.

A less subjective approach is to specify a time series model for x_t and use this model to generate forecasts of x_{T+i}. Suppose for the sake of argument that an AR(2) model is proposed for x_t. The bivariate system of equations to be estimated is then

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t, & (7.10)\\ x_t &= \phi_0 + \phi_1 x_{t-1} + \phi_2 x_{t-2} + v_t. & (7.11) \end{aligned}$$

To generate the first forecast at time $T+1$ the system of equations is written as

$$\begin{aligned} y_{T+1} &= \beta_0 + \beta_1 x_{T+1} + u_{T+1},\\ x_{T+1} &= \phi_0 + \phi_1 x_T + \phi_2 x_{T-1} + v_{T+1}. \end{aligned}$$

Replacing the unknowns with the best available guesses, yields

$$\begin{aligned} \widehat{y}_{T+1} &= \widehat{\beta}_0 + \widehat{\beta}_1 \widehat{x}_{T+1}, & (7.12)\\ \widehat{x}_{T+1} &= \widehat{\phi}_0 + \widehat{\phi}_1 x_T + \widehat{\phi}_2 x_{T-1}. & (7.13) \end{aligned}$$

Equation (7.13) is used to generate the forecast $\widehat{x}_{T+1}$, which is the substituted into equation (7.12) to generate a $\widehat{y}_{T+1}$

Alternatively, these calculations can be performed in one step by substituting (7.13) for $\widehat{x}_{T+1}$ into (7.12) to give

$$\begin{aligned} \widehat{y}_{T+1} &= \widehat{\beta}_0 + \widehat{\beta}_1(\widehat{\phi}_0 + \widehat{\phi}_1 x_T + \widehat{\phi}_2 x_{T-1})\\ &= \widehat{\beta}_0 + \widehat{\beta}_1\widehat{\phi}_0 + \widehat{\beta}_1\widehat{\phi}_1 x_T + \widehat{\beta}_1\widehat{\phi}_2 x_{T-1}. \end{aligned}$$

Of course, the case where there are multiple explanatory variables is easily handled by specifying a VAR to generate the required multivariate forecasts. The regression model may be used to forecast United States equity returns, r_t, using dividend returns, d_t. As in earlier illustrations, the data are from February 1871 to June 2004. Estimation of equations (7.10) and (7.11), in which for simplicity the latter is restricted to an AR(1) representation, gives

$$\begin{aligned} r_t &= 0.3353 + 0.0405 d_{1t} + \widehat{u}_t,\\ d_t &= 0.0309 + 0.8863 d_{t-1} + \widehat{v}_t. \end{aligned}$$

Based on these estimates, the forecasts for dividend returns in July and August are, respectively,

$$\begin{aligned}\widehat{d}_{T+1} &= 0.0309 + 0.8863\, d_T = 0.0309 + 0.8863 \times 1.0449 = 0.9570\%,\\ \widehat{d}_{T+2} &= 0.0309 + 0.8863\, d_{T+1} = 0.0309 + 0.8863 \times 0.9570 = 0.8791\%\,,\end{aligned}$$

so that in July and August the forecasted equity returns are

$$\begin{aligned}\widehat{r}_{T+1} &= 0.3353 + 0.0405\, d_{T+1} = 0.3353 + 0.0405 \times 0.9570 = 0.3741\%,\\ \widehat{r}_{T+2} &= 0.3353 + 0.0405\, d_{T+2} = 0.3353 + 0.0405 \times 0.8791 = 0.3709\%.\end{aligned}$$

7.9 Predictive Regressions

Forecasting in finance using regression models, or predictive regressions, as outlined in Section 7.8 is one that is currently receiving quite a lot of attention (Stambaugh, 1999). In a series of recent papers Goyal and Welch (2003; 2008) provide empirical evidence of the predictability of the equity premium, eqp_t (defined as the total rate of return on the S&P 500 index, rm_t, minus the short-term interest rate) in terms of the dividend-price ratio dp_t and the dividend yield ratio dy_t. What follows reproduces some of the results from Goyal and Welch (2003).

Table 7.2 provides summary statistics for the data. There are difficulties in reproducing all the summary statistics reported by Goyal and Welch in their papers because the data they provide is updated continuously. The summary statistics reported here are for slightly different sample periods than those listed in Goyal and Welch (2003), but the mean and standard deviation for the sample period 1927 to 2005 of 6.04% and 19.17%, respectively, are identical to those for the same period listed in Goyal and Welch (2008). Furthermore the plots of the logarithm of the equity premium and the logarithms of the dividend yield and dividend price ratio in Figure 7.5 are almost identical to the plots in Figure 1 of Goyal and Welch (2003).

The predictive regressions used in this piece of empirical analysis are, respectively,

$$eqp_t = \alpha_y + \beta_y dy_{t-1} + u_{y,t}, \tag{7.14}$$

$$eqp_t = \alpha_p + \beta_p dp_{t-1} + u_{p,t}\,. \tag{7.15}$$

The parameter estimates obtained from estimating these equations for two different sample periods, namely, 1926 to 1990 and 1926 to 2002, respectively, are reported in Table 7.3.

These results suggest that dividend yields and price dividend ratios had at least some forecasting power with respect to the equity premium for the period 1926 - 1990, at least for the S&P 500 index. It is noticeable however that the size of the coefficients on both dp_{t-1} and dy_{t-1} are substantially reduced when the sample size is increased to 2002. Although the results are not identical to those in Table 2 of Goyal and Welch (2003) because of data revisions, the

Table 7.2

Descriptive statistics for the annual total market return, the equity premium, the dividend price ratio and the dividend yield all defined in terms of the S&P 500 index. All variables are in percentages.

		Mean	St.dev.	Min.	Max.	Skew.	Kurt.
1926 - 2003	rm_t	9.79	19.10	-53.99	42.51	-0.82	3.69
	eqp_t	6.11	19.28	-55.13	42.26	-0.65	3.41
	dp_t	-3.28	0.44	-4.48	-2.29	-0.64	3.63
	dy_t	-3.22	0.42	-4.50	-2.43	-1.07	4.33
1946 - 2003	rm_t	10.52	15.58	-30.12	41.36	-0.46	2.66
	eqp_t	5.88	15.93	-37.64	40.43	-0.43	2.84
	dp_t	-3.37	0.42	-4.48	-2.63	-0.76	3.52
	dy_t	-3.30	0.43	-4.50	-2.43	-0.81	3.96
1927 - 2005	rm_t	9.69	18.98	-53.99	42.51	-0.80	3.71
	eqp_t	6.04	19.17	-55.13	42.26	-0.65	3.44
	dp_t	-3.30	0.45	-4.48	-2.29	-0.57	3.28
	dy_t	-3.24	0.43	-4.50	-2.43	-0.96	3.79

coefficients are similar and so is the pattern of size of the coefficient estimates decreasing as the sample size is increased.

This sub-sample instability of the estimated regression coefficients in Table 7.3 is further illustrated by considering the recursive plots of the slope coefficients on dp_{t-1} and dy_{t-1} from equations (7.14) and (7.15). Figure 7.6 reveals that although the coefficient on dy_{t-1} appears to be marginally statistically significant at the 5% level over long periods, the coefficient on dp_{t-1} increases over time while the coefficient on dy_{t-1} steadily decreases. In other words, as time progresses the forecaster would rely less on dy_t and more on dp_t despite the fact that the dy_t coefficient appears more reliable in terms of statistical significance. In fact, the dividend yield almost always produces an inferior forecast to the unconditional mean of the equity premium and the dividend-price ratio fares only slightly better. The point being made is that a trader relying on information available at the time a forecast was being made and not relying on information relating to the entire sample would have had difficulty in extracting meaningful forecasts.

The main tool for interpreting the performance of predictive regressions supplied by Goyal and Welch (2003) is a plot of the cumulative sum of squared one-step-ahead forecast errors of the predictive regressions expressed relative to the forecast error of the best current estimate of the mean of the equity premium. Let the one-step-ahead forecast errors of the dividend yield and dividend-price ratio models be $\widehat{u}_{y,t+1|t}$ and $\widehat{u}_{p,t+1|t}$, respectively, and let the forecast errors for the unconditional mean estimate be $\widehat{u}_{t+1|t} = eqp_t - \overline{eqp}_t$,

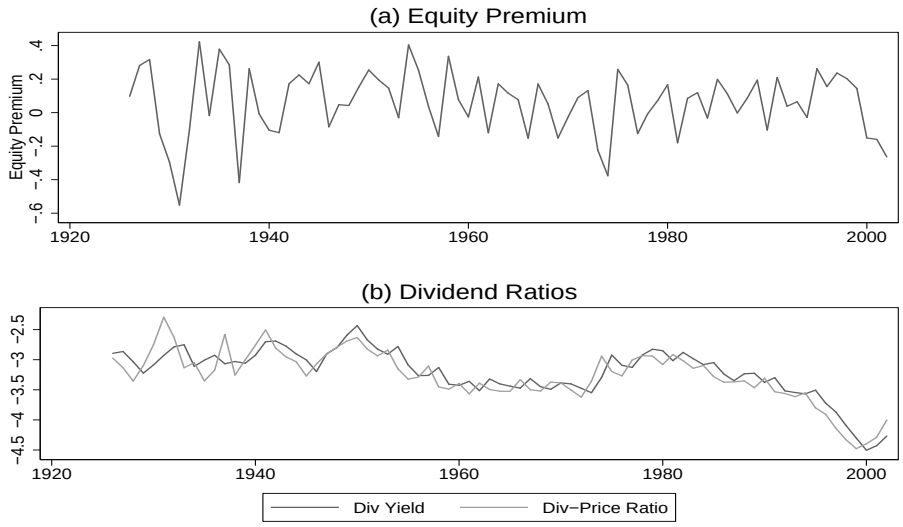


Figure 7.5: Plots of the time series of the logarithm of the equity premium, dividend yield, and dividend-price ratio.

then Figure 7.7 plots the two series

$$SSE(y) = \sum_{t=1946}^{2003} (\widehat{u}^2_{t+1|t} - \widehat{u}^2_{y,t+1|t}) \qquad \text{[Dividend Yield Model]}$$

$$SSE(p) = \sum_{t=1946}^{2003} (\widehat{u}^2_{t+1|t} - \widehat{u}^2_{p,t+1|t}) \qquad \text{[Dividend-Price Ratio Model].}$$

A positive value for SSE means that the model forecasts are superior to the forecasts based solely on the mean thus far. A positive slope implies that over the recent year the forecasting model performs better than the mean.
Figure 7.7 indicates that the forecasting ability of a predictive regression using the dividend yield is abysmal as $SSE(y)$ is almost uniformly less than zero. There are two years in the mid-1970s and two years around 2000 when $SSE(y)$ has a positive slope but these episodes are aberrations. The forecasting performance of the predictive regression using the dividend-price ratio is slightly better than the forecasts generated by the mean, $SSE(p) > 0$. This is not a conclusion that emerges naturally from Figure 7.6 which indicates that the slope coefficient from this regression is almost always statistically insignificant.
There are a few important practical lessons to learn from predictive regressions. The first of these is that good in-sample performance does not necessarily imply that the estimated equation will provide good ex ante forecasting

Table 7.3

Predictive regressions for the equity premium using the dividend price ratio, dp_t, and the dividend yield, dy_t, as explanatory variables.

	α	β	R^2	$\overline{R}^2$	Std. error	N
			Sample 1926 - 1990			
dp_t	0.5700	0.1630	0.0595	0.0446	0.1930	65
	(0.257)	(0.0818)				
	(0.030)	(0.050)				
dy_t	0.7380	0.2210	0.0851	0.0706	0.1903	65
	(0.282)	(0.0913)				
	(0.011)	(0.018)				
			Sample 1926 - 2002			
dp_t	0.3790	0.0984	0.0461	0.0334	0.1898	77
	(0.169)	(0.0517)				
	(0.028)	(0.061)				
dy_t	0.4670	0.1280	0.0680	0.0556	0.1876	77
	(0.176)	(0.0547)				
	(0.010)	(0.022)				

ability. As in the case of the performance of pooled forecasts, parameter instability is a problem for good predictive performance. Second, there is a fundamental problem using variables that are almost nonstationary processes as explanatory equations in predictive regressions which purport to explain stationary variables. So Stambaugh (1999) finds that dividend ratios are almost random walks while the equity premia are stationary. It may therefore be argued that dividend ratios are good predictors of their own future behaviour only and not of the future path of the equity premium.

7.10 Stochastic Simulation of Value-at-Risk

Forecasting need not necessarily be about point forecasts or best guesses. Sometimes important information is conveyed by the degree of uncertainty inherent in the best guess. One important application of this uncertainty in finance is the concept of Value-at-Risk which was introduced in Chapter 2. Stated formally, Value-at-Risk represents the losses that are expected to occur with probability α on an asset or portfolio of assets, P, after N days. The $N-day$ $(1-\alpha)\%$ Value-at-Risk is expressed as $VaR(P, N, 1-\alpha)$.
That Value-at-Risk is related to the uncertainty in the forecast of future values of the portfolio is easily demonstrated. Consider the case of US monthly data on equity prices. Suppose that the asset in question is one which pays

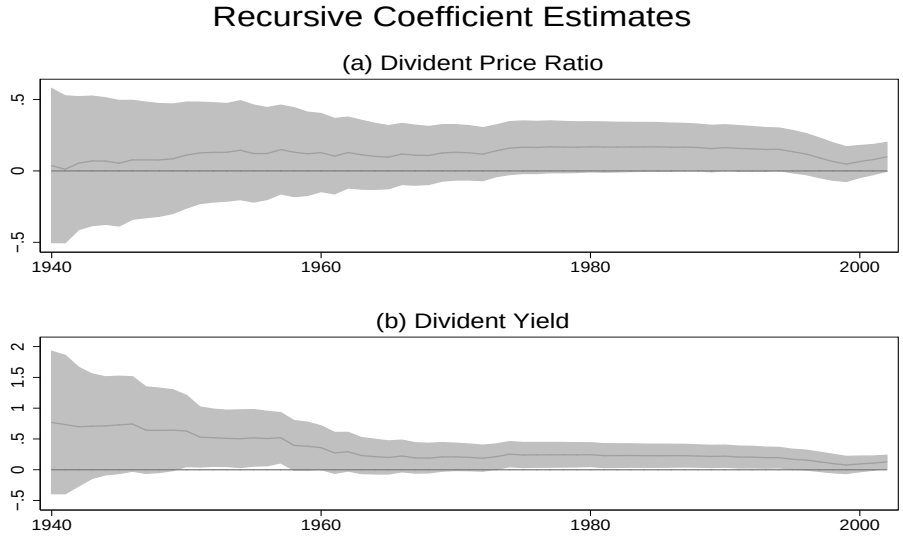


Figure 7.6: Recursive estimates of the coefficients on the dividend-price ratio and the dividend yield from (7.14) and (7.15).

the value of the index. An investor who holds this asset in June 2004, the last date in the sample, would observe that the value of the portfolio is $1132.76. The value of the portfolio is now forecast out for six months to the end of December 2004. In assessing the decision to hold the asset or liquidate the investment, it is not so much the best guess of the future value that is important as the spread of the distribution of the forecast. The situation is illustrated in Figure 7.8 where the shaded region captures the 90% confidence interval of the forecast. Clearly, the investor needs to take this spread of likely outcomes into account and this is exactly the idea of Value-at-Risk. It is clear therefore that forecast uncertainty and Value-at-Risk are intimately related.
Recall from Chapter 2 that Value-at-Risk may be computed by historical simulation, the variance-covariance method, or Monte Carlo simulation. Using a model to make forecasts of future values of the asset or portfolio and then assessing the uncertainty in the forecast is the method of Monte Carlo simulation. In general simulation refers to any method that randomly generates repeated trials of a model and seeks to summarise uncertainty in the model forecast in terms of the distribution of these random trials. The steps to perform a simulation are as follows:

Step 1: Estimate the model

Estimate the following (simple) AR(1) regression model

$$y_t = \phi_0 + \phi_1 y_{t-1} + v_t,$$

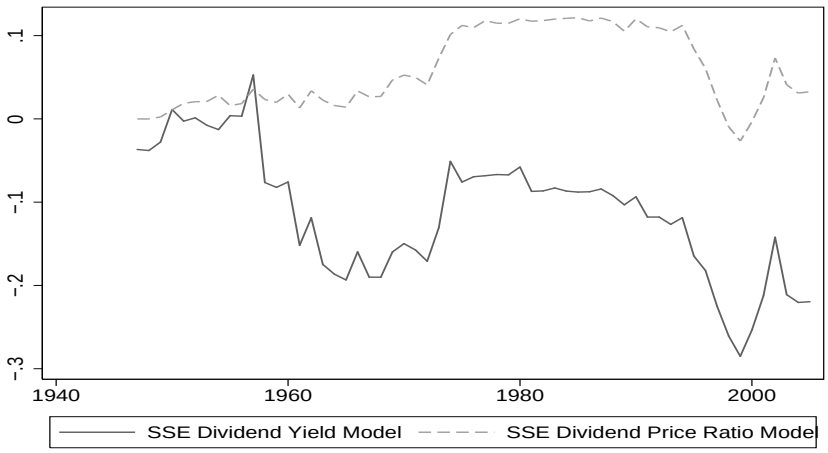


Figure 7.7: Plots of the cumulative squared relative one-step-ahead forecast errors obtained from the equity premium predictive regressions. The squared one-step-ahead forecast errors obtained from the models are subtracted from the squared one-step-ahead forecast errors based solely on the best current estimate of the unconditional mean of the equity premium.

and store the parameter estimates $\widehat{\phi}_0$ and $\widehat{\phi}_1$. Note that the AR(1) model is used for illustrative purposes only and any model of y_t could be used.

Step 2: Solve the model

For each available time period t in the model, use $\widehat{\phi}_0$ and $\widehat{\phi}_1$ to generate a one-step-ahead forecast

$$\widehat{y}_{t+1|t} = \widehat{\phi}_0 + \widehat{\phi}_1 y_t,$$

and then compute and store the one-step-ahead forecast errors

$$\widehat{v}_{t+1|t} = y_{t+1} - \widehat{y}_{t+1|t}\,.$$

Step 3: Simulate the model

Now forecast the model forward but instead of a forecast based solely on the best guesses for the unknowns, the uncertainty is explicitly accounted for by including an error term. The error term is obtained either by drawing from some parametric distribution (such as the normal distribution) or by taking a random draw from the estimated one-step-ahead forecast errors

$$\begin{aligned}
\widehat{y}^1_{T+1} &= \phi_0 + \phi_1 y_T + \tilde{v}_{T+1}, \\
\widehat{y}^1_{T+2} &= \phi_0 + \phi_1 \widehat{y}_{T+1} + \tilde{v}_{T+2}, \\
&\vdots \\
\widehat{y}^1_{T+H} &= \phi_0 + \phi_1 \widehat{y}_{T+H-1} + \tilde{v}_{T+H},
\end{aligned}$$

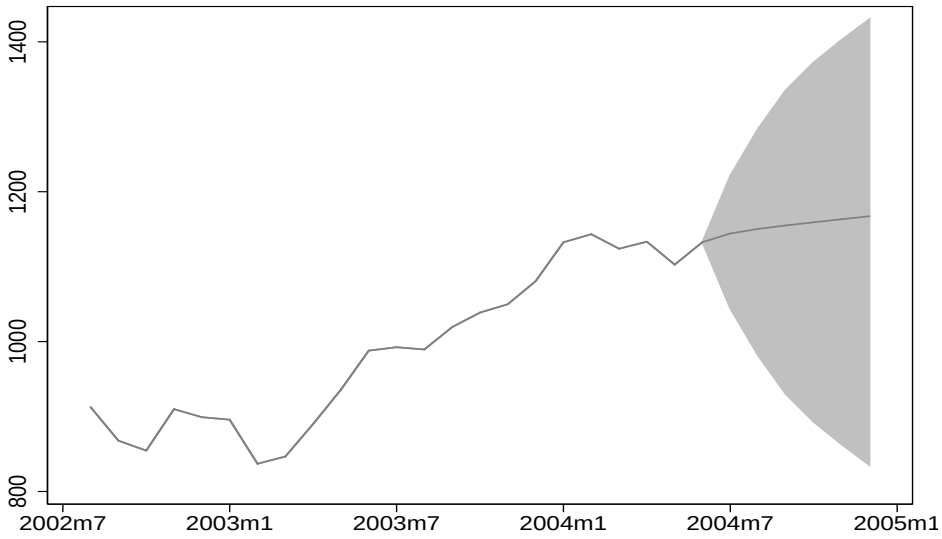


Figure 7.8: Stochastic simulation of the equity price index over the period July 2004 to December 2004. The ex ante forecasts are shown by the solid line while the confidence interval encapsulates the uncertainty inherent in the forecast.

where $\tilde{v}_{T+i}$ are all random drawings from $\widehat{v}_{t+1|t}$, the computed one-step-ahead forecast errors from Step 2. One repetition of a Monte Carlo simulation of the model is represented by the series of forecasts $\{\widehat{y}^1_{T+1}, \widehat{y}^1_{T+2} \cdots, \widehat{y}^1_{T+H}\}$.

Step 4: Repeat

Step 3 is now repeated S times to obtain an ensemble of forecasts

$$\begin{array}{cccccc}
\widehat{y}^1_{T+1} & \widehat{y}^2_{T+1} & \widehat{y}^3_{T+1} & \cdots & \widehat{y}^{S-1}_{T+1} & \widehat{y}^S_{T+1} \\
\widehat{y}^1_{T+2} & \widehat{y}^2_{T+2} & \widehat{y}^3_{T+1} & \cdots & \widehat{y}^{S-1}_{T+2} & \widehat{y}^S_{T+2} \\
\vdots & \vdots & \widehat{y}^3_{T+1} & \vdots & \vdots & \vdots \\
\widehat{y}^1_{T+H} & \widehat{y}^1_{T+H} & \widehat{y}^3_{T+1} & \cdots & \widehat{y}^{S-1}_{T+H} & \widehat{y}^S_{T+H}
\end{array}$$

Step 5: Summarise the uncertainty

Each column of this ensemble of forecasts is a representative of a possible outcome of the model and therefore collectively the ensemble captures the uncertainty of the forecast. In particular, the percentiles of these simulated forecasts for each time period $T + i$ give an accurate picture of the distribution of the forecast at that time. The disturbances used to generate the forecasts are drawn from the actual one-step-ahead prediction errors and not from a normal distribution and the forecast

uncertainty will then reflect any non-symmetry or fat tails present in the estimated prediction errors.

One practical item of importance concerns the reproduction of the results of the simulation. In order to reproduce simulation results it is necessary to use the same set of random numbers. To ensure this reproducibility it is important to set the seed of the random number generator before carrying out the simulations. If this is not done, a different set of random numbers will be used each time the simulation is undertaken. Of course as $S \rightarrow \infty$ this step becomes unnecessary, but in most practical situations the number of replications is set as a realistic balance between computational considerations and accuracy of results.

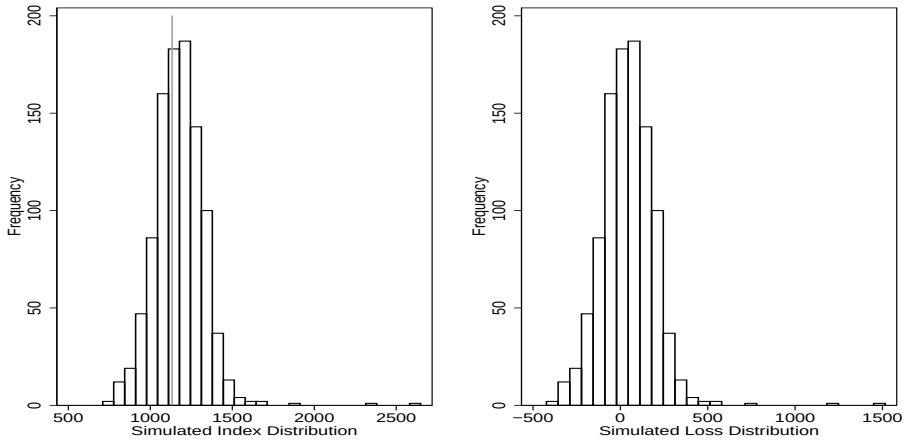


Figure 7.9: Simulated distribution of the equity index and the profit/loss on the equity index over a six month horizon from July 2004.

Consider now the problem of computing the 99% Value-at-Risk for the asset which pays the value of the United States equity index over a time horizon is six months. On the assumption that equity returns are generated by an AR(1) model, the estimated equation is

$$r_t = 0.2472 + 0.2853\, r_{t-1} + \widehat{v}_t,$$

which may be used to forecast returns for period $T+1$ but ensuring that uncertainty is explicitly introduced. The forecasting equation is therefore

$$\widehat{r}_{T+1} = 0.2472 + 0.2853\, r_T + \tilde{v}_{T+1},$$

where $\tilde{v}_{T+1}$ is a random draw from the computed one-step-ahead forecast errors computed by means of an in-sample static forecast. The value of the asset at $T+1$ in repetition s is computed as

$$\widehat{P}^s_{T+1} = P_T \exp\left[\widehat{r}_{T+1}/100\right],$$

where the forecast returns are adjusted so that they no longer expressed as percentages. A recursive procedure is now used to forecast the value of the asset out to $T + 6$ and the whole process is repeated S times. The distribution of the value of the asset at $T + 6$ after S repetitions of the is shown in panel (a) of Figure 7.9 with the initial value at time T of $P_T = \$1132.76$ superimposed. The distribution of simulated losses obtained by subtracting the initial value of the asset from the terminal value is shown in panel (b) of Figure 7.9. The first percentile value of this terminal distribution is \$833.54 so that six month 99% Value-at-Risk is $\$833.54 - \$1132.76 = \$299.13$, where by convention the minus sign is dropped when reporting Value-at-Risk.
Of course this approach is equally applicable to simulating Value-at-Risk for more complex portfolios comprising more than one asset and portfolios that include derivatives.

7.11 Exercises

1. **Recursive Ex Ante Forecasts of Real Equity Returns**

   pv.wf1, pv.dta, pv.xlsx

   Consider monthly data on the logarithm of real United States equity prices, p_t, and the logarithm of real dividend payments, d_t, from January 1871 to June 2004.

   (a) Estimate an AR(1) model of real equity returns, r_t, with the sample period ending in June 2004 . Generate forecasts of r_t from July to December of 2004.

   (b) Estimate an AR(2) model of real equity returns, r_t, with the sample period ending in June 2004. Generate forecasts of r_t from July to December of 2004.

   (c) Repeat parts (a) and (b) for real dividend returns, d_t.

   (d) Estimate a VAR(1) containing for r_t and d_t with the sample period ending in June 2004. Generate forecasts of real equity returns from July to December of 2004.

   (e) Estimate a VAR(2) for r_t and d_t with the sample period ending in June 2004. Generate forecasts of real equity returns from July to December of 2004.

   (f) Estimate a VECM(1) for r_t and d_t with the sample period ending in June 2004 and where the specification is based on Model 3, as set out in Chapter 6. Generate forecasts of real equity returns from July to December of 2004.

   (g) Repeat part (f) with the lag length in the VECM increasing from 1 to 2.

(h) Repeat part (g) with the VECM specification based on Model 2, as set out in Chapter 6.

(i) Now estimate a VECM(1) containing real equity returns, r_t, real dividend returns, d_t, and real earnings growth, ry_t, with the sample period ending in June 2004 and the specification is based on Model 3. Assume a cointegrating rank of 1. Generate forecasts of real equity returns from July to December of 2004.

(j) Repeat part (a) with the lag length in the VECM increasing from 1 to 2.

(k) Repeat part (i) with the VECM specification based on Model 2

2. **Recursive Ex Post Forecasts of Real Equity Returns**

pv.wf1, pv.dta, pv.xlsx

Consider monthly data on the logarithm of real United States equity prices, p_t, and the logarithm of real dividend payments, d_t, from January 1871 to June 2004.

(a) Estimate an AR(1) model of real equity percentage returns (y_{1t}) with the sample period ending December 2003, and generate ex post forecasts from January to June of 2004.

(b) Estimate a VAR(1) model of real equity percentage returns (y_{1t}) and real dividend percentage returns (y_{2t}) with the sample period ending December 2003, and generate ex post forecasts from January to June of 2004.

(c) Estimate a VECM(1) model of real equity percentage returns (y_{1t}) and real dividend percentage returns (y_{2t}) using Model 3, with the sample period ending December 2003, and generate ex post forecasts from January to June of 2004.

(d) For each set of forecasts generated in parts (a) to (c), compute the MSE and the RMSE. Which is the better forecasting model? Discuss.

3. **Regression Based Forecasts of Real Equity Returns**

pv.wf1, pv.dta, pv.xlsx

Consider monthly data on the logarithm of real United States equity prices, p_t, and the logarithm of real dividend payments, d_t, from January 1871 to June 2004.

(a) Estimate the following regression of real equity returns (y_{1t}) with real dividend returns (y_{2t}) as the explanatory variable, with the sample period ending in June 2004

$$y_{1t} = \beta_1 + \beta_2 y_{2t} + u_t,$$

(b) Estimate an AR(1) model of dividend returns

$$y_{2t} = \rho_0 + \rho_1 y_{2t-1} + v_t,$$

and combine this model with the estimated model in part (a) to generate forecasts of real equity returns from July to December of 2004.

(c) Estimate an AR(2) model of dividend returns

$$y_{2t} = \rho_0 + \rho_1 y_{2t-1} + \rho_2 y_{2t-2} + v_t,$$

and combine this model with the estimated model in part (a) to generate forecasts of real equity returns from July to December of 2004.

(d) Use the estimated model in part (a) to generate forecasts of real equity returns from July to December of 2004 assuming that real dividends increase at 3% per annum.

(e) Use the estimated model in part (a) to generate forecasts of real equity returns from July to December of 2004 assuming that real dividends increase at 10% per annum.

(f) Use the estimated model in part (a) to generate forecasts of real equity returns from July to December of 2004 assuming that real dividends increase at 3% per annum from July to September and by 10% from October to December.

4. **Pooling Forecasts**

This question is based on the EViews file **HEDGE.WF1** which contains daily data on the percentage returns of seven hedge fund indexes, from the 1st of April 2003 to the 28th of May 2010, a sample size of $T = 1869$.

R_CONVERTIBLE	:	Convertible Arbitrage
R_DISTRESSED	:	Distressed Securities
R_EQUITY	:	Equity Hedge
R_EVENT	:	Event Driven
R_MACRO	:	Macro
R_MERGER	:	Merger Arbitrage
R_NEUTRAL	:	Equity Market Neutral

(a) Estimate an AR(2) model of the returns on the equity market neutral hedge fund (y_{1t}) with the sample period ending on the 21st of May 2010 (Friday)

$$y_{1t} = \rho_0 + \rho_1 y_{1t-1} + \rho_2 y_{1t-2} + v_{1t}.$$

Generate forecasts of y_{1t} for the next working week, from the 24th to the 28th of May, 2010.

(b) Repeat part (a) for S&P500 returns (y_{2t}).

(c) Estimate a VAR(2) containing the returns on the equity market neutral hedge fund (y_{1t}) and the returns on the S&P500 (y_{2t}), with the sample period ending on the 21st of May 2010 (Friday)

$$\begin{aligned} y_{1t} &= \alpha_0 + \alpha_1 y_{1t-1} + \alpha_2 y_{1t-2} + \alpha_3 y_{2t-1} + \alpha_4 y_{2t-2} + v_{1t} \\ y_{2t} &= \beta_0 + \beta_1 y_{1t-1} + \beta_2 y_{1t-2} + \beta_3 y_{2t-1} + \beta_4 y_{2t-2} + v_{2t}. \end{aligned}$$

Generate forecasts of y_{1t} for the next working week, from the 24th to the 28th of May, 2010.

(d) For the AR(2) and VAR(2) forecasts obtained for the returns on the equity market neutral hedge fund (y_{1t}) and the S&P500 (y_{2t}), compute the RMSE (a total of four RMSEs). Discuss which model yields the superior forecasts.

(e) Let f_{1t}^{AR} be the forecasts form the AR(2) model of the returns on the equity market neutral hedge fund and f_{1t}^{VAR} be the corresponding VAR(2) forecasts. Restricting the sample period just to the forecast period, 24th to the 28th of May, estimate the following regression which pools the two sets of forecasts

$$y_{1t} = \phi_0 + \phi_1 f_{1t}^{AR} + \phi_2 f_{1t}^{VAR} + \eta_t,$$

where η_t is a disturbance term with zero mean and variance σ_η^2. Interpret the parameter estimates and discuss whether pooling the forecasts has improved the forecasts of the returns on the equity market neutral hedge fund.

5. **Evaluating Forecast Distributions using the PIT**

pv.wf1, pv.dta, pv.xlsx

(a) *(Correct Model Specification)* Simulate $y_1, y_2, \cdots, y_{1000}$ observations $(T = 1000)$ from the true model given by a $N(0,1)$ distribution. Assuming that the specified model is also $N(0,1)$, for each t compute the PIT

$$u_t = \Phi(y_t).$$

Interpret the properties of the histogram of u_t.

(b) *(Mean Misspecification)* Repeat part (a) except that the true model is $N(0.5, 1)$ and the misspecified model is $N(0, 1)$.

(c) *(Variance Misspecification)* Repeat part (a) except that the true model is $N(0, 2)$ and the misspecified model is $N(0, 1)$.

(d) *(Skewness Misspecification)* Repeat part (a) except that the true model is the standardised gamma distribution

$$y_t = \frac{g_t - br}{\sqrt{b^2 r}},$$

where g_t is a gamma random variable with parameters $\{b = 0.5, r = 2\}$ and the misspecified model is $N(0, 1)$.

(e) *(Kurtosis Misspecification)* Repeat part (a) except that the true model is the standardised Student t distribution

$$y_t = \frac{s_t}{\sqrt{\frac{\nu}{\nu - 2}}},$$

where s_t is a Student t random variable with degrees of freedom equal to $\nu = 5$, and the misspecified model is $N(0, 1)$.

6. Now estimate an AR(1) model of real equity returns, r_t, on monthly United States data for the period February 1871 to June 2004.

$$r_t = \phi_0 + \phi_1 r_{t-1} + v_t,$$

and compute the standard error of the residuals, $\widehat{\sigma}$. Use the PIT to compute the transformed time series

$$u_t = \Phi\left(\frac{\widehat{v}_t}{\widehat{\sigma}}\right).$$

Interpret the properties of the histogram of u_t.

7. **Predicting the Equity Premium**

goyal_annual.wf1, goyal_annual.dta, goyal_annual.xlsx

The data are annual observations on the S&P 500 index, dividends $d12_t$ and the risk free rate of interest, $rfree_t$, used by Goyal and Welch (2003; 2008) in their research on the determinants of the United States equity premium.

(a) Compute the equity premium, the dividend price ratio and the dividend yields as defined in Goyal and Welch (2003).

(b) Compute basic summary statistics for S&P 500 returns, rm_t, the equity premium, eqp_t, the dividend-price ratio dp_t and the dividend yield, dy_t.

(c) Plot eqp_t, dp_t and dy_t and compare the results with Figure **??**.

(d) Estimate the predictive regressions

$$\begin{aligned} eqp_t &= \alpha_y + \beta_y dy_{t-1} + u_{y,t} \\ eqp_t &= \alpha_p + \beta_p dp_{t-1} + u_{p,t} \end{aligned}$$

for two different sample periods, 1926 to 1990 and 1926 to 2002, and compare your results with Table 7.3.

(e) Estimate the regressions recursively using data up to 1940 as the starting sample in order to obtain recursive estimates of β_y and β_p together with 95% confidence intervals. Plot and interpret the results.

8. **Simulating VaR for a Single Asset**

pv.wf1, pv.dta, pv.xlsx

The data are monthly observations on the logarithm of real United States equity returns, r_t, from January 1871 to June 2004, expressed as percentages. The problem is to simulate 99% Value-at-Risk over a time horizon of six months for the asset that pays the value of the United States equity index

(a) Assume that the equity returns are generated by an AR(1) model

$$r_t = \phi_0 + \phi_1 r_{t-1} + v_t \,.$$

(b) Use the model to provide ex post static forecasts of the entire sample and thus compute the one-step-ahead prediction errors, $\widehat{v}_{t+1}$.

(c) Generate 1000 forecasts of the terminal equity price P_{T+6} using stochastic simulation by implementing the following steps.

i. Forecast $\widehat{rp}^s_{T+k}$ using the scheme

$$\widehat{rp}^s_{T+k} = \widehat{\phi}_0 + \widehat{\phi}_1 \widehat{rp}^s_{T+k-1} + \tilde{v}_{T+k} \,,$$

where $\tilde{v}_{T+k}$ is a random draw from the estimated one-step-ahead prediction errors, $\widehat{v}_{t+1}$.

ii. Compute the simulated equity price

$$\widehat{P}^s_{T+k} = \widehat{P}^s_{T+k} \exp(\widehat{rp}^s_{T+k}/100)$$

iii. Repeat (i) and (ii) for $k = 1, 2, \cdots 6$.

iv. Repeat (i), (ii) and (iii) for $s = 1, 2, \cdots 1000$.

(d) Compute the 99% Value-at-Risk based on the S simulated equity prices at $T + 6$, $\widehat{P}^s_{T+6}$.

Part III

Beyond Least Squares

Chapter 8

Instrumental Variables

8.1 Introduction

Consider the linear regression model introduced in Chapter 3 where the dependent variable y_t is expressed as a linear function of a regressors x_t, or a set of regressors in the case of the multiple regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t, \tag{8.1}$$

where $u_t \sim (0.\sigma_u^2)$ is a disturbance term. Under the conditions outlined in Chapter 3, the ordinary least squares estimator of the parameters of this model have very desirable properties. Paramount among these is arguably the property of consistency, which requires that the estimator converges in probability to the true population parameter as the sample size is increased. In particular, one assumption outlined in Chapter 3, that is fundamental in ensuring this convergence is that of independence. This condition requires that there is no correlation between the regressors and the disturbance term

$$\mathrm{E}(u_t x_t) = 0. \tag{8.2}$$

The independence condition in (8.2) is satisfied provided that the model in (8.1) is indeed the model that is estimated. However, there are a number of situations in financial econometrics where an alternative model is actually estimated with the implication that the regressors and the disturbance term of this model are not necessarily independent. Some examples of this situation occurring are as follows.

(i) Errors in variables

In models of the risk-return tradeoff, expected return is a function of the expected variance of the asset. As the variance of a risky asset is typically unobserved econometric investigations of this relationship typically use proxies for the variance, such as an index of volatility published on the Chicago Board Options Exchange known as the VIX. Given

that the VIX is an imprecise measure of the variance, the independence assumption between the proxy variable and the disturbance of the augmented equation is likely to be violated.

(ii) Omitted variables

The classic example in financial econometrics of omitted variables is the CAPM. Traditionally the CAPM is estimated as a one factor model with the excess return on the market as the only risk factor being priced. More recently a number of other potential risk factors, such as size, value and momentum, are proposed. Clearly if these additional factors are indeed important in the pricing of a risky asset, then the Beta-risk estimated from a single factor model is unlikely to be estimated consistently.

(iii) Simultaneity

In empirical corporate finance, a typical question is whether or not family-owned firms perform better than other firms. This question is usually answered by positing that family ownership is an important explanatory variable for a measure of firm performance. It is quite possible, however, that the reverse is true and that it is the firm's performance which explains the ownership structure. Any single equation study in which a given variable is modelled in terms of a number of regressors runs the risk of the simultaneous determination of the response variable and the regressors. This problem is particularly acute in empirical corporate finance where there is no strong theoretical underpinnings.

Despite the fact that there is more than one fundamental reason for the violation of the independence assumption in a linear regression model, regressors that are correlated with the disturbance term are generally referred to as endogenous variables while regressors that are uncorrelated with the disturbance term are referred to as being exogenous. For this reason the violation of the independence assumption is generally referred to as the endogeneity problem and testing a regressor for correlation with the disturbance term is known as testing for endogeneity.
This chapter explores the method known as instrumental variables estimation for obtaining consistent estimates of the parameters of a linear regression model where the independence assumption of this model is violated. The key requirement for implementing this method is the availability of one or more variables, known as *instrumental variables* or just *instruments*, denoted as z_t, which satisfy two properties: (i) z_t is correlated with the problematic (endogenous) regressor(s), but (ii) uncorrelated with the regression disturbance term.

8.2 Estimating the Risk-Return Tradeoff

A fundamental idea in finance is that investors require a larger risk premium at times when the stock market is riskier with the size of the premium related

to how risk averse investors are. The inter-temporal CAPM model (Merton, 1973, 1980) provides a formal statement of this relationship in which the expected excess return on the aggregate stock market at time t (r_t) is a linear function of the expected variance at time t (σ_t^2)

$$\mathrm{E}_{t-1}(r_t) = \gamma\, \mathrm{E}_{t-1}(\sigma_t^2)\,,$$

in which γ is a parameter capturing the degree of risk aversion of the representative investor and the notation $\mathrm{E}_{t-1}(\cdot)$ emphasises that the expectations are based on information at time $t-1$. In a linear regression framework, the inter-temporal CAPM can be written as

$$r_t = \alpha + \gamma\, \mathrm{E}_{t-1}(\sigma_t^2) + u_t\,, \tag{8.3}$$

in which $\alpha = 0$ and $\gamma > 0$ is hypothesised.
Equation (8.3) cannot be estimated in its present form as the conditional variance, $\mathrm{E}_{t-1}(\sigma_t^2)$ is unobservable. A common approach is to choose an observable proxy for the conditional variance. The Chicago Board Options Exchange (CBOE) publishes an index of the riskiness (volatility) of the S&P500 stock market index, known as the VIX Implied Volatility Index, which is constructed using the information contained in financial options written on the S&P500 Index (see Chapter 16). Although the VIX is quoted as an annualised conditional standard deviation, it is easily scaled to reflect a daily conditional variance. An alternative approach that is dealt with extensively in Chapters 11 and 12 is to specify a model of the conditional variance.
Denoting the observable proxy for the expected conditional variance as h_t, the relationship between provy variance h_t and the true unobserved variance $\mathrm{E}_{t-1}(\sigma_t^2)$ is

$$h_t = \mathrm{E}_{t-1}(\sigma_t^2) + \epsilon_t\,, \tag{8.4}$$

where ϵ_t represents the measurement which is assumed to be distributed as $(0, \sigma_\epsilon^2)$. Using this expression to substitute out the unobservable variable $\mathrm{E}_{t-1}(\sigma_t^2)$ in (8.3) yields the following augmented regression equation whereby all variables are now observable

$$r_t = \alpha + \gamma h_t + v_t\,. \tag{8.5}$$

The disturbance term v_t is

$$v_t = -\gamma\epsilon_t + u_t \tag{8.6}$$

which represents a composite of two disturbance terms: the disturbance term u_t in (8.3) which represents deviations from the inter-temporal CAPM relationship, and the measurement error ϵ_t in (8.6) which arises from using a proxy for the unobserved variance. Given the observable proxy h_t, it is tempting to estimate the risk aversion parameter, γ, in equation (8.5) by ordinary least squares.
Figure 8.1 provides a scatter plot of the relationship between daily returns to the S&P 500 Index and the proxy for the conditional variance based on the

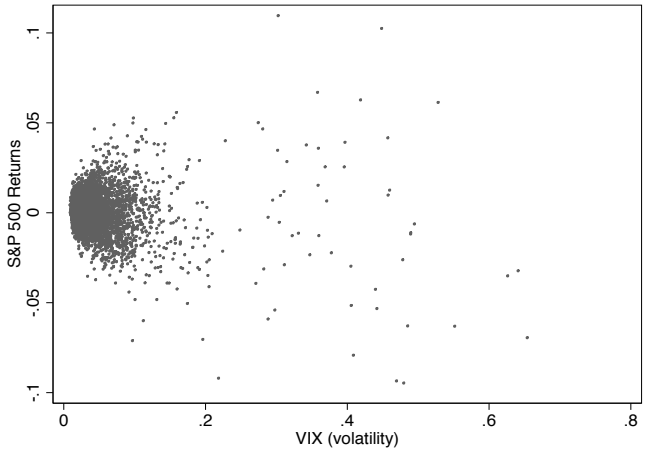


Figure 8.1: Scatter plot illustrating the relationship between returns to the S&P 500 Index and the proxy for the conditional variance based on the VIX Index.

VIX Index for the period 2 January 1990 to 1 June 2012 ($T = 5652$). It is apparent that the relationship is fairly noisy and their is no obvious positive relationship evident in the scatter. Estimating equation (8.5) by ordinary least squares using the returns to the S&P 500 Index and the VIX proxy for the conditional variance yields the following results

$$r_t = \underset{(0.0002)}{0.0018} - \underset{(0.9534)}{9.7202}\, h_t + \widehat{v}_t\,, \tag{8.7}$$

where standard errors are given in parentheses. Although the estimate of the constant term is not significantly different from zero, the estimate of the coefficient of risk aversion is negative, a result which is markedly at odds with the theory and suggests that something is amiss with the econometrics.
To understand the problems with estimating equation (8.5) by ordinary least squares consider whether the independence condition between the VIX and the measurement error ϵ_t, is satisfied. From (8.4) and (8.6) the covariance between the VIX and the measurement error is

$$\text{cov}(h_t, v_t) = \text{cov}[(\text{E}_{t-1}(\sigma_t^2) + \epsilon_t), (-\gamma\epsilon_t + u_t)] = -\gamma\sigma_\epsilon^2 \neq 0\,. \tag{8.8}$$

In this situation, ordinary least squares does not yield consistent parameter estimates. To gain insight into the effects of this violation on the least squares estimator of γ using the augmented regression model (8.5), from Chapter 3 consider rewriting the population slope parameter for this model as

$$\frac{\text{cov}(r_t, h_t)}{\text{var}(h_t)} = \frac{\text{cov}(\alpha + \gamma h_t + v_t, h_t)}{\text{var}(h_t)} = \gamma + \frac{\text{cov}(v_t, h_t)}{\text{var}(h_t)} = \gamma - \gamma\frac{\sigma_\epsilon^2}{\text{var}(h_t)}\,, \tag{8.9}$$

where the second step uses (8.5) and the last step uses (8.8). Only if there is no measurement error, $\sigma_\epsilon^2 = 0$, would the slope population parameter of (8.5) correspond to the risk aversion parameter γ in the true model given in (8.3). When there is measurement error, $\sigma_\epsilon^2 \neq 0$, the slope population parameter of (8.5) corresponds to a value biased downwards from the true value of γ. The implication of this result for estimation is that whilst the least squares estimator is a consistent estimator of the left hand-side of (8.9), it is an inconsistent estimator of γ.
To circumvent the problems with applying OLS to (8.5) directly, suppose now there is a variable z_t which satisfies two important conditions

(i) $\text{cov}(h_t, z_t) \neq 0$.

(ii) $\text{cov}(z_t, v_t) = 0$.

The first condition ensures that the variable is correlated with the proxy variable h_t, and the second condition ensures that z_t is uncorrelated with the disturbance term v_t. The covariance between returns, r_t, and the variable z_t, which satisfies these conditions, is given by

$$\text{cov}\,(r_t, z_t) = \text{cov}(\alpha + \gamma h_t + v_t, z_t) = \text{cov}\,(\gamma h_t, z_t) + \text{cov}\,(v_t, z_t) = \gamma\,\text{cov}(h_t, z_t)\,. \tag{8.10}$$

Rearrangement of equation (8.10) suggests that an alternative expression for the population parameter γ is

$$\gamma = \frac{\text{cov}(r_t, z_t)}{\text{cov}(h_t, z_t)}\,.$$

The variable z_t is known as an instrumental variable and when the population quantities are replaced by their sample counterparts the resulting estimator is known as the instrumental variables estimator

$$\widehat{\gamma}^{IV} = \frac{\frac{1}{T}\sum_{t=1}^{T}(r_t - \overline{r})(z_t - \overline{z})}{\frac{1}{T}\sum_{t=1}^{T}(h_t - \overline{h})(z_t - \overline{z})}\,. \tag{8.11}$$

Under certain conditions the instrumental variables estimator is consistent and asymptotically normally distributed. Situations where these properties do not hold are discussed in Section 8.5.
Of course, the difficulty in implementing the instrumental variables estimator in equation (8.11) is finding a suitable instrumental variable, z_t, that satisfies the two conditions outlined earlier. In the current context a solution is readily available because for the current sample the first order autocorrelation coefficient, defined in Chapter 2, for the conditional variance proxy, h_t turns out to be 0.9705 meaning that $\text{cov}(h_t, h_{t-1}) \neq 0$. Moreover, at time t the lagged value h_{t-1} may be taken as given and therefore $\text{cov}(h_{t-1}, v_t) = 0$. Consequently, an

instrumental variables estimator of γ using h_{t-1} as the instrument is a natural approach to take. Estimating equation (8.5) by instrumental variables yields

$$r_t = \underset{(0.0002)}{-0.0003} + \underset{(0.9975)}{3.0190}\, h_t + \widehat{v}_t\,, \tag{8.12}$$

where standard errors are given in parentheses.[1] The estimate of the risk-aversion parameter is now positive and significant as predicted by financial theory. In addition, the size of the estimated coefficient, 3.0190, is consistent with estimates in the published literature (Ghysels, Santa-Clara and Valkanov, 2005; Bali and Peng, 2006). Moreover, a comparison of the OLS and IV risk aversion parameter estimates in (8.7) and (8.12) respectively, show that the OLS estimate is biased downwards as predicted by the econometric theory (8.9).
In summary, the violation of the assumption of independence between the disturbance term and the explanatory variable(s) in a linear regression, which is known as the exogeneity assumption, results in problems for the the ordinary least squares estimator. Specifically the estimates of the coefficient on the variables which do not satisfy this assumption are inconsistent. The use of instrumental variables estimation to correct this problem is now explored in more detail.

8.3 The General IV Estimator

The risk-return tradeoff model with risk proxied by the VIX, represents a regression equation where the dependent variable is the return on an asset and the explanatory variable is the VIX which acts as a proxy for risk. This model is characterised by two endogenous variables which is estimated using IV by letting the VIX at time $t-1$ act as an instrument for the VIX at time t. This type of model structure is now extended to allow for additional explanatory variables in the model which are all exogenous. The aim now is to augment the IV estimator in such a way that information on all of the exogenous variables in the system (both included and excluded) are combined to provide an improved instrument for the endogenous explanatory variable.
To highlight how the general IV estimator is implemented consider the following multi-factor CAPM

$$r_{it} - r_{ft} = \alpha + \beta_1(r_{mt} - r_{ft}) + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{MOM}_t + v_t, \tag{8.13}$$

where $r_{it} - r_{ft}$ is the excess return on the asset i, $r_{mt} - r_{ft}$ is the market factor computed as the excess return on the market, SMB is *Size* factor measuring the performance of small stocks relative to big stocks, HML is the *Value* factor which measures the performance of value stocks relative to growth stocks,

[1] To compute the standard errors for the IV estimator, the residuals used to compute the standard errors at the second stage are defined by replacing the generated regressors with their actual values, while still using the instrumental variables parameter estimates. This adjustment is computed automatically in all econometric software packages.

MOM is a *Momentum* factor and v_t is a disturbance term. The contributions of SMB, HML and MOM are determined by the parameters β_2, β_3 and β_4 respectively. For convenience, the ordinary least squares parameter estimates of this model, using monthly data on a portfolio index for Nondurables in the United States from January 1927 to December 2013, are reproduced here from Chapter 3 (standard errors in parentheses):

$$\begin{aligned} r_{it} - r_{ft} = \underset{(0.0706)}{0.1786} + \underset{(0.0140)}{0.7674}\,(r_{mt} - r_{ft}) - \underset{(0.0223)}{0.0315}\,\text{SMB}_t \\ + \underset{(0.0211)}{0.0284}\,\text{HML}_t + \underset{(0.0162)}{0.0247}\,\text{MOM}_t + \widehat{v}_t. \end{aligned} \tag{8.14}$$

The price of market risk estimate is 0.7674 which suggests that Nondurables represent a conservative stock.

The market factor in the multi-factor CAPM in equation (8.13) is defined as the return on all equities. By definition the market factor must contain returns on the endogenous variable Nondurables thus making the market factor also endogenous. Moreover, in theory the market factor in the CAPM and multi-factor version of the CAPM represents the return on all wealth, not just equities. This suggests that the market return on equities also represents a proxy variable for the return on all wealth resulting in an errors in variables problem when estimating the model by ordinary least squares. These arguments suggest that for this model

$$\text{cov}[(r_{mt} - r_{ft}), v_t] \neq 0, \tag{8.15}$$

leading to a violation of the conditions needed for ordinary least squares estimator applied to (8.13) be consistent.

To re-estimate the multi-factor CAPM in equation (8.13) by instrumental variables, the broad strategy is to follow the IV approach used in Section 8.2 in the case of estimating the risk-return trade-off model. The approach is to choose as an instrument for the market factor, $(r_{mt} - r_{ft})$, the lagged returns on the market factor, $(r_{mt-1} - r_{ft-1})$. However, to incorporate information from the exogenous variables $\{SMB_t, HML_t, MOM_t\}$ in equation (8.13) so as to improve the overall quality of the instrument for the market factor, all of the exogenous variables are now combined together by specifying the following regression equation

$$r_{mt} - r_{ft} = \pi_0 + \pi_1(r_{mt-1} - r_{ft-1}) + \pi_2\,\text{SMB}_t + \pi_3\,\text{HML}_t + \pi_4\,\text{MOM}_t + e_t, \tag{8.16}$$

where e_t is a disturbance term which is independent of all of the exogenous variables in (8.16). This equation is often referred to as the reduced form which expresses an endogenous variable as a function of all of the exogenous variables in the system. Plots of all four factors are given in Figure 8.2. The ordinary least squares estimates of equation (8.16) with standard errors in paren-

theses, are

$$(r_{mt} - r_{ft}) = \underset{(0.1560)}{0.6806} + \underset{(0.0287)}{0.0113}\ (r_{m\,t-1} - r_{f\,t-1}) + \underset{(0.0484)}{0.4719}\ \text{SMB}_t + \underset{(0.0466)}{0.1223}\ \text{HML}_t\ \underset{(0.0347)}{-0.2970}\ \text{MOM}_t + \widehat{e}_t, \tag{8.17}$$

where $\widehat{e}_t$ is the ordinary least squares residual. All four instruments are statistically significant suggesting that all four exogenous variables are important explanatory variables of the market factor.

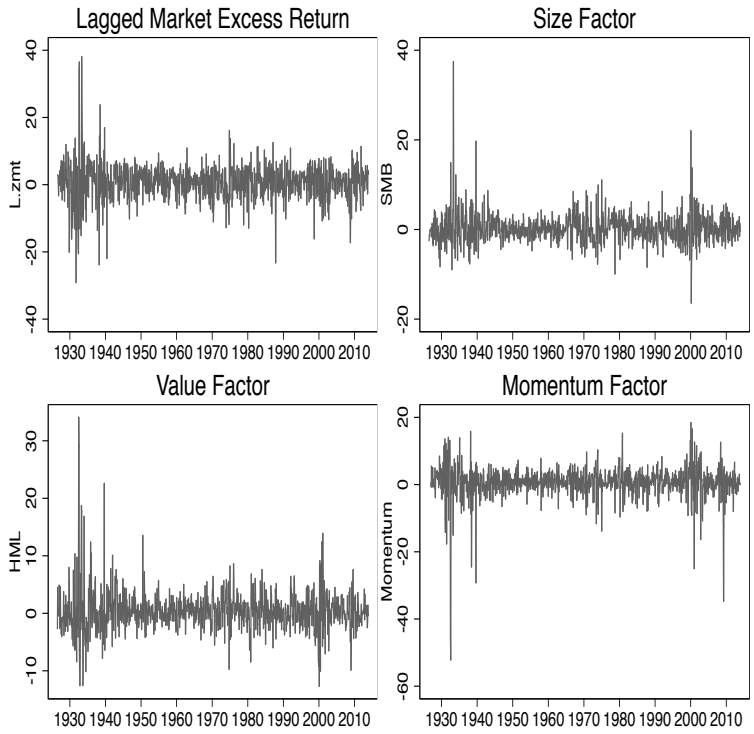


Figure 8.2: Instruments used to estimate the extended CAPM in equation (8.16). Data are monthly data on U.S. nondurables for the period January 1927 to December 2013.

The instrument of the market factor is computed as the predictor of this equation

$$\widehat{(r_{mt} - r_{ft})} = 0.6806 + 0.0113\ (r_{m\,t-1} - r_{f\,t-1}) + 0.4719\ \text{SMB}_t + 0.1223\ \text{HML}_t - 0.2970\ \text{MOM}_t. \tag{8.18}$$

The estimated reduced form equation represents a weighted average of all of the four exogenous variables in the system with the weights being deter-

mined optimally in the sense that the estimated model provides the best predictor of the endogenous variable $(r_{mt} - r_{ft})$ from a conditional expectations point of view. A plot of the constructed instrument in (8.18) is given in Figure 8.3.

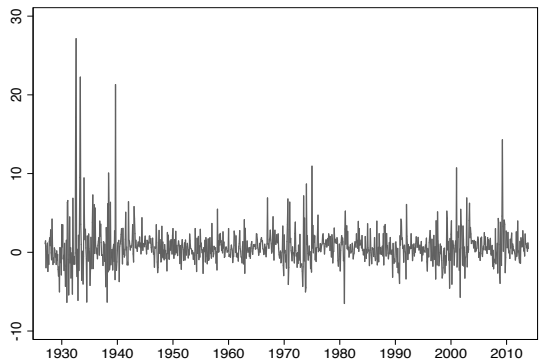


Figure 8.3: The composite instrument computed as a weighted average of all four exogenous variabes in the extended CAPM in equation (8.16). Data are monthly data on U.S. nondurables for the period January 1927 to December 2013.

Estimating the extended CAPM specified in equation (8.13) by IV is achieved by using ordinary least squares to regress $r_{it} - r_{ft}$ on $\{1, \widehat{r_{mt} - r_{ft}}, \text{SMB}_t, \text{HML}_t, \text{MOM}_t\}$ where $\widehat{r_{mt} - r_{ft}}$ is defined in (8.18). The estimated model with standard errors in parentheses, is

$$\begin{aligned}(r_{it} - r_{ft}) = \underset{(0.8750)}{0.0303} + \underset{(1.2681)}{0.9833}\,(r_{mt} - r_{ft}) - \underset{(0.6044)}{0.1343}\text{SMB}_t \\ + \underset{(0.1572)}{0.0020}\text{HML}_t + \underset{(0.3780)}{0.0890}\text{MOM}_t + \widehat{v}_t. \qquad (8.19)\end{aligned}$$

Again notice that the estimated model is expressed in terms of the original regressors in the model even though estimation is based on replacing $(r_{mt} - r_{ft})$ by its instrument. Unlike the parameter estimates in (8.14), the IV parameter estimates in (8.19) are statistically consistent as the requirement that all regressors used to estimate the model are independent of the disturbance term is now satisfied because

$$\text{cov}[(\widehat{r_{mt} - r_{ft}}), v_t] = 0, \qquad (8.20)$$

from the fact that $(\widehat{r_{mt} - r_{ft}})$ is simply a linear function of the exogenous variables $\{1, \text{SMB}_t, \text{HML}_t, \text{MOM}_t\}$ which by definition are individually independent of v_t and thus must be jointly independent of v_t as well. A comparison of the ordinary least squares parameter estimates in equation (8.14) and the IV estimates in (8.19) show that the estimate of the market price of risk has

increased from 0.7674 to 0.9833 suggesting that this asset is not a conservative stock but actually tracks the market nearly one-to one. A formal test of this hypothesis is given by the t statistic

$$t = \frac{0.9833 - 1.000}{1.2681} = -0.0132.$$

The p value is 0.9895 showing a failure to reject th null hypothesis that the market price of risk is unity at the 5% level.

The extended CAPM is characterised by a single endogenous regressor and multiple exogenous variables. This class of models is easily extended to the case where there are multiple endogenous regressors and multiple exogenous variables. Suppose that there are N endogenous regressors and K exogenous variables so that the model is specified as

$$y_t = \beta_0 + \sum_{i=1}^{N} \beta_i x_{it} + \sum_{i=k}^{K} \phi_k w_{kt} + v_t \,. \tag{8.21}$$

In order to identify the parameters $\beta_1, \cdots, \beta_N$, there must be at least L instruments $z_1, \cdots, z_L$ where $L \geq N$, that are not included in the model. This condition, known as the *order condition*, is a necessary, but not sufficient, condition to identify all of the parameters of the structural model in (**??**). If the number of instruments matches the number of endogenous variables, the model is referred to as just identified. If there are more instruments available than endogenous regressors, the model is said to be over-identified. If there are insufficient instruments the model is under-identified and cannot be estimated.

In general the instrumental variable estimator is computed as follows:

Step 1: Estimate the reduced form by regressing each of the N endogenous regressors on all of the L instruments and K exogenous variables in the following sequence of N ordinary least squares regressions

$$x_{it} = \pi_{i0} + \sum_{j=1}^{L} \pi_{ij} z_{jt} + \sum_{k=1}^{K} \varphi_{ik} w_{kt} + e_{it}, \qquad i = 1, 2, \cdots, N.$$

Let the predicted values from each of these N regressions be $\widehat{x}_{1t}, \cdots, \widehat{x}_{Nt}$.

Step 2: Regress y_t on the N predicted values of the endogenous regressors and the exogenous variables in the equation

$$y_t = \beta_0 + \sum_{i=1}^{N} \beta_i \widehat{x}_{it} + \sum_{k=1}^{K} \phi_k w_{kt} + v_t \,,$$

to obtain the instrumental variables estimator $\widehat{\theta}_{IV} = \{\widehat{\beta}_0, \cdots, \widehat{\beta}_N, \widehat{\phi}_1, \cdots, \widehat{\phi}_K\}$.

8.4 Testing for Endogeneity

The advantage of using an IV estimator is that it yields consistent parameter estimates even if:

(i) The regressors are measured with error.

(ii) There are some missing variables from the model.

(iii) Some of the regressors are endogenous as the model is part of a simultaneous system of equations.

If none of these conditions apply so OLS would yield consistent parameter estimates anyway, the IV estimator is still consistent. This suggests that the IV estimator is robust to various types of misspecification which nonetheless delivers consistent estimates of the parameters in many situations. However, there is a cost in using the IV estimator when it does not need to be used as there is a loss of information from using an instrument for a variable. As the instrument is necessarily an imprecise measure of the variable it is instrumenting there is a loss of efficiency from using the IV estimator which translates into standard errors that are larger than the ordinary least squares standard errors[2]

$$\text{var}(\theta_{OLS}) < \text{var}(\theta_{IV}). \tag{8.22}$$

To guard against an unnecessary loss of efficiency from using an IV estimator unnecessarily, it is appropriate to perform a preliminary test of endogeneity of the regressors. To construct a test of endogeneity consider the two variable regression model

$$y_t = \beta_0 + \beta_1 x_t + v_t, \qquad v_t \sim N(0, \sigma^2). \tag{8.23}$$

A test of the endogeneity of x_t is formulated in terms of the following hypotheses

$$\begin{aligned} H_0 &: \ \text{cov}(x_t, v_t) = 0 \qquad [x_t \text{ Exogenous}] \\ H_1 &: \ \text{cov}(x_t, v_t) \neq 0 \qquad [x_t \text{ Endogenous}]. \end{aligned}$$

If the null hypothesis is rejected then x_t is endogenous whereby an IV estimator is used to achieve consistency. If there is a failure to reject the null hypothesis, in this case x_t is exogenous and there is no need to use the IV estimator as the OLS estimator is now consistent.
To construct a test of endogeneity the approach is based on the auxiliary regression approach proposed by Davidson and MacKinnon (1989, 1993). Suppose that a valid instrument for x_t in (8.23) exists so that

$$x_t = \pi_0 + \pi_1 z_t + e_t, \qquad \pi_1 \neq 0\,. \tag{8.24}$$

[2]From Chapter 3 the variance of the OLS estimator is $\text{var}(\theta_{OLS}) = T^{-1}\sigma_u^2/\sigma_x^2$. In contrast the variance of the IV estimator is $\text{var}(\theta_{IV}) = T^{-1}\sigma_u^2/(\rho_{xz}^2\sigma_x^2)$, where $0 < \rho_{xz}^2 < 1$ is the squared correlation between the endogenous regressor x_t and the instrument z_t. As $0 < \rho_{xz}^2 < 1$, it immediately follows that $\text{var}(\theta_{OLS}) < \text{var}(\theta_{IV})$.

It follows that

$$\text{cov}(x_t, v_t) = \text{cov}(\pi_0 + \pi_1 z_t + e_t, v_t) = \text{cov}(e_t, v_t)\,,$$

using the fact that $\text{cov}(z_t, v_t) = 0$ by virtue of z_t being a valid instrument. It follows therefore that

$$\text{cov}(x_t, v_t) = 0 \Longrightarrow \text{cov}(e_t, v_t) = 0\,. \tag{8.25}$$

The condition in expression (8.25) may be written in a slightly different way by requiring that the coefficient α in the linear regression

$$v_t = \alpha e_t + \eta_t\,, \tag{8.26}$$

is zero. Of course because v_t and e_t are disturbance terms they are unobserved and equation (8.26) cannot be estimated as it stands and a test of $\alpha = 0$ cannot be conducted based on this equation alone. Substituting equation (8.26) for v_t into the original model in equation (8.23) gives

$$y_t = \beta_0 + \beta_1 x_t + \alpha e_t + \eta_t\,,$$

a result that suggests the following auxiliary regression based test of endogeneity of x_t.

Step 1: Estimate the regression

$$x_t = \pi_0 + \pi_1 z_t + e_t,$$

by ordinary least squares to obtain the estimates $\widehat{\pi}_0$ and $\widehat{\pi}_1$ and then compute the residuals $\widehat{e}_t$.

Step 2: Estimate the regression

$$y_t = \beta_0 + \beta_1 x_t + \alpha \widehat{e}_t + \eta_t,$$

by ordinary least squares and test $H_0 : \alpha = 0$ using a t test. Failure to reject the null hypothesis will indicate that condition (8.25) is not satisfied and that there is a potential endogeneity problem.

This simple regression based test of endogeneity may be illustrated by returning to the risk return tradeoff application estimated in Section 8.2. The first step is to regress the VIX proxy for the conditional variance on its own lag

$$h_t = \underset{(0.0000)}{0.0000} + \underset{(0.0032)}{0.9705}\, h_{t-1} + \widehat{e}_t\,,$$

in which $\widehat{e}_t$ are the residuals. The second stage regression yields

$$r_t = \underset{(0.0002)}{-0.0003} + \underset{(0.6869)}{3.0190}\, h_t - \underset{(2.8475)}{218.802}\, \widehat{e}_t + \widehat{\eta}_t\,.$$

The test statistic is now constructed as

$$t = -\frac{218.802}{2.8475} = -76.84\,,$$

which gives a p value of 0.000. The null hypothesis of no endogeneity is easily rejected, a result that validates the use of instrumental variables to estimate the risk-return relationship.

This test, known as the Durbin-Wu-Hausman test, is easily extended to the case of multiple instruments and multiple potential endogenous regressors. All that is required is that multiple estimated residual terms be included in the regression in Step 2 and each of these tested separately for a zero coefficient. Another of the main advantages of the auxiliary regression based test of endogeneity is that it can be used when the disturbances in the original regression model (8.11) are not identically and independently distributed. In this situation the t test used to test zero restrictions in Step 2 should be based on robust estimates of the standard errors.

8.5 Weak Instruments

One of the two conditions required for z_t to be a valid instrument of x_t is that the two variables are indeed correlated with each other. This condition is made explicit using the reduced form equation (8.24) where correlation between z_t and x_t requires the condition $\pi_1 \neq 0$. If this condition is not satisfied so $\pi_1 = 0$, the link between the two variables is broken whereby the instrument z_t, provides no information on the endogenous regressor x_t. In this situation the IV estimator of θ breaks down as it now provides no information on the unknown parameters of the true model.[3]

This discussion suggests two extreme situations where z_t is a good instrument $(\pi_1 \neq 0)$ and where it is a bad instrument $(\pi_1 = 0)$. However, there is a very important intermediate case where the instrument does exhibit some correlation with x_t so $\pi_1 \neq 0$, but that this correlation is relatively low. In this situation the instrument z_t is referred to as a weak instrument. In particular, it has been shown, for example by Bound, Jaeger, and Baker (1995), and Staiger and Stock (1997), that the weak-instruments problem can arise even when the correlations between endogenous regressors and the instruments are significant at conventional levels (5% or 1%) and the sample size is large. The main implication of using weak instruments to estimate θ is that the distribution of the instrumental variables estimator is no longer asymptotically normal.

[3]Since $\text{var}(\theta_{IV}) = T^{-1}\sigma_u^2/(\rho_{xz}^2\sigma_x^2)$, if $\rho_{xz} = 0$ so there is no relationship between the two variables, the variance of the IV estimator of θ approaches infinity.

8.5.1 Illustrating the Effect of Weak Instruments

The problem posed by weak instruments is easy to demonstrate by simulation. Consider the simple model

$$\begin{array}{ll} y_t = \beta x_t + v_t & \text{[Structural equation]} \\ x_t = \pi z_t + e_t & \text{[Reduced form equation]}, \end{array} \tag{8.27}$$

in which y_t and x_t are endogenous variables, z_t is an exogenous variable and

$$\begin{bmatrix} v_t \\ e_t \end{bmatrix} \sim iid \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_v^2 & \sigma_{ve} \\ \sigma_{ve} & \sigma_e^2 \end{bmatrix} \right).$$

The instrumental variables estimator of β is

$$\widehat{\beta}_{IV} = \frac{\text{cov}\,(y_t, z_t)}{\text{cov}\,(x_t, z_t)},$$

and if cov (x_t, z_t) is relatively small the denominator is nearly zero. The sampling distribution of $\widehat{\beta}_{IV}$ and its t statistic is not well approximated by a normal distribution. The intuition is that small changes in $\text{cov}(x_t, z_t)$ from one sample to the next can induce big changes in $\widehat{\beta}_{IV}$. So if the instruments are weak, the usual methods of inference are potentially unreliable.
The parameter π in (8.27) controls the strength of the instrument. A value of $\pi = 0$ means that there is no correlation between x_t and z_t, in which case z_t is not a valid instrument. The weak instrument problem occurs when the value of π is 'small' relative to σ_e^2, the variance of e_t. To highlight the properties of the instrumental variables estimator of β in the presence of a weak instrument, let the parameters of the model in (8.27) be $\beta = 0$, $\pi = 0.25$ and

$$\begin{bmatrix} v_t \\ e_t \end{bmatrix} \sim iid\, N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.99 \\ 0.99 & 1 \end{bmatrix} \right).$$

The sampling distribution of the ordinary least squares and instrumental variables estimator of β, computed by Monte Carlo methods for a sample of size $T = 5$ with 10000 replications, is shown in Figure 8.4. The sampling distribution of $\widehat{\beta}$ is far from being normal or centered on the true value of $\beta = 0$, despite the fact that $\pi \neq 0$. In fact, the sampling distribution is bimodal with neither of the two modes being located near the true value of β. By improving the quality of the instrument, as represented by higher values of π, the sampling distribution of the instrumental variable estimator approaches normality with its mean located at the true value of $\beta = 0$. The distribution of the ordinary least squares estimator is clearly biased but its variance is relatively small compared to the distribution of the instrumental variables estimator.

STAN: The figure has beta_1 when it should be beta

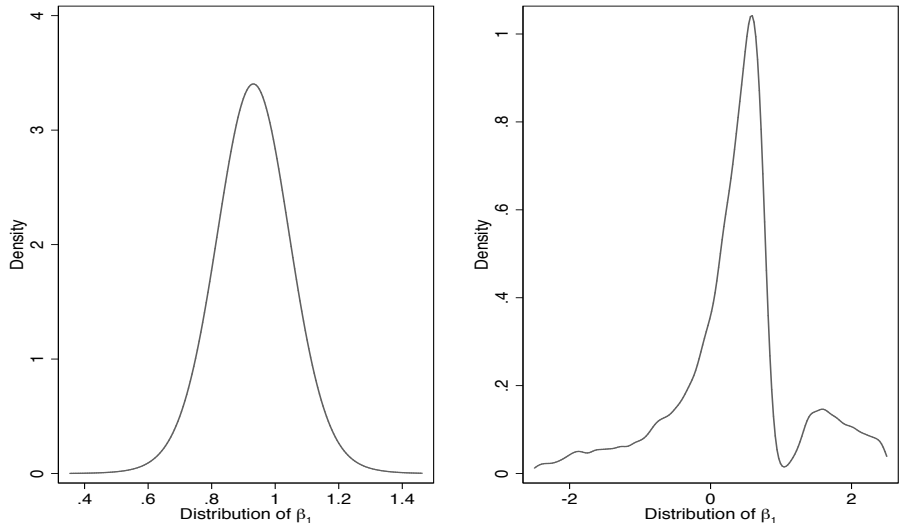


Figure 8.4: Sampling distribution of the instrumental variables estimator in the presence of a weak instrument. The distribution is approximated using a Gaussian kernel density estimator with bandwidth 0.07.

Staiger and Stock (1997), Staiger, D. Stock, J.H. show in the worst case scenario, weak instruments can result in the bias of the instrumental variables estimator being the same as the bias of the ordinary least squares estimator. The instrumental variables estimator with weak instruments becomes inconsistent and estimation by its use can actually aggravate the endogeneity problem.

8.5.2 Tests for Weak Instruments

A natural test of weak instruments is based on the reduced form equation linking the endogenous regressor x_t, the instrument z_t and the exogenous regressor in the model w_t. Defining the reduced form as

$$x_t = \pi_0 + \pi_1 z_{jt} + \varphi_1 w_t + e_t, \tag{8.28}$$

a test of weak instruments is based on using a F test of the joint restrictions

$$\begin{array}{lll} H_0: & \pi_1 = \varphi_1 = 0 & \text{[Weak instruments]} \\ H_1: & \text{at least one restriction fails} & \text{[Good instruments].} \end{array} \tag{8.29}$$

This testing framework is easily extended to the case of $N = 1$ endogenous regressor, L instruments and K exogenous regressors. To allow for N endogenous regressors the weak instrument test is based on a joint test of the quality of all instruments as instruments of all endogenous regressors. These tests are

known as reduced rank tests which is a multivariate analogue of the F test, which are available in most econometric software packages.
The crucial question, however, in implementing the weak instrument test is what critical values to adopt for these tests statistics, given the observation that weak instruments can still occur even if the under-identification hypothesis is rejected at conventional significance levels. A popular rule-of-thumb is to reject the null hypothesis of weak instruments if the F statistic is greater than 10, a critical value well in excess of the value required by conventional significance levels. This rule may not however be sufficiently strict and alternative critical values are provided by Stock and Yogo (2005), Stock, J.H. Yogo, M. based on two criteria.

(i) **Critical values based on maximum relative bias**:

Critical values are based on the ratio of the bias of the IV estimator to the bias of OLS. The null hypothesis is that weak instruments lead to an asymptotic relative bias greater than some value b. So for example, if the least squares estimator suffers a maximum bias of 10%, and if the relative bias is 0.1, then the maximum bias of the IV estimator is 1%.

(ii) **Critical values based on maximum size**:

The Wald test of the hypothesis $H_0 : \widehat{\beta}_i = 0$ for all i is likely to have a size distortion in the presence of weak instruments. Specifically, the actual actual rejection rate of a true null hypothesis will be greater than than the nominal rejection rate (nominal size). The second set of critical values depends on the maximum rejection rate of a true null hypothesis for a given nominal size. For example, we may be willing to accept a maximum rejection rate of 10% for a test at the 5% level, but we may not be willing to accept a rejection rate of 20% for a 5% level test.

Note that these critical values correspond to the i.i.d. error case and are therefore to be used with caution when applied to the statistics based on a robust estimator of the covariance matrix.

8.5.3 Robust Inference in the Structural Equation

Suppose that z_t is a valid instrument for x_t in the sense that in the reduced form regression, $\pi_1 \neq 0$ at conventional significance levels. As noted earlier, this conclusion does not preclude the problem of weak instruments. It turns out that it is still possible to make robust inference on the parameter β_1 in the structural equation, notwithstanding the presence of weak instruments.
Consider the simple model with $N = 1$ endogenous regressor, $K = 1$ exogenous regressor and $L = 1$ instrument

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + \phi_1 w_t + v_t, \\ x_t &= \pi_0 + \pi_1 z_t + \varphi_1 w_t + e_t \,. \end{aligned} \tag{8.30}$$

Substituting for x_t in the structural equation yields the reduced form equation for y_t given by

$$y_t = \widetilde{\pi}_0 + \widetilde{\pi}_1 z_t + \widetilde{\phi}_1 w_t + \widetilde{e}_t \,, \tag{8.31}$$

with

$$\widetilde{\pi}_0 = \beta_0 + \beta_1 \pi_0, \qquad \widetilde{\pi}_1 = \beta_1 \pi_1, \qquad \widetilde{\phi}_1 = \beta_1 \varphi_1 + \phi_1 \,.$$

To perform a test on the parameter β_1 in (8.30) the approach is to perform a test on $\widetilde{\pi}_1$ in (8.31) instead. This test is known as the Anderson-Rubin test which is an F test of the null hypothesis $H_0 : \ \widetilde{\pi}_1 = 0$, which is in fact a test of $\beta_1 = 0$ given that $\pi_1 \neq 0$ by virtue of the fact that z_t is an instrument for x_t and the identification condition has been met. If the hypothesis $\widetilde{\pi}_1 = 0$ cannot be rejected, then $\beta_1 = 0$ also cannot be rejected. The test is robust to weak instruments. Weak instruments imply that π_1 is small, so that $\widetilde{\pi}_1 = \beta_1 \pi_1$ is also small and hence rejecting the null hypothesis of $\widetilde{\pi}_1 = 0$ is less likely. In other words, weak instruments reduce the power of the test.

8.6 Consumption CAPM

One of the most celebrated examples of endogeneity and weak instruments in financial econometrics is provided by asset pricing models in which asset prices are related to the the consumption and savings decisions of investors. Since deferred consumption is used to finance future portfolio choices, the consumption decision is an important determinant of intertemporal asset prices. The consumption based Capital Asset Pricing Model (C-CAPM) assumes that a representative agent chooses current and future real consumption $\{C_t, C_{t+1}, C_{t+2}, \cdots\}$ to maximise the inter-temporal expected utility function

$$\mathrm{E}_t \left[\sum_{j=0}^{\infty} \delta^j U(C_{t+j}) \right], \tag{8.32}$$

subject to the wealth constraint

$$W_{t+1} = (1 + R_{t+1})(W_t - C_t),$$

where W_t is wealth, R_t is the simple return on an asset (more precisely on wealth), E_t is the conditional expectations operator based on information at time t and δ is the discount rate.
One of the first-order conditions (known as an Euler equation) describing the investor's optimal consumption decision is given by

$$U'(C_t) = \mathrm{E}_t[\delta U'(C_{t+1})(1 + R_{t+1})] \,, \tag{8.33}$$

in which the notation $U'(\cdot)$ refers to the first derivative of $U(\cdot)$. The Euler equation encapsulates the condition that the investor should consume to the point at which the the marginal utility of one real dollar of current consumption is equal to the discounted expected marginal utility of investing the real

dollar at the current interest rate and consuming the proceeds. Dividing equation (8.33) by $U'(C_t)$ gives

$$\mathrm{E}_t\left[\delta\frac{U'(C_{t+1})(1+R_{t+1})}{U'(C_t)}\right]=1\,,$$

in which the term $\delta U'(C_{t+1})/U'(C_t)$ is known as the stochastic discount factor.
In order to make progress with the investors consumption decision, a form for $U(C_t)$ must be proposed. A utility function often used in empirical research is the power utility function

$$U(C_t)=\frac{C_t^{1-\gamma}-1}{1-\gamma}\,,$$

in which γ is known as the coefficient of relative risk aversion. Utility functions are concave functions with the measure of curvature, $-U''(C_t)/U'(C_t)$ giving the degree of risk aversion. For this particular utility function

$$U'(C_t)=C_t^{-\gamma},\qquad U''(C_t)=-\gamma C_t^{-\gamma-1},\qquad -C_t\frac{U''(C_t)}{U'(C_t)}=\gamma\,,$$

so that this function has constant relative risk aversion, γ.
Using the power utility function, the Euler equation becomes

$$\mathrm{E}_t\left[\delta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}(1+R_{t+1})\right]=1. \tag{8.34}$$

Taking natural logarithms of this equation gives

$$\log\mathrm{E}_t\left[\delta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}(1+R_{t+1})\right]=0, \tag{8.35}$$

since $\log 1=0$. The left hand side of equation (8.35) is the logarithm of a conditional expectation which may be simplified if some additional assumptions are made.
Let the variable X follow a log-normal distribution, then by a property of this distribution

$$\log\mathrm{E}_t[X]=\mathrm{E}_t[\log X]+\frac{1}{2}\mathrm{var}_t(\log X)\,. \tag{8.36}$$

Now define

$$X=\delta(C_{t+1}/C_t)^{-\gamma}(1+R_{t+1}) \tag{8.37}$$

so that the task becomes one of finding the relatively straightforward expressions for the two terms on the right hand side of (8.36), based on the assumption that X does indeed follow a log-normal distribution. Taking the logarithm of the variable X in equation (8.37) yields

$$\log X=\log\delta-\gamma\Delta c_{t+1}+r_{t+1}\,. \tag{8.38}$$

in which $\Delta c_{t+1} = \log C_{t+1} - \log C_t$ and $r_{t+1} = \log(1 + R_{t+1})$. The two required terms on the right-hand side of (8.36) are then given by

$$\begin{aligned} \mathrm{E}_t[\log X] &= \log \delta - \gamma \, \mathrm{E}_t[\Delta c_{t+1}] + \mathrm{E}_t[r_{t+1}] \\ \mathrm{var}_t(\log X) &= \gamma^2 \sigma_c^2 + \sigma_r^2 - 2\gamma\sigma_{cr} \,, \end{aligned}$$

in which

$$\sigma_c^2 = \mathrm{var}_t(\Delta c_{t+1}), \quad \sigma_r^2 = \mathrm{var}_t(r_{t+1}), \quad \sigma_{cr} = \mathrm{cov}_t(\gamma \Delta c_{t+1}, r_{t+1}).$$

Using these results together with equation (8.36), equation (8.35) can be re-expressed as

$$\log \delta - \gamma \, \mathrm{E}_t[\Delta c_{t+1}] + \mathrm{E}_t[r_{t+1}] + \frac{1}{2}(\gamma^2 \sigma_c^2 + \sigma_r^2 - 2\gamma\sigma_{cr}) = 0 \,. \tag{8.39}$$

As it stands, equation (8.39) is of little help because it contains terms representing unobserved expectations. A common approach is to define the following expectations generating equations

$$\begin{aligned} r_{t+1} &= \mathrm{E}_t[r_{t+1}] + u_{1\,t+1} \\ \Delta c_{t+1} &= \mathrm{E}_t\,[\Delta c_{t+1}] + u_{2\,t+1} \,, \end{aligned}$$

in which u_{1t} and u_{2t} represent errors in forming conditional expectations. Using these expressions in (8.39) gives a linear regression model between the log returns of an asset and the growth rate in consumption

$$r_{t+1} = \beta_0 + \beta_1 \Delta c_{t+1} + v_{t+1}, \tag{8.40}$$

in which

$$\begin{aligned} \beta_0 &= -\log \delta - \frac{1}{2}(\gamma^2 \sigma_c^2 + \sigma_r^2 - 2\gamma\sigma_{cr}) \,, \\ \beta_1 &= \gamma \,, \\ v_{t+1} &= u_{1\,t+1} - \gamma u_{2\,t+1} \,. \end{aligned}$$

In this expression, the slope parameter of the regression equation is in fact the relative risk aversion coefficient, γ. The expression of the intercept term shows that β_0 is a function of a number of parameters including the relative risk aversion parameter γ, the discount rate δ, the variance of consumption growth σ_c^2, the variance of log asset returns σ_r^2 and the covariance between logarithm of asset returns and real consumption growth, σ_{cr}.
Note that v_{t+1} is a composite error term that comprises the rational forecasting errors on both r_{t+1} and Δc_{t+1}. In other words, a direct consequence of the theoretical model leading to equation (8.40) is that $\mathrm{cov}(v_{t+1}, \Delta c_{t+1}) \neq 0$. This means that the ordinary least squares estimate of β_1 in equation (8.40) and hence the estimate of the coefficient of risk aversion will be inconsistent.

As an illustration of the problems involved in estimating the risk aversion parameter in this framework, Table 8.1 provides estimates of γ based on equation (8.40) using the same dataset as that used by Ferson and Harvey (1992). Ferson, W.E. Campbell, R.H. The data are in quarterly observations on United States data from 1947:2 to 1987:4. Seasonally adjusted nondurables deflated by seasonally adjusted personal consumption deflators are used as the measure of consumption and real returns are nominal returns deflated by the price index corresponding to the consumption growth measure, Δc_t. Rates of return to to four portfolios considered, namely, value-weighted indices of the smallest, $d1$, and largest deciles, $d2$, of common stocks on the New York Stock Exchange, a long-term government bond, gb, and a long-term corporate bond, cb.

Table 8.1:

Estimating the risk aversion parameter using equation (8.40) using the data set of Ferson and Harvey (1992). Ferson, W.E. Campbell, R.H. The instruments used in the estimation are the lagged growth rate of real consumption, Δc_t, and the lagged return on a Treasury bill, r_t. R^2 refers to the coefficient of determination from the first stage regression and the F statistics refer to the tests of model significance also in the first stage regression.

	$\widehat{\gamma}$	p value	$\widehat{\gamma}^{IV}$ Inst: Δc_t	p value	$\widehat{\gamma}^{IV}$ Inst: Δc_t, r_t	p value
gb	-0.2362	0.963	206.48	0.980	27.908	0.430
cb	0.1429	0.783	260.52	0.980	27.491	0.668
$d1$	2.2861	0.029	290.62	0.980	30.519	0.667
$d10$	1.2879	0.021	286.79	0.980	24.693	0.664
p value	R^2	F	Robust F			
206.48	0.980	0.000	0.000	0.000		
260.52	0.980	0.000	0.000	0.000		
290.62	0.980	0.000	0.000	0.000		
286.79	0.980	0.000	0.000	0.000		

The estimates of γ obtained by using ordinary least squares are mainly positive as required by the underlying theory and the estimates are also statistically significant for $d1$ and $d10$. When instrumental variables estimation is used with lagged consumption growth, Δc_t, as the single instrument for Δc_{t+1}, the estimate blows up substantially. but all the estimates are not significantly different from zero. The problem is only slight better when the lagged Treasury Bill rate, r_t is added to the instrument list. The estimate of γ appears to be more realistic, but the p values indicate that the estimates are not significant.

The problem with this estimation procedure is that the endogenous regressor, Δc_{t+1}, is difficult to forecast using historical data and therefore the instru-

ments, Δc_t and r_t, are weak. For the single instrument case the R^2 and the F statistic (both the simple and robust forms) from the first stage regression all take the value 0.000, strongly indicative that there is a severe weak instrument problem. In this particular case, the critical value for the F statistic for a maximal size distortion relative to ordinary least squares of 20%, as tabulated by Stock and Yogo (2005) Stock, J.H. Yogo, M. is 19.93, which provides a graphic illustration of the scale of the problem. The values for these statistics in the two-instrument case are 0.002, 0.180 and 0.090 respectively, with a 20% maximal size critical value of 16.63. It appears that with this data set the use of instrumental variables to estimate the coefficient of relative risk aversion from the linearised Euler equation is to be avoided.

8.7 Endogeneity and Corporate Finance

Endogeneity is a particularly acute problem in empirical corporate finance, partly due to an absence of formal theory underpinning the models being estimated. Usually the applied researcher in empirical finance will have a prior view of the variables that may be interest as explanatory variables, but the regression specification, more often than not, is not derived from a specific theory. Take for example the question of whether or not family-owned firms perform better than other firms (Andersen and Reeb, 2003; Adams, Almeida and Ferreira, 2009). Anderson, R.C. Reeb, D.M. Adams, R. Almeida, H. Ferreira, D. On the one hand, there are a number of perfectly valid reasons why founding family firms may perform worse than publicly owned firms. Combining ownership and control allows shareholders to exchange profit for private rent or the extraction of private benefits from the firm. Executive management positions are limited to family and this constricts the talent pool from which management is drawn. One the other hand, there are also strong arguments in favour of the profitability of family-owned firms. Combining ownership and control enhances monitoring and control of the firm. Family firms may longer investment horizons which may enhance investment efficiency. So not only does theory provide no specific guidance on the sign of a family ownership variable in a regression model, more importantly theory is not clear on whether firms performance is driven by ownership structure or whether the ownership structure is as a consequence of the performance of the firm.
Some of the problems faced when examining this problem empirically will be illustrated using a subset of the data used in Adams, Almeida and Ferreira (2009). Adams, R. Almeida, H. Ferreira, D. The data consists of 2254 firm-year observations over the period 1992 to 1999. The logarithm of Tobin's Q is used as a measure of firm performance, which is to be explained in terms of the size of the firm, *assets*, the age of the firm, *age*, the volatility of the operating environment, *vol* as measured by the standard deviation of the previous 60 month returns and a dummy variable indicating whether or not the founder of the firm is also its chief executive officer, *CEO*.
Table 8.2 provides the relevant summary statistics stratified by whether or not

the firm is a family-owned firm. The evidence from Table 8.2 seems to support the hypothesis that family firms perform better than their public counterparts with mean $\log Q$ for the family owned firms being larger that that for the public companies.

Table 8.2:

Summary statistics for a subset of the data used in Adams, Almeida and Ferreira (2009) is their study on family firms. The data consists of 2254 firm-year observations over the period 1992 to1999.

	$\log Q$	$\log(assets)$	$\log(age)$	vol
Non-Family Firms				
Mean	0.560	8.748	3.941	0.279
Std. Dev.	0.431	1.098	0.814	0.103
Max.	2.772	12.912	4.990	1.052
Min.	−0.174	5.778	0.000	0.115
Family Firms				
Mean	0.749	8.078	2.772	0.390
Std. Dev.	0.577	0.879	0.747	0.121
Max.	2.953	10.271	4.304	0.810
Min.	−0.181	6.151	0.000	0.190
Total				
Mean	0.584	8.666	3.796	0.293
Std. Dev.	0.456	1.095	0.893	0.112
Max.	2.953	12.912	4.990	1.052
Min.	−0.181	5.778	0.000	0.115

In this application, all observations will be treated as independent draws from the distribution of all United States firms and the fact that there are repeated observations of the same firm in different years will be ignored. Dummy variables for each of the years will be included to account for time series differences in the performance of firms. The linear regression to be estimated is therefore

$$\log(Q_i) = \beta_0 + \beta_1 CEO_i + \beta_2 \log(assets_i) + \beta_3 \log(age_i) + \beta_4 vol_i + v_i \quad (8.41)$$

One of the points at issue here is whether or not the family firm variable, represented here by the binary variable, CEO, is endogenous or not. This may be tested using the Durbin-Wu-Hausman test outlined in Section 8.4 if at least one instrument for CEO can be found. The instrument suggested by Adams, Almeida and Ferreira (2009) Adams, R. Almeida, H. Ferreira, D. is the current age of the founder, $ageF$, regardless of whether the founder works for the company or not (if there are multiple founders, the average age is used). For simplicity the variable is measured only in 1994, but is used for the whole sample. The motivation for using this variable as an instrument stems from

the fact that the age of the founder is unlikely to be driven by firm performance and this then alleviates the endogeneity problem. A possible caveat is that founders age may very well be correlated with firm age, which could have direct effects on firm performance.
Note that in all the regressions required to implement the test, year dummies are used to capture different macroeconomic conditions in each of the years 1992 to 1998 but these dummies are omitted from the equations to economise on notation. The two regression equations required to implement the test of endogeneity are, respectively,

$$\begin{aligned} CEO_i &= \pi_0 + \pi_1 ageF_i + \pi_2 \log(assets_i) + \pi_3 \log(age_i) + \pi_4 vol_i + e_i \\ \log(Q_i) &= \beta_0 + \beta_1 CEO_i + \beta_2 \log(assets_i) + \beta_3 \log(age_i) + \beta_4 vol_i + \beta_5 \widehat{e}_i + v_i \,. \end{aligned}$$

An F test of the restriction that $\beta_5 = 0$ is $F(1, 2237) = 149.21$ with a p value of 0.000. There is therefore strong evidence of endogeneity and therefore the use of instrumental variables estimation is indicated. The regression of the potentially endogenous regressor on the instrument and the other explanatory variables simply ignores the fact that the dependent variable of this regression, CEO_i, is in fact a binary dependent variable. For the moment this problem is simply ignored but it will be returned to in Chapter **??** where problems of limited dependent variables are encountered where an adjustment to this simple instrumental variables estimation will be considered.
Table 8.3 reports the parameter estimates for equation (8.41) using both ordinary least squares regression and instrumental variables with *ageF* used as an instrument for *CEO*. A Breusch-Pagan test for heteroskedasticity using the fitted values of $\ln(Q_i)$ in the auxiliary regression yields a χ^2_4 test statistic of 77.68 with a p value of 0.000 and consequently corrected standard errors are reported.
Results are unequivocal. The *CEO* coefficient is significant and positive indicating that family owned firms perform significantly better than their public counterparts. The large increase in the size of the coefficient on this variable when using instrumental variables, 0.895 as opposed to 0.223, is strongly suggestive of an endogeneity bias in the ordinary least squares estimate. In this case it doesn't lead to rejection of the null hypothesis but the ordinary least squares results seriously understate the importance of this effect and provide a good illustration of the perils of endogeneity in empirical corporate finance.

8.8 Exercises

1. Risk Return Relationship

   pv.wf1, pv.dta, pv.xlsx

   The data are daily returns to the S&P 500 Index and a proxy for the conditional variance based on the VIX Index for the period 2 January 1990

Table 8.3:

Ordinary least squares and instrumental variables regressions of the logarithm of Tobin's Q, $\log(Q_i)$, on the explanatory variables shown. The potentially endogenous variable *CEO* is instrumented with the mean age of the founder in 1994, *ageF*, in the instrumental variables regression.

	OLS	OLS (Robust se.)	IV (Robust se.)
CEO	0.223	0.223	0.895
	(0.032)	(0.039)	(0.091)
$\log(assets)$	−0.026	−0.026	−0.017
	(0.009)	(0.009)	(0.009)
$\log(age)$	−0.036	−0.036	0.050
	(0.012)	(0.012)	(0.014)
vol	−0.816	−0.816	−1.186
	(0.099)	(0.094)	(0.115)
Constant	1.127	1.127	0.732
	(0.110)	(0.107)	(0.118)
N	2250	2250	2250

Standard errors in parentheses.

to 1 June 2012 ($T = 5652$). A simple statement of the intertemporal CAPM is

$$r_t = \alpha + \gamma \, \mathrm{E}_{t-1}(\sigma_t^2) + u_t \,,$$

in which $\alpha = 0$ and $\gamma > 0$ is hypothesised. Using an observable proxy, h_t, for $\mathrm{E}_{t-1}(\sigma_t^2)$ yields

$$\begin{aligned} r_t &= \alpha + \gamma h_t - \gamma \epsilon_t + u_t \\ &= \alpha + \gamma h_t + v_t \,, \end{aligned}$$

in which $v_t = -\gamma \epsilon_t + u_t$ is now a composite error term.

(a) Draw a scatter plot of the relationship between daily returns to the S&P 500 Index and the proxy for the conditional variance and thus reproduce Figure 8.1.

(b) Estimate the risk aversion parameter γ by ordinary least squares. Discuss the properties of the estimator.

(c) Now estimate γ by instrumental variables using instrumental variables using h_{t-1} as and instrument for h_t. Discuss the results.

(d) Test for the endogeneity of h_t in the original regression specification using the auxiliary regression approach. What do you conclude?

(e) Compute the instrumental variables estimator of γ in two stages by first constructing $\widehat{h}_t$ from a reduced from regression on its instrument and then substituting these fitted values into the structural equation. Compare the standard errors you obtain with those from part (c). Correct the standard error on γ obtained from the two stage procedure and hence reproduce the value obtained in (c).

---

2. Properties of the Instrumental Variables Estimator

This Monte Carlo experiment demonstrates the inconsistency of the ordinary least squares estimator and the efficacy of instrumental variables estimation for the errors in variables problem. Consider the simple linear model

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$
$$x_t = z_t + v_t,$$

with

$$\begin{bmatrix} u_t \\ x_t \end{bmatrix} \sim N \begin{bmatrix} \sigma_u^2 & \rho \\ \rho & \sigma_\epsilon^2 \end{bmatrix}.$$

(a) Simulate the model for $\rho = \{0.2, 0.5\}$ and $T = \{50, 100, 200, 400\}$ with $u_t \sim N(0,1)$, $z_t \sim N(0,1)$, $\beta_0 = 10$ and $\beta_1 = 2$. For each of 1000 repetitions of the simulation store the estimate of β_1 obtained by ordinary least squares estimator and instrumental variables using z_t as an instrument for x_t.

(b) Summarise the results for the ordinary least squares estimator. What do you conclude about the consistency of ordinary least squares in this problem.

(c) Repeat part (b) for the instrumental variables estimator. Discuss your results.

(d) Simulate the model just once with $T = 500$.

i. Estimate the model by instrumental variables.

ii. Estimate the model using the two-step least squares approach and compare the standard errors on the parameters to those obtained in (i). Explain why the two estimates of standard error are not the same.

---

3. Weak Instruments

   Consider the model

$$\begin{array}{rcll} y_t & = & \beta x_t + v_t & \\ x_t & = & \pi z_t + e_t, & \end{array} \qquad \left[\begin{array}{c} v_t \\ e_t \end{array} \right] \sim iid\, N\left(\left[\begin{array}{c} 0 \\ 0 \end{array} \right], \left[\begin{array}{cc} 1.00 & 0.99 \\ 0.99 & 1.00 \end{array} \right] \right).$$

   The sample size is $T = 5$ and 10000 replications are used to generate the sampling distribution of the estimator.

   (a) Generate the sampling distribution of the instrumental variables estimator for the parameter values $\beta = 0.0$ and $\pi = \{0.25, 0.5, 1.0\}$. Discuss the sampling properties of the instrumental variables estimator in each case.

   (b) Generate the sampling distribution of the instrumental variables estimator for the parameter values $\beta = 0$ and $\pi = 0$. Compare this sampling distribution to the three sampling distributions obtained in part (a). Also compute the sampling distribution of the ordinary least squares estimator for this case. Note that for this model the ordinary least squares estimator has the property (see Stock, Wright and Yogo, 2002) Stock, J.H. Wright, J.H. Yogo, M.

$$\text{plim}(\widehat{\beta}_{OLS}) = \frac{\sigma_{12}}{\sigma_{22}} = 0.99\,.$$

   (c) Repeat parts (a) and (b) for samples of size $T = \{50, 500\}$. Discuss whether the results in parts (a) and (b) are affected by asymptotic arguments.

---

4. Consumption CAPM

ccapm.wf1, ccapm.dta, ccapm.xlsx

   Linearising the Euler equation in the consumption CAPM model results in a a linear regression model between the log returns of a risky asset and the growth rate in consumption

$$r_{t+1} = \beta_0 + \beta_1 \Delta c_{t+1} + v_{t+1},$$

   in which

$$\begin{aligned} \beta_0 &= -\log\delta - \frac{1}{2}(\gamma^2\sigma_c^2 + \sigma_r^2 - 2\gamma\sigma_{cr}), \\ \beta_1 &= \gamma, \\ v_{t+1} &= u_{1\,t+1} - \gamma u_{2\,t+1}\,. \end{aligned}$$

   In this expression, the slope parameter of the regression equation is in fact the relative risk aversion coefficient, γ.

(a) Estimate γ by ordinary least squares. What is the problem with using this estimator to compute the parameter estimates of this model?

(b) Estimate γ by instrumental variables using

i. Δc_t as an instrument Δc_{t+1}; and

ii. Δc_t and r_t as instruments Δc_{t+1}.

Comment on the results.

(c) Perform a series of tests for weak instruments for the regressions in part (b). What do you conclude.

(d) Is it appropriate to implement the Anderson-Rubin test for robust inference on β_1 and hence γ in the structural equation.

---

5. Performance of Family Firms

familyfirms.wf1, familyfirms.dta, familyfirms.xlsx

The data consists of 2254 firm-year observations over the period 1992 to1999. The logarithm of Tobin's Q is used as a measure of firm performance, which is to be explained in terms of the size of the firm, *assets*, the age of the firm, *age*, the volatility of the operating environment, *vol* as measured by the standard deviation of the previous 60 month returns and a dummy variable indicating whether or not the founder of the firm is also its chief executive officer, *CEO*.

(a) Compute summary statistics for the variables in the data set stratified by whether or not the firm is a family-owned firm.

(b) In the linear regression given by

$$\log(Q_i) = \beta_0 + \beta_1 CEO_i + \beta_2 \log(assets_i) + \beta_3 \log(age_i) + \beta_4 vol_i + v_i,$$

test whether the family firm variable, represented here by the binary variable, *CEO*, is endogenous or not. Use the instrument *ageF*.

(c) Irrespective of your results in part (b) estimate the equation by ordinary least squares and instrumental variables (once again using *ageF* as the instrument for *CEO*) and compare your results.

Chapter 9

Generalised Method of Moments

9.1 Introduction

The generalised method of moments (GMM) estimator provides a powerful procedure that is applicable for estimating the parameters of a broad range of models in finance. This class of estimators also nests the least squares estimator widely used throughout Parts I and II of the book, as well as the instrumental variable estimator discussed in Chapter 8. The GMM estimator also nests the method of moments estimator which is discussed in this chapter and under certain conditions is related to the maximum likelihood estimator which is presented next in Chapter 10. Under general conditions the GMM estimator is consistent, satisfies an efficiency bound and is asymptotically normally distributed.

The main requirement for using the GMM estimator is where the model is expressed in terms of population moments, although it is not always necessary to know what is the exact form of the stochastic distribution that is characterised by these population moments. This requirement is satisfied by most, if not all models in finance, including the capital asset pricing model, models that are represented by the first order conditions of an intertemporal dynamic optimisation problem such as the consumption based capital asset pricing model, the Black-Scholes option price model, latent factor models, levels effects models of interest rates and models of trade durations. In implementing the GMM estimator it is appropriate to distinguish between financial models where analytical solutions of the GMM estimator are available and cases where numerical solutions are needed. The former case is dealt with initially while the latter case is discussed later on in the chapter.

9.2 Single Parameter Models

This section deals with financial models where there is a single unknown parameter that needs to be determined from the data. An important feature of the models considered is that they are characterised by one population moment and one unknown parameter. This matching of moments and parameters will be referred to as an exactly specified model.

9.2.1 Present Value Model

To motivate the GMM estimator consider the present value model where the price of an asset P_t is equated to the discounted flow of future dividend payments D_t, with discount rate δ

$$P_t = \mathrm{E}_t \sum_{i=1}^{\infty} \frac{D_{t+i}}{(1+\delta)^i}, \tag{9.1}$$

where E_t is the conditional expectations operator based on information available at time t.
Consider again the monthly data on United States equity prices and dividends used in Chapter 2. Figure 9.1 plots the equity price, dividend payments and the dividend yield, calculated as D_t/P_t, from January 1871 to June 2004, $T = 1602$. The aim is to estimate the unknown discount rate δ implied by the data on P_t and D_t.

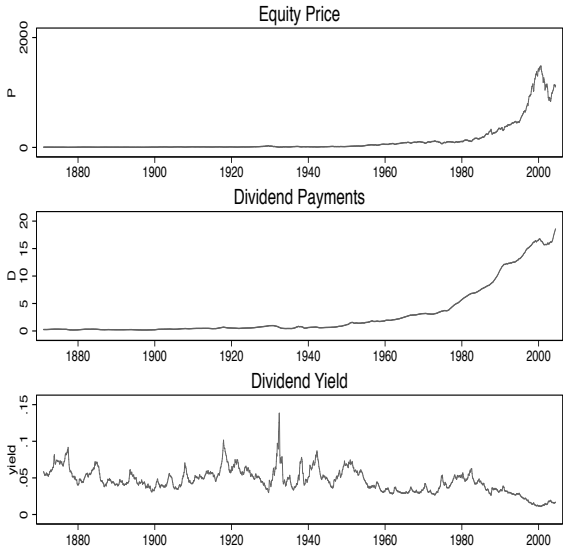


Figure 9.1: Time series plots of United States equity prices, dividend payments and the dividend yield for the period January 1871 to June 2004.

Adopting the approach of Chapter 2, dividend payments are assumed to follow a random walk. This implies that the conditional expectations of future dividend payments are then given by

$$\mathrm{E}_t\, D_{t+i} = D_t, \qquad i \geq 0. \tag{9.2}$$

Using the condition in (9.1) and rearranging, simplifies the present value relationship to

$$P_t = \frac{D_t}{\delta}, \tag{9.3}$$

or to

$$\frac{D_t}{P_t} = \delta. \tag{9.4}$$

This expression shows that the present value model is characterised by the equilibrium condition that the dividend-price ratio at time t, that is the dividend yield at time t, equals the discount parameter δ.
To derive the GMM estimator of $\theta = \{\delta\}$ equation (9.4) is re-expressed in terms of the GMM moment equation, m_t, according to

$$m_t = \frac{D_t}{P_t} - \theta. \tag{9.5}$$

A plot of m_t at each t with the unknown parameter chosen as $\theta = 0.05$ is given in Figure **??**. A comparison of Figure **??** and the dividend yield in Figure **??** clearly shows that the two time series display exactly the same fluctuations over time. The difference in the two series is caused by the shift in the level of m_t by setting $\theta = 0.05$, with the result that the dividend yield tends to fluctuate around 0.05 over the sample, whereas m_t tends to hover around zero.

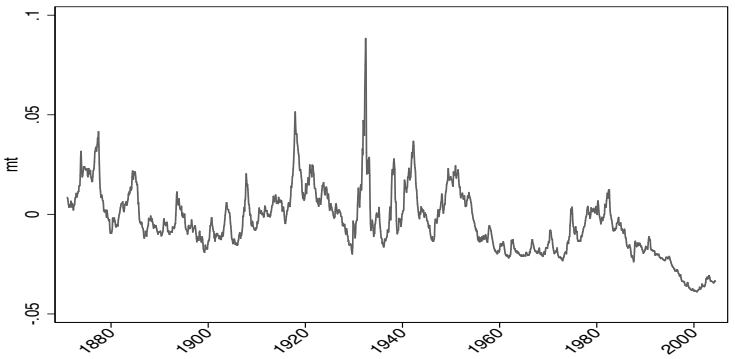


Figure 9.2: A time series plot of the GMM moment equation for the present value model in (9.5) with $\theta = 0.05$. The data are United States price and dividend data for the period January 1871 to June 2004.

To derive the GMM estimator of θ, consider taking the sample average of this moment equation

$$M(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t. \tag{9.6}$$

The GMM estimator $\widehat{\theta}$, is the solution of

$$M(\widehat{\theta}) = 0. \tag{9.7}$$

Using the moment expression for the present value model given by (9.5) in (9.7) shows that the GMM estimator is determined by solving

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{D_t}{P_t} - \widehat{\theta}\right) = 0. \tag{9.8}$$

Upon rearranging this expression, the GMM estimator is

$$\widehat{\theta} = \frac{1}{T}\sum_{t=1}^{T}\frac{D_t}{P_t}, \tag{9.9}$$

which is the sample mean of the dividend yields over the sample period. Using the United States price and dividend data for January 1871 to June 2004 gives a GMM estimate of the discount parameter of $\widehat{\theta} = \widehat{\delta} = 0.0460$, or 4.6% per annum.

9.2.2 Model of Trade Durations

As an alternative class of financial models consider the following model of durations y_t, based on the exponential distribution

$$f(y_t;\theta) = \frac{1}{\gamma}\exp\left(-\frac{y_t}{\gamma}\right), \tag{9.10}$$

where γ is an unknown parameter. Figure 9.3 gives the empirical distribution of the time between trades (in seconds) for American Airlines (AMR) on 1 August 2006 from 09:30 to 04:00, a total of $T = 23401$ observations. The empirical distribution provides strong support for the choice of an exponential distribution as it is positively skewed with a peak around zero.
A property of the exponential distribution in (9.10) is that the population mean of durations is

$$\mathrm{E}[y_t] = \gamma. \tag{9.11}$$

Letting $\theta = \{\gamma\}$ represent the unknown parameter, this suggests that the GMM moment equation is defined as

$$m_t = y_t - \theta. \tag{9.12}$$

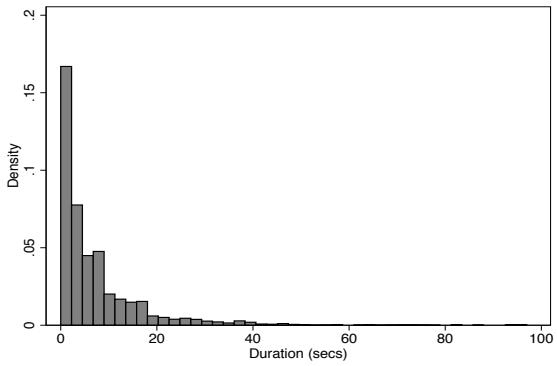


Figure 9.3: Time between trades (in seconds) for American Airlines (AMR) on 1 August 2006 from 09:30 to 04:00, $T = 23401$ observations.

In the case of the trade durations data in Figure **??**, this moment is plotted in Figure **??** for the case where $\theta = 10$.
To find the GMM estimator of θ the condition in (9.7) is used which shows that the estimator is determined by solving

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T}\left(y_t - \widehat{\theta}\right) = 0. \tag{9.13}$$

The GMM solution is the sample mean of the observed durations

$$\widehat{\theta} = \frac{1}{T}\sum_{t=1}^{T} y_t = \overline{y}. \tag{9.14}$$

For the trade durations in Figure **??** the GMM estimate is $\widehat{\theta} = \widehat{\gamma} = 7.2696$.

9.3 Multiple Parameter Models

The GMM framework is now extended to financial models with more than a single unknown parameter and where the model is represented by a set of moment conditions which just match the number of unknown parameters. Let the number of unknown parameters be K so θ is a $(K \times 1)$ vector θ. As there are K moments then m_t is now a $(K \times 1)$ vector given by

$$m_t = \begin{bmatrix} m_{1t} \\ m_{2t} \\ \vdots \\ m_{Kt} \end{bmatrix}, \tag{9.15}$$

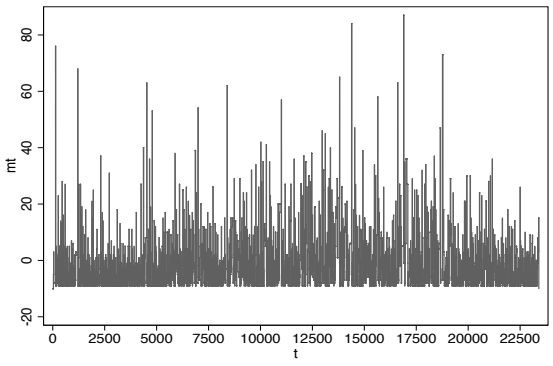


Figure 9.4: A time series plot of the GMM moment equation for the time between trades (in seconds) in equation (9.10) using $\theta = 10$. The data are for American Airlines (AMR) on 1 August 2006 from 09:30 to 04:00, $T = 23401$ observations.

where m_{it} represents the ith moment at time t. Correspondingly, the GMM condition in (9.7) is now represented as a $(K \times 1)$ vector which is again given here for convenience

$$M(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t, \tag{9.16}$$

with the difference that $M(\theta)$ is now a $(K \times 1)$ vector of sample moments. For this class of models the GMM estimator of θ, is obtained by solving

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} m_t(\widehat{\theta}) = 0. \tag{9.17}$$

for $\widehat{\theta}$ which represents a simultaneous set of K equations.

9.3.1 CAPM

Consider the capital asset pricing model where y_t is the excess return on an asset and x_t is the excess return on the market portfolio

$$y_t = \alpha + \beta x_t + u_t, \qquad u_t \sim iid(0, \sigma^2)\,. \tag{9.18}$$

The unknown parameters are

$$\theta = \{\alpha, \beta, \sigma\}, \tag{9.19}$$

where α is a measure of abnormal returns, β represents the beta-risk of the asset and σ is the disturbance standard deviation that measures the idiosyncratic risk of the asset. From Chapter 3, the CAPM is shown to be a special

case of the linear regression model which can be estimated by ordinary least squares. For the ordinary least squares estimator to have desirable properties the linear regression model needs to satisfy the following population moment conditions

$$\mathrm{E}(u_t) = 0, \qquad \mathrm{E}(u_t x_t) = 0, \qquad \mathrm{E}(u_t^2) = \sigma^2. \tag{9.20}$$

The first condition is that the mean of the idiosyncratic term u_t, is zero. The second condition is that the excess return on the market given by x_t, needs to be uncorrelated with the idiosyncratic risk term u_t. The third and final condition is that the variance of the idiosyncratic risk is constant and equals σ^2. The three population moments in (9.20) suggest the following GMM moment equations

$$\begin{aligned} m_{1t} &= u_t - 0 \\ m_{2t} &= u_t x_t - 0 \\ m_{3t} &= u_t^2 - \sigma^2. \end{aligned} \tag{9.21}$$

As $u_t = y_t - \alpha - \beta x_t$ from (9.18), the vector of moments m_t in (9.15) is the following (3×1) vector

$$m_t = \begin{bmatrix} m_{1,t} \\ m_{2,t} \\ m_{3,t} \end{bmatrix} = \begin{bmatrix} y_t - \alpha - \beta x_t \\ (y_t - \alpha - \beta x_t)x_t \\ (y_t - \alpha - \beta x_t)^2 - \sigma^2 \end{bmatrix}. \tag{9.22}$$

Using the GMM moment condition in (9.17) the GMM estimator $\widehat{\theta} = \{\widehat{\beta}_0, \widehat{\beta}_1, \widehat{\sigma}^2\}$, is obtained by solving the following (3×1) system of equations

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} y_t - \widehat{\alpha} - \widehat{\beta} x_t \\ (y_t - \widehat{\alpha} - \widehat{\beta} x_t)x_t \\ (y_t - \widehat{\alpha} - \widehat{\beta} x_t)^2 - \widehat{\sigma}^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \tag{9.23}$$

The GMM estimators are

$$\begin{aligned} \widehat{\beta} &= \frac{\sum_{t=1}^{T}(x_t - \overline{x})(y_t - \overline{y})}{\sum_{t=1}^{T}(x_t - \overline{x})^2} \\ \widehat{\alpha} &= \overline{y} - \widehat{\beta}\overline{x} \\ \widehat{\sigma}^2 &= \frac{1}{T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta} x_t)^2, \end{aligned} \tag{9.24}$$

which are equivalent to the solutions obtained for the ordinary least squares estimator of the CAPM in Chapter 3. This is an important result as it shows that not just for the CAPM, but for all linear regression models satisfying the population moment properties in (9.20), the GMM and ordinary least squares estimators are equivalent.

Data on the excess returns for Exxon and the market are given in Figure 9.5 for the period May 1990 to July 2004. The GMM estimates are computed using (9.24) with the result that

$$\widehat{\theta} = \{\widehat{\alpha}, \widehat{\beta}, \widehat{\sigma}\} = \{\widehat{\alpha}, \widehat{\beta}, \widehat{\sigma}\} = \{0.012, 0.502, 0.038\}. \tag{9.25}$$

The parameter estimates for α and β are identical to the estimates reported in Chapter 3. The estimate of σ is numerically very similar, with the difference being that the GMM estimate is not employ the degrees of freedom adjustment that is used in ordinary least squares.

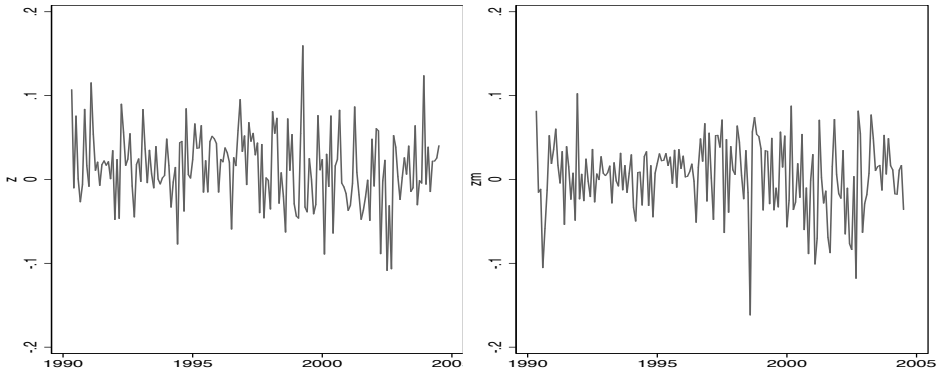


Figure 9.5: Monthly excess returns on Exxon and the S&P 500 Market Index for the period May 1990 to July 2004.

The three GMM moment equations in (9.22), evaluated at the GMM parameter estimates $\widehat{\theta}$ in (9.25), are plotted in Figure **??**. The sample means of $m_{1,t}, m_{2t}, m_{3t}$ are by construction

$$\overline{m}_1 = \frac{1}{T}\sum_{t=1}^{T} m_{1t}(\widehat{\theta}) = 0, \qquad \overline{m}_2 = \frac{1}{T}\sum_{t=1}^{T} m_{2t}(\widehat{\theta}) = 0, \qquad \overline{m}_3 = \frac{1}{T}\sum_{t=1}^{T} m_{3t}(\widehat{\theta}) = 0.$$

Moreover, the covariance matrix of all three series are

$$\text{cov}(m_t) = \frac{1}{T}\sum_{t=1}^{T} m_t m_t' = \begin{bmatrix} 0.001456 & 0 & 0 \\ 0 & 3.15 \times 10^{-6} & 0 \\ 0 & 0 & 5.22 \times 10^{-6} \end{bmatrix}$$

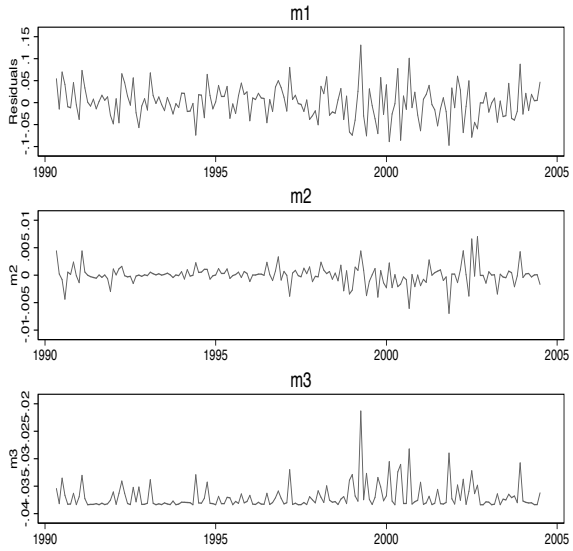


Figure 9.6: Time series plots of the three GMM moment conditions for the CAPM in equation (9.22) using monthly excess returns on Exxon and the S&P 500 Market Index for the period May 1990 to July 2004. The moment conditions are evaluated at the GMM parameter estimates $\widehat{\theta}$.

The bivariate CAPM can also be extended to allow for multiple factors as was done in Chapter 3 in which case the model is represented as a multiple linear regression model. GMM and ordinary least squares are equivalent for a bivariate linear regression model and this equivalence carries over to the multiple linear regression model.

9.3.2 A Gamma Model of Asset Prices

Table 9.1 provides annual average prices for General Electric over the period 1991 to 2003, $T = 13$ observations. Also given are the first and second uncentered moments of P_t, the first centered moment $P_t - \overline{P}$, and the inverse moment $1/P_t$. As the price of General Electric shares is constrained to be positive, the specficied price model is chosen to be the gamma distribution

$$f(P_t;\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} P_t^{\alpha-1} \exp(-\beta P_t), \qquad P_t > 0, \tag{9.26}$$

with $\theta = \{\alpha, \beta\}$ as the unknown parameters which satisfy the properties $\alpha, \beta > 0$. The first two population moments of the gamma distribution are

$$\mathrm{E}[P_t] = \frac{\alpha}{\beta}, \qquad \mathrm{E}[P_t^2] = \frac{\alpha(\alpha+1)}{\beta^2}. \tag{9.27}$$

This suggests that the GMM estimator is based on the following GMM moment equations

$$\begin{aligned} m_{1t} &= P_t - \frac{\alpha}{\beta} \\ m_{2t} &= P_t^2 - \frac{\alpha(\alpha+1)}{\beta^2}. \end{aligned} \tag{9.28}$$

Upon using (9.17) the GMM estimator $\widehat{\theta} = \{\widehat{\alpha}, \widehat{\beta}\}$, is the solution of

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} P_t - \dfrac{\widehat{\alpha}}{\widehat{\beta}} \\ P_t^2 - \dfrac{\widehat{\alpha}(\widehat{\alpha}+1)}{\widehat{\beta}^2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{9.29}$$

Upon solving for $\widehat{\theta}$ yields the following GMM estimators

$$\widehat{\alpha} = \frac{\left(T^{-1}\sum_{t=1}^{T} P_t\right)^2}{T^{-1}\sum_{t=1}^{T} P_t^2 - \left(T^{-1}\sum_{t=1}^{T} P_t\right)^2}, \quad \widehat{\beta} = \frac{T^{-1}\sum_{t=1}^{T} P_t}{T^{-1}\sum_{t=1}^{T} P_t^2 - \left(T^{-1}\sum_{t=1}^{T} P_t\right)^2}. \tag{9.30}$$

Using the data in Table 9.1 the GMM estimates are

$$\widehat{\alpha} = \frac{18.97^2}{591.2467 - 18.97^2} = 1.5552, \qquad \widehat{\beta} = \frac{18.97}{591.2467 - 18.97^2} = 0.0820\,. \tag{9.31}$$

As $\widehat{\alpha} = 1.5552 > 1$, this implies that the price distribution is hump-shaped with positive skewness.
The population moments of the gamma distribution given in (9.27) are not the only moments of the gamma distribution. Another population moment for example, is

$$\mathrm{E}\left[\frac{1}{P_t}\right] - \frac{\beta}{\alpha - 1} = 0\,. \tag{9.32}$$

In which case if this population moment is used with the first population moment in (9.27) then the GMM estimator is now based on the following two moments

$$\begin{aligned} m_{1t} &= P_t - \frac{\alpha}{\beta} \\ m_{2t} &= \frac{1}{P_t} - \frac{\beta}{\alpha - 1}. \end{aligned} \tag{9.33}$$

The GMM estimator $\widehat{\theta} = \{\widehat{\alpha}, \widehat{\beta}\}$, is obtained by using (9.17) to solve

$$M(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} P_t - \dfrac{\widehat{\alpha}}{\widehat{\beta}} \\ \dfrac{1}{P_t} - \dfrac{\widehat{\beta}}{\widehat{\alpha} - 1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{9.34}$$

Table 9.1

Average annual stock price of General Electric, P_t: 1991 to 2003, $T = 13$.

Year	P_t	P_t^2	P_t^{-1}
1991	1.12	1.2544	0.8928
1992	1.76	3.0976	0.5681
1993	3.07	9.4249	0.3257
1994	4.30	18.4900	0.2325
1995	6.29	39.5641	0.1589
1996	10.52	110.6704	0.0950
1997	16.88	284.9344	0.0592
1998	24.44	597.3136	0.0409
1999	34.55	1193.7020	0.0289
2000	47.27	2234.4530	0.0211
2002	40.03	1602.4010	0.0249
2002	29.07	845.0649	0.0344
2003	27.31	745.8361	0.0366
Average:	18.97	591.2467	0.1938

which yield the solutions

$$\widehat{\alpha} = \frac{T^{-1}\sum_{t=1}^{T} P_t}{T^{-1}\sum_{t=1}^{T} P_t - \left(T^{-1}\sum_{t=1}^{T} 1/P_t\right)^{-1}}, \qquad \widehat{\beta} = \frac{\widehat{\alpha}}{T^{-1}\sum_{t=1}^{T} P_t}. \tag{9.35}$$

Using the data in Table 9.1 yields the method of moments estimates

$$\widehat{\alpha} = \frac{18.97}{18.97 - 0.1938^{-1}} = 1.3736, \qquad \widehat{\beta} = \frac{1.3736}{18.97} = 0.0724.$$

A comparison of the gamma distribution parameter estimates in (9.31) and (9.35) shows that the estimates are not unique. In fact, a third set of GMM estimates could be obtained by using another combination of two population moments. By choosing different moments to estimate the unknown parameters results in different estimates. A natural problem that immediately arises is in determining which of the two sets of estimators are the more reliable from the point of view of estimating the population parameters of θ. Part of the answer comes from identifying which of the moments m_t, are relatively more important in modelling the price distribution. The solution to this problem is not just to choose a subset of m_t to derive the GMM estimator, but rather to use all of them. However, to reflect the relative importance of different moments it is necessary to weight each in terms of its precision in estimating the parameters of the price distribution.

9.4 Over-Identified Models

The GMM estimators discussed so far are obtained using a system of equations where the number of moment equations of the model perfectly matches the number of unknown parameters. These models are referred to as just-identified models. However, in the case of the gamma asset price model discussed immediately above where there were two unknown parameters, three population moments were identified which produced different estimators depending upon the choices of population moments. Whilst within each set of moments the models were just identified, by combining all moments together would result in a system where there were more equations than unknown parameters. Models having this property are referred to as over-identified models.[1] This section is devoted to generalising the GMM estimation procedure for just-identified models based on (9.17) to over-identified models where there are N moment equations and K unknown parameters so the degree of over-identification is $N - K > 0$.

In estimating the unknown parameters of over-identified models by GMM the trick is to use all of the moments when deriving the GMM estimator. Formally this requires weighting each of the moments in terms of their relative precision, with the moments having the greatest precision given the greatest influence when computing the GMM estimates. To motivate the approach reconsider the trade durations model in Section 9.2.2 where there are now $N = 2$ population moments given respectively by the mean and the variance

$$\mathrm{E}[y_t] = \gamma, \qquad \mathrm{var}(y_t) = \gamma^2, \tag{9.36}$$

with a single unknown parameter γ. Letting $\theta = \{\gamma\}$ be the $K = 1$ unknown parameter, the corresponding moments are

$$\begin{array}{rcl} m_{1t} & = & y_t - \theta \\ m_{2t} & = & (y_t - \theta)^2 - \theta^2, \end{array} \tag{9.37}$$

resulting in the GMM moment condition

$$M(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t = \frac{1}{T}\sum_{t=1}^{T} \left[\begin{array}{c} y_t - \theta \\ (y_t - \theta)^2 - \theta^2 \end{array} \right] = \left[\begin{array}{c} \overline{y} - \theta \\ s^2(\theta) - \theta^2 \end{array} \right], \tag{9.38}$$

where

$$\overline{y} = \frac{1}{T}\sum_{t=1}^{T} y_t, \qquad s^2(\theta) = \frac{1}{T}\sum_{t=1}^{T} (y_t - \theta)^2, \tag{9.39}$$

are respectively the sample mean and the sample variance of y_t for an unknown θ. All of the GMM estimators discussed so far require finding $\widehat{\theta}$ by

[1] Models where the number of moment equations is less than the number of parameters are referred to as under-identified models. In these case there is insufficient information to be able to identify all of the unknown parameters. In this chapter the focus is on financial models that are either just-identified or over-identified.

solving (9.17). For the trade durations model this involves solving

$$\begin{bmatrix} \overline{y} - \widehat{\theta} \\ s^2 - \widehat{\theta}^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{9.40}$$

simultaneously. Even if the true distribution is exponential so the population moments in (9.36) are indeed the appropriate moments, from a sampling point of view the conditions in (9.39) such that $\widehat{\theta} = \overline{y} = s$ hold jointly is unlikely to be satisfied.
The GMM solution of the over-identified model in (9.40) is not to view the two moment conditions separately, but to combine these moments into a single condition resulting in an exactly identified system with $N = 1$ and $K = 1$. The method of combining moment equations formally involves weighting each moment.

9.4.1 Choosing the Weights

Each moment in m_t will have some role to play in identifying the unknown parameters θ. However, some moments will be relatively more important than other moments as they will tend to reveal greater features of the data thereby yielding greater precision in identifying the underlying parameters of the model. A natural approach to assign weights representing the relative importance of each moment is to use variances of each element of m_t: moments with smaller variances display greater precision and thus are assigned higher weight in computing the GMM estimator.

Diagonal Weights

Reconsider the present value model in Section 9.2.1 where from equation (9.5) the GMM moment equation is

$$m_t = \frac{D_t}{P_t} - \theta. \tag{9.41}$$

As the sample mean of m_t evaluated at the GMM estimator $\widehat{\theta}$ is zero from the GMM condition in (9.17), the variance of this moment simplifies as

$$\text{var}(m_t) = \frac{1}{T}\sum_{t=1}^{T} m_t^2 = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{D_t}{P_t} - \widehat{\theta}\right)^2. \tag{9.42}$$

Using the price-dividend ratio data in Figure **??** and the GMM estimate of the discount parameter of $\widehat{\delta} = \widehat{\theta} = 0.0460$, the variance of this moment is computed as

$$\text{var}(m_t) = 0.000255.$$

For the CAPM in Section 9.3.1, there are three parameters $\theta = \{\alpha, \beta, \sigma^2\}$ with respective moments

$$m_t = \begin{bmatrix} m_{1,t} \\ m_{2,t} \\ m_{3,t} \end{bmatrix} = \begin{bmatrix} y_t - \alpha - \beta x_t \\ (y_t - \alpha - \beta x_t)x_t \\ (y_t - \alpha - \beta x_t)^2 - \sigma^2 \end{bmatrix}, \tag{9.43}$$

where y_t is the monthly excess return on Exxon and x_t is the monthly excess return on the market based on the S&P500 stock index. Using the GMM parameter estimates of $\widehat{\theta}$ given in (9.25), the sample variances of the GMM moment equations are

$$\begin{aligned} \text{var}(m_{1t}) &= \frac{1}{T}\sum_{t=1}^{T} m_{1t}^2 = \frac{1}{T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta}x_t)^2 = 0.001456 \\ \text{var}(m_{2t}) &= \frac{1}{T}\sum_{t=1}^{T} m_{2t}^2 = \frac{1}{T}\sum_{t=1}^{T}((y_t - \widehat{\alpha} - \widehat{\beta}x_t)x_t)^2 = 3.15 \times 10^{-6} \\ \text{var}(m_{3t}) &= \frac{1}{T}\sum_{t=1}^{T} m_{3t}^2 = \frac{1}{T}\sum_{t=1}^{T}((y_t - \widehat{\alpha} - \widehat{\beta}x_t)^2 - \widehat{\sigma}^2)^2 = 5.22 \times 10^{-6}. \end{aligned} \tag{9.44}$$

Letting $W(\theta)$ represent the weighting matrix corresponding to the N GMM moments in m_t based on the unknown parameter vector θ, the diagonal form of this matrix is

$$W(\theta) = \begin{bmatrix} \text{var}(m_{1t}) & 0 & \cdots & 0 \\ 0 & \text{var}(m_{2t}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{var}(m_{Nt}) \end{bmatrix}. \tag{9.45}$$

Evaluating this matrix at the GMM estimator $\widehat{\theta}$, gives

$$W(\widehat{\theta}) = \begin{bmatrix} \text{var}(m_{1t}) & 0 & \cdots & 0 \\ 0 & \text{var}(m_{2t}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{var}(m_{Kt}) \end{bmatrix}. \tag{9.46}$$

Using the variance estimates in (9.44) for the CAPM model, $W(\widehat{\theta})$ is computed as

$$W(\widehat{\theta}) = \begin{bmatrix} 0.001456 & 0 & 0 \\ 0 & 3.15 \times 10^{-6} & 0 \\ 0 & 0 & 5.22 \times 10^{-6} \end{bmatrix}.$$

Covariance Weights: Heteroskedasticity

The diagonal form of the weighting matrix in (9.45) focusses just on the variances, but in general there are covariances amongst the moments which can

also be taken into account in weighting moments. For this more general case the weighting matrix is now defined as

$$\begin{aligned} W(\theta) &= \begin{bmatrix} \text{var}(m_{1t}) & \text{cov}(m_{1t}, m_{2t}) & \cdots & \text{cov}(m_{1t}, m_{Nt}) \\ \text{cov}(m_{2t}, m_{1t}) & \text{var}(m_{2t}) & \cdots & \text{cov}(m_{2t}, m_{Nt}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(m_{Nt}, m_{1t}) & \text{cov}(m_{Nt}, m_{2t}) & \cdots & \text{var}(m_{Nt}) \end{bmatrix} \\ &= \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} m_{1t}^2 & m_{1t}m_{2t} & \cdots & m_{1t}m_{Kt} \\ m_{2t}m_{1t} & m_{2t}^2 & \cdots & m_{2t}m_{Kt} \\ \vdots & \vdots & \ddots & \vdots \\ m_{Kt}m_{1t} & m_{Kt}m_{2t} & \cdots & m_{Kt}^2 \end{bmatrix} \\ &= \frac{1}{T}\sum_{t=1}^{T} m_t m_t'. \end{aligned} \tag{9.47}$$

In the case of the gamma asset price model, the weighting matrix evaluated at the GMM parameter estimates is computed as the (3×3) matrix

$$W(\widehat{\theta}) = \begin{bmatrix} 0.001456 & 6.47 \times 10^{-6} & 9.38 \times 10^{-6} \\ 6.47 \times 10^{-6} & 3.15 \times 10^{-6} & 3.65 \times 10^{-8} \\ 9.38 \times 10^{-6} & 3.65 \times 10^{-8} & 5.22 \times 10^{-6} \end{bmatrix}.$$

Covariance Weights: Heteroskedasticity and Autocorrelation

The form of the weighting matrix in (9.47) allows for the moments to be related to each other contemporaneously but not over time. To allow for autocorrelation in m_t, the weighting matrix is defined as

$$W(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t m_t' + \sum_{i=1}^{P} w_i \left(\frac{1}{T}\sum_{t=i+1}^{T} \left(m_t m_{t-i}' + m_{t-i} m_t' \right) \right). \tag{9.48}$$

The first term on the right hand-side represents the heteroskedastic weighting matrix given in (9.47). The second term allows for autocorrelation of length P. The w_i terms represent weights which control the contribution of autocorrelation at each lag resulting in a positive definite weighting matrix $W(\theta)$. A common choice of the weights is

$$w_i = 1 - \frac{i}{P+1}, \qquad i = 1, 2, \cdots, P, \tag{9.49}$$

which are knowns as Newey-West weights, although other choices exist. These weights have the effect of dampening the contribution of autocorrelation on $W(\theta)$ from longer lags. In the special case where there is no autocorrelation

$$w_1 = w_2 = \cdots = w_P = 0, \tag{9.50}$$

the weighting matrix in (9.48) resorts to (9.47).

An an example, if there is autocorrelation of order 2, so $P = 2$, the weighting matrix is

$$\begin{aligned} W(\theta) &= \frac{1}{T}\sum_{t=1}^{T} m_t m_t' + w_1\left(\frac{1}{T}\sum_{t=2}^{T}\left(m_t m_{t-1}' + m_{t-1}m_t'\right)\right) \\ &\quad + w_2\left(\frac{1}{T}\sum_{t=3}^{T}\left(m_t m_{t-2}' + m_{t-2}m_t'\right)\right), \end{aligned} \tag{9.51}$$

while the weights are chosen as

$$w_1 = 1 - \frac{1}{3} = \frac{2}{3}, \qquad w_2 = 1 - \frac{2}{3} = \frac{1}{3}.$$

The choice of the maximum autocorrelation lag P, can be determined by theory in some cases where the value of P follows directly from the properties of the model. An example of this is in models of forward market efficiency where an overlapping data problem arises where the expectations horizon is of a different frequency to the data (see Martin, Hurn and Harris, 2013). If there are no necessary guidelines from the theoretical model, an alternative approach based on more statistical methods is to choose P according to the rule

$$P = \text{int}\left[4\left(\frac{T}{100}\right)^{2/9}\right], \tag{9.52}$$

where $[\cdot]$ denotes the smallest integer of the term in brackets. This rule is based on asymptotic arguments and has been found to perform well in Monte Carlo simulations.

9.4.2 Objective Function

To circumvent the problem of over-identification while at the same time recognising the need for weights to determine the relative importance of each moment, the strategy is to derive the GMM estimator as the solution of the following quadratic equation

$$\widehat{\theta} = \arg\min_{\theta} Q(\theta) = M(\theta)' W^{-1} M(\theta)\,, \tag{9.53}$$

where θ is a $(K \times 1)$ vector of unknown parameters, $M(\theta)$ is a $(N \times 1)$ vector of sample moments given by

$$M(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t, \tag{9.54}$$

and $W(\theta)$ is a $(N \times N)$ weighting matrix that measures the precision of each moment in m_t. To derive the GMM estimator the first order condition is obtained by using the properties of matrix differentiation

$$G(\theta) = \frac{\partial Q}{\partial \theta} = 2D(\theta)' W^{-1} M(\theta), \tag{9.55}$$

where

$$D(\theta) = \frac{\partial M(\theta)}{\partial \theta'}, \tag{9.56}$$

is a $(N \times K)$ matrix of derivatives of the GMM moment function $M(\theta)$. The GMM estimator $\widehat{\theta}$ is given by setting the derivative in (9.55) to zero at $\theta = \widehat{\theta}$, which after simplifying is obtained as the solution of

$$D(\widehat{\theta})' W^{-1} M(\widehat{\theta}) = 0. \tag{9.57}$$

Equation (9.57) represents in general, a nonlinear set of K equations in $\widehat{\theta}$ which needs numerical methods to compute the GMM estimator: the dimension of this system is K as $D(\widehat{\theta})'$ is a $(K \times N)$ which combines with the $(N \times N)$ weighting matrix W, and then, in turn, with the $(N \times 1)$ vector $M(\widehat{\theta})$ to yield a $(K \times 1)$ vector which just matches the number of unknown parameters in θ. This contrasts with the over-identified system of equations in (9.54) where $M(\widehat{\theta})$ is a $(N \times 1)$ vector containing K unknown parameters. The weighting matrix $W(\theta)$ plays the role of reducing the over-identified system where there are N equations and K unknown parameters, to a just-identified system of K equations and K unknowns. In essence this is achieved by constructing a new set of K equations from the initial $N > K$ equations from weighted sums of all of the full set of GMM equations in m_t.
To see this formally, consider the trade durations model where the sample moments are given by (9.39). To derive the GMM estimator using (9.57), the following terms are needed

$$M(\widehat{\theta}) = \begin{bmatrix} \overline{m}_1 \\ \overline{m}_2 \end{bmatrix} = \begin{bmatrix} \overline{y} - \widehat{\theta} \\ s^2(\widehat{\theta}) - \widehat{\theta}^2 \end{bmatrix}, \tag{9.58}$$

and

$$D(\theta) = \frac{\partial M(\theta)}{\partial \theta'} = \frac{\partial}{\partial \theta'} \begin{bmatrix} \overline{y} - \theta \\ s^2(\theta) - \theta^2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2(\overline{y} - \theta) - 2\theta \end{bmatrix} = \begin{bmatrix} -1 \\ -2\overline{y} \end{bmatrix}, \tag{9.59}$$

as $\partial s^2(\theta)/\partial \theta = -2(\overline{y} - \theta)$ which evaluated at $\widehat{\theta}$ becomes

$$D(\widehat{\theta}) = \begin{bmatrix} -1 \\ -2\overline{y} \end{bmatrix}. \tag{9.60}$$

Substituting (9.58) and (9.60) in (9.57) and for simplicity using a diagonal weighting matrix, the GMM estimator is the solution of

$$\begin{aligned} D(\widehat{\theta})' W^{-1} M(\widehat{\theta}) &= \begin{bmatrix} -1 \\ -2\overline{y} \end{bmatrix}' \begin{bmatrix} \text{var}(m_{1t}) & 0 \\ 0 & \text{var}(m_{2t}) \end{bmatrix}^{-1} \begin{bmatrix} \overline{y} - \widehat{\theta} \\ s^2 - \widehat{\theta}^2 \end{bmatrix} \\ &= -\frac{(\overline{y} - \widehat{\theta})}{\text{var}(m_{1t})} - \frac{2\overline{y}(s^2 - \widehat{\theta}^2)}{\text{var}(m_{2t})} = 0. \end{aligned} \tag{9.61}$$

This expression shows that the GMM criterion compresses the two moment conditions into a single equation by taking a weighted average of these two moment conditions with the weights based on the weighting matrix. In effect, the over-identified model of two equations and one unknown parameter is converted into a just-identified model with one equation and one unknown parameter by using the variances of the respective moments m_{1t} and m_{2t}, as the weights.
Explicit expressions of the weights in (9.61) as given by $\text{var}(m_{1t})$ and $\text{var}(m_{2t})$, are obtained for the durations model by evaluating the variances at the GMM estimator $\widehat{\theta}$. The resulting expressions are

$$\begin{aligned} \text{var}(m_{1t}) &= \frac{1}{T}\sum_{t=1}^{T} m_{1t}^2 = \frac{1}{T}\sum_{t=1}^{T}\left(y_t - \widehat{\theta}\right)^2 = s^2 \\ \text{var}(m_{2t}) &= \frac{1}{T}\sum_{t=1}^{T} m_{2t}^2 = \frac{1}{T}\sum_{t=1}^{T}\left((y_t - \widehat{\theta})^2 - \widehat{\theta}^2\right)^2. \end{aligned} \tag{9.62}$$

The first weight represents the sample variance of y_t evaluated at $\widehat{\theta}$, while the second weight represents the fourth moment or the kurtosis coefficient of y_t, again evaluated at $\widehat{\theta}$. Upon inspection of (9.61) shows that the smaller is this sample variance the greater is the weight placed on the first moment m_{1t} relative to the second moment m_{2t}.

9.4.3 Estimation

To compute the GMM estimates of an over-identified model, in general it is necessary to use an iterative algorithm as analytical solutions are invariably not available. In fact, the estimation procedures suggested in this section are also appropriate for just-identified models especially where the moment equations are nonlinear functions of the unknown parameters in θ.
To gain insight into the features of an iterative algorithm to compute GMM parameter estimates, consider the weighted moment equation in (9.61) corresponding to the trade durations model. Let the starting parameter value be

$$\theta_{(0)} = \gamma_{(0)} = 0.0.$$

Using this value in the GMM moment equations in (9.37) gives

$$\begin{aligned} m_{1t} &= y_t - \theta_{(0)} = y_t \\ m_{2t} &= (y_t - \theta_{(0)})^2 - \theta_{(0)}^2 = y_t^2 \end{aligned}$$

From (9.37) with $\theta = \theta_{(0)} = 0$, gives

$$M(\theta_{(0)}) = \begin{bmatrix} \overline{m}_1 \\ \overline{m}_2 \end{bmatrix} = \frac{1}{T}\sum_{t=1}^{T}\begin{bmatrix} y_t - \theta_{(0)} \\ (y_t - \theta_{(0)})^2 - \theta_{(0)}^2 \end{bmatrix} = \begin{bmatrix} 7.270 \\ 138.308 \end{bmatrix}.$$

Also from (9.62)

$$\begin{aligned}\text{var}(m_{1t}) &= \frac{1}{T}\sum_{t=1}^{T}\left(y_t-\theta_{(0)}\right)^2 = \frac{1}{T}\sum_{t=1}^{T}y_t^2 = 138.308\\ \text{var}(m_{2t}) &= \frac{1}{T}\sum_{t=1}^{T}\left(\left(y_t-\theta_{(0)}\right)^2-\theta_{(0)}^2\right) = \frac{1}{T}\sum_{t=1}^{T}y_t^2 = 246,632.091,\end{aligned}$$

so

$$W(\theta_{(0)}) = \begin{bmatrix} \text{var}(m_{1t}) & 0 \\ 0 & \text{var}(m_{2t}) \end{bmatrix} = \begin{bmatrix} 138.308 & 0 \\ 0 & 246,632.091 \end{bmatrix}.$$

Finally, from (9.60)

$$D(\theta_{(0)}) = \begin{bmatrix} -1 \\ -2\overline{y} \end{bmatrix} = \begin{bmatrix} -1 \\ -2\times 7.280 \end{bmatrix} = \begin{bmatrix} -1 \\ -14.560 \end{bmatrix}.$$

Using all of these tersm in the gradient function (9.55) gives

$$\begin{aligned}G(\theta_{(0)}) &= 2D(\theta_{(0)})'W^{-1}M(\theta_{(0)})\\ &= 2\begin{bmatrix} -1 \\ -14.540 \end{bmatrix}'\begin{bmatrix} 138.308 & 0 \\ 0 & 246,632.091 \end{bmatrix}^{-1}\begin{bmatrix} 7.270 \\ 138.308 \end{bmatrix}\\ &= -0.121.\end{aligned}$$

Since the gradient is negative, the initial guess of $\theta_{(0)} = 0$ is below the GMM estimate $\widehat{\theta}$ that minimises the objective function (9.53). Table 9.2 summarises the calculations for $\theta_{(0)} = 0$ as well as giving the results for increasing values of θ in steps of unity.

By increasing the parameter value in the next step to $\theta_{(1)} = 1$, shows that the gradient is now smaller suggesting that the updated value is closer to $\widehat{\theta}$. Increasing the values in the grid of θ reveals that $\widehat{\theta}$ falls within the range of $\theta = \{7, 8\}$ as this is where the gradient changes sign from negative to positive. This range is consistent with the estimate of $\widehat{\theta} = 7.2696$ given in (**??**) using just the moment condition m_{1t} in (**??**). The value of the gradient function for values of θ beginning at $\theta_{(0)} = 0$ and ending at $\theta_{(0)} = 10$ are also given in Figure 9.7. The gradient function exhibits an increasing slope for lower values of θ suggesting that this function is nonlinear, at least over certain ranges of θ.

The estimation method outlined above for the trade durations model provides a fairly crude updating method based on the sign of the gradient with the step lengths arbitrarily chosen to be equal to unity. More computationally efficient procedures exist which make use of the gradient and the Hessian of (9.53) to provide optimal step lengths. Appendix D provides a discussion of some of the more well-known numerical optimisation algorithms. When applying these algorithms to minimise the objective function in (9.53) some choices are available that depend upon how the weighting matrix $W(\theta)$ is treated during the iterations. In fact, there exist four commonly adopted methods which are summarised below.

EndExpansion

Table 9.2

GMM parameter estimates of the over-identified trade durations model based on a grid search algorithm, using the moments in equation (9.37) and a diagonal weighting matrix.

θ	$\overline{m}_1$	$\overline{m}_2$	$\text{var}(m_{1t})$	$\text{var}(m_{2t})$	G
0.000	7.270	138.308	138.308	246632.091	−0.121
1.000	6.270	123.769	124.769	227946.079	−0.116
2.000	5.270	109.229	113.229	210366.531	−0.108
3.000	4.270	94.690	103.690	193893.448	−0.097
4.000	3.270	80.151	96.151	178526.830	−0.081
5.000	2.270	65.612	90.612	164266.676	−0.062
6.000	1.270	51.072	87.072	151112.986	−0.039
7.000	0.270	36.533	85.533	139065.761	−0.014
8.000	−0.730	21.994	85.994	128125.001	0.012
9.000	−1.730	7.454	88.454	118290.705	0.037
10.000	−2.730	−7.085	92.915	109562.874	0.061

1. **One-step estimator**:

   The one step GMM estimator, which is denoted as $\widehat{\theta}_{(1)}$, is to set $W(\theta)$ equal to the identity matrix I_N.

   $$\widehat{\theta}_{(1)} = \arg\min_{\theta} M(\theta)' M(\theta), \tag{9.63}$$

   However, there is a lack of precision suffered by the one-step estimator which stems from the fact that the moment conditions are not receiving the appropriate weights in constructing the GMM criterion function $Q(\theta)$.

2. **Two-step estimator**:

   An improvement over the one-step approach is to use $\widehat{\theta}_{(1)}$, obtained from a one-step estimator and then compute the covariance matrix of the moment conditions using these consistent estimates, $W(\widehat{\theta}_{(1)})$. The two-step GMM estimator is then computed using $W(\widehat{\theta}_{(1)})^{-1}$ as the weighting matrix and the resultant parameter estimates are denoted $\widehat{\theta}_{(2)}$.

   $$\widehat{\theta}_{(2)} = \arg\min_{\theta} M(\theta)' W(\widehat{\theta}_{(1)})^{-1} M(\theta), \tag{9.64}$$

3. **Iterative estimator**:

   A natural extension of the two-step estimator is to update the weighting matrix as $W(\widehat{\theta}_{(2)})^{-1}$ and recomputing the GMM estimator. This would

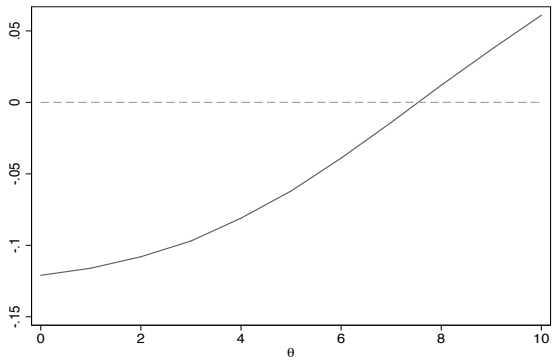


Figure 9.7: The gradient function for the over-identified trade duration model.

then give rise to a sequence of estimators, commonly referred to as the iterative estimator, of the form

$$\widehat{\theta}_{(j+1)} = \arg\min_{\theta} M(\theta)' W(\widehat{\theta}_{(j)})^{-1} M(\theta), \tag{9.65}$$

for $j = 1, 2, \cdots$ until the estimators and the weighting matrix converge to within some tolerance.

4. **Continuous updating**:

   Rather than switching between the separate estimation of θ and $W(\theta)$, an alternative strategy is to estimate them jointly using

$$\widehat{\theta} = \arg\min_{\theta} M(\theta)' W(\theta)^{-1} M(\theta). \tag{9.66}$$

   This is referred to as the continuous updating estimator. Hansen, Heaton and Yaron (1996) provide a useful comparison on the performance of these variants of the GMM estimator.

Table 9.3 gives the results of applying the continuous updating estimator to the over-identified durations trade model in (9.37) based on the full weighting matrix allowing for heteroskedasticty. The point estimate is

$$\widehat{\theta} = 6.7038,$$

which is slightly smaller than the estimate obtained using the diagonal weighting matrix given in Table 9.2 where the estimate was found to lie within the range of $\theta = \{7, 8\}$. Also given in the Table 9.2 is the GMM estimate based on the just-identified model using m_{1t} as the moment. This estimate is $\widehat{\theta} = 7.2696$, which from (**??**) corresponds to the sample mean of y_t. For completeness the GMM estimate based on just using the moment m_{2t} is also given.

Table 9.3

GMM parameter estimates of the trade durations model using the continuous updating algorithm. The moments are based on (9.37) and a heteroskedastic weighting matrix: $T = 23,401$.

Moments	Estimate	SE	t-stat.	p-value	Q	$J = TQ$
m_{1t}, m_{2t}	6.7038	0.0510	131.4197	0.0000	0.0192	449.3000
m_{1t}	7.2696	0.0604	120.2952	0.0000	0.0000	0.0000
m_{2t}	9.5127	0.1516	62.7519	0.0000	0.0000	0.0000

This estimate is $\widehat{\theta} = 9.5127$ which is higher than the estimate just based on m_{1t} and the estimate based on the over-identified model using m_{1t} and m_{2t} as the moments.

An important feature of the numerical results in Table 9.2 is that for the just identified model based on the moment m_{1t}, this result is equivalent to the earlier GMM result given in (**??**) where the weighting matrix W was not used to compute the GMM estimate. The reason for this is that for a just-identified model with $N = K$, the GMM estimator $\widehat{\theta}$ is independent of W. This property is highlighted by inspecting the first order condition in (9.57). In the case of just-identification, $D(\widehat{\theta})'$ and W are both $(K \times K)$ matrices. This means that $D(\widehat{\theta})'W^{-1}$ is also a $(K \times K)$ matrix which from (9.53) must also be negative semi-definite. In which case for the first order condition to be satisfied then $M(\widehat{\theta}) = 0$, as is the basis of the GMM solution used earlier.

This property of an exactly identified model is further highlighted in Table 9.2 where the value of the GMM objective function $Q(\theta)$ is zero for the two just-identified cases as $M(\widehat{\theta}) = 0$ must hold. However, for the over-identified model in Table 9.2 this does not hold as the value of the objective function is $Q(\widehat{\theta}) = 0.0192$, suggesting that $M(\widehat{\theta}) \neq 0$ in this case.

9.5 Sampling Properties of the GMM Estimator

The properties of the GMM estimator are now discussed without proof (for more detail see Martin, Hurn and Harris, citeMartin13). Letting θ_0 represent the population parameter, under certain conditions known as regularity conditions, the GMM estimator $\widehat{\theta}$ of θ_0 satisfies the following four conditions as the sample size T increases without limit. The first represents the definition of the population moment. The second relates to the mean of the asymptotic distribution, the third to the variance of the asymptotic distribution and the fourth refers to the shape of the asymptotic distribution.

9.5.1 Population Moment

Let $\mathrm{E}[m_t(\theta)]$ represent the GMM population moment. The relationship between the sample and population moments is based on the (weak) law of large numbers which states that the sample mean of $m_t(\theta)$ approaches the population mean as the sample size T increases without limit

$$\overline{m} = \frac{1}{T}\sum_{t=1}^{T} m_t(\theta) \xrightarrow{p} \mathrm{E}[m_t(\theta)], \tag{9.67}$$

where $\xrightarrow{p}$ represents convergence in probability, often denoted as p lim and written as $\mathrm{p\,lim}(\overline{m}) = \mathrm{E}[m_t(\theta)]$. By increasing the sample size ad infinitum, the sample mean is said to be 'close enough' to the population mean provided that it is within a small band around $\mathrm{E}[m_t(\theta)]$.

An important property of the population moment $\mathrm{E}[m_t(\theta)]$, is that by evaluating it at the population parameter θ_0 then

$$\mathrm{E}[m_t(\theta_0)] = 0. \tag{9.68}$$

A comparison of (9.68) and the GMM condition in (9.17) suggests that the latter represents the sample analogue of former, but with θ_0 replaced by $\widehat{\theta}$. The justification for replacing θ_0 by $\widehat{\theta}$ is given by the next condition.

9.5.2 Consistency

If the model is correctly specified so that the moment conditions $M(\theta)$ provide a good summary of the population model, the GMM estimator $\widehat{\theta}$ is a consistent estimator of θ_0 as it satisfies the property

$$\mathrm{plim}(\widehat{\theta}) = \theta_0\,. \tag{9.69}$$

This result reflects that the GMM estimator $\widehat{\theta}$ is centered on the population parameter θ_0, asymptotically. An example of consistency is given in Figure **??** where the GMM estimator $\widehat{\theta}$ is shown to approach the true population parameter $\theta_0 = 10$ with the variability between $\widehat{\theta}$ and θ_0 decreasing as T continually increases.

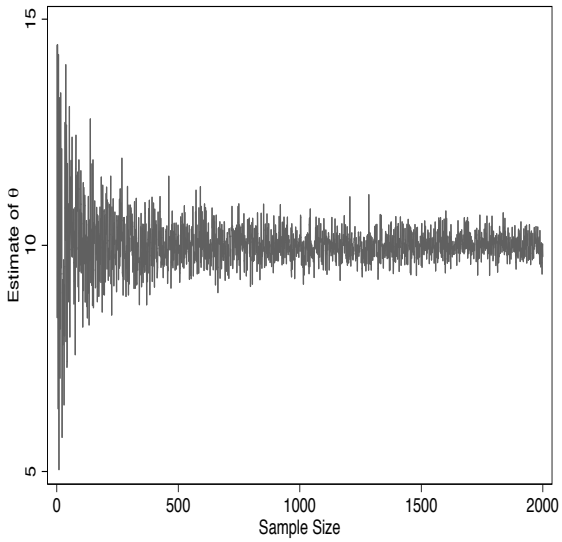


Figure 9.8: Demonstration of the consistency property of GMM estimator $\widehat{\theta}$. The population distribution is exponential with parameter $\theta_0 = 10$ and the GMM estimator is the sample mean.

The consistency property follows from (9.67) whereby the sample moments converge to the true moments as the sample size grows. Consistency also implies that the GMM criterion function $Q(\theta)$ approaches its true population value $Q(\theta_0)$. This result does not depend on the choice of the weighting matrix $W(\theta)$, provided only that this matrix is positive definite in the limit. For this reason the 1-step GMM estimator in (9.63) where the weighting matrix is not used in the updating of the estimator provides a consistent estimator of θ_0.

9.5.3 Efficiency

The asymptotic covariance matrix of $\widehat{\theta}$ is

$$\Omega = \frac{1}{T}\left[D(\theta_0)'W(\theta_0)^{-1}D(\theta_0)\right]^{-1}, \tag{9.70}$$

where

$$D\left(\theta_0\right) = \frac{1}{T}\sum_{t=1}^{T}\left.\frac{\partial m_t\left(\theta\right)}{\partial \theta'}\right|_{\theta=\theta_0}, \tag{9.71}$$

is the $(N \times K)$ matrix of first-derivatives of the moment conditions with respect to the parameters evaluated at the population parameter vector θ_0, and

$W(\theta_0)$ is the weighting matrix evaluated at θ_0 which is defined as the covariance matrix of the chosen moments m_t according to

$$W(\theta_0) = \frac{1}{T}\sum_{t=1}^{T} m_t(\theta_0)m_t'(\theta_0). \tag{9.72}$$

This choice of weighting matrix is optimal in the sense that it achieves the smallest variance based on the given set of moments in m_t, compared to a covariance matrix Ω^* based on any other weighting matrix. Formally this result means that $\Omega^* - \Omega$ represents a positive semi-definite matrix (Hansen, (1982)).

In interpreting this form of asymptotic efficiency it is important to remember that it is restricted to the set of moments chosen in m_t. By changing this set of moments will result in a different asymptotically efficient GMM estimator. Moreover, as the 2-step, iterative and continuous GMM estimators in (9.64) to (9.66) do use the weighting matrix in the updating schemes, these estimators are both consistent and asymptotically efficient compared with the 1-step estimator in (9.63) which is just consistent but not asymptotically efficient as it is not based on the weighting matrix.[2]

$$\text{var}(\widehat{\theta}_{(1)}) > \{\text{var}(\widehat{\theta}_{(2)}), \text{var}(\widehat{\theta}_{(j+1)}), \text{var}(\widehat{\theta})\}. \tag{9.73}$$

9.5.4 Asymptotic Normality

The asymptotic distribution of the GMM estimator is

$$\sqrt{T}\left(\widehat{\theta} - \theta_0\right) \xrightarrow{d} N\left(0, \Omega\right), \tag{9.74}$$

where $\xrightarrow{d}$ denotes convergence in distribution and $\Omega = \Omega(\theta_0)$ is given by (9.70). The asymptotic normality property follows from assuming that a uniform weak law of large numbers holds and that the GMM moments m_t satisfy a central limit theorem. Figure **??** provides a demonstration of asymptotic normality which gives the sampling distributions of the 1-step and 2-step estimators. This figure also demonstrates the asymptotic efficiency of the 2-step estimator over the 1-step estimator given in (9.73) as the former exhibits a smaller variance.

[2]From an asymptotic point of view the 2-step, iterative and continuous estimators are all equivalent. The potential gains from from adopting the iterative and continuous estimators are that these estimators tend to exhibit better small sample properties than the 2-step estimator.

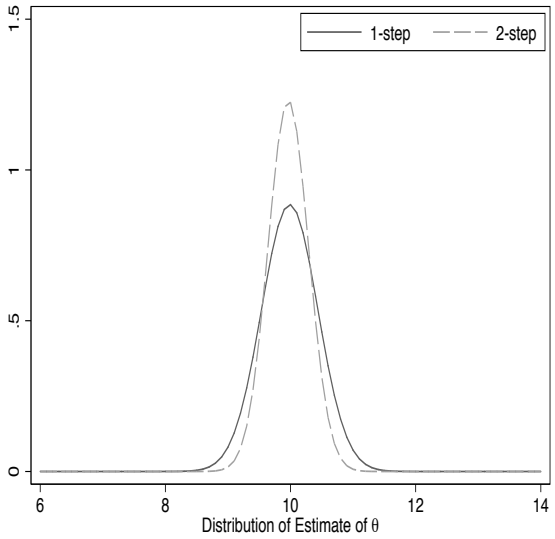


Figure 9.9: Demonstration of the asymptotic normality property of GMM estimator, $\widehat{\theta}$. The population distribution is exponential with parameter $\theta_0 = 10$ and the GMM estimators are the one-step and two-step estimators.

An important implication of establishing asymptotic normality of the GMM estimator is that hypothesis tests can now be performed with p-values based on the asymptotic normal distribution. To perform hypothesis tests in practice however, the covariance matrix of $\widehat{\theta}$, is evaluated by replacing the unknown population parameter θ_0 by the GMM estimator $\widehat{\theta}$ in Ω, yielding

$$\text{cov}(\widehat{\theta}) = \frac{1}{T}\widehat{\Omega} = \frac{1}{T}\left[D(\widehat{\theta})'W(\widehat{\theta})^{-1}D(\widehat{\theta})\right]^{-1}, \tag{9.75}$$

where $\widehat{\Omega} = \Omega(\widehat{\theta})$. The standard errors are computed as the square roots of the diagonal elements of $\text{cov}(\widehat{\theta})$. For the trade durations model the pertinent standard errors are reported in Table 9.3.

9.6 Testing

The focus so far has been on specifying a theoretical model as characterised by a set of moments and then estimating the unknown parameters of the model by GMM. This of course presupposes that the model is correct in that the specified moments provide an accurate and complete description of the underlying variable y_t. To test the adequacy of the model specification three broad classes of tests are investigated. The first provides an overall test of the

adequacy of the model, the second identifies the role of the explanatory variables in the model and the third represents a diagnostic test to identify if any features of y_t have been excluded from the model.

9.6.1 Overall Test

An overall test of the adequacy of the specified model is given by the The Hansen-Sargan J test. The null and alternative hypotheses are respectively

$$\begin{array}{ll} H_0: & \text{Model correctly specified} \\ H_1: & \text{Model not correctly specified.} \end{array} \tag{9.76}$$

The J test is based on scaling the GMM objective function evaluated at $\widehat{\theta}$ by the sample size

$$J = T\, Q(\widehat{\theta})\,. \tag{9.77}$$

Under the null hypothesis the model is assumed to be specified correctly. This implies that the sample moments based on y_t match the population moments of the specified model. As the GMM moment condition $M(\theta)$ provides a measure of the distance between the sample and population moments then under the null hypothesis this distance should be small in a statistical sense, which in turn, would yield a small value of the GMM objective function $Q(\theta)$. If, however, the model is not properly specified there will be a mismatch between the sample and the population moments resulting in a large value of $M(\theta)$ and hence $Q(\theta)$.

To implement the J test the distribution of the test statistic is needed. Under the null hypothesis the J statistic in (9.77) is distributed asymptotically as χ^2_{N-K}, where the number of over-identifying restrictions $N - K$, represents the number of degrees of freedom. The null hypothesis H_0 is rejected for values of J larger than the critical value from the χ^2_{N-K} distribution for a chosen level of significance.

A test of the overall specification of the trade durations model is given in Table 9.3. The value of the J statistic to test the 2-moment version of the model specification is $J = 449.3000$. As there are $N = 2$ moments and $K = 1$ unknown parameter the excess moment condition is $N - K = 1$. From the chi-square distribution with 1 degree of freedom and a size of 5 the critical value is 3.8410. As J clearly exceeds the critical value the model specification is strongly rejected at the 5% level.

Table 9.3 also provides the GMM estimates of the trade durations model based on alternative single moment GMM equations. In both cases the objective functions equal $Q(\widehat{\theta}) = 0$, with corresponding values of the over-identification test of $J = 0$. However, for this case this result does not imply that the chosen model is necessarily correctly specified as both models are just-identified so a value of zero for the GMM objective function is obtained by construction. The implication of this result is that the J test can only be implemented for over-identified models.

9.6.2 Wald Tests

Instead of testing the overall specification of the model, Wald tests can be used to test specific parts of the model. The simplest form of a Wald test is a $t-test$ which provides a test of a single parameter. For example, suppose that the null and alternative hypotheses are

$$\begin{array}{ll} H_0: & \theta = \theta_0 \\ H_1: & \theta \neq \theta_0. \end{array} \tag{9.78}$$

The t-statistic is constructed as

$$t = \frac{\widehat{\theta} - \theta_0}{\text{se}(\widehat{\theta})}, \tag{9.79}$$

which under the null hypothesis is distributed asymptotically as $N(0,1)$. For the trade durations model t-statistics for the case of $\theta_0 = 0$ are given in Table 9.3 together with p-values based on asymptotic normality.[3]
Other types of Wald tests can be performed which involve joint testing of subsets of the parameters. Under the null hypothesis these Wald tests are distributed asymptotically as χ^2_R where R represents the number of joint restrictions being tested.

9.6.3 Diagnostic Tests

Diagnostic tests aim to detect model misspecification by identifying potential patterns in the disturbance terms. In the case of the GMM model the disturbance is represented by the gap between the sample moments and the population moments. For a correctly specified model this suggests that $E[M(\theta_0)] = 0$, which implies that a test of the model can be based on the following respective null and alternative hypotheses

$$\begin{array}{ll} H_0: & \text{E}[M(\theta_0)] = 0 \\ H_1: & \text{E}[M(\theta_0)] \neq 0. \end{array} \tag{9.80}$$

This test effectively means that the disturbance corresponding to each moment should have zero mean.
To implement the diagnostic test reconsider the trade durations model based on the $N = 2$ moments in (9.37). Evaluating these moments at the GMM estimator $\widehat{\theta}$ gives the moment equations

$$\begin{array}{lcl} m_{1t} & = & y_t - \widehat{\theta} \\ m_{2t} & = & (y_t - \widehat{\theta})^2 - \widehat{\theta}. \end{array} \tag{9.81}$$

[3] Technically the t-statistics in Table 9.3 do not have an asymptotic normal distribution in this case as the null hypothesis falls on the boundary of the feasible parameter space which is $\theta = 0$ for the exponential distribution.

In the present context these are also interpreted as "residuals". If the model is correctly specified the sample means should be zero under the null hypothesis. To test this hypothesis in the case of the first moment consider estimating the following regression equation

$$m_{1t} = \beta + u_t, \tag{9.82}$$

where u_t is a disturbance term distributed as $(0, \sigma_u^2)$. The null hypothesis is $\beta = 0$, which is tested using a t-test. The results from estimating this regression equation by OLS with standard errors in parentheses, are

$$\underset{(0.0604)}{m_{1t} = 0.5659} + \widehat{u}_t,$$

A t-test is based on

$$t = \frac{0.5659 - 0.0000}{0.0604} = 9.3634,$$

which is distributed asymptotically as $N(0,1)$ under the null hpothesis. The p-value is 0.0000 resulting in a rejection of the null at the 5% level and hence providing evidence of misspecification.
Repeating the diagnostic test for the second moment the regression equation is now specified as

$$m_{2t} = \beta + u_t, \tag{9.83}$$

where u_t is again a disturbance term distributed as $(0, \sigma_u^2)$. Estimating this regression equation by OLS gives

$$\underset{(2.4534)}{m_{2t} = 79.0767} + \widehat{u}_t.$$

The t-statistic is computed as

$$t = \frac{79.0767 - 0.0000}{2.4534} = 32.2312.$$

The p-value is 0.0000, resulting in evidence of misspecification at the 5% level in this moment as well.
The diagnostic tests performed immediately above relate to the mean of the moments m_t, however other types of tests can be designed. For example a test of excess kurtosis can be performed by setting the dependent variable in the test equation in (9.82) to be $v_t^4 - 3$ where v_t is the standardised moment of m_t which has zero mean and unit variance. Alternatively, a test of first order autocorrelation is based on defining the dependent variable as $v_t v_{t-1}$. In both cases the dependent variable is regressed on a constant which is tested to be zero under the null hypothesis.
Another class of tests known as conditional moment tests is based on augmenting the test equation in (9.82) by a set of variables which are already included in the model. The test is still about testing for a zero intercept in the presence of a set of conditioning variables. Conditional moment tests are also investigated in the next chapter where an alternative class of estimators based on maximum likelihood methods is presented where augmentation is now based on the gradients of what is referred to as the log-likelihood function.

9.7 Relationships to Other Estimators

The GMM estimator nests a number of well known estimators used in empirical finance. It has already been shown that the GMM estimator is equivalent to the OLS estimator of the linear regression model in the context of the capital asset pricing model for the case where the population moments are given by (9.20). Now the GMM estimator is shown to be related to two other estimators known as method of moments and instrumental variables which was discussed in Chapter 8. The GMM estimator can also be shown to be related to the maximum likelihood estimator which will be discussed in the next Chapter.

9.7.1 Method of Moment Estimators

The method of moments estimator is based on the (weak) law of large numbers whereby the sample mean approaches the population mean as the sample size T increases without limit. In the case of the sample mean of the kth moment y_t^k, this requires the result

$$\frac{1}{T}\sum_{t=1}^{T} y_t^k \xrightarrow{p} E[y_t^k] = \mu_k. \tag{9.84}$$

The aim of the method of moments estimator is to choose the parameters of the population moments which equate the sample and population moments. To derive the method of moments estimator it is necessary that the number of unknown parameter just matches the number of sample moments. But this is equivalent to the GMM estimators derived for the just-identified models investigated in Sections 9.2 and 9.3 as the moment conditions used to derive the GMM estimator are based on replacing the population moment by the sample moment. It is for over-identified models however, that the method of moments and GMM estimators differ as the latter uses all of the information from all moments, whereas the method of moments estimator requires choosing necessarily a restricted set of moments. To create a just-identified system. The innovation in the GMM estimator is to use a weighting matrix to circumvent this problem.

9.7.2 Instrumental Variables

The establishment of the relationship between the GMM and OLS estimators for the parameters of the linear regression model can be extended to include the instrumental variable estimator of Chapter 8. In particular, for the IV estimator the moment conditions given in (9.20) are replaced by

$$\mathrm{E}[u_t] = 0, \qquad \mathrm{E}[u_t z_t] = 0, \qquad \mathrm{E}[u_t^2] = \sigma^2, \tag{9.85}$$

where z_t is the instrument. For the CAPM this suggests the GMM moments

$$m_t = \begin{bmatrix} m_{1,t} \\ m_{2,t} \\ m_{3,t} \end{bmatrix} = \begin{bmatrix} y_t - \alpha - \beta x_t \\ (y_t - \alpha - \beta x_t) z_t \\ (y_t - \alpha - \beta x_t)^2 - \sigma^2 \end{bmatrix}, \tag{9.86}$$

with the GMM estimator being the solution of the system of equations

$$M(\widehat{\theta}) = \frac{1}{T} \sum_{t=1}^{T} \begin{bmatrix} y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t \\ (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t) z_t \\ (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t)^2 - \widehat{\sigma}^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \tag{9.87}$$

Solving for $\widehat{\theta} = \{\widehat{\alpha}, \widehat{\beta}, \widehat{\sigma}^2\}$ yields the GMM estimators

$$\begin{aligned} \widehat{\beta}_1 &= \frac{\sum_{t=1}^{T}(z_t - \overline{z})(y_t - \overline{y})}{\sum_{t=1}^{T}(z_t - \overline{z})(x_t - \overline{x})} \\ \widehat{\beta}_0 &= \overline{y} - \widehat{\beta}_1 \overline{x} \\ \widehat{\sigma}^2 &= \frac{1}{T} \sum_{t=1}^{T} (y_t - \widehat{\beta}_0 - \widehat{\beta}_1 x_t)^2, \end{aligned} \tag{9.88}$$

which are equivalent to the IV estimators presented in Chapter 8.

9.8 Decomposing International Equity Returns

An important class of models in finance is where the dependent variables are determined by a set of factors that are not directly observable, that is latent. This class of models is discussed at greater detail in Chapter 14, but is investigated here not only as an application of the GMM estimator, but also as a vehicle for highlighting the advantages of using this estimation framework in the context of latent factor models compared to other estimators such as the ordinary least squares estimator.
Consider the following trivariate latent factor model of returns on international equity markets

$$\begin{aligned} r_{1t} &= \lambda_1 w_t + \sigma_1 v_{1t} \\ r_{2t} &= \lambda_2 w_t + \sigma_2 v_{2t} \\ r_{3t} &= \lambda_3 w_t + \sigma_3 v_{3t}. \end{aligned} \tag{9.89}$$

The returns on the left hand-side are given by $r_t = \{r_{1t}, r_{2t}, r_{3t}\}$ which for convenience are assumed to be centred to have zero mean so as to avoid having to specify intercept terms in (9.89). The model contains four factors, all of which are latent. The first factor w_t, is a world factor representing external shocks which simultaneously impact upon all three country asset markets with the impact measured by the parameters $\lambda_1, \lambda_2, \lambda_3$. The other factors are

v_{1t}, v_{2t}, v_{3t}, which represent idiosyncratic factors as they capture shocks solely occurring within a country with the effects of these shocks measured by the parameters $\sigma_1, \sigma_2, \sigma_3$. In total there are $K = 6$ unknown parameters

$$\theta = \{\lambda_1, \lambda_2, \lambda_3, \sigma_1, \sigma_2, \sigma_3\}. \tag{9.90}$$

An important feature of this model is that only the terms on the left hand-side of each equation in (9.89) are measurable, whereas all of the terms on the right hand-side are not: $\{w_t, v_{1t}, v_{2t}, v_{3t}\}$ are factors that are latent and hence not measurable. This class of models is used by Dungey, Fry, González-Hermosillo and Martin (2010) to model the transmission of contagion asit has the advantage of circumventing the need to construct an ad hoc proxy variables of contagion from observable variables. This is especially true in those situations where high frequency returns are available, but contagion proxy variables are only available on a lower frequency, such as monthly or even quarterly.

For the special case where the world factor w_t is observable then the parameters of the model can be simply obtained by regressing each return on w_t by ordinary least squares to obtain the λ_i parameters, with the idiosyncratic parameters estimated as the standard deviations of the OLS residuals. Without the requirement that w_t is observable estimation by ordinary least squares infeasible. Despite this supposed level of difficulty in specifying a model where all of the terms of the right hand-side are undefined the parameter vector θ in (9.89) can nonetheless be estimated by GMM using just data on asset returns r_t, provided that some additional structure is imposed on the model. This structure consists of the following three sets of conditions. The first set is that all factors have zero mean

$$\mathrm{E}[w_t] = \mathrm{E}[v_{1t}] = \mathrm{E}[v_{2t}] = \mathrm{E}[v_{3t}] = 0. \tag{9.91}$$

The second set is that all factors have unit variance

$$\mathrm{E}[w_t^2] = \mathrm{E}[v_{1t}^2] = \mathrm{E}[v_{2t}^2] = \mathbb{E}[v_{3t}^2] = 1. \tag{9.92}$$

The third and final set is that the factors are independent of each other

$$\mathrm{E}[w_t v_{it}] = 0, \quad \forall i, \qquad \mathrm{E}[v_{it} v_{jt}] = 0, \quad \forall i \neq j. \tag{9.93}$$

The first two conditions represent normalization conditions while the third effectively means that the latent factors represent structural shocks which can be classified as world and country idiosyncratic shocks.[4]

To derive the GMM estimator of θ consider the ith equation in (9.89). Squaring this equation and taking unconditional expectations gives

$$\begin{aligned} \mathbb{E}[r_{it}^2] &= \mathbb{E}[(\lambda_i w_t + \sigma_i v_{it})^2] \\ &= \mathbb{E}[\lambda_i^2 w_t^2 + \sigma_i^2 v_{it}^2 + 2\lambda_i \sigma_i w_t v_{it}] \\ &= \lambda_i^2 \mathbb{E}[w_t^2] + \sigma_i^2 \mathbb{E}[v_{it}^2] + 2\lambda_i \sigma_i \mathbb{E}[w_t v_{it}]. \end{aligned}$$

[4] In particular, if the third condition (**??**) is not satisfied, then w_t and v_{it} are correlated so w_t no longer represents external shocks and v_{it} internal shocks.

This expression simplifies by using the conditions (9.91) to (9.93)

$$\text{var}(r_{it}) = \mathbb{E}[r_{it}^2] = \lambda_i^2 + \sigma_i^2, \qquad \forall i. \tag{9.94}$$

This equation shows that the (squared) volatility of the return in country i can be decomposed in terms of the contributions from the world factor (λ_i^2) and the idiosyncratic factor (σ_i^2).
Now consider taking the unconditional expectation of the cross-product of returns

$$\begin{aligned}
\mathbb{E}[r_{it}r_{jt}] &= \mathbb{E}[(\lambda_i w_t + \sigma_i v_{it})(\lambda_j w_t + \sigma_j v_{jt})] \\
&= \mathbb{E}[(\lambda_i\lambda_j w_t^2 + \lambda_i\sigma_j w_t v_{jt} + \lambda_j\sigma_i w_t v_{it} + \sigma_i\sigma_j v_{it}v_{jt})] \\
&= \lambda_i\lambda_j\mathbb{E}[w_t^2] + \lambda_i\sigma_j[\mathbb{E}[w_t v_{jt}] + \lambda_j\sigma i[\mathbb{E}[w_t v_{it}] + \sigma_i\sigma_j\mathbb{E}[v_{it}v_{jt}].
\end{aligned}$$

Using the conditions in (9.91) to (9.93) yields the covariance equation

$$\text{cov}(r_{it}) = \mathbb{E}[r_{it}r_{jt}] = \lambda_i\lambda_j, \qquad \forall i \neq j. \tag{9.95}$$

This expression shows that the co-volatility of the return in country i is determined solely by the global factor w_t as any common movements observed in country returns must be solely determined by the world factor w_t. How the world factor jointly affects the returns of countries is determined by the signs of the parameters λ_i. If the signs of these parameters are the same the returns exhibit positive covariance, whereas if they are of the opposite sign the returns then exhibit negative covariances.
Given the population moments in (9.94) and (9.95) this suggests the following $N = 6$ GMM moment equations

$$\begin{aligned}
m_{1t} &= r_{1t}^2 - \lambda_1^2 - \sigma_1^2 \\
m_{2t} &= r_{2t}^2 - \lambda_2^2 - \sigma_2^2 \\
m_{3t} &= r_{3t}^2 - \lambda_3^2 - \sigma_3^2 \\
m_{4t} &= r_{1t}r_{2t} - \lambda_1\lambda_2 \\
m_{5t} &= r_{1t}r_{3t} - \lambda_1\lambda_3 \\
m_{6t} &= r_{2t}r_{3t} - \lambda_2\lambda_3.
\end{aligned} \tag{9.96}$$

Given that there are $K = 6$ unknown parameters in (9.90) this is a just-identified model. Whilst this is a just-identified model the system of equations is nonlinear in the parameters which nonetheless requires estimation to be based on an iterative gradient algorithm.
The data file contains daily equity prices (P_t) on the SP500, FTSE100 and the EURO50, from 29 July 2004 to 3 March 2009, $T = 1198$. Let r_{it} be the percentage demeaned log-returns expressed as a percentage. The returns are given in Figure 9.10. The sample covariance matrix of the returns is

$$\text{cov}(r_t) = \begin{bmatrix} 1.8079 & 1.4600 & 1.5893 \\ 1.4600 & 1.7967 & 1.6078 \\ 1.5893 & 1.6078 & 1.8730 \end{bmatrix}. \tag{9.97}$$

The EURO50 exhibits the highest volatility over the period followed by the S&P500 and the FTSE100. All equity markets move in the same direction on average as they have positive covariances.

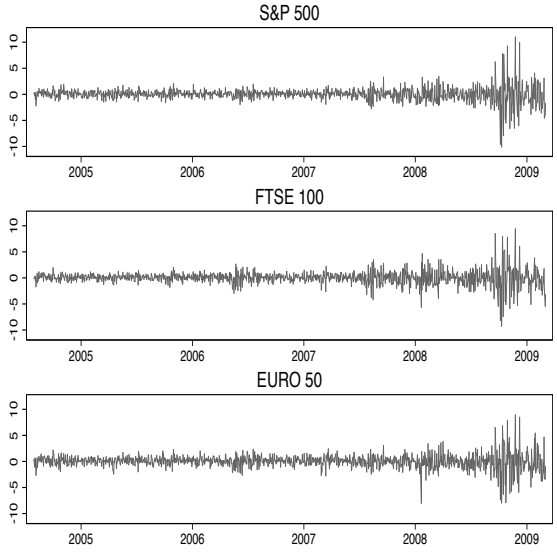


Figure 9.10: Daily percentage entered log returns on the SP500, FTSE100 and the EURO50, from 29 July 2004 to 3 March 2009.

To provide further insight into the nature of the shocks driving the three asset markets Table 9.4 gives the GMM estimates of the latent factor model using the continuous updating algorithm. As the model is just-identified the GMM objective function is

$$Q(\widehat{\theta}) = 0.0.$$

All of the parameter estimates associated with the world factor w_t are of the same sign suggesting that the covariances amongst the returns are all positive, a result which is consistent with the returns covariance matrix in (9.95). A comparison of the global factor and idiosyncratic parameter estimates suggests that external shocks are relatively more important than domestic shocks in affecting equity markets by at least a factor of 2 as $\lambda_i > \sigma_i$ in all three asset markets. To formalise these calculations, from (9.94) it follows that the proportionate share of volatility arising from global and idiosyncratic shocks are respectively

$$\frac{\lambda_i^2}{\lambda_i^2 + \sigma_i^2}, \qquad \frac{\sigma_i^2}{\lambda_i^2 + \sigma_i^2}. \tag{9.98}$$

Using the point estimates in Table 9.4 the proportionate contribution to total

Table 9.4

GMM parameter estimates of the latent factor model of equities using the moments in (9.96): 29 July 2004 to 3 March 2009, $T = 1198$. Based on the continuous updating algorithm and the heteroskedastic weighting matrix.

Parameter	Estimate	Standard Error	t-statistic	p-value
λ_1	1.2013	0.0921	13.0440	0.0000
λ_2	1.2153	0.0758	16.0314	0.0000
λ_3	1.3230	0.0683	19.3821	0.0000
σ_1	0.6040	0.0354	17.0789	0.0000
σ_2	0.5654	0.0253	22.3252	0.0000
σ_3	0.3504	0.0468	7.4813	0.0000

volatility from global shocks are

$$\begin{aligned}
\text{S\&P500} \quad &: \quad \frac{1.2013^2}{1.2013^2 + 0.6040^2} = 0.7982 \\
\text{FTSE100} \quad &: \quad \frac{1.2153^2}{1.2153^2 + 0.5654^2} = 0.8220 \\
\text{EURO50} \quad &: \quad \frac{1.3230^2}{1.3230^2 + 0.3504^2} = 0.9340.
\end{aligned}$$

These results suggest that the European asset market is more open to external shocks than the other two asset markets with 93.40% of volatility the result of common shocks from the world factor w_t. Nonetheless, common shocks also play a large role in determining the volatility in the US and the UK asset markets where the contribution is just under \$80 for the US and just over \$80 for the UK.

The analysis regarding the decomposition of volatility into external and internal shocks demonstrates an important advantage of the latent factor model in that it reveals additional insight into the properties of the asset returns covariance matrix in (9.97) that are not necessarily transparent from inspection of this matrix. To highlight this property further, consider reconstructing the empirical covariance matrix in (9.97) using the GMM parameter estimates given

in Table 9.4 as follows

$$\begin{aligned}
&\begin{bmatrix} \widehat{\lambda}_1^2+\widehat{\sigma}_1^2 & \widehat{\lambda}_1\widehat{\lambda}_2 & \widehat{\lambda}_1\widehat{\lambda}_3 \\ \widehat{\lambda}_2\widehat{\lambda}_1 & \widehat{\lambda}_2^2+\widehat{\sigma}_2^2 & \widehat{\lambda}_2\widehat{\lambda}_3 \\ \widehat{\lambda}_3\widehat{\lambda}_1 & \widehat{\lambda}_3\widehat{\lambda}_2 & \widehat{\lambda}_3^2+\widehat{\sigma}_3^2 \end{bmatrix} \\
&= \begin{bmatrix} 1.2013^2+0.6040^2 & 1.2013\times 1.2153 & 1.2013\times 1.3230 \\ 1.2153\times 1.2013 & 1.2153^2+0.5654^2 & 1.2153\times 1.3230 \\ 1.3230\times 1.2013 & 1.3230\times 1.2153 & 1.3230^2+0.3504^2 \end{bmatrix} \\
&= \begin{bmatrix} 1.8079 & 1.4600 & 1.5893 \\ 1.4600 & 1.7967 & 1.6078 \\ 1.5893 & 1.6078 & 1.8730 \end{bmatrix},
\end{aligned}$$

which agrees with the covariance matrix in (9.97) computed directly from the asset returns. The fact that the empirical covariance matrix is perfectly recovered in this case follows from the property that the model is just-identified. One feature of the empirical results of the latent factor model is that the parameter estimates associated with the global factor are very similar. To test that a global factor shock has the same simultaneous impact on all asset markets, the following restrictions are tested

$$\begin{array}{ll} H_0: & \lambda_1=\lambda_2=\lambda_3=\lambda \\ H_1: & \text{at least one restrictions fails.} \end{array} \tag{9.99}$$

As the number of parameters to be estimated under the null reduces from $K=6$ to $K=4$, the model is now over-identified with the degree of over-identification equal to $N-K=2$. Reestimating the latent factor model subject to the 2 restrictions under the null hypothesis yields the following parameter estimates

$$\widehat{\theta}=\{\widehat{\lambda}=1.2897,\widehat{\sigma}_1=0.565040,\widehat{\sigma}_2=0.553986,\widehat{\sigma}_3=0.384174\}.$$

The value of the GMM objective function under the null hypothesis is

$$Q(\widehat{\theta})=0.008032.$$

As the value of the GMM objective function is $Q(\widehat{\theta})=0$ for the unrestricted model, a test of the restrictions embedded in (9.99) is simply given by

$$J=TQ(\widehat{\theta}_0)=1198\times 0.008032=9.6223.$$

Under the null hypothesis the test statistic is distributed asymptotically as χ^2 with 2 degrees of freedom. The critical value at the 5% level is 5.991 showing that the restrictions are rejected at the 5% level.

9.9 Consumption CAPM

The consumption based capital asset pricing model (C-CAPM) as already discussed in Chapter 8 represents an important application of GMM in empirical

finance as it one of the vary early applications of this estimation framework. The C-CAPM model is based on the assumption that a representative agent maximises the inter-temporal utility function

$$\sum_{i=0}^{\infty} \delta^i \, \mathrm{E}_t \left[\frac{C_{t+i}^{1-\gamma} - 1}{1-\gamma} \right], \tag{9.100}$$

subject to the wealth constraint

$$W_{t+1} = (1 + R_{t+1})(W_t - C_t), \tag{9.101}$$

where C_t is real consumption, W_t is wealth, R_t is the simple return on an asset (more precisely on wealth), and E_t represents expectations conditional on all relevant variables available at time t. The unknown parameters are the discount rate δ, and the relative risk aversion parameter γ, which are summarised by the $K = 2$ parameter vector θ, as

$$\theta = \{\delta, \gamma\}.$$

The first-order condition is

$$\mathrm{E}_t \left[\delta \Big(\frac{C_{t+1}}{C_t} \Big)^{-\gamma} (1 + R_{t+1}) - 1 \right] = 0. \tag{9.102}$$

It is illuminating to reexpress the model by defining

$$u_{t+1} = \delta \Big(\frac{C_{t+1}}{C_t} \Big)^{-\gamma} (1 + R_{t+1}) - 1, \tag{9.103}$$

which is also interpreted as a disturbance term. Using (9.103) in (9.102) now simplifies the population moment of the model as

$$\mathrm{E}_t \left[u_{t+1} \right] = 0. \tag{9.104}$$

Equation (9.104) represents the population moment characterizing the C-CAPM with $K = 2$ unknown parameters. Before deriving the GMM parameter estimators for θ, two issues need to be addressed. The first is that the expectation in (9.104) is a conditional expectation whereas the population expectations used so far are expressed as unconditional expectations.[5] The second is that there is just one moment equation and two unknown parameters resulting in an under-identified model without further information.
The first issue is solved by using the law of iterated expectations which converts a conditional expectation into an unconditional expectation. In the case of (9.104)

$$\mathbb{E}[\mathbb{E}_t \left[u_{t+1} \right]] = \mathrm{E} \left[u_{t+1} \right]. \tag{9.105}$$

[5] An exception is the present value model in Section 9.2.1 where the pertinent moment condition is expressed in terms of a conditional expectation of future dividends. There the solution is to assume that dividends follow a random walk so the conditional expectation based on information at time t of future dividends simply equals the currrent dividend D_t.

The intuition behind this result is that by definition a conditional expectation implies that the expected value of u_{t+1} varies across all possible values of the information set. By averaging across all of these conditional expectations corresponding to each potential value of the information set "averages out" the role of the information set thereby yielding the unconditional expectation of u_{t+1}.[6]

The second issue is solved by using the properties of instrumental variables as discussed in Chapter 8 as well as Section 9.7.2. Here the instruments are chosen to be the information set underlying the conditional expectation in (9.100) or (9.104). For example, let the information set be represented by a constant and C_t/C_{t-1}, the lagged real consumption ratio

$$z_t = \{1, C_t/C_{t-1}\}. \tag{9.106}$$

From the properties of the IV estimator, u_{t+1} is independent of z_t as the following condition is satisfied

$$\mathrm{E}\left[u_{t+1} z_t\right] = 0,$$

or upon using (9.106), this is equivalent to

$$\mathrm{E}\left[u_{t+1}\right] = 0, \qquad \mathrm{E}\left[u_{t+1} C_t/C_{t-1}\right] = 0. \tag{9.107}$$

There are now two population moments characterizing the C-CAPM from which it is possible to derive the GMM estimators for the two population parameters $\theta = \{\delta, \gamma\}$. As there are $N = 2$ population moments and $K = 2$ unknown parameters, this form of the model is just-identifed.

To derive the GMM estimator of θ, the two population moments in (9.107) imply the following GMM moment equations

$$\begin{aligned} m_{1t} &= \delta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}(1+R_{t+1}) - 1 \\ m_{2t} &= \left(\delta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}(1+R_{t+1}) - 1\right)\frac{C_t}{C_{t-1}}. \end{aligned} \tag{9.108}$$

[6] To prove the law of iterated expectations let y be the random variable and x represent the information set. Using the definitions of unconditional and conditional expectations

$$E_x\left[E\left[y|\,x\right]\right] = \int \left[E\left[y|\,x\right]\right] f\left(x\right) dx = \int \left[\int y f\left(y|\,x\right) dy\right] f\left(x\right) dx,$$

where the subscript on $E_x\left[\cdot\right]$ emphasises that the expectation is taken with respect to x. From the definition of a conditional distribution $f\left(y|\,x\right) = f\left(y,x\right)/f\left(x\right)$ this expression becomes after rearranging

$$E_x\left[E\left[y|\,x\right]\right] = \int \left[\int y \frac{f\left(y,x\right)}{f\left(x\right)} dy\right] f\left(x\right) dx = \int y \left[\int f\left(y,x\right) dx\right] dy.$$

From the definition of marginal probability $\int f\left(y,x\right) dx = f\left(y\right)$ so $\mathrm{E}_x\left[\mathrm{E}\left[y|\,x\right]\right] = \int y f\left(y\right) dy = \mathrm{E}_y\left[y\right]$.

Upon using (9.17) the GMM estimator $\widehat{\theta} = \{\widehat{\delta}, \widehat{\gamma}\}$, satisfies

$$\begin{array}{rcl} \frac{1}{T}\sum_{t=1}^{T}\Big(\widehat{\delta}\Big(\frac{C_{t+1}}{C_t}\Big)^{-\widehat{\gamma}}(1+R_{t+1})-1\Big) & = & 0 \\ \frac{1}{T}\sum_{t=1}^{T}\Big(\widehat{\delta}\Big(\frac{C_{t+1}}{C_t}\Big)^{-\widehat{\gamma}}(1+R_{t+1})-1\Big)\frac{C_t}{C_{t-1}} & = & 0. \end{array} \tag{9.109}$$

This is a nonlinear system of two equations with two unknowns which is solved using an iterative algorithm.
The C-CAPM is estimated using monthly data for the United States beginning February 1959 and ending December 1978, $T = 238$. The data consist of seasonally adjusted real consumption C_t, the real Treasury bill rate R_t, and real value weighted returns r_t. Forming the consumption ratio C_t/C_{t-1} means that the effective sample begins in March 1959, $T = 159$. The data are a revised version of the original data used by Hansen and Singleton (1982). The GMM parameter estimates are given in Table 9.5 for various choices of instruments.

Table 9.5

GMM parameter estimates of the C-CAPM using the GMM moment equations in (9.108) for alternative instrument sets: March 1959 to December 1978, $T = 159$. Based on the continuous updating algorithm and the heteroskedastic weighting matrix. Standard errors are in parentheses.

Parameter	Instrument Set (z_t)		
	$\{1, C_t/C_{t-1}\}$	$\{1, C_t/C_{t-1}, R_t\}$	$\{1, C_t/C_{t-1}, R_t, r_t\}$
δ	0.9983	0.9982	0.9977
	(0.0045)	(0.0045)	(0.0043)
γ	0.5542	1.0234	0.7144
	(1.9390)	(1.8770)	(1.7755)
Q	0.0000	0.0044	0.0053
J	0.0000	0.7099	0.8494
p-value		0.3995	0.6540

The first instrument set consists of $z_t = \{1, C_t/C_{t-1}\}$, which yields a just-identified model resulting in the GMM objective function equaling $Q(\widehat{\theta}) = 0$. The second instrument set is $z_t = \{1, C_t/C_{t-1}, R_t\}$, resulting in an over-identified system with the degree of overidentification equal to $N - K = 1$. The third and final instrument set is $z_t = \{1, C_t/C_{t-1}, R_t, r_t\}$, which leads to $N - K = 2$ overidentifying restrictions. An overall test of the model using the J test shows support for the C-CAPM specification as the overidentifying restrictions in the case of the second and third instrument sets, are not rejected

at the 5% level.
The discount parameter estimates Table 9.5 are robust across all instrument sets with values of around 0.998. As $\delta = 1/(1+r)$ where r is the constant real discount rate, this implies that

$$r = \frac{1}{\widehat{\delta}} - 1 = \frac{1}{0.998} - 1 = 0.002\,,$$

or 0.2%, which appears to be quite low. The estimates of the relative risk aversion parameter, γ, range from 0.554 to 1.0234. These estimates suggest that relative risk aversion is also low over the sample period considered. However, the standard errors are relatively large to the point estimates suggesting that γ is being estimated relatively imprecisely compared with the discount parameter.

9.10 Testing a CKLS Model of Interest Rates

As observed in Chapter 6, the short term interest rate (or short maturity bond yield) plays an important role in financial markets because of the expectations hypothesis which posits that long rates are simply a weighted average of expected future short rates. Consequently, modelling the dynamics of the short term interest rate has received a lot of attention in the financial econometrics literature. One of the most popular models of the short term interest rate is the CKLS model due to Chan, Karolyi, Longstaff and Saunders (1992), given by

$$\Delta r_{t+1} = \alpha + \beta r_t + \sigma r_t^{\gamma} z_{t+1}\,, \qquad z_{t+1} \sim iid(0,1)\,, \tag{9.110}$$

in which r_t is the short term rate of interest rate, $\Delta r_{t+1} = r_{t+1} - r_t$, z_{t+1} is a disturbance term and $\theta = \{\alpha, \beta, \sigma, \gamma\}$ are parameters. Although this model relates the current change in interest rates in a linear fashion to the level of the interest rate in period t, it also allows the variance of the model to be heteroskedastic. In fact, the variance of the change in interest rates is also related to the level of the interest rate in period t because z_{t+1} is scaled by a factor σr_t^{γ}. Consequently the parameter γ is commonly referred to as the levels effect parameter.
Figure 9.11 plots the levels and differences of the 1, 3 and 6 month United States zero coupon bond yields for the period December 1946 to February 1991. It is apparent from the behaviour of the yields particularly in the early 1980s, a period known as the Volcker experiment because interest rates to fluctuate freely, that there is a relationship between the high levels of interest rates and increased volatility in the interest rate changes.
Of course it is possible to go ahead and estimate α and β by ordinary least squares and correct the standard errors for heteroskedasticity using White's method as outlined in Chapter 3. This approach is sub-optimal, however, because the ordinary least squares estimate of σ will be be biased and perhaps

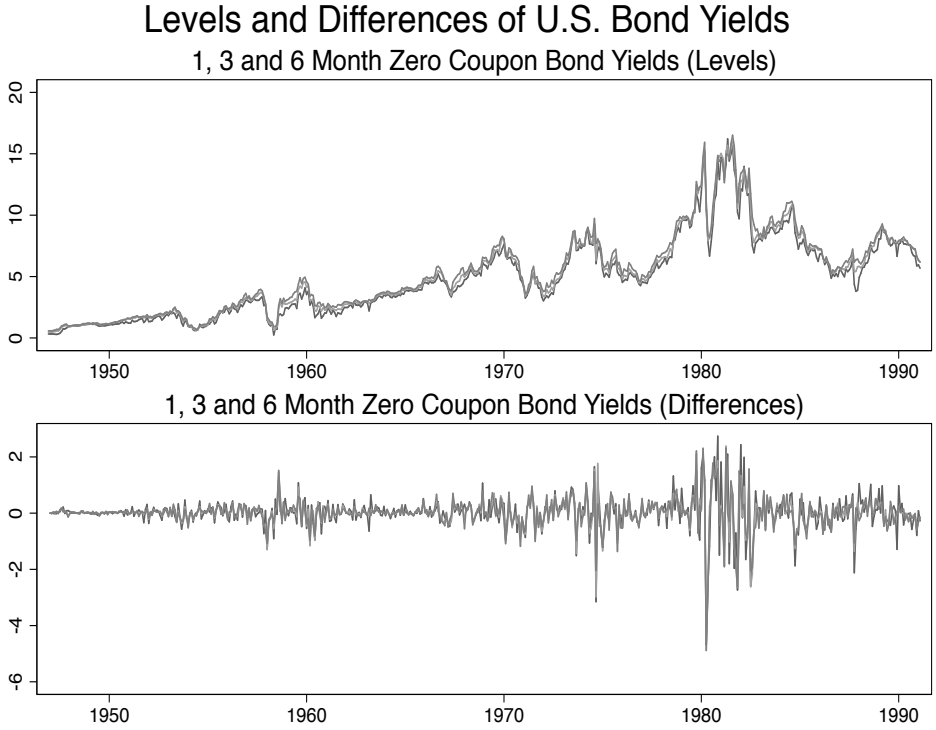


Figure 9.11: Monthly United States zero coupon bond yields for the period December 1946 to February 1991. The top panel shows the levels of 1, 3 and 6 month bond yields, while the lower panel shows the differences in these yields.

more important, no estimate of γ can be obtained. This is particularly important because testing hypotheses on the level parameter is of particular interest in this model.
Instead of using ordinary least squares, it is possible to estimate all the parameters of the model by GMM. The full set of moment conditions for estimating the parameters of the model (9.110) are

$$m_t = \begin{bmatrix} m_{1t} \\ m_{2t} \\ m_{3t} \\ m_{4t} \end{bmatrix} = \begin{bmatrix} (\Delta r_{t+1} - \alpha - \beta r_t) \\ (\Delta r_{t+1} - \alpha - \beta r_t) r_t \\ (\Delta r_{t+1} - \alpha - \beta r_t)^2 - \sigma^2 r_t^{2\gamma} \\ ((\Delta r_{t+1} - \alpha - \beta r_t)^2 - \sigma^2 r_t^{2\gamma}) r_t \end{bmatrix}, \tag{9.111}$$

where the first two moment conditions relate to the mean of Δr_{t+1} and the second two moment conditions relate to the variance of Δr_{t+1}. As there are $\theta = \{\alpha, \beta, \gamma, \sigma^2\}$ four parameters and four moment conditions, the model is just identified.

Solving the system of equations for θ yields the method of moments estimates of the parameters of the CKLS interest rate model in equation (9.110). The method of moments estimators for θ obtained when using the United States zero coupon bond yield data from Figure 9.11 for maturities of 1, 3 and 6 months, together with data for maturities of 9 months and 10 years, are reported in Table 9.6. The model is estimated using a one-step estimator with a a heteroskedastic-consistent weighting matrix used to compute the standard errors. The results also show a strong levels effect in United States interest rates that changes over the maturity of the asset. The parameter estimates of γ increase in magnitude as maturity increases from 0 to 6 months, reach a peak at 6 months, and then taper off thereafter.

Table 9.6: GMM estimation of the CKLS interest rate model. A one-step estimator is used and robust standard errors are reported in parentheses. Data are monthly United States zero coupon bond yields (with maturities of 1, 3, 6 and 9 months and 10 years) for the period December 1946 to February 1991 ($T = 53$).

	1 month	3 months	6 months	9 months	12 months
α	0.1057	0.0896	0.0894	0.0903	0.0462
	(0.05872)	(0.05379)	(0.05677)	(0.05629)	(0.02669)
β	−0.0198	−0.0154	−0.0147	−0.0146	−0.0056
	(0.01591)	(0.01373)	(0.01369)	(0.01332)	(0.005638)
σ	0.0494	0.0353	0.0260	0.0267	0.0286
	(0.01849)	(0.01673)	(0.01531)	(0.01542)	(0.006727)
γ	1.3512	1.4222	1.5333	1.5145	1.1779
	(0.1883)	(0.2320)	(0.2833)	(0.2783)	(0.1136)

Returning to the estimation of the CKLS model in equation (9.110), there are two important constraints to be tested on the value of γ.

1. $H_0 : \gamma = 0.5$:
   The CKLS model with $\gamma = 0.5$ corresponds to the square-root or CIR model proposed by (Cox, Ingersoll and Ross, 1985). The importance of this restriction stems from the fact that the CIR model is more analytically tractable that the CKLS model and allows the development of some important theoretical results relating to the term structure of interest rates and the pricing of bonds.

2. $H_0 : \gamma = 1.0$:
   The CKLS model with $\gamma = 1.0$ corresponds to the model proposed by Vasicek (1977) This restriction is important because it is essentially a test of whether or not there is a levels effect at all.

One way to proceed would be to test the hypothesis directly using the the estimated parameters and associated standard errors from Table **??**. Another approach is to estimate the model imposing the restriction and then test the over identifying restrictions using the Hansen-Sargan J test. The tests statistics from this latter approach, together with the estimate of γ in the unrestricted model for comparative purposes, are reported in Table 9.7. These estimates are produced using a two-step GMM with a heteroskedastic consistent weighting matrix.

Table 9.7: GMM tests of the restriction $\gamma = 0.5$ and $\gamma = 1.0$ imposed on the CKLS model. A two-step estimator is used with a weighting matrix that is robust to heteroskedasticity. Data are monthly United States zero coupon bond yields (with maturities of 1, 3, 6 and 9 months and 10 years) for the period December 1946 to February 1991 ($T = 53$).

	Zero Coupon Bond Maturity				
	1 month	3 month	6 month	9 month	10 years
γ	1.350	1.424	1.525	1.510	1.178
	(0.188)	(0.232)	(0.285)	(0.279)	(0.114)
$H_0 : \gamma = 0.5$:					
$Q_T(\widehat{\theta})$	0.021	0.017	0.013	0.014	0.045
J_{HS}	11.04	8.859	7.043	7.159	23.84
p value	0.001	0.003	0.008	0.008	0.000
$H_0 : \gamma = 1.0$:					
$Q_T(\widehat{\theta})$	0.005218	0.004930	0.004732	0.004691	0.004467
J_{HS}	2.7657	2.6131	2.5081	2.4862	2.3675
p value	0.09631	0.1060	0.1133	0.1148	0.1239

On the evidence provided by these results, there is a strong rejection of the CIR square-root model, $H_0 : \gamma = 0.5$. Simple inspection of the point estimates of γ and their standard errors indicate that this restriction will not be supported and this intuition is borne out in every case by the value of the Hansen-Sargent J test. A completely different picture emerges for the Vasicek model, $H_0 : \gamma = 1.0$. In every case the null hypothesis cannot be rejected, a set of results that is strongly supportive of the model for these data.

9.11 Exercises

1. **Method of Moments Estimation**

Program files `gmm_mom.*`

This exercise is based on the following data.

Table 9.8:

Calculation of various empirical moments for the $T = 10$ observations on the variable y_t.

t	y_t	y_t^2	y_t^3	y_t^4	$y_t - m_1$	$(y_t - m_1)^2$	$\ln(y_t)$	$1/y_t$
1	2.0	4.0	8.0	16.0	−2.90	8.410	0.693	0.500
2	7.0	49.0	343.0	2401.0	2.10	4.410	1.946	0.143
3	5.0	25.0	125.0	625.0	0.10	0.010	1.609	0.200
4	6.0	36.0	216.0	1296.0	1.10	1.210	1.792	0.167
5	4.0	16.0	64.0	256.0	−0.90	0.810	1.386	0.250
6	8.0	64.0	512.0	4096.0	3.10	9.610	2.079	0.125
7	5.0	25.0	125.0	625.0	0.10	0.010	1.609	0.200
8	5.0	25.0	125.0	625.0	0.10	0.010	1.609	0.200
9	4.0	16.0	64.0	256.0	−0.90	0.810	1.386	0.250
10	3.0	9.0	27.0	81.0	−1.90	3.610	1.099	0.333
Moment:	4.9	26.9	160.9	1027.70	0.000	2.890	1.521	0.237

The method of moments estimator of θ for a just identified model is based on the solution of

$$M_T(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} m_t(y_t;\widehat{\theta}) = 0\,,$$

where $m_t = m_t(y_t;\theta)$ is the moment at time t.

(a) The normal distribution

$$f(y;\theta) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{1}{2}\Big(\frac{y-\mu}{\sigma}\Big)^2\right],$$

has moments $\mathrm{E}[y_t] = \mu_0$, $\mathrm{E}[y_t^2] = \sigma_0^2 + \mu_0^2$ and $\mathrm{E}[(y_t - \mu)^4] = 3\sigma_0^4$.

i. Estimate μ and σ^2 using the moment conditions

$$m_t = \begin{bmatrix} y_t - \mu \\ y_t^2 - \sigma^2 - \mu^2 \end{bmatrix}.$$

ii. Estimate μ and σ^2 using the moment conditions

$$m_t = \begin{bmatrix} y_t - \mu \\ (y_t - \mu)^4 - 3\sigma^4 \end{bmatrix}.$$

Compare the two sets of point estimates of μ and σ^2.

(b) The Student t distribution

$$f(y;\theta) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\Gamma[\nu/2]}\left[1+\frac{(y-\mu)^2}{\nu}\right]^{-(\nu+1)/2},$$

has moments $\mathrm{E}[y_t] = \mu_0$, $\mathrm{E}[(y_t-\mu_0)^2] = \nu_0/(\nu_0-2)$, and $\mathrm{E}[(y_t-\mu_0)^4] = 3\nu_0^2/(\nu_0-2)(\nu_0-4)$.

i. Estimate μ and ν using the moment conditions

$$m_t = \begin{bmatrix} y_t-\mu \\ (y_t-\mu)^2 - \dfrac{\nu}{\nu-2} \end{bmatrix}.$$

ii. Estimate μ and ν using the moment conditions

$$m_t = \begin{bmatrix} y_t-\mu \\ (y_t-\mu)^4 - \dfrac{3\nu^2}{(\nu-2)(\nu-4)} \end{bmatrix}.$$

Compare the two sets of point estimates of μ and ν.

(c) The gamma distribution

$$f(y;\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\exp[-\beta y]y^{\alpha-1}, \qquad y \geqslant 0, \alpha > 0, \beta > 0,$$

has moments $\mathrm{E}[y_t] = \alpha_0/\beta_0$, $\mathrm{E}[y_t^2] = \alpha_0(\alpha_0+1)/\beta_0^2$, and $\mathrm{E}[1/y_t] = \beta_0/(\alpha_0-1)$.

i. Estimate α and β using the moment conditions

$$m_t = \begin{bmatrix} y_t - \dfrac{\alpha}{\beta} \\ \dfrac{1}{y_t} - \dfrac{\beta}{\alpha-1} \end{bmatrix}.$$

ii. Estimate α and β using the moment conditions

$$m_t = \begin{bmatrix} y_t - \dfrac{\alpha}{\beta} \\ y_t^2 - \dfrac{\alpha(\alpha+1)}{\beta^2} \end{bmatrix}.$$

Compare the two sets of point estimates for α and β. Briefly discuss the potential problems associated with the estimates based on the moment conditions in part (ii).

(d) The Pareto distribution

$$f(y) = \frac{\alpha y_{\min}}{y^{\alpha+1}}, \qquad y > y_{\min},$$

has moments $\mathrm{E}[y_t] = \alpha_0 y_{\min}/(\alpha_0-1)$ and $\mathrm{E}[y_t^2] = \alpha_0 y_{\min}^2/((\alpha_0-1)^2(\alpha_0-2))$, where $y_{\min}$ represents the minimum value.

i. Choosing $y_{\min} = 2$, estimate α using the moment condition

$$m_t = \left[y_t - \frac{\alpha y_{\min}}{\alpha - 1} \right].$$

ii. Choosing $y_{\min} = 2$, estimate α using the moment condition

$$m_t = \left[y_t^2 - \frac{\alpha y_{\min}^2}{(\alpha - 1)^2(\alpha - 2)} \right].$$

2. **Estimating a Gamma Distribution**

Program files `gmm_gamma.*`

Example 9.1 (gmm_gamma.*)**. Estimating the Gamma Distribution**
Consider estimating α in the gamma distribution using the $T = 10$ observations on y_t in Table 9.8 based on the over-identified model in Example **??**. The moments are $m_t = [y_t - \alpha,\ y_t^2 - \alpha(\alpha+1)]'$ and

$$\begin{aligned} M_T(\alpha) &= \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha+1) \end{bmatrix}, \\ W_T(\alpha) &= \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha+1) \end{bmatrix} \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha+1) \end{bmatrix}'. \end{aligned}$$

Choosing the first moment as the starting value $\alpha_{(0)} = T^{-1}\sum_{t=1}^{T} y_t = 4.9$, then

$$M\left(\alpha_{(0)}\right) = \begin{bmatrix} -0.0001 \\ -2.0106 \end{bmatrix}, \qquad W\left(\alpha_{(0)}\right) = \begin{bmatrix} 2.8900 & 29.0901 \\ 29.0901 & 308.1327 \end{bmatrix}$$

resulting in the value of the objective function of

$$\begin{aligned} Q\left(\alpha_{(0)}\right) &= \frac{1}{2} M'\left(\alpha_{(0)}\right) W\left(\alpha_{(0)}\right)^{-1} M\left(\alpha_{(0)}\right) \\ &= \frac{1}{2} \begin{bmatrix} -0.0001 \\ -2.0106 \end{bmatrix}' \begin{bmatrix} 2.8900 & 29.0901 \\ 29.0901 & 308.1327 \end{bmatrix}^{-1} \begin{bmatrix} -0.0001 \\ -2.0106 \end{bmatrix} = 0.1319. \end{aligned}$$

The first and second numerical derivatives of (**??**) evaluated at $\alpha_{(0)}$ are

$$G_{(0)} = \left.\frac{dQ_T(\alpha)}{d\alpha}\right|_{\alpha=4.9} = 0.0709, \quad H_{(0)} = \left.\frac{d^2 Q_T(\alpha)}{d\alpha^2}\right|_{\alpha=4.9} = 0.3794,$$

which yield the first Newton-Raphson iteration of

$$\alpha_{(1)} = \alpha_{(0)} - H_{(0)}^{-1} G_{(0)} = 4.9 - \frac{0.0709}{0.3794} = 4.7130.$$

Iterating a second time gives

$$\alpha_{(2)} = \alpha_{(1)} - H_{(1)}^{-1} G_{(1)} = 4.7130 - \frac{-0.0021}{0.3905} = 4.7184,$$

which produces a first derivative of $Q_T(\alpha)$ at $\alpha_{(2)}$ of 1.6×10^{-6}, suggesting that the algorithm has converged. As $H_{(2)} = 0.3912$, $\text{var}(\widehat{\alpha}) = 1/(10 \times 0.3912) = 0.2556$ and the standard error is $\text{se}(\widehat{\alpha}) = \sqrt{0.2556} = 0.5056$. □

This exercise is based on the data in Table 9.8. The generalised method of moments estimator is based on the solution of

$$\widehat{\theta} = \arg\min_{\theta} Q_T(\theta), \tag{9.112}$$

where

$$Q_T(\theta) = \frac{1}{2} M_T'(\theta) W_T^{-1} M_T(\theta), M_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t, W_T(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} m_t m_t',$$

and where $m_t = m_t(y_t; \theta)$ is the moment at time t.

(a) The first two uncentered moments of the gamma distribution with $\beta_0 = 1$ are $\text{E}[y_t] = \alpha_0$, $\text{E}[y_t^2] = \alpha_0(\alpha_0 + 1)$.

i. Using the starting value $\theta_{(0)} = \left\{\alpha_{(0)} = \overline{y}\right\}$ compute the following

$$M_T(\theta_{(0)}),\ W_T(\theta_{(0)}),\ Q_T(\theta_{(0)}),\ G_{(0)} = \frac{\partial Q_T(\theta_{(0)})}{\partial \theta},\ H_{(0)} = \frac{\partial^2 Q_T(\theta_{(0)})}{\partial \theta \partial \theta'},$$

where the derivatives are computed numerically.

ii. Use the results in part (i) to compute the Newton-Raphson update

$$\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1} G_{(0)},$$

and compare $Q_T(\theta_{(0)})$ and $Q_T(\theta_{(1)})$. Iterate until convergence to find the GMM parameter estimate of α and its standard error using

$$se(\widehat{\alpha}) = \sqrt{\frac{1}{T} H^{-1}(\widehat{\alpha})}.$$

(b) Repeat part (a) using $\text{E}[y_t] = \alpha_0$, $\text{E}[y_t^2] = \alpha_0(\alpha_0 + 1)$, $\text{E}[1/y_t] = 1/(\alpha_0 - 1)$.

(c) Repeat part (a) with α and β unknown using $\text{E}[y_t] = \alpha_0/\beta_0$, $\text{E}[y_t^2] = \alpha_0(\alpha_0 + 1)/\beta_0^2$, and $\text{E}[1/y_t] = \beta_0/(\alpha_0 - 1)$.

3. **Estimating a Student t Distribution**

Program files `gmm_student.*`

This exercise is based on the data in Table 9.8. The generalised method of moments estimator is based on the solution of

$$\widehat{\theta} = \arg\min_{\theta} Q_T(\theta) \,, \tag{9.113}$$

where

$$Q_T(\theta) = \frac{1}{2} M_T'(\theta) W_T^{-1} M_T(\theta) \,, M_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} m_t \,, W_T(\widehat{\theta}) = \frac{1}{T}\sum_{t=1}^{T} m_t m_t' \,,$$

and where $m_t = m_t(y_t; \theta)$ is the moment at time t.

(a) Consider the following moments of the Student t distribution $\mathrm{E}[y_t] = \mu_0$, $\mathrm{E}[(y_t - \mu_0)^2] = \nu_0/(\nu_0 - 2)$.

i. Using the starting value $\theta_{(0)} = \left\{\mu_{(0)} = \overline{y}, \nu_{(0)} = 5\right\}$ compute the following

$$M_T(\theta_{(0)}),\ W_T(\theta_{(0)}),\ Q_T(\theta_{(0)}),\ G_{(0)} = \frac{\partial Q_T(\theta_{(0)})}{\partial \theta},\ H_{(0)} = \frac{\partial^2 Q_T(\theta_{(0)})}{\partial \theta \partial \theta'} \,,$$

where the derivatives are computed numerically.

ii. Use the results in part (i) to compute the Newton-Raphson update

$$\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1} G_{(0)} \,,$$

and compare $Q_T(\theta_{(0)})$ and $Q_T(\theta_{(1)})$. Iterate until convergence to find the GMM parameter estimate of θ and its standard error using

$$se(\widehat{\theta}) = \sqrt{\frac{1}{T} H^{-1}(\widehat{\theta})} \,.$$

(b) Repeat part (a) using the moments $\mathrm{E}[y_t] = \mu_0$, $\mathrm{E}[(y_t - \mu_0)^2] = \nu_0/(\nu_0 - 2)$, $\mathrm{E}[(y_t - \mu_0)^4] = 3\nu_0^2/(\nu_0 - 2)(\nu_0 - 4)$.

4. **The Consumption Capital Asset Pricing Model**

Program files `gmm_ccapm.*`
Data files `ccapm.*`

The data are 238 observations on the real United States consumption ratio c_{t+1}/c_t $(CRATIO)$, the real Treasury bill rate r_{t+1} (R), and the real value weighted returns e_{t+1} (E). This is the adjusted Hansen and Singleton (1982) data set used in their original paper. Consider the first-order condition of the C-CAPM

$$\mathrm{E}_t[\beta_0(c_{t+1}/c_t)^{-\gamma_0}(1+r_{t+1})-1]=0\,,$$

where c_t is real consumption and r_t is the real interest rate. The parameters are the discount factor, β, and the relative risk aversion coefficient γ.

(a) Estimate the parameters $\theta=\{\beta,\gamma\}$ by GMM using $w_t=\{1,c_t/c_{t-1}\}$ as instruments and starting values $\theta_{(0)}=\{\beta=1.0,\gamma=1.0\}$. Interpret the parameter estimates and test the number of over-identifying restrictions.

(b) Repeat part (a) using $w_t=\{1,c_t/c_{t-1},r_t\}$ as instruments.

(c) Repeat part (a) using $w_t=\{1,c_t/c_{t-1},r_t,e_t\}$, as instruments.

(d) Repeat part (a) using $w_t=\{1,c_t/c_{t-1},r_t,e_t,e_{t-1}\}$, as instruments.

(e) Compare the parameter estimates across the four sets of instruments in parts (a) to (d).

5. **Decomposing International Equity returns**

Program files	`gmm_equity.*`
Data files	`equity_decomposition.*`

The data file contains daily equity prices (P) on the SP500, FTSE100 and the EURO50, from 29 July 2004 to 3 March 2009. Let $e_{i,t}=r_{i,t}-\overline{r}_i$ represent the centered daily percentage equity returns where $r_{i,t}=100(\ln P_{i,t}-\ln P_{i,t-1})$.

(a) Compute the covariance matrix of $e_{i,t}$ and interpret the empirical moments.

(b) Consider the latent factor model

$$e_{i,t}=\lambda_i s_t+\phi_i z_{i,t},\qquad i=1,2,3\,,$$

where $\{s_t,z_{1,t},z_{2,t},z_{3,t}\}$ are *iid* $(0,1)$. Show that the theoretical moments of $e_{i,t}$ are

$$\begin{aligned}\mathrm{E}[e_{i,t}^2] &= \lambda_i^2+\phi_i^2, \qquad i=1,2,3\,,\\ \mathrm{E}[e_{i,t}e_{j,t}] &= \lambda_i\lambda_j, \qquad i\neq j\,.\end{aligned}$$

(c) Using the moment structure in part (b) estimate the parameters $\theta = \{\lambda_1, \lambda_2, \lambda_3, \phi_1, \phi_2, \phi_3\}$ by GMM. Interpret the parameter estimates by computing the relative contributions of the common factor (s_t) and the idiosyncratic factors $(z_{1,t}, z_{2,t}, z_{3,t})$ given by

$$\frac{\lambda_i^2}{\lambda_i^2 + \phi_i^2}, \qquad \frac{\phi_i^2}{\lambda_i^2 + \phi_i^2}, \qquad i = 1, 2, 3\,.$$

(d) Show that the factor decomposition in part (b) gives an exact decomposition of the empirical covariance matrix of $e_{i,t}$ computed in (a).

6. **Modelling Contagion in the Asian Crisis**

Program files	`gmm_contagion.*`
Data files	`contagion.*`

The data file contains daily data on the exchange rate $(s_{i,t})$ of the following seven countries: South Korea, Indonesia, Malaysia, Japan, Australia, New Zealand and Thailand. The sample period is 2 June 1997 to 31August 1998, a total of 319 observations. Let $e_{i,t} = r_{i,t} - \bar{r}_i$, represent the zero-mean daily percentage currency returns where $r_{i,t} = 100(\ln s_{i,t} - \ln s_{i,t-1})$.

(a) Estimate the latent factor model of the exchange rate

$$e_{i,t} = \lambda_i s_t + \phi_i z_{i,t} + \gamma_i z_{7,t}, \qquad i = 1, 2, \cdots, 7\,,$$

by GMM with $\gamma_7 = 0$ and where the factors $\{s_t, z_{1,t}, \cdots, z_{7,t}\}$ are all *iid* $(0, 1)$

(b) For each country, estimate the proportion of volatility arising from contagion by evaluating

$$\frac{\gamma_i^2}{\lambda_i^2 + \phi_i^2 + \gamma_i^2}, \qquad i = 1, 2, ..., 6\,.$$

(c) Perform a test of contagion $\gamma_1 = \gamma_2 = \cdots, = \gamma_6 = 0$.

7. **Consistency of GMM**

Program files	`gmm_consistency.*`

(a) Simulate $T = 100000$ observations from the gamma distribution with parameters $\theta_0 = \{\alpha_0 = 10, \beta_0 = 1\}$. Compute the GMM population objective function

$$Q_T(\alpha) = \frac{1}{2} M_T'(\alpha) W_T^{-1}(\alpha) M_T(\alpha),$$

for values of $\alpha = \{6, \cdots, 14\}$ and $\beta = 1$, where

$$M_T(\alpha) = \frac{1}{T}\sum_{t=1}^{T} m_t, W_T(\alpha) = \frac{1}{T}\sum_{t=1}^{T} m_t m_t', m_t = \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha+1) \\ y_t^{-1} - (\alpha-1)^{-1} \end{bmatrix}.$$

(b) Repeat part (a) for the finite samples of size $T = 10, 100, 200, 400$ and discuss the consistency property of the GMM estimator of α.

(c) Repeat parts (a) and (b) with $m_t = \begin{bmatrix} y_t - \alpha & y_t^2 - \alpha(\alpha+1) \end{bmatrix}'$.

(d) Repeat parts (a) and (b) with $m_t = \begin{bmatrix} y_t - \alpha \end{bmatrix}$.

8. **Monte Carlo Evidence for the Gamma Model**

Program files `gammasim.*`

Let y_t have an *iid* gamma distribution for $t = 1, \cdots, T$ with shape parameter $\alpha_0 = \{1, 2, 3, 4, 5\}$ and scale parameter $\beta_0 = 1$.

(a) Investigate the finite sample bias and variance of the following estimators of α_0 assuming that $\beta_0 = 1$ is known, for sample sizes of $T = \{50, 100, 200, 400\}$ with 10000 replications.

i. The maximum likelihood estimator.

ii. The GMM estimator based on the first moment of the gamma distribution $m_t = [y_t - \alpha]'$.

iii. The GMM estimator based on the first two moments of the gamma distribution $m_t = [y_t - \alpha,\ y_t^2 - \alpha(\alpha+1)]'$.

(b) Compute the finite sample size of the t statistic $(\widehat{\alpha} - \alpha_0)/se(\widehat{\alpha})$ based on the three estimators in part (a) as well as the J_{HS} statistic of misspecification.

(c) Compute the finite sample power of the t statistic $(\widehat{\alpha} - 1)/se(\widehat{\alpha})$ based on the three estimators in part (a) for parameter values of $\alpha_0 = \{1.05, 1.10, \cdots, 1.30\}$.

(d) Suppose that the data generating process is now an exponential distribution with parameter values of $\alpha_0 = \{1.0, 1.2, \cdots, 2.0\}$ whereas the estimators in part (a) are still based on the gamma distribution. Redo parts (a) to (c) and discuss the effects of misspecification on the sampling properties of the three estimators, their associated test statistics and the J_{HS} statistic of misspecification.

9. **Level Effects in United States Interest Rates**

Program files	`gmm_level.*`
Data files	`level.*`

The data are monthly and cover the period December 1946 to February 1991. The zero coupon bonds have maturities of 0, 1, 3, 6, 9 months and 10 years.

(a) For each yield, estimate the following interest rate equation by GMM

$$r_{t+1} - r_t = \alpha + \beta r_t + \sigma r_t^{\gamma} z_{t+1},$$

where z_t is *iid* $(0, 1)$, and the instrument set is $w_t = \{1, r_t\}$;

(b) For each yield test the following restrictions: $\gamma = 0.0$, $\gamma = 0.5$ and $\gamma = 1.0$.

(c) If the level effect model of the interest rate captures time-varying volatility and $\alpha, \beta \simeq 0$, then

$$\mathrm{E}\left[\left(\frac{r_{t+1} - r_t}{r_t^{\gamma}}\right)^2\right] \simeq \sigma^2 .$$

Plot the series

$$\frac{r_{t+1} - r_t}{r_t^{\gamma}},$$

for $\gamma = 0.0, 0.5, 1.0, 1.5$, and discuss the properties of the series.

10. **Risk Aversion and the Equity Premium Puzzle**

Program files	`gmm_risk_aversion.*`
Data files	`equity_mp.*`

In this exercise, the risk aversion parameter is estimated by GMM using the data originally used by Mehra and Prescott (1985) in their work on the equity premium puzzle. The data are annual for the period 1889 to 1978, a total of 91 observations on the following United States variables: the real stock price, S_t; real dividends, D_t; real per capita consumption, C_t; the nominal risk free rate on bonds, expressed as a per annum percentage, R_t; and the price of consumption goods, P_t.

(a) Compute the following returns series for equities, bonds and consumption, respectively,

$$\begin{aligned} R_{s,t+1} &= \frac{S_{t+1} + D_t - S_t}{S_t} \\ R_{b,t+1} &= (1 + R_t)(\frac{P_t}{P_{t+1}}) - 1 \\ R_{c,t+1} &= \frac{C_{t+1} - C_t}{C_t} . \end{aligned}$$

(b) Consider the first-order conditions of the C-CAPM model

$$\begin{aligned} \mathrm{E}_t[\beta(1 + R_{c,t+1})^{-\gamma}(1 + R_{b,t+1}) - 1] &= 0 \\ \mathrm{E}_t[\beta(1 + R_{c,t+1})^{-\gamma}(1 + R_{s,t+1}) - 1] &= 0 , \end{aligned}$$

where the parameters are the discount factor, β, and the relative risk aversion coefficient, γ. Estimate the parameters $\theta = \{\beta, \gamma\}$ by GMM with instruments $w_t = \{1, R_{c,t}\}$. Interpret the parameter estimates and test the number of over-identifying restrictions.

(c) Repeat part (b) with instruments $w_t = \{1, R_{c,t}, R_{b,t}\}$.

(d) Repeat part (b) with instruments $w_t = \{1, R_{c,t}, R_{b,t}, R_{s,t}\}$.

(e) Discuss the robustness properties of the parameter estimates of θ in parts (b) to (d).

Chapter 10

Maximum Likelihood

10.1 Introduction

The models in Part **??** are linear and estimation of the unknown parameters given by θ based on ordinary least squares. Chapters 8 and **??** have examined two alternative methods to ordinary least squares. In this chapter the maximum likelihood estimator of θ is introduced. Maximum likelihood estimation is a general method for estimating the parameters of financial econometric models both linear and nonlinear. Maximum likelihood plays a central role in both estimation and inference: maximum likelihood estimators posses a number of desirable properties and three important test procedures are based on the likelihood principal.

Maximum likelihood estimation of θ requires that the following conditions are satisfied.
(1) The probability distribution of the observed variable y_t is known.
(2) The specifications of the moments of the distribution of y_t are known.
(3) The probability distribution of y_t can be evaluated for all values of the parameters, θ.

10.2 Distributions in Finance

10.2.1 Returns

A common assumption adopted in finance is that the returns on an asset, r_t, are normally distributed with mean μ and variance σ^2. The returns distribution is formally written as

$$f(r_t; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(r_t - \mu)^2}{2\sigma^2}\right]$$

This assumption is stated more compactly as

$$r_t \sim N(\mu, \sigma^2)$$

Very often the assumption of a normal distribution refers to the disturbance term of a financial model from which it may be derived that returns are also distributed as a normal distribution. Two important models in finance which adopt the assumption of normally distributed disturbance terms are as follows.

i **Constant mean model**

$$r_t = \mu + u_t\,, \qquad u_t \sim iid\, N(0, \sigma^2)$$

It follows directly from the distributional assumption on the disturbance term that the distribution of r_t is

$$f(r_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(r_t - \mu)^2}{2\sigma^2}\right],$$

where $\theta = \{\mu, \sigma^2\}$ is the parameter vector.

ii **CAPM**

$$r_t - r_{ft} = \alpha + \beta\left(r_{mt} - r_{ft}\right) + u_t\,, \qquad u_t \sim iid\, N(0, \sigma^2)$$

In this instance the distribution of r_t is

$$f(r_t|\, r_{mt}, r_{ft}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(r_t - r_{ft} - \alpha - \beta\left(r_{mt} - r_{ft}\right))^2}{2\sigma^2}\right],$$

with parameter vector $\theta = \{\alpha, \beta, \sigma^2\}$.

Figure 10.1 gives histograms for the monthly log returns on six assets. Superimposed on each histogram is the normal distribution with μ and σ^2 the sample estimates. All six log-return distributions are reasonably well represented by the normal distribution. The main distributional characteristic not captured by the normal distribution is the peakedness of the distribution with the normal distribution underestimating the observed mode.
A formal test of normality is given in following Table 10.1 based on the Jarque-Bera test. Four of the six log-returns pass the normality test whereas gold and Microsoft do not. To capture potential nonnormal asset returns the assumption of a normal distribution is sometimes replaced by the assumption that returns follow a Student t distribution given by

$$f(r_t; \mu, \sigma^2, \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\sigma^2\nu}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1 + \frac{(r_t - \mu)^2}{\sigma^2\nu}\right)^{-\left(\frac{\nu+1}{2}\right)}$$

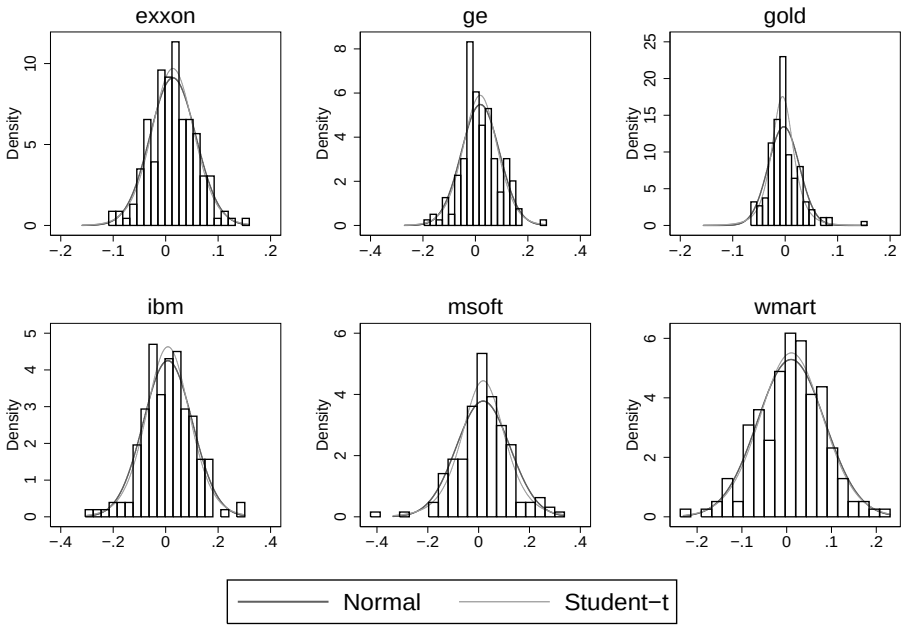


Figure 10.1: Histogram of the monthly log returns to five United States stocks and the commodity gold for the period April 1990 to July 2004. Overlaid on the histograms are the normal distribution and the t distribution.

where ν represents the degrees of freedom parameter.
The parameter ν provides additional flexibility to model the empirical distribution, but note, however, that in the special case of $\nu \to \infty$, the t distribution becomes the normal distribution. The plots in Figure 10.1 illustrates how the Student t distribution does a better job of capturing the distributional features of returns than the normal distribution, particularly in terms of the peakedness of the distribution.

10.2.2 Prices

By definition, log-returns are computed as the change over time in the natural logarithm of the price P_t, of an asset

$$r_t = \log P_t - \log P_{t-1}$$

Assuming that r_t is normally distributed with mean μ and variance σ^2, implies that P_t conditional on the lagged price P_{t-1}, is log-normally distributed

$$f(P_t | P_{t-1}; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2} P_t} \exp\left[-\frac{(\log P_t - (\mu + \log P_{t-1}))^2}{2\sigma^2} \right]$$

Table 10.1

Jarque-Bera test of normality on the monthly log returns to five United States stocks and gold for the period April 1990 to July 2004.

Stock	Mean	St. Dev.	JB	pv
Exxon	0.0172	0.0441	1.8094	0.4047
General Electric	0.0233	0.0736	2.3042	0.3160
Gold	0.0004	0.0296	186.4526	0.0000
IBM	0.0121	0.0940	4.3729	0.1123
Microsoft	0.0209	0.1068	17.5927	0.0001
Walmart	0.0135	0.0761	1.1493	0.5629

A plot of the distribution of P_t based on log-normality is given in Figure 10.2 with parameters $\mu = 1$, $\sigma^2 = 0.4$ and $P_{t-1} = 1$. An important feature of this distribution is that it is just defined over the positive region resulting in the distribution exhibiting positive skewness.

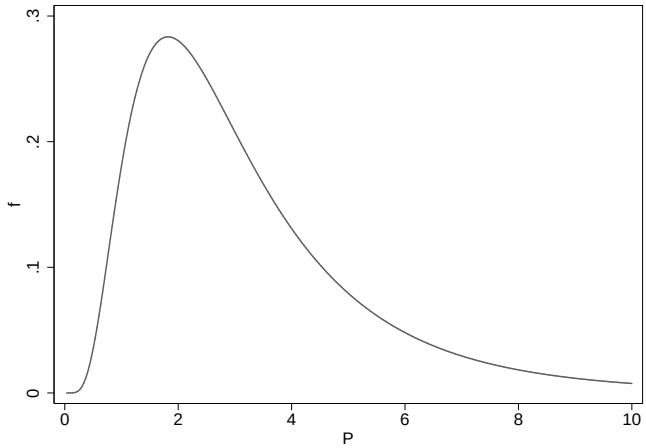


Figure 10.2: A plot of the lognormal distribution for equity prices, P_t, with parameters $\mu = 1$, $\sigma^2 = 0.4$ and $P_{t-1} = 1$.

Two important examples of the incidence of the log-normal distribution in finance are the following.

i **Simple gross returns to an asset,** R_{gt}:
Log-returns are defined as

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log\left(1 + \frac{P_t - P_{t-1}}{P_{t-1}}\right) = \log(1 + R_t) = \log(R_{gt})$$

It follows that if r_t is normally distributed with mean, μ, and variance, σ^2, then $R_{gt} = P_t/P_{t-1}$ is log-normally distributed

$$f(R_{gt};\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}R_{gt}} \exp\left[-\frac{(\log R_{gt}-\mu)^2}{2\sigma^2}\right]$$

ii **Black-Scholes option pricing model**:
In the Black-Scholes model it is assumed that the price of an asset is given by

$$\log P_t - \log P_{t-1} = \mu + u_t\,, \qquad u_t \sim N(0,\sigma^2)$$

The use of cumulative normal distribution functions in computing the option price directly come from the lognormality assumption.

10.2.3 Yields and Interest Rates

An important distributional property of yields or the interest rate on a bond, y_t, is that it must be positive. This suggests that the assumption of normality is potentially inappropriate, as it can assign a non-zero probability to the possibility of negative yields. This is especially the case when yields are relatively close to the zero boundary, as is the case during the recent global financial crisis.
Assuming a normal distribution results in a positive probability of negative yields.To capture the fact that yields and nominal interest rates are positive random variables two possible distributions are:

i Lognormal distribution with parameter vector $\theta = \{\mu, \sigma^2\}$

$$f(y_t;\theta) = \frac{1}{\sqrt{2\pi\sigma^2}y_t} \exp\left[-\frac{(\log y_t-\mu)^2}{2\sigma^2}\right]$$

ii Gamma distribution with parameter vector $\theta = \{\alpha, \beta\}$

$$f\left(y_t;\theta\right) = \frac{\alpha^\beta}{\Gamma\left(\beta\right)} y_t^{\beta-1} e^{-\alpha y_t}$$

where α, β are parameters controlling the shape of the distribution and $\Gamma(\nu)$ represents the gamma function

$$\Gamma(\nu) = \int_0^\infty s^{\nu-1}e^{-s}ds$$

Figure 10.3 is a plot of the histogram for daily observations on the monthly Eurodollar rate from 4 January 1971 to the 31 December 1991, $T = 5477$. The leakage of density into the negative region under the assumption that the interest rates are normally distributed is clearly shown. The gamma distribution on the other hand is a more appropriate distributional assumption in this case.The assumption of a gamma distribution underlies the continuous time model of interest rates proposed by Cox, Ingersoll and Ross (1985).

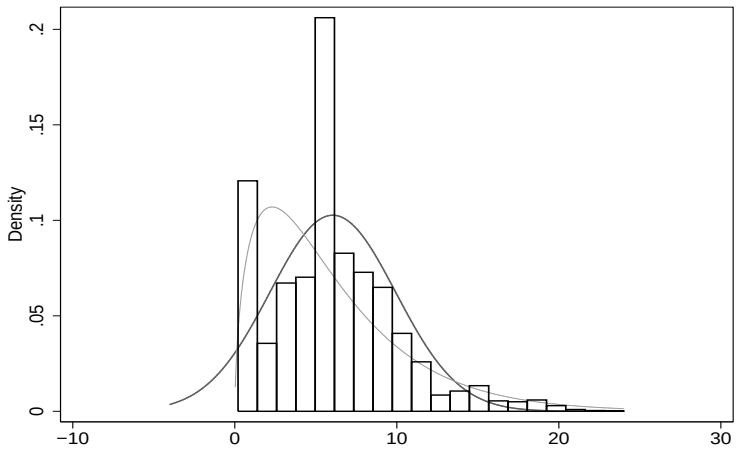


Figure 10.3: Histogram of the monthly Eurodollar rate from 4 January 1971 to the 31 December 1991. Superimposed on the histogram is a plot of the best fitting normal and gamma distributions.

10.2.4 Durations

Figure 10.4 gives a histogram of the duration between trades on the United States stock AMR, the parent company of American Airlines. The data are recorded at second intervals from 9.30am to 4.00pm on 1 August 2006, a total of 23368 observations with a sample mean of 7.2799 seconds between trades over the day. The shape of the empirical distribution suggests an exponential distribution

$$f(y_t;\alpha) = \alpha e^{-\alpha y_t}, \qquad \alpha > 0$$

This is verified by superimposing the exponential distribution over the histogram with α chosen as 0.1374.
The choice of α is based on the maximum likelihood estimator which is derived below. A number of generalizations of the exponential distribution to model the time duration between trades can be specified.

1. Weibull distribution

$$f(y_t;\alpha,\beta) = \alpha\beta y_t^{\beta-1} e^{-\alpha y_t^{\beta}}, \qquad \alpha,\beta > 0\,.$$

   A special case of the Weibull distribution is the exponential which occurs by imposing the restriction $\beta = 1$.

2. Gamma distribution

$$f(y_t;\mu,\sigma,\nu) = \frac{\sigma^{-\nu}}{\Gamma(\nu)} (y_t-\mu)^{\nu-1} e^{-(y_t-\mu)/\sigma}$$

   A special case is the exponential distribution where $\mu = 0$, $\sigma = 1/\alpha$ and $\nu = 1$.

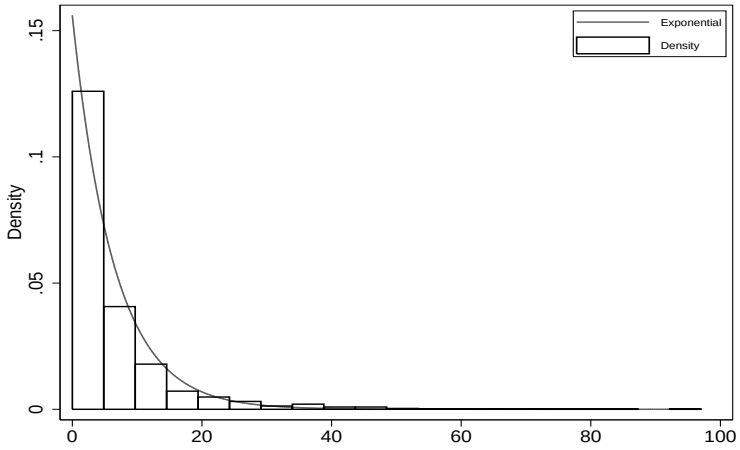


Figure 10.4: Histogram of the durations between ARM trades with an exponential distribution superimposed.The data are durations between AMR trades measured in intervals of one second from 9.30am to 4.00pm on 1 August 2006.

10.3 Estimation by Maximum Likelihood

The previous section showed that financial variables at each point in time t, are summarized by their probability distributions, $f(y_t;\theta)$. In this section the maximum likelihood estimator of an unknown set of parameters θ related to this distribution is introduced.

10.3.1 The Log-Likelihood Function

For each distribution there is a set of parameters, θ, which are to be estimated from a sample of T observations $\{y_1, y_2, \cdots, y_T\}$ by the method of maximum likelihood. The objective function of maximum likelihood estimation is the sample average of the logarithm of the probability distribution of y_t evaluated at each point in time, formally known as the log-likelihood function. Four types of (average) log-likelihoods are presented depending upon the type of probability distribution that governs y_t.

1. Identically and Independently Distributed Observations

$$\log L(\theta) = \frac{1}{T}\sum_{t=1}^{T} \log f(y_t;\theta)$$

Case 1 is the simplest of the four cases where the distribution of y_t is identical at each point in time as well as being independent of its lags. In this case y_t is identically and independently distributed, abbreviated as *iid*.

2. Non-identically Distributed Obsevations

$$\log L(\theta) = \frac{1}{T}\sum_{t=1}^{T} \log f(y_t|x_t;\theta)$$

Case 2 allows for the distribution of y_t to be conditional on a set of explanatory variables x_t, which has the effect of shifting the distribution of y_t over time.

3. Dependently Distributed Observations

$$\log L(\theta) = \frac{1}{T-1}\sum_{t=2}^{T} \log f(y_t|y_{t-1};\theta)$$

Case 3 allows for the distribution of y_t to be dependent upon its past value, y_{t-1}. This case can also be generalized to allow for several lags, $\{y_{t-1}, y_{t-2}, \cdots, y_{t-p}\}$, in which case the averaging of the probability distribution of y_t at each point in time is now restricted to the last $T-p$ observations.

4. Non-identically and Dependently Distributed Observations

$$\log L(\theta) = \frac{1}{T-1}\sum_{t=2}^{T} \log f(y_t|x_t, y_{t-1};\theta)$$

Case 4 is the most general model where y_t is conditional on both a set of explanatory variables x_t as well as its past values y_{t-1}.
The maximum likelihood estimator of θ, denoted $\widehat{\theta}$, is found where the log-likelihood function, $\log L(\theta)$, is at its maximum

$$\widehat{\theta} = \arg\max_{\theta} \log L(\theta)\,.$$

The motivation of maximum likelihood estimation is to regard $\{y_1, y_2, \cdots, y_T\}$ as a realised dataset and search over potential parameter values of θ that are 'most likely' to generate the observed data.
As the log-likelihood is equivalent to the average of the logarithm of the joint probability distribution of $\{y_1, y_2, \cdots, y_T\}$, maximum likelihood estimation reverses the roles of y_t and θ, where in the joint probability context y_t is random and θ is fixed, whereas in the likelihood context y_t is fixed and θ is random.

10.3.2 The Maximum Likelihood Estimator

The analytical solution of the maximum likelihood problem is found using basic calculus. The $\log L(\theta)$ function is maximised by taking the first derivative with respect to θ which is known as the gradient (or score) and denoted by $G(\theta)$. In the case of a single parameter $(K = 1)$ the gradient scalar is based on the derivative

$$G(\theta) = \frac{d \log L(\theta)}{d\theta}$$

In the case of K parameters the $(K \times 1)$ gradient vector is based on partial derivatives

$$G(\theta) = \frac{\partial \log L(\theta)}{\partial \theta}$$

The maximum likelihood estimator of θ, denoted $\widehat{\theta}$, occurs where all of the gradients are zero

$$G(\widehat{\theta}) = \left. \frac{\partial \log L(\theta)}{\partial \theta} \right|_{\theta=\widehat{\theta}} = 0\,.$$

To establish that the maximum likelihood estimator maximizes $\log L(\theta)$ (as opposed to finding a turning point which is not the maximum), the second derivative of the log-likelihood function, known as the Hessian and denoted $H(\theta)$, is needed. For the single $(K = 1)$ parameter case

$$H(\theta) = \frac{d^2 \log L(\theta)}{\partial \theta^2}$$

For the K-parameter case the $(K \times K)$ matrix of second derivatives is

$$H(\theta) = \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'}$$

The condition required for the maximum likelihood estimator to maximise $\log L(\theta)$, is that the Hessian when evaluated at $\widehat{\theta}$ is negative definite, or

$$H(\widehat{\theta}) = \left. \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\widehat{\theta}}\,.$$

The conditions for negative definiteness are

$$\begin{array}{lcl} \text{1-parameter} & : & H_{11} < 0 \\ \text{2-parameters} & : & H_{11} < 0, H_{11}H_{22} - H_{12}H_{21} > 0 \end{array}$$

where H_{ij} is the ij^{th} element of $H(\widehat{\theta})$.

10.4 Maximum Likelihood Estimators of Financial Models

Maximum likelihood estimation is not only applicable to the parameters of financial distributions. In this section, the principal of maximum likelihood estimation is illustrated for a series of important models in financial econometrics.

10.4.1 Duration Model of Trades

The duration model of trades, y_t, is given by the exponential distribution

$$f(y_t;\alpha) = \alpha e^{-\alpha y_t}, \qquad \alpha > 0$$

where y_t is assumed to be *iid*. To derive the maximum likelihood estimator of $\theta = \{\alpha\}$, the construction of the log-likelihood function, $\log L(\theta)$, is based on Case 1

$$\begin{aligned}
\log L(\theta) &= \frac{1}{T}\sum_{t=1}^{T} \log f(y_t;\theta) \\
&= \frac{1}{T}\sum_{t=1}^{T} \log\left(\theta \exp\left[-\theta y_t\right]\right) \\
&= \frac{1}{T}\sum_{t=1}^{T} \log\theta - \frac{1}{T}\sum_{t=1}^{T} \theta y_t \\
&= \log\theta - \theta\frac{1}{T}\sum_{t=1}^{T} y_t
\end{aligned}$$

Using the durations between trades data for the company AMR measured at one second intervals on 1 August 2006, $\log L(\theta)$ is plotted in Figure 10.5 for $\theta = \alpha$ in the range $0 < \theta \leq 1$. The $\log L(\theta)$ function appears to be highest for values of θ in the range $0.1 \leq \theta \leq 0.2$.

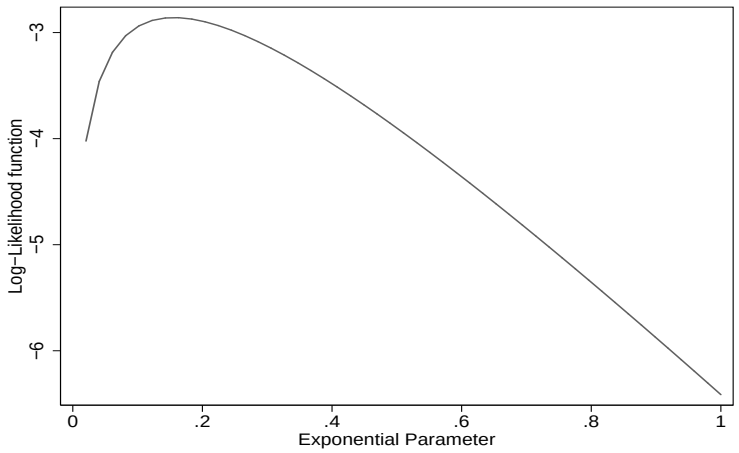


Figure 10.5: Log-likelihood function with respect to the parameter of the exponential model of durations, θ. The data are durations between AMR trades measured in intervals of one second from 9.30am to 4.00pm on 1 August 2006.

To derive the maximum likelihood estimator formally the gradient function is

$$G(\theta) = \frac{d \log L(\theta)}{d\theta} = \frac{1}{\theta} - \frac{1}{T}\sum_{t=1}^{T} y_t$$

Setting $G(\widehat{\theta}) = 0$ gives

$$\frac{1}{\widehat{\theta}} - \frac{1}{T}\sum_{t=1}^{T} y_t = 0$$

Solving for $\widehat{\theta}$ gives the maximum likelihood estimator as

$$\widehat{\theta} = \frac{T}{\sum_{t=1}^{T} y_t} = \frac{1}{\overline{y}}$$

the reciprocal of the sample mean of the durations data.
Since the sample mean is $\overline{y} = 7.2699$, the maximum likelihood estimator is

$$\widehat{\theta} = \frac{1}{7.2799} = 0.1374$$

confirming the intuition provided by the Figure of $\log L(\theta)$.
To check the second-order condition for a maximum, the Hessian is

$$H(\theta) = \frac{d^2 \log L(\theta)}{d\theta^2} = -\frac{1}{\theta^2}\,.$$

Evaluating this expression at $\widehat{\theta}$ shows that this condition is satisfied

$$H(\widehat{\theta}) = -\frac{1}{\widehat{\theta}^2} = -\frac{1}{0.1374^2} = -52.9700 < 0\,.$$

As the Hessian for this model is negative for all values of θ the log-likelihood function is globally concave.

10.4.2 Constant Mean Model

The constant mean model of returns is

$$r_t = \mu + u_t\,, \qquad u_t \sim iid\, N(0, \sigma^2). \tag{10.1}$$

Although the distributional assumption made here applies to the disturbance term, u_t, the distribution of interest rates, r_t, may be deduced. It follows by simple inspection of (10.1) that

$$u_t = (r_t - \mu) \sim iid\, N(0, \sigma^2) \Longrightarrow r_t \sim iid\, N(\mu, \sigma^2)$$

so that the sample data, r_t, can simply be regarded as independent drawings from a normal distribution normally with mean μ and variance σ^2. The distribution of r_t with parameter vector $\theta = \{\mu, \sigma^2\}$ is therefore given by

$$f(r_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(r_t - \mu)^2}{2\sigma^2}\right].$$

To derive the maximum likelihood estimator of θ, the log-likelihood function based on Case 1 is

$$\begin{aligned}
\log L(\theta) &= \frac{1}{T}\sum_{t=1}^{T}\log f(r_t;\theta) \\
&= \frac{1}{T}\sum_{t=1}^{T}\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(r_t-\mu)^2}{2\sigma^2}\right]\right) \\
&= \frac{1}{T}\sum_{t=1}^{T}\log\frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{T}\sum_{t=1}^{T}\frac{(r_t-\mu)^2}{2\sigma^2} \\
&= -\frac{1}{2}\log 2\pi - \frac{1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\frac{1}{T}\sum_{t=1}^{T}(r_t-\mu)^2
\end{aligned}$$

with (2×1) gradient vector

$$G(\theta) = \begin{bmatrix} \dfrac{\partial \log L(\theta)}{\partial \mu} \\ \dfrac{\partial \log L(\theta)}{\partial \sigma^2} \end{bmatrix} = \begin{bmatrix} \dfrac{1}{\sigma^2 T}\sum_{t=1}^{T}(y_t-\mu) \\ -\dfrac{1}{2\sigma^2} + \dfrac{1}{2\sigma^4 T}\sum_{t=1}^{T}(y_t-\mu)^2 \end{bmatrix}.$$

The maximum likelihood estimator is found by setting $G(\widehat{\theta}) = 0$

$$\begin{aligned}
\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T}(y_t-\widehat{\mu}) &= 0 \\
-\frac{1}{2\widehat{\sigma}^2} + \frac{1}{2\widehat{\sigma}^4 T}\sum_{t=1}^{T}(y_t-\widehat{\mu})^2 &= 0,
\end{aligned}$$

and solving for $\widehat{\mu}$ and $\widehat{\sigma}^2$.
Solving for the maximum likelihood estimator of μ yields

$$\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T}(y_t-\widehat{\mu}) = \sum_{t=1}^{T}(y_t-\widehat{\mu}) = \sum_{t=1}^{T}y_t - T\widehat{\mu} = 0$$

so that the maximum likelihood estimator is given by

$$T\widehat{\mu} = \sum_{t=1}^{T}y_t \Rightarrow \widehat{\mu} = \frac{1}{T}\sum_{t=1}^{T}y_t = \overline{y},$$

which is the sample mean of y_t.
From the second expression in $G(\widehat{\theta})$ the maximum likelihood estimator of σ^2

is the sample variance (with T in the denominator) because

$$\begin{aligned} -\frac{1}{2\widehat{\sigma}^2}+\frac{1}{2\widehat{\sigma}^4 T}\sum_{t=1}^{T}(y_t-\widehat{\mu})^2 &= 0 \\ \frac{1}{2\widehat{\sigma}^2} &= \frac{1}{2\widehat{\sigma}^4 T}\sum_{t=1}^{T}(y_t-\widehat{\mu})^2 \\ \frac{2\widehat{\sigma}^4}{2\widehat{\sigma}^2} &= \frac{1}{T}\sum_{t=1}^{T}(y_t-\widehat{\mu})^2 \\ \widehat{\sigma}^2 &= \frac{1}{T}\sum_{t=1}^{T}(y_t-\widehat{\mu})^2 \end{aligned}$$

Differentiating $G(\theta)$ with respect to θ now yields a (2×2) Hessian matrix given by

$$\begin{aligned} H(\theta) &= \begin{bmatrix} \dfrac{\partial^2 \ln L(\theta)}{\partial\mu^2} & \dfrac{\partial^2 \ln L_T(\theta)}{\partial\mu\partial\sigma^2} \\ \dfrac{\partial^2 \ln L_T(\theta)}{\partial\sigma^2\partial\mu} & \dfrac{\partial^2 \ln L_T(\theta)}{\partial(\sigma^2)^2} \end{bmatrix} \\ &= \begin{bmatrix} -\dfrac{1}{\sigma^2} & -\dfrac{1}{\sigma^4 T}\sum_{t=1}^{T}(y_t-\mu) \\ -\dfrac{1}{\sigma^4 T}\sum_{t=1}^{T}(y_t-\mu) & \dfrac{1}{2\sigma^4}-\dfrac{1}{\sigma^6 T}\sum_{t=1}^{T}(y_t-\mu)^2 \end{bmatrix} \end{aligned}$$

Evaluating the Hessian at $\widehat{\theta}$

$$H(\widehat{\theta})=\begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}=\begin{bmatrix} -\dfrac{1}{\widehat{\sigma}^2} & 0 \\ 0 & -\dfrac{1}{2\widehat{\sigma}^4} \end{bmatrix}$$

which uses

$$\frac{1}{T}\sum_{t=1}^{T}(y_t-\widehat{\mu})=0, \qquad \frac{1}{T}\sum_{t=1}^{T}(y_t-\mu)^2=\widehat{\sigma}^2$$

where the first condition is based on $G(\widehat{\theta})$ and the second condition is based on the maximum likelihood estimator of σ^2.
The relevant conditions for a maximum are satisfied as

$$\begin{aligned} H_{11} &= -\frac{1}{\widehat{\sigma}^2}<0 \\ H_{11}H_{22}-H_{12}H_{21} &= \left(-\frac{1}{\widehat{\sigma}^2}\right)\left(-\frac{1}{2\widehat{\sigma}^4}\right)-(0)\,(0)=\frac{1}{2\widehat{\sigma}^6}>0 \end{aligned}$$

10.4.3 CAPM

The excess returns to and asset and the market may be defined as

$$\begin{aligned} y_t &= r_t - r_{ft} \qquad \text{[Excess return to asset]} \\ x_t &= r_{mt} - r_{ft} \qquad \text{[Excess return to market]}, \end{aligned}$$

respectively. Written in terms of these excess returns, the CAPM becomes

$$y_t = \alpha + \beta x_t + u_t\,, \qquad u_t \sim iid\ N(0,\sigma^2)\,, \tag{10.2}$$

in which once more the distribution assumption refers to the disturbance term u_t. If follows from (10.2) that

$$y_t - \alpha - \beta x_t \sim iid\, N(0,\sigma^2) \quad \Longrightarrow \quad y_t \sim iid\, N(\alpha + \beta x_t, \sigma^2)\,.$$

Taking r_{mt} as given, the distribution of r_t with parameter vector $\theta = \{\alpha, \beta, \sigma^2\}$ is

$$f(r_t|\, r_{mt}, r_{ft}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(r_t - r_{ft} - \alpha - \beta\left(r_{mt} - r_{ft}\right)\right)^2}{2\sigma^2} \right].$$

To derive the maximum likelihood estimator of $\theta = \{\alpha, \beta, \sigma^2\}$, $\log L(\theta)$ is based on Case 2

$$\begin{aligned} \log L(\theta) &= \frac{1}{T}\sum_{t=1}^{T} \log f(y_t|\, x_t; \theta) \\ &= \frac{1}{T}\sum_{t=1}^{T} \log\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_t - \alpha - \beta x_t)^2}{2\sigma^2} \right] \right) \\ &= \frac{1}{T}\sum_{t=1}^{T} \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{T}\sum_{t=1}^{T} \frac{(r_t - \alpha - \beta x_t)^2}{2\sigma^2} \\ &= -\frac{1}{2}\log 2\pi - \frac{1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\frac{1}{T}\sum_{t=1}^{T}(r_t - \alpha - \beta x_t)^2 \end{aligned}$$

The (3×1) gradient vector is

$$G(\theta) = \begin{bmatrix} \dfrac{\partial \log L(\theta)}{\partial \alpha} \\ \dfrac{\partial \log L(\theta)}{\partial \beta} \\ \dfrac{\partial \log L(\theta)}{\partial \sigma^2} \end{bmatrix} = \begin{bmatrix} \dfrac{1}{\sigma^2 T}\sum_{t=1}^{T}(y_t - \alpha - \beta x_t) \\ \dfrac{1}{\sigma^2 T}\sum_{t=1}^{T}(y_t - \alpha - \beta x_t)\, x_t \\ -\dfrac{1}{2\sigma^2} + \dfrac{1}{2\sigma^4 T}\sum_{t=1}^{T}(y_t - \alpha - \beta x_t)^2 \end{bmatrix}$$

Setting $G(\widehat{\theta}) = 0$ requires that

$$G(\widehat{\theta}) = \begin{bmatrix} \frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta}x_t) \\ \frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta}x_t)x_t \\ -\frac{1}{2\widehat{\sigma}^2} + \frac{1}{2\widehat{\sigma}^4 T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta}x_t)^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This is a linear system of three equations and three unknowns with solution

$$\begin{aligned} \widehat{\alpha} &= \overline{y} - \widehat{\beta}\overline{x} \\ \widehat{\beta} &= \frac{\sum_{t=1}^{T}(y_t - \overline{y})(x_t - \overline{x})}{\sum_{t=1}^{T}(x_t - \overline{x})^2} \\ \widehat{\sigma}^2 &= \frac{1}{T}\sum_{t=1}^{T}(y_t - \widehat{\alpha} - \widehat{\beta}x_t)^2 \end{aligned}$$

The expressions for $\widehat{\alpha}$ and $\widehat{\beta}$ are in fact the ordinary least squares estimators from Chapter 3, demonstrating that for this class of models the ordinary least squares estimator is equivalent to the maximum likelihood estimator. The expression for $\widehat{\sigma}^2$ shows that maximum likelihood estimator is equivalent to the ordinary least squares estimator apart from the degrees of freedom correction given by $T - 2$ for the bivariate model in the case of ordinary least squares.

Differentiating $G(\theta)$ with respect to θ now yields a (3×3) Hessian matrix given by

$$\begin{aligned} H(\theta) &= \begin{bmatrix} \frac{\partial^2 \log L(\theta)}{\partial \alpha^2} & \frac{\partial^2 \log L(\theta)}{\partial \alpha \partial \beta} & \frac{\partial^2 \log L(\theta)}{\partial \alpha \partial \sigma^2} \\ \frac{\partial^2 \log L(\theta)}{\partial \beta \partial \alpha} & \frac{\partial^2 \log L(\theta)}{\partial \beta^2} & \frac{\partial^2 \log L(\theta)}{\partial \beta \partial \sigma^2} \\ \frac{\partial^2 \log L(\theta)}{\partial \sigma^2 \partial \alpha} & \frac{\partial^2 \log L(\theta)}{\partial \sigma^2 \partial \beta} & \frac{\partial^2 \log L(\theta)}{\partial (\sigma^2)^2} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{1}{\sigma^2} & -\frac{1}{\sigma^2 T}\sum_{t=1}^{T} x_t & -\frac{1}{\sigma^4 T}\sum_{t=1}^{T} u_t \\ -\frac{1}{\sigma^2 T}\sum_{t=1}^{T} x_t & -\frac{1}{\sigma^2 T}\sum_{t=1}^{T} x_t^2 & -\frac{1}{\sigma^4 T}\sum_{t=1}^{T} u_t x_t \\ -\frac{1}{\sigma^4 T}\sum_{t=1}^{T} u_t & -\frac{1}{\sigma^4 T}\sum_{t=1}^{T} u_t x_t & \frac{1}{2\sigma^4} - \frac{1}{\sigma^6 T}\sum_{t=1}^{T} u_t^2 \end{bmatrix} \end{aligned}$$

where $u_t = y_t - \alpha - \beta x_t$ is the disturbance term.

Evaluating this expression at $\widehat{\theta}$ gives

$$H(\widehat{\theta}) = \begin{bmatrix} -\frac{1}{\widehat{\sigma}^2} & -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t & -\frac{1}{\widehat{\sigma}^4 T}\sum_{t=1}^{T} \widehat{u}_t \\ -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t & -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t^2 & -\frac{1}{\widehat{\sigma}^4 T}\sum_{t=1}^{T} \widehat{u}_t x_t \\ -\frac{1}{\widehat{\sigma}^4 T}\sum_{t=1}^{T} \widehat{u}_t & -\frac{1}{\widehat{\sigma}^4 T}\sum_{t=1}^{T} \widehat{u}_t x_t & \frac{1}{2\widehat{\sigma}^4} - \frac{1}{\widehat{\sigma}^6 T}\sum_{t=1}^{T} \widehat{u}_t^2 \end{bmatrix},$$

where $\widehat{u}_t = y_t - \widehat{\alpha} - \widehat{\beta} x_t$ represents the residual.
The Hessian simplifies to

$$H(\widehat{\theta}) = \begin{bmatrix} -\frac{1}{\widehat{\sigma}^2} & -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t & 0 \\ -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t & -\frac{1}{\widehat{\sigma}^2 T}\sum_{t=1}^{T} x_t^2 & 0 \\ 0 & 0 & -\frac{1}{2\widehat{\sigma}^4} \end{bmatrix}$$

which uses the results

$$\sum_{t=1}^{T} \widehat{u}_t = 0, \qquad \sum_{t=1}^{T} \widehat{u}_t x_t = 0, \qquad \widehat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T} \widehat{u}_t^2$$

where the first two are based on the first order conditions and third based on the solution of $\widehat{\sigma}^2$.
It is easy to verify that this matrix is negative definite, thereby satisfying the second order conditions for a maximum.

10.4.4 Vasicek Interest Rate Model

The discrete time model of the interest rate, r_t, proposed by Vasicek (1977) assumes that the dynamics of the interest rate obey the AR(1) equation

$$r_t = \alpha + \rho r_{t-1} + u_t\,, \qquad u_t \sim iid\, N\left(0, \sigma^2\right)$$

resulting in the distribution of r_t with parameter vector $\theta = \left\{\alpha, \rho, \sigma^2\right\}$ given by

$$f\left(r_t \,|\, r_{t-1}; \alpha, \rho, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(r_t - \alpha - \rho r_{t-1})^2}{2\sigma^2}\right]$$

Examples of the conditional distribution of r_t are given in Figure 10.6 with conditional mean $\alpha + \rho r_{t-1}$ of 5%, 10% and 15%, and setting $\sigma^2 = 0.5$.
To derive the maximum likelihood estimator of $\theta = \{\alpha, \rho, \sigma^2\}$, $\log L(\theta)$ is based on Case 3

$$\log L(\alpha, \rho, \sigma^2) = -\frac{1}{2}\log 2\pi - \frac{1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\frac{1}{T-1}\sum_{t=2}^{T}(r_t - \alpha - \rho r_{t-1})^2$$

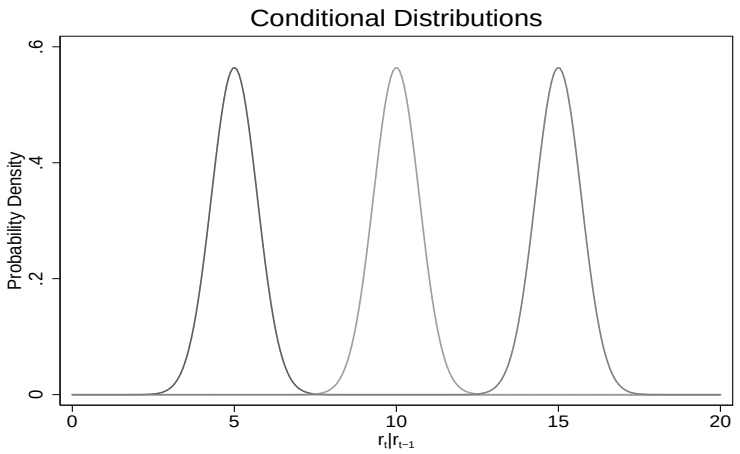


Figure 10.6: Conditional distributions of the Vasicek model, taking $\alpha + \rho r_{t-1}$ to be 5%, 10% and 15% respectively.

This expression is equivalent to the form of $\log L(\theta)$ for the CAPM with r_{t-1} replaced by x_t. This observation suggests that the maximum likelihood estimators are

$$\begin{aligned}
\widehat{\alpha} &= \overline{r}_t - \widehat{\rho}\overline{r}_{t-1} \\
\widehat{\rho} &= \frac{\sum_{t=2}^{T}(r_t - \overline{r}_t)(r_{t-1} - \overline{r}_{t-1})}{\sum_{t=2}^{T}(r_{t-1} - \overline{r}_{t-1})^2} \\
\widehat{\sigma}^2 &= \frac{1}{T-1}\sum_{t=2}^{T}(r_t - \widehat{\alpha} - \widehat{\rho} r_{t-1})^2
\end{aligned}$$

where

$$\overline{r}_t = \frac{1}{T-1}\sum_{t=2}^{T} r_t, \qquad \overline{r}_{t-1} = \frac{1}{T-1}\sum_{t=2}^{T} r_{t-1}$$

10.5 Maximum Likelihood Estimation by Numerical Methods

In many important cases of interest in financial econometrics, the maximum likelihood estimator, $\widehat{\theta}$, cannot be solved for analytically. In other words, the system of first order conditions for a maximum of the log-likelihood function, $G(\widehat{\theta}) = 0$, does not have an analytical solution and $\widehat{\theta}$ cannot be expressed in terms of the data.
An important case in point concerns relates to the CAMP and specifically the potential problem with CAPM based on the assumption of normal dis-

turbances. Specifically, is there are outliers in the disturbance distribution, these can skew the distribution resulting in distorted estimates of the beta-risk parameter. One potential solution is to replace the normality assumption by specifying a fat-tailed distribution that captures the outliers and helps to reduce distortion of the estimates of beta-risk parameter due to outliers. This class of models is referred to as a robust CAPM as the parameters of the model are invariant to the presence of outliers.
In specfying the robust version of the CAPM, the normal distribution with zero mean and variance σ^2, is replaced by the Student t distribution which is standardised to have zero mean and variance σ^2. The robust CAPM, once again expressed in terms of excess returns as in equation (10.2), is

$$y_t = \alpha + \beta x_t + u_t\,, \qquad u_t \sim iid\ St(0, \sigma^2, \nu)$$

where $y_t = r_t - r_{ft}$ is the excess return on the asset and $x_t = r_{mt} - r_{ft}$ is the excess return on the market. The notation $St(0, \sigma^2, \nu)$ represents the standardised Student t distribution given by

$$f(u_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\sigma^2(\nu-2)}\Gamma\left(\frac{\nu}{2}\right)}\left(1 + \frac{u_t^2}{\sigma^2(\nu-2)}\right)^{-\left(\frac{\nu+1}{2}\right)}$$

This form of the Student t distribution is slightly different to the expression given before in the lecture to ensure that u_t has variance given by σ^2. The parameter ν is the degrees of freedom parameter which captures the effects outliers in the tails of the distribution.
Expressing this distribution in terms of y_t gives

$$f(y_t|\, x_t; \theta) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\sigma^2(\nu-2)}\Gamma\left(\frac{\nu}{2}\right)}\left(1 + \frac{(y_t - \alpha - \beta x_t)^2}{\sigma^2(\nu-2)}\right)^{-\left(\frac{\nu+1}{2}\right)}$$

To derive the maximum likelihood estimator of $\theta = \{\alpha, \beta, \sigma^2, \nu\}$, $\log L(\theta)$ is based on Case 2

$$\begin{aligned}
\log L(\theta) &= \frac{1}{T}\sum_{t=1}^{T} \log f(y_t|\, x_t; \theta) \\
&= \log\Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2}\log\sigma^2 - \frac{1}{2}\log(\pi(\nu-2)) \\
&\quad - \log\Gamma\left(\frac{\nu}{2}\right) - \left(\frac{\nu+1}{2}\right)\frac{1}{T}\sum_{t=1}^{T}\log\left(1 + \frac{(y_t - \alpha - \beta x_t)^2}{\sigma^2(\nu-2)}\right)
\end{aligned}$$

Since this expression for the log-likelihood function is nonlinear in the parameters, a numerical solution is adopted to compute $\widehat{\theta}$.

In these instances, a numerical procedure is used which takes some starting value for θ, denoted $\theta_{(0)}$, and proceeds iteratively until convergence. Convergence occurs where $G(\widehat{\theta}_{(k)}) \simeq 0$, as then $\widehat{\theta}_{(k+1)} \simeq \widehat{\theta}_{(k)}$. This process is known as numerical optimisation and full description of the problem and the various optimisation algorithms often used in this context is provided in Appendix D. The results of estimating the robust version of the CAPM by maximum likelihood methods are given Table 10.2. The robust estimates correspond to the case where the distribution is Student t, while for comparison the CAPM estimates based on normality are also presented. The estimates of the degrees of freedom parameter, $\widehat{\nu}$, suggest the presence of outliers especially for Gold $(\widehat{\nu} = 4.251)$, IBM $(\widehat{\nu} = 5.681)$ and Microsoft $(\widehat{\nu} = 4.710)$. Despite evidence of outliers, a comparison of the estimates of the beta-risk $(\widehat{\beta})$, for both normal and Student t disturbances suggests that none of the estimates are particularly badly distorted by outliers in the returns.

10.6 Properties of Maximum Likelihood Estimators

10.6.1 Consistency

An important property of the maximum likelihood estimator is that under certain conditions, known as regularity conditions, it is consistent. Consistency refers to the distance between between $\widehat{\theta}_T$ and θ_0 as the sample size T increases. A consistent estimator is one where more information, in terms of a larger sample size, results in it being closer to the true population parameter θ_0. Figure 10.7 demonstrates the situation in which $\widehat{\theta}_T$ is is the sample mean used to estimate the population mean $\theta_0 = 1$ for samples of increasing size. For sample less than $T < 50$, $\widehat{\theta}_T$ is rather erratic. For samples greater than $T > 100$, $\widehat{\theta}_T$ becomes closer and closer to θ_0. Notice that even though $\widehat{\theta}_T$ approaches θ_0 it never quite equals θ_0 because $\widehat{\theta}_T$ is a random variable and not a deterministic process. This idea is expressed formally as

$$\lim_{T\to\infty} \widehat{\theta}_T \neq \theta_0$$

To reflect the difference in convergence between a deterministic process and a stochastic process, consistency in the case of the random variable $\widehat{\theta}_T$, is written as

$$\text{plim}(\widehat{\theta}_T) = \theta_0$$

where 'plim' denotes convergence in probability.

10.6.2 Efficiency

Efficiency is about the amount of scatter (variance) of $\widehat{\theta}_T$ around θ_0 as the sample size T increases. Inspection of the Figure 10.7 shows that for each T $\widehat{\theta}_T$

Table 10.2

Estimating the robust version of the CAPM for the monthly excess log-returns to five United States financial assets and the commodity gold for the period May 1990 to July 2004. Standard errors are in parentheses.

Stock	Distribution	$\widehat{\alpha}$	$\widehat{\beta}$	$\widehat{\sigma}$	$\widehat{\nu}$	$\log L$
Exxon	Normal	0.0120	0.5020	0.0380		1.8470
		(0.003)	(0.063)	(0.002)		
	Student t	0.0120	0.5020	0.0380	11.7210	1.8520
		(0.003)	(0.063)	(0.003)	(12.017)	
GE	Normal	0.0160	1.1440	0.0550		1.4890
		(0.004)	(0.083)	(0.003)		
	Student t	0.0140	1.1630	0.0550	13.6500	1.4920
		(0.004)	(0.087)	(0.004)	(17.233)	
Gold	Normal	−0.0030	−0.0980	0.0290		2.1050
		(0.002)	(0.047)	(0.001)		
	Student t	−0.0050	−0.1110	0.0300	4.2510	2.1710
		(0.002)	(0.042)	(0.003)	(1.355)	
IBM	Normal	0.0040	1.2050	0.0780		1.1280
		(0.006)	(0.143)	(0.003)		
	Student t	0.0060	1.2140	0.0780	5.6810	1.1600
		(0.005)	(0.131)	(0.007)	(2.510)	
Microsoft	Normal	0.0120	1.4470	0.0870		1.0280
		(0.007)	(0.176)	(0.003)		
	Student t	0.0130	1.3540	0.0870	4.7100	1.0710
		(0.006)	(0.137)	(0.009)	(1.792)	
Walmart	Normal	0.0070	0.8680	0.0660		1.3000
		(0.005)	(0.105)	(0.003)		
	Student t	0.0080	0.8920	0.0660	11.6880	1.3030
		(0.005)	(0.109)	(0.004)	(11.546)	

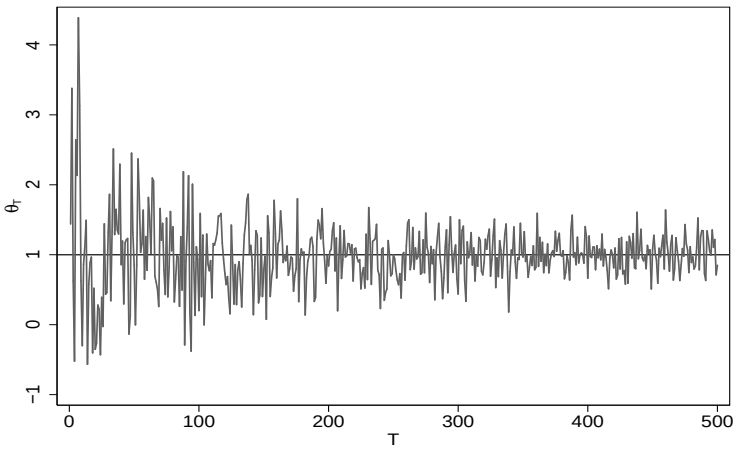


Figure 10.7: .

is scattered around θ_0 reflecting that it has a variance. The spread of the scatter decreases as T increases. This property are summarised by the covariance matrix

$$E[(\widehat{\theta}_T - \theta_0)(\widehat{\theta}_T - \theta_0)'] = \frac{1}{T}\Omega(\theta_0)$$

As $\Omega(\theta_0)$ is a finite matrix, the term T^{-1} shows that for increasing sample sizes this covariance matrix becomes smaller.
An important property of the maximum likelihood estimator is that under certain conditions (the regularity conditions again) it is relatively more efficient than any other estimator. Achieving this efficiency level corresponds to achieving the possible smallest variance, commonly known as the Cramer-Rao lower bound.
In practice, there are two choice to estimate $\Omega(\theta_0)$.

i An estimate based on the Hessian matrix

$$\Omega(\widehat{\theta}_T) = -H(\widehat{\theta}_T)^{-1}$$

where

$$H(\widehat{\theta}_T) = \frac{1}{T}\sum_{t=1}^{T} \left.\frac{\partial^2 \log f(y_t;\theta)}{\partial\theta\partial\theta'}\right|_{\theta=\widehat{\theta}_T}$$

is the Hessian evaluated at the maximum likelihood estimator $\widehat{\theta}_T$.

(ii) An estimate based on the Outer Product of Gradients matrix

$$\Omega(\widehat{\theta}_T) = J(\widehat{\theta}_T)^{-1}$$

where

$$J(\widehat{\theta}_T) = \frac{1}{T}\sum_{t=1}^{T} \frac{\partial \log f(y_t;\theta)}{\partial \theta} \frac{\partial \log f(y_t;\theta)}{\partial \theta'}\bigg|_{\theta=\widehat{\theta}_T}$$

is the outer product of gradients matrix evaluated at the maximum likelihood estimator, $\widehat{\theta}_T$.

Using these two choices of estimators for $\Omega(\theta_0)$, the covariance matrix of $\widehat{\theta}_T$ is then estimated as

$$\text{cov}(\widehat{\theta}_T) = \begin{cases} -\frac{1}{T}H(\widehat{\theta}_T)^{-1} & : \text{ Hessian} \\ \frac{1}{T}J(\widehat{\theta}_T)^{-1} & : \text{ OPG} \end{cases} \tag{10.3}$$

Standard errors of $\widehat{\theta}_T$ are given by the square roots of the diagonal elements of this matrix.

10.6.3 Normality

Consistency is about the mean of the distribution of $\widehat{\theta}_T$, efficiency is about the variance of the distribution of $\widehat{\theta}_T$ and normality is about the form of this distribution. Formally, the asymptotic distribution of $\widehat{\theta}_T$ is written as

$$\widehat{\theta}_T \overset{a}{\sim} N(\theta_0, \frac{1}{T}\Omega(\theta_0))\,, \qquad \sqrt{T}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega(\theta_0))$$

where the symbol $\overset{a}{\sim}$ signifies the asymptotic distribution. This is an important result as it facilitates statistical tests on the unknown parameter vector θ which is performed in practice by using $\text{cov}(\widehat{\theta}_T)$ as defined above.

10.6.4 Invariance

For any arbitrary nonlinear function, $\tau(\cdot)$, the maximum likelihood estimator of $\tau(\theta_0)$ is given by $\tau(\widehat{\theta}_T)$. The invariance property is particularly useful in situations when an analytical expression for the maximum likelihood estimator is not available but can be computed by substitution.
A measurement of fundamental concept in financial econometrics, volatility, relies upon the property of invariance The population measure of risk of an asset is given by the variance of the returns σ^2. The maximum likelihood estimator is

$$\widehat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2$$

where r_t is the return with sample mean $\bar{r}$. As volatility is represented by the population standard deviation σ, the maximum likelihood estimator of volatility is

$$\widehat{\sigma} = \sqrt{\widehat{\sigma}^2}$$

where $\tau(\cdot)$ corresponds to the square root function.
Similarly, the construction of the minimum portfolio relies on the property of invariance of maximum likelihood estimators. Let the population variances of the returns on two assets be given by σ_1^2 and σ_2^2, while the covariance is $\sigma_{1,2}$. The corresponding maximum likelihood estimators for a sample of size T are

$$\begin{aligned}
\widehat{\sigma}_1^2 &= \frac{1}{T}\sum_{t=1}^{T}(r_{1,t}-\overline{r}_1)^2 \\
\widehat{\sigma}_2^2 &= \frac{1}{T}\sum_{t=1}^{T}(r_{2,t}-\overline{r}_2)^2 \\
\widehat{\sigma}_{1,2} &= \frac{1}{T}\sum_{t=1}^{T}(r_{1,t}-\overline{r}_1)(r_{2,t}-\overline{r}_2)
\end{aligned}$$

where r_{it} is the return on the ith asset with sample mean $\overline{r}_i$.
Since the optimal weight on the first asset is

$$w_1 = \frac{\sigma_2^2 - \sigma_{1,2}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{1,2}}$$

the maximum likelihood estimator of the population parameter w_1 is

$$\widehat{w}_1 = \frac{\widehat{\sigma}_2^2 - \widehat{\sigma}_{1,2}}{\widehat{\sigma}_1^2 + \widehat{\sigma}_2^2 - 2\widehat{\sigma}_{1,2}}$$

10.7 Hypothesis Testing

In the estimation problem, $\widehat{\theta}$ is the value of θ that maximises the log-likelihood function where to simplify notation, the T subscript on $\widehat{\theta}$ is dropped. The discussion is now extended to determining if the population parameter has a certain hypothesised value, θ_0. If this value differs from $\widehat{\theta}$, then by definition, it must correspond to a lower value of the log-likelihood function and the crucial question is then how significant this decrease is. Determining the significance of this reduction of the log-likelihood function represents the basis of hypothesis testing.
The general statement of a hypothesis test is of the form

$$H_0 : \theta = \theta_0, \quad H_1 : \theta \neq \theta_0$$

where H_0 and H_1 are known as the null and alternative hypotheses respectively, and M represents the number of restrictions.
In testing based on the principle of maximum likelihood, there types of test, namely the Likelihood Ratio test (LR), the Wald test (WD) and the Lagrange Multiplier test (LM). These tests are distinguished by whether estimation

takes place under the null hypothesis or the alternative hypothesis or both so that there are two types of estimators to consider, namely,

$$\begin{array}{rcl} \widehat{\theta}_0 & : & \text{parameters estimated under } H_0 \\ \widehat{\theta}_1 & : & \text{parameters estimated under } H_1 \end{array}$$

If estimation is under the null hypothesis, the estimated parameters are denoted $\widehat{\theta}_0$. If estimation is under the alternative hypothesis, the estimated parameters are denoted $\widehat{\theta}_1$.
The forms of the three test procedures are:

$$\begin{array}{lrcl} \text{Likelihood ratio :} & LR & = & -2T\big(\log L(\widehat{\theta}_0) - \log L(\widehat{\theta}_1)\big) \\ \text{Wald :} & WD & = & T[\widehat{\theta}_1 - \theta_0]'[\Omega(\widehat{\theta}_1)]^{-1}[\widehat{\theta}_1 - \theta_0] \\ \text{Lagrange multiplier :} & LM & = & TG(\widehat{\theta}_0)'[\Omega(\widehat{\theta}_0)]G(\widehat{\theta}_0) \end{array}$$

where the choices of the matrix $\Omega(\widehat{\theta})$ are as given in expression (10.3). An important feature of all three tests is that in large samples, they are distributed as chi-squared with degrees of freedom equal to the number restrictions, M under the null hypothesis, χ^2_M.

10.7.1 Likelihood Ratio Test

The LR test measures the distance between the value of the log-likelihood under the null hypothesis, $\log L(\widehat{\theta}_0)$, and the alternative hypothesis, $\log L(\widehat{\theta}_1)$ as illustrated on the vertical axis in Figure 10.8. A large difference in the two log-likelihood values constitutes a rejection of the null hypothesis.

10.7.2 Wald Test

The WD test is based on the distance between the parameter estimates under the null hypothesis, $\widehat{\theta}_0$ and the estimates under the alternative hypothesis, $\widehat{\theta}_1$, illustrated on the horizontal axis of Figure **??**. In computing the test, this distance is weighted by the inverse of the covariance evaluated under the alternative hypothesis, $\Omega(\widehat{\theta}_1)$, which provides a measure of the curvature of log-likelihood function. In particular, for sharper log-likelihood functions which correspond to relatively smaller variances provides tighter inference.

10.7.3 Lagrange Multiplier Test

The LM test is based on the distance between the gradient of the log-likelihood function under the null hypothesis, $G(\widehat{\theta}_0)$, and the gradient of the log-likelihood function under the alternative hypothesis, $G(\widehat{\theta}_1) = 0$ as illustrated in Figure 10.9.

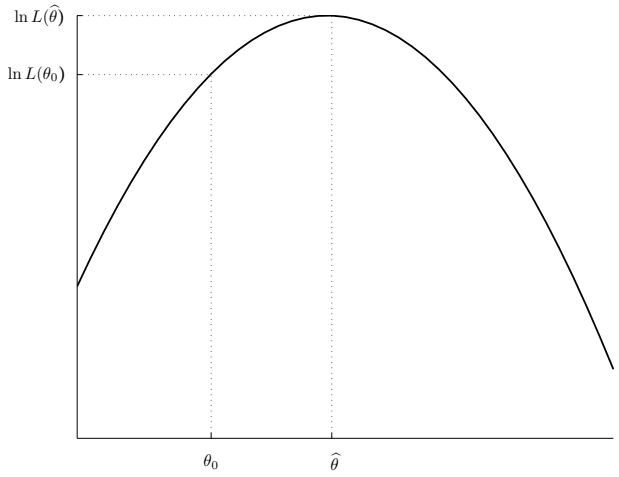


Figure 10.8: Comparison of the value of the log-likelihood function under the null hypothesis, θ_0, and under the alternative hypothesis, $\widehat{\theta}$.

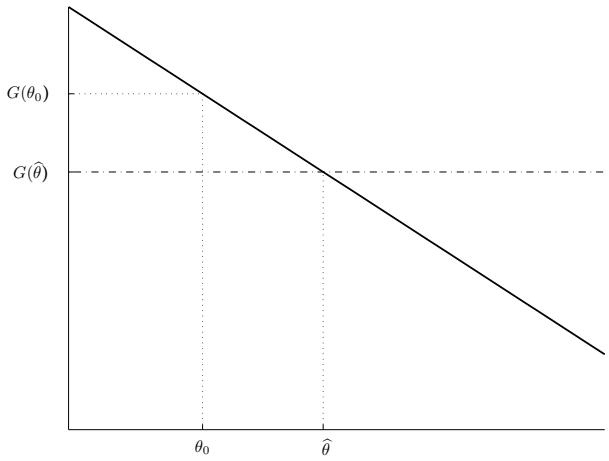


Figure 10.9: Comparison of the value of the gradient of the log-likelihood function under the null hypothesis, θ_0, and under the alternative hypothesis, $\widehat{\theta}$.

The test statistic is weighted by the covariance matrix $\Omega(\widehat{\theta}_0)$ which is the inverse of the variance of $G(\widehat{\theta}_0)$, as defined previously. A convenient form for computing the LM statistic is where $\Omega(\widehat{\theta}_0)$ is based on the OPG, $\Omega(\widehat{\theta}_0) = T^{-1}J(\widehat{\theta})^{-1}$, so

$$LM = TG(\widehat{\theta}_0)'[T^{-1}J(\widehat{\theta}_0)^{-1}]G(\widehat{\theta}_0)$$

where

$$\begin{aligned} G(\widehat{\theta}_0) &= \frac{1}{T}\sum_{t=1}^{T} g_t \\ J(\widehat{\theta}_0) &= \frac{1}{T}\sum_{t=1}^{T} g_t g_t' \end{aligned}$$

and all gradients are evaluated at $\widehat{\theta}_0$.
Or, alternatively,

$$\begin{aligned} LM &= T\left[\frac{1}{T}\sum_{t=1}^{T} g_t\right]' \frac{1}{T}\left[\frac{1}{T}\sum_{t=1}^{T} g_t g_t'\right]^{-1}\left[\frac{1}{T}\sum_{t=1}^{T} g_t\right] \\ &= \left[\sum_{t=1}^{T} g_t\right]' \left[\sum_{t=1}^{T} g_t g_t'\right]^{-1}\left[\sum_{t=1}^{T} g_t\right] \end{aligned}$$

This expression is equivalent to the explained sum of squares from an auxiliary regression of a vector of ones on the regressors in g_t (without an intercept).
As yet another form that will also be useful in implementing the LM test is to note that the total sum of squares (TSS) from this regression equals the sample size T. This follows from the property that as the auxiliary regression equation does not contain an intercept, TSS is simply the sum of squares of the observations. But as the dependent variable is simply a vector of ones, then

$$TSS = 1^2 + 1^2 + \cdots + 1^2 = T\,.$$

This means that the LM test can be computed as

$$LM = T - RSS$$

where RSS is the residual sum of squares from the auxiliary regression. This form of the LM test can be rewritten as

$$LM = T\frac{T - RSS}{T} = TR^2$$

where R^2 is the coefficient of determination from the auxiliary regression.
Computing the LM test involves the following steps:

Step 1: Estimate the restricted model and compute $\widehat{\theta}_0$.

Step 2: Estimate the auxiliary regression where the dependent variable is a vector of ones and the explanatory variables are the gradients in g_t, all evaluated at $\widehat{\theta}_0$.

Step 3: Compute the statistic $LM = TR^2$ where R^2 is the coefficient of determination in Step 2.

A number of diagnostic tests used to determine the validity of estimated regression models are in fact LM tests.

(i) The autocorrelation test for involves estimating the model without autocorrelation (the restricted model) and then estimating an auxiliary regression equation with the test statistic based on TR^2.

(ii) The White test for heteroskedasticity involves estimating the model without heteroskedasticity (the restricted model) and then estimating an auxiliary regression equation with the test statistic based on TR^2.

10.8 Testing the Duration Model of Trades

Consider the $T = 23368$ trades that took place on August 1 in 2006 on the AMR stock. The duration of times between trades is assumed to follow the Weibull distribution

$$f(y_t;\alpha,\beta) = \alpha\beta y_t^{\beta-1} e^{-\alpha y_t^{\beta}}, \qquad \alpha,\beta > 0$$

A natural test of this model is represented by the hypotheses

$$H_0 : \beta = 1, \quad H_1 : \beta \neq 1$$

as under the null hypothesis the Weibull distribution reduces to an exponential distribution.

10.8.1 Likelihood Ratio Test

To perform a likelihood ratio test both the unrestricted and restricted models are estimated by maximum likelihood methods. Using the Weibull distribution the unrestricted log-likelihood function based on Case 1 is

$$\log L(\theta) = \log(\alpha) + \log(\beta) + (\beta-1)\frac{1}{T}\sum_{t=1}^{T}\log y_t - \alpha\frac{1}{T}\sum_{t=1}^{T} y_t^{\beta}$$

The gradients are

$$\begin{aligned}
\frac{\partial \log L(\theta)}{\partial \alpha} &= \frac{1}{\alpha} - \frac{1}{T}\sum_{t=1}^{T} y_t^{\beta} \\
\frac{\partial \log L(\theta)}{\partial \beta} &= \frac{1}{\beta} + \frac{1}{T}\sum_{t=1}^{T}\log y_t - \alpha\frac{1}{T}\sum_{t=1}^{T}\log(y_t) y_t^{\beta}
\end{aligned}$$

where the second expression uses logarithmic differentiation.

Setting these two derivatives to zero the maximum likelihood estimator $\widehat{\theta} = \{\widehat{\alpha}, \widehat{\beta}\}$ is the solution of

$$\begin{aligned} 0 &= \frac{1}{\widehat{\alpha}} - \frac{1}{T}\sum_{t=1}^{T} y_t^{\widehat{\beta}} \\ 0 &= \frac{1}{\widehat{\beta}} + \frac{1}{T}\sum_{t=1}^{T} \log y_t - \widehat{\alpha}\frac{1}{T}\sum_{t=1}^{T} \log(y_t) y_t^{\widehat{\beta}} \end{aligned}$$

This is a nonlinear system of equations which is solved using an iterative algorithm.
The unrestricted maximum likelihood estimates, $\widehat{\theta}_1 = \{\widehat{\alpha}_1, \widehat{\beta}_1\}$, are

$$\begin{aligned} \widehat{\alpha}_1 &= 0.159744 \\ \widehat{\beta}_1 &= 0.939682\,. \end{aligned}$$

The unrestricted log-likelihood value is $\log L(\widehat{\theta}_1) = -2.9813$.
The restricted maximum likelihood estimates, $\widehat{\theta}_0 = \{\widehat{\alpha}_0, \widehat{\beta}_0\}$, are

$$\begin{aligned} \widehat{\alpha}_0 &= \frac{1}{7.279913} = 0.137365 \\ \widehat{\beta}_0 &= 1.000000\,. \end{aligned}$$

Note the restricted estimate for α is also obtained directly by using the analytical result that the parameter estimate is the reciprocal of the sample mean of durations. The restricted log-likelihood value is

$$\log L(\widehat{\theta}_0) = -2.9851$$

The likelihood ratio statistic is computed as

$$\begin{aligned} LR &= -2T\big(\log L(\widehat{\theta}_0) - \log L(\widehat{\theta}_1)\big) \\ &= -2 \times 23368(-2.9851 + 2.9813) = 176.6000 \end{aligned}$$

Under the null hypothesis LR is distributed as χ^2_1, resulting in a p value of 0.0000. There is a strong rejection at the 5% level that the durations between trades are exponentially distributed.

10.8.2 Wald Test

To perform a Wald test of the hypotheses estimate the unrestricted model and test the restriction $\alpha = 1$. The value of the Wald statistic is $WD = 133.2459$ and under the null hypothesis WD is distributed as χ^2_1, resulting in a p value of 0.0000 indicating a clear rejection of the null hypothesis at the 5% level. This is the same qualitative results as obtained using the LR test.

Since the Wald test involves just the one restriction, an alternative way of performing the test is to perform a simple t test. The t statistic is

$$t = \frac{0.939682 - 1.0000}{0.005225} = -11.544\,.$$

Squaring this value gives the value of the Wald statistic computed earlier $WD = (-11.544)^2 = 133.2459$. This result highlights a more general result, namely that t tests are in fact Wald tests because the parameters are estimated under the alternative hypothesis.

10.8.3 Lagrange Multiplier Test

To perform a Lagrange multiplier test of the restrictions the gradients of the log-likelihood function must be evaluated under the null hypothesis. The gradients at each t are

$$\begin{aligned} g_{1t} &= \frac{\partial \log f(y_t;\alpha,\beta)}{\partial \alpha} = \frac{1}{\alpha} - y_t^{\beta} \\ g_{2t} &= \frac{\partial \log f(y_t;\alpha,\beta)}{\partial \beta} = \frac{1}{\beta} + \log y_t - \alpha \log(y_t) y_t^{\beta} \end{aligned}$$

Evaluating these expressions at the restricted parameter estimates, $\widehat{\theta}_0 = \{0.137365, 1.000000\}$, gives

$$\begin{aligned} g_{1,t} &= \frac{1}{0.137365} - y_t \\ g_{2,t} &= 1 + \log y_t - 0.137365 \log(y_t) y_t \end{aligned}$$

A plot of the two constructed series is given in Figure 10.10.
Performing the OLS regression of a vector of ones as the dependent variable and $g_{1,2}$ and $g_{2,t}$ as the explanatory variables yields a coefficient of determination given by

$$R^2 = \frac{T - RSS}{T} = \frac{23368 - 23166.74}{23368} = 8.612\,6 \times 10^{-3}\,.$$

The value of the LM statistic is

$$LM = TR^2 = 23368 \times 8.612\,6 \times 10^{-3} = 201.2600$$

Under the null hypothesis LM is distributed as χ_1^2 resulting in a p value of 0.0000, once again indicating strong rejection of the null hypothesis at the 5% level. This is the same qualitative result as those obtained using the LR and WD tests.
An important feature of the LM test in testing for durations is that it is not necessary to estimate the model using an iterative optimisation algorithm.

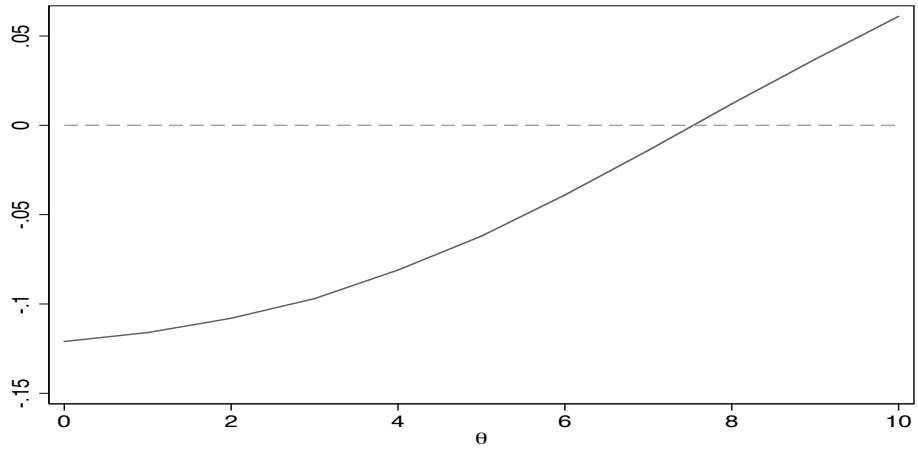


Figure 10.10: A plot of the gradients, g_{1t} and g_{2t}, of the log-likelihood function evaluated at the restricted parameter estimates, θ_0.

This is because the test is constructed under the null hypothesis resulting in analytical expressions for the estimators. In this instance, α is estimated as the inverse of the sample mean of the durations data. This contrasts with the LR and WD tests which both require using an iterative algorithm because both tests require estimation of the model under the alternative hypothesis where no analytical expressions for the estimators exist.
As all three testing procedures are equivalent in large samples, the choice is a matter of convenience which for the present example would be the LM test. In the application presented next, it turns out that the Wald test is the more convenient form of the test to adopt.

10.9 Testing the CAPM

The Capital Asset Pricing Model is based on the linear regression equation where the dependent variable is the excess return on an asset $(r_{it} - r_{ft})$ and the explanatory variable is the excess return on the market $(r_{mt} - r_{ft})$

$$r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + u_t\,,$$

where u_t is a disturbance term. Consider testing the joint restrictions

$$H_0 : \alpha = 0, \beta = 1, \quad H_1 : \text{ at least one restriction is not satisfied}$$

Under the null hypothesis the model becomes

$$r_{it} - r_{ft} = 0 + (r_{mt} - r_{ft}) + u_t\,,$$

Table 10.3

Wald tests of restrictions on the CAPM equations for the monthly excess log-returns to five United States financial assets and the commodity gold for the period May 1990 to July 2004, with p values in parentheses.

Stock	Null Hypothesis		
	$H_0 : \alpha = 0$	$H_0 : \beta = 1$	$H_0 : \alpha = 0, \beta = 1$
Exxon	16.6479	52.3976	64.5067
	(0.0000)	(0.0000)	(0.0000)
GE	13.9334	2.1270	17.1048
	(0.0002)	(0.1447)	(0.0000)
Gold	1.4016	426.2581	434.9162
	(0.2365)	(0.0000)	(0.0000)
IBM	0.5154	2.1061	2.8182
	(0.4728)	(0.1467)	(0.2444)
Microsoft	3.3414	8.2005	12.5180
	(0.0676)	(0.0042)	(0.0012)
Waalmart	1.8494	1.2177	2.8333
	(0.1739)	(0.2698)	(0.2425)

or more simply

$$r_{it} - r_{mt} = u_t \,,$$

so that the test of the restrictions is equivalent to testing that the excess return of the asset relative to the market, is random.

The unrestricted model is easily estimated by ordinary least squares and it is therefore convenient to perform a Wald test of the restrictions. The Wald test of the hypotheses may use either the Hessian $H(\widehat{\theta}_1)$ or the $J(\widehat{\theta}_1)$ matrix to compute the covariance matrix of the estimates. The three sets of Wald tests on the CAPM for the six assets are summarised in Table 10.3. The parameter estimates are the unrestricted maximum likelihood estimates based on the assumption that the disturbances normally distributed (equivalent to the ordinary least squares estimates) which are reported in Table 10.2. The covariance matrix of the disturbances is computed using the Hessian matrix of the unrestricted model, $H(\widehat{\theta}_1)$.

In the case of Exxon the value of the Wald statistic is $WD = 64.5067$. Under the null hypothesis WD is distributed as χ^2_2 resulting in a p value of $pv = 0.0000$ showing strong rejection of the null hypothesis at the 5% level. As the null hypothesis is rejected the two restrictions in the null hypothesis are tested separately. A test that the intercept is zero is represented by the hypotheses

$$H_0 : \alpha = 0, \quad H_1 : \alpha \neq 0$$

The value of the Wald statistic is

$$WD = \left(\frac{0.012018 - 0.0}{0.002945}\right)^2 = 16.6479$$

Using χ^2_1, the p value is

$$pv = 0.0000$$

showing strong rejection of the null hypothesis at the 5% level. Taking the square root of the Wald statistic gives

$$\text{t-stat } = \frac{0.012018 - 0.0}{0.002945} = 4.0802$$

which is the corresponding t statistic.
A test that the slope is unity is represented by the hypotheses

$$H_0 : \beta = 1, \quad H_1 : \beta \neq 1$$

The value of the Wald statistic is

$$WD = \left(\frac{0.501768 - 1.0}{0.068830}\right)^2 = 52.3976$$

Using χ^2_1, the p value is

$$pv = 0.0000$$

showing strong rejection of the null hypothesis at the 5% level. Taking the square root of the Wald statistic gives

$$t = \frac{0.501768 - 1.0}{0.068830} = -7.2386$$

which is the corresponding t statistic.

10.10 Exercises

1. **Equity Prices, Dividends and Returns**

pv.wf1, pv.dta, pv.xlsx

   (a) Plot the equity price over time and interpret its time series properties. Compare the result with Figure 2.1.
   (b) Plot the natural logarithm of the equity price over time and interpret its time series properties. Compare this graph with Figure 2.2.
   (c) Plot the return on equities over time and interpret its time series properties. Compare this graph with Figure 2.3.

(d) Plot the price and dividend series using a line chart and compare the result in Figure 2.4.

(e) Compute the dividend yield and plot this series using a line chart. Compare the graph with Figure 2.5.

(f) Compare the graphs in parts (a) and (b) and discuss the time series properties of equity prices, dividend payments and dividend yields.

(g) The present value model predicts a one-to-one relationship between the logarithm of equity prices and the logarithm of dividends. Use a scatter diagram to verify this property and compare the result with Figure **??**.

(h) Compute the returns on United States equities and then calculate the sample mean, variance, skewness and kurtosis of these returns. Interpret the statistics.

2. **Yields**

zero.wf1, zero.dta, zero.xlsx

(a) Plot the 2, 3, 4, 5, 6 and 9 months United States zero coupon yields using a line chart and compare the result in Figure 2.6.

(b) Compute the spreads on the 3-month, 5-month and 9-month zero coupon yields relative to the 2-month yield and and plot these spreads using a line chart. Compare the graph with Figure 2.6.

(c) Compare the graphs in parts (a) and (b) and discuss the time series properties of yields and spreads.

3. **Computing Betas**

capm.wf1, capm.dta, capm.xlsx

(a) Compute the monthly excess returns on the United States stock Exxon and the market excess returns.

(b) Compute the variances and covariances of the two excess returns. Interpret the statistics.

(c) Compute the Beta of Exxon and interpret the result.

(d) Repeat parts (a) to (c) for General Electric, Gold, IBM, Microsoft and Wal-Mart.

4. **Duration Times Between American Airline (AMR) Trades**

amr.wf1, amr.dta, amr.xlsx

(a) Use a histogram to graph the empirical distribution of the duration times between American Airline trades. Compare the graph with Figure 2.9.

(b) Interpret the shape of the distribution of durations times.

5. **Exchange Rates**

hour.wf1, hour.dta, hour.xlsx

(a) Draw a line chart of the $\$/£$ exchange rate and discuss its time series characteristics.

(b) Compute the returns on $\$/£$ pound exchange rate. Draw a line chart of this series and discuss its time series characteristics.

(c) Compare the graphs in parts (a) and (b) and discuss the time series properties of exchange rates and exchange rate returns.

(d) Use a histogram to graph the empirical distribution of the returns on the $\$/£$. Compare the graph with Figure 2.11.

(e) Compute the first 10 autocorrelations of the returns, squared returns, absolute returns and the square root of the absolute returns.

(f) Repeat parts (a) to (e) using the $\mathbb{DM}/\$$ exchange rate and comment on the time series characteristics, empirical distributions and patterns of autocorrelation for the two series. Discuss the implications of these results for the efficient markets hypothesis.

6. **Value-at-Risk**

bankamerica.wf1, bankamerica.dta, bankamerica.xlsx

(a) Compute summary statistics and percentiles for the daily trading revenues of Bank of America. Compare the results with Table 2.2.

(b) Draw a histogram of the daily trading returns and superimpose a normal distribution on top of the plot. What do you deduce about the distribution of the daily trading revenues.

(c) Plot the trading revenue together with the historical 1% VaR and the reported 1% Var. Compare the results with Figure 2.12.

(d) Now assume that a weekly VaR is required. Repeat parts (a) to (c) for weekly trading revenues.

Part IV

Modelling Volatility

Chapter 11

Modelling Variance I: Univariate Analysis

11.1 Introduction

An important feature of many of the previous chapters is on specifying and estimating financial models of expected returns. Formally these models are based on the conditional mean of the distribution where conditioning is based on either lagged values of the dependent variable, or additional explanatory variables, or a combination of the two. From a financial perspective however, modelling the variance of financial returns is potentially more interesting because it is an important input into many aspects of financial decision making. Examples include portfolio management, the construction of hedge ratios, the pricing of options and the pricing of risk in general. In implementing these strategies, practitioners soon released that the variance, or the square root of the variance known as *volatility*, was time varying.
The traditional approach to modelling conditional variance is the autoregressive conditional heteroskedasticity class of models (ARCH), originally developed by Engle (1982) and extended by Bollerslev (1986) and Glosten, Jagannathan and Runkle (1993). This is a flexible class of volatility models that can capture a wide range of features that characterise time-varying risk and which generalise to multivariate settings in which time-varying models of variances and covariances are dealt with. This class of models is particularly important in modelling time-varying hedge ratios, and spillover risk.

11.2 Volatility Clustering

To investigate the econometrics of modelling time-varying volatility in both a univariate and multivariate environment, the returns on five international stock markets are investigated, namely,

SPX : Standard and Poors index from the United States;
DJX : Dow Jones index from the United States;
HSX : Hang Seng index from Hong Kong;
NKX : Nikkei index from Japan;
DAX : Deutscher Aktien Index from Germany; and
UKX : FTSE index from the United Kingdom.

The data are daily beginning 4 January 1999 and ending 2 April 2014, $T = 3978$. This period covers a range of crises including the dotcom bubble in early 2000, the sub-prime crisis from mid 2007 to late 2008, the Great Recession from 2008 to 2010, and the European debt crisis from 2010.

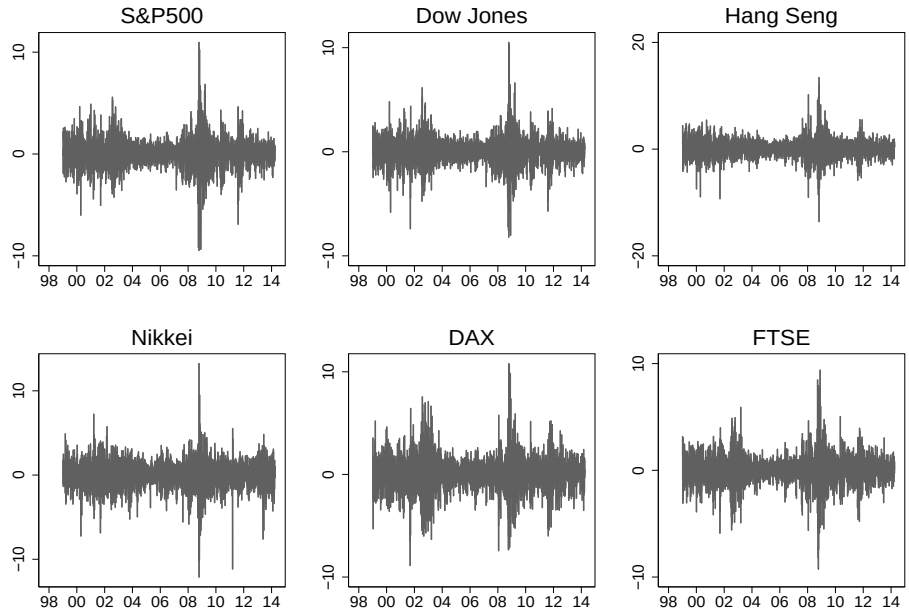


Figure 11.1: Annualised daily returns to five international stock market indices for the period 4 January 1999 to 2 April 2014 which have been standardised to have zero mean and unit variance.

One of the most documented features of financial asset returns is the tendency for large changes in asset prices to be followed by further large changes (market turmoil) or for small changes in prices to be followed by further small changes (market tranquility). This phenomenon is known as volatility clustering which highlights the property that the variance of financial returns is not constant over time, but appears to come in bursts. Figure 11.4 plots the annualised daily returns on the five international stock indices after standardisation to have zero mean and unit variance. The tendency for volatility to cluster is clearly demonstrated, particularly during the crisis periods in July of

2007 and the second half of 2008. There are also periods of tranquility when the magnitude of movements in the returns is relatively small.

A further implication of volatility clustering is that unconditional returns to the asset do not follow a normal distribution. This result is highlighted in Figure 11.2 which plots the histograms of the daily returns for each of the five stock market indices. In each case, the distribution of r_t is leptokurtic, because it has a sharper peak and fatter tails than the best-fitting normal distribution, which is overlaid on the histogram in Figure 11.2.

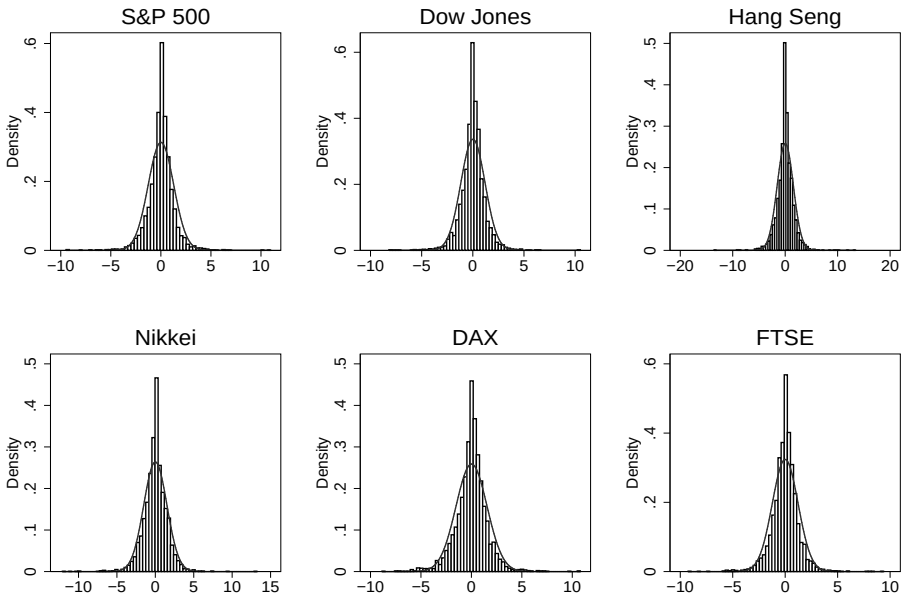


Figure 11.2: The distribution of the daily returns to five international stock indices over the 4 January 1999 to 2 April 2014. Superimposed on the histogram is a normal distribution with mean and variance equal to the sample mean and sample variance of the respective index returns.

To understand the relationship between volatility clustering and leptokurtosis consider a model of returns, r_t, which are characterised by two regimes where the variance is low in the tranquil regime and high in the turbulent regime, $h_{tranquil} < h_{turbulent}$ Assuming that the means in the two regimes are the same,

$$\mu_{tranquil} = \mu_{turmoil} = \mu$$

and that the returns in both these regimes are normally distributed, then

$$r_t \sim \begin{cases} N\left(\mu, h_{tranquil}\right) & : \text{ Tranquil regime} \\ N\left(\mu, h_{turbulent}\right) & : \text{ Turbulent regime.} \end{cases}$$

The tranquil regime is characterised by returns being close to their mean μ whereas for the turbulent regime there are large positive and negative returns which are relatively far from their mean of μ. Averaging the two distributions over the sample yields a leptokurtic distribution with the sharp peak primarily corresponding to the returns from the returns distribution during the tranquil periods. The leptokurtotic distribution is computed as the mixture distribution given by

$$f(r) = wN\left(\mu, h_{tranquil}\right) + (1-w)N\left(\mu, h_{turbulent}\right) ,$$

in which the weight ω is the proportion of returns coming from each period. The parameters of the distributions in each of the regimes are estimated for the returns on the merger hedge fund index as

$$\mu = 0.02, \quad h_{tranquil} = 0.1, \quad h_{turbulent} = 2.0 .$$

There is thus a 20-fold increase in the volatility during the turbulent period. where the weight is $w = 0.7$ representing that 70% of returns come from the tranquil period and 30% from the period of turbulence. A plot of the two distributions is given in Figure 11.3. The fat-tails largely (if not all) corresponding to the returns from the returns distribution during the turbulent periods.

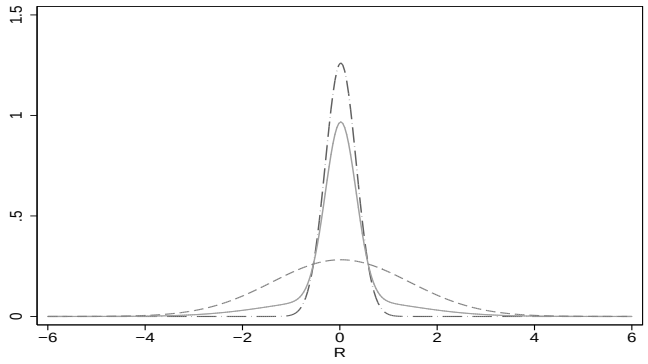


Figure 11.3

11.3 Simple Models of Time Varying Variance

By convention, the notation used for the time-varying variance in the volatility literature is h_t and not σ_t^2 as one might perhaps expect. This choice of h

is thought to be motivated by '*h*eteroskedasticity' which represents time-varying variance. The simplest model of time-varying variance is the historical variance, given by

$$h_t = \frac{1}{k}\sum_{i=0}^{k-1} r_{t-k}^2 \tag{11.1}$$

where k is the window over which the variance is computed and for simplicity is simply assumed that the mean return over the period may be set to zero. The advantages of this measure are that it is easy to compute involving the choice of only one parameter, namely, the window length k. The choice of k is critical – if the it is too long then the estimate is not dynamic enough, but if it is too short the estimate will be very noisy – but the model does not offer any prescription for how k is to be chosen.
The exponentially weighted moving average model (EWMA) is another simple model of time-varying variance which differs from historical volatility primarily insofar as it allows a higher weight to be attached to more recent observations. The EWMA model of variance is given by

$$\begin{aligned} h_t &= (1-\lambda)\sum_{j=1}^{\infty}\lambda^j r_{t-j-1}^2 \\ &= (1-\lambda)r_{t-1}^2 + \left[(1-\lambda)\lambda r_{t-2}^2 + (1-\lambda)\lambda^2 r_{t-3}^2 + \cdots\right] \\ &= (1-\lambda)r_{t-1}^2 + \lambda\left[(1-\lambda)r_{t-2}^2 + (1-\lambda)\lambda r_{t-3}^2 + \cdots\right] \\ &= (1-\lambda)r_{t-1}^2 + \lambda h_{t-1}\,, \end{aligned} \tag{11.2}$$

where λ is known as the decay parameter which governs how recent observations are weighted relative to more distant observations. The model depends crucially on the decay parameter λ, although the model does not indicate how the crucial parameter λ is to be estimated. In many cases a value is simply imposed with $\lambda = 0.94$ as suggested by RiskMetrics Group being a popular choice.
There are perhaps two fundamental problems with both these simple models of time-varying variance.

1. Neither model offers any prescription as to how to estimate the crucial parameters from historical data.

2. In terms of forecasting the future value of the time-varying variance, both these models suggest that the best forecast is the current estimate, h_t, and moreover, that this estimate is also the forecast for all future periods. This is a very undesirable feature of the models because it is to be expected that the variance will tend to revert to its long-run mean.

In order to address these fundamental flaws, an explicit dynamic model of variance is required whose parameters may be estimated from the data on historical returns.

11.4 The ARCH Model

One important implication of volatility clustering is that it should be possible to predict the evolution of the variance of returns because evidence of clustering implies autocorrelation in the variance of returns. This property may be demonstrated by realising that the square of the centered returns, r_t^2, provides an estimate of the variance of the returns at each t and the autocorrelation function (ACF) and partial autocorrelation function (PACF) computed using r_t^2 should show positive positive and statistically significant autocorrelation. This expectation is in contrast with returns, r_t, which are unpredictable, at least according to the efficient markets hypothesis. Figure **??** plots the ACF and PACF for both the daily returns, r_t, and squared daily returns, r_t^2, to the DAX index. The results of this exercise are as expected with the autocorrelations of r_t being statistically insignificant from zero, while the autocorrelations of r_t^2 indicate strong autocorrelation structure in r_t^2.

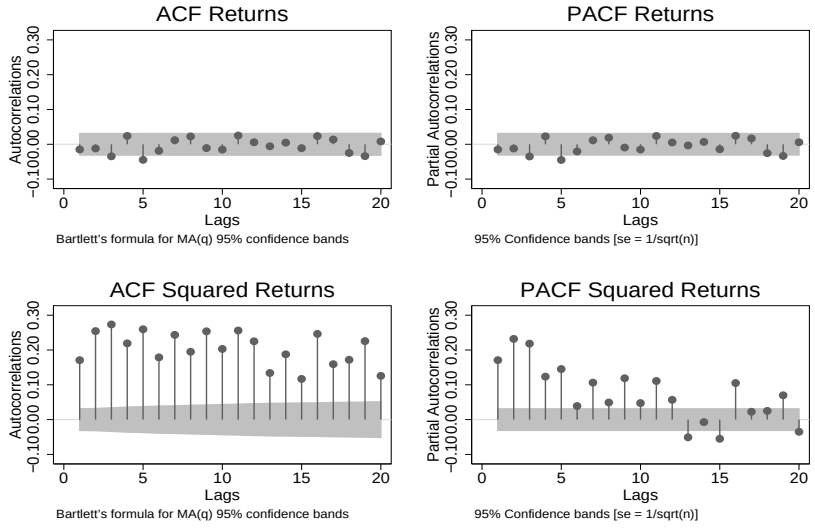


Figure 11.4: Autocorrelation function (ACF) and partial autocorrelation function (PACF) computed for 20 lags on the annualised daily returns to the DAX index for the period 4 January 1999 to 2 April 2014.

A natural test of time-variation in the variance of a variable would be to estimate an AR(p) model for the squared returns and to perform a joint test on the parameters of the lags. The idea of specifying a model that allows for time-varying variance in which an AR(p) model is estimated with the variables expressed in squares instead of in levels is now developed more formally which leads to the AutoRegressive Conditional Heteroskedasticity (ARCH) class of model introduced by of Engle (1982).

To motivate the structure of the GARCH model consider the following AR(1) model of returns

$$r_t = \phi_0 + \phi_1 r_{t-1} + u_t$$

where u_t is a disturbance term. The slope parameter ϕ_1 is the first-order autocorrelation coefficient for the returns. The conditional mean given information up to time $t-1$ is

$$\mathbb{E}_{t-1}(r_t) = \phi_0 + \phi_1 r_{t-1}$$

This is the conditional mean of returns which is time-varying because it is a function of lagged returns r_{t-1}.
Now consider replacing r_t, by r_t^2, so the AR(1) model becomes

$$r_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + v_t$$

where v_t is another disturbance term. The slope parameter α_1 is now the first-order autocorrelation coefficient of squared returns. The conditional expectation of r_t^2 given information at time $t-1$ is

$$\mathbb{E}_{t-1}(r_t^2) = \alpha_0 + \alpha_1 r_{t-1}^2 \,.$$

Assuming that the mean of returns is zero, or that the mean has been subtracted from returns, this expression also represents the conditional variance. It is the use of lagged squared returns to model the (conditional) variance that is the key property underlying ARCH models. Moreover, the conditional variance of returns is time-varying (heteroskedastic) as it is a function of variables at time $t-1$.
The ARCH model proposes a weighted average of past squared returns, similar to the historical volatility estimate as in equation (11.1), with the important improvement that the weights on the past variances are estimated from historical data. The ARCH(q) model is

$$\begin{array}{lcll} r_t & = & \phi_0 + \phi_1 r_{t-1} + u_t & \text{[Mean]} \\ h_t & = & \alpha_0 + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 & \text{[Variance]} \\ u_t & \sim & N(0, h_t) & \text{[Distribution]} \end{array}$$

where the q represents the length of the lag in the conditional variance equation given by h_t. The disturbance term u_t is commonly referred to as the 'news' because it represents the unanticipated movements in returns in excess of the conditional mean. In the special case of a constant variance $\alpha_1 = \alpha_2 = \cdots \alpha_q = 0$, and the variance of u_t and hence y_t, reduces to $h_t = h = \alpha_0$.
This observation suggests that a relatively simple test for ARCH can be performed by testing that $\alpha_i = 0$ for all i in a regression of the form

$$r_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i r_{t-i}^2 + v_t \,.$$

Under the null hypothes $\mathbb{E}_{t-1}(r_t^2)$ will be the constant value α_0. The null and alternative hypotheses are

$$\begin{array}{lcll} H_0 & : & \alpha_i = 0 & \text{for all } i \quad \text{[No ARCH]} \\ H_1 & : & \alpha_i \neq 0 & \text{for some} i \quad \text{[ARCH]}\,. \end{array}$$

The LM test (see Chapter 10) of these hypothesis is commonly used since it simply involves estimating an ordinary least squares regression equation and performing a goodness-of-fit test. The ARCH(q) test is implemented using the following steps.

Step 1: Estimate the regression equation

$$r_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i r_{t-i}^2 + v_t\,,$$

by ordinary least squares, where v_t is a disturbance term.

Step 2: Compute TR^2 from this regression and the corresponding p-value using the χ_q^2 distribution. A p-value less than 0.05 is evidence of ARCH in r_t at the 5% level.

11.5 The GARCH Model

A generalisation of the ARCH(q) model is the Generalised ARCH model or GARCH(p,q) model (Bollerslev, 1986), which allows for lags of the conditional variance to affect the conditional variance at time t. The model is

$$\begin{array}{lcll} r_t & = & \phi_0 + \phi_1 r_{t-1} + u_t & \text{[Mean]} \\ h_t & = & \alpha_0 + \sum\limits_{i=1}^{q} \alpha_i u_{t-i}^2 + \sum\limits_{i=1}^{p} \beta_i h_{t-i} & \text{[Variance]} \\ u_t & \sim & N\left(0, h_t\right) & \text{[Distribution].} \end{array} \tag{11.3}$$

Although the conditional mean is here specified to be an AR(1) model, any other specification for the mean is allowed. The key feature of the model which makes it a GARCH model is the fact that the conditional variance is given by q lags of the squared disturbance term u_t^2 and p lags of the conditional variance h_t.
An important special case of this model is the GARCH(1,1) model in which the conditional variance is specified as

$$h_t = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1}. \tag{11.4}$$

The GARCH(1,1) model is now easily interpreted as a generalisation of the EWMA model of equation (11.2) where instead of the just one parameter (the delay parameter λ) there are now three unknown parameters, α_0, α_1 and β_1. The GARCH model allows for two types of dynamics to affect the variance.

(i) Lagged shocks due to the news, $\{u_{t-1}^2, \cdots u_{t-q}^2\}$, have a finite effect of q periods on the conditional variance h_t. The effect of a shock on h_t is finite, equal to q periods.

(ii) Lagged terms in the conditional variance, $\{h_{t-1}^2 \cdots h_{t-p}^2\}$ allow shocks to the conditional variance to have a memory longer than p periods. For example in the GARCH(1,1) model in equation (11.4), the the dynamic effects of a shock on h_t are

$$\begin{array}{lcl} \text{Period 1} & : & \alpha_1 \\ \text{Period 2} & : & \alpha_1\beta_1 \\ \vdots & & \vdots \\ \text{Period n} & : & \alpha_1\beta_1^{n-1} \end{array}$$

If $\beta_1 = 0$, the memory is 1-period. The bigger is β_1, the longer is the memory of the shock.

A simple extension to the GARCH model is to allow for the effects of additional explanatory variables, $x_{1,t}, x_{2,t} \cdots x_{K,t}$, on the conditional moments. The GARCH model then becomes

$$\begin{array}{lcll} r_t & = & \phi_0 + \phi_1 r_{t-1} + \sum_{k=1}^{K} \gamma_k x_{k,t} + u_t & \text{[Mean]} \\ h_t & = & \alpha_0 + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i} + \sum_{k=1}^{K} \psi_k x_{k,t} & \text{[Variance]} \\ u_t & \sim & N(0, h_t) & \text{[Distribution]} \end{array}$$

Examples of potential explanatory variables are trade volumes, and dummy variables to model day-of-the-week effects and policy announcements.

11.6 Estimating Univariate (G)ARCH Models

GARCH models are estimated by maximum likelihood estimation which was dealt with in Chapter 10. The GARCH model in equation (11.3) specifies that the distribution of u_t is normal with zero mean and (conditional) variance h_t. From this it may be deduced that the conditional distribution of r_t is

$$f(r_t | r_{t-1}, r_{t-2}, \cdots; \theta) = \frac{1}{\sqrt{2\pi h_t}} \exp\left(-\frac{(r_t - \phi_0 - \phi_1 r_{t-1})^2}{2h_t}\right)$$

Based on this conditional distribution the log-likelihood function for an observation at time t is

$$\begin{aligned} \log L_t(\theta) &= \log f(r_t | r_{t-1}, r_{t-2}, \cdots; \theta) \\ &= -\frac{1}{2}\log 2\pi - \frac{1}{2}\log h_t - \frac{1}{2}\frac{u_t^2}{h_t}, \end{aligned} \tag{11.5}$$

where

$$\begin{aligned} u_t &= r_t - \phi_0 - \phi_1 r_{t-1} \\ h_t &= \alpha_0 + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i} \end{aligned}$$

and $\theta = \{\phi_0, \phi_1, \alpha_1, \alpha_2, \cdots, \alpha_q, \beta_1, \beta_2, \cdots, \beta_p\}$.
To estimate the GARCH model using an iterative optimisation algorithm, a set of starting values are needed for the parameters, θ_0, and also some initial values for computing the conditional variance. In the case of the GARCH(1,1) model the specification at observation $t = 1$ is

$$h_1 = \alpha_0 + \alpha_1 u_0^2 + \beta_1 h_0$$

so that starting values for u_0 and h_0 are required in order to compute h_1 . For u_0 the mean of its distribution can be used ($u_0 = 0$). For h_0 the unconditional variance can be used which is simply the sample variance of r_t.
Given these starting values the evaluation of the log-likelihood function proceeds as follows.

(i) The disturbance term, u_t, is evaluated for all observations using the starting values θ_0.

(ii) Given the starting values θ_0 and the initial values u_0 and h_0, the conditional variance h_t is evaluated recursively at all observations by using the computed values of u_t in the previous step.

(iii) Given values of u_t and h_t at each observation the log-likelihood function $\log L(\theta_{(0)})$ is evaluated.

The recursive computation of the log-likelihood function is embedded in a numerical optimisation routine which then solves iteratively for the maximum likelihood estimates of the parameters, $\widehat{\theta}$.
An important aspect of the estimation is that the conditional variance, h_t, must always be positive at all observations both from a theoretical perspective – h_t is a variance – and from a practical perspective – the value $\log h_t$ is computed in equation (11.5). To restrict h_t to be positive one strategy is to restrict all parameters to be positive by expressing h_t as

$$h_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i^2 u_{t-i}^2 + \sum_{i=1}^{p} \beta_i^2 h_{t-i}$$

Ensuring that the constraint $h_t > 0$ is enforced is one of the major issues faced by the various specifications of multivariate GARCH models which are introduced in Chapter 12.
The GARCH model specified so far assumes that the distribution of shocks is normal. It has already been noted that the combination of conditional normality and GARCH variance yields an unconditional distribution of financial

returns that is leptokurtotic. In practice, however, a simple GARCH model specified with normal disturbances is sometimes not able to model all of the leptokurtosis in the data. Consequently, two leptokurtotic distributions are commonly used to construct the log-likelihood function for GARCH models.

Standardised t distribution

Adopting the assumption that $u_t \sim St(0, h_t, \nu)$, where $\nu > 2$ is the degrees of freedom' parameter, implies that the conditional distribution for the GARCH(1,1) model is now

$$f(r_t|r_{t-1}, r_{t-2}, \cdots; \theta) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi h_t (\nu-2)}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{(r_t - \phi_0 - \phi_1 r_{t-1})^2}{h_t (\nu-2)}\right)^{-\left(\frac{\nu+1}{2}\right)}$$

where $\theta = \{\phi_0, \phi_1, \alpha_1, \alpha_2, \cdots, \alpha_q, \beta_1, \beta_2, \cdots, \beta_p, \nu\}$. The log-likelihood function for observation t is

$$\log L_t(\theta) = -\frac{1}{2}\log(\pi(\nu-2)) - \frac{1}{2}\log(h_t) + \log\left(\Gamma\left(\frac{\nu+1}{2}\right)\right) - \log\left(\Gamma\left(\frac{\nu}{2}\right)\right) - \left(\frac{\nu+1}{2}\right)\log\left(1 + \frac{(y_t - \phi_0 - \phi_1 r_{t-1})^2}{h_t(\nu-2)}\right),$$

with

$$\begin{aligned} u_t &= r_t - \phi_0 - \phi_1 r_{t-1} \\ h_t &= \alpha_0 + \sum_{i=1}^{q} \alpha_i u_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}. \end{aligned}$$

As before, an iterative optimisation algorithm is needed to estimate the parameters of the model by maximum likelihood.
The degrees of freedom parameter ν must be constrained to be positive and greater in value than 2. For $\nu < 2$ the variance of the t distribution does not exist in which case trying estimate the conditional variance is counterintuitive.

Generalised Error Distribution (ged)

Adopting the assumption that $u_t \sim ged(0, h_t, s)$, where s is the shape parameter, implies that the conditional distribution for the GARCH(1,1) model is now

$$f(r_t|r_{t-1}, r_{t-2}, \cdots; \theta) = \frac{s \exp\left(-\frac{1}{2}\left|\frac{r_t - \phi_0 - \phi_1 r_{t-1}}{\lambda\sqrt{h_t}}\right|^s\right)\frac{1}{h_t}}{\lambda 2^{(1+1/s)}\Gamma(\frac{1}{s})}$$

in which $s > 0$ and

$$\lambda = \left[\frac{\Gamma(1/v)}{2^{2/s}\Gamma(3/v)}\right]^{1/2}.$$

Table 11.1

Parameter estimates of GARCH(1,1) models for the daily returns to five international stock indices for log-likelihood functions based on the normal, t and generalised error distributions. The sample period is 4 January 1999 to 2 April 2014.

	SPX	DJX	HSX	NKX	DAX	UKX
Mean			Normal Distribution			
ϕ_0	0.045	0.050	0.046	0.044	0.073	0.037
	(0.014)	(0.013)	(0.018)	(0.019)	(0.018)	(0.013)
Variance						
α_0	0.014	0.013	0.012	0.041	0.024	0.014
	(0.002)	(0.002)	(0.003)	(0.007)	(0.004)	(0.003)
α_1	0.077	0.082	0.056	0.090	0.086	0.095
	(0.006)	(0.006)	(0.004)	(0.006)	(0.006)	(0.007)
β_1	0.913	0.907	0.939	0.893	0.903	0.896
	(0.006)	(0.007)	(0.005)	(0.007)	(0.007)	(0.007)
Mean			Student-t Distribution			
ϕ_0	0.059	0.057	0.051	0.053	0.085	0.047
	(0.013)	(0.013)	(0.017)	(0.019)	(0.017)	(0.013)
Variance						
α_0	0.010	0.010	0.008	0.030	0.016	0.013
	(0.003)	(0.003)	(0.003)	(0.008)	(0.004)	(0.003)
α_1	0.079	0.082	0.050	0.074	0.084	0.095
	(0.009)	(0.009)	(0.006)	(0.008)	(0.009)	(0.010)
β_1	0.917	0.913	0.948	0.914	0.912	0.898
	(0.009)	(0.009)	(0.006)	(0.010)	(0.009)	(0.010)
ν	6.557	6.605	6.251	7.806	8.218	9.500
	(0.730)	(0.710)	(0.711)	(0.893)	(1.072)	(1.383)
Mean			Generalized Error Distribution			
ϕ_0	0.051	0.043	0.021	0.019	0.083	0.042
	(0.013)	(0.012)	(0.015)	(0.018)	(0.016)	(0.013)
Variance						
α_0	0.011	0.011	0.008	0.035	0.019	0.014
	(0.003)	(0.003)	(0.004)	(0.010)	(0.005)	(0.003)
α_1	0.079	0.083	0.052	0.079	0.085	0.096
	(0.009)	(0.010)	(0.007)	(0.010)	(0.009)	(0.010)
β_1	0.915	0.911	0.946	0.907	0.908	0.896
	(0.010)	(0.010)	(0.007)	(0.011)	(0.010)	(0.010)
s	1.277	1.270	1.172	1.258	1.391	1.464
	(0.038)	(0.035)	(0.037)	(0.037)	(0.041)	(0.045)

Standard errors in parentheses

The log-likelihood function for observation t is

$$\log L_t(\theta) = \log\left(\frac{s}{\lambda}\right) - \left(1+\frac{1}{s}\right)\log 2 - \log\Gamma(\frac{1}{s}) - \frac{1}{2}\log h_t - \frac{1}{2}\left|\frac{(r_t-\phi_0-\phi_1 r_{t-1})^2}{\lambda^2 h_t}\right|^{\frac{s}{2}}$$

The increasing use of the ged distribution in estimating GARCH models derives from its versatility. In particular

$s > 2$ ged has thinner tails than the normal distribution

$s = 2$ ged is identical to the normal distribution

$s < 2$ ged has fatter tails than the normal distribution

While the ged is capable of generating very fat tails it cannot match the t distribution. When $\nu < 4$ in the t distribution the the tails are so fat that the kurtosis is infinite. So in general for financial data, it is to be expected that parameter values of $\nu > 4$ will result when using the t distribution, while the ged distribution should yield estimates $s < 2$. A GARCH(1,1) model is fitted to the the six international stock index returns for the period 4 January 1999 to 2 April 2014 using each of the three distributions discussed in this section. The results are reported in Table 11.1.

An interesting feature of the empirical results is the consistency of the parameter estimates across all five equity returns and also across the three distributions. The estimates of β_1 are all in the vicinity of 0.9 and the estimates of α_1 being between 0.05 and 0.095. A pleasing result is that the parameters nu and s are both in the expected range and consistently indicate that the normal distribution is perhaps not appropriate in these examples. On the other hand, the similarity of the estimated parameters suggests that choice of appropriate distribution will need to be made on the basis of features other than point estimates of the parameters, such as forecast performance or the distribution of the standardised residuals.

Very often in GARCH(1,1) models of equity returns $\widehat{\alpha}_1 + \widehat{\beta}_1 \simeq 1$. The sum $\widehat{\alpha}_1 + \widehat{\beta}_1$ for these indices appears to be in keeping with this general observation, although the sum in each case is less than one. If $\widehat{\alpha}_1 + \widehat{\beta}_1 = 1$ then the volatility series is nonstationary, a situation known as IGARCH, where the 'I' stands for integrated following the discussion of nonstationary models in Part **??**.

A simpler approach to dealing with the problem of leptokurtosis is to recognise that the assumption of normally distributed disturbances results in the misspecification of the log-likelihood function. Despite the shape of the distribution being incorrect, the mean and variance of this distribution are, however, correctly specified. It turns out that estimation of the parameters of the conditional mean and the conditional variance are still consistent but the standard errors require correction. The corrected standard errors of the maximum likelihood estimators of θ that are consistent are known as Bollerslev-Wooldridge standard errors (Bollerslev and Wooldridge, 1992). Computation of these standard errors involves using information in both the Hessian matrix and the outer product of gradients matrix.

11.7 Asymmetric Volatility Effects

Consider the GARCH(1,1) model

$$\begin{aligned} r_t &= \phi_0 + u_t \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} \,. \end{aligned}$$

For no news-days, $u_{t-1} = 0$ the conditional variance has a minimum value at $h_t = \alpha_0$. An important property of this GARCH(1,1) specification is that shocks of the same magnitude, positive or negative, result in the same increase in volatility h_t. That is, positive news, $u_{t-1} > 0$, has the same effect on the conditional variance as negative news $u_{t-1} < 0$ because it is only the absolute size of the news that matters since u_{t-1}^2 enters the equation. In the case of stock markets, an asymmetric response to the news in which negative shocks $u_{t-1} < 0$ have a larger effect on conditional variance is supported by theory. A negative shock raises the debt-equity ratio,) thereby increasing leverage and consequently risk and this so-called leverage effect therefore suggests that bad news causes a greater increase in conditional variance than good news.
There are two popular specifications in the GARCH class of model that relax the restriction of a symmetric response to the news.

1. **Threshold GARCH (TGARCH)**:
   The TGARCH specification of the conditional variance is

   $$h_t = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 I_{t-1}$$

   where I_{t-1} is an indicator variable defined as

   $$I_{t-1} = \begin{cases} 1 & : \quad u_{t-1} \geq 0 \\ 0 & : \quad u_{t-1} < 0 \end{cases}$$

   To make the asymmetry in the effect of news on the conditional variance explicit, this model can also be written as

   $$h_t = \begin{cases} \alpha_0 + (\alpha_1 + \lambda) u_{t-1}^2 + \beta_1 h_{t-1} & : \quad u_{t-1} \geq 0 \\ \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} & : \quad u_{t-1} < 0 \end{cases}$$

   If $\widehat{\lambda} > 0$ then positive news, $u_{t-1} \geq 0$, has a greater effect on volatility than negative news. The leverage effect in equity markets would lead us to expect $\widehat{\lambda} < 0$ so that negative news, $u_{t-1} < 0$, is associated with a higher effect on volatility than positive news.

2. **Exponential GARCH (EGARCH)**:
   The EGARCH specification of the conditional variance is

   $$\log h_t = \alpha_0 + \sum_{i=1}^{q} \left(\alpha_i \left| \frac{u_{t-i}}{\sqrt{h_{t-i}}} \right| + \lambda_i \frac{u_{t-i}}{\sqrt{h_{t-i}}} \right) + \sum_{j=1}^{p} \beta_j \log(h_{t-j})$$

An important advantage of the EGARCH specification is that the conditional variance is guaranteed to be positive at each point in time because the variance is expressed in terms of $\log h_t$ so that the actual variance obtained by exponentiation.

ARCH(1), GARCH(1,1), TARCH(1,1) and EGARCH(1,1) models were fitted to daily returns from 4 January 1999 to 2 April 2014 on the returns to six international stock indices. The results are reported in Table 11.2.

Table 11.2

Parameter estimates of TARCH(1,1) models for the daily returns to five international stock indices expressed as percentages. The sample period is 4 January 1999 to 2 April 2014.

	SPX	DJX	HSX	NKX	DAX	UKX
Mean						
ϕ_0	0.003	0.013	0.024	0.015	0.026	−0.001
	(0.014)	(0.013)	(0.018)	(0.020)	(0.018)	(0.014)
Variance						
α_0	0.014	0.013	0.016	0.053	0.029	0.017
	(0.001)	(0.001)	(0.003)	(0.008)	(0.003)	(0.002)
α_1	0.129	0.137	0.081	0.131	0.139	0.136
	(0.009)	(0.010)	(0.006)	(0.010)	(0.010)	(0.009)
β_1	0.938	0.926	0.939	0.889	0.911	0.920
	(0.005)	(0.006)	(0.005)	(0.008)	(0.007)	(0.007)
γ	−0.157	−0.149	−0.057	−0.090	−0.133	−0.143
	(0.010)	(0.010)	(0.006)	(0.008)	(0.009)	(0.010)

Standard errors in parentheses

As expected $\widehat{\lambda} < 0$ for each of the returns series considered and the parameter is statistically significant indicating the presence of the leverage effect in these markets. A plot of h_t against u_{t-1}, is known as the news impact curve. The news impact curve illustrates quite sharply the differences between the various specifications of the conditional variance. To demonstrate this point an ARCH(1), GARCH(1,1), TARCH(1,1) and EGARCH(1,1) model is fitted to the returns to the S&P 500 and for each model the news impact curve is plotted in Figure **??**.

The major point to note is that the simple ARCH and GARCH models impose a symmetric news impact curve whereas the TARCH and EGARCH models relax this assumption and allow for asymmetric adjustment to the news. For the models estimated here, the news impact curve is much flatter for positive shocks than it is for negative shocks, indicating that negative news has a much larger impact on volatility than positive news. The situation depicted here of the the news impact curve actually decreasing as positive shocks get larger is not typical of many applications for stock market returns.

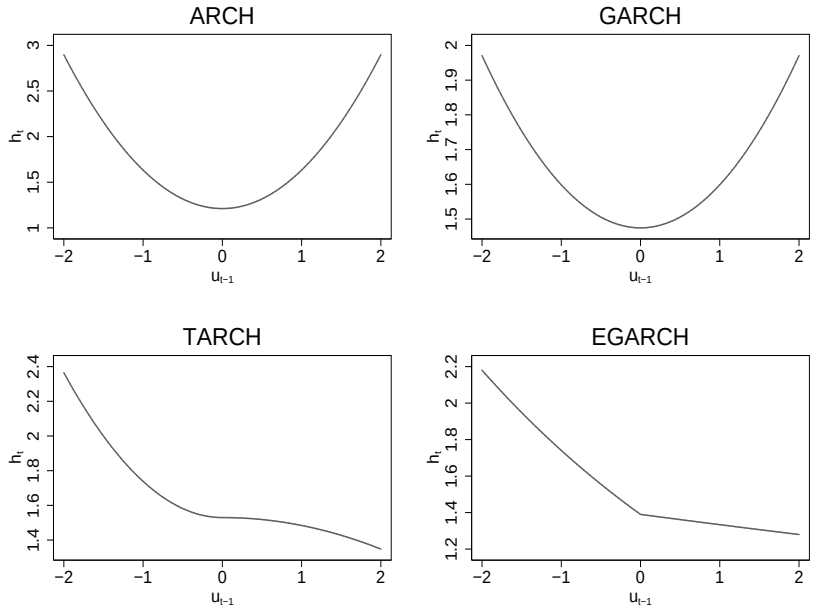


Figure 11.5: News Impact Curves for ARCH(1), GARCH(1,1), TARCH(1,1) and EGARCH(1,1) models fitted to daily returns data from January 2003 to May 2010 on a merger hedge fund index. The conditional mean contains an AR(1) term.

11.8 The Risk-Return Trade-off

The standard deviation is commonly used as a measure of risk in portfolio theory as it represents the deviations of actual returns from their conditional mean. The larger the deviation the larger is the risk of the portfolio. To compensate an investor for bearing more risk, an investor should receive a higher expected return, resulting in a positive relationship between the mean and the risk of the portfolio.

Letting $\mu_t = \mathbb{E}_{t-1}[r_t]$ represent the conditional mean of the portfolio and $\mathbb{E}_{t-1}\left[r_t^2\right] = h_t$ represent the conditional variance of the portfolio. The fundamental relationship in finance between risk and return is specified as

$$\mu_t = \phi_0 + \phi_1 h_t^{\omega} = \phi_0 + \phi_1 \sigma_t^{2\omega}$$

where, for notational convenience the conditional standard deviation or risk of the portfolio is denoted $\sqrt{h_t} = \sigma_t$ and $\phi_1 > 0$ to allow for a positive relationship between the expected return and risk.

The compensation an investor wishes to receive from bearing higher risk is

given by

$$\frac{d\mu_t}{d\sigma_t} = 2\omega\phi_1\sigma_t^{2\omega-1}\,,$$

giving rise to two special cases.

(i) **Case 1:** $\omega = 0.5$
There is a linear relationship between the mean, μ_t, and the conditional standard deviation of the portfolio, σ_t. Compensation for bearing more risk increases at the constant rate given by

$$\frac{d\mu_t}{d\sigma_t} = \phi_1\,.$$

(ii) **Case 2:** $\omega = 1.0$
There is a nonlinear relationship between the mean, μ_t, and the conditional standard deviation of the portfolio, σ_t. Compensation for bearing more risk (increase in σ_t) increases at an increasing rate

$$\frac{d\mu_t}{d\sigma_t} = 2\phi_1\sigma_t$$

To illustrate the risk-return trade-off data for the 10 Fama-French industry portfolios, which comprises monthly data beginning January 1927 and ending December 2013, are used. Define the excess returns to the portfolios and the market, respectively as

$$z_{it} = r_{it} - r_{ft}\,, \qquad z_{mt} = r_{mt} - r_{ft}\,,$$

The then the relationship between risk and return may be captured in the GARCH(1,1) framework by the specifying the conditional mean as a function of the conditional variance as follows

$$\begin{aligned} z_{it} &= \phi_0 + \phi_1 h_t^{\omega} + \phi_2 z_{mt} + u_t \\ u_t &\sim N\left(0, h_t\right) \\ h_t &= \alpha_0 + \sum_{i=1}^{q}\alpha_i u_{t-i}^2 + \sum_{i=1}^{p}\beta_i h_{t-i}\,. \end{aligned} \tag{11.6}$$

The model is an extension of the CAPM with an allowance for time-varying risk to identify the trade-off preferences of investor between risk and return and the specification is known as GARCH-M.
The critical parameters reflecting the risk preferences of investors are ϕ_1 and ω. The GARCH-M augmented version of the CAPM is estimated for 10 industry portfolios using values $\omega = \{0.5, 1.0\}$ and Table 11.3 summarises the results relating to the trade-off parameter ϕ_1 for all 10 portfolios. Inspection of the results for $\omega = 0.5$ shows that the Nondurables, Durables and Retail portfolios exhibit the greatest trade-off, all with $\widehat{\phi})1 \simeq 0.3$. The Utilities and

Manufacturing portfolio have the smallest positive trade-offs, while the Energy and Health portfolios even exhibit a negative trade-off. The issue of a negative trade-off is investigated further below by testing the strength of the risk-return relationships.

Table 11.3

Estimation of parameter ϕ_1 the GARCH-M version of the CAPM in (11.6) with $\omega = \{0.5, 1.0\}$ for 10 Fama-French industry portfolios. The data are monthly excess returns for the period January 1927 to December 2013.

		$\omega = 0.5$		
Portfolio	ϕ_1	t test	p value	AIC
Nondurables	0.286	1.859	0.063	4468.639
Durables	0.378	2.861	0.004	5718.193
Manufacturing	0.135	1.068	0.285	3925.937
Energy	−0.053	−0.430	0.667	5680.519
Technology	0.052	0.357	0.721	5188.485
Telecom.	0.131	0.931	0.352	5156.026
Retail	0.301	1.886	0.059	4999.628
Health	−0.276	−1.881	0.060	5388.758
Utilities	0.011	0.120	0.904	5407.391
Other	0.115	1.170	0.242	4362.822
		$\omega = 1.0$		
Nondurables	0.064	2.053	0.040	4467.797
Durables	0.039	2.403	0.016	5718.994
Manufacturing	0.044	1.299	0.194	3924.829
Energy	−0.010	−0.599	0.549	5680.353
Technology	0.001	0.062	0.950	5188.598
Telecom.	0.016	0.741	0.459	5156.408
Retail	0.055	1.964	0.050	4998.995
Health	−0.029	−1.504	0.133	5390.037
Utilities	0.000	0.006	0.995	5407.404
Other	0.037	2.170	0.030	4360.769

A test of a trade-off between risk and return is based on the hypotheses

$$\begin{aligned} H_0: &\quad \phi_1 = 0 \quad \text{[No tradeoff]} \\ H_1: &\quad \phi_1 \neq 0 \quad \text{[Tradeoff]} \end{aligned}$$

The Wald test of these hypothesis is conveniently given by the t statistic. The p values of the models when $\omega = 0.5$ indicate that only the Durables portfolio has a statistically significant trade-off between risk and return at the conventional level of 5%. The results for the Nondurables and Retail portfolios are marginal, as is the result for the Health portfolio which has the counterintuitive negative relationship. The situation is more promising for the ability of

this model to capture the risk-return relationship when $\omega = 1.0$. The Nondurables, Durables, Retail and Other portfolios are all indicate a significant risk-return tradeoff and the anomalous result for the Health portfolio is resolved because $\widehat{\phi}_1$ is not significant.
Table 11.3 also reports the Akaike Information Criteria (AIC) which may be used as a test to determine what type of risk preferences are most consistent with the portfolios because the models based on $\omega = 0.5$ and $\omega = 1.0$, are nonnested. As discussed in Chapter 4, the AIC statistic is computed as

$$AIC = -2\log L(\widehat{\theta}) + \frac{2K}{T}$$

where $K = 7$ is the number of estimated parameters which applies to both models.
A comparison of the AICs for the two models associated with all 10 portfolios reveals an even split between the portfolios for which the statistic is minimised when $\omega = 0.5$ (Nondurables, Technology, Telecom., Health and Utilities) and $\omega = 1.0$ (Durables, Manufacturing, Energy, Retail and Other).

11.9 Forecasting

Forecasting GARCH models is similar to forecasting ARMA models discussed in Chapter 7. The only difference is that with ARMA forecasts the focus is on the level of the series whereas with GARCH forecasts it is on the variance of the series. To highlight the process of forecasting GARCH conditional variances, consider the GARCH(1,1) model. To forecast volatility at time $T+1$, the conditional variance is written at $T+1$

$$h_{T+1} = \alpha_0 + \alpha_1 u_T^2 + \beta_1 h_T$$

Taking conditional expectations based on information at time T, the one-step ahead forecast of h_t is

$$\begin{aligned} h_{T+1|T} &= \mathbb{E}_T\left[h_{T+1}\right] = \mathbb{E}_T\left[\alpha_0 + \alpha_1 u_T^2 + \beta_1 h_T\right] \\ &= \alpha_0 + \alpha_1 u_T^2 + \beta_1 h_T \end{aligned}$$

since $\mathbb{E}_T\left[u_T^2\right] = u_T^2$ and $\mathbb{E}_T\left[h_T\right] = h_T$. Similarly, to forecast volatility at time $T+2$, the conditional variance is written at $T+2$

$$h_{T+2} = \alpha_0 + \alpha_1 u_{T+1}^2 + \beta_1 h_{T+1}$$

Taking conditional expectations based on information at time T, the two-step ahead forecast of h_{T+2} is

$$\begin{aligned} h_{T+2|T} &= \mathbb{E}_T\left[h_{T+2}\right] = \mathbb{E}_T\left[\alpha_0 + \alpha_1 u_{T+1}^2 + \beta_1 h_{T+1}\right] \\ &= \alpha_0 + \alpha_1 \mathbb{E}_T\left[u_{T+1}^2\right] + \beta_1 \mathbb{E}_T\left[h_{T+1}\right] \\ &= \alpha_0 + \alpha_1 h_{T+1|T} + \beta_1 h_{T+1|T} \\ &= \alpha_0 + (\alpha_1 + \beta_1)\, h_{T+1|T} \end{aligned} \tag{11.7}$$

as by definition $\mathbb{E}_T\left[u_{T+1}^2\right] = h_{T+1|T}$ and $\mathbb{E}_T[h_{T+1}] = h_{T+1|T}$. By extending this argument to $T+k$ gives

$$h_{T+k|T} = \alpha_0 + (\alpha_1 + \beta_1)\, h_{T+k-1|T} \tag{11.8}$$

Recursive substitution for the term $h_{T+k-1|T}$ in (11.8), using the results of the form (11.7), the conditional forecast of volatility for k periods ahead is

$$\begin{aligned} h_{T+k|T} &= \alpha_0 + (\alpha_1 + \beta_1)\,\alpha_0 + \cdots + (\alpha_1 + \beta_1)^{k-2}\,\alpha_0 \\ &\quad + (\alpha_1 + \beta_1)^{k-1}\, h_{T+1|T} \end{aligned}$$

In summary, the forecasts for the GARCH(1,1) model are summarised as

$$\begin{aligned} h_{T+1|T} &= \alpha_0 + \alpha_1 u_T^2 + \beta_1 h_T \\ h_{T+k|T} &= \alpha_0 + (\alpha_1 + \beta_1)\, h_{T+k-1|T}, \qquad k \geqslant 2 \end{aligned}$$

In practice, these forecasts for the GARCH(1,1) model are computed by replacing the unknown parameters α_0, α_1 and β_1 and the unknown quantities u_T^2 and h_T by their respective sample estimates. The forecasts are computed recursively staring with

$$\widehat{h}_{T+1|T} = \widehat{\alpha}_0 + \widehat{\alpha}_1 \widehat{u}_T^2 + \widehat{\beta}_1 \widehat{h}_T$$

Given this estimate, $\widehat{h}_{T+2|T}$ is computed from (11.7) as

$$\widehat{h}_{T+2|T} = \widehat{\alpha}_0 + (\widehat{\alpha}_1 + \widehat{\beta}_1)\widehat{h}_{T+k-1|T}$$

which, in turn, is used to compute $\widehat{h}_{T+3|T}$ etc. To forecast higher order GARCH models the same recursive approach is adopted.
One of the main issues highlighted in Section **??** with forecasting time-varying variances using either a historical average or an exponentially weighted moving average, is that the current forecast of the variance is also the expected future value of the variance. In other words, if variance is at historically high levels when the EWMA estimate is computed then this high value for the variance is forecast to continue indefinitely. By contrast, the forecast from a GARCH(1,1) model will converge relatively quickly to the long-term average volatility implied by the model, which is given by

$$\overline{h} = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}\,.$$

Figure 11.6 demonstrates this convergence for S&P 500 returns. A GARCH(1,1) model is fitted and then out-of-sample predictions are made for two different periods, the first starting on 1 January 2010 and the second on 1 July 2010. The forecasts in both cases converge to the long-term mean despite the fact the forecast starts below the long-term mean for the January forecast and

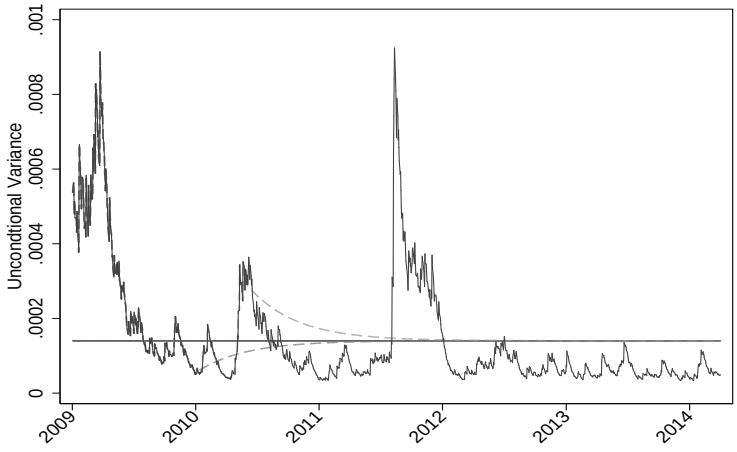


Figure 11.6: Forecasts of the conditional variance of S&P 500 returns obtained from a GARCH(1,1) model. Also shown are the estimated conditional variance prior to the forecast and the long-term mean of the variance implied by the models. Both the forecasts beginning on 1 January 2010 and 1 July 2010 converge to the long-term mean.

above that the long-term mean for the July forecast. The fact that the converge occurs over a 12 month period indicates that the conditional volatility series is quite persistent. Notice that for the forecast starting in July 2010, the actual estimated conditional variance series drops off a lot more quickly than the forecast.

One of the distinguishing features of the conditional variance literature has been the rapid proliferation of types of model available. Factors which have to chosen range from the specification of the mean process through the choice of specification for the conditional variance to the selection of the appropriate error distribution on which to base the construction of the log-likelihood function. Given this overwhelming choice, one of the more interesting results to emerge is that despite its simplicity, when it comes to forecasting the conditional variance, the simple GARCH(1,1) model is difficult to beat (Hansen and Lunde, 2005).

The claimed efficacy of the GARCH(1,1) for forecasting conditional variance then naturally leads to the question of assessing the accuracy of variance forecasts. In theory, determining the accuracy of the forecasts of the conditional variance can be accomplished using any of the statistical measures outlined in Chapter 7. In practice, however, this proves difficult because it is possible to compare the forecasts with the actual value of the conditional variance with its forecast because the former is never directly observed.

The standard method to assess volatility models, therefore, is to evaluate the forecast using a volatility proxy and such as the squared return, r_t^2. Early at-

tempts are forecast evaluation were based on Mincer-Zarnowitz regressions (Mincer and Zarnowitz, 1969) in which the realisation of the variable of interest is regressed on the forecast

$$r_t^2 = \delta_0 + \delta_1 \widehat{h}_t + u_t \,.$$

The null and alternative hypotheses are

$$\begin{aligned} H_0: &\quad \delta_0 = 0 \text{ and } \delta_1 = 1 \\ H_1: &\quad \delta_0 \neq 0 \text{ or } \quad \delta_1 \neq 1 \,. \end{aligned}$$

The use of r_t^2 as a proxy is problematic, however, as returns that are large in absolute value may have a large impact on the estimation results. Two examples of alternative specifications that have been tried are

$$\begin{aligned} |r_t| &= \delta_0 + \delta_1 \sqrt{\widehat{h}_t} + u_t \\ \log r_t^2 &= \delta_0 + \delta_1 \log \widehat{h}_t + u_t \,, \end{aligned}$$

which use transformations of the volatility proxy to reduce the impact of large returns.
In general, the trend has been to move away from Mincer-Zarnowitz regressions and to use the measures of forecast performance outlined in Chapter 7 together with the Diebold-Mariano test (Diebold and Mariano, 1995) to assess volatility forecasts. This has been a particularly fertile area of research and has seen the development of new loss functions, such as the quasi-likelihood loss function, which is defined for observation t as

$$QLIKE = \log \widehat{h}_t + \frac{r_t^2}{\widehat{h}_t} \,.$$

The name QLIKE is derived from the similarity to the (negative) Gaussian log-likelihood and its use as a quasi-likelihood in mis-specified models. Specified in this way the QLIKE function can become negative when dealing with very small returns because the term in $\log h_t$ will be negative and dominate the other term in the expression. To avoid this, an equivalent alternative specification (see Christoffersen, 2012) which is always positive is

$$QLIKE = \frac{r_t^2}{\widehat{h}_t} - \log\left(\frac{r_t^2}{\widehat{h}_t}\right) - 1 \,.$$

The QLIKE function has become very popular in evaluating variance forecasts. The major reason for this popularity is the fact that the QLIKE criterion is that it is not symmetric. Figure 11.7 plots the RMSE and the QLIKE measures for forecasts ranging from 0.5 to 3 when the true value is 2. Unlike the RMSE, the QLIKE penalises underestimating the volatility more heavily than overestimating it. This may be a desirable characteristic in a loss function if

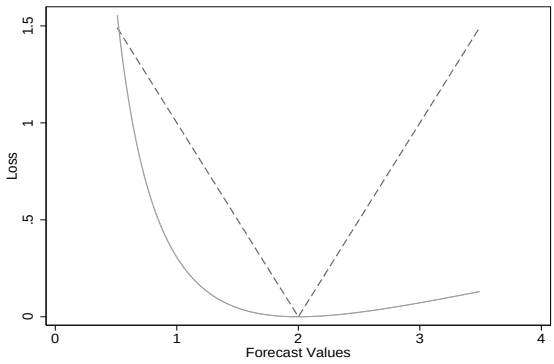


Figure 11.7: The RMSE (dashed line) and QLIKE loss functions plotted for a forecasts ranging from 0.5 to 3.5 for the true value 2.

Table 11.4

Evaluation of forecasts of the S%P 500 conditional variance based on a GARCH(1,1) model and an EWMA model with weighting parameter 0.94. Forecasts begin on 1 July 2010 and are made for the subsequent 500 days. For each evaluation criterion, the Diebold-Mariano test and associated p value are also reported.

Criterion	GARCH(1,1)	EWMA	DM t test	p value
MAE	0.000	0.000	-17.895	0.000
MAPE	1982.110	2946.970	-1.464	0.144
RMSE	0.000	0.000	-3.460	0.001
QLIKE	1.986	2.090	-1.968	0.049

the risk manager is particularly conservative. Other loss functions based on economic notions of loss such as expected utility have also been proposed.

Table 11.4 provides a comparison of the forecasts generated by a GARCH(1,1) model fitted to S&P 500 returns against a forecast of the conditional variance using an exponentially weighted moving average of squared returns. The forecasts start at 1 July 2010 and are made for 500 days after this date. The EWMA forecast is computed using the weighting parameter 0.94 as suggested by RiskMetrics. The MAE, MAPE and RMSE as discussed in Chapter 7 are reported together with the Diebold-Mariano test of equal predictive accuracy. The QLIKE metic and associated Diebold-Mariano test are also reported. It is quite clear from these results that the GARCH(1,1) model dominates the EWMA in terms of forecasting accuracy. It is only in the MAPE case that the Diebold-Mariano test of equal predictive ability cannot be rejected at the 5% level and in each case the t statistic is negative indicating that the GARCH(1,1) loss function has a smaller value than the EWMA loss function.

11.10 Exercises

1. **Time-variation in Hedge Funds**

   This question is based on the EViews file **HEDGE.WF1** which contains daily data on the percentage returns of seven hedge fund indexes, from the 1st of April 2003 to the 28th of May 2010, a sample size of $T = 1869$.

R_CONVERTIBLE	:	Convertible Arbitrage
R_DISTRESSED	:	Distressed Securities
R_EQUITY	:	Equity Hedge
R_EVENT	:	Event Driven
R_MACRO	:	Macro
R_MERGER	:	Merger Arbitrage
R_NEUTRAL	:	Equity Market Neutral

   (a) Using the returns on the Merger hedge fund estimate the constant mean model

   $$R_MERGER_t = \gamma_0 + u_t,$$

   and interpret the time series properties of $\widehat{u}_t$ and $\widehat{u}_t^2$, where $\widehat{u}_t$ is the demeaned return.

   (b) Compute the empirical distribution of $\widehat{u}_t$. Perform a test of normality and interpret the result.

   (c) Test for ARCH of orders $p = 1, 2, 5, 10$, in the Merger hedge fund returns.

   (d) Repeat parts (a) and (b) for the other six hedge funds.

2. **Time-variation in Stock Market Indexes**

   This question is based on the EViews file **HEDGE.WF1** which contains daily data on percentage returns of the S&P500, DOW and NASDAQ, from the 1st of April 2003 to the 28th of May 2010, a sample size of $T = 1869$.

   (a) Using the returns on the S&P500 index estimate the constant mean model

   $$R_SP500_t = \gamma_0 + u_t,$$

   and interpret the time series properties of $\widehat{u}_t$ and $\widehat{u}_t^2$, where $\widehat{u}_t$ is the demeaned return.

   (b) Compute the empirical distribution of $\widehat{u}_t$. Perform a test of normality and interpret the result.

   (c) Test for ARCH of orders $p = 1, 2, 5, 10$, in the S&P500 returns.

   (d) Repeat parts (a) and (b) for the DOW and NASDAQ stock market indexes.

3. **GARCH Models of Hedge Funds**

   This question is based on the EViews file **HEDGE.WF1** used in Question 1.

   (a) Using the returns on the Merger hedge fund estimate the GARCH(1,1) model

$$\begin{aligned} R_MERGER_t &= \gamma_0 + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1}. \end{aligned}$$

   Sketch the news impact curve.

   (b) Extend part (a) by estimating the TARCH(1,1) model

$$\begin{aligned} R_MERGER_t &= \gamma_0 + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

   Sketch the news impact curve. The parameter λ is commonly interpreted as the leverage effect in models of equity returns. Perform a test of symmetry by testing the restriction $\lambda = 0$. Interpret the results of the test.

   (c) Expand the TARCH model in part (b) to allow for day-of-the-week effects in both the conditional mean and the conditional variance. Perform a test of these effects in the

      i. Conditional mean
      ii. Conditional variance
      iii. Conditional mean and the conditional variance.

   (d) Given the results of part (c), reestimate the model by replacing the conditional normality distribution by the conditional standardized Student t distribution. Interpret the "degrees of freedom" parameter estimate.

   (e) Repeat parts (a) to (d) for the other size hedge funds. For each model compare the parameter estimates associated with each of the seven hedge funds.

4. **GARCH Models of Stock Market Indexes**

   This question is based on the EViews file **HEDGE.WF1** used in Question 2.

   (a) Using the daily percentage returns on the S&P500 index, estimate the GARCH(1,1) model

$$\begin{aligned} R_SP500_t &= \gamma_0 + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1}. \end{aligned}$$

Sketch the news impact curve.

(b) Extend part (a) by estimating the TARCH(1,1) model and sketch the news impact curve

$$\begin{aligned} R_SP500_t &= \gamma_0 + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

Sketch the news impact curve. Perform a test of symmetry by testing the restriction $\lambda = 0$. Interpret the results of the test.

(c) Expand the TARCH model in part (b) to allow for day-of-the-week effects in both the conditional mean and the conditional variance. Perform a test of these effects in the

i. Conditional mean

ii. Conditional variance

iii. Conditional mean and the conditional variance.

(d) Given the results of part (c), reestimate the model by replacing the conditional normality distribution by the conditional Standardized Student t. Interpret the "degrees of freedom" parameter estimate.

(e) Repeat parts (a) to (d) for the DOW and the NASDAQ. For each model compare the parameter estimates associated with each of the three stock market indexex.

(f) Suggest an alternative conditional volatility model to the TARCH model that allows for shocks to have asymmetric effects on volatility. Briefly discuss the advantages of the two alternative specifications.

5. **Time-varying Risk in Hedge Funds**

This question is based on the EViews file **HEDGE.WF1** used in Question 1.

(a) Estimate the following model

$$\begin{aligned} R_MERGER_t &= \gamma_0 + \theta h_t^{\beta} + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 d_{t-1}, \end{aligned}$$

with $\beta_1 = 1$. Test for a time-varying risk premium by testing the restriction $\theta = 0$. Interpret the result.

(b) Repeat part (a) with $\beta = 0.5$.

(c) Estimate the following model

$$\begin{aligned} R_MERGER_t &= \gamma_0 + \theta \log h_t + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

Test for a time-varying risk premium by testing the restriction $\theta = 0$. Interpret the result.

(d) Briefly discuss the differences in the risk preferences associated with the models in parts (a) to (c).

(e) Repeat parts (a) to (d) for the other six hedge funds. Compare the estimates of time-varying risk premium of the seven indexes.

6. **Time-varying Risk in Stock Market Indexes**

This question is based on the EViews file **HEDGE.WF1** used in Question 2.

(a) Estimate the following model

$$\begin{aligned} R_SP500_t &= \gamma_0 + \theta h_t^{\beta} + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_t + \lambda u_{t-1}^2 d_{t-1}, \end{aligned}$$

with for $\beta = 1$. Test for a time-varying risk premium by testing the restriction $\theta = 0$. Interpret the result.

(b) Repeat part (a) with $\beta = 0.5$.

(c) Estimate the following model

$$\begin{aligned} R_SP500_t &= \gamma_0 + \theta \log h_t + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1} + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

Test for a time-varying risk premium by testing the restriction $\theta = 0$. Interpret the result.

(d) Briefly discuss the differences in the risk preferences associated with the models in parts (a) to (c).

(e) Repeat parts (a) to (d) for the DOW and NASDAQ. Compare the estimates of time-varying risk premium of the three indexes.

7. **Capital Asset Pricing Model of Hedge Funds**

This question is based on the EViews file **HEDGE.WF1** used in Question 1.

(a) Estimate the following CAPM with constant variance for the Merger hedge fund

$$\begin{aligned} R_MERGER_t &= \gamma_0 + \gamma_1 R_SP500_t + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0. \end{aligned}$$

Interpret the parameter estimates and compute estimates of the idiosyncratic risk and the systematic risk.

(b) Estimate the following CAPM with time-varying variance for the Merger hedge fund

$$\begin{aligned} R_MERGER_t &= \gamma_0 + \gamma_1 R_SP500_t + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

Interpret the parameter estimates and sketch the news impact curve. Test the significance of the threshold parameter λ and interpret the result.

(c) Estimate the following CAPM with time-varying variance for the Merger hedge fund

$$\begin{aligned} R_MERGER_t &= \gamma_0 + \gamma_1 R_SP500_t + \theta h_t + u_t \\ u_t &\sim N(0, h_t) \\ h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \lambda u_{t-1}^2 d_{t-1}. \end{aligned}$$

Conduct a test of time variation in the risk on the hedge fund.

(d) Repeat parts (a) to (c) for the other six hedge funds. How successful are the hedge funds in minimizing exposure from systematic risk to the market? Discuss.

8. **Estimating Foreign Exchange Market Volatility**

This question is based on the file **HOUR.WF1** which contains hourly data on the British pound (BP) and the Deutchmark (DM) exchange rates, relative to the US, over the period 0.00am January 1st, 1986 to 11.00am July 15th 1986.

(a) For each exchange rate E, compute the returns as

$$R_E = LOG(E) - LOG(E(-1))$$

(b) Test for an $ARCH(1)$ effect in foreign exchange returns.

(c) Test for an $ARCH(2)$ effect in foreign exchange returns.

(d) Provided that there is a significant ARCH effect estimate a $GARCH(1,1)$ model.

(e) Provided that there is a significant ARCH effect estimate a $GARCH(1,1) - M$ model where the conditional mean is a function of the conditional standard deviation.

(f) Provided that there is a significant ARCH effect estimate a $GARCH(1,1) - M$ model where the conditional mean is a function of the conditional variance.

(g) Discuss the volatility of foreign exchange returns by comparing the estimated $GARCH$ standard deviation from the various models with actual returns.

9. **Minimum Variance Portfolio Model with Time-Varying Weights**

This question is based on the EViews file **HEDGE.WF1** used in Question 1. Let $R1$ and $R2$ be respectively the percentage continuously compounded daily returns on the Convertible Arbitrage hedge fund and the Distressed hedge fund

$$\begin{aligned} R1 &= R_CONVERTIBLE \\ R2 &= R_DISTRESSED \end{aligned}$$

(a) Calculate the variance-covariance matrix of $R1$ and $R2$ and interpret the elements.

(b) Consider a portfolio containing shares on the Convertible Arbitrage hedge fund and the Distressed hedge fund with respective weights w_1 and $w_2 = 1 - w_1$. The variance of the portfolio is given by

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{1,2},$$

where σ_1^2 and σ_2^2 are the respective Convertible Arbitrage hedge fund and the Distressed hedge fund return variances, and $\sigma_{1,2}$ is the covariance between the returns on the two funds. The minimum variance portfolio is given by setting the weight on the Convertible Arbitrage hedge fund in the portfolio to

$$w_1 = \frac{\sigma_2^2 - \sigma_{1,2}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{1,2}}.$$

Using the estimates of the variances and the covariance obtained in part (a), estimate w_1 and hence w_2. Interpret the results.

(c) Estimate the following regression equation

$$R2_t = \beta_0 + \beta_1 (R2_t - R1_t) + u_t,$$

where u_t is an error term and β_0 and β_1 are parameters. Hence identify the relationship between the estimate of β_1 obtained in this question and the estimate of w_1 obtained in part (b). Using this information, test the statistical significance of minimizing risk through portfolio diversification.

(d) Consider the minimum variance portfolio model with time-varying weights

$$R2_t = \beta_0 + \beta_{1,t-1}\left(R2_t - R1_t\right) + u_t,$$

where

$$\beta_{1,t-1} = \frac{Cov(R2_t, R2_t - R1_t|\Omega_{t-1})}{Var(R2_t - R1_t|\Omega_{t-1})},$$

and Ω_{t-1} is the information set at time $t-1$.

(e) Compute the following

$$\begin{aligned} E1 &= R1 - @MEAN\left(R1\right) \\ E2 &= R2 - @MEAN\left(R2\right) \end{aligned}$$

What is the relationship between the sample mean of $R1$ and $E1$? What is the relationship between the sample variance of $R1$ and $E1$?

(f) Estimate the following regression equations

$$\begin{aligned} \left(E2_t - E1_t\right)^2 &= \alpha_0 + \alpha_1\left(E2_{t-1} - E1_{t-1}\right)^2 + v_{1,t}, \\ \left(E2_t\right)\left(E2_t - E1_t\right) &= \beta_0 + \beta_1\left(E2_{t-1}\right)\left(E2_{t-1} - E1_{t-1}\right) + v_{2,t}, \end{aligned}$$

and use these results to estimate the time-varying weight $\beta_{1,t-1}$. Compare the estimate of $\beta_{1,t-1}$ with the constant weight estimate obtained in part (c).

Chapter 12

Modelling Variance II: Multivariate Models

Investing in financial securities is risky because of the variability of future returns. As seen in Chapter 11, the time-varying variance of financial asset returns is relatively easy to model for the case of an individual asset. Understanding the co-movements of financial returns is also of great practical importance. For example, asset pricing depends on the covariance of the assets in a portfolio and risk management and asset allocation relate for instance to finding and updating optimal hedging positions. In addition, forecasts of conditional covariance matrices of financial returns continue to influence the management of large amounts of funds under management. A 2012 survey of 139 North American investment managers representing $12 trillion worth of assets under management reported that the majority of managers use volatility and correlation forecasts to construct equity portfolios (Amenc, Goltz, Tang and Vaidyanathan, 2012). It follows therefore that specifying, estimating and forecasting using multivariate models of the conditional covariance matrix of a portfolio of assets are of great practical importance.
Let r_t denote a vector of time series $r_t = \{r_{1t}, \cdots r_{Nt}\}$ and let I_{t-1} represent the information set available at time $t-1$, the fundamental multivariate time-varying conditional variance problem addressed in this chapter is

$$\begin{aligned} r_t &= \mathrm{E}(r_t|I_{t-1}) + u_t \\ \mathrm{var}(u_t|I_{t-1} &= H_t. \end{aligned} \tag{12.1}$$

The focus here will be on the specification of H_t and, as a consequence, the mean equation will largely be ignored by invoking that simplifying assumption that r_t has been recentered to have zero mean. The natural approach to modelling a time-varying conditional covariance matrix is to extend the univariate GARCH framework to a multivariate version in which both variances and covariances are allowed to be time-varying. In addition to the the multivariate GARCH (MGARCH) models more recent developments focussing on

the dynamics of correlations between asset will be also be discussed. The central feature relating to volatility from Chapter 11, namely that it is regarded as unobservable, is maintained in these multivariate extensions. The relatively new areas of research in which realised volatility and realised covariance, which serve observable measures for volatility and covariance, are proposed are dealt with in Chapter **??**. The problem of forecasting multivariate volatility will be postponed until these new observable proxies for volatility have been introduced.

12.1 Motivation

12.1.1 Time-Varying Beta Risk

In Chapter 3 the beta risk of asset i, is defined as

$$\beta = \frac{\text{cov}(r_{it} - r_{ft}, r_{mt} - r_{ft})}{\text{var}(r_{mt} - r_{ft})} = \frac{\text{E}[(r_{it} - r_{ft})(r_{mt} - r_{ft})]}{\text{E}[(r_{mt} - r_{ft})^2]}$$

where $r_{it} - r_{ft}$ is the excess return on the asset relative to the risk-free rate given by r_{ft} and $r_{mt} - r_{ft}$ is the corresponding excess return on the market portfolio. Using the monthly data set on United States stocks for the period April 1990 to July 2004 ($T = 172$) introduced in Chapter 3, the constant beta risk for the stock Microsoft is easily estimated using the CAPM least squares regression. The estimate of the constant beta risk is $\widehat{\beta} = 1.447$.
The key restriction of constant beta risk may, however, be unrealistic. For example, the early 2000s was the period of the DotCom bubble and the beta risk of a technology stock like Microsoft could have been affected. Consequently, it would be desirable to be able to relax this restriction and allow beta to be time-varying. The specification of beta then becomes

$$\beta_t = \frac{\text{E}_{t-1}[(r_{1t} - r_{ft})(r_{mt} - r_{ft})]}{\text{E}_{t-1}[(r_{mt} - r_{ft})^2]} \tag{12.2}$$

where $\text{E}_{t-1}[(r_{1t} - r_{ft})(r_{mt} - r_{ft})]$ is the conditional covariance between the excess returns of Microsoft and the market based on information at time $t-1$ and $\text{E}_{t-1}[(r_{mt} - r_{ft})^2]$ is the conditional variance of the market excess return. The problem is how to operationalise the estimate of β_t in equation (12.2). From Chapter 11 the conditional covariance of Microsoft returns can be estimated using a simple GARCH model. Similarly, the same approach can be used to estimate the conditional covariance of the returns to the market portfolio, in this case proxied by the returns to the S&P 500 index. The problem is the conditional covariance term $\text{E}_{t-1}[(r_{1t} - r_{ft})(r_{mt} - r_{ft})]$. Clearly what is required is a bivariate model in which conditional variances of the constituent assets and their conditional covariance are all modelled simultaneously.
One approach would be to adopt the the multivariate versions of the historical variance and the exponentially weighted moving average estimate

discussed in Chapter 11. It the excess returns are collected into the vector $r_t = [r_{1t}\ r_{mt}]'$ then these measures are given as follows.

(i) **Historical Variance**:
The multivariate version of the historical estimate of the conditional covariance matrix of $\boldsymbol{r}_t$ is

$$H_t = \frac{1}{M}\sum_{j=1}^{T} r_{t-j} r'_{t-j}\,.$$

The unconditional covariance matrix results if $M = T$. The forecast of volatility for k-periods ahead is simply given by the current value H_t, irrespective of the value of k.

(ii) **Exponential Weighted Moving Average**:
The multivariate version of the exponentially weighted moving average estimate of conditional covariance is given by

$$\begin{aligned} H_t &= (1-\lambda)^j \sum_{j=0}^{\infty} \lambda^j r_{t-j} r'_{t-j} \\ &= (1-\lambda) r_t r'_t + \lambda H_t \end{aligned}$$

where, as before, λ is the delay parameter.

These two estimates suffer from exactly the same shortcomings as their univariate counterparts discussed in Chapter 11 and will not be pursued here. The search for a viable approach to the estimation of time-varying conditional covariance matrices starts by estimating simple GARCH(1,1) models for the excess returns to Microsoft, $r_{1t} - r_{ft}$, and the excess returns to the S&P 500 index, $r_{mt} - r_{ft}$. This exercise yields the following estimated equations for the conditional variances of Microsoft, h_{1t}, and the S&P 500, h_{mt}, respectively,

$$\begin{aligned} h_{1t} &= 5.955 + 0.141\,\widehat{u}_{1t}^2 + 0.802\,h_{t-1} \\ h_{mt} &= 0.573 + 0.112\,\widehat{u}_{2t}^2 + 0.858\,h_{mt-1}, \end{aligned}$$

in which $\widehat{u}_{1t}$ and $\widehat{u}_{2t}$ are the residuals from the respective mean equations which contain only a constant term. The conditional variance equations describe time variation in the variance of the individual assets but do not capture time variation in the conditional covariance between the two assets, h_{1mt}. From Chapter 2, recall that this the correlation between these two random variables is given by

$$\rho = \frac{h_{1m}}{\sqrt{h_m}\sqrt{h_1}}$$

where h_{1m} is the conditional covariance between them. This unconditional measure can be made into a time-varying one by allowing the variances and covariances to vary so that

$$\rho_t = \frac{h_{1mt}}{\sqrt{h_{mt}}\sqrt{h_{1t}}}.$$

If the additional restriction of a constant correlation is imposed, $\rho_t = \overline{\rho}$ then an estimate of the time-varying conditional covariance is given by

$$h_{1mt} = \overline{\rho}\sqrt{h_{mt}}\sqrt{h_{1t}}.$$

A reasonable choice for $\overline{\rho}$ is the sample correlation between the excess returns to Microsoft and the excess returns on the market. Using the estimated equations for h_{mt} and h_{1t} and $\overline{\rho} = 0.5804$, the sample correlation coefficient, a series for the conditional covariance, h_{1mt}, can be computed. The two conditional variances and the conditional covariance computed in this way are shown in Figure 12.1. Although this a very simple approach, the major insight it provides has proved important in developing workable multivariate GARCH models which are discussed in Section 12.4.

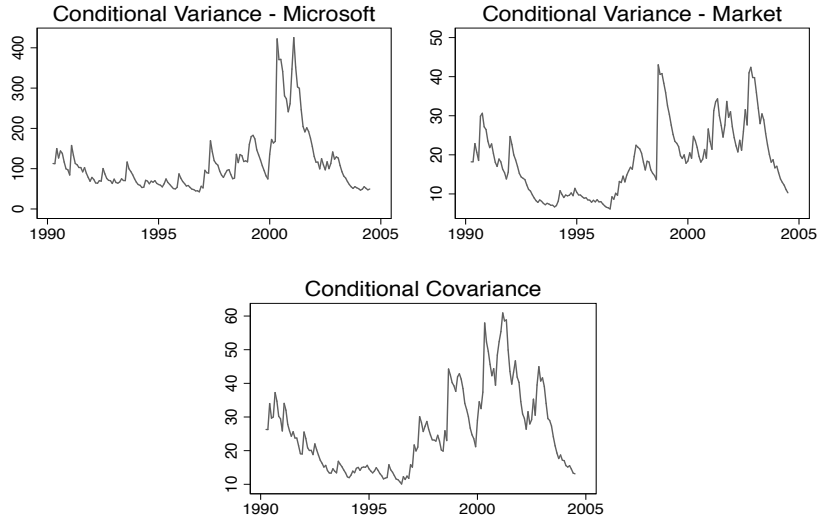


Figure 12.1: Conditional variances (top panel) and covariance (bottom panel) of Microsoft and the S&P500 index. The data are monthly for the period April 1990 to July 2004 ($T = 172$).

It is apparent that the conditional variances change over the sample with Microsoft showing an marked increase in volatility at the time of the DotCom bubble in the early 2000s. Figure 12.1 also shows that the covariance and the variance of the market tend to decrease in-step with each other in the first half of the sample, but appear to be out of alignment in the second half of the sample.

Using these estimates for the conditional covariance, h_{1mt}, and the conditional variance of the market, h_{mt}, an estimate of time-varying beta risk is plotted in Figure 12.2 and superimposed on the constant estimate of beta risk. There are

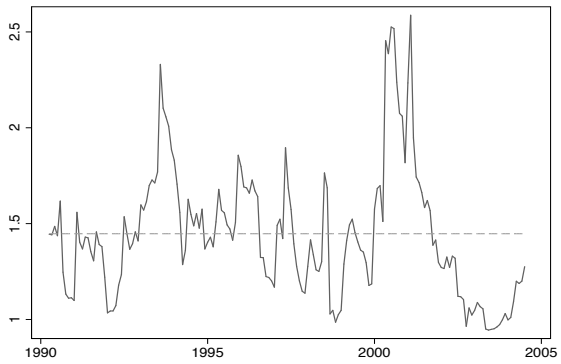


Figure 12.2: Estimate of time-varying beta risk for Microsoft based on the assumption of a constant correlation between Microsoft and the S&P 500 index. The constant beta risk estimated from a CAPM model of 1.447 is shown as the dashed line. The data are monthly for the period April 1990 to July 2004 ($T = 172$).

some very large changes in the beta risk of Microsoft, ranging from around 0.2 at the end of 1995 to nearly 2.5 at the time of the DotCom bubble in the middle of 2000. The sample average is 1.196 which is a little lower than the constant estimate of beta risk given by 1.447.

12.1.2 Time-varying Portfolio Weights

Another simple ($N = 2$) but important example where knowledge of the the entire time-varying conditional covariance matrix is required is the construction of the optimal portfolio. The minimum variance portfolio containing two assets with returns r_{1t} and r_{2t} has optimal weights given by

$$w_1 = \frac{\text{var}(r_{2t}) - \text{cov}(r_{1t}, r_{2t})}{\text{var}(r_{1t}) + \text{var}(r_{2t}) - 2\,\text{cov}(r_{1t}, r_{2t})}$$
$$w_2 = \frac{\text{var}(r_{1t}) - \text{cov}(r_{1t}, r_{2t})}{\text{var}(r_{1t}) + \text{var}(r_{2t}) - 2\,\text{cov}(r_{1t}, r_{2t})}$$

where w_1 is the optimal weight allocated to asset 1 in the portfolio and w_2 is the corresponding weight on asset 2.
Even in this simple case of two assets, $N = 2$, construction of a model based estimate of the covariance matrix is not entirely straightforward. There are two time-varying variances and one time-varying covariance

$$\begin{aligned}
\text{var}(r_{1t}) &: h_{11t} = \text{E}_{t-1}[(r_{1t} - \text{E}_{t-1}[r_{1t}])^2] \\
\text{var}(r_{2t}) &: h_{22t} = \text{E}_{t-1}[(r_{2t} - \text{E}_{t-1}[r_{2t}])^2] \\
\text{cov}(r_{12}, r_{2t}) &: h_{12t} = \text{E}_{t-1}[(r_{1t} - \text{E}_{t-1}[r_{1t}])(r_{2t} - \text{E}_{t-1}[r_{2t}])],
\end{aligned}$$

so that the conditional covariance matrix is

$$H_t = \begin{bmatrix} h_{11t} & h_{12t} \\ h_{21t} & h_{22t} \end{bmatrix} = \begin{bmatrix} h_{11t} & h_{12t} \\ h_{12t} & h_{22t} \end{bmatrix} \tag{12.3}$$

because of symmetry.

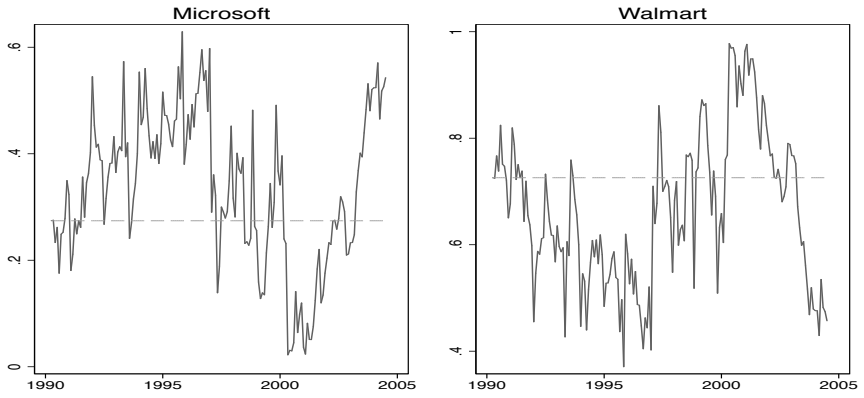


Figure 12.3: Estimates of time-varying weights for portfolio comprising Microsoft and Walmart. The weights are constructed based on the assumption of a constant correlation between Microsoft and Walmart. The data are monthly for the period April 1990 to July 2004 ($T = 172$).

The time-varying portfolio weights, together with the optimal constant weights are shown in Figure 12.3. Obviously as the weights in this two asset portfolio sum to 1, the time varying weights are mirror images of each other. The one important feature of these weights is that the constant values obtained from the regression approach in Chapter 3 are completely dominated by the DotCom crisis. The time-varying versions show that Microsoft should have received the higher weight in the portfolio for the entire 10-year period leading up to 2000. This result demonstrates the usefulness of modelling time-variation in the variances and covariances of financial assets explicitly rather than relying on the simplifying assumption of constant relationships.

12.2 Heatwaves and Meteor Showers

An early empirical application aimed at extending the GARCH framework into multi-dimensions, while still using the tools developed in the univariate context, examined how volatility is transmitted through different regions of the world during the course of a global financial trading day (Ito, 1987: Ito and Roley, 1987; Engle, Ito and Lin (1990). The approach is to partition each 24 hour period (calendar day) into a three major trading zones, namely, Japan

(12am to 7am GMT), Europe (7am to 12:30pm GMT) and the United States (12:30pm to 9pm GMT), which may be illustrated as follows:

$$\underbrace{\overbrace{12\text{am}\cdots 7\text{am}}^{\text{Japan}}\ \overbrace{7\text{am}\cdots 12\text{pm}}^{\text{Europe}}\ \overbrace{12\text{pm}\cdots 9\text{pm}}^{\text{U.S.}}}_{\text{One Trading Day}}$$

Note that there are other ways of carving up the global trading day, see for example Dungey, Fakhrutdinova and Goodhart (2009), but the main thrust of the argument remains the same irrespective of the minor adjustments to this definition.

The calendar structure implied by the global trading day defines a number of restrictions on a three equation system which uses a simple GARCH(1,1) model for modelling the conditional variance in each of the trading zones. Define r_{1t}, r_{2t} and r_{3t} as the daily returns to the Japanese zone, the European zone and the United States zone, respectively. The model is

$$\begin{bmatrix} r_{1t} \\ r_{2t} \\ r_{3t} \end{bmatrix} = \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix}, \qquad \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} h_{1t} & 0 & 0 \\ 0 & h_{2t} & 0 \\ 0 & 0 & h_{3t} \end{bmatrix} \right)$$

$$\begin{bmatrix} h_{1t} \\ h_{2t} \\ h_{3t} \end{bmatrix} = \begin{bmatrix} \alpha_{10} \\ \alpha_{20} \\ \alpha_{30} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ \alpha_{21} & 0 & 0 \\ \alpha_{31} & \alpha_{32} & 0 \end{bmatrix} \begin{bmatrix} u_{1t}^2 \\ u_{2t}^2 \\ u_{3t}^2 \end{bmatrix} + \begin{bmatrix} \beta_{11} & 0 & 0 \\ 0 & \beta_{22} & 0 \\ 0 & 0 & \beta_{33} \end{bmatrix} \begin{bmatrix} h_{1\,t-1} \\ h_{2\,t-1} \\ h_{3\,t-1} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} \\ 0 & \gamma_{22} & \gamma_{23} \\ 0 & 0 & \gamma_{33} \end{bmatrix} \begin{bmatrix} u_{1\,t-1}^2 \\ u_{2\,t-1}^2 \\ u_{3\,t-1} \end{bmatrix}. \tag{12.4}$$

The calendar structure of the global trading day is now apparent. New developments at the start of the global trading day in Japan, u_{1t}^2, can potentially influence volatility in Europe and the United States via the coefficients α_{21} and α_{31}. Similarly news from Europe, u_{2t}^2, can influence volatility in the United States on the same global trading day, α_{32}. The natural calendar structure, however, implies that events in the United States will be transmitted to Japan only on the following day. The restrictions on the γ_{ij} on the lagged innovations, which require the matrix to be upper diagonal, imply that all information originates during United States trading times.

While this model looks very much like a multivariate GARCH model, there is no contemporaneous conditional covariance because the regions are defined to be non-overlapping. For this reason single equation estimation of the model by the maximum likelihood can be performed on each zone using the estimation methods outlined in Chapter 11. The aim is to examine international linkages in volatility between these regions and investigate in partic-

ular two patterns as possible descriptors of international volatility transmission.

(i) **Heatwave**
Volatility in any one region is primarily a function of the previous day's volatility in the same region.

(ii) **Meteor Shower**
Volatility in one region is driven primarily by volatility in the region immediately preceding it in terms of calendar time.

To test this model continuously-traded, high-frequency data is used to construct returns to the Euro-Dollar exchange rate futures contracts traded on the Chicago Mercantile Exchange for period 3 January 2005 to 28 February 2013 ($T = 1976$ trading days). The return in each zone is calculated as the difference between the last and the first transaction price for the time period in which the zone trades, normalised by the number of hours for which the zone trades, given by

$$r_{it} = (\log P_{it}^c - \log P_{it}^o)/\sqrt{nh_i}, \qquad i = 1, 2, 3 \tag{12.5}$$

in which, P_{it}^c is a closing price of the futures contract in zone i on day t, and P_{it}^o is the opening price of the contract in zone i on day t and nh_i is the number of hours for which zone i trades. Descriptive statistics for the returns from each zone are presented in Table 12.1. While none of the returns series from the three zones exhibit large degrees of skewness, they all exhibit excess kurtosis. Formal testing reveals that all the series strong ARCH effects at the 5% level.

Table 12.1

Descriptive statistics for daily returns (annualised percentages) on the Euro-Dollar exchange rate futures contract traded on the Chicago Mercantile Exchange for the Japanese, European and the United States trading zones. The sample period is 3 January 2005 to 28 February 2013 ($T = 1976$).

	Japan	Europe	U.S.
Mean	0.038	-0.086	0.056
Std. dev.	1.214	1.873	1.876
Max.	6.707	8.147	11.547
Min.	-5.821	-12.754	-9.307
Skewness	-0.014	-0.293	0.049
Kurtosis	5.640	5.874	5.378

The estimation results for equation (12.4) based on the foreign exchange market futures data are reported in Table 12.2. Note that the constant term in

the variance equation is suppressed. There are two general conclusions that emerge from inspection of these results.

1. All the lagged conditional variance terms, h^i_{t-1}, are statistically significant and of the order of 0.9, an estimate which is commonly obtained in univariate GARCH models applied to financial returns. The statistical significance of these terms is consistent with the heatwave hypothesis being part of the explanation of the patterns in global volatility.

2. The meteor shower effect (diurnal effect of the news) is also important: Japanese news affects Europe and European news affects the United States on the same trading day. Note that in the case of Japan, the meteor shower effect shows up in the significance of the lagged influence of the United States innovations, $u^2_{3\,t-1}$, on Japan.

It seems clear, therefore, that the pattern of volatility interaction in global foreign markets is a combination of both heat waves and meteor showers. There is no support for the conclusion that either one of these patterns dominates.

Table 12.2

Coefficient estimates of the equation (12.4) for the Euro-Dollar foreign exchange futures data for each of the three trading zones. Coefficients that are significant at the 5% level are marked (*).

	Japan	Europe	United States
u^2_{1t}	-	0.0356*	-0.0320
u^2_{2t}	-	-	0.0646*
u^2_{3t}	-	-	-
$u^2_{1\,t-1}$	0.0850*	-	-
$u^2_{2\,t-1}$	-0.0258	0.0850*	-
$u^2_{3\,t-1}$	0.0764*	-0.0027*	0.0814*
h^i_{t-1}	0.8832*	0.9024*	0.8992*

This pattern of interaction suggests the transmission of news between different regions of the world on the same trading day is a potentially important explanation of volatility. This result simply adds to the compelling argument to support the importance of being able to estimate the full covariance matrix of a system of M financial asset returns, $r_t = \{r_{1t}, \cdots, r_{Mt}\}$, given that continuous trading on many important financial exchanges now makes the artefact of dividing the global day into different trading zones largely irrelevant.

12.3 Multivariate Conditional Covariance

Consider a set of N asset returns, $\boldsymbol{r}_t = \{r_{1t}, \cdots r_{Nt}\}$, where without loss of generality the returns are assumed to be centered to have zero mean. A time-varying estimate of the conditional covariance matrix requires construction of the matrix

$$H_t = \begin{bmatrix} h_{11t} & h_{12t} & \cdots & h_{1Nt} \\ h_{21t} & h_{22t} & \cdots & h_{2Nt} \\ \vdots & \vdots & \ddots & \vdots \\ h_{N1t} & h_{N2t} & \cdots & h_{NNt} \end{bmatrix},$$

which is a symmetric matrix because $h_{ijt} = h_{jit}$ so that there are $N(N+1)/2$ unique elements. A good estimate of H_t satisfies two conditions.

(i) **Positive Definiteness**
All conditional variances and covariances must be positive for all t. This restriction requires that the conditional covariance matrix H_t be positive definite. For $N = 2$ in equation, the conditions for positive definiteness are

$$\begin{aligned} h_{11t} &> 0 \\ h_{11t}h_{22t} - h_{12t}^2 &> 0 \end{aligned}$$

The second condition may be rewritten as

$$-1 < \frac{h_{12t}}{\sqrt{h_{11t}h_{22t}}} < 1$$

so the correlation needs to be between -1 and 1 at every t. Ensuring that this condition is met is not straightforward, particularly as the dimension grows.

(ii) **Parameter Dimension**
Consider a multivariate version of the simple AR(1) model of squared returns introduced in Chapter 11 to motivate the ARCH model. In theory there is no reason why conditional variances and covariances should not be functions of shocks from all other assets, an observation that suggests the following specification

$$\begin{aligned} r_{1t}^2 &= \alpha_0 + \alpha_1 r_{1t-1}^2 + \alpha_2 r_{2t-1}^2 + \alpha_3 r_{1t-1}r_{2t-1} + v_{1t} \\ r_{2t}^2 &= \beta_0 + \beta_1 r_{1t-1}^2 + \beta_2 r_{2t-1}^2 + \beta_3 r_{1t-1}r_{2t-1} + v_{2t} \\ r_{1t}r_{2t} &= \gamma_0 + \gamma_1 r_{1t-1}^2 + \gamma_2 r_{2t-1}^2 + \gamma_3 r_{1t-1}r_{2t-1} + v_{3t} \end{aligned}$$

This simple specification for two returns r_{1t} and r_{2t} already has 12 parameters to estimate. For large portfolios, say $N = 50$ or 100, the model becomes very difficult to estimate as there are potentially too many parameters.

Dealing with these two problems has seen the development of a number of multivariate GARCH specifications which are designed to provide a specification that is flexible enough to model the dynamics of volatility and covolatility over time, while ensuring both that the covariance matrix is positive definite and controlling the dimension of the parameter space.

12.3.1 The VECH Model

This is the first MGARCH specification proposed in the literature. The specification is based on using just the $N(N+1)/2$ unique elements of the covariance matrix H_t, which this is achieved by using the vech $(\cdot)$ matrix operator. For $N = 2$, the vech operator is defined as follows:

$$H_t = \begin{bmatrix} h_{11t} & h_{12t} \\ h_{12t} & h_{22t} \end{bmatrix} \implies \text{vech}(H_t) = \begin{bmatrix} h_{11t} \\ h_{12t} \\ h_{22t} \end{bmatrix}$$

For $N = 3$ the operation of the vech operator yields

$$H_t = \begin{bmatrix} h_{11t} & h_{12t} & h_{13t} \\ h_{12t} & h_{22t} & h_{23t} \\ h_{13t} & h_{23t} & h_{33t} \end{bmatrix} \implies \text{vech}(H_t) = \begin{bmatrix} h_{11t} \\ h_{12t} \\ h_{13t} \\ h_{22t} \\ h_{23t} \\ h_{33t} \end{bmatrix}.$$

The equation governing the dynamics of the conditional variance for an MARCH(1,1) model using the VECH specification is

$$\text{vech}(H_t) = C + A\ \text{vech}(u_{t-1}u'_{t-1}) + B\ \text{vech}(H_{t-1})$$

which, in the case of $N = 2$ becomes

$$\begin{bmatrix} h_{11t} \\ h_{12t} \\ h_{22t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} u^2_{1\,t-1} \\ u_{1\,t-1}u_{2\,t-1} \\ u^2_{2\,t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{3,1} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} h_{11\,t-1} \\ h_{12\,t-1} \\ h_{22\,t-1} \end{bmatrix},$$

after application of the vech$(\cdot)$ operator.
The VECH model represents the first attempt at building an MGARCH model but it has two important drawbacks. The first is that H_t is not necessarily positive definite, even if $c_i > 0$ for $i = 1, 2, 3$. The second is that the number of parameters to be estimated increases exponentially as the dimension increases. For example, in the case of $N = 2$ there are

$$\frac{N(N+1)}{2} + 2\Big(\frac{(N(N+1)}{2}\Big)^2 = \frac{2(2+1)}{2} + 2\Big(\frac{(2(2+1)}{2}\Big)^2 = 21,$$

variance parameters to be estimated. If $N = 3$ it is easily verified that there are 78 variance parameters to be estimated. Consequently the VECH version of the MGARCH model is not used much in practice, although a diagonal version, in which all the off-diagonal elements of the matrices C, A and B are zero, has proved more popular.

12.3.2 The BEKK Model

Letting H_t be the conditional covariance matrix at time t, and u_t the vector of disturbances, the BEKK specification for a MGARCH(1,1) model is

$$H_t = CC' + A\, u_{t-1}u'_{t-1}\, A' + B\, H_{t-1}B'$$

where C is a $(N \times N)$ lower triangular matrix of unknown parameters, and A and B are $(N \times N)$ matrices each containing N^2 unknown parameters associated with the lagged disturbances and the lagged conditional covariance matrix respectively.
There are three different flavours of BEKK model, each of which is now spelled out for the case $N = 2$.

(i) **Asymmetric BEKK Model**

The parameter matrices are

$$C = \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

This is the asymmetric BEKK model as $a_{12} \neq a_{21}, b_{12} \neq b_{21}$.

(ii) **Symmetric BEKK Model**

The restrictions $a_{12} = a_{2,1}$ and $b_{12} = b_{21}$, are imposed so

$$C = \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \end{bmatrix}$$

(iii) **Diagonal BEKK Model**

The restrictions $a_{12} = a_{21} = 0$ and $b_{12} = b_{21} = 0$, are imposed so

$$C = \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & 0 \\ 0 & b_{22} \end{bmatrix}$$

A special case of the BEKK model is where there is just the one variable $(N = 1)$, so the parameter matrices become scalars

$$C = [\ c_{11}\ ], \quad A = [\ a_{11}\ ], \quad B = [\ b_{11}\ ].$$

The conditional covariance matrix now reduces to a scalar given by

$$H_t = h_{11t} = c_{11}^2 + a_{11}^2 u_{t-1}^2 + b_{11}^2 h_{11\,t-1}$$

which is simply the univariate GARCH(1,1) model discussed earlier with the difference that the parameters are squared.
This last feature of the BEKK model highlights the motivation behind the choice of the specification. In the univariate case the conditional variance is constrained to be positive because all terms on the RHS are positive, that is,

$$u_{t-1}^2 > 0, \quad \left\{c_{11}^2, a_{11}^2, b_{11}^2\right\} > 0, \quad h_{11\,t-1} > 0.$$

The BEKK specification has the advantage that it solves the first problem of positive definiteness, but not necessarily the second problem when the dimension of the model is relatively large. For most empirical work using this model is based on N less than 10 and often it is $N = 2$ or $N = 3$. In addition, the unrestricted model contains parameters that do not represent directly the impact of u_{t-1} or H_{t-1} on the elements of H_t and this makes it hard to interpret the parameters of a BEKK model.

12.4 Multivariate Conditional Correlation

In Section 12.1 the simple assumption of constant correlation between two financial assets was made in order to develop an estimate of the conditional covariance between them. This assumption has proved to be a good starting point for the development of multivariate models of conditional correlation which have become the preferred way in which to model the conditional covariance between systems of financial assets.

12.4.1 Constant Conditional Correlation (CCC) Model

As established in Chapter 2, a covariance is equal to correlation times the respective standard deviations. The conditional covariance matrix is therefore defined as

$$H_t = S_t\, R\, S_t,$$

where S_t is a $(N \times N)$ diagonal matrix of time-varying standard deviations and R is a $(N \times N)$ matrix of *constant* correlations. This generalises the simple assumption made in Section 12.1 from the bivariate to the multivariate case. The standard deviation matrix S_t is a diagonal matrix with the time-varying standard deviations down the main diagonal

$$S_t = \begin{bmatrix} \sqrt{h_{1t}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{h_{Nt}} \end{bmatrix}$$

which are the square roots of univariate GARCH specifications

$$h_{it} = \alpha_{0i} + \alpha_{1i} u_{i\,t-1}^2 + \beta_{1i} h_{i\,t-1}, \qquad i = 1, 2, \cdots N.$$

The constant correlation matrix R is defined as

$$R = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1N} \\ r_{21} & 1 & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \cdots & 1 \end{bmatrix} = \text{diag}(\overline{Q})^{-1/2}\, \overline{Q}\, \text{diag}(\overline{Q})^{-1/2},$$

which is a symmetric matrix. The matrix $\overline{Q}$ is defined as

$$\overline{Q} = \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} z_{1t}^2 & z_{1t}z_{2t} & \cdots & z_{1t}z_{Nt} \\ z_{2t}z_{1t} & z_{2t}^2 & \cdots & z_{2t}z_{Nt} \\ \vdots & \vdots & \ddots & \vdots \\ z_{Nt}z_{1t} & z_{Nt}z_{2t} & \cdots & z_{Nt}^2 \end{bmatrix} \tag{12.6}$$

and z_{it} represents the standardised disturbances from the GARCH models

$$z_{it} = \frac{u_{it}}{\sqrt{h_{it}}}.$$

For illustrative purposes, it is useful to outline the construction of the conditional covariance matrix of the constant model for the 2 asset case. The usual notation to denote estimates will be suppressed for simplicity. Use the stationary univariate GARCH(1, 1) models for the two assets

$$\begin{aligned} h_{1t} &= \alpha_{01} + \alpha_{11}u_{1\,t-1}^2 + \beta_{11}h_{1\,t-1} \\ h_{2t} &= \alpha_{02} + \alpha_{12}u_{2\,t-1}^2 + \beta_{12}h_{2\,t-1}, \end{aligned}$$

construct the standard deviation matrix

$$S_t = \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix}.$$

Compute the standardised disturbances

$$\begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} = S_t^{-1}u_t = \begin{bmatrix} \frac{1}{\sqrt{h_{1t}}} & 0 \\ 0 & \frac{1}{\sqrt{h_{2t}}} \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix} = \begin{bmatrix} \frac{u_{1t}}{\sqrt{h_{1t}}} \\ \frac{u_{2t}}{\sqrt{h_{2t}}} \end{bmatrix}$$

and $\overline{Q}$

$$\overline{Q} = \begin{bmatrix} \overline{q}_{11} & \overline{q}_{12} \\ \overline{q}_{12} & \overline{q}_{22} \end{bmatrix} = \frac{1}{T}\sum_{t=1}^{T} \begin{bmatrix} z_{1t}^2 & z_{1t}z_{2t} \\ z_{2t}z_{1t} & z_{2t}^2 \end{bmatrix}.$$

Use $\overline{Q}$ to estimate the constant correlation matrix as

$$R = \begin{bmatrix} 1 & \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}} \\ \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}} & 1 \end{bmatrix}.$$

Finally, construct the conditional covariance matrix as

$$\begin{aligned} H_t &= \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix} \begin{bmatrix} 1 & \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}} \\ \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix} \\ &= \begin{bmatrix} h_{1t} & \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}}\sqrt{h_{1t}h_{2t}} \\ \frac{\overline{q}_{12}}{\sqrt{\overline{q}_{11}\overline{q}_{22}}}\sqrt{h_{1t}h_{2t}} & h_{2t} \end{bmatrix} \end{aligned}$$

The CCC model solves both the positive definiteness and the dimensionality issues. However, the assumption of correlations being constant over the sample is potentially restrictive. Consequently subsequent developments have attempted to relax this assumption.

12.4.2 Dynamic Conditional Correlation (DCC) Models

The Dynamic Conditional Correlation (DCC) model relaxes the assumption that the correlations are constant over time and introduces some simple dynamics to the time evolution of the matrix. The conditional covariance matrix is now specified as

$$H_t = S_t \, R_t \, S_t$$

where R_t is now a $(N \times N)$ *conditional* correlation matrix. As before S_t is a diagonal matrix containing the conditional standard deviations

$$S_t = \begin{bmatrix} \sqrt{h_{1t}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{h_{N,t}} \end{bmatrix}$$

where the conditional variances have univariate GARCH representations

$$h_{it} = \alpha_{0i} + \alpha_{1i} u_{i\,t-1}^2 + \beta_{1i} h_{i\,t-1}, \qquad i = 1, 2, \cdots N.$$

The conditional correlation matrix R_t is given by

$$R_t = \text{diag}(Q_t)^{-1/2} \, Q_t \, \text{diag}(Q_t)^{-1/2},$$

where Q_t has a GARCH(1,1) type specification

$$Q_t = (1 - \alpha - \beta)\overline{Q} + \alpha z_{t-1} z_{t-1}' + \beta Q_{t-1}.$$

The definition of the intercept in the dynamics of Q_t as $(1 - \alpha - \beta)\overline{Q}$ corresponds to the idea of variance targeting in introduced in Engle and Mezrich (1996), in which specifying the intercept in this way reduces the difficulty of

the estimation problem because the intercept no longer has to be estimated separately. In fact, this specification of the dynamics of the conditional correlation matrix is particularly parsimonious because it contains just two unknown scalar parameters, α and β (with $\alpha + \beta < 1$), the standardised residuals

$$z_{it} = \frac{u_{it}}{\sqrt{h_{it}}},$$

and the quasi-correlation matrix (Aielli, 2009), $\overline{Q}$, given by

$$\overline{Q} = \frac{1}{T}\sum_{t=1}^{T}\begin{bmatrix} z_{1t}^2 & z_{1t}z_{2t} & \cdots & z_{1t}z_{Nt} \\ z_{2t}z_{1t} & z_{2t}^2 & \cdots & z_{2t}z_{Nt} \\ \vdots & \vdots & \ddots & \vdots \\ z_{Nt}z_{1t} & z_{Nt}z_{2t} & \cdots & z_{Nt}^2 \end{bmatrix}. \tag{12.7}$$

A special case of the DCC model is the constant correlation model that arises when $\alpha = \beta = 0$ resulting in $Q_t = \overline{Q}$ and $R_t = R$. Note that there are no tests of constancy of correlations directly against the DCC model because the DCC model is only identified if correlations are changing (see Silvennoinen and Teräsvirta, 2009).

An alternative specification of the evolution of the correlation matrix, R_t, is provided by the varying correlation model (VCC) of Tse and Tsui (2002). In this model the dynamics are specified directly for the matrix R_t and are given by

$$R_t = (1 - \alpha - \beta)S + \alpha S_{t-1} + \beta R_{t-1},$$

where S is symmetric positive definite matrix with ones on the main diagonal and S_{t-1} is a sample correlation matrix of the past M standardised residuals. The typical element of S_{t-1} is given by

$$s_{ij\,t-1} = \frac{\sum_{m=1}^{M} z_{i\,t-m}z_{j\,t-m}}{\sqrt{\left(\sum_{m=1}^{M} z_{i\,t-m}^2\right)\left(\sum_{m=1}^{M} z_{j\,t-m}^2\right)}}.$$

A necessary condition for S_{t-1} to be positive definite is that the size of the window M be greater than the number of assets in the system being estimated.

The estimation of the DCC model for the $N = 2$ asset case proceeds as follows. The stationary univariate GARCH(1, 1) models for the two assets

$$\begin{aligned} h_{1t} &= \alpha_{01} + \alpha_{11}u_{1\,t-1}^2 + \beta_{11}h_{1\,t-1} \\ h_{2t} &= \alpha_{02} + \alpha_{12}u_{2\,t-1}^2 + \beta_{12}h_{2\,t-1}, \end{aligned}$$

are used to construct the standard deviation matrix

$$S_t = \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix}.$$

Compute the standardised residuals

$$\begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} = S_t^{-1} u_t = \begin{bmatrix} \frac{1}{\sqrt{h_{1t}}} & 0 \\ 0 & \frac{1}{\sqrt{h_{2t}}} \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix} = \begin{bmatrix} \frac{u_{1t}}{\sqrt{h_{1t}}} \\ \frac{u_{2t}}{\sqrt{h_{2t}}} \end{bmatrix},$$

and $\overline{Q}$

$$\overline{Q} = \begin{bmatrix} \overline{q}_{11} & \overline{q}_{12} \\ \overline{q}_{12} & \overline{q}_{22} \end{bmatrix} = \frac{1}{T} \sum_{t=1}^{T} \begin{bmatrix} z_{1t}^2 & z_{1t} z_{2t} \\ z_{2t} z_{1t} & z_{2t}^2 \end{bmatrix}.$$

The matrix Q_t is now given by

$$\begin{aligned} Q_t &= (1 - \alpha - \beta) \begin{bmatrix} \overline{q}_{11} & \overline{q}_{12} \\ \overline{q}_{12} & \overline{q}_{22} \end{bmatrix} \\ &\quad + \alpha \begin{bmatrix} z_{1\,t-1}^2 & z_{1\,t-1} z_{2\,t-1} \\ z_{1\,t-1} z_{2\,t-1} & z_{2\,t-1}^2 \end{bmatrix} + \beta \begin{bmatrix} q_{11\,t-1} & q_{12\,t-1} \\ q_{12\,t-1} & q_{22\,t-1} \end{bmatrix}, \end{aligned}$$

and the correlation matrix is

$$R_t = \begin{bmatrix} 1 & \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} \\ \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & \rho_{12t} \\ \rho_{12t} & 1 \end{bmatrix}.$$

The final step is the construction of the conditional covariance matrix, which is given by

$$\begin{aligned} H_t &= \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix} \begin{bmatrix} 1 & \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} \\ \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{h_{1t}} & 0 \\ 0 & \sqrt{h_{2t}} \end{bmatrix} \\ &= \begin{bmatrix} h_{1t} & \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} \sqrt{h_{1t} h_{2t}} \\ \frac{q_{12t}}{\sqrt{q_{11t} q_{22t}}} \sqrt{h_{1t} h_{2t}} & h_{2t} \end{bmatrix}. \end{aligned}$$

Notice that the covariance (off-diagonal terms of H_t are identical) is now time-varying as a result of the two GARCH models h_{1t} and h_{2t} and the time-varying correlation $r_{12t} = q_{12t} / \sqrt{q_{11t} q_{22t}}$.

12.4.3 Dynamic Equicorrelation (DECO) Model

The key assumption underlying the Dynamic Equicorrelation (DECO) model of Engle and Kelly (2009) is that the unconditional correlation matrix of systems of financial returns has entries of roughly similar magnitude. The contemporaneous correlations of the DCC model are therefore assumed to be

equal across all N variables, but not over time. The pertinent restrictions on the correlation matrix are

$$R_t = \begin{bmatrix} 1 & r_{12t} & \cdots & r_{1Nt} \\ r_{21t} & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & r_{N-1\,Nt} \\ r_{N1t} & \cdots & r_{N\,N-1\,t} & 1 \end{bmatrix} = \begin{bmatrix} 1 & \bar{r}_t & \cdots & \bar{r}_t \\ \bar{r}_t & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & \bar{r}_t \\ \bar{r}_t & \cdots & \bar{r}_t & 1 \end{bmatrix},$$

where $\bar{r}_t$ represents the average of the $N(N-1)/2$ correlations at time t

$$\bar{r}_t = \frac{2}{N(N-1)} \sum_{i>j} r_{ijt}\,.$$

Rewriting these restrictions shows that

$$R_t = (1-\bar{r}_t) \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & 1 \end{bmatrix} + \bar{r}_t \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & 1 \\ 1 & 1 & \cdots & 1 \end{bmatrix} = (1-\bar{r}_t) I_N + \bar{r}_t O_N\,,$$

where I_N is the identity matrix and O_N is a $(N \times N)$ matrix of ones. This form of the correlation matrix, R_t, has the following analytical properties

$$|R_t| = (1-\bar{r}_t)^{N-1}(1+(N-1)\bar{r}_t) \tag{12.8}$$

$$R_t^{-1} = \frac{1}{1-\bar{r}_t} I_N - \frac{\bar{r}_t}{(1-\bar{r}_t)(1+(N-1)\bar{r}_t)} O_N\,. \tag{12.9}$$

As will become apparent, these expressions greatly simplify the construction of the log-likelihood function when the model is estimated.

12.5 Estimation

Following the methods introduced in Chapter 10, for a sample of $t = 1, 2, ..., T$, observations, the log-likelihood function of a multivariate GARCH model is given by

$$\log L = \frac{1}{T} \sum_{t=1}^{T} \log L_t = \frac{1}{T} \sum_{t=1}^{T} \log f(r_{1t}, r_{2t}, \cdots, r_{N,t})$$

where $f(r_{1t}, r_{2t}, \cdots, r_{N,t})$ is a N-dimensional multivariate probability distribution. To implement the estimation by maximum likelihood methods, it is necessary to specify the functional form of the N-dimensional probability distribution. There are two popular choices in empirical applications, namely, the multivariate normal distribution and the multivariate t distribution.

12.5.1 Log-likelihood Functions

Multivariate Normal Distribution

The multivariate normal distribution is given by

$$f(r_{1t}, r_{2t}, \cdots, r_{N,t}) = \left(\frac{1}{2\pi}\right)^{N/2} |H_t|^{-1/2} \exp\left(-0.5\, u_t' H_t^{-1} u_t\right)$$

where u_t is the $(N \times 1)$ vector of disturbances at time t. The disturbances are obtained from

$$\begin{bmatrix} u_{1t} \\ u_{2t} \\ \vdots \\ u_{N,t} \end{bmatrix} = \begin{bmatrix} r_{1t} \\ r_{2t} \\ \vdots \\ r_{N,t} \end{bmatrix} - \begin{bmatrix} \mu_{1t} \\ \mu_{2t} \\ \vdots \\ \mu_{N,t} \end{bmatrix}$$

$$u_t = r_t - \mu_t.$$

in which r_t as a $(N \times 1)$ vector of returns, μ_t is the $(N \times 1)$ vector of conditional means of the returns series and H_t is the conditional covariance matrix of the returns.
In the case of the multivariate conditional normality, the log-likelihood function at time t takes the form

$$\begin{aligned} \log L_t &= \log f(r_{1t}, r_{2t}, \cdots, r_{N,t}) \\ &= -0.5N \log(2\pi) - 0.5 \log|H_t| - 0.5 u_t' H_t^{-1} u_t. \end{aligned} \tag{12.10}$$

Multivariate (Standardised) Student t Distribution

The standardised multivariate t distribution is given by

$$\begin{aligned} f(r_{1t}, r_{2t}, \cdots, r_{N,t}) = &\frac{\Gamma\left(\frac{\nu+N}{2}\right)}{(\pi(\nu-2))^{N/2}\,\Gamma\left(\frac{\nu}{2}\right)} \\ &\times |H_t|^{-1/2} \left(1 + \frac{u_t' H_t^{-1} u_t}{\nu - 2}\right)^{-\left(\frac{\nu+N}{2}\right)}. \end{aligned}$$

As for case of the normal distribution,

$$u_t = r_t - \mu_t$$

in which r_t as a $(N \times 1)$ vector of returns, μ_t as a $(N \times 1)$ vector of conditional means and H_t is the conditional covariance matrix. The parameter ν is the degrees of freedom parameter whereby small values represent fat-tails.

In this case, the log-likelihood function at time t is

$$\begin{aligned}\log L_t &= \log f(r_{1t}, r_{2t}, \cdots, r_{N,t}) \\ &= \log \Gamma\left(\frac{\nu+N}{2}\right) - \frac{N}{2}\log\left(\pi\left(\nu-2\right)\right) - \log \Gamma\left(\frac{\nu}{2}\right) \\ &- \frac{1}{2}\log|H_t| - \left(\frac{\nu+N}{2}\right)\left(1+\frac{u_t' H_t^{-1} u_t}{\nu-2}\right) \end{aligned} \tag{12.11}$$

The log-likelihood function in (12.10) or (12.11) is then maximised using standard optimisation algorithms. In the case of VECH and BEKK models in Section 12.3 the matrix H_t is constructed directly. For the correlation models in Section 12.4.1 H_t is constructed as either as $H_t = S_t R S_t$ (constant correlations) or $H_t = S_t R_t S_t$ (dynamic correlations). The advantage of the results in equations (12.8) and (12.9) for the DECO model are now clear because the matrix computations in the log-likelihood function requiring the determinant and the inversion of H_t are greatly simplified.

12.5.2 Estimating the BEKK Model

Consider estimating a bivariate diagonal BEKK model for the percentage excess returns to Microsoft, r_{1t}, and the S&P 500 index, r_{mt}, used in Section 12.1. The model is given by

$$\begin{aligned} r_{1t} &= \gamma_1 + u_{1t} \\ r_{mt} &= \gamma_2 + u_{2t} \end{aligned}$$

with

$$\begin{aligned} u_t &= \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix} \sim N\left(0, H_t\right) \\ H_t &= \begin{bmatrix} h_{11t} & h_{12t} \\ h_{12t} & h_{22t} \end{bmatrix} = CC' + Au_{t-1}u_{t-1}'A' + BH_{t-1}B' \\ C &= \begin{bmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & 0 \\ 0 & b_{22} \end{bmatrix}. \end{aligned}$$

The maximum likelihood parameter estimates of the bivariate BEKK model using a log-likelihood function based on the normal distribution are

$$\begin{aligned} r_{1t} &= 2.170 + \widehat{u}_{1t} \\ r_{mt} &= 0.454 + \widehat{u}_{2t}, \end{aligned}$$

with estimated conditional covariance matrix, $\widehat{H}_t$, given by

$$\begin{aligned}
\begin{bmatrix} \widehat{h}_{11t} & \widehat{h}_{12t} \\ \widehat{h}_{12t} & \widehat{h}_{22t} \end{bmatrix} &= \begin{bmatrix} 9.784 & 0.000 \\ 1.279 & 0.795 \end{bmatrix} \begin{bmatrix} 9.784 & 1.279 \\ 0.000 & 0.795 \end{bmatrix} \\
&+ \begin{bmatrix} 0.365 & 0.000 \\ 0.000 & 0.248 \end{bmatrix} \begin{bmatrix} \widehat{u}^2_{1\,t-1} & \widehat{u}_{1\,t-1}\widehat{u}_{2\,t-1} \\ \widehat{u}_{2\,t-1}\widehat{u}_{1\,t-1} & \widehat{u}^2_{2\,t-1} \end{bmatrix} \begin{bmatrix} 0.365 & 0.000 \\ 0.000 & 0.248 \end{bmatrix} \\
&+ \begin{bmatrix} 0.875 & 0.000 \\ 0.000 & 0.941 \end{bmatrix} \begin{bmatrix} \widehat{h}_{11\,t-1} & \widehat{h}_{12\,t-1} \\ \widehat{h}_{12\,t-1} & \widehat{h}_{22\,t-1} \end{bmatrix} \begin{bmatrix} 0.875 & 0.000 \\ 0.000 & 0.941 \end{bmatrix}
\end{aligned}$$

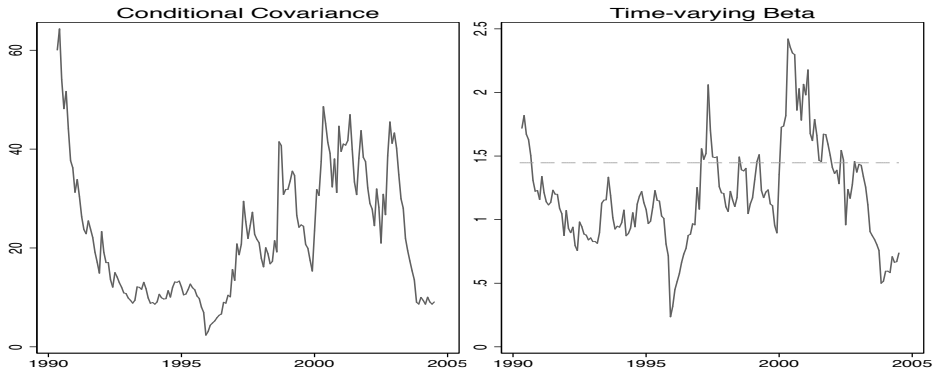


Figure 12.4: Estimates of the time-varying covariance and associated beta estimated using a diagonal BEKK model for the percentage excess returns to Microsoft and the S&P 500 index. The data are monthly for the period April 1990 to July 2004 ($T = 172$).

Figure 12.4 plots the conditional covariance between the excess returns on the market and Microsoft and the associated time-varying estimate of beta risk. These results should be compared with those in Figures 12.1 and 12.2 in Section 12.1, which were generated using the constant correlation assumption. The BEKK results produce a much steeper fall in the conditional covariance at the beginning of the sample than do the earlier constant correlation results. This major difference causes the time-variation in beta risk to be quite different in the first half of the sample period. However, during the second half of the period, the effect of the DotCom bubble is shown in the sharp increase in beta risk of Microsoft and this effect is common to both sets of results.

The early multivariate GARCH models are known to have their problems: the VECH model suffers because of the dimension of the parameter space; the BEKK model has parameters which are difficult to interpret. It is probably fair to say, therefore, that recent empirical work has concentrated more on the multivariate correlation class of models.

12.5.3 Estimating Multivariate Correlation Models

Maximum likelihood estimation of this class of model is illustrated using daily returns data on United States industry portfolios for the period 1 January 1990 to 31 December 2008. The industries considered are: Consumer Durables, NonDurables, Wholesale, Retail, and Services (Csnmr); Manufacturing, Energy, and Utilities (Manuf); Business Equipment, Telephone and Television Transmission (HiTec); and Healthcare, Medical Equipment, and Drugs (Hlth).

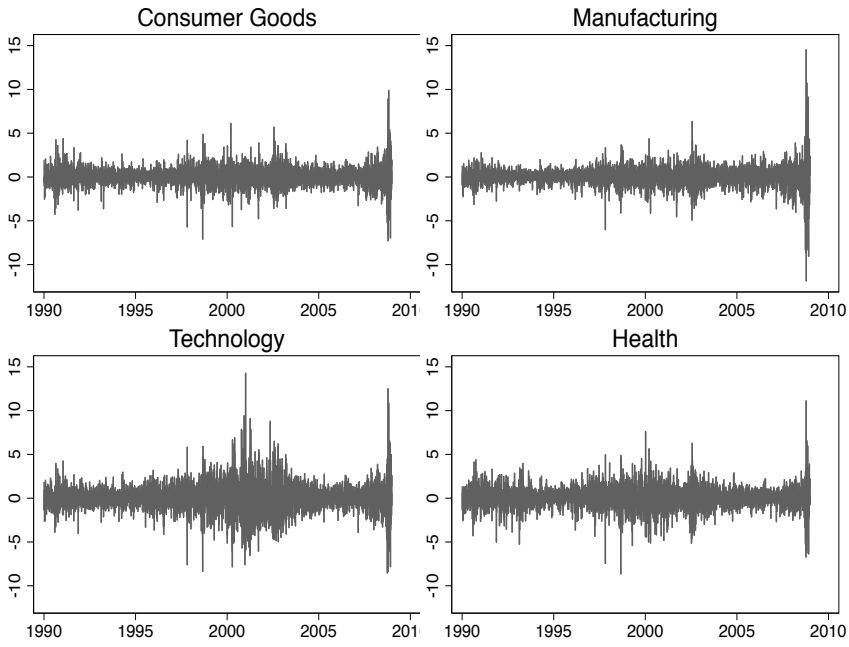


Figure 12.5: Daily returns to 4 industry portfolios for the period 1 January 1990 to 31 December 2008.

The returns to the industry portfolios are plotted in Figure 12.5. It is immediately apparent that all the stocks experience an increase in volatility at the end of the sample period as the global financial crisis begins. There is also evidence that the DotCom bubble had a much greater influence on the volatility of the technology industry than on the others, an observation that confirms the potential advantages of estimating multivariate GARCH models allow for volatility spillovers from one industry to another. To provide a comparison of the correlation models discussed in Section 12.4.1, the CCC and DCC models will be estimated using likelihood functions based on both the multivariate normal and multivariate standardised Student t distribution.

Although the estimation of the multivariate correlation models appears quite

complex, reasonable starting values for the optimisation can be obtained in a systematic approach as follows.

1. The starting values for the parameters in the mean equations can be obtained by simple ordinary least squares regression.

2. The starting values for the parameters in the variance equations can be found by fitting univariate GARCH models to each of the series.

3. The estimates of the conditional variances can be used to standardise the residuals and obtain starting values for correlations.

4. This leaves only the parameters α and β the govern the dynamics of the correlation to be provided. These parameters must satisfy the constraints $\alpha, \beta \geq 0$ and $0 \leq \alpha + \beta < 1$, so a fairly safe guess at starting values would be to choose values for these parameters that are similar to values obtained from univariate GARCH models with α taken to be of the order of the coefficient on the lagged squared residuals (news) and β being of the order of the coefficient on the lagged conditional variance. A more formal procedure would be to perform a crude two-dimensional grid search.

Once these starting parameters are provided, the log-likelihood function is optimised using one a standard iterative optimisation algorithm.
The parameter estimates for the various models are reported in Table 12.3. The sums of the parameters on the ARCH, α_{1i}, and GARCH, β_{1i}, terms in the conditional variance part of models are all close to unity and indeed in a couple of instances marginally exceeds unity. This indicates that the volatility of these portfolios is fairly persistent. The adjustment parameters α and β on the DCC model are statistically significant and this suggests that there is statistical evidence to support the claim that correlations are time varying, despite the fact note previously that it is difficult to test this hypothesis formally. There is some scope however for exploring the hypothesis of constant correlations using a residual-based method suggested by Bollerslev (1990). If the multivariate GARCH model in equation (12.1) is correctly specified then conditional on information set at $t-1$, it follows that $\mathrm{E}(u_{it}u_{jt}) = h_{ijt}$. Bollerslev (1990) suggests pairwise residual based regression diagnostic tests for the adequacy of a constant correlation specification. The procedure is as follows:

$$i = j: \quad \text{Regress } \frac{\widehat{u}_{it}^2}{\widehat{h}_{iit}} \text{ on } \quad \frac{1}{\widehat{h}_{iit}}, \frac{\widehat{u}_{it-1}^2}{\widehat{h}_{iit}}, \cdots, \frac{\widehat{u}_{it-q}^2}{\widehat{h}_{iit}}$$

$$i \neq j: \quad \text{Regress } \frac{\widehat{u}_{it}\widehat{u}_{jt}}{\widehat{h}_{ijt}} \text{ on } \quad \frac{1}{\widehat{h}_{ijt}}, \frac{\widehat{u}_{it-1}^2}{\widehat{h}_{iit}}, \frac{\widehat{u}_{jt-1}^2}{\widehat{h}_{iit}}, \frac{\widehat{u}_{it-1}\widehat{u}_{jt-1}}{\widehat{h}_{iit}}, \cdots, \frac{\widehat{u}_{it-q}\widehat{u}_{jt-q}}{\widehat{h}_{ijt}},$$

and test the null hypothesis that all the regressors are zero using a conventional F test.

Table 12.3

Coefficient estimates for the CCC and DCCC models for likelihood functions based on the multivariate normal and the multivariate standardised student t distribution. The data are daily returns for 4 industry portfolios (Cnsmr=1, Manuf=2, Hitec=3, Hlth=4). The sample period is 1 January 1990 to 31 December 2008. The parameter ν represents the degrees of freedom parameter when estimation is based on the t distribution. Standard errors are in parentheses.

	Normal Distribution		Student t Distribution	
	ccc	dcc	ccc	dcc
μ_{01}	0.054	0.052	0.049	0.053
	(0.010)	(0.011)	(0.010)	(0.010)
α_{01}	0.005	0.007	0.003	0.005
	(0.001)	(0.001)	(0.001)	(0.001)
α_{11}	0.041	0.058	0.033	0.049
	(0.004)	(0.004)	(0.003)	(0.004)
β_{11}	0.955	0.936	0.966	0.946
	(0.004)	(0.004)	(0.004)	(0.004)
μ_{02}	0.063	0.056	0.062	0.059
	(0.010)	(0.010)	(0.009)	(0.009)
α_{02}	0.003	0.004	0.002	0.003
	(0.001)	(0.001)	(0.001)	(0.001)
α_{12}	0.039	0.053	0.033	0.046
	(0.004)	(0.004)	(0.003)	(0.004)
β_{12}	0.959	0.943	0.968	0.951
	(0.004)	(0.004)	(0.003)	(0.004)
μ_{03}	0.063	0.057	0.063	0.064
	(0.014)	(0.013)	(0.013)	(0.013)
α_{03}	0.003	0.006	0.002	0.005
	(0.001)	(0.001)	(0.001)	(0.001)
α_{13}	0.043	0.050	0.037	0.044
	(0.004)	(0.004)	(0.003)	(0.004)
β_{13}	0.958	0.947	0.966	0.954
	(0.004)	(0.004)	(0.003)	(0.004)
μ_{04}	0.060	0.057	0.052	0.055
	(0.013)	(0.013)	(0.012)	(0.012)
α_{04}	0.005	0.009	0.003	0.006
	(0.002)	(0.002)	(0.001)	(0.002)
α_{14}	0.039	0.051	0.031	0.041
	(0.004)	(0.004)	(0.004)	(0.004)
β_{14}	0.958	0.943	0.969	0.954
	(0.005)	(0.005)	(0.004)	(0.005)
α		0.035		0.034
		(0.002)		(0.002)
β		0.952		0.952
		(0.002)		(0.002)
ν			6.605	9.089
			(0.298)	(0.499)

These pairwise test statistics cannot provide evidence in favour of a DCC specification over a the CCC model, neither can they distinguish between misspecification due to the incorrect treatment of correlations as opposed to the use of an incorrect GARCH model for the conditional covariances. However, the tests do shed some light on the adequacy of the assumption of constant correlations and also their pairwise construction helps to gain an understanding as to where any problems with the constant correlation assumption are to be found. In the case of the industry portfolios, when $i = j$ the F statistics are 2.127, 3.324, 2.784 2.785 which all reject the null hypothesis (although for the consumer portfolio the rejection is only at the 10% level). These results are not as interesting as the $i \neq j$ cases and in particular the tests involving the portfolio of high technology stocks. The F tests for constant correlations between high technology and consumer goods, high technology and manufacturing and high technology and health, are 216.642, 7.486 and 94.915, respectively. These test values are orders of magnitude higher than any other recorded test statistic. This indicates very strongly that high technology stocks behaved differently to the others over the sample period in the context of the constant correlation assumption. Of course this behaviour is clearly seen in Figure 12.5 where the volatility of high technology stocks during the DotCom crisis is clearly different to those of the other industries. Once again, these results should be regarded merely as preliminary evidence and certainly does not provide an argument in favour of using the DCCC model. Indeed testing either the assumption of dynamic conditional correlation model or event the adequacy of the constant correlation assumption is a matter of ongoing research (Harvey and Thiele, 2015; Silvennoinen and Teräsvirta, 2015).

The symmetric quasi-correlation matrix for the DCC model in equation (12.7), for estimation based on the normal distribution is given by

$$\begin{bmatrix} 1 & & & \\ \underset{(0.0169)}{0.7948} & 1 & & \\ \underset{(0.0175)}{0.7841} & \underset{(0.0214)}{0.7319} & 1 & \\ \underset{(0.0224)}{0.7063} & \underset{(0.0271)}{0.6325} & \underset{(0.0271)}{0.6322} & 1 \end{bmatrix}.$$

The standard errors shown in parentheses are obtained using the delta method. It is fairly clear from these results that the quasi-correlations would not support the restriction that they are all equal and hence the DECO model is not indicated in this instance.

There is also an argument to be made in favour of the multivariate t distribution over the normal distribution as the basis for the construction of the log-likelihood function. The degrees of freedom parameter of the t distribution, ν, is estimated quite precisely in both the CCC and DCC models. Once again a formal test of this hypothesis must be carefully formulated as the null hypothesis is not $\widehat{\nu} = 0$ as in the traditional t test, neither is it $\widehat{\nu} = \infty$ which would be the case if a normal distribution were the appropriate choice. In

practice, the normal distribution and the t distribution become indistinguishable for $\nu > 30$ and it is quite clear from the point estimates $\widehat{\nu}$ and the estimated standard errors that the hypothesis $\nu = 30$ would be strongly rejected.

12.6 Capital Ratios and Financial Crises

The global financial crisis of 2007– 2009 heightened awareness of the importance of systemic risk to financial policymaking. Broadly speaking the systemic risk of a financial institution is the contribution the institution would make to the deterioration of the entire financial system in a crisis period. In a recent paper, Brownlees and Engle (2012) introduce a method for determining the marginal expected shortfall (MES) of a financial institution, defined as the expected loss an investor in the financial firm would experience if the market declined substantially. In order to estimate MES, a multivariate volatility model must be specified, estimated and simulated in order to provide a view of the potential evolution of market and firm returns.
The actual capital ratio of a firm at time t is defined as

$$ACF_{it} = \frac{W_{it}}{W_{it} + D_{it}}, \tag{12.12}$$

where W_{it} is the firm's equity value, D_{it} is the book value of its debt and $W_{it} + D_{it}$ represents the total value of the firm's assets. Table 12.4 gives the actual capital ratios of 18 financial institutions in the United States at the end of 2014, with the institutions sorted in terms of their equity values, W_{it}. The important question is whether these captial ratios are sufficient for the firms to remain solvent during periods of financial distress in the future. To tackle this question, Brownlees and Engle (2012) derive a safe measure of capital using a multivariate GARCH model of asset returns.
To derive the safe capital ratio, define the working capital of a firm at time t to be the difference between its equity value W_t and a proportion k of its total assets $W_t + D_t$,

$$K_{it} = W_{it} - k(W_{it} + D_{it}). \tag{12.13}$$

During a financial crisis there is a large fall in the market return, r_{mt+h}, in excess of some threshold, c, which can result in a capital shortfall, CS, in the future given by

$$CS_{it+h} = -\,\mathrm{E}(\,K_{it+h}|\,r_{mt+h} < c). \tag{12.14}$$

Using (12.13) the capital shortfall is rewritten as

$$\begin{aligned} CS_{it+h} &= -\,\mathrm{E}(\,W_{it+h} - k(W_{it+h} + D_{t+h})|\,r_{mt+h} < c) \\ &= \mathrm{E}(\,D_{it+h}|\,r_{mt+h} < c) - (1-k)\,\mathrm{E}(\,W_{it+h}|\,r_{mt+h} < c). \end{aligned} \tag{12.15}$$

Assuming that debt cannot be renegotiated $\mathrm{E}(\,D_{it+h}|\,r_{mt+h} < c) = D_{it}$ and using the result that the future equity value of the firm is $W_{it+h} = W_{it}(1 -$

Table 12.4

Capital ratios.of selected financial institutions as at 31 December 2014. Actual capital ratios are defined as in equation (12.12).

Institution	Equity Value (E)	Actual Capital Ratios (ACR)
Wells Fargo	284385548	0.163666
JPMorgan Chase	233935868	0.092507
Bank of America	188139291	0.090744
Citigroup	163925596	0.089366
American Express	96266348	0.418410
Goldman Sachs Group	84421881	0.097466
Morgan Stanley	75947236	0.092937
Capital One Financial	45895407	0.151976
Bank of New York Mellon	45670055	0.116009
State Street	32773358	0.114548
SunTrust Banks	21849006	0.117233
Fifth Third Bank	16789143	0.123762
Northern Trust	15873037	0.134048
Regions Financial	14535576	0.124688
KeyCorp	12041918	0.131752
Huntington Bank	8568056	0.128700
Comerica	8416539	0.120482
Zions Bank	5785591	0.107296

r_{it+h}), where r_{it+h} is the future return on the firm, this expression is rewritten as

$$\begin{aligned} CS_{it+h} &= D_{it} - (1-k)W_{it}E(\,(1-r_{it+h})|\,r_{mt+h} < c) \\ &= D_{it} - (1-k)W_{it}(1-MES_{it}), \end{aligned} \tag{12.16}$$

where

$$MES_{it+h} = \mathrm{E}(\,r_{it+h}|\,r_{mt+h} < c), \tag{12.17}$$

represents the marginal expected shortfall.
The safe capital ratio is defined as that ratio for which it is not necessary to raise any additional external capital during a crisis. By setting $CS_{it+h} = 0$ in (12.15) and rearranging, the safe capital ratio for the firm at time t is a function of the marginal expected shortfall, MES, and the prudential parameter k given by

$$SCR_{it} = \frac{k}{1-(1-k)MES_{it+h}}. \tag{12.18}$$

Consider estimating the $h = 1$ day marginal expected shortfall in (12.17) for Morgan Stanley, r_{1t}, where the market returns, r_{mt}, given by the value

weighted returns of the S&P 500 index. The first step is to estimate the bivariate CCC model

$$\begin{aligned} r_{mt} &= \gamma_m + u_{mt} \\ r_{1t} &= \gamma_i + u_{1t}, \end{aligned} \tag{12.19}$$

where

$$u_t = \begin{bmatrix} u_{mt} \\ u_{1t} \end{bmatrix} \sim N(0, H_t)$$

and the conditional variance matrix is

$$H_t = S_t R S_t = \begin{bmatrix} \sqrt{h_{mt}} & 0 \\ 0 & \sqrt{h_{1t}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{m1} \\ \rho_{m1} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{h_{mt}} & 0 \\ 0 & \sqrt{h_{1t}} \end{bmatrix} \tag{12.20}$$

in which ρ_{m1} is the constant correlation between market excess returns and Morgan Stanley excess returns. The conditional variances, h_{mt} and h_{it}, are generated from univariate GARCH(1,1) models

$$\begin{aligned} h_{mt} &= \alpha_{0m} + \alpha_{1m} u^2_{m\,t-1} + \beta_{1m} h_{m\,t-1} \\ h_{1t} &= \alpha_{01} + \alpha_{11} u^2_{1\,t-1} + \beta_{11} h_{1\,t-1}. \end{aligned} \tag{12.21}$$

The parameter estimates of the bivariate CCC model in equations (12.19), (12.20) and (12.21) using daily returns data for the period 14 December 2001 to 31 December 2014 are given in Table 12.5. The GARCH parameter estimates for the two returns display typical empirical features of near integrated conditional volatility models, with the estimates on the lagged squared residuals around 0.1 and the estimates of the lagged conditional variances around 0.9, with the sum of the parameter estimates for each asset being close to unity. The estimated correlation coefficient is $\widehat{\rho}_{mi} = 0.7290$.

Table 12.5

Parameter estimates for a bivariate CCC model for returns to Morgan Stanley and returns to the value weighted S&P 500 index. The data are daily returns for the period 14 December 2001 to 31 December 2014.

	Market		Morgan Stanley	
	Coef.	Std. Err.	Coef.	Std. Err.
γ_i	0 .069366	0.0143008	0.1011702	0.030781
α_{0i}	0.0222742	0.0033543	0.0707545	0.0133622
α_{1i}	0.0775348	0.0073557	0.0825115	0.0083559
β_{1i}	0.899839	0.0090589	0.9058241	0.0091302
ρ_{1m}	0.7290	0.0082132		

The second step is to estimate the 1-day MES given the conditional variance estimates. The approach is to follow Brownlees and Engle (2012) by using

stochastic simulation methods. This involves simulating the estimated bivariate CCC model in (12.19) and (12.20) where u_t is drawn with replacement from the estimated residuals $\widehat{u}_t$. This is repeated S times which generates $s = 1, 2, \cdots, S$ simulated returns r^s_{mt} and r^s_{it}. The estimated MES is then computed by extracting the simulated values r^s_{it} for those cases where the simulated market returns r^s_{mt} are less than c. Formally, the $h = 1$ day MES is computed as

$$MES = \frac{\sum_{s=1}^{S} r^s_{it} I\left(r^s_{mt} < c\right)}{\sum_{s=1}^{S} I\left(r^s_{mt} < c\right)}, \tag{12.22}$$

where $I\left(r^s_{mt} < c\right)$ is an indicator function defined as

$$I\left(r^s_{mt} < c\right) = \begin{cases} 1 & : \quad r^s_{mt} < c \\ 0 & : \quad r^s_{mt} \geqslant c. \end{cases} \tag{12.23}$$

The estimated 1-day MES for Morgan Stanley with $c = -0.02$ and $S = 10000$ simulations, is $MES = 0.056$. The safe capital ratio with a prudential parameter of $k = 0.08$ is

$$SCR = \frac{k}{1 - (1 - k)MES} = \frac{0.02}{1 - (1 - 0.02)MES} = 0.013.$$

This value is much smaller than the actual capital ratio of $ACR = 0.092937$ for this company given in Table 12.4. This of course accords with intuition as it is to be expected that the banks would be well situated to deal with any MES over a 1-day horizon. Of course, the real question of the adequacy of the capital structure of Morgan Stanley can only be gauged under simulation if much longer time horizons are used.

12.7 Optimal Hedge Ratios

Another important empirical application of multivariate GARCH models has been in the area of the estimation of dynamic hedge ratios (Baillie and Myers, 1991). Consider an investor who sells futures contracts to hedge against movements in the spot price of an asset rate. The return on the hedged portfolio, r_{ht}, is

$$r_{ht} = r_{st} - \eta r_{ft}$$

where r_{st} is the return in the spot market, r_{ft} is the return on the futures contract and η is the number of contracts the hedger sells for each unit of spot commodity, known as the hedge ratio. The expected return on the portfolio is

$$\mu_h = \mathrm{E}[r_{s,t} - \eta r_{ft}] = \mathrm{E}[r_{s,t}] - \eta \mathrm{E}[r_{ft}] \quad \text{or} = \mu_s - \eta \mu_f,$$

and the variance of the portfolio is

$$\begin{aligned}\sigma_h^2 &= \mathrm{E}[(r_{ht} - \mu_h)^2] = \mathrm{E}[((r_{s,t} - \mu_s) - \eta(r_{ft} - \mu_f))^2] \\ &= \mathrm{E}[(r_{st} - \mu_s)^2] + \eta^2 \mathrm{E}[(r_{ft} - \mu_f)^2] \\ &- 2\eta \mathrm{E}[(r_{st} - \mu_s)(r_{ft} - \mu_f)] \\ &= \sigma_s^2 + \eta^2 \sigma_f^2 - 2\eta\sigma_{sf}. \end{aligned} \tag{12.24}$$

The optimal minimum variance portfolio is found by minimising σ_h^2 by choice of η. Differentiating expression (12.24) with respect to η gives

$$\frac{d\sigma_h^2}{d\eta} = 2\eta\sigma_f^2 - 2\sigma_{sf}.$$

Setting this derivative to zero and solving for η gives the optimal hedge ratio

$$\eta = \frac{\sigma_{sf}}{\sigma_f^2}, \tag{12.25}$$

which is the ratio of the covariance of the returns on the spot and futures contracts to the variance of the return on futures. The objective of variance minimisation assumes a high degree of risk aversion on the part of economic agents. However, Baillie and Myers (1989) show that if the expected returns to holding futures are zero, the minimum variance hedging rule is also the expected utility-maximising rule.

The expression forf the optimal hedge ratio in (12.25) assumes that the covariance and the variance are constant. This, in turn, results in a hedge ratio that is also constant, implying that the hedger never rebalances the portfolio in response to shocks in the spot and futures markets. To relax the restriction of a constant hedge ratio, the covariance of the returns on the spot and futures contracts and the variance of the return on futures contract must be specified as time-varying. The resultant dynamic hedge ratio is then

$$\eta_t = \frac{\sigma_{sft}}{\sigma_{ft}^2}$$

which is now the time-varying ratio of the conditional covariance of the returns on the spot and futures contracts to the conditional variance of the return on futures. To model the time-variation in the conditional covariance and variance, a bivariate GARCH model is required.

Consider the problem of hedging the returns to the 4 United States industry portfolios using the daily returns for the period is 1 January 1990 to 31 December 2008 using the futures contract on the S&P 500 index for the same period. The constant hedge ratio for each of the industry portfolios is found by estimating the regression

$$r_{jst} = \beta_{j0} + \beta_{j1} r_{ft} + u_{jt}, \qquad j = 1, 2, 3, 4,$$

with r_{jst} representing the returns to the relevant industry portfolio and r_{ft} represents the returns to the three month S&P 500 futures contract. The constant hedge ratios are estimated to be

$$\text{Cnsmr} = 0.773, \qquad \text{Manuf} = 0.787, \qquad \text{Hitec} = 1.124, \qquad \text{Hlth} = 0.768.$$

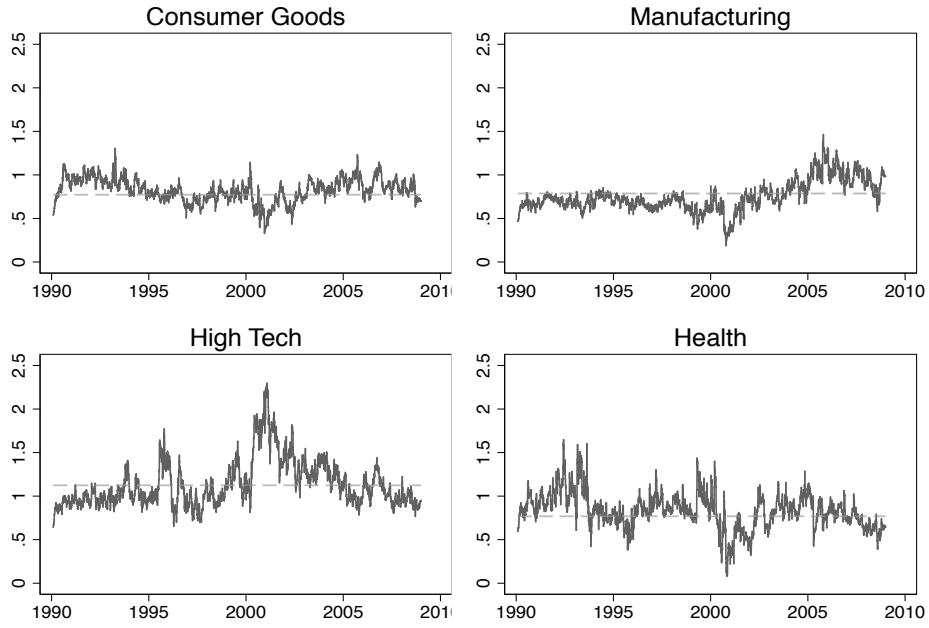


Figure 12.6: Dynamic hedge ratios (solid line) for 4 United States industry portfolios hedged using the 3-month S&P index futures contract. The relevant time-varying variances and covariances are computed using a DCC model using daily data for the period 1 January 1990 to 31 December 2008. The optimal constant hedge ratio is shown as dashed line.

The dynamic hedge ratios are computed using bivariate dynamic conditional correlation (DCC) models specified for each industry portfolio relative to the S&P500 futures contract. The model is estimated using the normal distribution. The parameter estimates for the bivariate are not reported but are not vastly different from the multivariate DCC models reported in Table 12.3. The dynamic hedge ratios computed in this manner for each industry portfolio are plotted in Figure 12.6.

For the consumer goods industry the constant hedge ratio looks like a reasonable strategy and the value of the dynamic ratio is seldom strays too far away from this constant value. The same conclusion cannot be made about the other three portfolios and particularly the manufacturing and high technology industries. The dynamic hedge ratio for manufacturing is below the

constant value for most of the early part of the sample and then switches to being above it after the DotCom bubble unwinds in the early 2000s. The effect of the DotCom bubble on high technology stocks is very clear and the advantage of using dynamic hedging is obvious. Finally, while the dynamic hedge ratio for the health portfolio fluctuates around the constant hedge ratio, the deviations during the early 1990s (above the constant ratio) and the DotCom bubble (below the line) suggest that dynamic hedging would provide a substantial reduction in risk exposure.

12.8 Exercises

1. **Bivariate Constant Correlation Models**

   capm.csv, capm.dta, capm.wf1

   The data are monthly observations for the period April 1990 to July 2004 ($T = 172$) on the share prices of five United States stocks and also the price of the commodity gold as well as the S&P 500 index.

   (a) Compute the excess returns to the market portfolio (S&P 500 index) and the excess returns to Microsoft. Estimate Microsoft's constant beta risk using the CAPM model.

   (b) Estimate a GARCH(1,1) model for the excess returns to the market and excess returns to Microsoft, respectively. Comment on your results.

   (c) Based on the assumption of a constant correlation between Microsoft and the market, compute an estimate of the conditional covariance between Microsoft and the market using the conditional variances obtained in (b). Hence provide an estimate of time-varying beta risk for Microsoft.

   (d) Is the estimate of the time-varying beta risk significantly affected by the introduction of a leverage effect in the univariate GARCH models?

   (e) Is the estimate of time-varying beta risk significantly affected by the use of a multivariate diagonal VECH or BEKK model?

   (f) Compute the monthly excess returns to Walmart and estimate a GARCH(1,1) model using these returns. Based on the assumption of constant correlation between Microsoft and Walmart compute the optimal time-varying portfolio weights for this two asset portfolio. Contrast your results with the optimal constant portfolio weights.

   (g) Now estimate a multivariate GARCH model (either a diagonal VECH or BEKK) model for Microsoft and Walmart and re-compute

the optimal weights. Are the results significantly different to those obtained in (d)?

2. **Time-varying Correlation in Hedge Funds**

hedgefunds.csv, hedgefunds.dta, hedgefunds.wf1

The data are daily returns to various hedge funds for period 1 April 2003 to 28 May 2010 ($T = 1869$) obtained from Hedge Fund Research, Inc. ("HFR").

(a) Estimate a bivariate DCC model for Merger hedge fund returns and returns to the S&P500 index. Compute and plot an estimate of the time varying correlation between the Merger fund returns and the market returns. Comment on your result.

(b) Repeat part (a) for the other six hedge funds. Discuss how successful the hedge funds were in minimising exposure to systematic risk from the market during the global financial crisis from mid 2007 to the end of 2009?

(c) Now estimate a DCC model which deals with all 7 hedge fund returns (Convertible, Distressed, Equity Event, Macro, Merger and Neutral). Comment on whether or not a DECO model would be appropriate for this system.

3. **Industry Portfolios**

hedgeratio.csv, hedgeratio.dta, hedgeratio.wf1

The data are daily returns on United States industry portfolios for the period 1 January 1990 to 31 December 2008. The industries considered are: Consumer Durables, NonDurables, Wholesale, Retail, and Services (Csnmr); Manufacturing, Energy, and Utilities (Manuf); Business Equipment, Telephone and Television Transmission (HiTec); and Healthcare, Medical Equipment, and Drugs (Hlth).

(a) Plot the returns to the 4 industry portfolios and comment on their time series properties.

(b) For a system comprising the 4 industry portfolio returns estimate the parameters of CCC, DCC and DECO specifications using log-likelihood functions based on the multivariate normal and the multivariate t distributions, respectively. Comment on the results.

(c) Estimate the linear regression

$$r_t = \alpha + \beta r_{mt} + u_t$$

where r_t is the return to the Consumer industry portfolio and r_{mt} is the return to the three month S&P futures index. What interpretation can be given to $\widehat{\beta}$ in this particular case.

(d) Estimate a bivariate DCC model for the returns to the Consumer industry portfolio and three month S&P futures index returns. Use the results of the DCC model to provide an estimate of a dynamic hedge ratio for the consumer portfolio. Interpret your results.

(e) Repeat parts (c) and (d) for the remaining three industry portfolios and comment on your results.